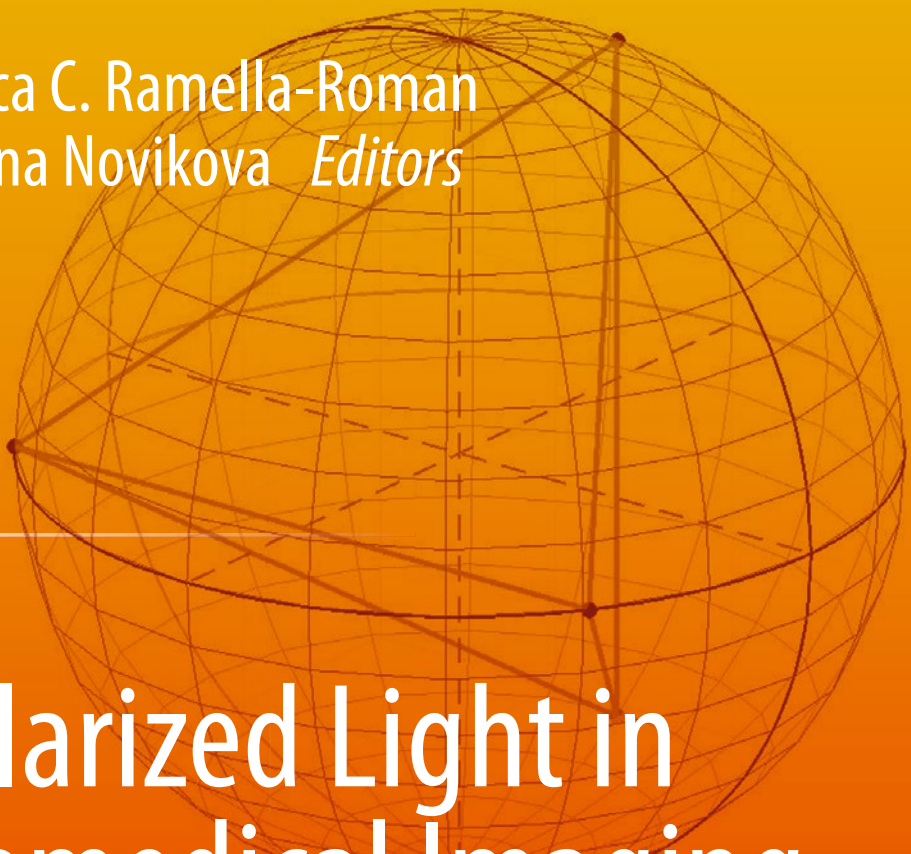
Jessica C. Ramella-Roman
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Polarized Light in Biomedical Imaging and Sensing

Clinical and Preclinical Applications

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Polarized Light



Tatiana Novikova and Jessica C. Ramella-Roman

1 Introduction

Human vision is not sensitive to the polarization of the light. We detect the intensity of the visible light and its wavelengths (as a color), but the vector nature of the light—polarization—escapes human eyes, contrary to some animals and insects [1].

When polarized probing light interacts with a sample, the polarization of reflected, transmitted, and scattered light will change, and any sample's polarimetric properties can be investigated through specialized instrumentation. The polarized light microscope, introduced in the early 1800 [2], is an example of polarized light application toward the investigation of material properties.

Recently, new focus has been given to polarized light imaging and sensing for biomedical applications, with studies that go beyond microscopy and include measurements in bulk tissue; this is the focus of our book. Although early work introduced the use of polarized light in imaging of cells, bacteria, and other small biological particles, [3–5] Anderson in 1991 [6] was the first to focus on bulk tissue reporting on the use of polarized sunglasses and cross-polarization imaging for dermatological applications, particularly for enhancement of vascular contrast in the study of rosacea and to minimize surface glare. Jacques et al. [7] continued this effort developing a clinic-ready tool and focusing on skin cancer imaging enhancing

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the borders of the lesion that were not visible to the naked eye, so did Morgan et al. [8, 9] investigating circular polarization rather than linear states.

Demos et al. [10, 11] investigated the spectral dependence of polarized light penetration in biological media; spectral polarization difference imaging was utilized to demonstrate subsurface imaging at different depths. Intravital polarized microscopy for measurement of perfusion was studied by Groner et al. [12–17].

The expansion of research in the use of polarized light for biomedical applications in the last 30 years has been tremendous and has been propelled by advances in both instrumentation and methodologies. Several articles have recently reviewed the state of the field [18–22]. Here we hope to expand on this work by covering in-depth some of the most interesting applications of polarized lights.

In this chapter, we briefly review the theoretical descriptions of the light polarization properties, namely, the Jones formalism for totally polarized light and the Stokes–Mueller formalism, which is more general and can adequately account for any polarization state.

1.1 Jones Formalism

The electromagnetic field propagating as a wave in different media is a spatiotemporal field which is vectorial by its nature, coupling the vector of electric field strength $\mathbf{E}(\mathbf{r}, t)$ and the vector of magnetic field strength $\mathbf{H}(\mathbf{r}, t)$. $\mathbf{r}$ is the vector of spatial position, and t is the time. Sometimes one can ignore the vector nature of the light and treat it within the frame of scalar theory, like it is done in the geometrical ray optics theory.

Nevertheless, the study of polarization of light requires the full vector description. If the temporal evolution of a vector of electromagnetic field at the given point of a space is *identical* for all the space points, we call this electromagnetic field polarized. Otherwise, we deal with a partially or even completely depolarized state of an electromagnetic field.

The light propagation in time and space can be fully described by Maxwell's equations [23]:

$$\nabla \cdot \mathbf{D}(\mathbf{r}, t) = \rho, \qquad \nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial \mathbf{B}(\mathbf{r}, t)}{\partial t} \tag{1}$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0, \qquad \nabla \times \mathbf{H}(\mathbf{r}, t) = \mathbf{J} + \frac{\partial \mathbf{D}(\mathbf{r}, t)}{\partial t} \tag{2}$$

$$\mathbf{D} = \varepsilon\, \mathbf{E}, \qquad \mathbf{B} = \mu\, \mathbf{H}, \qquad \mathbf{J} = \sigma\, \mathbf{E} \tag{3}$$

where t-time, **D**-dielectric displacement vector, ρ-charge density, **E**-vector of electric field strength, **B**-magnetic induction or magnetic flux density vector, **H**-vector of magnetic field strength, **J**-current density, ε-dielectric constant (or dielectric permittivity) of the medium, μ-magnetic permeability of the medium (in the most general case, ε and μ are the tensors), σ-electrical conductivity of the medium.

The values of ε and μ are constant in a homogeneous medium. Moreover, for a homogeneous linear optical material without any charge and current densities, $\rho = 0$; J $= 0$; $\sigma = 0$. Equations (1), (2), and (3) can be written as

$$\nabla \cdot \mathbf{D}\,(\mathbf{r}, t) = 0, \qquad \nabla \times \mathbf{E}\,(\mathbf{r}, t) = -\frac{\partial \mathbf{B}\,(\mathbf{r}, t)}{\partial t} \tag{4}$$

$$\nabla \cdot \mathbf{B}\,(\mathbf{r}, t) = 0, \qquad \nabla \times \mathbf{H}\,(\mathbf{r}, t) = \frac{\partial \mathbf{D}\,(\mathbf{r}, t)}{\partial t} \tag{5}$$

$$\mathbf{D} = \varepsilon\,\mathbf{E}, \qquad \mathbf{B} = \mu\,\mathbf{H} \tag{6}$$

Maxwell's equations can be further simplified and formulated for electric and magnetic fields separately by taking the curl ($\nabla\times$) of (4) and (5). The electric and magnetic fields then obey the well-known wave equations:

$$\nabla^2\mathbf{E}\,(\mathbf{r}, t) - \mu\varepsilon\frac{\partial^2\mathbf{E}\,(\mathbf{r}, t)}{\partial t^2} = 0 \tag{7}$$

$$\nabla^2\mathbf{H}\,(\mathbf{r}, t) - \mu\varepsilon\frac{\partial^2\mathbf{H}\,(\mathbf{r}, t)}{\partial t^2} = 0 \tag{8}$$

Solving the problem for a propagation of the electric wave **E**, one can further find the solution for the magnetic wave **H,** and finally obtain the complete solution of Maxwell's equations.

It was shown that any electromagnetic field may be decomposed into an infinite number of monochromatic waves with angular frequency ω_i, any of which in turn can be decomposed into an infinite number of plane waves with the wavevector $\mathbf{k_j}$ [24]. Fourier integral theory describes this by the following equation:

$$\mathbf{D}\,(\mathbf{r}, t) = \int_{\omega} d\omega \int_{\mathbf{k}} \mathbf{D}\,(\mathbf{k}, \omega)\exp\left[j\,(\omega\,t - \mathbf{kr})\right]\,d\mathbf{k} \tag{9}$$

Hence, for the studies of light polarization properties, we can consider an elementary component of this decomposition, namely, a plane monochromatic wave characterized by its angular frequency ω and wavevector **k.** If the electromagnetic fields associated with the elementary components have the same vectorial behavior, the problem can be easily solved.

When a plane homogeneous optical wave travels through an infinite isotropic medium, the electric field **E** is confined to oscillate in the plane of the wave, which is orthogonal to the wave vector **k.** Without the loss of generality, we can define the Oz axis as the direction of wave propagation ($\mathbf{k} = k\mathbf{z}$, **z**-unit vectors along Oz axis). The oscillation of electric field **E** in the plane orthogonal to Oz axis and intersecting Oz axis at point (0, 0, z) is the superposition of two independent simple harmonic oscillations E_x' and E_y' [25].

$$\mathbf{E}(z,t) = \begin{bmatrix} \tilde{E}_x \cos(\omega t - kz + \delta_x) \\ \tilde{E}_y \cos(\omega t - kz + \delta_y) \end{bmatrix} = \begin{bmatrix} E_x' \\ E_y' \end{bmatrix} \tag{10}$$

where $\tilde{E}_i$ and δ_i are the amplitude and the phase of the oscillation along the i-axis, respectively, ω–angular frequency of temporal harmonic oscillations of the components of monochromatic field. Using a complex number representation, a monochromatic wave propagating through an infinite isotropic medium can be written in a form:

$$\mathbf{E}(z,t) = \mathrm{Re}\{\mathbf{E}(z)\exp(j\omega t)\} \tag{11}$$

where

$$\mathbf{E}(z) = \exp(-jkz)\begin{bmatrix} \tilde{E}_x \exp(j\delta_x) \\ \tilde{E}_y \exp(j\delta_y) \end{bmatrix} \tag{12}$$

To exclude the spatial dependence on z, we define the vector $\mathbf{E}(0)$ as

$$\mathbf{E}(0) = \begin{bmatrix} \tilde{E}_x \exp(j\delta_x) \\ \tilde{E}_y \exp(j\delta_y) \end{bmatrix} = \begin{bmatrix} E_x \\ E_y \end{bmatrix} \tag{13}$$

It follows from (11), (12), and (13) that if we know the oscillations of electric field at time $t = 0$ and plane $z = 0$, we can reconstruct the electromagnetic field at any time and any point of the space just by multiplying $\mathbf{E}(0)$ by the factor $\exp(-j(kz - \omega t))$ and taking the real part of the resulting expression.

$$\mathbf{E}(z,t) = \mathrm{Re}\{\mathbf{E}(0)\exp(-j(kz - \omega t))\} \tag{14}$$

The vector $\mathbf{E}(0)$ (see Eq. 13) is called the Jones vector. It provides a compact description of a homogeneous monochromatic harmonic wave. It is worth mentioning that the elements E_x and E_y of Jones vector are complex numbers; hence, Jones vector is not a vector in a real space.

At wave plane $z = 0$, the vector of electric field (see Eq. 10) described the locus of points in Oxy plane, and the curve generated by those points represents the Lissajous ellipse—superposition of two harmonic oscillations along Ox and Oy axes with the same frequency and different amplitudes and phase shift. It follows from (10) that [26].

$$\frac{E_x{}'}{\tilde{E}_x} = \cos(\omega t)\cos(\delta_x) - \sin(\omega t)\sin(\delta_x) \tag{15a}$$

$$\frac{E_y{}'}{\tilde{E}_y} = \cos(\omega t)\cos(\delta_y) - \sin(\omega t)\sin(\delta_y) \tag{15b}$$

Hence

$$\frac{E_x{}'}{\tilde{E}_x}\sin(\delta_y) - \frac{E_y{}'}{\tilde{E}_y}\sin(\delta_x) = \cos(\omega t)\sin(\delta_x - \delta_y) \tag{16a}$$

$$\frac{E_x{}'}{\tilde{E}_x}\cos(\delta_y) - \frac{E_y{}'}{\tilde{E}_y}\cos(\delta_x) = \sin(\omega t)\sin(\delta_x - \delta_y) \tag{16b}$$

Taking square of (16a) and (16b) and adding them gives

$$\frac{E_x{}'^2}{\tilde{E}_x^2} + \frac{E_y{}'^2}{\tilde{E}_y^2} - 2\frac{E_x{}'}{\tilde{E}_x}\frac{E_y{}'}{\tilde{E}_y}\cos(\delta_x - \delta_y) = \sin^2(\delta_x - \delta_y) \tag{17}$$

This is an equation of ellipse, which is called the *polarization ellipse*. The presence of the "cross term" in (17) indicates that, in general, the polarization ellipse is rotated by the angle θ (the azimuth of its major axis). Another important parameter ε is the ellipticity angle (see Fig. 1).

It can be shown that the parameters $\tilde{E}_x$, $\tilde{E}_y$, and $\delta = \delta_x - \delta_y$ of polarization ellipse from (17) are related to an azimuth angle θ and ellipticity angle ε as

$$\tan 2\theta = 2\frac{\tilde{E}_x\tilde{E}_y}{\tilde{E}_x^2 - \tilde{E}_y^2}\cos\delta, \tag{18a}$$

$$\tan\varepsilon = \pm\frac{b}{a}, \quad -\frac{\pi}{4} \le \varepsilon \le \frac{\pi}{4} \tag{18b}$$

$$a^2 = \tilde{E}_x^2\cos^2\theta + \tilde{E}_y^2\sin^2\theta + 2\tilde{E}_x\tilde{E}_y\cos\theta\sin\theta\cos\delta \tag{18c}$$

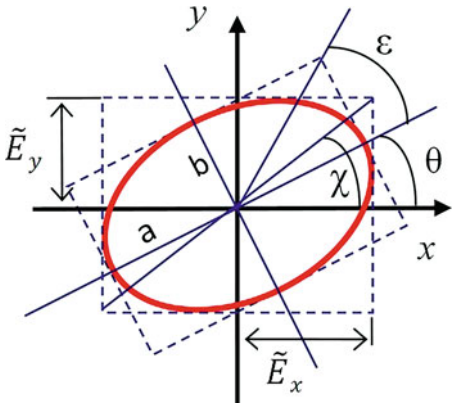


Fig. 1 Example of electric field vector trajectory in the plane perpendicular to the propagation direction for the fully elliptically polarized light wave

$$b^2 = \tilde{E}_x^2 \sin^2\theta + \tilde{E}_y^2 \cos^2\theta - 2\tilde{E}_x \tilde{E}_y \cos\theta \sin\theta \cos\delta \tag{18d}$$

$$a^2 + b^2 = \tilde{E}_x^2 + \tilde{E}_y^2, \quad \pm ab = \tilde{E}_x \tilde{E}_y \sin\delta, \tag{18e}$$

For certain values of $\tilde{E}_x$, $\tilde{E}_y$, and δ, the polarization ellipse takes special forms:

- $\tilde{E}_y = 0$. In this case, the field oscillates only along x direction. The light is called linear *horizontally polarized* light. When $\tilde{E}_x = 0$ we speak of linear *vertically polarized* light, because the field oscillates along the y direction.
- $\delta = 0$ or π. It follows from (17) that $E_y = \pm\left(\tilde{E}_y/\tilde{E}_x\right) E_x$, which is the equation of the straight line. Hence, we have *linearly* polarized light. When the amplitudes $\tilde{E}_y = \tilde{E}_x$, we observe that $E_y = \pm E_x$. For $\delta = 0$, the light is said to be *linear $-45°$ polarized*, while for $\delta = \pi$ the slope is positive, and we speak of *linear $+45°$ polarized* light.
- $\delta = \pi/2$ or $3\pi/2$, and $\tilde{E}_y = \tilde{E}_x = E_0$. The polarimetric ellipse reduces to a circle. It is said that the light is *right ($\delta = \pi/2$) or left ($\delta = 3\pi/2$) circularly polarized.*

Partially polarized states (see Fig. 2) correspond to more disordered motions of the electric field, which can be properly described only statistically, from cross-correlation functions, as discussed below.

Jones vector contains the different phase factors for each of vector components (see Eq. 13). It may be important in some cases when the polarized beam under study interferes with another beam. However, if we deal with a single-beam polarimetry, only the relative shift of the phase $\Delta\delta = \delta_x - \delta_y$ is relevant. So, without the loss of generality, we can, for example, put $\delta_y = 0$.

In the absence of depolarization, the interaction with a sample transforms the Jones vector of the incident beam into another Jones vector, by a linear transformation:

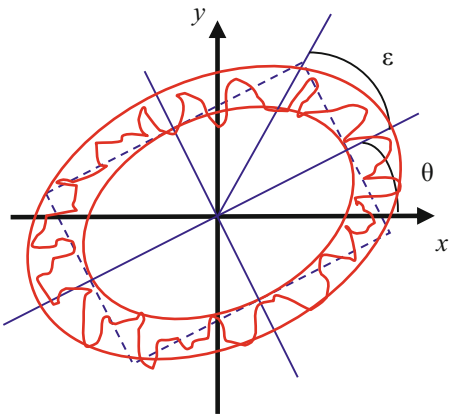


Fig. 2 Analogous to Fig. 1 for partially polarized light wave

$$\begin{pmatrix} E_x^{out} \\ E_y^{out} \end{pmatrix} = \begin{pmatrix} t_{xx} & t_{xy} \\ t_{yx} & t_{yy} \end{pmatrix} \begin{pmatrix} E_x^{in} \\ E_y^{in} \end{pmatrix} \tag{19}$$

where t_{ij} are the elements of complex 2×2 Jones matrix.

1.2 Stokes Formalism

The representation of light by the polarization ellipse allows us to describe various polarization states by a single equation. However, the tip of the electric field vector of polarized light traces out the locus of the points defined by (17) in a time interval of order 10^{-15} seconds, which makes it impossible to observe the polarization ellipse experimentally in real-time.

Another limitation is that for partially depolarized states, the disordered motion of the electric field in the *xy* plane can be properly described only by its *statistical properties*. One could think that a full probability distribution of **E** would be needed to characterize such states. However, normally, the amplitude of the optical field cannot be observed. In fact, what one can observe and measure is an intensity, which is obtained by time-averaging the square of the wave amplitude.

Hence, to predict the results of any classical measurements of *intensity,* one needs to deal with the *second moments* (quadratic quantities) of the electric field distributions. This assumption holds for most of the optical instruments considered in this book but would not be true for nonlinear optics experiments, such as second harmonic generation (SHG), for example.

The intensities related to the components of the electric field along the *x* and *y* axes are defined as

$$I_x = \left\langle E_x^2(t) \right\rangle, \quad I_y = \left\langle E_y^2(t) \right\rangle \tag{20}$$

Assuming that the waves are quasi-monochromatic (see Eq. 10), let us discuss the averaging process for the partially polarized states (see Eq. 20).

The wave amplitudes $\tilde{E}_x$, $\tilde{E}_y$, and their phases δ_x, δ_y are not constant, but vary *much slower* than the optical angular frequency ω.

At each spatial location, the average of a factor $\cos^2(\omega\, t - kt + \delta_i)$ is taken, which oscillates at frequency 2ω. At this scale of time (*very short*), the amplitude $\tilde{E}_i$ $(i = x, y)$ can be considered as constant at chosen point. As a result of averaging of a $\cos^2(\omega\, t - kt + \delta_i)$ factor, the "instantaneous" values of $\tilde{E}_i^2$ have to be divided by 2. Then an average of amplitudes $\tilde{E}_i^2$ over much longer period is taken.

$$I_i = \left\langle E_i^2(t) \right\rangle = \frac{1}{2} \left\langle \tilde{E}_i^2 \right\rangle \tag{21}$$

Other types of averaging can be performed as well, depending on the various experimental conditions:

- If spatial coherence domains at the detector location are smaller than the size of a detector (different points of detector receive different polarization states), a spatial averaging may be needed.
- If polarization varies within the spectral range, then a spectral averaging may be performed.

If we consider a coherent superposition of two waves, characterized by their Jones vectors $\mathbf{J}_1$ and $\mathbf{J}_2$ defined in (13), the Jones vector of the resulting wave is defined as $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$. The corresponding intensity I is given by

$$I = \frac{1}{2} \mathrm{Re}\left(\mathbf{J}^* \mathbf{J}\right) = \frac{1}{2} \mathrm{Re}\left[(\mathbf{J}_1 + \mathbf{J_2})^* \cdot (\mathbf{J}_1 + \mathbf{J_2})\right] = \frac{1}{2} (I_1 + I_2) + \mathrm{Re}\left(\mathbf{J}_1^* \mathbf{J_2}\right). \tag{22}$$

The last term of (22) describes the interference of the two waves. Two polarization states with Jones vectors $\mathbf{J}_1$ and $\mathbf{J}_2$ are said to be *orthogonal* if this interference term vanishes, independently of the relative phase shift of the two waves.

If this phase shift is changed by δ, the intensities I_1 and I_2 remain constant, while the scalar product $\mathbf{J}_1{}^* \mathbf{J}_2$ is multiplied by the factor $\exp(j\delta)$. Consequently, two polarization states with Jones vectors $\mathbf{J}_1$ and $\mathbf{J}_2$ are *orthogonal* if and only if the scalar product $\mathbf{J}_1{}^* \mathbf{J}_2$ is equal to zero.

The correlation matrix of two components of the electric field written in a complex form $E_i^c = \tilde{E}_i \exp\left(j\left(\omega\, t - kz + \delta_i\right)\right)$ $(i = x, y)$ can be expressed as

$$\mathbf{C} = \begin{bmatrix} \left\langle E_x^c E_x^{c*} \right\rangle & \left\langle E_x^c E_y^{c*} \right\rangle \\ \left\langle E_y^c E_x^{c*} \right\rangle & \left\langle E_y^c E_y^{c*} \right\rangle \end{bmatrix} = \begin{bmatrix} \left\langle \tilde{E}_x^2 \right\rangle & \left\langle \tilde{E}_x \tilde{E}_y \exp\left(-j\delta\right) \right\rangle \\ \left\langle \tilde{E}_x \tilde{E}_y \exp\left(j\delta\right) \right\rangle & \left\langle \tilde{E}_y^2 \right\rangle \end{bmatrix} \tag{23}$$

because the time average of the terms oscillating at optical frequencies is zero. Now we can define four real parameters using the correlation matrix $\mathbf{C}$:

$$S_0 = \frac{1}{2}\left(C_{xx} + C_{yy}\right) = \frac{1}{2}\left\langle \tilde{E}_x^2 + \tilde{E}_y^2 \right\rangle \tag{24a}$$

$$S_1 = \frac{1}{2}\left(C_{xx} - C_{yy}\right) = \frac{1}{2}\left\langle \tilde{E}_x^2 - \tilde{E}_y^2 \right\rangle \tag{24b}$$

$$S_2 = \frac{1}{2}\left(C_{xy} + C_{yx}\right) = \frac{1}{2}\left\langle 2\tilde{E}_x \tilde{E}_y \cos\left(\delta\right) \right\rangle \tag{24c}$$

$$S_3 = \frac{j}{2}\left(C_{xy} - C_{yx}\right) = \frac{1}{2}\left\langle 2\tilde{E}_x \tilde{E}_y \sin\left(\delta\right) \right\rangle \tag{24d}$$

The parameter S_0 is the total intensity, S_1 is the difference of intensities I_x and I_y measured after an ideal polarizer aligned along either x or y axes, respectively. The interpretation of S_2 involves the difference of intensities which would be measured through an ideal linear polarizer oriented along either $+45°$ or $-45°$, respectively, in the plane perpendicular to the direction of propagation. The physical meaning of the parameter S_3 is the difference of I_L and $\mathrm{I_R}$, which are the intensities transmitted by a left or right circular polarizer, respectively [24].

Finally, for the orthogonal axes (x,y), any partially polarized field in linear optics is completely characterized by its Stokes vector $\mathbf{S}$ defined as

$$\mathbf{S} = \begin{pmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{pmatrix} = \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix} = \begin{pmatrix} I_x + I_y \\ I_x - I_y \\ I_{45°} - I_{-45°} \\ I_L - I_R \end{pmatrix} \tag{25}$$

Contrary to the Jones vector formalism involving the electric field amplitudes and relative phase shift, the Stokes formalism deals with directly measurable intensities.

The Stokes vector components for *fully polarized* states (described by Jones vectors, see Eq. 13) are simply given by

$$\mathbf{S} = \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix} = \begin{pmatrix} I_x + I_y \\ I_x - I_y \\ I_{45°} - I_{-45°} \\ I_L - I_R \end{pmatrix} = \begin{bmatrix} \tilde{E}_x^2 + \tilde{E}_y^2 \\ \tilde{E}_x^2 - \tilde{E}_y^2 \\ 2\tilde{E}_x \tilde{E}_y \cos\delta \\ 2\tilde{E}_x \tilde{E}_y \sin\delta \end{bmatrix} \tag{26}$$

without any need of averaging. In the most general case of partially polarized light, the averaging of Eq. 24 (right-hand side) takes into account all possible ways of averaging (e.g., *spatial, spectral, or temporal*), depending on the sample and the conditions of the measurements. Consequently, the partially polarized states can

be considered as *incoherent superpositions of fully polarized states with different polarizations*.

From (24) and Cauchy–Schwarz inequality ($\langle ab\rangle^2 \leq \langle a\rangle^2\langle b\rangle^2$) for any random variables a and b, we can deduce that

$$\left\langle \tilde{E}_x \tilde{E}_y \cos\delta \right\rangle^2 + \left\langle \tilde{E}_x \tilde{E}_y \sin\delta \right\rangle^2 \leq \left\langle \tilde{E}_x^2 \tilde{E}_y^2 \right\rangle \left\langle \cos^2\delta + \sin^2\delta \right\rangle = \left\langle \tilde{E}_x^2 \tilde{E}_y^2 \right\rangle \tag{27}$$

$$S_1^2 + S_2^2 + S_3^2 \leq \left\langle \left(\tilde{E}_x^2 - \tilde{E}_y^2 \right)^2 \right\rangle + 4\left\langle \tilde{E}_x^2 \tilde{E}_y^2 \right\rangle = \left\langle \left(\tilde{E}_x^2 + \tilde{E}_y^2 \right)^2 \right\rangle = S_0^2 \tag{28}$$

When light is totally polarized, the values of amplitudes $\tilde{E}_x$ and $\tilde{E}_x$ are constant and an inequality (28) becomes equality. Within the Stokes formalism, the *degree of polarization* ρ related to a given Stokes vector $\mathbf{S}$ is defined as

$$\rho = \frac{\sqrt{Q^2 + U^2 + V^2}}{I} \quad (0 \leq \rho \leq 1) \tag{29}$$

and this quantity varies between 0 for totally depolarized (fully disordered) states and 1 for totally polarized states. We can also introduce the degree of linear (ρ_l) and circular (ρ_c) polarizations:

$$\rho_l = \frac{\sqrt{Q^2 + U^2}}{I}, \quad \rho_c = \frac{\sqrt{V^2}}{I} \tag{30}$$

It is worth mentioning that an arbitrary four-dimensional real vector, for which $\rho > 1$, is not a physical Stokes vector and cannot represent any polarization state. Moreover, Stokes vectors do not form vectorial space, contrary to the Jones vectors. For example, one can notice that the Stokes vector of linear polarized light at $+45°$ is not a sum of Stokes vectors of light polarized linearly at $0°$ and $90°$.

1.3 Poincaré Sphere Representation

We will first consider the Stokes vector of totally elliptically polarized light, normalized by the value of total intensity S_0. Using the conventions about the angles θ, ε, and χ (see Fig. 1), one can write $\tan\chi = \tilde{E}_y/\tilde{E}_x$, $\tan\varepsilon = b/a$, and obtain the following relations:

$$\cos 2\chi = \cos^2\chi - \sin^2\chi = \frac{\tilde{E}_x^2 - \tilde{E}_y^2}{\tilde{E}_x^2 + \tilde{E}_y^2} \tag{31a}$$

$$\sin 2\chi = 2\cos\chi \sin\chi = \frac{2\tilde{E}_x \tilde{E}_y}{\tilde{E}_x^2 + \tilde{E}_y^2} \tag{31b}$$

$$\cos 2\varepsilon = \cos^2\varepsilon - \sin^2\varepsilon = \frac{a^2 - b^2}{a^2 + b^2} \tag{31c}$$

$$\sin 2\varepsilon = 2\cos\varepsilon \sin\varepsilon = \frac{2ab}{a^2 + b^2} \tag{31d}$$

$$\tan 2\chi = \frac{2\tilde{E}_x \tilde{E}_y}{\tilde{E}_x^2 - \tilde{E}_y^2} \tag{31e}$$

$$\tan 2\varepsilon = \frac{2ab}{a^2 + b^2} \tag{31f}$$

It follows from (18a) that

$$\tan 2\theta = 2\frac{\tilde{E}_x \tilde{E}_y}{\tilde{E}_x^2 - \tilde{E}_y^2}\cos\delta = \frac{2\tan\chi}{1 - \tan^2\chi}\cos\delta, \tag{32}$$

Using the Eqs. (31) and (18), it may be shown that

$$\begin{cases} \cos 2\varepsilon \cos 2\theta = \cos 2\chi \\ \cos 2\varepsilon \sin 2\theta = \sin 2\chi \cos\delta \\ \sin 2\varepsilon = \pm \sin 2\chi \ \ \sin\delta \\ \tan 2\varepsilon = \pm \sin 2\theta \ \ \tan\delta \end{cases} \tag{33}$$

Useful representation of Stokes vector can be made by expressing the amplitudes $\tilde{E}_x$ and $\tilde{E}_y$ in (26) in terms of angle χ (see Fig. 1):

$$S_0 = \tilde{E}_x^2 + \tilde{E}_y^2 = E_0^2, \ \ \tilde{E}_x = E_0\cos\chi, \ \ \tilde{E}_y = E_0\sin\chi, \ \ 0 \le \chi \le \frac{\pi}{2}, \tag{34}$$

Substituting (34) into (26) leads to the following presentation of Stokes vector for elliptically polarized light:

$$\mathbf{S} = E_0^2 \begin{bmatrix} 1 \\ \cos 2\chi \\ \sin 2\chi \cos \delta \\ \pm \sin 2\chi \sin \delta \end{bmatrix} \tag{35}$$

Using the relations (33) we can re-write the expression (35) in the form of Stokes vector expressed in terms of total intensity S_0 and angles ε and θ:

$$\mathbf{S} = E_0^2 \begin{bmatrix} 1 \\ \cos 2\varepsilon \cos 2\theta \\ \cos 2\varepsilon \sin 2\theta \\ \sin 2\varepsilon \end{bmatrix} \tag{36}$$

It is common to write the Stokes vector in normalized form by setting $S_0 = E_0^2 = 1$. The interpretation of the relations (36) is quite straightforward, if we recall that spherical coordinates r, θ_s, and φ_s (θ_s – polar angle, φ_s – azimuthal angle) are related to Cartesian coordinates x, y, and z by

$$\begin{aligned} x &= r \sin\theta_s \cos\phi_s \\ y &= r \sin\theta_s \sin\phi_s \\ z &= r \cos\theta_s \end{aligned} \tag{37}$$

We see that (36) and (37) are identical if the radius $r = 1$ and the angles are related by

$$\theta_s = 90^\circ - 2\varepsilon, \qquad \varphi_s = 2\theta \tag{38}$$

Consequently, one can represent a totally polarized state by a point on the surface of a sphere with unit radius, which is called the *Poincaré* sphere: the position (x, y, z) of this point is determined by its longitudinal coordinate 2θ and latitudinal coordinate 2ε (see Fig. 3).

- The perfectly linear polarized states are represented as the points on the "equator." In particular, the linear polarization states are aligned along the x and y axes (identified as L_{0° and L_{90° in Fig. 3), the linear polarization states oriented at $\pm 45^\circ$ (identified as L_{+45° and L_{-45°) are the points of interception of ζ_2 axis with the "equator."
- The right and left circularly polarized states (marked as RC and LC) are the points along the ζ_3 axis at the north and south poles, respectively. In general, all

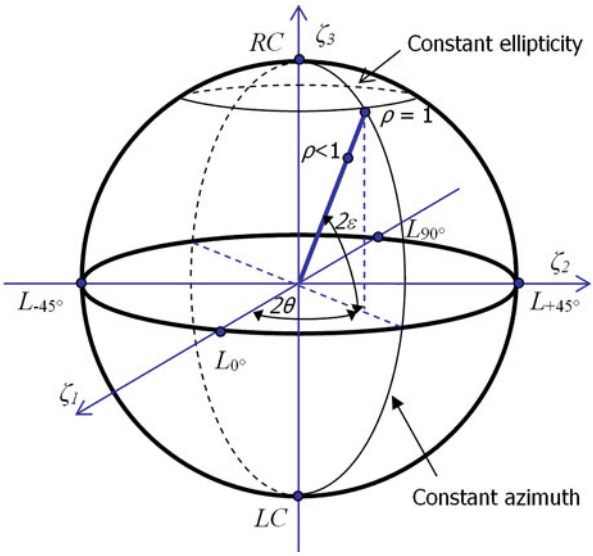


Fig. 3 The Poincaré sphere

Table 1 The Stokes vectors of the fundamental states of polarization

Linear at 0°	Linear at 90°	Linear at +45°	Linear at θ°	Left circular	Right circular	Elliptical
$I_0 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$	$I_0 \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$	$I_0 \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$	$I_0 \begin{bmatrix} 1 \\ \cos 2\theta \\ \sin 2\theta \\ 0 \end{bmatrix}$	$I_0 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$	$I_0 \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$	$I_0 \begin{bmatrix} 1 \\ \cos 2\varepsilon \cos 2\theta \\ \cos 2\varepsilon \sin 2\theta \\ \sin 2\varepsilon \end{bmatrix}$

right-handed elliptical states are situated at the northern hemisphere, while the opposite holds for the left-handed states.

- The states with constant ellipticity are situated on "parallels," the states with constant azimuth on "meridians."
- Orthogonal polarization states are the points of intercept of a diameter with the surface of the Poincaré sphere.

As it follows from Eq. (29)

$$\varsigma_1^2 + \varsigma_2^2 + \varsigma_3^2 = \rho^2, \tag{39}$$

where ρ is the degree of polarization. For partially polarized states $\rho < 1$, hence the corresponding points are situated *inside* the sphere at distance ρ from the center of the sphere.

The Stokes vectors of fundamental states of polarization are shown in Table 1.

1.4 Mueller Matrix

Upon interaction with any sample the Stokes vector, which is directly related to intensities, undergoes a linear transformation, described by a 4 × 4 real-valued matrix **M**, called the Mueller matrix of a sample. It was introduced by Hans Mueller in 1940s [24, 27]:

$$\begin{bmatrix} S_0^{\text{out}} \\ S_1^{\text{out}} \\ S_2^{\text{out}} \\ S_3^{\text{out}} \end{bmatrix} = \begin{bmatrix} M_{00} & M_{01} & M_{02} & M_{03} \\ M_{10} & M_{11} & M_{12} & M_{13} \\ M_{20} & M_{21} & M_{22} & M_{23} \\ M_{30} & M_{31} & M_{32} & M_{33} \end{bmatrix} \cdot \begin{bmatrix} S_0^{\text{in}} \\ S_1^{\text{in}} \\ S_2^{\text{in}} \\ S_3^{\text{in}} \end{bmatrix} \tag{40a}$$

$$\mathbf{S}^{\text{out}} = \mathbf{M} \cdot \mathbf{S}^{\text{in}} \tag{40b}$$

The Mueller matrix depends on the sample properties only, while the output Stokes vector of light depends also on the Stokes vector of the incident light. The 16 coefficients of the matrix M are real and measurable.

It was previously shown that the Stokes vectors can describe *any* polarization state, hence, the Mueller matrix can fully describe the polarimetric properties of *any* sample, be it depolarizing or not. A complex optical system defined by n sequential elementary optical systems with corresponding Mueller matrices $\mathbf{M}_i$, $(i = 1, n)$ is characterized by the matrix product:

$$\mathbf{S}^{\text{out}} = \mathbf{M}_n \mathbf{M}_{n-1} \ldots \ldots \mathbf{M}_1 \mathbf{S}^{\text{in}} \tag{41}$$

The rotation of reference frame by an angle φ transforms the Stokes vector by the rotation matrix $\mathbf{R}_\mathbf{M}(\varphi)$:

$$\mathbf{S}' = \begin{bmatrix} S_0' \\ S_1' \\ S_2' \\ S_3' \end{bmatrix} = \mathbf{R}_\mathbf{M}(\phi) \cdot \mathbf{S} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\phi & -\sin 2\phi & 0 \\ 0 & \sin 2\phi & \cos 2\phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} \tag{42}$$

Hence, the Mueller matrix $\mathbf{M}'$ in the rotated frame can be expressed as

$$\mathbf{S}^{\text{out}\prime} = \mathbf{M}'\mathbf{S}^{\text{in}\prime} \tag{43a}$$

$$\mathbf{S}^{out\prime} = \mathbf{R}_\mathbf{M}(\phi)\left(\mathbf{S}^{\text{out}}\right) = \mathbf{R}_\mathbf{M}(\phi)\left(\mathbf{M} \cdot \mathbf{S}^{\text{in}}\right) = \mathbf{M}'\mathbf{S}^{\text{in}\prime} = \mathbf{M}'\left(\mathbf{R}_\mathbf{M}(\phi)\,\mathbf{S}^{\text{in}}\right) \tag{43b}$$

$$\mathbf{R}_\mathbf{M}(\phi)\,\mathbf{M} = \mathbf{M}'\,\mathbf{R}_\mathbf{M}(\phi), \quad \Rightarrow \quad \mathbf{M}' = \mathbf{R}_\mathbf{M}(\phi)\,\mathbf{M}[\mathbf{R}_\mathbf{M}(\phi)]^{-1} \tag{43c}$$

In contrast with Jones matrix, the Mueller matrix does not carry any information about the overall optical phase shift introduced by a sample. The coefficient M_{11} represents the overall transmissivity (or reflectivity) of a sample. Depending on this quantity is of interest or not the Mueller matrix may be considered in its original non-normalized or normalized forms, where the normalization implies the division of all Mueller matrix elements by M_{00}.

As it was mentioned above, the Jones vectors form the vectorial space, so in principle, any 2×2 complex matrix may be an acceptable Jones matrix. Contrary to that, any real-valued 4×4 matrix is not necessarily a physically realizable Mueller matrix. Any acceptable Stokes vector (i.e., with its degree of polarization between 0 and 1) must be transformed into another acceptable Stokes vector, i.e., a realizable Mueller matrix *never overpolarizes* the incoming light. This is an intrinsic necessary condition for the real-valued 4×4 matrix to be a physically realizable Mueller matrix.

However, this condition is not sufficient, and another criterion can be defined, from the so-called 4×4 Hermitian target *coherency matrix* $\mathbf{T}$ that is a tensor product (or Kronecker product) of Pauli matrices related to the Mueller matrix $\mathbf{M}$ of interest:

$$\mathbf{T} = M_{ij}\sigma_i \otimes \sigma_j, \tag{44}$$

where M_{ji} are the elements of Mueller matrix, σ_i are the Pauli matrices:

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 0 & -j \\ j & 0 \end{pmatrix} \tag{45}$$

$\mathbf{M}$ is physically acceptable if and only the eigenvalues of $\mathbf{T}$ are non-negative [28, 29].

A sample is *non-depolarizing* if and only if its Mueller matrix can be derived from a sample Jones matrix, and a coherency matrix $\mathbf{T}$ exhibits only one strictly positive eigenvalue (the others vanish). It is worthy of note that a depolarization depends both on a sample and the instrument used to characterize a sample. Another criterion for sample polarization properties, namely, the quadratic depolarization index P, was proposed in [30]

$$P = \sqrt{\frac{\sum_{i,j=0}^{3} {M_{ij}}^2 - {M_{00}}^2}{3{M_{00}}^2}} \tag{46}$$

The quadratic depolarization index P varies from 0, for a perfect depolarizer (only M_{00} element is different from zero) to 1, for non-depolarizing matrices, and criterion (46) is much easier to put into the practice.

It can be easily shown that The Mueller matrix may be expressed in terms of second moments of the probability distribution function of the elements of the Jones matrix $\mathbf{t}$

$$
\begin{aligned}
M_{00} &= \tfrac{1}{2}\left(|t_{xx}|^2 + |t_{yy}|^2 + |t_{xy}|^2 + |t_{yx}|^2\right) \\
M_{01} &= \tfrac{1}{2}\left(|t_{xx}|^2 - |t_{yy}|^2 - |t_{xy}|^2 + |t_{yx}|^2\right) \\
M_{10} &= \tfrac{1}{2}\left(|t_{xx}|^2 - |t_{yy}|^2 + |t_{xy}|^2 - |t_{yx}|^2\right) \\
M_{11} &= \tfrac{1}{2}\left(|t_{xx}|^2 + |t_{yy}|^2 - |t_{xy}|^2 - |t_{yx}|^2\right) \\
M_{02} &= \mathrm{Re}\left(t_{xx}^* t_{xy} + t_{yx}^* t_{yy}\right) \\
M_{03} &= -\mathrm{Im}\left(t_{xx}^* t_{xy} + t_{yx}^* t_{yy}\right) \\
M_{12} &= \mathrm{Re}\left(t_{xx}^* t_{xy} - t_{yx}^* t_{yy}\right) \\
M_{13} &= -\mathrm{Im}\left(t_{xx}^* t_{xy} - t_{yx}^* t_{yy}\right) \\
M_{20} &= \mathrm{Re}\left(t_{xx}^* t_{yx} + t_{xy}^* t_{yy}\right) \\
M_{21} &= \mathrm{Re}\left(t_{xx}^* t_{yx} - t_{xy}^* t_{yy}\right) \\
M_{22} &= \mathrm{Re}\left(t_{xx}^* t_{yy} + t_{xy}^* t_{yx}\right) \\
M_{23} &= -\mathrm{Im}\left(t_{xx}^* t_{yy} - t_{xy}^* t_{yx}\right) \\
M_{30} &= \mathrm{Im}\left(t_{xx}^* t_{yx} + t_{xy}^* t_{yy}\right) \\
M_{31} &= \mathrm{Im}\left(t_{xx}^* t_{yx} - t_{xy}^* t_{yy}\right) \\
M_{32} &= \mathrm{Im}\left(t_{xx}^* t_{yy} + t_{xy}^* t_{yx}\right) \\
M_{33} &= \mathrm{Re}\left(t_{xx}^* t_{yy} - t_{xy}^* t_{yx}\right)
\end{aligned}
\tag{47}
$$

2 Basic Polarimetric Properties of Any Sample

The polarimetric response of a given sample describes the changes of the polarization of incident light due to the interaction with a sample. Despite obvious complexity of this interaction process, this response can be expressed in terms of three basic polarimetric properties, namely, the sample *diattenuation*, *retardance*, and *depolarization*.

When samples possess all polarimetric properties simultaneously, these properties are not uniquely defined but rather depend on the procedure of matrix decomposition. However, two polarimetric properties, namely, *diattenuation*, which describes the variation of output intensity when changing the polarization of constant input intensity, and *polarizance*, which describes the polarization of totally unpolarized beam by the system are directly accessible from Mueller matrix elements. Any Mueller matrix M may be presented in the following form

$$\mathbf{M} = T_0 \begin{bmatrix} 1 & \mathbf{D}^{\mathrm{T}} \\ \mathbf{P} & \mathbf{m} \end{bmatrix} \tag{48}$$

where T_0 is the transmission for the incident non-polarized light, $\mathbf{D}$ is a diattenuation vector, $\mathbf{P}$ is a polarizance vector, and $\mathbf{m}$ is 3×3 real-valued matrix.

If we define the normalized Stokes vector of incident light as $\mathbf{S}^{\mathrm{in}} = [1, \mathbf{s}^{\mathrm{in}}]^{\mathrm{T}}$, it is clear that an emerging total intensity, given by the first component S_0^{out} of output Stokes vector $\mathbf{S}^{\mathrm{out}}$, varies within the interval

$$T_{\min} = T_0 (1 - D) \leq S_0^{\mathrm{out}} \leq T_0 (1 + D) = T_{\max} \tag{49}$$

and reaches the maximum and minimum transmission values $T_{\max}$ and $T_{\min}$ when vector $\mathbf{s}^{\mathrm{in}} = \pm \mathbf{D}/\|\mathbf{D}\|$. We can also determine the value of scalar diattenuation D as

$$D = \frac{T_{\max} - T_{\min}}{T_{\max} + T_{\min}}, \quad 0 \leq D \leq 1 \tag{50}$$

Scalar diattenuation characterizes the variation of transmission independently of the input polarization states.

For totally depolarized input Stokes vector $\mathbf{S}^{\mathrm{in}} = [1, 0]^{\mathrm{T}}$ the output Stokes vector is defined as $\mathbf{S}^{\mathrm{out}} = [1, \mathbf{P}]^{\mathrm{T}}$. It shows that polarizance quantifies the capability of the system to polarize an incident totally depolarized beam.

The optical element is called diattenuator or retarder, when it exhibits only diattenuation or retardance property, respectively. These elements are usually used as "building blocks" of the measurements system to characterize more complex samples. These homogeneous elements are characterized by pair of fully polarized *orthogonal eigenstates* (elliptical in the most general case), shown in Fig. 4. Each state of the pair can be obtained from the other state by simply exchanging the major and minor axes of the ellipses and reversing the handedness of the considered state. As it was mentioned above, the orthogonal states do not interfere, hence their intensities are just simply added to each other.

Generally, for many usual polarization optical components, such as retardation plates or polarizers, these states are actually linearly polarized.

Simple *diattenuators* transmit (or reflect) each of their eigenstates with *different intensities* but without introducing any phase shift between them (see Fig. 5).

The eigenvalues P_1 and P_2 of Jones matrix of diattenuator are both real ($P_1 > P_2$) and related to the transmitted (reflected) intensities as

$$T(R)_{\max} = P_1^2, \quad T(R)_{\min} = P_2^2 \tag{51}$$

The Mueller matrix of such diattenuator has the following form:

$$\mathbf{M_D} = T_0 \begin{bmatrix} 1 & \mathbf{D}^{\mathrm{T}} \\ \mathbf{D} & \mathbf{m_D} \end{bmatrix}, \tag{52}$$

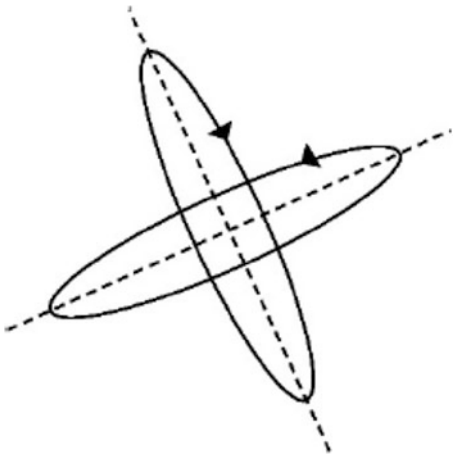


Fig. 4 Elliptically polarized orthogonal states

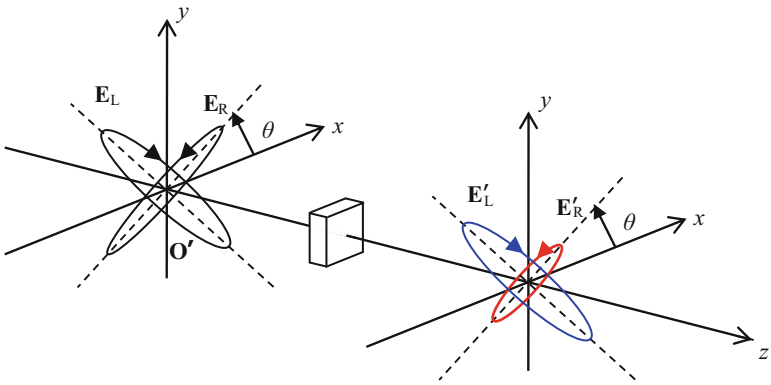


Fig. 5 Transformation of polarization eigenstates by a homogeneous diattenuator

$$\mathbf{m_D} = \sqrt{1 - D^2}\mathbf{I_3} + \left(1 - \sqrt{1 - D^2}\right)\hat{\mathbf{D}}\hat{\mathbf{D}}^{\mathrm{T}} \tag{53}$$

where D is a scalar diattenuation, $\hat{\mathbf{D}}$ is a unit vector parallel to the diattenuation vector $\mathbf{D}$, and $\mathbf{I_3}$ is 3×3 identity matrix. The Mueller matrix of diattenuator describes a rotation on the Poincaré sphere of incident polarization state toward the location of the highest transmission axis in the plane containing the points corresponding to both transmission axes and incident polarization state. The linear polarizers are characterized by the linear eigenstates and scalar diattenuation $D = 1$.

Contrary to diattenuators, the simple *retarders* transmit their eigenstates with the same intensity but different *phases* (or optical paths). This effect is illustrated in Fig. 6.

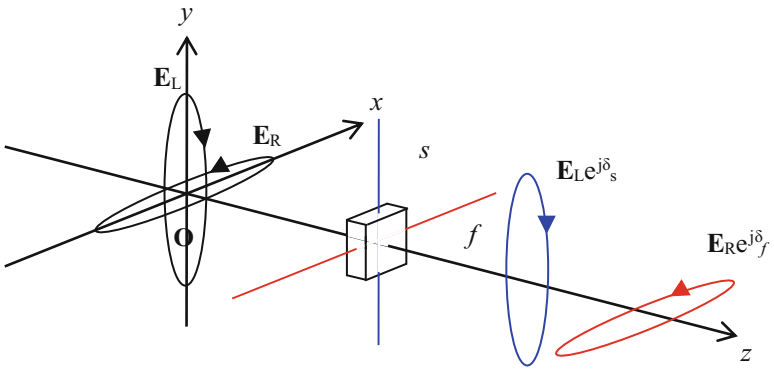


Fig. 6 Transformation of polarization eigenstates by a homogeneous phase retarder (s is for slow axis, f is for fast axis)

The Jones matrix of a homogeneous retarder is a unitary matrix. The eigenvalues of this matrix have the following form:

$$\lambda_1 = \exp(j\delta_s), \quad \lambda_2 = \exp(j\delta_f), \quad \delta_R = \delta_s - \delta_f \tag{54}$$

The Mueller matrix of such retarder has the following form:

$$\mathbf{M_R} = \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & \mathbf{m_R} \end{bmatrix} \tag{55}$$

where $\mathbf{0}^{\mathrm{T}}$ is zero vector, because the pure retarder does not exhibit any diattenuation, $\mathbf{m_R}$ is the reduced 3×3 matrix of retarder. This matrix describes the rotation by angle δ_R on the Poincaré sphere about the diameter defined by the two orthogonal polarization eigenstates.

In contrast with diattenuators and retarders, *depolarizers* do not feature any polarization eigenstates (i.e., no polarization state is left invariant after interaction of light with a depolarizer). Depolarizers reduce the degree of light polarization ρ defined in (29). In particular, these optical components transform totally polarized states into partially polarized ones, as illustrated in Fig. 7. This depolarization depends on the incident polarization state of the light.

It was shown that Mueller matrix of a pure depolarizer in the most general form is the following one [31]:

$$\mathbf{M_\Delta} = \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{0} & \mathbf{m_\Delta} \end{bmatrix} \tag{56}$$

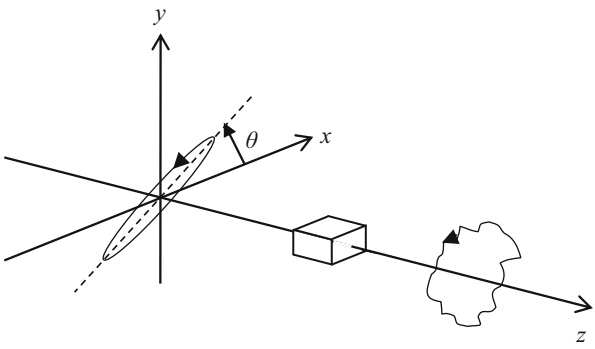


Fig. 7 Transformation of polarization eigenstate by a typical depolarizer

where $\mathbf{m_\Delta}$ is 3×3 symmetric matrix. The eigenvalues of $\mathbf{m_\Delta}$ provide three principal depolarization factors while the eigenvectors of $\mathbf{m_\Delta}$ ($\mathbf{v}_i$, i = 1, 2, 3) give three orthogonal principal axes. By unitary transformation, the matrix of depolarizer can be diagonalized:

$$\mathbf{M}_{\Delta\,\mathbf{d}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & b & 0 \\ 0 & 0 & 0 & c \end{bmatrix}, \quad |a|, \quad |b|, \quad |c| < 1 \tag{57}$$

When the incident polarized light characterized by Stokes vector of the form $\mathbf{S}^{in} = [1, \mathbf{v}^{in}]^T$ will interact with the pure depolarizer, its degree of polarization will be reduced by a factor equal to the relevant eigenvalue among *a*, *b,* or *c*. This matrix describes the migration of a corresponding point on the Poincaré sphere from the surface toward the center of sphere.

As mentioned above, partially polarized states can be seen as incoherent superposition of totally polarized states with different polarizations. As a result, a depolarizer can be described as an optical system with different non-depolarizing responses which add up incoherently. In practice, this occurs for samples featuring *spatial, spectral, or temporal inhomogeneities* resulting in different output polarization states contributing to the intensity signals which are summed at the detector.

2.1 Decomposition of Mueller Matrices

Despite the last century invention of Mueller matrix formalism (1940s), the physical interpretation of Mueller matrices is an actively evolving topic nowadays. In 1996, Lu and Chipman [31] proposed a three-factor polar decomposition of an arbitrary (non-depolarizing or depolarizing) Mueller matrix **M** containing a diattenuator $\mathbf{M_D}$, a retarder $\mathbf{M_R}$, and a depolarizer $\mathbf{M_\Delta}$:

$$\mathbf{M} = \mathbf{M}_{\boldsymbol{\Delta}}\mathbf{M}_{\mathbf{R}}\mathbf{M}_{\mathbf{D}} \tag{58}$$

It has to be pointed out that depolarizer matrix $\mathbf{M}_{\boldsymbol{\Delta}}$ has the following form (so-called polarizing depolarizer):

$$\mathbf{M} = \begin{bmatrix} 1 & \mathbf{0}^{\mathrm{T}} \\ \mathbf{P}_{\boldsymbol{\Delta}} & \mathbf{m}_{\boldsymbol{\Delta}} \end{bmatrix} \tag{59}$$

where $\mathbf{P}_{\boldsymbol{\Delta}}$ denotes the polarizance vector of this depolarizer and the polarizance is not zero. This choice of a polarizing depolarizer was necessary to always assure the physically realizable matrices for the elementary components of polar decomposition. However, the implementation of this polar decomposition for real systems has a logical limitation because of the imposed serial order of diattenuator, retarder, and depolarizer. The matrix product is not invariant under commutation. Hence, if elementary components of a system occur in different order (e.g., depolarizing part acts before diattenuating part), the phenomenological parameters of Lu–Chipman decomposition may be inappropriate.

The first studies of Mueller matrix polar decomposition for the system with a reverse serial arrangement of elementary components have been done by Morio and Goudail [32]. Later, Ossikovski et al. [33] introduced a new type of depolarizer with zero polarizance and non-zero diattenuation. It was shown theoretically and experimentally that for a system of components in serial reverse order (with a depolarizer before a diattenuator), this decomposition assures the physical realizability of the elementary matrices and the convergence of method. The symmetrical decomposition proposed in [34] makes use of five factor product of Mueller matrix

$$\mathbf{M} = \mathbf{M}_{\mathbf{D2}}\mathbf{M}_{\mathbf{R2}}\mathbf{M}_{\boldsymbol{\Delta}\,\mathbf{d}}\mathbf{M}_{\mathbf{R1}}\mathbf{M}_{\mathbf{D1}} \tag{60}$$

where **D1** and **D2** are two homogeneous diattenuators, **R1** and **R2** are two homogeneous retarders, and $\boldsymbol{\Delta}\mathbf{d}$ is diagonal depolarizer, with zero polarizance and diattenuation. The latest advances in this field include both differential and integral Mueller matrix decompositions [35, 36].

In complex samples (e.g., biological tissues), all three basic polarimetric properties—diattenuation, retardance, and depolarization—may not be localized but rather mixed and may appear simultaneously. Thus, depending on the nature of studied samples and adopted phenomenological model, the best suited decomposition for the system under study has to be chosen in order to characterize the basic polarimetric properties of a sample. The detailed survey of the existing Mueller matrix decomposition methods is provided in the next Chapter with relevant examples.

3 Summary

The brief introduction to the Jones and Stokes–Mueller formalisms for polarized light description was conducted to facilitate the reading and to define the basic concepts and notations used throughout the book. It was shown that Jones formalism which describes the polarization states and their transformation by propagation through optical systems in terms of electric fields is well adapted for the totally polarized light and non-depolarizing samples.

The Stokes parameters that were deduced from description of partially polarized states in terms of electric field correlation functions, allow us to introduce the important parameter—degree of polarization. The use of Stokes–Mueller formalism has proven to be necessary when dealing with the partially polarized light and depolarizing samples. At these conditions, the Jones matrix of a system is defined only statistically. Hence, only the second-order correlation functions of various Jones matrix elements are useful.

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Mueller Matrix Analysis, Decompositions, and Novel Quantitative Approaches to Processing Complex Polarimetric Data



Oriol Arteaga and Razvigor Ossikovski

1 Introduction

This chapter is devoted to data analysis methods for biomedical Mueller matrix imaging techniques. The objective is to gain qualitative and/or quantitative optical contrast of the tissue architecture or of the cellular structure and dynamics. Elastic scattering interactions between light and matter are highly sensitive to subtle alterations in tissue morphology and depend on the polarization of the incident field. It should be emphasized that this work assumes that the Mueller matrix data to analyze come from elastic scattering interactions, but not from inelastic ones, such as those taking place in the field of fluorescence imaging, the latter requiring a work on its own.

In a polarimetric experiment, the full Mueller matrix, **M**, is typically measured. This matrix is the medium transfer function: a 4×4 matrix of real elements that relate the input and output Stokes vectors by $\mathbf{S}_{out} = \mathbf{M}\mathbf{S}_{in}$. When compared to other applications of Mueller matrix polarimetry in fields such as material science, biomedical imaging presents two particularities that we must highlight from the beginning: (1) biological tissues often have a highly scattering nature that makes that the light collected by the instrument does not correspond to a single type of scattering event but is rather a statistical average of many different events [1], leading to extensive depolarization. (2) as a result of this highly scattering nature, the majority of bulk tissue analysis is made in a backscattering configuration (or in a close-to-backscattering one).

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In addition to bulk tissue analysis, there is another common form of biomedical experimentation based on studying thin slices (histological cuts) of the relevant tissues. This analysis, requiring more complex sample preparation and most often not applicable in vivo, is typically done with transmission microscopy equipment, which can also be supplemented with a Mueller matrix imaging option. Compared with bulk tissue analysis, the extent of depolarization in polarimetric measurements on thin tissue slices tends to be much smaller, which often simplifies the data analysis.

For years, the Lu-Chipman decomposition [2] (or forward polar decomposition) has been the most popular analysis method for extracting phenomenological information from polarimetric measurements on biological tissues [3]. Its popularity likely comes from the fact that it was one of the earliest methods, proposed almost 25 years ago, and that it can be applied to any Mueller matrix with a relatively easy algebraic implementation. However, the result of the decomposition, given by a sequence of three order-depending factors, limits its practical usefulness to media where retarding, diattenuating, and depolarizing properties can be considered as separated in three ordered effective layers. Over the last ∼15 years, many other different data analysis methods have been proposed that, in certain circumstances, are better suited than the Lu-Chipman decomposition. This chapter does not pretend to be an exhaustive and complete compilation of all available methods for polarimetric data analysis (a comprehensive monograph is available in [4]) but rather a review of those approaches which we think are the more relevant and useful for researchers applying the polarimetric technique to the biomedical field. Rather than focusing on detailed theoretical aspects, we will prioritize the practical aspects of each method, showing how the data analysis algorithms can be implemented and discussing their main context of application.

In line with the two main configurations (bulk tissue in backscattering and thin tissue in transmission) used for biomedical polarimetric analysis, Sect. 2 starts by studying the particularities that these two configurations impose on the form and the symmetries of the Mueller matrices. In the next sections, several different data analysis methods are reviewed. These are grouped in three main blocks: methods based on a serial model (Sect. 3), methods based on a differential model (Sect. 4), and methods based on polarimetric indices (Sect. 5). Finally, Sect. 6 illustrates all discussed methods on an experimental example.

2 General Considerations About the Form and Structure of Mueller Matrices in Biological Tissues

As a result of refractive index non-uniformity at the microscale arising from, for example, extracellular proteins like collagen fibrils, cell membranes, cytoplasms, nuclei, organelles, etc., elastic scattering is the major optical interaction mechanism between polarized light and bulk biological tissues [5]. Photons undergoing multiple

scattering events result in longer average pathlengths and can provide information from deeper portions of the tissue. Multiple scattering randomizes not only the pathlength and the direction of propagation of the photons but also their polarization states, resulting in depolarization [5].

The retardance observable in tissues mainly originates from photons undergoing a single or a limited number of scattering events from non-randomly oriented scatterers. It is characteristic of regions with anisotropic fibrous tissue structures like collagen fibrils and elastin fibers. Besides scattering, tissue absorption also has a significant influence on depolarization. In general, multiply scattered photons are more likely to be absorbed due to longer propagation pathlengths. As a result, absorption, if present, reduces the average propagation pathlengths or the average number of scattering events, leading to a weaker depolarization effect. However, absorption and diattenuation are usually less pronounced than depolarization and retardance in bulk tissues.

In the most general case, a Mueller matrix depends on 16 independent parameters and can be written as

$$\mathbf{M} = \begin{pmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \\ m_{30} & m_{31} & m_{32} & m_{33} \end{pmatrix}. \tag{1}$$

The objective of the Mueller matrix data analysis is to provide quantitative tools for the label-free characterization of tissues, offering new forms of image contrast or enhancing existing ones. This permits correlating measured optical effects with characteristic structural or pathological features of the tissues [6–8]. Before reviewing different data analysis methods, it is useful to consider the particularities imposed by the two main polarimetric configurations used in experiments.

2.1 Bulk Tissue Backscattering

The backscattering configuration (scattering angle of $\sim 180°$) limits the degrees of freedom of the experiment. As a result, the most general Mueller matrix features a number of symmetries and has only 10 independent parameters:

$$\mathbf{M} = \begin{pmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{01} & m_{11} & m_{12} & m_{13} \\ -m_{02} & -m_{12} & m_{22} & m_{23} \\ m_{03} & m_{13} & -m_{23} & m_{33} \end{pmatrix}. \tag{2}$$

A matrix with these symmetries remains the same if the input and output beams are interchanged, as it is to be expected in the backscattering configuration. This matrix

is called a *reciprocal* Mueller matrix and, as such, it satisfies the following identity:

$$\mathbf{M} = \mathbf{O}\mathbf{M}^T\mathbf{O}^{-1}, \tag{3}$$

where the superscript ^{T} indicates the matrix transpose and $\mathbf{O} = \mathrm{diag}(1, 1, -1, 1)$. The backscattering configuration is the most common configuration used for in-vivo measurements [9, 10]. In practice, as many polarimetric systems are not operated in strict backscattering but rather at a small angle of incidence and furthermore, are inevitably affected by experimental noise, Eq. (2) is satisfied only approximately.

For some authors it has become customary to left-multiply experimental Mueller matrices measured in the backscattering configuration by the diagonal matrix $\mathrm{diag}(1, 1, -1, -1)$ to "assimilate" them to those measured in forward scattering, which are better handled by some data analysis methods. This multiplication flips the signs of the two last rows of the Mueller matrix, altering the matrix symmetries shown in Eq. (2). In general, it is preferable to preserve the original form of the measured backscattering Mueller matrices when presenting experimental raw data and, in case such sign flips are applied to data analysis, it is recommended to mention them explicitly.

The Mueller matrix from Eq. (2) may fulfill one additional condition depending on whether the collected backreflected light originates from a collection of single scattering events or rather, from multiply scattered photons. In the first case, each individual backscattering event can be described by 2×2 scattering matrix of complex amplitudes (a Jones matrix) with the following reciprocal symmetry [11]:

$$\mathbf{T} = \begin{pmatrix} S_2 & S_3 \\ -S_3 & S_1 \end{pmatrix}. \tag{4}$$

The nondepolarizing Mueller matrix $\mathbf{M}_J$ associated with this scattering matrix is:

$$\mathbf{M}_J = \frac{1}{2}\left(\begin{array}{cccc}
S_1S_1^* + S_2S_2^* & -S_1S_1^* & S_3(S_2^* - S_1^*) & iS_3(S_2^* + S_1^*) \\
+2S_3S_3^* & +S_2S_2^* & +S_3^*(-S_1 + S_2) & -iS_3^*(S_1 + S_2) \\
\hline
-S_1S_1^* & S_1S_1^* + S_2S_2^* & S_3(S_2^* - S_1^*) & iS_3(S_2^* - S_1^*) \\
+S_2S_2^* & -2S_3S_3^* & +S_3^*(S_1 + S_2) & iS_3^*(S_1 - S_2) \\
\hline
-S_3(S_2^* - S_1^*) & -S_3(S_2^* - S_1^*) & S_2S_1^* + S_1S_2^* & iS_1S_2^* \\
-S_3^*(-S_1 + S_2) & -S_3^*(S_1 + S_2) & -2S_3S_3^* & -iS_2S_1^* \\
\hline
iS_3(S_2^* + S_1^*) & iS_3(S_2^* - S_1^*) & -iS_1S_2^* & S_2S_1^* + S_1S_2^* \\
-iS_3^*(S_1 + S_2) & +iS_3^*(S_1 - S_2) & +iS_2S_1^* & +2S_3S_3^*
\end{array}\right) \tag{5}$$

and has the symmetry given in Eq. (2). Moreover, the diagonal elements of this matrix comply with the additional condition

$$m_{00} - m_{11} + m_{22} - m_{33} = 0. \tag{6}$$

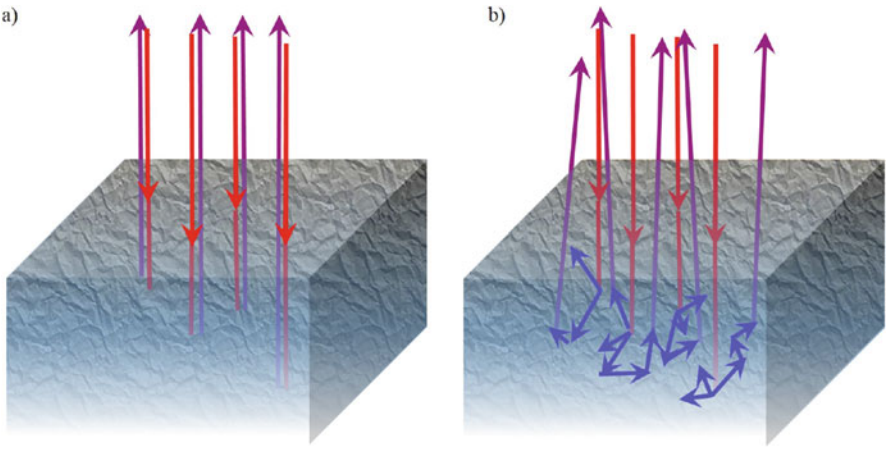


Fig. 1 Schematic representations of single backscattering (**a**) and multiple backscattering (**b**)

An ensemble of (not necessarily identical) single scatterers, illustrated in Fig. 1a, is described by the sum of their nondepolarizing Mueller matrices

$$\mathbf{M^1} = \sum_i \mathbf{M}_{J,i}. \tag{7}$$

If each scatterer has different polarization properties, the ensemble Mueller matrix, $\mathbf{M^1}$, will be depolarizing but it will still fulfill the condition from Eq. (6). The situation is different if multiple scattering is present, as we shall discuss now. This case implies that there exists an ensemble of scattering paths that connect each point of the sample to the illuminating photons [12], as it is shown in Fig. 1b. The detailed theory of multiple scattering is well beyond the scope of this chapter, but it is useful to recall the result found by Mishchenko [13, 14] who showed that the total Mueller matrix in the exact backscattering configuration can be presented in the form of a sum,

$$\mathbf{M} = \mathbf{M^1} + \mathbf{M}^L + \mathbf{M}^C, \tag{8}$$

which includes the single scattering contribution, $\mathbf{M^1}$, that we have previously described, the incoherent contribution of second or higher order scattering, $\mathbf{M}^L$, which can be calculated by using the radiative transfer equation [15], and the coherent contribution $\mathbf{M}^C$. Mishchenko showed [13] that, in the backscattering configuration, the last term is related to the incoherent contribution one by

$$
\begin{aligned}
m_{ij}^C &= m_{ij}^L, \quad i \neq j, \\
m_{00}^C &= \frac{1}{2}(m_{00}^L + m_{11}^L - m_{22}^L + m_{33}^L), \\
m_{11}^C &= \frac{1}{2}(m_{00}^L + m_{11}^L + m_{22}^L - m_{33}^L), \\
m_{22}^C &= \frac{1}{2}(-m_{00}^L + m_{11}^L + m_{22}^L + m_{33}^L), \\
m_{33}^C &= \frac{1}{2}(m_{00}^L - m_{11}^L + m_{22}^L + m_{33}^L).
\end{aligned} \tag{9}
$$

From these equations, it is evident that, in the multiple scattering case, the diagonal elements of the Mueller matrix will not satisfy Eq. (6), contrary to what happens for single scattering. Therefore, Eq. (6) can be used to discriminate between single and multiple order scattering in backscattering configuration measurements.

Another interesting preliminary check for any Mueller matrix in the backscattering configuration consists in verifying if the studied system exhibits rotational symmetry. In this configuration, this means that the measured Mueller matrix must satisfy [16]:

$$
\mathbf{M} = \mathbf{R}(\theta)\mathbf{M}\mathbf{R}(\theta), \tag{10}
$$

where the rotation matrix is given by

$$
\mathbf{R}(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(2\theta) & \sin(2\theta) & 0 \\ 0 & -\sin(2\theta) & \cos(2\theta) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{11}
$$

When Eq. (10) is satisfied we shall call the system *isotropic*; otherwise, it will be termed *anisotropic* implying that its optical response depends on orientation. In polarimetry, anisotropy is associated with the existence of linear birefringence and/or linear dichroism which, in tissue polarimetry, may be indicative of collagen organization, muscular hypertrophy, or cellular orientation, for instance. An isotropic system in the backscattering configuration is easily recognized because its Mueller matrix obeys the following symmetry:

$$
\mathbf{M} = \begin{pmatrix} m_{00} & 0 & 0 & m_{03} \\ 0 & m_{11} & 0 & 0 \\ 0 & 0 & -m_{11} & 0 \\ m_{03} & 0 & 0 & m_{33} \end{pmatrix}. \tag{12}
$$

In most common isotropic situations, the element m_{03} usually vanishes. A non-vanishing m_{03} element is most often indicative of the lack of mirror symmetry

(e.g., the existence of chirality) in the scattering system, which is not a common phenomenon in biological tissues.

2.2 Transmission in a Thin Tissue

The transmission configuration (scattering angle of $\sim 0^\circ$) does not limit the degrees of freedom of the Mueller matrix which is still represented by the general form from Eq. (1). Thin tissues are, in most occasions, best studied within the uniform medium picture by means of methods based on the differential decomposition discussed in Sect. 4. The extent of depolarization is typically small but it can become more significant if the contribution of multiple scattering is predominant. The latter largely depends on the thickness of the medium as well as on the acceptance angle of the detector, so that the specifications of the measurement equipment can play an important role in the amount of the experimentally observed depolarization.

A rotation of the medium is given by

$$\mathbf{M}' = \mathbf{R}(-\theta)\mathbf{M}\mathbf{R}(\theta). \tag{13}$$

Therefore, the Mueller matrix of medium with rotational symmetry (the matrix equivalent to Eq. (12) but for transmitted light) is

$$\mathbf{M} = \begin{pmatrix} m_{00} & 0 & 0 & m_{03} \\ 0 & m_{11} & m_{12} & 0 \\ 0 & -m_{21} & m_{11} & 0 \\ m_{30} & 0 & 0 & m_{33} \end{pmatrix}. \tag{14}$$

3 Methods Based on a Serial Model

Serial models assume that light passes through consecutive polarization elements arranged along the direction of propagation. The Mueller matrix of a serial combination is given by the ordered product of the Mueller matrices of the different elements (components).

3.1 Polar Decompositions of Depolarizing Mueller Matrices

The forward (or Lu and Chipman [2]) and reverse [17] polar decompositions of depolarizing Mueller matrices have become widely used algebraic tools in interpreting experimental polarimetry data of complex media and structures. Both

polar decompositions factor any depolarizing Mueller matrix down to the product of diattenuator, retarder, and depolarizer matrices allowing for straightforward interpretation of the polarimetric properties of the measured sample in terms of diattenuation, retardance, and depolarization. Because of the serial nature of the decompositions, the choice of the order of the different polarimetric effects that occur in the sample matters and the whole range of combinations has been studied [18]. It must be stressed that the decoupling of the different polarimetric effects is physically realistic only in media where the particular model based on a sequence of three distinct layers is applicable [19]. If this is not the case for the system under study, then the serial model decompositions rather describe alternative systems that are polarimetrically equivalent to the studied one.

The forward decomposition, due to Lu and Chipman, factors $\mathbf{M}$ down to the product

$$\mathbf{M} = \mathbf{M}_{\Delta}\mathbf{M}_{R}\mathbf{M}_{D}, \tag{15}$$

in which $\mathbf{M}_D$ is a diattenuator, $\mathbf{M}_R$ is a retarder, and $\mathbf{M}_\Delta$ is a depolarizer with zero diattenuation. The standard algebraic procedure for this decomposition starts by considering the following partitioned form of a Mueller matrix $\mathbf{M}$ [20]:

$$\mathbf{M} = m_{00}\begin{pmatrix} 1 & \vec{D}^T \\ \vec{P} & \mathbf{m} \end{pmatrix}, \tag{16}$$

where m_{00} is the first element of $\mathbf{M}$, $\vec{D}$, and $\vec{P}$ are, respectively, the diattenuation and polarizance vectors, and $\mathbf{m}$ is a 3×3 matrix. The diattenuator matrix can be directly found from $\vec{D}$ as follows:

$$\mathbf{M}_D = T_u\begin{pmatrix} 1 & \vec{D}^T \\ \vec{D} & \mathbf{m}_D \end{pmatrix}, \tag{17}$$

in which

$$\mathbf{m}_D = \sqrt{1-D^2}\,\mathbf{I} + \left(1-\sqrt{1-D^2}\right)\frac{\vec{D}\vec{D}^T}{D^2}, \tag{18}$$

where $\mathbf{I}$ is the 3×3 identity matrix, $D = |\vec{D}|$ and T_u is the transmission of the diattenuator for unpolarized light (if the overall transmission or reflection of the sample is disregarded, then one can set $T_u = 1$). The diattenuator matrix is symmetric by construction, i.e., $\mathbf{M}_D^T = \mathbf{M}_D$. If the diattenuation vector is expressed in component form, $\vec{D} = (D_H \quad D_{45} \quad D_C)^T$, then $\mathbf{M}_D$ is given by

$$\mathbf{M}_D = T_u \begin{pmatrix} 1 & D_H & D_{45} & D_C \\ D_H & a + bD_H^2 & bD_H D_{45} & bD_H D_C \\ D_{45} & bD_H D_{45} & a + bD_{45}^2 & bD_{45} D_C \\ D_C & bD_H D_C & bD_{45} D_C & a + bD_C^2 \end{pmatrix}, \tag{19}$$

where $a = \sqrt{1 - D^2}$, $b = (1 - a)/D^2$ and $D = \sqrt{D_H^2 + D_{45}^2 + D_C^2}$. If $\mathbf{M}_D$ is not singular, then a new matrix $\mathbf{M}'$ is defined as

$$\mathbf{M}' = \mathbf{M}\mathbf{M}_D^{-1} = \frac{m_{00}}{T_u} \begin{pmatrix} 1 & \vec{0}^T \\ \vec{P}_\Delta & \mathbf{m}' \end{pmatrix}, \tag{20}$$

where $\vec{P}_\Delta$ is the polarizance vector of the depolarizer, given by $\vec{P}_\Delta = \frac{\vec{P} - \mathbf{m}\vec{D}}{1 - D^2}$ and $\mathbf{m}'$ is a 3×3 block given by $\mathbf{m}' = \frac{\mathbf{m}\mathbf{m}_D - \vec{P}\vec{D}^T}{1 - D^2}$.

The depolarizer matrix $\mathbf{M}_\Delta$ with zero diattenuation is

$$\mathbf{M}_\Delta = \begin{pmatrix} 1 & \vec{0}^T \\ \vec{P}_\Delta & \mathbf{m}_\Delta \end{pmatrix}, \tag{21}$$

where the 3×3 block $\mathbf{m}_\Delta$ is given by

$$\begin{aligned} \mathbf{m}_\Delta = \varepsilon[\mathbf{m}'(\mathbf{m}')^T + (\sqrt{\lambda_1\lambda_2} + \sqrt{\lambda_2\lambda_3} + \sqrt{\lambda_3\lambda_1})\mathbf{I}]^{-1} \\ \times[(\sqrt{\lambda_1} + \sqrt{\lambda_2} + \sqrt{\lambda_3})\mathbf{m}'(\mathbf{m}')^T + \sqrt{\lambda_1\lambda_2\lambda_3}\mathbf{I}] \end{aligned}, \tag{22}$$

where $\varepsilon = \text{sign}[\det(\mathbf{m}')] = \text{sign}[\det(\mathbf{M})]$, $\mathbf{I}$ is the 3×3 identity matrix, and λ_1, λ_2, and λ_3 are the (non-negative) eigenvalues of the (positive semi-definite) matrix $\mathbf{m}'(\mathbf{m}')^T$. Note that the form of the depolarizer $\mathbf{M}_\Delta$ is peculiar because it is not a diagonal one and, therefore, combines both depolarization and polarizance.

Finally, the retarder matrix, $\mathbf{M}_R$, is obtained from

$$\mathbf{M}_R = \mathbf{M}_\Delta^{-1}\mathbf{M}'. \tag{23}$$

$\mathbf{M}_R$ can be also expressed in the usual partitioned form

$$\mathbf{M}_R = \begin{pmatrix} 1 & \vec{0}^T \\ \vec{0} & \mathbf{m}_R \end{pmatrix} \tag{24}$$

in which $\mathbf{m}_R$ is a 3×3 rotation matrix that, given a retardation vector $\vec{R} = (R_H \quad R_{45} \quad R_C)^T$, can be expressed as:

$$\mathbf{m}_R = \begin{pmatrix} \cos R + R_H^2 C & R_H R_{45} C + R_C S & R_H R_C C - R_{45} S \\ R_H R_{45} C - R_C S & \cos R + R_{45}^2 C & R_{45} R_C C + R_H S \\ R_H R_C C + R_{45} S & R_{45} R_C C - R_H S & \cos R + R_{45}^2 C \end{pmatrix} \tag{25}$$

with $C = (1 - \cos R)/R^2$, $S = \sin R/R$ and

$$R = \sqrt{R_H + R_{45} + R_C} = \cos^{-1}\left(\frac{\mathrm{tr}(\mathbf{M}_R)}{2} - 1\right), \tag{26}$$

One practical way to obtain the reverse decomposition in which the depolarizer is set before the diattenuator ($\mathbf{M} = \mathbf{M}_R^r \mathbf{M}_D^r \mathbf{M}_\Delta^r$) is to apply the forward decomposition presented above to $\mathbf{M}^T$ (instead of to $\mathbf{M}$) and re-transpose the resulting matrix product.

This algebraic procedure has to be generalized if $\mathbf{M}$ has a negative determinant ($\det(\mathbf{M}) < 0$)—which is not an uncommon situation in strongly depolarizing systems—through the following substitution rules [21]:

$$\mathbf{M}_R \longrightarrow \mathbf{S}\mathbf{M}_R = \begin{pmatrix} 1 & \vec{0}^T \\ \vec{0}^T & \mathbf{s}\mathbf{m}_R \end{pmatrix}, \tag{27a}$$

$$\mathbf{M}_\Delta \longrightarrow \mathbf{M}_\Delta \mathbf{S} = \begin{pmatrix} 1 & \vec{0}^T \\ \vec{P}_\Delta & \mathbf{m}_\Delta \mathbf{s} \end{pmatrix}, \tag{27b}$$

where $\mathbf{S}$ is a 4×4 sign matrix defined as

$$\mathbf{S} = \begin{pmatrix} 1 & \vec{0}^T \\ \vec{0}^T & \mathbf{s} \end{pmatrix}. \tag{28}$$

The 3×3 block matrix $\mathbf{s}$ is given by $\mathbf{s} = \mathrm{diag}(\varepsilon_1 \quad \varepsilon_2 \quad -\varepsilon_1\varepsilon_2)$ where $\varepsilon_{1,2} = \pm 1$ are "selectable" sign factors. The sign-flip matrix $\mathbf{S}$ changes the signs of the determinants of $\mathbf{M}_R$ and $\mathbf{M}_\Delta$ and ensures that $\det(\mathbf{M}_R) = +1$ in order for $\mathbf{M}_R$ to physically represent a retarder. This algebraic procedure for the polar decomposition of Mueller matrices with negative determinants is sufficient for diagonal depolarizer matrices, which are the most common form of experimentally occurring depolarizer matrices. Procedures handling the most general case of nondiagonal depolarizer matrix factors have been recently published [22, 23] and will not be discussed here.

3.2 *Normal Form and Symmetric Decomposition*

The normal form of a depolarizing Mueller matrix [24] is based on the fact that any Mueller matrix can be expressed as

$$\mathbf{M} = \mathbf{M}_{J2}\mathbf{M}_{\Delta}\mathbf{M}_{J1}, \tag{29}$$

where $\mathbf{M}_{J1}$ and $\mathbf{M}_{J2}$ are nondepolarizing Mueller matrices and $\mathbf{M}_{\Delta}$ is a depolarizer. The two nondepolarizing components can be further decomposed down to products of diattenuators and retarders by applying the polar decomposition, thus getting a sequence of five factors that is known as the symmetric decomposition [25],

$$\mathbf{M} = \mathbf{M}_{D2}\mathbf{M}_{R2}\mathbf{M}_{\Delta}\mathbf{M}_{R1}^{T}\mathbf{M}_{D1}, \tag{30}$$

where $\mathbf{M}_{D1}$ and $\mathbf{M}_{D2}$ are two diattenuators and $\mathbf{M}_{R1}$ and $\mathbf{M}_{R2}$ are two retarders. In the normal form, as well as in the symmetric decomposition, the depolarizer matrix $\mathbf{M}_{\Delta}$ usually takes the familiar diagonal form:

$$\mathbf{M}_{\Delta} = \begin{pmatrix} d_0 & 0 & 0 & 0 \\ 0 & d_1 & 0 & 0 \\ 0 & 0 & d_2 & 0 \\ 0 & 0 & 0 & d_3 \end{pmatrix}. \tag{31}$$

Most generally, not all Mueller matrices admit a decomposition with a diagonal depolarizer. Specifically, this is not possible for Mueller matrices that do not depolarize a single totally polarized input state (while depolarizing all the rest). Such non-diagonalizable matrices are commonly called type-II Mueller matrices [4]. Since this is not the case for the vast majority of Mueller matrices of biomedical interest, type-II matrices will not be considered in this section; they can be treated with a mixed algebraic-numerical method that has been recently proposed [26]. Here, we discuss only the algebraic procedure for the symmetric decomposition given by Eq. (30), from which one can obtain the normal form in Eq. (29) by direct multiplication of the diattenuator and retarder matrix factors in the order given in (30).

The inner matrix terms of Eq. (30) can be rewritten as:

$$\mathbf{M}' = \mathbf{M}_{R2}\mathbf{M}_{\Delta}\mathbf{M}_{R1}^{T} = \mathbf{M}_{D2}^{-1}\mathbf{M}\mathbf{M}_{D1}^{-1}. \tag{32}$$

Notice that, since the depolarizer matrix $\mathbf{M}_{\Delta}$ has zero polarizance and diattenuation vectors ($\vec{P} = \vec{D} = 0$), then $\mathbf{M}'$ is of the block-diagonal form

$$\mathbf{M}' = \begin{pmatrix} d_0 & \vec{0}^{T} \\ \vec{0}^{T} & \mathbf{m}' \end{pmatrix}. \tag{33}$$

By using this fact, it is easy to show that the diattenuator matrices $\mathbf{M}_{D1}$ and $\mathbf{M}_{D2}$ can be obtained from the following pair of eigenvalue-eigenvector equations

$$(\mathbf{M}^{T}\mathbf{G}\mathbf{M}\mathbf{G})S_1 = d_0^2 S_1, \quad (\mathbf{M}\mathbf{G}\mathbf{M}^{T}\mathbf{G})S_2 = d_0^2 S_2, \tag{34}$$

where $\mathbf{G}$ is the Minkowski metric matrix, $\mathbf{G} = \mathrm{diag}(1, -1, -1, -1)$ and the eigenvectors S_1 and S_2 are

$$S_1 = \begin{pmatrix} 1 \\ \vec{D}_1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 1 \\ \vec{D}_2 \end{pmatrix}. \tag{35}$$

Knowing $\vec{D}_1$ and $\vec{D}_2$, one can, respectively, obtain $\mathbf{M}_{D1}$ and $\mathbf{M}_{D2}$ from Eq. (17) and (18) (T_u can be set to $T_u = 1$). Notice that it is not necessary to solve both eigenvalue-eigenvector equations in Eq. (34); it is sufficient to solve, for instance, just the first one and get S_2 from $S_1 = \mathbf{MG}S_1/(\mathbf{MG}S_1)_0$ where $(\mathbf{MG}S_1)_0$ denotes the zeroth component of the vector. This is possible because the two matrices have the property $\mathbf{MG}(\mathbf{M}^T\mathbf{GMG}) = (\mathbf{MGM}^T\mathbf{G})\mathbf{MG}$.

Once $\mathbf{M}_{D1}$ and $\mathbf{M}_{D2}$ are found, $\mathbf{M}'$ is calculated from the last expression in Eq. (32). The next step is to apply a singular value decomposition (or SVD) on $\mathbf{m}'$,

$$\mathbf{m}' = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T, \tag{36}$$

where $\mathbf{U}$ and $\mathbf{V}^T$ are 3×3 orthogonal matrices and $\boldsymbol{\Sigma}$ is a diagonal matrix that contains the (non-negative) singular values of $\mathbf{m}'$, ordered in decreasing order by convention. From the matrices $\mathbf{U}$ and $\mathbf{V}^T$ one can, respectively, obtain the submatrices $\mathbf{m}_{R2}$ and $\mathbf{m}_{R1}^T$ that form the retarder matrices $\mathbf{M}_{R2}$ and $\mathbf{M}_{R1}^T$. Likewise, from $\boldsymbol{\Sigma}$ one can directly identify the depolarizer elements d_1, d_2, and d_3 (and $d_0 = m'_{00}$). By convention, the SVD decomposition will generate a diagonal depolarizer with $d_0 \geq d_1 \geq d_2 \geq d_3$, which may not necessarily correspond to the physical reality. Moreover, a direct application of the SVD may provide $\mathbf{m}_{R2}$ and $\mathbf{m}_{R1}^T$ matrices with negative determinants (even in the case where the determinant of $\mathbf{m}'$ is positive), in contradiction with the fact that determinant of a physically realizable retarder must equal $+1$. As a result, additional treatment is needed, as explained below.

Uniqueness and physical realizability of the SVD decomposition can be warranted by applying the so-called minimum retardance principle to $\mathbf{M}_{R1}$. These can be achieved by permuting the columns of $\mathbf{m}_{R1}$ and by changing their signs until $\mathbf{m}_{R1}$ attains the largest possible trace which corresponds to minimum retardance. To preserve the invariance of the SVD, all column permutations and sign changes applied to $\mathbf{m}_{R1}$ must be also applied to $\mathbf{m}_{R2}$. Furthermore, it is also necessary to permute the affected diagonal elements of $\mathbf{m}_\Delta$.

A common occurrence for Mueller matrices of biomedical interest is that the depolarizer matrix has partially degenerate elements with two out of the three coefficients being equal. For forward scattering this usually implies $d_1 = d_2 \neq d_3$. The equivalent situation for backward scattering is $d_1 = -d_2 \neq d_3$ but if one applies the sign flip of the last two rows of the Mueller matrix discussed in Sect. 2.1, the same situation as for forward scattering is found. Under such condition, the decomposition of $\mathbf{M}'$ through the SVD becomes non-unique [27] because it is invariant upon rotation of the depolarizer $\mathbf{M}_\Delta$ by an arbitrary angle γ:

$$\mathbf{M}' = \mathbf{M}_{R2}\mathbf{M}_{\Delta}\mathbf{M}_{R1}^{T} = \mathbf{M}_{R2}\mathbf{R}(\gamma)\mathbf{M}_{\Delta}\mathbf{R}(-\gamma)\mathbf{M}_{R1}^{T}. \tag{37}$$

Due to the presence of the rotation matrices with a value of γ set arbitrarily by the SVD, the retarder matrices $\mathbf{M}_{R2}$ and $\mathbf{M}_{R1}^{T}$ obtained from the SVD may exhibit a strong circular retardance (R_C) contribution that has no real physical meaning, as discussed in [27]. Only the linear retardances (R_H and R_{45}) have physical meaning. The spurious circular retardance "added" by the SVD is efficiently removed from the two retarders by decomposing them down to linear and circular factors [27].

4 Methods Based on a Differential Model

The physical model on which the differential decomposition is based assumes that the polarization properties of the analyzed medium have uniform distribution along the optical path. As a result, the differential methods presented in this section are particularly suitable for the analysis of *transmission experiments* such as transmission Mueller matrix microscopy. The differential Stokes–Mueller matrix formalism [28–30] (or its equivalent roots decomposition [31]) expresses the local evolution of the Mueller matrix or of the Stokes vector for light propagating through a homogeneous optical medium. Over the last decade, the methods based on a differential analysis of the Mueller matrix have gained increasing popularity but, in a purely historical aspect, the basic idea of a differential treatment of the propagation of polarized light in a continuous medium dates back to 1929 with the pioneering work of Soleillet [32, 33].

The differential Mueller formalism is generally introduced with the equation

$$\frac{d\mathbf{M}}{dz} = \mathbf{m}\mathbf{M}, \tag{38}$$

where the differential matrix $\mathbf{m}$ relates the Mueller matrix of the homogeneous anisotropic medium to its spatial derivative along the direction of light propagation (z). This differential equation can be alternatively rewritten in terms of the Stokes vector

$$\frac{d\mathbf{S}}{dz} = \mathbf{m}\mathbf{S}, \tag{39}$$

explicitly describing the evolution of the Stokes vector $\mathbf{S}$ along the propagation direction z, resulting from its interaction with the anisotropic and generally depolarizing medium.

A nondepolarizing Mueller matrix, $\mathbf{M}_J$, satisfying the differential Eq. (38), can be written as the following matrix exponential:

Table 1 Symbols used and definitions for the spectroscopic polarization properties

Effect	Symbol	Definition[a]
Isotropic amplitude absorption	k	$2\pi\kappa l/\lambda_0$
(x–y) linear dichroism	LD	$\frac{2\pi}{\lambda_0}(\kappa_x - \kappa_y)l$
(x–y) linear birefringence	LB	$\frac{2\pi}{\lambda_0}(n_x - n_y)l$
45° linear dichroism	LD′	$\frac{2\pi}{\lambda_0}(\kappa_{45} - \kappa_{135})l$
45° linear birefringence	LB′	$\frac{2\pi}{\lambda_0}(n_{45} - n_{135})l$
Circular dichroism	CD	$\frac{2\pi}{\lambda_0}(\kappa_- - \kappa_+)l$
Circular birefringence	CB	$\frac{2\pi}{\lambda_0}(n_- - n_+)l$

[a] n stands for refractive index, κ for extinction coefficient, l for path length through the medium, and λ_0 for the vacuum wavelength of light. Subscripts specify the polarization of light as, x, y, 45° to the x axis, 135° to the x axis, circular right $+$, or left $-$

$$\mathbf{M}_J = e^{\mathbf{m}d}, = e^{\mathbf{L}_{nd}}, \quad \mathbf{L} = \begin{pmatrix} -\kappa & -\mathrm{LD} & -\mathrm{LD}' & \mathrm{CD} \\ -\mathrm{LD} & -\kappa & \mathrm{CB} & \mathrm{LB}' \\ -\mathrm{LD}' & -\mathrm{CB} & -\kappa & -\mathrm{LB} \\ \mathrm{CD} & -\mathrm{LB}' & \mathrm{LB} & -\kappa \end{pmatrix}. \tag{40}$$

Parameters κ, LD, LD′, CD, LB, LB′, and CB shall be referred as the *spectroscopic polarization properties* [34]; they are defined in Table 1. The definitions shown in the table connect the polarization properties with the (anisotropic) refractive indices and absorption coefficients of the medium. They hold for normal-incident-light transmission in a uniform medium of path length l. In other configurations (e.g., in reflection mode) LD, LD', and CD must be regarded as describing diattenuation (instead of dichroism) whereas LB, LB', and CB must be regarded as describing retardation (instead of birefringence).

Table 2 shows the correspondence between the polarization properties in the spectroscopic notation and in the polarimetric notation that we have previously used in Sect. 3.1 and that is generally used for the Lu-Chipman decomposition [19]. In general, the spectroscopic notation is preferred over the polarimetric one because, besides leading to the simple and physically meaningful definitions given in Table 1, it allows for grouping the polarization properties by their nature,

$$\mathrm{L} = \mathrm{LB} - i\mathrm{LD}, \tag{41a}$$

$$\mathrm{L}' = \mathrm{LB}' - i\mathrm{LD}', \tag{41b}$$

$$\mathrm{C} = \mathrm{CB} - i\mathrm{CD}, \tag{41c}$$

that will be useful in the next sections.

Table 2 Correspondence between the polarimetric notation used by Lu-Chipman[a] and the spectroscopic notation used in the differential calculus

Retardances	Absorbances
$\mathrm{LB} = -R_H$	$\mathrm{LD} = -(\mathrm{arctanh}D/D)D_H$
$\mathrm{LB}' = -R_{45}$	$\mathrm{LD}' = -(\mathrm{arctanh}D/D)D_{45}$
$\mathrm{CB} = R_C$	$\mathrm{CD} = (\mathrm{arctanh}D/D)D_C$
$\mathrm{T}_R = \sqrt{\mathrm{LB}^2 + \mathrm{LB}'^2 + \mathrm{CB}^2} = R$	$\mathrm{T}_D = \sqrt{\mathrm{LD}^2 + \mathrm{LD}'^2 + \mathrm{CD}^2} = \mathrm{arctanh}D$

[a] The notation of Lu-Chipman is based on the diattenuation vector $\vec{D} = (D_H, D_{45}, D_C)^T$ and on the retardation vector $\vec{R} = (R_H, R_{45}, R_C)^T$. The present table gives the relations between the components of these two vectors and the polarization properties defined in Table 1

Instead of reporting the values of LB, LB′, LD, and LD′, it is usually advisable to calculate the following four related quantities that allow for a more straightforward physical interpretation of these optical effects [35]:

$$\mathrm{LB}_m = \sqrt{\mathrm{LB}^2 + \mathrm{LB}'^2}, \tag{42a}$$

$$\mathrm{LB}_\theta = \frac{1}{2}\mathrm{atan2}\left(\frac{\mathrm{LB}'}{\mathrm{LB}}\right), \tag{42b}$$

$$\mathrm{LD}_m = \sqrt{\mathrm{LD}^2 + \mathrm{LD}'^2}, \tag{42c}$$

$$\mathrm{LD}_\theta = \frac{1}{2}\mathrm{atan2}\left(\frac{\mathrm{LD}'}{\mathrm{LD}}\right), \tag{42d}$$

where atan2 stands for the four-quadrant inverse tangent. LB_m and LD_m, respectively, denote the magnitudes of linear retardation and linear diattenuation, while LB_θ and LD_θ indicate their orientation angles, which will be given in the range between $-\pi/2$ and $\pi/2$ radians. Note that in the general case LB_θ can be fully independent from LD_θ, but it is common to find experimental situations where both orientation angles are either parallel ($\mathrm{LB}_\theta = \mathrm{LD}_\theta$) or perpendicular ($\mathrm{LB}_\theta = \mathrm{LD}_\theta \pm \pi/2$), which usually indicate that linear diattenuation and linear retardation have a common origin in the sample, for example, if they are produced by the same fibrous structure.

A generally depolarizing Mueller matrix satisfying the differential Eq. (38) also admits a differential decomposition which can be written as:

$$\mathbf{M} = e^{\mathbf{m}d} = e^{\mathbf{L}}, \quad \mathbf{L} = \begin{pmatrix} l_{00} & l_{01} & l_{02} & l_{03} \\ l_{10} & l_{11} & l_{12} & l_{13} \\ l_{20} & l_{21} & l_{22} & l_{23} \\ l_{30} & l_{31} & l_{32} & l_{33} \end{pmatrix}. \tag{43}$$

In this case, the differential matrix $\mathbf{L}$ does not have the highly symmetric form shown in Eq. (40). The interpretation of this matrix is given in the next section.

4.1 Differential Analysis Based on the Matrix Logarithm

Given a generally depolarizing experimental Muller matrix $\mathbf{M}$, the differential Mueller matrix $\mathbf{L}$ of Eq. (43) can be obtained by taking the matrix logarithm of $\mathbf{M}$:

$$\mathbf{L} = \ln\mathbf{M}. \tag{44}$$

Algorithms for the numerical calculation of a matrix logarithm are available in all modern programming software packages, but not all Mueller matrices have a logarithm and the logarithm may be multi-valued [36, 37]. In the biomedical imaging practice, these limitations typically manifest in that for most depolarizing measurements in the backscattering configuration there exists no real matrix logarithm unless the sign of the last two rows is flipped, as discussed in Sect. 2.1.

$\mathbf{L}$ can be further decomposed into two terms as $\mathbf{L} = \mathbf{L}_m + \mathbf{L}_u$ [28], where the G-antisymmetric and G-symmetric matrices $\mathbf{L}_m$ and $\mathbf{L}_u$ are given by:

$$\mathbf{L}_m = \frac{1}{2}(\mathbf{L} - \mathbf{G}\mathbf{L}^T\mathbf{G}) = \frac{1}{2}\begin{pmatrix} 0 & l_{01}+l_{10} & l_{02}+l_{20} & l_{03}+l_{30} \\ l_{01}+l_{10} & 0 & l_{12}-l_{21} & l_{13}-l_{31} \\ l_{02}+l_{20} & l_{21}-l_{12} & 0 & l_{23}-l_{32} \\ l_{03}+l_{30} & l_{31}-l_{13} & l_{32}-l_{23} & 0 \end{pmatrix}, \tag{45a}$$

$$\mathbf{L}_u = \frac{1}{2}(\mathbf{L} + \mathbf{G}\mathbf{L}^T\mathbf{G}) = \frac{1}{2}\begin{pmatrix} 2l_{00} & l_{01}-l_{10} & l_{02}-l_{20} & l_{03}-l_{30} \\ l_{01}-l_{10} & 2l_{11} & l_{12}+l_{21} & l_{13}+l_{31} \\ l_{02}-l_{20} & l_{21}+l_{12} & 2l_{22} & l_{23}+l_{32} \\ l_{03}-l_{30} & l_{31}+l_{13} & l_{32}+l_{23} & 2l_{33} \end{pmatrix}, \tag{45b}$$

where $\mathbf{G} = \mathrm{diag}(1, -1, -1, -1)$ is the Minkowski metric. The matrix $\mathbf{L}_m$ can be interpreted as containing the mean values of the polarization properties, according to the parameterization given in Eq. (40). $\mathbf{L}_u$ is a matrix that contains the uncertainties or fluctuations of the polarization properties, which can be used to calculate their variances and covariances within a statistical interpretation picture [38, 39]. In this differential model, the physical interpretation of the depolarizing Mueller matrix is thus given by:

$$\mathbf{L}_m = \begin{pmatrix} 0 & -\langle \mathrm{LD} \rangle & -\langle \mathrm{LD}' \rangle & \langle \mathrm{CD} \rangle \\ -\langle \mathrm{LD} \rangle & 0 & \langle \mathrm{CB} \rangle & \langle \mathrm{LB}' \rangle \\ -\langle \mathrm{LD}' \rangle & -\langle \mathrm{CB} \rangle & 0 & -\langle \mathrm{LB} \rangle \\ \langle \mathrm{CD} \rangle & -\langle \mathrm{LB}' \rangle & \langle \mathrm{LB} \rangle & 0 \end{pmatrix}, \tag{46}$$

where the brackets $\langle \ldots \rangle$ denote spatial, temporal, and/or frequency averaging. The variances and covariances are found from:

$$\langle |\Delta \mathrm{L}|^2 \rangle = l_{u00} + l_{u11} - l_{u22} - l_{u33}, \tag{47a}$$

$$\langle |\Delta \mathrm{L}|'^2 \rangle = l_{u00} - l_{u11} + l_{u22} - l_{u33}, \tag{47b}$$

$$\langle |\Delta \mathrm{C}|^2 \rangle = l_{u00} - l_{u11} - l_{u22} + l_{u33}, \tag{47c}$$

$$\langle \Delta \mathrm{L} \Delta \mathrm{L}'^* \rangle = 2(l_{u12} + i l_{u03}), \tag{47d}$$

$$\langle \Delta \mathrm{L} \Delta \mathrm{C}^* \rangle = 2(l_{u13} - i l_{u02}), \tag{47e}$$

$$\langle \Delta \mathrm{L}' \Delta \mathrm{C}^* \rangle = 2(l_{u23} + i l_{u01}). \tag{47f}$$

High values of the variances in Eqs. (47a), (47b), and (47c) indicate strong depolarization, while the covariances in Eqs. (47d), (47e), and (47f) express the correlations between different types of anisotropies. Examples of the application of the differential analysis to biological tissue measurements are given in [40, 41].

Eventually, it should be noted that, for the above statistical interpretation of the matrix logarithm to be valid, the differential Mueller matrix must be physically realizable. Differential physical realizability is guaranteed if the lower 3×3 block of the coherency matrix (defined in the next section) associated with the matrix $\mathbf{L}_u$ is positive semi-definite [36]. Note also that the (global) physical realizability of $\mathbf{M}$ does not imply its (local) differential physical realizability; the latter must be checked additionally. The only case where global physical realizability entails differential one is the (trivial) case of a nondepolarizing $\mathbf{M}$; all nondepolarizing and nonsingular Mueller matrices allow matrix logarithms.

4.2 Differential Analysis Based on an Analytic Inversion

The differential decomposition has a tractable analytic solution when the medium or system described is nondepolarizing. This analytic solution becomes particularly simple when it is given in terms of a 2×2 Jones matrix $\mathbf{J} = \begin{pmatrix} j_{00} & j_{01} \\ j_{10} & j_{11} \end{pmatrix}$ [42, 43]. It is written as:

$$\kappa = -\mathrm{Im}(2i\ln(1/K)), \tag{48a}$$

$$\mathrm{L} = i\Omega(j_{00} - j_{11}), \tag{48b}$$

$$\mathrm{L}' = i\Omega(j_{01} + j_{10}), \tag{48c}$$

$$\mathrm{C} = i\Omega(j_{01} - j_{10}), \tag{48d}$$

where

$$K = 1/\sqrt{\det(\mathbf{J})}, \tag{49a}$$

$$\Omega = \frac{K(T + 2\pi n)}{2\sin(T/2)}, \tag{49b}$$

$$T = 2\mathrm{acos}(K(j_{00} + j_{11})/2). \tag{49c}$$

The extra factor $2\pi n$, where n is an integer, accounts for the multi-valued nature of the trigonometric functions which makes possible that samples with increasing amounts of polarization properties have the same Jones/Mueller matrices. Experimentally, this is typical of samples with large linear birefringence where values of n must progress in the order $[0, -1, 1, -2, 2, \ldots]$ to account for changes of order of birefringence. In samples with birefringence within the range $0 \leq \sqrt{\mathrm{LB}^2 + \mathrm{LB}'^2 + \mathrm{CB}^2} \leq \pi$ rad one can safely set $n = 0$. The spectroscopic polarization properties are found from the real and imaginary parts of L, L′, and C, as shown by Eq. (48).

In order to extend this analytic inversion method to depolarizing Mueller matrices, one can make use of one of several available methods for determining the nondepolarizing estimate of a depolarizing Mueller matrix [44–46], a process sometimes termed matrix filtering. It is also possible to combine the analytic inversion with serial decompositions, for example, by applying it to each one of the nondepolarizing factors contained in the normal form decomposition from Eq. (29).

Here, we review in detail only the most commonly used nondepolarizing estimation based on the Cloude sum decomposition [46]. It allows for obtaining the closest nondepolarizing Matrix, $\mathbf{M}_J^{est}$, to an experimentally determined depolarizing one. This method determines the nondepolarizing matrix that minimizes the sum-of-squares estimator:

$$\delta^2 = \sum_{i,j} (\mathbf{M}_{ij} - \mathbf{M}_{Jij}^{est})^2 \rightarrow \min. \tag{50}$$

As shown by Cloude [46], a depolarizing $\mathbf{M}$ can be represented by a linear combination of up to four nondepolarizing matrix terms,

$$\mathbf{M} = \lambda_0 \mathbf{M}_{J0} + \lambda_1 \mathbf{M}_{J1} + \lambda_2 \mathbf{M}_{J2} + \lambda_3 \mathbf{M}_{J3}, \tag{51}$$

where the scalars λ_i obey $\lambda_0 \geq \lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$. The nondepolarizing estimate is obtained from the first term having the largest contribution to the sum decomposition: $\mathbf{M}_J^{est} = \lambda_0 \mathbf{M}_{J0}$. The calculation process for this nondepolarizing estimate starts by reorganizing the information available in the Mueller matrix to construct the following 4×4 Hermitian positive semi-definite coherency matrix:

$$\mathbf{C} = \frac{1}{4} \begin{pmatrix} m_{00} + m_{11} & m_{01} + m_{10} & m_{02} + m_{20} & m_{03} + m_{30} \\ m_{22} + m_{33} & -i(m_{23} - m_{32}) & +i(m_{13} - m_{31}) & -i(m_{12} - m_{21}) \\ m_{01} + m_{10} & m_{00} + m_{11} & m_{12} + m_{21} & m_{13} + m_{31} \\ +i(m_{23} - m_{32}) & -m_{22} - m_{33} & +i(m_{03} - m_{30}) & -i(m_{02} - m_{20}) \\ m_{02} + m_{20} & m_{12} + m_{21} & m_{00} - m_{11} & m_{23} + m_{32} \\ -i(m_{13} - m_{31}) & -i(m_{03} - m_{30}) & +m_{22} - m_{33} & +i(m_{01} - m_{10}) \\ m_{03} + m_{30} & m_{13} + m_{31} & m_{23} + m_{32} & m_{00} - m_{11} \\ +i(m_{12} - m_{21}) & +i(m_{02} - m_{20}) & -i(m_{01} - m_{10}) & -m_{22} + m_{33}. \end{pmatrix} \tag{52}$$

The coefficients λ_i from Cloude's expansion are the eigenvalues of $\mathbf{C}$. Cloude's nondepolarizing estimate is found from the eigenvector corresponding to the largest eigenvalue (λ_0) of $\mathbf{C}$. If this eigenvector is $\mathbf{c} = (c_0 \quad c_1 \quad c_2 \quad c_3)^T$, then the Jones matrix that represents the closest nondepolarizing estimate is given by:

$$\mathbf{J} = \begin{pmatrix} j_{00} & j_{01} \\ j_{10} & j_{11} \end{pmatrix} = \begin{pmatrix} c_0 + c_1 & c_2 - ic_3 \\ c_2 + ic_3 & c_0 - c_1 \end{pmatrix}. \tag{53}$$

Finally, it is enough to use Eq. (48) on Eq. (53) to find the elementary polarization properties of the nondepolarizing estimate.

Comparisons have shown that the results obtained by the matrix logarithm and by the analytic inversion method applied on a nondepolarizing estimate are very close, provided the depolarization is low and the multi-valued nature of the solutions is properly handled [47]. (If the original Mueller matrix is nondepolarizing, then both methods produce identical results.) The analytic method is computationally

faster and can process singular matrices that are not tractable by the logarithm-based method. Conversely, the logarithm-based method allows obtaining not only the elementary polarization properties but also their variances and covariances. Furthermore, the logarithm-based method is applicable to both weakly and strongly depolarizing Mueller matrices unlike the analytic inversion method which implicitly assumes a sufficiently weakly depolarizing Mueller matrix for its nondepolarizating estimate to be physically meaningful.

4.3 The Case of a Weakly Anisotropic Depolarizing Medium

The experimentally relevant case of both weakly anisotropic and weakly depolarizing medium, often encountered when studying certain biological tissues, allows for a more straightforward analysis [48]. It is applicable when the elementary polarization properties listed in Table 1 as well as their fluctuations are small ($\ll 1$).

In this particular case, the mean values of the elementary polarization properties can be directly obtained from

$$\langle \mathrm{LB} \rangle = 2(m_{32} - m_{23})/t, \tag{54a}$$

$$\langle \mathrm{LD} \rangle = -2(m_{01} + m_{10})/t, \tag{54b}$$

$$\langle \mathrm{LB}' \rangle = 2(m_{13} - m_{31})/t, \tag{54c}$$

$$\langle \mathrm{LD}' \rangle = -2(m_{02} + m_{20})/t, \tag{54d}$$

$$\langle \mathrm{CB} \rangle = 2(m_{12} - m_{21})/t, \tag{54e}$$

$$\langle \mathrm{CD} \rangle = -2(m_{03} + m_{30})/t, \tag{54f}$$

where t is the trace of the Mueller matrix ($t = m_{00} + m_{11} + m_{22} + m_{33}$).

In a weakly depolarizing medium the polarization properties exhibit only small fluctuations. In this case, the variances that quantify the extent of the different forms of depolarization, and the covariances that indicate the correlation between the different types of anisotropy, are given by

$$\langle |\Delta \mathrm{L}|^2 \rangle = 4(m_{00} + m_{11} - m_{22} - m_{33})/t - (\langle \mathrm{LB} \rangle^2 + \langle \mathrm{LD} \rangle^2), \tag{55a}$$

$$\langle|\Delta \mathrm{L}'|^2\rangle = 4(m_{00} - m_{11} + m_{22} - m_{33})/t - (\langle \mathrm{LB}'\rangle^2 + \langle \mathrm{LD}'\rangle^2), \tag{55b}$$

$$\langle|\Delta \mathrm{C}|^2\rangle = 4(m_{00} - m_{11} - m_{22} + m_{33})/t - (\langle \mathrm{CB}\rangle^2 + \langle \mathrm{CD}\rangle^2), \tag{55c}$$

$$\langle \Delta \mathrm{L} \Delta \mathrm{L}'^*\rangle = 4[m_{12} + m_{21} + i(m_{03} - m_{30})]/t - \langle \mathrm{L}\rangle\langle \mathrm{L}'^*\rangle, \tag{55d}$$

$$\langle \Delta \mathrm{L} \Delta \mathrm{C}^*\rangle = 4[m_{13} + m_{31} - i(m_{02} - m_{20})]/t - \langle \mathrm{L}\rangle\langle \mathrm{C}^*\rangle, \tag{55e}$$

$$\langle \Delta \mathrm{L}' \Delta \mathrm{C}^*\rangle = 4[m_{23} + m_{32} + i(m_{01} - m_{10})]/t - \langle \mathrm{L}'\rangle\langle \mathrm{C}^*\rangle. \tag{55f}$$

To summarize, in the case of a weakly anisotropic and depolarizing medium, one can safely simplify and speed up the data analysis by using the above equations.

5 Methods Based on Polarimetric Indices

The methods described in the previous sections are based on the assumption of a physical model for the studied system. Thus, methods based on product decompositions assume a particular sequence of basic optical elements, while methods based on differential analysis assume a continuous propagation in a uniform medium. The methods based on polarimetric indices that are discussed in this section do not originate from a certain model but rather rely entirely on more fundamental polarimetric principles. Owing to this fact, they have a very wide range of applicability, basically covering any kind of polarimetric measurement; furthermore, their algebraic implementation is very simple. Their main drawback is the smaller number of physically meaningful parameters they provide (with respect to that provided by the decomposition and differential approaches).

5.1 Depolarization Index

The depolarization index, commonly denoted DI (also known as the degree of polarimetric purity), provides a straightforward measure of how depolarizing the medium represented by **M** is. It is defined as [49]:

$$\mathrm{DI} = \sqrt{\frac{1}{3m_{00}^2}\left(\sum_{i,j=0}^{3} m_{ij}^2 - m_{00}^2\right)}. \tag{56}$$

The DI is a dimensionless parameter that is invariant with respect to changes of the reference frame and is restricted to the interval $0 \leq \mathrm{DI} \leq 1$. When $\mathrm{DI} = 1$ then $\mathbf{M}$ is nondepolarizing (or pure); when $\mathrm{DI} = 0$ then the medium is a perfect depolarizer meaning that there is no polarimetric preference and that any incoming light is fully depolarized (implying $m_{ij} = 0$ except for m_{00}).

5.2 Anisotropy Coefficients

The anisotropy coefficients [50] evaluate the extent to which the medium exhibits a different response to two incoming mutually orthogonal polarization states. Three anisotropy coefficients, α_L, $\alpha_{L'}$, and α_C, can be naturally defined along the three Stokes axes that quantify the amount of anisotropy between pairs of orthogonal states: horizontal (H) and vertical (V) linear polarizations; 45° and −45° linear polarizations and left and right-circular polarization. They are defined as:

$$\alpha_L = \sqrt{\frac{1}{\Sigma}[(m_{01} + m_{10})^2 + (m_{23} - m_{32})^2]}, \tag{57a}$$

$$\alpha_{L'} = \sqrt{\frac{1}{\Sigma}[(m_{02} + m_{20})^2 + (m_{13} - m_{31})^2]}, \tag{57b}$$

$$\alpha_C = \sqrt{\frac{1}{\Sigma}[(m_{03} + m_{30})^2 + (m_{12} - m_{21})^2]}, \tag{57c}$$

where only the positive signs of the roots are considered and

$$\begin{aligned} \Sigma = {} & 3m_{00}^2 - (m_{11}^2 + m_{22}^2 + m_{33}^2) \\ & + 2(m_{01}m_{10} + m_{02}m_{20} + m_{03}m_{30}) \\ & - 2(m_{23}m_{32} + m_{13}m_{31} + m_{12}m_{21}). \end{aligned} \tag{58}$$

Note that, as a result of the matrix symmetries in Eq. (2), a backscattering measurement will always result in $\alpha_{L'} = 0$. Moreover, any system with rotational symmetry, either in backscattering (Eq. (12)) or in transmission (Eq. (14)), will always feature $\alpha_L = \alpha_{L'} = 0$.

The anisotropy coefficients can be grouped into an anisotropy vector $\vec{\alpha} = (\alpha_L, \alpha_{L'}, \alpha_C)$, whose absolute value can be understood as a degree of anisotropy, DoA, [4] given by:

$$\mathrm{DoA} = |\boldsymbol{\alpha}| = \sqrt{\alpha_L^2 + \alpha_{L'}^2 + \alpha_C^2}. \tag{59}$$

Table 3 Nondepolarizing Mueller matrices having only one non-vanishing anisotropy coefficient

Nondepolarizing **M**		
$\alpha_L = 1, \alpha_{L'}{=}\alpha_C{=}0$	$\alpha_{L'} = 1, \alpha_L{=}\alpha_C{=}0$	$\alpha_C = 1, \alpha_L{=}\alpha_{L'}{=}0$
$\begin{pmatrix} 1 & m_{01} & 0 & 0 \\ m_{01} & 1 & 0 & 0 \\ 0 & 0 & m_{22} & m_{23} \\ 0 & 0 & -m_{23} & m_{22} \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & m_{02} & 0 \\ 0 & m_{11} & 0 & m_{13} \\ m_{02} & 0 & 1 & 0 \\ 0 & -m_{13} & 0 & m_{11} \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & m_{03} \\ 0 & m_{11} & m_{12} & 0 \\ 0 & -m_{12} & m_{11} & 0 \\ m_{03} & 0 & 0 & 1 \end{pmatrix}$
with $m_{01}^2 + m_{22}^2 + m_{23}^2 = 1$	with $m_{02}^2 + m_{11}^2 + m_{13}^2 = 1$	with $m_{03}^2 + m_{11}^2 + m_{12}^2 = 1$

The following inequality holds:

$$\mathrm{DoA} \leq \mathrm{DI} \leq 1. \tag{60}$$

The maximum value $\mathrm{DoA} = \mathrm{DI} = 1$ is reached if and only if **M** is a nondepolarizing Mueller matrix, indicating that DoA can be understood as a metric for depolarization. One can also define the degree of linear anisotropy, DoLA, as $\mathrm{DoLA} = \sqrt{\alpha_L^2 + \alpha_{L'}^2}$.

Table 3 reports the three nondepolarizing Mueller matrices having only one non-vanishing anisotropy coefficient (i.e., only α_L, only α'_L or only α_C), with their special symmetries.

If the parameter Σ vanishes, then the coefficients α_L, $\alpha_{L'}$, and α_C become undefined. This happens only for the following four diagonal Mueller matrices: $\mathrm{diag}(1, 1, 1, 1)$, $\mathrm{diag}(1, 1, -1, -1)$, $\mathrm{diag}(1, -1, 1, -1)$, and $\mathrm{diag}(1, -1, -1, 1)$. Any other diagonal matrix will produce vanishing anisotropy coefficients ($\alpha_L{=}\alpha_{L'}{=}\alpha_C{=}0$) indicating the absence of any anisotropy.

The information provided by each anisotropy coefficient can be summarized as follows:

1. α_L indicates a horizontal linear anisotropy: the medium exhibits distinct responses to H and V linear polarizations.
2. $\alpha_{L'}$ indicates a 45° linear anisotropy: the medium exhibits distinct responses to 45° and −45° linear polarizations.
3. α_C indicates a circular anisotropy: the medium exhibits distinct responses to left- and right-circular polarizations.

5.3 Indices of Polarimetric Purity

The indices of polarimetric purity (IPPs) [51, 52] are based on Cloude's decomposition reported in Eq. (51). They are invariant dimensionless quantities that contain information about the depolarization properties of the medium. They are defined as:

$$P_1 = \hat{\lambda}_0 - \hat{\lambda}_1, \tag{61a}$$

$$P_2 = \hat{\lambda}_0 + \hat{\lambda}_1 - 2\hat{\lambda}_2, \tag{61b}$$

$$P_3 = \hat{\lambda}_0 + \hat{\lambda}_1 + \hat{\lambda}_2 - 3\hat{\lambda}_3, \tag{61c}$$

where $\hat{\lambda}_i = \lambda_i/(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)$, λ_i being the eigenvalues of the coherency matrix $\mathbf{C}$ (Eq. (52)), ordered as $\lambda_0 \geq \lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$. The values of the IPPs are restricted by the series of inequalities

$$0 \leq P_1 \leq P_2 \leq P_3 \leq 1. \tag{62}$$

The DI can be expressed in terms of the indices of polarimetric purity,

$$\mathrm{DI}^2 = \frac{1}{3}\left(P_1^2 + \frac{2}{3}P_2^2 + \frac{1}{3}P_3^2\right). \tag{63}$$

For nondepolarizing Mueller matrices one has $P_1 = P_2 = P_3 = 1$ whereas for the perfect depolarizer $P_1 = P_2 = P_3 = 0$. The additional advantage of the IPPs with respect to the DI is that they provide direct information about the relative amount of equivalent nondepolarizing components. A list of simple phenomenological rules for their straightforward understanding is the following:

1. If the three IPPs are equal to 1, the Mueller matrix is nondepolarizing.
2. If only two IPPs are equal to 1 ($P_3 = P_2 = 1$), the system is equivalent to an incoherent composition of two nondepolarizing media.
3. If only one IPP is equal to 1 ($P_3 = 1$), the system is equivalent to an incoherent composition of three nondepolarizing media.
4. If none of the IPPs reaches 1, the system is equivalent to an incoherent composition of four nondepolarizing media.

It has been shown that the IPPs can be used to distinguish between different depolarization mechanisms in biological samples [53, 54], as for instance, those depending on the different spatial organization, concentration, or size of the scattering structures present in the tissues.

5.4 *Average Degrees of Polarization*

Given a Stokes vector $\mathbf{S}$ with components $\mathbf{S} = (S_0, S_1, S_2, S_3)^T$ the degree of polarization (DoP) is defined as

$$\mathrm{DoP}(\mathbf{S}) = \frac{\sqrt{S_1^2 + S_2^2 + S_3^3}}{S_0}. \tag{64}$$

The average degree of polarization (AverageDoP) associated with a Mueller matrix $\mathbf{M}$ can be defined as the arithmetic mean of the degrees of polarization of the exiting light when averaged over all incident polarization states. It can be calculated by numerical integration of the DoP of the product $\mathbf{MS}$ as the incident state $\mathbf{S}$ varies over the entire Poincaré sphere [55]. However, this calculation approach involves the use of numerical integration routines that can be difficult to implement and that often are too time-consuming for high-resolution image processing.

A more practical definition can be obtained by considering the following set of four Stokes vectors,

$$\mathbf{S}_A = \begin{pmatrix} 1 \\ 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}, \ \mathbf{S}_B = \begin{pmatrix} 1 \\ -1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}, \ \mathbf{S}_C = \begin{pmatrix} 1 \\ 1/\sqrt{3} \\ -1/\sqrt{3} \\ -1/\sqrt{3} \end{pmatrix}, \ \mathbf{S}_D = \begin{pmatrix} 1 \\ -1/\sqrt{3} \\ 1/\sqrt{3} \\ -1/\sqrt{3} \end{pmatrix} \tag{65}$$

that define a regular tetrahedron whose vertices lie on the surface of the Poincaré sphere. One can then define the AverageDoP of an experimental Mueller matrix $\mathbf{M}$ as the following discrete average:

$$\mathrm{AverageDoP}(\mathbf{M}) = \frac{\mathrm{DoP}(\mathbf{MS}_A) + \mathrm{DoP}(\mathbf{MS}_B) + \mathrm{DoP}(\mathbf{MS}_C) + \mathrm{DoP}(\mathbf{MS}_D)}{4}. \tag{66}$$

The AverageDoP can take values between 0 and 1 and, in most cases, is very close to the DI from Eq. (56), although it is not strictly equivalent to it. Nevertheless, it can be shown that for a diagonal depolarizer [Eq. (31)] the AverageDoP = DI.

An analogous approach can be used to define the average degree of linear depolarization (AverageDoLP). In this case, one can use the following Stokes vectors that are uniformly distributed on the equator of the Poincaré sphere:

$$\mathbf{S}_A^L = \begin{pmatrix} 1 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}, \ \mathbf{S}_B^L = \begin{pmatrix} 1 \\ -1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{pmatrix}, \ \mathbf{S}_C^L = \begin{pmatrix} 1 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{pmatrix}, \ \mathbf{S}_D^L = \begin{pmatrix} 1 \\ -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}, \tag{67}$$

$$\mathrm{AverageDoLP}(\mathbf{M}) = \frac{\mathrm{DoP}(\mathbf{MS}_A^L) + \mathrm{DoP}(\mathbf{MS}_B^L) + \mathrm{DoP}(\mathbf{MS}_C^L) + \mathrm{DoP}(\mathbf{MS}_D^L)}{4}. \tag{68}$$

Likewise, it is also possible to define the average degree of circular polarization (AverageDoCP) to evaluate the extent of depolarization with respect to circularly polarized light. In this case, it is enough to consider two Stokes vectors, one for each pole of the Poincaré sphere:

$$\mathbf{S}_A^C = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \ \mathbf{S}_B^C = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}, \tag{69}$$

$$\text{AverageDoCP}(\mathbf{M}) = \frac{\text{DoP}(\mathbf{M}\mathbf{S}_A^C) + \text{DoP}(\mathbf{M}\mathbf{S}_B^C)}{2}. \tag{70}$$

The AverageDoLP and the AverageDoCP likewise take values between 0 and 1. Their calculation (together with the AverageDoP or the DI) can be useful, as different scattering regimes lead to different depolarization effects [56]:

- AverageDoLP > AverageDoCP is typical of Rayleigh scattering,
- AverageDoLP < AverageDoCP is typical of Mie scattering,
- AverageDoLP $\sim$ AverageDoCP is typical of scattering in the Rayleigh–Mie transition regime.

6 An Experimental Example

In this section, we illustrate the data analysis methods described in the previous sections on a unique experimental example. Consider the Mueller matrix

$$\mathbf{M} = \begin{pmatrix} 1.0000 & -0.044 & -0.0019 & -0.0011 \\ -0.0312 & 0.1992 & 0.0010 & -0.0003 \\ 0.0007 & -0.0152 & -0.1911 & -0.0101 \\ 0.0001 & -0.0008 & 0.0159 & -0.1561 \end{pmatrix}, \tag{71}$$

which corresponds to a healthy pig skin sample measured in the backscattering configuration [57]. All algebraic calculations reported in the next subsections have been done on this matrix, however, without implying that the results provided by all of them are strictly necessary for the description of the physical picture behind this particular example.

The matrix given in Eq. (71) is close to a diagonal depolarizer matrix and satisfies, within the experimental error, the symmetries given in Eq. (2). Note also that the condition for single scattering given in Eq. (6) is far from being satisfied, which is a clear indication that multiple scattering is the predominant effect in this measurement.

6.1 Polar Decomposition

The forward polar decomposition [Eq. (15)] results in

$$\mathbf{M}_{\Delta} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -0.0225 & 0.1980 & -0.0082 & -0.0004 \\ 0.0004 & -0.0082 & 0.1919 & -0.0042 \\ 0.0002 & -0.0004 & -0.0042 & 0.1570 \end{pmatrix}, \tag{72a}$$

$$\mathbf{M}_{R} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.9993 & -0.0359 & -0.0064 \\ 0 & -0.0363 & -0.9965 & -0.0746 \\ 0 & -0.0036 & -0.0748 & -0.9972 \end{pmatrix}, \tag{72b}$$

$$\mathbf{M}_{D} = \begin{pmatrix} 1 & -0.0440 & 0.0019 & 0.0011 \\ -0.0440 & 1.0000 & 0.0000 & 0.0000 \\ 0.0019 & 0.0000 & 0.9990 & 0.0000 \\ 0.0011 & 0.0000 & 0.0000 & 0.9990 \end{pmatrix}. \tag{72c}$$

The diattenuator exhibits very weak vertical diattenuation ($-m_{D01} \gg m_{D02} \simeq m_{D03} \simeq 0$ within the experimental uncertainty). The retarder has weak elliptical retardance and is close to a horizontal half-wave plate. This is a direct consequence of the backscattering measurement configuration; if the signs of the last two rows of the retarder are flipped (together with those of the last two columns of the depolarizer, to preserve the decomposition), then the retarder would become close to the identity, like in a forward scattering measurement. Finally, the depolarizer features very strong depolarization ($m_{\Delta 00} \gg m_{\Delta 11} \sim m_{\Delta 22} > m_{\Delta 33}$) together with some weak vertical polarizance ($-m_{\Delta 10} > m_{\Delta 20} \simeq m_{\Delta 30} \simeq 0$). If the depolarizer is further subject to sign flipping (together with the retarder), then it will take the form of a strongly depolarizing mirror (with some weak vertical polarizance).

6.2 Symmetric Decomposition and Normal Form

The symmetric decomposition of Eq.(30) is:

$$\mathbf{M}_{\Delta} = \begin{pmatrix} 0.9998 & 0 & 0 & 0 \\ 0 & 0.2039 & 0 & 0 \\ 0 & 0 & -0.1867 & 0 \\ 0 & 0 & 0 & -0.1565 \end{pmatrix}, \tag{73a}$$

$$\mathbf{M}_{R1}=\begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 0.8364 & -0.5475 & -0.0245\\ 0 & 0.5480 & 0.8349 & 0.0508\\ 0 & -0.0073 & -0.0560 & 0.9984\end{pmatrix},\ \mathbf{M}_{D1}=\begin{pmatrix}1 & -0.0394 & 0.0020 & 0.0011\\ -0.0394 & 1.0000 & 0.0000 & 0.0000\\ 0.0020 & 0.0000 & 0.9992 & 0.0000\\ 0.0011 & 0.0000 & 0.0000 & 0.9992\end{pmatrix} \tag{73b}$$

$$\mathbf{M}_{R2}=\begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 0.8160 & 0.5771 & 0.0329\\ 0 & -0.5762 & 0.8078 & 0.1242\\ 0 & 0.0451 & -0.1204 & 0.9917\end{pmatrix}\ \mathbf{M}_{D2}=\begin{pmatrix}1 & -0.0234 & 0.0005 & 0.0002\\ -0.0234 & 1.0000 & 0.0000 & 0.0000\\ 0.0005 & 0.0000 & 0.9997 & 0.0000\\ 0.0002 & 0.0000 & 0.0000 & 0.9997\end{pmatrix}. \tag{73c}$$

Both diattenuators exhibit weak vertical diattenuation ($-m_{D01} \gg m_{D02} \simeq m_{D03} \simeq 0$ within the experimental uncertainty). The retarders contain spurious circular retardance, as seen from $m_{R12} = -m_{R21} \neq 0$. It results from the degeneracy of the diagonal depolarizer ($d_1 = -d_2$ within the experimental uncertainty). The removal of the spurious circular component in accordance with the backscattering analog of Eq. (37) ,

$$\mathbf{M}' = \mathbf{M}_{R2}\mathbf{M}_{\Delta}\mathbf{M}_{R1}^{T} = \mathbf{M}_{R2}\mathbf{R}(\gamma)\mathbf{M}_{\Delta}\mathbf{R}(\gamma)\mathbf{M}_{R1}^{T} \tag{74}$$

following the procedure from Ref. [27] yields

$$\mathbf{M}_{\Delta}=\begin{pmatrix}0.9998 & 0 & 0 & 0\\ 0 & 0.1984 & 0.0080 & 0\\ 0 & -0.0080 & -0.1922 & 0\\ 0 & 0 & 0 & -0.1565\end{pmatrix}, \tag{75a}$$

$$\mathbf{M}_{R1}=\begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 0.9995 & 0.0204 & -0.0245\\ 0 & -0.0191 & 0.9985 & 0.0508\\ 0 & 0.0256 & -0.0503 & 0.9984\end{pmatrix},\ \mathbf{M}_{R2}=\begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 0.9993 & 0.0155 & 0.0329\\ 0 & -0.0195 & 0.9921 & 0.1242\\ 0 & -0.0307 & -0.1248 & 0.9917\end{pmatrix}. \tag{75b}$$

(The two diattenuators are unaffected.) Now, both retarders are close to the identity and exhibit only very weak linear retardance ($m_{R23} = -m_{R32}$). The degeneracy of the depolarizer ($d_1 = -d_2$), following from its rotational invariance, is seen even more clearly; the appearance of two opposite off-diagonal terms does not question its diagonal nature since their magnitude is of the order of the experimental uncertainty. Physically, the diagonal depolarizer matrix with its last two negative elements represents a strongly depolarizing mirror with rotational symmetry.

Finally, the two normal form [Eq. (29)] nondepolarizing matrices, obtained by multiplying the respective pairs of diattenuators and retarders, are:

$$\mathbf{M}_{J1} = \begin{pmatrix} 1 & -0.0394 & -0.0019 & -0.0011 \\ -0.0393 & 0.9995 & -0.0191 & 0.0256 \\ -0.0027 & 0.0204 & 0.9978 & -0.0503 \\ -0.0003 & -0.0245 & 0.0508 & 0.9976 \end{pmatrix}, \tag{76a}$$

$$\mathbf{M}_{J2} = \begin{pmatrix} 1 & -0.0234 & -0.0006 & -0.0009 \\ -0.0234 & 0.9993 & 0.0155 & 0.0329 \\ -0.0003 & -0.0195 & 0.9918 & 0.1242 \\ -0.0001 & -0.0307 & -0.1248 & 0.9914 \end{pmatrix}. \tag{76b}$$

They are both close to vertical diattenuating retarders (or retarding diattenuators) with weak diattenuation and retardance.

6.3 Differential Methods

The differential methods assume propagation through the volume of the medium; usually, they are best suited to Mueller matrices measured in forward scattering. Consequently, we left-multiply the backscattering matrix from Eq. (71) by the diagonal matrix diag(1, 1, −1, −1) to assimilate it to a forward scattering matrix, as discussed in Sect. 2.1. In this backscattering configuration it is also understood that the polarization properties LD, LD', and CD describe diattenuation (instead of dichroism) and the properties LB, LB', and CB describe retardation (instead of birefringence).

6.3.1 Matrix Logarithm

The matrix logarithm yields

$$\mathbf{L} = \begin{pmatrix} -0.0017 & -0.0889 & 0.0041 & 0.0023 \\ -0.0629 & -1.6188 & 0.0053 & -0.0017 \\ 0.0004 & 0.0777 & -1.6527 & 0.0584 \\ 0.0000 & 0.0082 & -0.0928 & -1.8544 \end{pmatrix}. \tag{77}$$

Next, Eq. (45) are used to calculate $\mathbf{L}_m$ and $\mathbf{L}_u$. From $\mathbf{L}_m$ we get the mean values,

$$\langle \mathrm{CD} \rangle = 0.0011, \tag{78a}$$

$$\langle \mathrm{CB} \rangle = -0.0362, \tag{78b}$$

$$\langle \mathrm{LD} \rangle = 0.0760, \tag{78c}$$

$$\langle \mathrm{LB} \rangle = -0.0751, \tag{78d}$$

$$\langle \mathrm{LD}' \rangle = -0.0023, \tag{78e}$$

$$\langle \mathrm{LB}' \rangle = -0.0050, \tag{78f}$$

and, from $\mathbf{L}_u$, the variances,

$$\langle |\Delta \mathrm{L}|^2 \rangle = 1.8865, \tag{79a}$$

$$\langle |\Delta \mathrm{L}|'^2 \rangle = 1.8188, \tag{79b}$$

$$\langle |\Delta \mathrm{C}|^2 \rangle = 1.4154 \tag{79c}$$

of the spectroscopic polarization properties. The mean values of all properties are weak, with the linear retardation and diattenuation prevailing. According to Eq. (42) these linear optical effects they can be also presented as: $\langle \mathrm{LD}_m \rangle = 0.0760$, $\langle \mathrm{LB}_m \rangle = 0.0753$, $\langle \mathrm{LD}_\theta \rangle = -0.0151$, and $\langle \mathrm{LB}_\theta \rangle = -1.5376$. The variances of the three groups of properties are quite important, which is a direct consequence of the strong depolarization exhibited by the sample (the larger the depolarization, the larger the uncertainties of the properties). The covariances are rather small, indicating independently fluctuating properties.

6.3.2 Analytic Inversion

To apply the analytic inversion [Eq. (48)], it is necessary to calculate first a nondepolarizing estimate. Here, we calculate Cloude's nondepolarizing estimate. The largest eigenvalue of the associated coherency matrix $\mathbf{C}$ is $\lambda_1 = 0.3889$. The corresponding eigenvector is

$$\mathbf{c} = \begin{pmatrix} 0.9935 \\ -0.1063 + 0.0368i \\ -0.0010 + 0.0027i \\ 0.0010 - 0.0182i \end{pmatrix}, \tag{80}$$

which, according to Eq. (53), corresponds to the following Jones matrix:

$$\mathbf{J} = \begin{pmatrix} 0.8872 + 0.0368i & -0.0192 + 0,0016i \\ 0.0172 + 0.0036i & 1.0998 - 0.0368i \end{pmatrix}. \tag{81}$$

The polarization properties of this nondepolarizing estimate are

$$\mathrm{CD} = 0.0020, \tag{82a}$$

$$\mathrm{CB} = -0.0368, \tag{82b}$$

$$\mathrm{LD} = 0.2144, \tag{82c}$$

$$\mathrm{LB} = -0.0748, \tag{82d}$$

$$\mathrm{LD}' = 0.0020, \tag{82e}$$

$$\mathrm{LB}' = -0.0053. \tag{82f}$$

With respect to the matrix logarithm analysis, the mean values of all polarization properties are similar, except for the linear diattenuation that is significantly overestimated. This is due to the fact that, the experimental matrix being a strongly depolarizing one, its nondepolarizing estimate describes only partially its structure. This is seen from the relatively small magnitude (0.3889) of the largest normalized eigenvalue of the associated coherency matrix; for a nondepolarizing estimate to be able to substitute effectively for the initial depolarizing Mueller matrix, this magnitude should be close to the unit.

6.3.3 Weak Anisotropy Approximation

By using the weakly anisotropic depolarizing approximation [Eq. (54)] the following results for the mean values of the polarization properties are obtained:

$$\langle \mathrm{CD} \rangle = 0.0013, \tag{83a}$$

$$\langle \mathrm{CB} \rangle = -0.0184, \tag{83b}$$

$$\langle \mathrm{LD} \rangle = 0.0973, \tag{83c}$$

$$\langle \mathrm{LB} \rangle = -0.0336, \tag{83d}$$

$$\langle \mathrm{LD}' \rangle = -0.0016, \tag{83e}$$

$$\langle \mathrm{LB}' \rangle = -0.0014, \tag{83f}$$

whereas the variances are:

$$\langle |\Delta \mathrm{L}|^2 \rangle = 2.1932, \tag{84a}$$

$$\langle |\Delta \mathrm{L}|'^2 \rangle = 2.1619, \tag{84b}$$

$$\langle |\Delta \mathrm{C}|^2 \rangle = 1.9805. \tag{84c}$$

The mean values of the properties are clearly low which justifies the use of the approximation. However, their magnitudes are different from those found by the matrix logarithm method; in particular, the linear diattenuation is somewhat larger whereas the linear retardation is substantially smaller. Similarly, the magnitudes of all three variances slightly exceed those determined from the matrix logarithm. These discrepancies are due to the strong depolarization of the initial Mueller matrix lowering the accuracy of the approximation.

6.4 Depolarization Index

From Eq. (56):

$$\mathrm{DI} = 0.1863. \tag{85}$$

The DI value is substantially smaller than the unit, accounting for the strongly depolarizing nature of the sample.

6.5 Anisotropy Coefficients

By using Eq. (57):

$$\alpha_L = 0.0467, \tag{86a}$$

$$\alpha_{L'} = 0.0015, \tag{86b}$$

$$\alpha_C = 0.0095, \tag{86c}$$

$$\mathrm{DoA} = 0.04769. \tag{86d}$$

All anisotropy coefficients feature small values, in full accordance with the weak values of the polarization properties determined from the previous methods. In particular, linear anisotropy clearly prevails, in agreement with the predominance of the previously determined linear diattenuation and retardation.

6.6 Indices of Polarimetric Purity

According to Eq. (61):

$$P_1 = 0.1747, \tag{87a}$$

$$P_2 = 0.1909, \tag{87b}$$

$$P_3 = 0.2373. \tag{87c}$$

All three indices are much smaller than the unit, indicating a strongly depolarizing sample. Their values are relatively similar, revealing the essentially isotropic nature of the depolarization.

6.7 Average Degrees of Polarization

The average degrees of polarization of Eqs. (66), (68), and (70) are, respectively:

$$\mathrm{AverageDoP} = 0.1846, \tag{88a}$$

$$\mathrm{AverageDoLP} = 0.1964, \tag{88b}$$

$$\mathrm{AverageDoCP} = 0.1595. \tag{88c}$$

All three average degrees of polarization are weak and of relatively similar magnitudes. Like with the indices of purity, the last observation is an indication for a substantially isotropic depolarization.

7 Summary and Outlook

The ever growing interest for Mueller matrix polarimetry to biomedical applications is stimulating the development of novel methods for the interpretation and the analysis of experimental Mueller matrices. There is actually a clear demand for the development of analysis methods that are quantitative, noise resilient, of simple implementation and of straightforward phenomenological interpretation. Ideally, these methods should be used to relate the retrieved polarimetric parameters to clinically meaningful information that, for example, could be used in surgical and diagnostic applications.

Methods compatible with the study of optically thick tissues in backscattering mode are possibly among the ones having the greatest interest in future developments, as they offer compatibility with in-vivo measurements used in diagnostics or in optical biopsy. Among those, methods based on the polarimetric indices potentially have a wider field of application since they do not rely on specific physical models of the measured sample and are of straightforward applicability. However, most of them describe only the depolarizing properties of the sample.

Polarimetric (or Mueller matrix) imaging for biomedical applications is still in its infancy but definitely represents one of the most promising analysis tools in the this field. The future efforts on the clinical validation of the results provided by the various methods of analysis will be the key to qualify Mueller matrix polarimetry as an important quantitative tool for medical diagnostics.

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Mueller Matrix Polarimetry in Biomedicine: Enabling Technology, Biomedical Applications, and Future Prospects



Michael D. Singh, Nirmalya Ghosh, and I. Alex Vitkin

1 Introduction

The prospect of evaluating media without destroying or otherwise disturbing its properties has long been sought and readily adopted in biomedicine and beyond. Low-intensity light has played a particularly important role in the field of non-destructive evaluation (NDE) due to its rather unobtrusive interaction with matter, along with its numerous versatile properties that can carry important information about interrogated media. Indeed, the field of diagnostic biomedical optics has seen powerful exploitations of light, relying on its intensity (brightness) for imaging and wavelength (colour) for spectroscopy. Somewhat less prevalent but nevertheless actively researched and occasionally accepted into the NDE arsenal are field properties of the light wave, namely, its amplitude and phase, coherence, and polarization. In fact, these field properties, coherence and polarization, are more fundamental and have played important roles in our understanding of the transverse character of the electromagnetic wave, angular momentum of light or photons, and a number of non-trivial optical phenomena related to the interference of light waves.

The polarization property of light, which describes the oscillatory nature of its electric or magnetic fields, however, has far from reached its potential in

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biomedicine. This may come as a surprise since polarimetry – the science of measuring light's polarization and its interaction with matter – has been actively studied for more than two centuries [1] and has already enjoyed successful deployment in the semiconductor industry as an NDE tool for many decades [2]. The powerful potential of polarimetry in biomedicine arises from the well-known polarimetric effects of tissues, namely, linear retardance and diattenuation due to linear birefringent and dichroic microstructures, optical activity (circular retardance and circular diattenuation) due to chiral structures, and depolarization primarily due to multiple scattering and other randomization effects – all of which have been demonstrated to carry important tissue information and have found important early applications [3]. However, depolarization is both a blessing and a curse for tissue polarimetry: while potentially of diagnostic value, its effect can be overwhelming due to the complex heterogeneous nature of tissues such that polarized light signals are effectively extinguished (bulk tissue depolarization), thereby seriously hindering polarimetry's adoption in biomedicine. There are, however, recent promising developments of new technologies and methodologies that can detect the often faint polarimetric signals, and thus extract fruitful tissue information. These advances have especially revived the interest in measuring Mueller matrices, which comprehensively describe the polarization properties of interrogated materials.

The Rise of Mueller Matrix Polarimetry in Biomedicine

In 1852, Sir George Gabriel Stokes introduced four measurable *intensity* quantities, now known as Stokes parameters, which completely described and enabled the observation of the *polarization behaviour of light* [4]. The Stokes parameters of both incident and scattered (post-interaction) light from a medium can then be related through a 4×4 matrix which fully encapsulate *the medium's polarization response*. These matrices were first formulated by Paul Soleillet in his 1929 thesis [5]; the calculus of the matrices was introduced by Francis Perrin in 1942 [6], and then developed in considerable detail by Hans Mueller beginning in 1943 [7], thus earning the name 'Mueller matrix' (MM) [8]. In parallel, Robert Clark Jones also developed a polarization matrix calculus in 1941 in a series of papers based on a field formulation (*amplitude* and *phase*) [9–12]. Jones calculus has remained fundamentally important, specifically in the context of establishing a link between the classical description of the polarization of electromagnetic waves and the fundamentally much richer quantum mechanical description of the polarization state vector for the photon. It is however limited to completely polarized light interactions and is not readily adaptable for describing depolarization events that occur in most heterogeneous materials such as biological tissues; thus, Jones matrix calculus has been largely supplanted by the more powerful and experimentally applicable Mueller matrix calculus [8].

Early uses of MM polarimetry included compositional analysis of atmosphere [13] and seawater [14], as well as astronomical observation [15, 16]. In biology, a seminal paper in 1976 by Bickel et al. [17] reported on a 'new biophysical tool' referring to their Mueller measurements of cells suspended in liquid, whose

elements provided information related to cellular structure and size. These early Mueller applications used previously existing technologies adapted to polarimetric implementations, such as nephlometers and photometers, or new technologies altogether. For example, the latter cellular study was enabled by a sensitive new device known as the photoelastic modulator (PEM) [18, 19], used in tandem with lock-in amplification for synchronous high-SNR detection of polarized light. It became obvious that Mueller polarimetry, with the help of developing technology, afforded useful NDE information about scattering media, potentially suitable for biomedical diagnostics.

Interest then arose in using MM polarimetry towards assessment of biological tissues, particularly for pathological detection; however, the complexity of biological media and severe polarization loss were problematic, and thus many initial biomedical studies were limited to cell suspensions [20] and turbid phantoms which mimicked tissue scattering (e.g. microsphere suspensions) [21–24]. One of the first MM measurements of actual tissue were those of human eyes *post-mortem* by Dreher et al. in 1992 [25], benefitting from the reduced multiple scattering and less complex polarization effects in such nearly transparent ocular media. The authors discussed the detection of glaucoma and other optic neuropathies by measuring the linear retardance, optic-axis orientation, and linear dichroism of retinal nerve fibres. More complex tissues which (1) cause extensive polarization loss and (2) exhibit multiple polarimetric effects simultaneously (e.g. depolarization, linear/circular birefringence, and dichroism) posed a greater challenge for MM polarimetry on both measurement (1) and analysis (2) fronts. With respect to the latter, since the MM is a polarization transfer function which effectively 'lumps' all the polarimetric effects of the interacting media in a complex way into its 16 elements, additional interpretation methods must be applied to the resultant MMs to tease these out individually. Numerous interpretation methods have been introduced in recent decades, based on various forms of matrix decomposition [26–29]. The polar decomposition approach proposed by Lu and Chipman [30] in 1996 was the first method applied to MM polarimetry of tissue in a 2000 study by Smith et al. [31] which identified cancerous and lupus lesions of the skin. It remains arguably the most common MM decomposition method in tissue polarimetry at present, but the matter is not settled and active research into this interesting avenue of inquiry continues to this day [32, 33].

Since these early studies, technologies and methodologies have progressed to enable more advanced demonstrations of tissue assessment through MM polarimetry in various biomedical scenarios, for example, (pre)cancer [34], bladder obstruction [35, 36], infarcted and stem-cell-regenerated myocardium [37], and glucose sensing [38–41]. Both thin-tissue transmission methods and bulk-tissue reflection geometries have been attempted, with the latter being more challenging due to greater depolarizing noise but also more biomedically relevant for eventual in-vivo clinical deployment. These demonstrations have mostly used free-space optics, since performing MM polarimetry through flexible fibre optics (e.g. medical endoscopes) introduces the significant artefact of large and variable fibre-induced strain birefringence. Using rigid endoscopes prevents such probe noise from drowning out

the tissue polarization signature, but rigid endoscopy is limited in its clinical scope [42]. Recent developments in advanced normalization methods for flexible MM endoscopy have also shown some promise [43–45]; if clinically realized, this fibre-optic-based approach may prove impactful by opening up previously inaccessible anatomical sites in the context of surgical guidance, early detection, diagnostics, and surveillance.

An interesting overall system design consideration is whether polarimetry (or any NDE diagnostic technique for that matter) should be used as a stand-alone modality or be combined with another interrogation technique that carries additional (and ideally complimentary) tissue information. The 'correct' decision on stand-alone versus hybrid approaches is clearly dependent on the biomedical information being sought, potential complementarity of the candidate techniques, technological complexity, and so on. This is particularly relevant to polarized light NDE, as some polarimetric implementations are technologically simple, and thus can be readily placed into the existing optical paths of other optical modalities, adding rich polarimetric information at 'minimal cost'. Thus, hybrid optical approaches have been demonstrated which implement MM polarimeters in combination with multispectral imaging [46–48], optical coherence tomography (OCT) [49–52], microscopy [53], mass spectrometry [54, 55], and possibly others.

MM polarimetry is driven by unmet scientific and clinical needs and is very much dependent on emerging state-of-the-art technologies and methodologies to enable various exciting applications in biomedicine, with promise to join the rank of the other more established biophotonics-based diagnostic techniques. This chapter reviews the current and prospective biomedical applications of MM polarimetry, along with the enabling technological advances that can further improve and realize potential exciting uses in the life sciences.

2 Mueller Polarimetry: Concepts and Methods

2.1 What Is Polarized Light?

In classical electromagnetic theory, polarization of light is described through the evolution of transverse electric ($\boldsymbol{E}$) and magnetic ($\boldsymbol{B}$) fields as a function of time at a given point in space. The $\boldsymbol{E}$ and the $\boldsymbol{B}$ fields sinusoidally oscillate in the xy plane, while the wave vector ($\boldsymbol{\beta}$) is directed along the propagation direction, z-axis. Since the triplet of vectors $\boldsymbol{E}$, $\boldsymbol{B}$, and $\boldsymbol{\beta}$ are mutually orthogonal, it suffices to look only at the behaviour of the vector $\boldsymbol{E}$. The wave may be considered as the superposition of two waves having orthogonal components E_x and E_y as

$$\boldsymbol{E}\,(z,t) = \boldsymbol{E_x}\,(z,t) + \boldsymbol{E_y}\,(z,t)\,, \tag{1}$$

with

$$\boldsymbol{E_x}(z,t) = E_{0x}\cos(\omega t - \beta z + \phi_x)\,\hat{\boldsymbol{x}}, \tag{2}$$

$$\boldsymbol{E_y}(z,t) = E_{0y}\cos\left(\omega t - \beta z + \phi_y\right)\hat{\boldsymbol{y}}, \tag{3}$$

and

$$\beta = \frac{2\pi n}{\lambda} \text{ and } \omega = 2\pi f, \tag{4}$$

where λ is the wavelength of light, n is the refractive index of the host medium, f is the oscillation frequency of the monochromatic electric field, and ϕ is the phase. If the trace of the vector extremity $\boldsymbol{E}$ follows a stationary curve in their temporal evolution, the wave is said to be completely polarized. Accordingly, the shape of the curve traced out by $\boldsymbol{E}$ vector defines the polarization state of the wave. The electric field vector, $\boldsymbol{E}$, traces out an ellipse through space whose shape may vary between perfectly linear or perfectly circular forms. Note that in the corresponding quantum mechanical description, it is assumed that each individual photon is polarized, and its associated state vector corresponds to one of the classical polarization states described by the Jones vector. In the limit of a large number of photons, their collective behaviour is consistent with the classical description of polarization ellipse. Though this polarization ellipse may describe light's polarization, it becomes impractical due to its dependency on the amplitude of $\boldsymbol{E}$ which is traced at the frequency of the light's electric field oscillation, on the order of 10^{-15} s. Furthermore, the polarization ellipse is applicable only to completely polarized and perfectly coherent light waves – an exception rather that the rule in most practical applications – and thus in order to describe partially or completely unpolarized light, one needs to incorporate the decoherence effect through time averaged and practically measurable quantities such as intensity of light. A more suitable formalism thus describes the *degree of polarization* (DOP) of light which ranges from completely unpolarized to completely polarized using directly observable and measurable polarization resolved intensities.

2.2 Stokes Vectors: Describing Polarized Light

Polarization behaviour is observable by using polarization modulating elements in certain configurations to measure the resultant time-averaged intensity (the square of the electric field) rather than the rapidly revolving electric field amplitude and the associated phase. Stokes vectors do just that by using linear and circular polarizers/analysers. Specifically, the difference in intensities between orthogonal linear and circular polarization states is measured and compared to the values of

total light intensity in order to quantify the linear, circular, and total DOPs. Note that Stokes vectors are not 'magnitude + direction' vectors in the conventional sense but are simply 4×1 arrays of real-valued numbers. The Stokes vector may however be represented as a directional vector in the polarization state space, in the so-called Poincaré sphere (not discussed here). They consist of four measurable quantities (Stokes parameters) which are the total intensity I, the amount of linear horizontal or vertical polarization Q, the amount of linear $+45°$ or $-45°$ polarization U, and the amount of right-circular or left-circular polarization V:

$$S = \begin{bmatrix} I \\ Q \\ U \\ V \end{bmatrix} = \begin{bmatrix} I_{\parallel} + I_{\perp} \\ I_{\mathrm{H}} - I_{\mathrm{V}} \\ I_{+45} - I_{-45} \\ I_{\mathrm{R}} - I_{\mathrm{L}} \end{bmatrix}, \tag{5}$$

where I_{H}, I_{V}, I_{+45}, I_{-45}, I_{R}, and I_{L} are the light intensities measured by a horizontal linear analyser, a vertical linear analyser, a linear analyser oriented at $+45°$, a linear analyser oriented at $-45°$, a right circular analyser, and a left circular analyser before reaching the detector, respectively. Additionally, $I_{\parallel}$ and $I_{\perp}$ are any two mutually orthogonal polarization states.

The degrees of linear, circular, and total light polarization can be calculated using the Stokes parameters according to

$$\mathrm{DOP_L} = \frac{\sqrt{Q^2 + U^2}}{I}, \tag{6}$$

$$\mathrm{DOP_C} = \frac{\sqrt{V^2}}{I}, \tag{7}$$

$$\mathrm{DOP} = \frac{\sqrt{Q^2 + U^2 + V^2}}{I}. \tag{8}$$

Importantly, Stokes vectors describe the polarization state of the light, but they do not describe the polarization properties of an interacting material.

2.3 Mueller Matrix: Describing Material Polarization Properties

The polarization states of light before and after their interaction with a medium are represented by Stokes vectors. A Mueller matrix, M, describes the polarimetric characteristics of the interacting medium and relates the incident and post-interaction Stokes vectors through

$$S_{\text{out}} = \begin{bmatrix} I \\ Q \\ U \\ V \end{bmatrix}_{\text{out}} = \begin{bmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{34} & M_{44} \end{bmatrix} \times \begin{bmatrix} I \\ Q \\ U \\ V \end{bmatrix}_{\text{in}} = M \cdot S_{\text{in}}. \tag{9}$$

Encapsulated within the MM are all the polarization properties of a sample, namely diattenuation, polarizance, retardance, and depolarization. As such, when tissue is interrogated by polarized light, the measured MM will contain all possible obtainable information about it. Such information includes the material properties described below.

Polarizance refers to a material's ability to increase the DOP of light, whereas *diattenuation* occurs when an interacting medium attenuates the orthogonal components of light's electric field unequally, resulting in preferential transmittance of a certain polarization orientation. The former is described in the Mueller matrix's first column $[M_{11}\ M_{21}\ M_{31}\ M_{41}]^T$ and the latter by the first row $[M_{11}\ M_{12}\ M_{13}\ M_{14}]$. These effects generally occur at material interfaces where the transmittance and reflectance are dependent upon the incident polarization state or in anisotropic materials that preferentially absorb certain polarization states such as linearly or circularly polarized light (referred to as *linear* and *circular dichroism*, respectively). Single scattering is also known to impart polarizance and diattenuation signatures on polarized light when observed at certain angles in a scattering plane.

Retardance occurs when a medium alters the phase-difference between the orthogonal components of light's electric field, represented by the sub-matrix $[M_{22}\ M_{23}\ M_{24}; M_{32}\ M_{33}\ M_{34}; M_{42}\ M_{43}\ M_{44}]$. Materials that produce this effect are known as *birefringent*. Note that retardance and diattenuation are the result of differences in the refractive indices for different polarization states. These are described through the ordinary and extraordinary axes and indices. Difference in the real parts of the said indices result in linear and circular birefringence (retardance or phase difference between orthogonal polarizations), while the difference in the imaginary parts gives rise to the linear and circular dichroism (diattenuation or differential attenuation of orthogonal polarizations). Generally, fibrous tissues such as collagen and tendons are linearly birefringent. As noted above, circular retardance, commonly referred to as *optical activity*, then affects the phase-difference between those circular states. This process maintains the light's linear ellipticity, rotating its plane about the propagation axis by an angle proportional to the *rotatory power* of the material. Glucose – an essential molecule for life – is *chiral* and thus optically active, allowing polarized light to probe its molecular concentrations (as has been routinely done in the sugar industry (glucometry) [56, 57]).

Depolarization refers to the scrambling of the photon polarizations which comprise a light beam. Materials generally depolarize light via multiple scattering which occurs at inter-medium interfaces that bring about abrupt changes to the refractive index, possibly randomizing the polarization orientation of the interacting photons. Scattering occurs in virtually all biotissues due their heterogeneous structure and composition. Depolarization may also occur in birefringent media with randomly

oriented microdomains that act as incremental retarders to randomize the phases of incident photons [58]. Depolarization can thus be a useful metric in probing tissues due to its prevalence, albeit when its effects are manageable. A material that depolarizes light is known as a depolarizer. An ideal depolarizer has MM of the form:

$$M_{\mathrm{D}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & b & 0 \\ 0 & 0 & 0 & c \end{bmatrix}, \tag{10}$$

where $|a|$, $|b|$, $|c|$ are all ≤ 1. Thus, the Stokes vector of light, S_{in}, incident upon the depolarizer will be transformed to S_{out} as

$$S_{\mathrm{out}} = M_{\mathrm{D}} \cdot S_{\mathrm{in}} = \begin{bmatrix} I \\ aQ \\ bU \\ cV \end{bmatrix}_{\mathrm{out}}, \tag{11}$$

where $\mathrm{DOP}_{\mathrm{out}} \leq \mathrm{DOP}_{\mathrm{in}}$.

These polarimetric effects occur simultaneously in complex media such as biological tissues and are thus 'lumped' into a single Mueller matrix. Therefore, suitable interpretation methods must be employed to decouple the effects and provide meaningful biophysical information. Polar decomposition proposed by Lu and Chipman [30] has been used extensively in Mueller bio-polarimetry whereby it serially decomposes a resultant Mueller matrix into an ordered sequence of depolarizer, retarder, and diattenuation sub-matrices, enabling separate extraction of these tissue properties. Other alternatives such as differential decomposition [26] model polarimetric effects more realistically as continuously occurring during light propagation rather than sequentially. It however relies on logarithmic operations which are formidable when employing reflectance configurations due to resultant negative eigenvalues of Mueller matrix elements (e.g. helicity-flipping [21]). There are also emerging interpretation methods such as Mueller transformation [59, 60] and machine learning [61–63] that may enable improved extraction of novel polarimetric parameters.

2.4 Mueller Matrix Measurement

In order to measure the Mueller matrix of a sample, the incident and post-interaction light must be modulated with certain polarization states using a polarization state generator (PSG) and polarization state analyser (PSA), respectively, before impinging onto a detector. To determine the complete 4×4 matrix, four polarization

states must be generated and analysed, out of the six possible ones: vertical (V) linear, horizontal (H) linear, $+45°$ (P^+) and $-45°$ (P^-) linear, and right- (R), and left-circular (L). For example, using the four states H, V, P^+, and R, a 'data matrix' (not a Mueller matrix) may be acquired [58]:

$$\begin{bmatrix} HH & HV & HP^+ & HR \\ VH & VV & VP^+ & VR \\ P^+H & P^+V & P^+P^+ & P^+R \\ RH & RV & RP^+ & RR \end{bmatrix}, \tag{12}$$

where the first and second symbols, respectively, indicate the generated and analysed polarization states; for example, VH is the signal when vertical linear polarization generated by the PSG illuminates the sample and the remitted light is sampled through horizontal linear polarization orientation of the PSA. From the data matrix, the Mueller matrix may be determined by an algorithm, such as the one introduced by Yao and Wang [49].

A schematic view of the general MM measurement scheme using four such states is shown in Fig. 1. The classical (and simplest) method to generate and analyse polarization states is by discrete settings of linear polarizers and phase retarders (e.g. quarter wave plates). This is however slow, cumbersome, and susceptible to error when adjusted manually. Thus, more advanced devices have been developed that operate as a function of time (dynamic modulation) or in the spectral/spatial domain. The former is generally achieved through dynamic retarding components

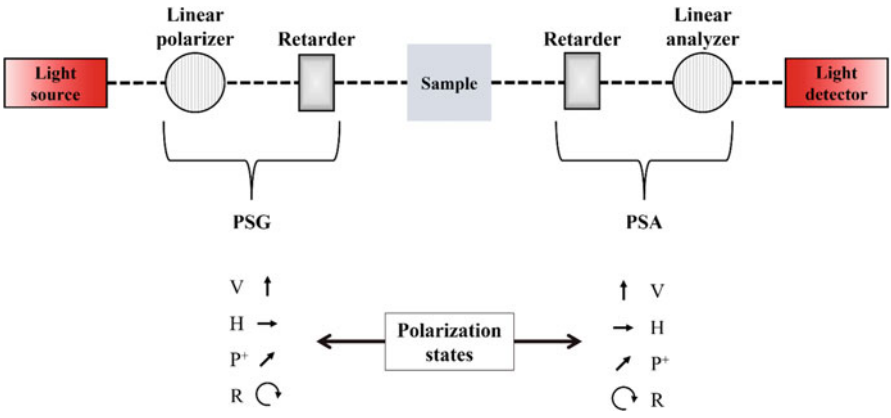


Fig. 1 General setup for measuring the Mueller matrix of a sample. A polarization state generator (PSG) generates different polarization states of light that are incident on the sample. The post-interaction light is analysed using a polarization state analyser with its components in the opposite order of the PSG before transmitting to a light intensity detector. In this example, four polarization states are generated and analysed to produce a data matrix (see Eq. 12) from which the Mueller matrix is determined using a suitable algorithm. Transmission geometry is shown for illustrative purposes only

such as motor-rotated quarter wave plates, PEMs, and liquid crystal retarders. PEMs typically consist of transparent slabs of quartz which are compressed and stretched periodically to produce a time-varying strain birefringence; thus, incident +45° linearly polarized light will have its polarization state periodically and continuously varying between unaltered +45° linear (when PEM strain = 0) and elliptical/circular (when PEM strain is set to its user-controlled maximum). Importantly, a lock-in amplifier can then be used to 'lock-in' on the PEM modulation frequency and its harmonic to synchronously detect polarized light signals amid significant backgrounds of depolarized 'noise' (thus enabling high SNR detection). Liquid crystal cells are also controlled electronically whereby the orientation of its molecules are modulated by an applied voltage, changing its birefringence, and thus dynamically retarding the phase of incident polarized light. These two devices are advantageous for tissue polarimetry since they are electronically controlled with no moving parts (robust) and exhibit millisecond response times (minimizing motion artefacts). Alternatively, so-called snapshot systems modulate polarized light using diffractive elements to generate interference fringes which can be analysed via Fourier analysis in the frequency domain. Comparatively speaking, snapshot systems perform the fastest Mueller measurements, however at the cost of limited image resolution [64].

Spectroscopic polarimeters yield polarization signatures as a function of wavelength and may offer additional insights in biomedicine such as early detection of epithelial cancers (based on a wavelength-dependent oscillatory component of backscattered light) [65, 66]. An interesting extension of this approach is to also measure polarization of the fluorescence emission which can provide biochemical information [67]. A representative system is illustrated in Fig. 2. For elastic scattering measurements, a broadband light source, such as a Xe lamp is used, while for fluorescence spectroscopic measurements, laser sources having desirable excitation wavelengths are used. As per above, the Mueller matrix measurement strategy is based on 16 (spectrally resolved) intensity measurements performed by sequentially generating and analysing four elliptical polarization states. This approach enables recording of spectroscopic Mueller matrices MM(λ) over a broad spectral range (e.g. $\lambda = 400$–800 nm) with high accuracy and sensitivity. For MM fluorescence spectroscopy, the samples are excited with a chosen wavelength (e.g. $\lambda_{ex} = 405$), and the emitted fluorescence spectra from the sample are recorded by a spectrometer (e.g. $\lambda_{em} = 450 - 750$ nm). An important point is that the generator matrix corresponding to the excitation wavelength is used and the analyser matrices are taken for the entire emission wavelengths. Results of fluorescence spectroscopic measurements from biological tissues using this experimental system are presented and discussed later.

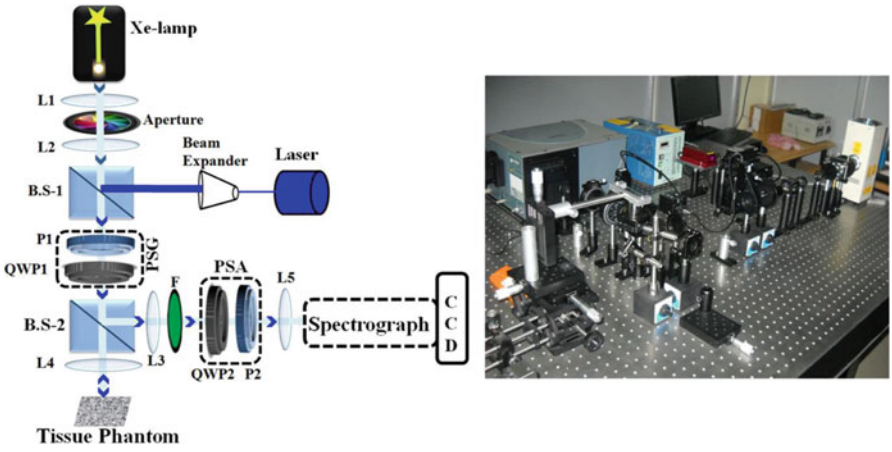


Fig. 2 Schematic of the experimental systems for elastic scattering and fluorescence Mueller matrix measurements in the exact backscattering configuration. P1 and P2 are linear polarizers; QWP1 and QWP2 are rotatable achromatic quarter wave plates; L1, L2, L3, L4, and L5 are lenses; BS-1 and BS-2 are beam splitters. P1, QWP1 form the PSG unit and P2, QWP2 form the PSA unit. The Xe-lamp is used as the excitation source for recording elastic scattering spectral Mueller matrices, whereas the 405 nm line of a diode laser is used as the excitation source for recording fluorescence spectral Mueller matrices. A long pass filter F is used to block the laser line for fluorescence measurements. A photograph of the experimental system is shown on the right

2.5 Experimental Design and Information Content Considerations

Polarimetry is a flexible NDE technique that can be implemented in many ways to extract useful information about biological tissues, with pros and cons associated with each implementation. For example, one must consider the polarization modulation/detection system and measurement geometry. Polarized light signals typically become unmeasurable after several millimetres of propagation in most tissues (~4–6 mm [68–70]) due to strong tissue scattering and depolarization effects. Generally speaking, transmission-type geometries are limited to thin tissue samples, whereas reflectance-based geometries are the only available configuration for bulk tissues. For the latter, it may be useful to determine the effective polarimetric sampling depth, for example through modelling and simulations [71, 72]. The other detection consideration is whether spatial polarimetric information is desired. If so, an *imaging* mode must be employed using a grid of photosensitive elements such as a CCD detector, otherwise, the light may be detected with a *point-sensor* (e.g. using a photodiode). Imaging may be preferable in many biomedical applications, however, it is slower than point-sensing and often challenging to perform while maintaining high SNR detection. In the case of the PEM, for example, its harmonic frequencies are synchronized with the intensity signal from a single photosensitive

element for high SNR synchronous detection, but major synchronization challenges arise when introducing a grid of imaging photodetectors.

On the information side, the major consideration is technique simplicity/complexity versus the resultant information content; the optimum compromise on this continuum is often task-specific, that is driven by the specific biological or clinical question to be addressed. In general, MM approaches are on the complex end of the polarimetry measurement/analysis spectrum, requiring at least 16 separate experimental polarization configurations to be recorded such that all elements of the MM can be derived. The benefit of this added complexity is that one obtains the complete polarization response of the interrogated material, and the resultant biophysical properties derived from MMs – for example retardance, optical rotation, diattenuation, and depolarization – are indicative of material biophysics only. This is in contrast with simpler polarimetric approaches such as crossed linear polarizers and various forms of Stokes polarimetry/ellipsometry, generally yielding fewer measured parameters that now depend on both material properties *and* (generally uninteresting) polarization optics of the measurement system. For each biomedical application then, one should consider the sufficiency of the obtained information content and whether the 'contamination' of results with experimental system details is acceptable. It is important to point out that measurement complexity considerations (and associated speed and cost) may become moot as technology advances to yield simpler/faster/cheaper MM measurements and analysis systems. We revisit these topics in the next section after reviewing various biomedical examples.

3 Biomedical Applications of Mueller Polarimetry

Many adverse physiological processes are associated with morphological and compositional changes in tissues which affect their polarization properties and enable polarimetric detection. Depolarization and linear retardance have found the most diagnostic use thus far since they are the strongest occurring bio-polarimetric effects due to the strong scattering effects of most tissues, along with the presence of birefringent fibrous structures. Diattenuation and dichroism appear much weaker and have not yet been of much biomedical use. Optical rotation (circular retardance) is also rather weak in biotissues due to modest concentrations of chiral molecules, but may eventually prove useful for non-invasive glucose monitoring of diabetic patients which remains one of the most important unsolved problems in clinical medicine [38, 73].

Polarimetric signals are cumulatively sampled over depth (as most polarimeters do not offer depth discrimination capabilities) and this complicates the interpretation of the resultant measurements in heterogeneous tissues (especially in the bulk-tissue reflectance geometry) [70]. The measurements may be further conflated when employing the abovementioned *non-Mueller* techniques due to their admixing of experimental parameters. MM polarimetry, with its ability to exclusively describe

sample properties and to decouple the 'lumped' polarization effects into their biophysically interpretable 'basis' properties, is thus beginning to demonstrate exciting and promising biomedical diagnostic capabilities. Heart infarctions, for example, cause significant remodelling of the myocardium, as do corresponding stem cell regenerative treatments, enabling both to be identified polarimetrically via Mueller retardance imaging [74]. Bladder obstruction can also be assessed through retardance imaging due to the induced strain on the bladder walls (strain birefringence) [35, 36]. But oncology has been the most actively explored medical field by the polarimetry community, driven by the importance of cancer detection/diagnosis/staging problems, and the prominent polarimetric changes due primarily to cellular and stromal (pre)malignant transformations associated with cancer development and growth [75–77]. In this section, we thus focus on detection and characterization of pre-cancer and cancer via several illustrative examples. A more complete list of polarimetry's presence in biomedicine can be found in the recently published comprehensive review [78].

3.1 Mueller Polarimetry and Pre-cancer/Cancer

Pre-cancer-to-cancer progression is marked by well-known changes to cellular concentration (cellular proliferation) which affects light-scattering [65, 66] and hence depolarization [20, 79, 80], as well as changes to collagen organization which affects both linear birefringence [81, 82] and depolarization [83]. Polarimetric contrast can thus be generated between (pre)cancerous and non-cancerous tissues. Early Mueller polarimetry experiments in the 2000s demonstrated this and highlighted its potential for clinical use, such as the study by Smith et al. [31], which identified malignant moles and lupus lesions of skin in vivo via retardance and depolarization images. Similar work was performed by Chung et al. [84] which demonstrated polarimetric diagnosis of oral pre-cancers in hamster cheek pouches in vivo. Since then, there have been several studies in various configurations to investigate and advance MM diagnostics of pre-cancer and cancer.

Mueller Imaging of Thin Tissues: Transmission-Based Configurations
Although limited in clinical scope relative to bulk tissues studies, thin tissues enable more detailed polarimetric assessment through higher SNR signals (minimal scattering), the ability to section tissue to isolate regions of interest without depth interference and availability of correlative ground-truth histology [85]. Notably, the latter is actually expensive and often subjective and difficult to reproduce [86, 87]. Polarimetry may thus improve upon current methods with its label-free, cost-effective, and quantitative (objective) characteristics. An important transmission-based thin tissue study was performed by Shukla et al. [82] which investigated the polarization responses of epithelial and stromal tissue of cervical dysplasia (pre-cancer). As seen in Fig. 3, depolarization is sensitive to cellular dysplasia of the epithelium (top row) and retardance is sensitive to stromal changes (bottom

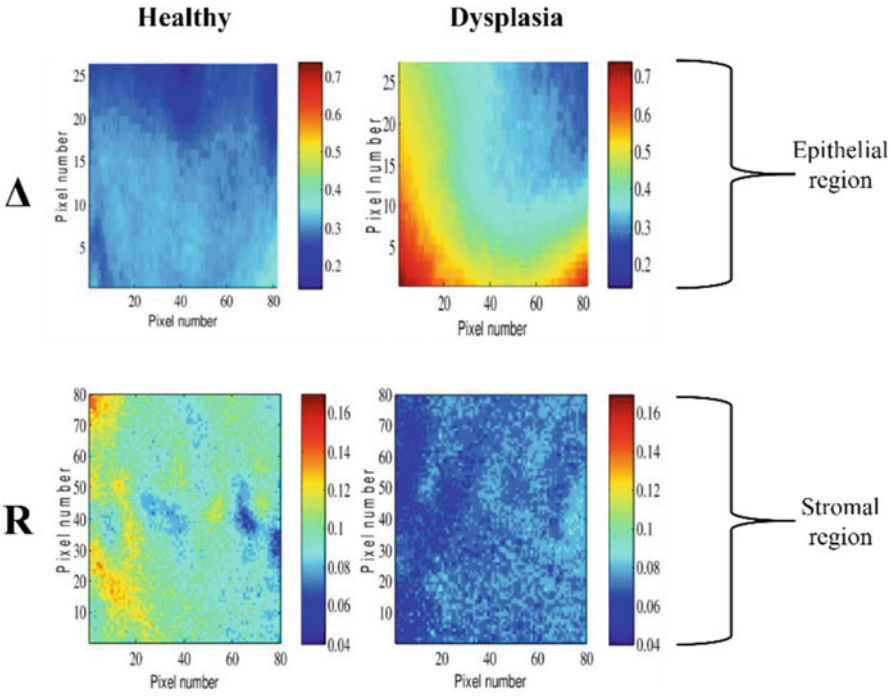


Fig. 3 Depolarization images (Δ, top row) of healthy and dysplastic epithelium regions of cervical tissue; note the increasing depolarization in the latter. Corresponding retardance images (R, bottom row) of healthy and dysplastic tissues in the stromal regions of cervix; decreasing retardance in dysplastic stroma is clearly seen. The tissue dimensions were 4 mm × 6 mm and 5 μm thick. (Adapted from [82])

row). Since epithelia are typically devoid of fibrous (birefringent) structures, the depolarization contrast arises from the pleomorphic and proliferative alterations to cells which affect the scattering anisotropy (related to scatterer size) and scattering coefficient (essentially the *turbidity*, related to scatterer concentration). In the connective tissue compartment, healthy stroma exhibits strong birefringence due to the ordered structure of its well- aligned collagen fibres, which is diminished during precancerous remodelling of more chaotic collagen networks [81]. These distinct polarimetric microstructural biomarkers have recently been used to assess breast cancer tissue slides at different pathology stages [88]; these promising results necessitate further studies in this area towards establishing a polarimetric clinical niche.

Mueller Imaging of Bulk Tissues Ex Vivo: Reflectance-Based Configurations

In light of the useful MM polarimetric probing ability of (pre)cancers shown in thin tissues, there is a push towards its in-vivo diagnostics to offer an informative alternative to the invasive tissue biopsies currently employed for (pre)cancer detection.

In-vivo assessment of course involves bulk tissues which must be examined in the backscatter direction. Bulk tissues are much more challenging than thin tissues to assess polarimetrically, since (1) there is a major increase in depolarization yielding overall weaker signals, (2) backscattered polarized light responds differently and more complexly than forward scattered light (e.g. helicity flipping and multiple backscattering pathways) [21, 22, 89], and (3) depth-effects (e.g. tissue layers which exhibit different polarization responses [70]) are introduced which can lead to unexpected results. It is thus important to assess the feasibility of bulk tissue characterization ex vivo before introducing further challenges of in-vivo examination.

Pierangelo et al. [90] presented an interesting polarimetric assessment and potential staging of bulk tissue cervical cancers ex vivo. Here, the freshly excised cervical samples were imaged *en face* (i.e. stroma beneath the epithelium), in contrast with the Shukla study [82] which isolated the epithelia and stroma through thin-slice sectioning. Dysplastic lesions, referred to as Cervical Interepithelial Neoplasia (CIN), are staged as the malignant cells approach the surface of the epithelium; the progressive stages CIN1, CIN2, and CIN3 indicate when one-third, two-thirds, or the entire epithelium is transformed, respectively. These cervical dysplasia stages along with their affected regions can be identified by MM polarimetry, as shown in Fig. 4a. As seen, depolarization decreased from healthy to CIN1 to CIN3 to glandular tissue and retardance disappeared in the dysplastic stages. But in fact one would expect depolarization to behave oppositely, instead increasing in dysplastic regions due the higher cellular concentrations (resulting in higher turbidity) as previously observed [82] (see Fig. 3); bulk tissue and backscatter geometry effects are likely responsible. Novikova et al. [91] proposed that *increased* backscattered polarization preservation from cancerous tissue may stem from entrapment of multiply scattered depolarized light and increased absorption arising from the strong vascularization in the tumour region, with both of these mechanisms yielding higher DOP of the detected light as observed. Further investigation is needed, but nonetheless significant polarimetric contrast was observed. Also unexpected was the large decrease in retardance of dysplastic regions as epithelia is largely absent of birefringent tissue; the authors attributed this to connective tissue lying beneath the epithelium, since dysplasia has been found to disrupt the organization of neighbouring stroma [81, 82, 92]. Rehbinder et al. [93] then applied thresholding to the retardance images of cervical tissue samples to separate healthy (above the retardance threshold) from abnormal regions (pre-cancerous or cancerous, below the retardance threshold), as shown in Fig. 4b. These interesting MM polarimetric responses have shown their potential in diagnosing cervical pre-cancers; however, the bulk tissue retroreflection imaging geometry and the resultant depth-cumulative polarimetric sampling introduce curious effects. These effects may be elucidated by depth-resolved hybrid modalities which are discussed in Sect. 4.1.

Mueller Imaging In Vivo

The ex-vivo studies of both thin and bulk tissues show that the information content afforded by MM polarimetry is promising for assessing pre-cancerous

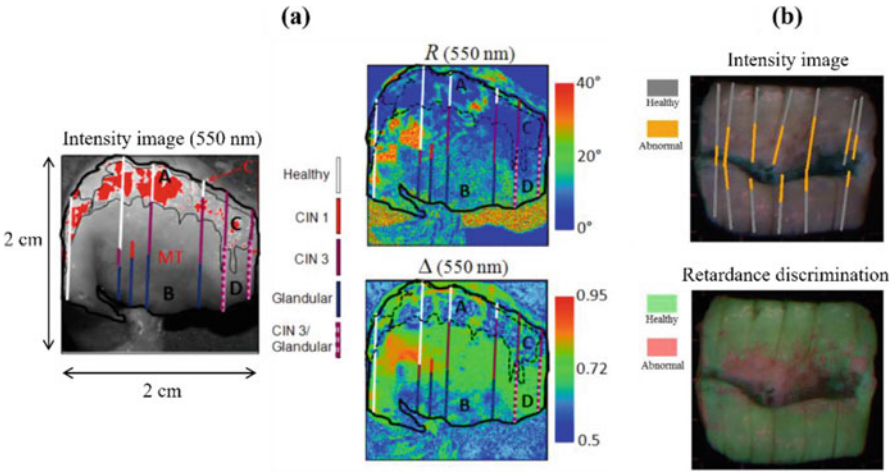


Fig. 4 Polarimetric staging of cervical cancer pathology. (**a**) Intensity image of surgically excised human cervical specimen with overlayed histological characterizations denoted by coloured lines (see legend); histology was performed after polarimetric imaging. The interior of the black solid line is malpighian (squamous epithelium) denoted by 'MT' and the exterior is stroma denoted by 'C'. The thin solid line delineates a region where polarimetric data may be inaccurate and the red coloured areas indicate saturation of the CCD. Retardance R (*top*) and depolarization Δ (*bottom*) images of the specimen exhibiting healthy, CIN1, CIN3, glandular, and CIN3/glandular (glandular buried beneath CIN3 tissue) regions. Depolarization appears to exhibit promising staging capability where each stage (except for CIN3/glandular) is marked by different values (decreasing from healthy to CIN3). Retardance compliments the depolarization image with clearer identification of healthy zones (strong retardance) and slight identification of CIN3/glandular tissue. (Adapted from [90]). (**b**) Intensity image (*top*) of surgically excised human cervical specimen with overlayed histological characterization denoted by coloured lines (see legend). Applied to the intensity image, retardance (*bottom*) is used as a discriminating parameter where above-threshold values are coloured green to indicate healthy tissue and below-threshold values are untouched to indicate diseased tissue. Retardance appears to be sufficient in assessing cervical pre-cancer/cancer and in delineating pathological margins. (Adapted from [93])

and cancerous tissues. This has recently led to efforts towards in-vivo imaging with a focus on technology development to address various clinical engineering challenges. In-vivo Mueller polarimeters must be robust, rapid (to minimize motion artefacts), accurate, and ideally cost-effective to successfully function in realistic clinical conditions [94]. Several such studies have focused on in-vivo cervical (pre)cancer assessment to improve upon current screening methods. Currently, colposcopy (low magnification examination of the cervix) of high grade dysplasia only yields 60–70% sensitivity and 50% specificity [95]; there is also low inter-operator reproducibility ($\kappa = \sim 0.4$) [96], thus suggesting highly operator-dependent results. Towards this goal, Ramella-Roman's group [97, 98] recently demonstrated a portable hybrid colposcope in the form of a snapshot MM polarimeter (Fig. 5i). The PSG consisted of a set of light-emitting diodes to provide the three linear states (vertical, horizontal, 45°) and right circularly polarized light. The novel

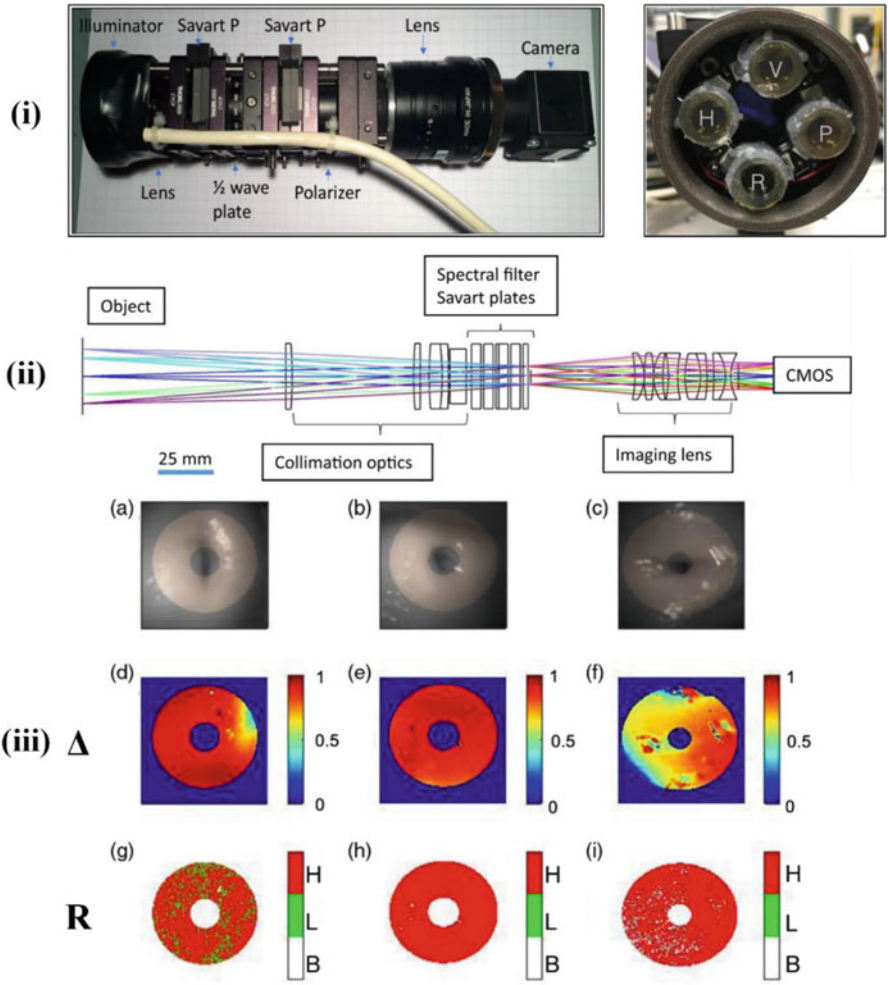


Fig. 5 Portable colposcope 'Snapshot' MM polarimeter. (Figures from [98]). (**i**) The portable snapshot Mueller polarimeter shown on the left panel; the illuminator (PSG) output end is shown on the right. The compactness, speed, and lack of moving parts are attractive features for clinical adoption. (**ii**) Schematic of the PSA portion of the system. (**iii**) In-vivo images of three healthy cervices (field of view = 30 mm diam). (a)–(c): intensity images; (d)–(f): depolarization images; (g)–(i): retardance images with corresponding intensity bars, where H and L indicate high and low retardance (threshold at 25°) and B indicates zero background. The trend of high and high retardance of the healthy cervices agrees with previously reported data (whereas precancerous regions are associated with lower values of these two metrics [90]). The non-uniformity of the depolarization image in (f) may be due to the heterogeneity of the cervix

PSA comprised of a quarter wave retarder and linear polarizer on either side of four Savart plates which divided the incident light into four separate paths. An imaging lens recombined the four separate channels onto the camera to create a

spatial interference pattern, where the fringes are a modulated transfer function of the optical elements and the polarimetric properties of the illuminated sample. Either Fourier transform or so-called sliding reconstruction was used to recover the sample's polarization properties (Fig. 5ii). The total acquisition time was approximately 1 s, suitable for in-vivo measurements. Furthermore, the cost of the polarimeter was estimated to be a modest ~$200 USD, a significant reduction compared to the typical cost of current colposcopes (~$2000 USD), and the system was fully powered by a laptop computer, eliminating the need for electrical outlets. Both of these practical advantages may enable wider deployment in lower resource settings.

In-vivo images were acquired of healthy cervices, three of which are shown in Fig. 5iii in the form of raw intensity, depolarization, and retardance images (the latter two derived from polar decomposition of the measured MM). The depolarization and retardance results were in agreement with previously reported data on healthy uterine cervices. One patient had a polyp that was marked by lower depolarization (data not shown), also in agreement with previous observations [95]. The imaging results of healthy cervical tissue are thus in line with the literature; the next step will be to validate the imaging accuracy in pre-cancerous and cancerous tissues. Overall then, this study demonstrates the feasibility of in-vivo cervical Mueller imaging in a clinical setting, using a cost-effective and easy-to-employ (laptop powered) polarimeter to enhance its clinical adoptability.

Another promising approach for in-vivo assessment of uterine cervix pathology is that by Vizet et al. [95] who mounted a miniaturized Mueller polarimetric imager onto a conventional colposcope (a stereoscopic binocular microscope used to illuminate and observe uterine cervix under low magnification of ~4–6X). This approach exploits the powerful advantage of Mueller polarimeters (and of most polarimetric systems in general), in that they are highly compatible with existing photonic devices so they may be rather simply placed in a beam path to furnish additional polarimetric information. The silica fibre bundle used for light delivery in the colposcope was replaced by a liquid light guide to increase the illumination intensity and ensure a measurable backscattered signal. The mounted polarimeter was sealed within an airtight metallic box measuring $3 \times 5 \times 10$ cm^3 that could be toggled between 'OFF' or 'ON' to enable regular colposcopic or MM imaging. The PSA of the Mueller polarimeter consisted of a linear polarizer followed by two tuneable ferroelectric liquid crystal cells that sandwiched a quarter wave plate: the PSG was the reverse order version of these elements and the CCD camera detected 550 nm light using a bandpass dichroic filter. The polarimeter was capable of acquiring a single polarization configuration image in 100 ms, resulting in a full MM (16 images) acquisition time of 1.6 s, suitable for in vivo imaging to reduce patient motion artefacts.

Before in-vivo imaging, the performance of the polarimetric colposcope was tested on ex-vivo cervical specimens, which compared the novel rapid 1.6 s acquisition Mueller images to (1) averaged Mueller images acquired over ~25 s (16 images) and (2) Mueller images from a previously demonstrated and routinely used Mueller polarimeter [47, 90]. Reasonable agreement between the three methods was

demonstrated. The thus-validated 1.6 s Mueller polarimeter was then used to image healthy uterine cervices in vivo (single images, not averaged), one of which also contained a polyp. The polyp served as a test case for tissue structure distinct from normal healthy epithelium, approximating a pathological (pre-cancerous/cancerous) lesion; polyps are layered with glandular (columnar) epithelium, whereas the exocervix is layered with Malpighian (squamous) epithelium. The healthy Malpighian tissue exhibits high retardance as expected due to its well-ordered collagen, whereas the polyp is well delineated by its very low retardance. Additionally, the polyp appears to be less depolarizing compared to the exocervix. The polarimeter is thus capable of discriminating between glandular and squamous tissue types, demonstrating its potential for pre-cancer/cancer detection of cervices. Moreover, this was performed using a validated, robust, rapid, and accurate in-vivo Mueller imaging tool, which was easily operated by the surgeon as reported. As in the previous example, the next step in using the device will be to image cervical pre-cancers and cancers to determine the polarimetric discrimination capabilities at such rapid speed in practical situations, particularly to determine the feasibility of differentiating between polyps and pre-cancerous lesions which both seem to exhibit low retardance and low depolarization.

Section 2.5 Revisited: Is the Full Mueller Matrix Approach Really Necessary? For all its advantages, Mueller polarimetry is a rather complex technique that requires several measurements (at least 16) and often relies on advanced technologies to enable practical and robust systems. As touched upon in Sect. 2.5, an important question then arises: Are Mueller matrices really worth measuring? Can adequate polarimetric information be attained in a simpler manner? Indeed, the added complexity of the Mueller approach, despite furnishing its associated rich information content, may not be necessary in all applications, and simpler polarimeters have thus been proposed and demonstrated. For example, Jacques et al. [83, 99] and Backman and Gurjar et al. [65, 66] have employed simple cross-linear and co-linear polarizer setups to differentiate between superficially scattered and deeper penetrating polarized light in epithelial tissues based on light depolarization rates. DOP depth discrimination using rotating polarizers and retarders has also been recently reported by Da Silva et al. [71, 72, 100–103] for probing epithelia using the *ellipticity* of light. Stokes DOP approaches have been applied to skin cancer detection and characterization with promising early results [104, 105], as well as towards carcinomas of the lung [106] and colon [107, 108]. Tissue birefringence has also been studied in the recent works of Vitkin et al. [109–111] which used simple rotating crossed polarizers to characterize morphological stromal features of human breast cancer slides.

Such simpler approaches – crossed- and co-linear polarizers and Stokes polarimeters – have primarily found applications in tissues where *only* depolarization or retardance is sufficient for characterization (e.g. epithelia contain little birefringent materials, leading to non-Mueller successes via DOP-based measurements). MM polarimetry, with its multiple polarization measurement configurations and matrix interpretation/decomposition analysis, is likely an

'overkill' if only one polarimetric response is satisfactory. The success of these non-Mueller approaches on epithelia can also be attributed, at least in part, to Mie theory [112] which has played a significant role in the understanding of the light scattering physics in such media where cell nuclei can be modelled as discrete scattering particles, having facilitated the use of microsphere suspensions as tissue phantoms and Monte Carlo computer modelling of polarized light propagation in turbid materials [21–23]. Indeed, one of the first proposals for depth discrimination using polarized light [24] relied on both of these tools.

However, many tissues, particularly those in bulk form, are both turbid and birefringent (as well as possibly optically active, linearly/circularly dichroic, and other second-order effects), which plays to the strengths of MM polarimetry since these can be retrieved and decoupled via decomposition and all used for pathological assessment as clearly demonstrated in preceding illustrations. An interesting case in point was recently provided by Pierangelo et al. [90] who attempted to analyse cervical dysplasia using orthogonal state contrast (OSC) imaging (DOP information) which showed some discrimination between the two tissue types. However, as the authors remarked, 'OSC imaging is not adequate for such samples, as it does not provide enough information to extract the relevant polarimetric parameters for healthy and anomalous tissues. To properly characterize the polarimetric response of healthy and pathological cervical tissue, full Mueller polarimetry is clearly needed'. Another example is the Stokes-DOP-based study by Anastasiadou et al. [107] which showed detection capability of cervical cancer and some visualization of margins, however to a far weaker extent than that shown by the MM depolarization and retardance images of Shukla et al. [82]. Moreover, the attractive simplicity of non-Mueller approaches may become a less important design consideration since technological advances are enabling easier acquisition of Mueller matrices through automation such as the snapshot polarimeter by Ramella-Roman et al. [98] and the ferroelectric liquid crystal cell-based polarimeter by Vizet et al. [95]. Therefore, with the advent of new instruments and approaches to measure MMs and faster computers to more rapidly analyse/decompose them for useful biomarker extraction, the use of this 'more complete' variant of polarimetry may prove readily feasible, and thus more widespread. Indeed, biomedical technologies tend to develop by clinical necessity driven by a biomedical pull: if an application calls for better instruments, engineers promptly take on the challenge. Such has been the story of MM polarimetry as an emerging technique in biomedicine.

3.2 Probing Sub-Micron Structural Anisotropy for Pre-cancer Detection

Probing the tissue structural anisotropy through MM measurements have been promising, however there is still much room for improvements, for example if the spatial sensitivity of the polarimetric techniques is improved down to the scale of

microscopic anisotropy of tissue architecture (micron/sub-micron length scales). It has been suggested that pre-cancer involves altered interactions between superficial epithelial cells and the underlying connective tissue, and the resulting changes in the *anisotropic* connective tissue morphology are rather subtle. These typically manifest optically as small changes at the sub-micron scale. Indeed, initial findings on sub-micron scale anisotropy in tissue multifractality derived from spectral Mueller matrices have shown that *microscopic* refractive index asymmetries are a much better metric of subtle pre-malignant transformation [113, 114] compared to the 'conventional macroscopic' polarimetry biomarkers discussed earlier. Here, a brief account of this promising research direction is provided.

Multifractals are a special class of complex self-similar processes, observed in diverse natural systems encompassing biology, material structures, physiological time series, turbulence, etc. [115–118]. Such complex signals typically consist of many interwoven fractal (self-similar) subsets, each of which are characterized by their own local scaling exponents (known as Hurst exponent, $0 < H < 1$) [115, 119]. Initial findings on tissue multifractality were based on *forward* multi-resolution analysis of differential interference contrast images of tissue [92] and *inverse* analysis on tissue light scattering spectra [120]. A state-of-the-art algorithm for multifractal research, the multifractal detrended fluctuation analysis (MFDFA), was optimized to extract and quantify the tissue multifractal parameters, namely, the generalized Hurst exponent ($h(q)$, q is the order of the moment) and the width of the singularity spectrum ($f(\alpha)$, α is the singularity strength) [119]. Interestingly, it was observed that the spatial refractive index fluctuations of tissue exhibit *microscopic* multifractal anisotropy that manifests as an intriguing wavelength-dependent linear diattenuation effect in the scattering Mueller matrix elements [113, 114]. The corresponding inverse problem was also addressed of extracting information on the microscopic multifractal anisotropy from the scattering Mueller matrix. This analysis is based on processing the wavelength dependence of selected Mueller matrix elements (that encode the diattenuation effect) in the Fourier domain using Born approximation followed by multifractal analysis. The model was applied on ex-vivo tissues of human cervix of different pathology grades. The experimental system of Fig. 2 was employed to record backscattering spectroscopic Mueller matrices for these studies, and the results are summarized below.

For the inverse analysis, it was identified that the multifractal anisotropy manifests most prominently in the wavelength variation of the $M_{12}(\lambda)$ and $M_{13}(\lambda)$ elements [113, 121] as these carry information on spectral linear diattenuation effect ($d(\lambda) = \frac{\sqrt{{M_{12}}^2+{M_{13}}^2}}{M_{11}}$). The spectral variations of these elements were analysed to extract information on the refractive index in homogeneity distributions along two orthogonal linear polarization directions ($\eta_{\parallel/\perp}(\rho)$ with ρ being the length scale) through Fourier domain analysis using the Born approximation as [113].

$$\begin{aligned}\eta_{\parallel/\perp}(\rho) &\sim \int k^{-2}\left\{M_{11}(K) \pm \sqrt{{M_{12}}^2(K) + {M_{13}}^2(\beta K)}\right\}^{\frac{1}{2}} e^{-i(\mathbf{K}\ldots\boldsymbol{r})} d^3\beta \\ &= \int \left|s_{2/1}(K = 2\pi\nu)\right| e^{-i(\mathbf{K}\ldots\boldsymbol{r})} d^3\beta\end{aligned} \tag{13}$$

Here, $\boldsymbol{K}$ is the scattering vector with modulus $K = 2k \sin\left(\frac{\theta}{2}\right) = 2\pi\nu$, ν is the spatial frequency, $k = 2\pi/\lambda$ is the wave vector, λ and θ are the wavelength and the scattering angle, and $s_{2/1}$ are the scattered field amplitudes for the two orthogonal polarizations. The extracted $\eta_{\parallel/\perp}(\rho)$ were then analysed via MFDFA to quantify the multifractal anisotropy parameters.

MFDFA is an advanced statistical tool for multifractal analysis as described in detail elsewhere [119]. Using this approach, any multifractal signal can be characterized via two sets of scaling exponents: (i) the generalized Hurst exponent $h(q)$ and (ii) the classical multifractal scaling exponent $\tau(q)$ along with the singularity spectrum $f(\alpha)$ [119]. These are inter-related as

$$\tau(q) = qh(q) - 1, \tag{14}$$

$$\alpha = \frac{d\tau}{dq}, \quad f(\alpha) = q\alpha - \tau q, \tag{15}$$

where α is the singularity strength and σ is the full width of $f(\alpha)$ is a quantitative measure of multifractality. In monofractal approximation, the generalized Hurst exponent $h(q = 2)$ is equivalent to fractal dimension. The variation of $h(q)$ or $\tau(q)$ for other values of q indicates multifractality or multi-resolutions (different q values) fractal dimension. The refractive index asymmetry distributions for orthogonal linear polarizations, $\eta_{\parallel}(\rho)$ and $\eta_{\perp}(\rho)$ were analysed separately using the MFDFA. The multifractal anisotropy was quantified via the differential classical scaling exponent, $\Delta\tau = |\tau(q = 2)_{\parallel} - \tau(q = 2)_{\perp}|$, and differential width of singularity spectrum $\Delta\sigma = |\sigma_{\parallel} - \sigma_{\perp}|$ for orthogonal linear polarizations.

With the theoretical context thus provided, Fig. 6 demonstrates multifractal anisotropy in the spectral Mueller matrix elements recorded from the connective tissue region of a Grade 1 pre-cancerous cervical tissue [113]. The M_{12} (λ) and M_{13} (λ) elements (Fig. 6a) and the resulting linear diattenuation parameter $d(\lambda)$ (Fig. 6b) exhibit considerable magnitude and wavelength variation (overall decay with λ), a potential signature of fractal (multifractal) anisotropy. This is characteristically manifested as a power law dependence of the $|s_1(\nu)|^2$ and $|s_2(\nu)|^2$ light-scattering parameters (Fig. 6c) which were fitted with fractal power law ($|s_2(\nu)|^2 \approx \nu^{-(2H_{\parallel}+D_E)}$ and $|s_1(\nu)|^2 \approx \nu^{-(2H_{\perp}+D_E)}$). Here, $H_{\parallel/\perp} \in (0,1)$ are the Hurst scaling exponents and $D_E = 3$ is the Euclidean dimension of the scattering object. The fitting yielded multiple power law exponents at different ν-ranges, suggesting multifractality. Importantly, fitting over broad ν-range yielded differences in average Hurst exponents: $H = 0.55$ and 0.65 derived from $|s_2(\nu)|^2$ and $|s_1(\nu)|^2$, respectively ($\Delta H = 0.1$). These results suggest self-similarity anisotropy in spatial refractive index variations.

Figure 6d and e summarizes the results of the more detailed multifractal analysis. Significant magnitudes of the differential classical multifractal scaling exponent [$\Delta\tau = |\tau(q = 2)_{\parallel} - \tau(q = 2)_{\perp}| = 0.16$] and differential width of singularity spectrum [$\Delta\sigma = |\sigma_{\parallel} - \sigma_{\perp}| = 0.85$] provided conclusive evidence of multifractal

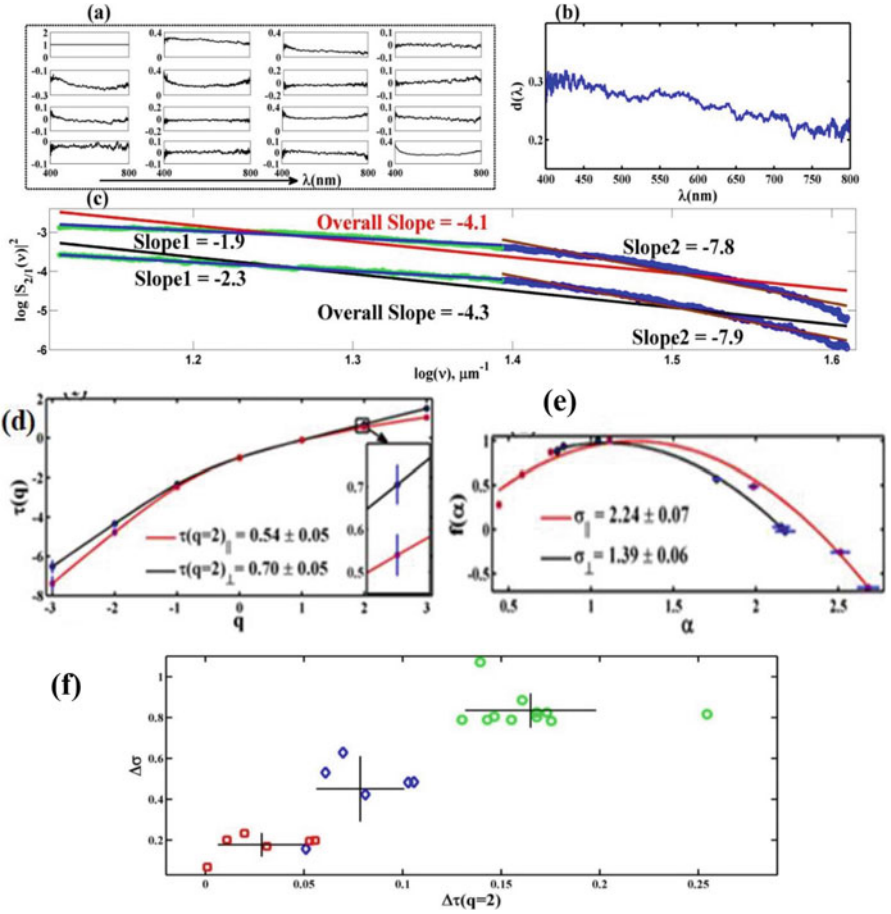


Fig. 6 Manifestation of multifractal anisotropy in the wavelength variation of scattering Mueller matrix elements of a Grade I precancerous tissue. Wavelength variation of (**a**) the normalized Mueller matrix elements and (**b**) linear diattenuation parameter $d(\lambda)$. (**c**) The spatial frequency (ν) distribution of the Mueller matrix-derived parameters $|s_2(\nu)|^2$ and $|s_1(\nu)|^2$ (log-log plot) via Eq. (13). Fitting at two selected ν-ranges (lower [blue] and higher [brown]) and overall fitting (red for $|s_2(\nu)|^2$ and black for $|s_1(\nu)|^2$) are shown and the corresponding values for the slopes are noted. (**d**) The moment (q)-dependence of the classical multifractal scaling exponent $\tau(q)$ (inset highlights the anisotropy or difference in τ around $q = 2$) and (**e**) the corresponding singularity spectra $f(\alpha)$ derived from $\eta_{\parallel}(\rho)$ (red square) and $\eta_{\perp}(\rho)$ (black circle). The values for $\tau(q = 2)$ and σ are noted. In (d) and (e) lines are guide for eye and the error bars represent standard deviations of the parameters for measurements on 10 non-overlapping spots. (**f**) The three different precancerous grades (Grade I – green circle, II – blue diamond and III – red square) are mapped by their differential classical multifractal scaling exponent, $\Delta\tau = |\tau(q = 2)_{\parallel} - \tau(q = 2)_{\perp}|$ and differential width of singularity spectrum $\Delta\sigma = |\sigma_{\parallel} - \sigma_{\perp}|$ for orthogonal linear polarizations. Higher grades of pre-cancers are associated with decrease of both Δh and $\Delta\sigma$ parameters, nicely separating different pathologies and implying reduction in multifractal anisotropy. (Adapted from [113])

anisotropy. Most importantly in the biomedical context, the values of $\Delta\tau(q=2)$ and $\Delta\sigma$ (1) cluster and (2) are observed to decrease with increasing pathology grades (Fig. 6f), suggesting a reduction in the multifractal anisotropy. This decrease in multifractal anisotropy was attributed to the disorganization of locally anisotropic microscopic domains (the collagen molecules and micro-fibrils) and/or reduction in local microscopic birefringence with increasing pathology grades [113, 121]. Importantly, none of the above trends could be gleaned from Mueller matrix-derived *macroscopic* linear retardance R δ parameters, which were generally low and did not exhibit appreciable difference between different grades (corresponding birefringence values at 600 nm were 0.08 ± 0.03, 0.04 ± 0.02 and 0.09 ± 0.05 rad for Grades I, II and III tissues, respectively; thus, no inter-grade separation was evident). These results suggest that the *microscopic* refractive index asymmetries are more sensitive to subtle pre-malignant transformation [113, 114, 121–123], as also often posited in the literature.

3.3 Fluorescence Mueller Polarimetry: A Spectroscopic Diagnostic Tool

Most biomedical polarimetry studies examine elastically scattered light from tissue that mainly carries morphological information. Conversely, polarization properties of fluorescence may provide valuable biochemical information on the changes in chemical structure, molecular organization, and local environment of the tissue fluorophores. However, the fluorescence polarization anisotropy parameter [124] traditionally measured reflects 'lumped' effects and is dominated by fluorescence depolarization in addition to the 'true' *intrinsic* tissue anisotropy that originate from the anisotropic molecular orientation and organization. Hence, a more encompassing fluorescence MM polarimetric approach has recently been developed [125, 126]. One can then polarimetrically probe and quantify the molecular orientation and organization of endogenous tissue fluorophores using this approach. The experimental system shown in Fig. 2 was employed to record fluorescence spectroscopic Mueller matrices ($\lambda_{ex} = 405$ nm, $\lambda_{em} = 450 - 750$ nm) in the exact backscattering configuration from connective tissue regions of precancerous human cervical tissues of different pathology grades [126–129]. These were then subjected to inverse analysis to yield the *intrinsic* fluorescence anisotropy parameters, namely fluorescence spectral diattenuation (differential excitation of fluorescence by orthogonal polarizations) and fluorescence polarizance (differential emission of fluorescence by orthogonal polarizations).

The analysis is based on writing the fluorescence Mueller matrix $\mathbf{R}$ as the product of three matrices such that $\mathbf{R} = \mathbf{M_1} \cdot \mathbf{M_d} \cdot \mathbf{M_0}$. Here, the matrix $\mathbf{M_d}$ is the dipole scattering Mueller matrix which takes care of the depolarization effects (both intrinsic and extrinsic) associated with fluorescence emission [125, 126]. The other two matrices $\mathbf{M_0}$ and $\mathbf{M_1}$ account for the absorption anisotropy of the ground molecular state (representing differential excitation of fluorescence with orthogonal

polarization) and emission anisotropy of the excited molecular state (representing differential emission of orthogonally polarized light), respectively. Using standard form of these matrices and with appropriate approximations [125, 126], it can be shown that the elements R_{12}, R_{13}, R_{14} (first row of fluorescence Mueller matrix) encode excitation anisotropy of the ground molecular state and R_{21}, R_{31}, R_{41} (first column) reflect emission anisotropy of the excited molecular state (cf. discussion following Eq. (9)). The intrinsic anisotropy effects of fluorescence are subsequently parameterized and quantified via the fluorescence diattenuation and polarizance parameters as [125, 126, 130]

$$\alpha_T = \frac{\sqrt{R_{12}^2 + R_{13}^2 + R_{14}^2}}{R_{11}}, \qquad \alpha_L = \frac{\sqrt{R_{12}^2 + R_{13}^2}}{R_{11}}, \qquad \alpha_C = \frac{R_{14}}{R_{11}}, \tag{16}$$

$$\beta_T = \frac{\sqrt{R_{21}^2 + R_{31}^2 + R_{41}^2}}{R_{11}}, \qquad \beta_L = \frac{\sqrt{R_{21}^2 + R_{31}^2}}{R_{11}}, \qquad \beta_C = \frac{R_{41}}{R_{11}} \tag{17}$$

where α_T, α_L, and α_C are the total, linear, and circular fluorescence diattenuations, respectively, β_T, β_L, and β_C are the total, linear, and circular polarizance, respectively.

Figure 7 shows typical fluorescence Mueller matrix $\mathbf{R}(\boldsymbol{\lambda})$ recorded from the connective tissue (stroma) region of a dysplasia grade III pre-cancerous cervical tissue [130]. The fluorescence spectra peaking around 500 nm with 405 nm excitation is primarily contributed by fluorescence from collagen and other endogenous tissue fluorophores elastin and NADH. While strong depolarization of fluorescence is evident from the small magnitudes of the diagonal elements (R_{22}, R_{33}, R_{44}), relatively higher magnitudes of R_{12} and R_{21} elements indicated the presence of strong excitation and emission (respectively) *intrinsic* anisotropies of fluorescence. The corresponding fluorescence linear diattenuation (excitation anisotropy – α_L) and fluorescence linear polarizance (emission anisotropy – β_L) parameters from typical grade I and grade III tissues samples showed significant magnitudes and prominent differences (Fig. 7b and c). The origin of these intrinsic linear anisotropies of fluorescence was linked to the crossed-linked collagen molecular structure in the connective tissue [130]. The rather weak but non-zero circular anisotropies (α_C and β_C) arises due to the presence of chiral moiety or helical molecular organization in the collagen structure. The reduction in the α_L and β_L anisotropy parameters in grade III tissue was attributed to the destruction of the collagen cross-links and the resulting loss of intrinsic linear anisotropic organization in the collagen molecular structure. These results suggest the sensitivity of the fluorescence Mueller polarimetry towards changes in the intrinsic molecular organization and orientation of tissue fluorophores. The actual potential of these intrinsic fluorescence anisotropy parameters as biomarkers for pre-cancer detection remains to be rigorously evaluated.

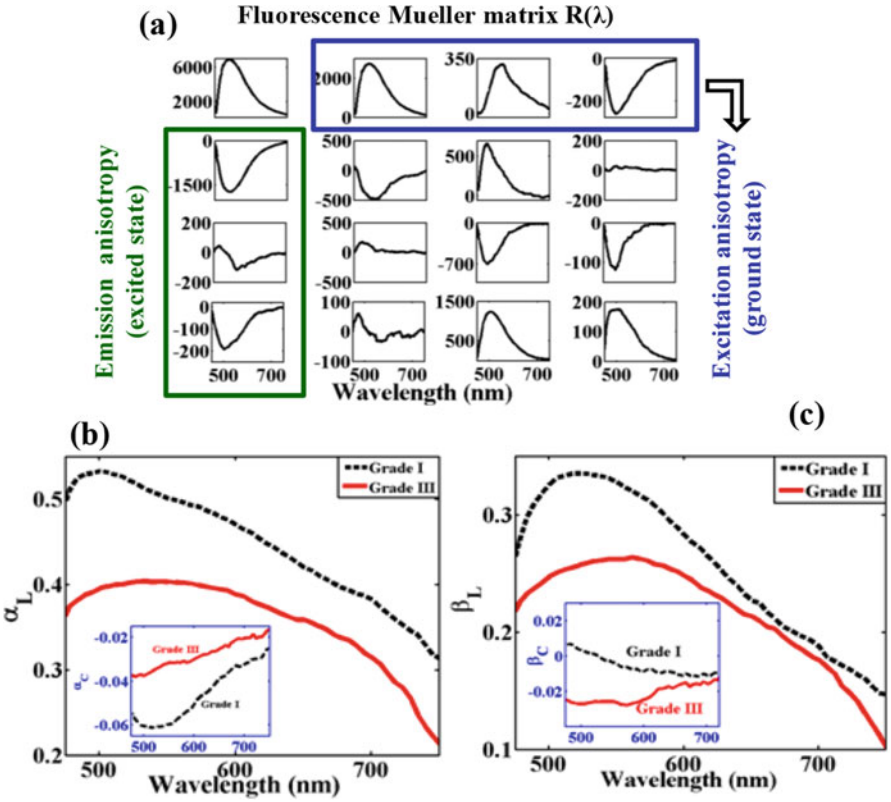


Fig. 7 (**a**) Typical fluorescence spectroscopic Mueller matrix **R** (λ_{em} = 450–750 nm) recorded with λ_{ex} = 405 nm excitation from the connective tissue region of a dysplasia grade III precancerous cervix. The first element (R_{11}) represents the polarization-independent fluorescence spectra. The other three elements in the first row (blue) represent excitation anisotropy related to the ground molecular state; the three elements in the first column (green) represent emission anisotropy related to the excited molecular state. The spectral variation of (**b**) fluorescence linear diattenuation (α_L) and (**c**) fluorescence linear polarizance (β_L) parameters for typical grade I and grade III precancerous cervical tissues. The insets show the variation of fluorescence circular diattenuation (α_C) and circular polarizance (β_C) for the same tissues. (Adapted from [130])

4 Hybrid and Endoscopic Mueller Approaches

4.1 Depth-Resolved Hybrid Modalities: Elucidating Depth Effects

Most tissues are heterogeneous throughout, where certain sub-regions may exhibit different polarization responses [70]. Since MM systems are typically not depth-resolved but instead gather tissue polarimetric information throughout their entire sampling depth, ambiguous and unexpected effects can arise (e.g. findings of

Pierangelo et al. [90] discussed above). It can therefore become important to elucidate depth effects; for example, this may help in differentiating tissue regions that exhibit similar depth-cumulative polarimetric responses such as in the aforementioned case of polyps and dysplastic lesions of cervix [95]. There has thus been an interest in combining/integrating Mueller polarimeters with depth-resolved modalities. Mueller polarimeters are advantageous in this regard due to their ease-of-integration with existing optical systems (e.g. the above-discussed colposcopic study by Vizet et al. [95]). OCT and confocal microscopy both offer depth-resolved imaging, thus prompting the polarimetry community to combine or integrate these technologies with Mueller polarimeters.

Optical Coherence Tomography Combined with Mueller Polarimetry

Optical coherence tomography (OCT) is a well-established depth-resolved tomographic imaging method [131, 132] routinely employed for in-vivo imaging; polarization-sensitive OCT (PS-OCT) has also been investigated for decades (for a detailed review see [133]). The time domain version of PS-OCT was first developed by Hee et al. [134], followed by the more sensitive, and thus more widely adopted frequency-domain versions (swept source and spectral-domain) [135]. PS-OCT relies on Jones vectors which describe fully polarized light interactions, and thus material depolarization cannot be determined. Instead, a parameter known as 'degree of polarization uniformity' (DOPU) [136] is invoked: the polarization state of each *speckle* is taken as unity (totally polarized), however in the case of a depolarizing medium, the polarization states of multiple speckles become randomized and uncorrelated, giving rise to the DOPU parameter. Mueller polarimeters may thus compliment PS-OCT systems by measuring actual material depolarization, and further PS-OCT systems can in return compliment Mueller polarimeters by providing information on axial tissue heterogeneity. To form this hybrid and complimentary imager, Chue-Sang et al. [51] have demonstrated a co-registration technique using a relatively simple PS-OCT system based on free-space optics and a Mueller polarimeter in a separate beam path. This enabled Mueller identification of burned regions of bovine tendon through diattenuation and depolarization images, as well as structural depth information of the bovine tendon via PS-OCT, shown in Fig. 8. This complimentary hybrid modality may be useful, for example, in identifying skin tumours through Mueller imaging, followed by assessment of their depth-of-invasion through PS-OCT imaging which can be an important prognostic factor [137, 138]. Furthermore, this approach may lead to better models for the recovery of local retardation using OCT-based imaging, rather than cumulative retardation [70], since many current recovery models rely on the measurement of surface retardance as a starting point [139].

Second Harmonic Generation Confocal Microscope and Mueller Polarimeter Hybrid

Instead of placing a Mueller polarimeter in a separate beam path as in the preceding Mueller-OCT co-registration example, it may be possible instead to integrate it directly in the beam path of a depth-resolved modality to furnish 3D

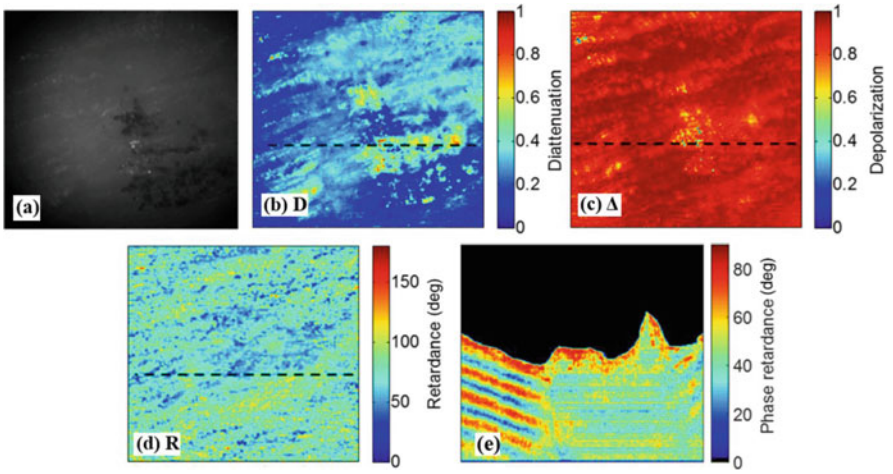


Fig. 8 Mueller matrix imaging and polarization-sensitive OCT of bovine tendon with a burned region (tendon dimensions $4 \times 4 \times 2$ mm^3). (**a**) CCD image showing the burned region (darker parts at the centre to bottom right portion of the image). (**b–d**) show diattenuation, depolarization, and retardance images acquired by the Mueller polarimeter. The diattenuation image appears to best demarcate the burned region. (**e**) PS-OCT B-scan of tendon (at the location of the horizontal line in the *en face* MM images of (**b**)–(**d**)), showing the burn boundary well delineated by its low retardance due to the thermal disruption of the stromal organization. The region to the left exhibits an oscillatory phase retardance consistent with healthy tendon. Using such complimentary information from each modality may be useful, for example, in identifying skin tumours through Mueller polarimetry and then assessing their depth of invasion through PS-OCT which can be an important prognostic factor [137, 138]

depth-resolved Mueller polarimetric volumes. This may be done by hybridizing a confocal microscope with a Mueller polarimeter. Confocal microscopy performs optical depth sectioning using a pinhole at a confocal plane to beam focus at the sample, thus eliminating backscattered photons from out-of-focus sample depths which cause blurring. This combined modality may be further enhanced with the integration of non-linear microscopy to form a second harmonic generation (SHG) confocal MM microscope. SHG microscopy is an excellent method for imaging collagenous tissue due to its preferential SHG response [140] which can thus provide additional corroborative information alongside Mueller decomposed images to better understand the polarization response of tissue, as shown in Fig. 9a for benign and malignant breast tissues [141]. Through depth-scanning, a 3D MM polarimetric volume may be generated alongside associated SHG images at various depths, such as in the approach by Satyashev et al. [142] with corresponding results shown in Fig. 9b. This type of hybrid Mueller microscopy approach may indeed prove useful in gaining true depth-resolved polarimetric insight or assisting histopathological diagnosis. However, long acquisition times (on the order of tens of minutes at best) currently hinder its use beyond the laboratory.

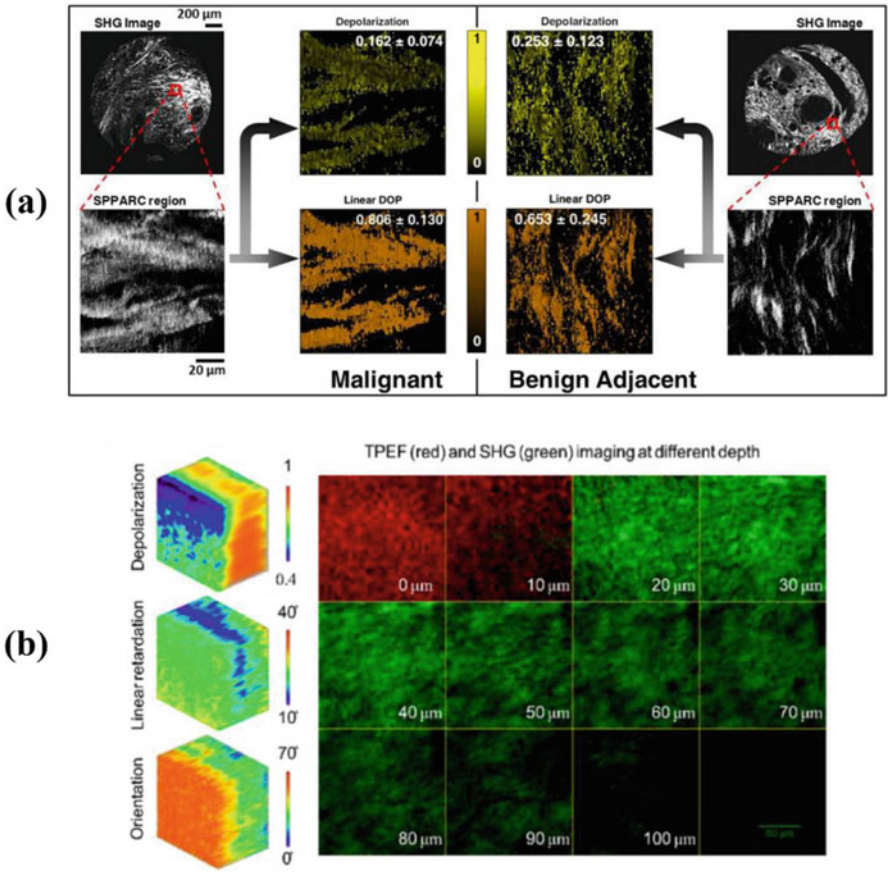


Fig. 9 SHG confocal Mueller matrix microscopy. (**a**) The images on the left and right sides correspond to malignant and benign breast tissues, respectively. SHG microscopic images are shown on the top left and top right. Shown as insets below the SHG images are 'second-harmonic patterned polarization-analysed reflection confocal microscopy' (SPPARC) images of selected regions. Corresponding to the SPPARC images are depolarization and linear degree of polarization spatial maps (see arrows) (along with inset values indicating the mean and standard deviation over all pixels in each image). This technique enables high resolution polarimetric images of tissue samples to analyse microstructural differences between tissue types and their corresponding polarimetric responses. (Adapted from [141]). (**b**) Full-depth two-photon and Mueller matrix confocal imaging of rat cornea (field of view: $120 \times 120\ \mu m^2$). The 3D polarimetric volumes on the left show depth-resolved determination of polarization responses in tissue which are generated by acquiring Mueller images confocally at incremental depths and then producing a 3D volume. This may help in differentiating between tissues which exhibit similar depth-unresolved cumulative polarimetric responses. The green SHG images on the right side can corroborate the Mueller decomposition images by providing information on collagen presence to assist in pathological detection. Two-photon excitation fluorescence was also imaged (TPEF, first two red panels in top row). (Adapted from [142])

4.2 *Mueller Polarimetry as a Guidance Tool for Mass Spectrometry*

Mass spectrometry is a very accurate technique for characterizing the chemical composition of materials, including biological tissues, by analysing the mass-to-charge ratios of constituent ionized molecules and atoms. This technique has undergone promising developments to enable highly sensitive and specific measurements, leading to impressive demonstrations including that of definitive tissue identification, cancer diagnosis, and tumour margin delineation [143, 144]. The disadvantage however is that typically large tissue regions must be raster scanned point-by-point to gain spatial information (i.e. imaging in a spatially untargeted fashion) which requires numerous measurements and long acquisition times. Thus, mass spectrometers may benefit significantly from a targeted approach to broaden their clinical scope.

As discussed previously, existing modalities can be significantly enhanced with the hybridization of Mueller polarimeters. In a similar vein, Mueller measurements afford useful additional contrast mechanisms for cancer detection, and thus may be used to guide mass spectrometers to pathological areas, thereby avoiding unnecessary measurements and (importantly) significantly reducing examination times. Thus, Woolman et al. [55] have demonstrated Mueller polarimetric guidance of mass spectrometry, employing both transmission imaging of thin tissues and reflectance imaging of thick tissues, the latter being a precursor for in-vivo assessment. In their study, they examined lymph nodes with breast cancer involvement which exhibited both viable and necrotic regions. For validation reference, the thin tissue was histologically stained by H&E, revealing the two regions shown in Fig. 10f. Mueller-derived depolarization images of the thin (10 μm) and thick (50 μm) tissues are shown in Fig. 10a and b, exhibiting excellent contrast identification of the viable and necrotic regions in agreement with the H&E slide. The thin tissue polarimetric image was used to determine pathological and heterogeneous regions of interest (Fig. 10c) where guided mass spectrometry was performed (Fig. 10d). For comparison, much slower untargeted mass spectrometry of the entire thin tissue was performed (Fig. 10e), using two mass-to-charge ratios corresponding to viable (red, m/z 572.48) and necrotic (green, m/z 391.25) cancers. The latter took ~6 h, while the polarimetry-guided mass spectrometry took only ~8 min, a tremendous reduction in time. This is of considerable importance in the clinic, for example in intraoperative assessment of excised tumours to ensure all cancerous tissue has been removed (e.g. breast cancer surgical lumpectomy procedure with its rather short margin assessment window before the surgeon closes the wound). This study thus demonstrated the feasibility of polarimetric-guided mass spectrometry of thin and thick tissues. Further research on the latter should be performed to investigate the prospects of in vivo polarimetric mass-spec guidance. If successful, this approach may have significant impacts on surgical oncology and other clinical applications.

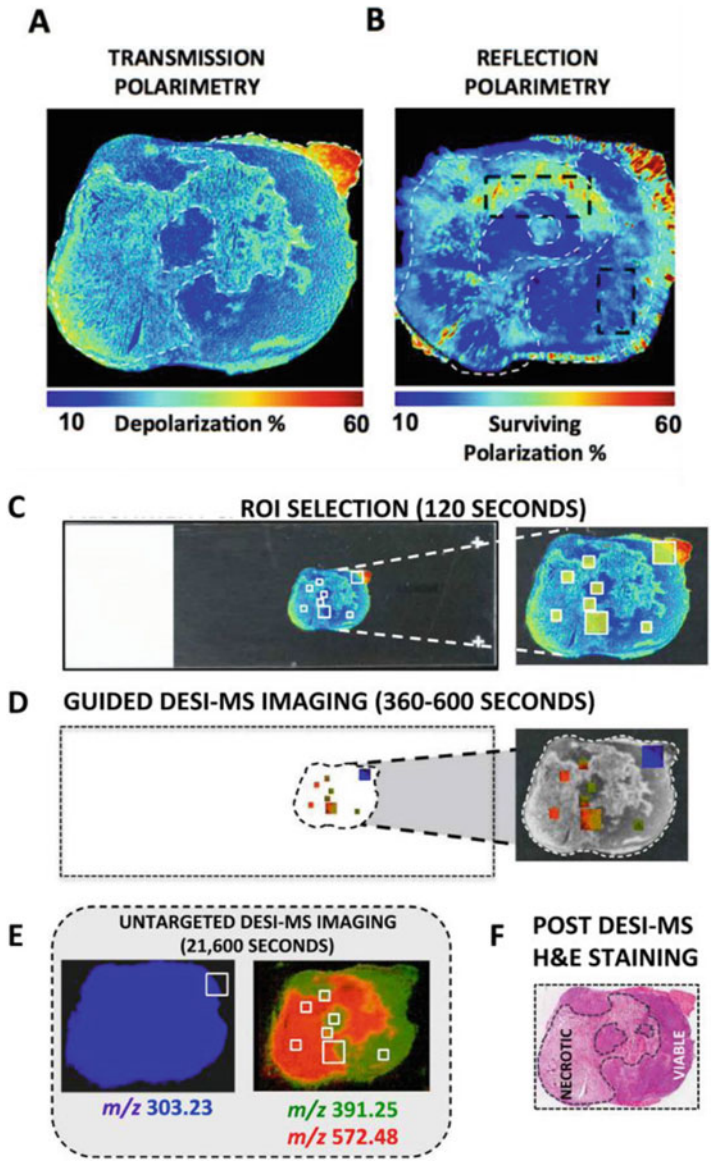


Fig. 10 Optimization of mass spectrometry using Mueller polarimetric guidance. (From [55]). The examined tissue was a cancer infiltrated lymph node with viable and necrotic regions. Mueller depolarization image of thinly sliced tissue (10 μm) using transmission polarimetry (**a**) and of thick tissue (50 μm) using reflectance polarimetry (**b**). The latter is a feasibility test for polarimetric guidance on thick tissues, important for clinical applications such as margin assessment in breast lumpectomy patients. The depolarization images of both tissues show excellent contrast identification of viable and necrotic regions (see (**f**)). (**c**) Suspected regions of pathology and tissue heterogeneity are selected on the thin tissue sample using the depolarization image. (**d**) Mass spectrometry is performed on those polarimetrically targeted regions, with an acquisition time of ~8 min. (DESI-MS = Desorption Electrospray Ionization mass spectrometry) (**e**) Untargeted mass spectrometry is performed on the whole slide using two mass-to-charge ratios corresponding to viable (red, m/z 572.48) and necrotic (green, m/z 391.25) cancers. This was done in a much longer acquisition time of ~6 h. (**f**) H&E staining of the thin tissue sample as a validation reference, performed after the mass spectrometry step

4.3 Endoscopic Mueller Polarimetry

The demonstrated tissue characterization potential of Mueller imaging, with its rich information content, non-invasive nature, and endogenous contrast, has prompted proposals for its use through optical fibres for endoscopic applications and surgical guidance. An excellent review of this topic has been written by Qi et al. [42] to which the reader may refer for details. Although in-vivo Mueller diagnostics can be done using free-space optics as in some of the preceding examples, the majority of bio-polarimetry applications have been performed ex vivo thus far. A key enabler towards wider in-vivo use will be fibre optic polarimetry, to gain significant access to human biology in various internal body cavities and orifices where many pathologies arise. The main challenge of this proposition is to overcome spurious polarization distortion artefacts introduced by (flexible) optical fibres. There also exists the 'usual' clinical engineering challenges of polarization generation and analysis in a practical manner (robustness, accuracy, speed, low complexity and cost, etc.).

To combat polarization distortion introduced by optical fibres (strain birefringence), the most effective solution thus far involves placing the polarization modulating components on the fibre's distal end. Qi et al. [145] used this approach to measure the linear 3 × 3 sub-matrix; this obviated the need for retarding components related to circular polarization generation and analysis, thereby simplifying the system. The information content of the linear sub-matrix, although less than that offered by the complete 4 × 4 MM, can still offer useful descriptions of sample properties such as scatterer concentration and size, as well as linear birefringence [146, 147]. To extract this information, the authors carried out the polar decomposition presented by Swami et al. [146] for 3 × 3 matrices. They then modified a rigid commercial endoscope consisting of an illumination channel with a linear polarizer affixed to its distal end, enabling the three linear states (horizontal, vertical, and 45°) to be generated by rotation of the entire fibre (motorized), and an imaging channel with a linear analyser placed before a CCD in the form of a rotating wheel containing three linear polarizers. The authors acquired depolarization and retardance images within a rat abdomen ex vivo in ~11.6 s, showing good discrimination between different organs as seen in Fig. 11a–c. It is thus clear that fibre optic MM polarimetry is feasible, at least in the context of a rigid endoscope. More information about this system will be provided in Chap. 7: Polarimetric Endoscopy.

Though the incomplete 3 × 3 Mueller polarimeter simplified the system design and reduced acquisition time, while also offering useful linear polarization-based information, measurement of the complete 4 × 4 MM is preferred since circular polarization also affords biomedically relevant information [91, 106, 148]. Qi and Elson [149] thus followed up with a complete fibre optic Mueller polarimeter using another rigid endoscope design utilizing a 'sheath' with motorized rotation. Its

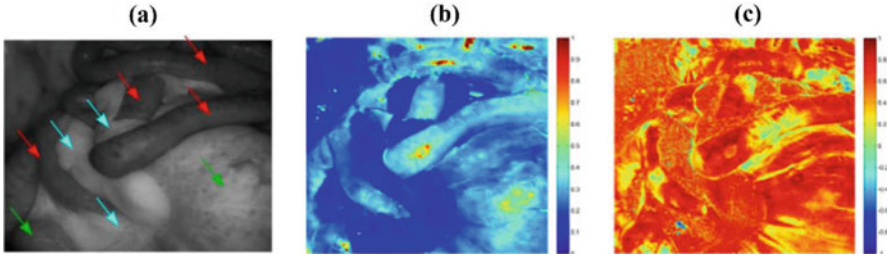


Fig. 11 (**a**) unpolarized image of rat abdomen with red, green, and blue arrows indicating the small bowel, large bowel, and fat, respectively. (**b**) linear depolarization image showing distinction between bowel and fat. (**c**) linear retardance image showing weak(er) discrimination between organs. The 3 × 3 polar decomposition affords useful information through linear depolarization. The images were taken with 546 nm light with field of view of 5.5 × 5.5 cm^2. (From [145])

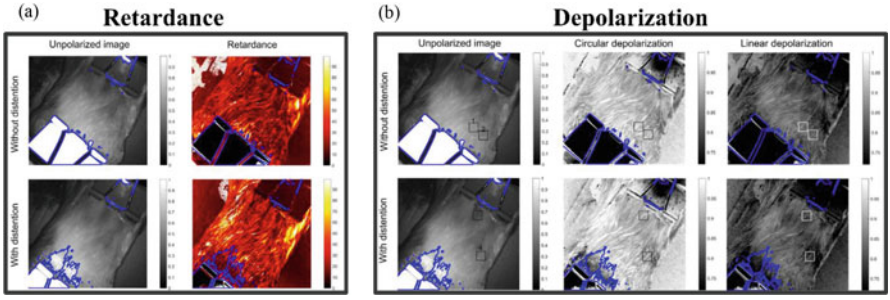


Fig. 12 Retardance (right column) and unpolarized (left column) images of porcine bladders. The bladders were normal (top row) or distended (bottom row) in both (**a**) and (**b**). There is clear increase in retardance with distention of the bladder, arising from induced birefringence. (**b**) unpolarized (first column from the left, same as in (**a**)), circular depolarization (second column), and linear depolarization (third column) images of porcine bladders. Regions enclosed by the blue contours indicate pixel saturation in at least one of the raw images used for Mueller image retrieval. The field of view of images in (**a**) and (**b**) are 7.8 × 7.8 cm^2

distal end contained a ring-shaped retarding film affixed to the rotatable sheath which covered the illumination channel and a stationary linear polarizer covering both channels. Such a design is analogous to a free-space MM polarimeter with motor-rotated retarders and stationary linear polarizers. The acquisition time of this system was 30 s, currently limited by the rotation speed of the sheath. The resultant rigid Mueller polarimetric endoscope was used to image normal and stretched porcine bladders with high fidelity, shown in Fig. 12a and b. The bladder distention does not appear to affect linear or circular depolarization; however, the latter is evidently much stronger than the former. Comparison of linear and circular depolarization can be informative of the scattering properties of the material, for

example, Mie scatterers tend to preserve linear polarization better than circular, whereas the opposite occurs in the Rayleigh scattering regime [21, 80]. There is thus an advantage of measuring the complete 4 × 4 Mueller matrix rather than just its linear 3 × 3 sub-matrix counterpart.

For all the advantages of rigid endoscopes, including more manageable polarized-light artefacts, longer shelf life, and lower chromatic aberrations [44], they are limited to relatively easy-to-access clinical sites. Flexible endoscopes offer a clear advantage in this regard with their ability to 'snake' into many more inner orifices and cavities of the body, thus gaining diagnostic access to a wider array of pathologies. Flexible fibres however introduce the much more challenging variable strain birefringence when bending and twisting which causes significant polarization distortion. To address this issue, Forward et al. [44] placed stationary linear polarizers on the distal ends of the flexible optical fibres. Thus, any strain birefringence would manifest as changes in illumination intensity (i.e. light intensity after the polarizers), which, importantly, could be accounted for with their careful calibration procedure. In total, six fibres were used (fashioned into a single probe), the illumination and detection channels consisted of three fibres each to form a PSA and PSG, respectively, with three linear polarizer orientations for each channel: horizontal, vertical, and −45° (see Fig. 13). Pairwise combinations of the PSG and PSA components enabled the measurement of the linear 3 × 3 MM. The authors tested the flexible fibre's performance in point-sensing mode on simple optical elements as well as ex-vivo tissues. The next steps will be to measure the complete

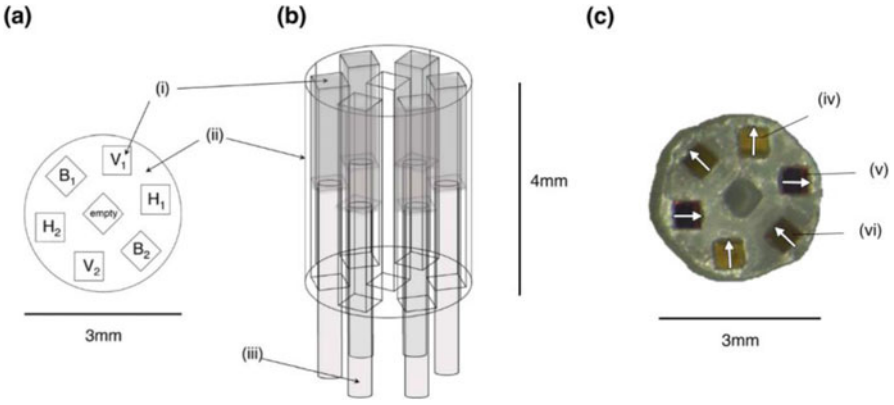


Fig. 13 Flexible Mueller polarimeter with six internal fibres for measuring the 3 × 3 linear MM. (From [44]). (**a**) Top view schematic showing the six internal fibres. Three polarizer orientations are used on the distal ends of each fibre: horizontal ($H_{1,2}$), vertical ($V_{1,2}$), −45° ($B_{1,2}$). There were two of each polarization orientation, one set as the PSG and the other set as the PSA. (**b**) Side view three-dimensional schematic showing the design of the housing for the fibres. (**c**) Top view photograph of the complete probe with the six internal fibres labelled with arrows indicating their polarizer orientation (labels (i)–(vi) can be ignored here)

4×4 MM, in imaging rather than point-sensing mode, to harness the full Mueller capabilities as illustrated previously. Optical fibre engineering advancements such as in-fibre dynamic polarization control [150] will likely accelerate progress towards these goals.

5 Summary and Future Outlook

Light has played an increasingly important role in biomedicine as a non-to-minimally invasive diagnostic tool by exploiting its many versatile properties. Practical utilization of these properties (e.g. modulation and detection) has relied heavily on the technologies of the day. Polarization is no exception; in fact, polarimetry is an excellent example of technology dependence to enable its deployment in the biomedical field. The major challenges of bio-polarimetry include (1) the weak signal effects from depolarization that must be overcome and (2) the complex and heterogeneous structure of tissues that must be deciphered. The information-rich Mueller matrix approach offers clear advantages with regard to (1) through various enabling technological advances and with regard to (2) through decomposition methods which can isolate the simultaneously occurring polarimetric effects. These advantages however come at a cost of measurement complexity (and associated lengthy acquisition times). Both the weak signal and measurement complexity challenges are being successfully addressed by enabling technologies and methodologies. Indeed, depolarization is becoming less of a hindrance and more of a diagnostic utility with the advent of more sensitive, robust, automated, and faster MM measurement systems.

With the challenges of MM tissue polarimetry steadily being addressed, its overall biomedical utility is finally at the stage of being properly evaluated. MM depolarization and retardance imaging presents strong contrast mechanisms between cancerous and non-cancerous tissues and may find a clinical niche in evaluating thin tissue slides to supplement current (expensive and often subjective) gold standard histopathological assessment. Additionally, rather invasive tissue biopsies for cancer detection may be reduced with the help of in-vivo polarimetric diagnostics, relying on bulk tissue retroreflection geometry deployments. In addition, although diattenuation has been thought to be of relatively little diagnostic value, it has recently shown some utility (e.g. burned bovine tendon study discussed previously). Optical activity, also amongst the weaker bio-polarimetric effects, may eventually be of significant clinical use in non-invasive glucose monitoring, which remains one of medicine's most pressing unsolved clinical needs. Overall then, MM polarimetry offers a suite of diagnostics options that are steadily finding their way into the biomedical arena.

It has also been shown that Mueller polarimeters can synergize with existing modalities through hybridization as seen in the combined OCT, SHG confocal microscopy, fluorescence microscopy, and mass spectrometry systems. Furthermore, the depth-resolved modalities can benefit Mueller polarimeters by providing

information on the often-problematic heterogeneous structure of tissue. The clinical scope of MM polarimetry may also be greatly extended through endoscopic techniques by enabling access to the interior of the various body cavities and orifices and hard-to-access anatomical sites where many pathologies arise. Additionally, the elusive prospect of detecting pre-cancer using polarized light [65, 66] may be significantly enhanced by probing sub-micron tissue structural anisotropy through multifractal analysis using MMs.

It seems that Mueller polarimetry is inextricably tied to both enabling technologies and novel methodologies which are increasingly extracting further biomedical diagnostic value from MMs. Far from having reached its potential in biomedicine (and other fields), MM polarimetry's true clinical prospects are arguably just beginning to emerge. Polarization – a fundamental property of light as well as one of the oldest to be studied – is thus beginning to earn its place amongst the more established optical diagnostic modalities in biomedicine, either alone or in combination, with exciting promise of selected clinical deployment in the near future.

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Polarization-Sensitive Monte Carlo



Tatiana Novikova and Jessica C. Ramella-Roman

Introduction

The propagation of polarized wave through scattering media has been studied by many groups, particularly from the atmospheric optics, astronomy, and oceanography communities. In 1968, Kattawar and Plass solved the vector radiative transfer equation (VRTE) for multiple scattered light in haze and clouds using the Monte Carlo method [1].

In biomedical optics, Monte Carlo programs have been used for the modeling of polarized light interaction with tissue. Measuring the intensities and polarization properties of the reflected/transmitted light, scientists extracted the unknown optical properties of biological sample (scattering coefficient μ_s, absorption coefficient μ_a and anisotropy factor g) by solving an inverse scattering problem. It is expected that the optical properties of pathological tissues differ from that of healthy one. These markers could be used for the optical diagnosis and pave the way for new non-invasive diagnostic tools.

An unpolarized Monte Carlo program for the solution of RTE in multilayered tissues was developed by Jacques and Wang [2] Later this program was generalized for the calculations of polarized light propagation in birefringent scattering samples [3] and modeling of time-resolved experiments [4]. Three Monte Carlo programs for the calculations of backscattering Mueller matrix, including the effect of skewed

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illumination were developed by Ramella-Roman et al. [5, 6] and shared freely. In these programs, the necessary tracking of the polarization reference frame was conducted in three different ways. Another freely available Monte Carlo code was proposed by D. Côté and A. Vitkin [7]. The generalized Monte Carlo model for simulation of coherent effects of multiple scattering, including Coherent Back-Scattering (CBS) and temporal intensity fluctuation of polarized laser radiation scattered within the random inhomogeneous turbid medium was developed by I. Meglinski. An online GPU-accelerated multipurpose Monte Carlo simulation tool for the needs of biophotonics and biomedical optics was developed by I. Meglinski and A. Doronin [8, 9].

A polarized Monte Carlo program was developed in Ecole polytechnique (France) for metrological applications [10, 11] and later generalized for modeling of polarized light propagation in biological samples [12]. In most Monte Carlo programs, photons emitted by a source are characterized by their position, direction of propagation, and given polarization (Stokes vector), propagating as ballistic photons till the next scattering location. Elastic scattering changes the direction and polarization of the photons, while absorption by the medium gradually reduces the statistical weight of a photon. Photons can be reflected or refracted crossing interfaces according to Fresnel's law. Photons will follow a trajectory within defined computational domain comprising regions with different optical properties. This process is repeated for the "photon packages" with different initial states of polarization. Depending on the characteristics of the detectors (position, sensitivity to the polarization, etc.), the detected photons are collected into bins. The photon intensities are averaged incoherently within each bin.

In this chapter, we cover the fundamentals of Monte Carlo modeling starting with the VRTE and its solutions. We further give some examples of polarized light Monte Carlo applications for biological media and conclude with a Matlab-based algorithm that we hope will aid researchers starting in this field.

1 Vector Radiative Transfer Equation (VRTE)

In the framework of the radiative transfer theory, the interaction of a polarized light beam with a dispersion medium can be described by the solutions of the vector integro-differential radiative transfer equation. We can derive this equation using the energy conservation law.

1.1 Integro-differential Form

We start first with the derivation of scalar RTE, neglecting the polarization effects. At every time moment, t and at each point $\mathbf{r}$ inside a scattering medium the intensity dI of light propagating in a solid angle $d\Omega$ around the direction defined by the unit

vector $\hat{\mathbf{\Omega}}$ is given by

$$dI\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) = L\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) d\Omega \tag{1}$$

where L is the radiance. The propagation of light directed into the solid angle $d\Omega$ over the period Δt leads to the variation of radiance L. The attenuation of light due to the absorption and scattering may be characterized by the extinction coefficient $\mu_{\mathrm{e}} = \mu_{\mathrm{s}} + \mu_{\mathrm{a}}$. The light beam can be deflected from their initial direction $\hat{\mathbf{\Omega}}$ and scattered into solid angle $d\Omega$, thus increasing the radiance L. The presence of internal sources (if any) continuously distributed within the volume and characterized by their radiance $\Sigma\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right)$ will also increase the radiance L. Light propagates within the medium with the velocity $v = c/n_m$ where n_m is the refractive index of the medium.

The variation of radiance along the direction $\hat{\mathbf{\Omega}}$ over a distance $v\Delta t$ is given by

$$\begin{aligned} &L\left(\mathbf{r} + v\Delta t\hat{\mathbf{\Omega}}, \hat{\mathbf{\Omega}}, t + \Delta t\right) - L\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) \\ &\quad = -\mu_e L\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) v\Delta t + \int_{\mathbf{\Omega}\prime} \mu_s v\Delta t \; p\left(\hat{\mathbf{\Omega}}' \to \hat{\mathbf{\Omega}}\right) L\left(\mathbf{r}, \hat{\mathbf{\Omega}}', t\right) d\Omega' + \Sigma\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) v\Delta t \end{aligned} \tag{2}$$

where $p\left(\hat{\mathbf{\Omega}}' \to \hat{\mathbf{\Omega}}\right)$ is the phase function with the norm

$$\int_{4\pi} p\left(\hat{\mathbf{\Omega}}' \to \hat{\mathbf{\Omega}}\right) d\Omega = \varpi \tag{3}$$

where optical albedo Ω of scattering and absorbing medium is defined as a ratio of scattering and extinction coefficients $\mu_{\mathrm{s}}/(\mu_{\mathrm{s}} + \mu_{\mathrm{a}})$. We have assumed that both scattering and extinction coefficients do not depend on time. Adding and subtracting from the left side of (2), the term $L\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t + \Delta t\right)$ and dividing it by $v\Delta t \to 0$ we express the evolution of radiance as

$$\begin{aligned} &\frac{1}{v}\frac{\partial L\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right)}{\partial t} + \hat{\mathbf{\Omega}}\nabla L\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) \\ &\qquad = -\mu_e L\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) + \int_{\mathbf{\Omega}\prime} \mu_s p\left(\hat{\mathbf{\Omega}}' \to \hat{\mathbf{\Omega}}\right) L\left(\mathbf{r}, \hat{\mathbf{\Omega}}', t\right) d\Omega' + \Sigma\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) \end{aligned} \tag{4}$$

To derive the vector RTE, we define a vector radiance $\mathbf{L}$ with the components of Stokes vector replacing the scalar intensity I. It will allow taking into account the polarization effects in RTE.

$$dS_i\left(\mathbf{r},\hat{\mathbf{\Omega}},t\right)=L_i\left(\mathbf{r},\hat{\mathbf{\Omega}},t\right)d\Omega,\quad (\mathrm{i}=0,..3) \tag{5}$$

Hence, the vector radiance **L** is the probability density function of Stokes vector components in Ω space. The scalar phase function $p\left(\hat{\mathbf{\Omega}}'\rightarrow\hat{\mathbf{\Omega}}\right)$ has also to be modified for the description of the vector radiance. The new phase function **P** is an angular dependent 4 × 4 real-valued matrix. Matrix **P** depends on the input vector radiance **L** because of normalization conditions.

In the reference frame of incident wave characterized by its Stokes vector $\mathbf{S}^0$ and traveling along $\hat{\mathbf{\Omega}}'$ direction ($\mathbf{e}^0{}_z=\hat{\mathbf{\Omega}}'$), the new direction $\hat{\mathbf{\Omega}}$ can be defined by polar and azimuthal angles (θ, φ). If scattering event happens and deflects the incident wave into direction $\hat{\mathbf{\Omega}}$, the following relation holds

$$\mathbf{S}^s(\theta,\varphi)=\mathbf{M}(\theta)\,\mathbf{R}_\mathbf{M}(\varphi)\,\mathbf{S}^0 \tag{6}$$

where $\mathbf{M}(\theta)$ is the Mueller matrix of a single spherical scatterer [13], $\mathbf{R}_\mathbf{M}(\varphi)$ is a the rotational change of basis matrix for Stokes vectors and Mueller matrices. Both incident and scattered wave are continuously distributed in the space according to their radiances. Hence, using Eqs. (5) and (6) we can write

$$\mathrm{d}^2\mathbf{S}^s\left(\mathbf{r},\hat{\mathbf{\Omega}},t\right)=\mathbf{L}\left(\mathbf{r},\hat{\mathbf{\Omega}},t\right)\mathrm{d}\Omega'\mathrm{d}\Omega=\frac{1}{I^s(\mathbf{L})}\mathbf{M}(\theta)\,\mathbf{R}_\mathbf{M}(\varphi)\,\mathbf{L}\left(\mathbf{r},\hat{\mathbf{\Omega}}',t\right)\mathrm{d}\Omega'\mathrm{d}\Omega \tag{7}$$

where $I^s(\mathbf{L})$ is defined as

$$I_s(\mathbf{L})=\mathrm{d}\Omega'\int_{4\pi}\left[\mathbf{M}(\theta)\,\mathbf{R}_\mathbf{M}(\varphi)\,\mathbf{L}\left(\mathbf{r},\mathbf{\Omega},t\right)\right]_0\mathrm{d}\Omega \tag{8}$$

Using (7) and (8), we can define the vector phase function **P** as

$$\mathbf{P}\left(\hat{\mathbf{\Omega}}'\rightarrow\hat{\mathbf{\Omega}},\mathbf{L}\right)=\frac{1}{I^s(\mathbf{L})}\mathbf{M}(\theta)\,\mathbf{R}_\mathbf{M}(\varphi) \tag{9}$$

Finally, the VRTE takes the following integro-differential form

$$\begin{aligned}&\frac{1}{v}\frac{\partial\mathbf{L}\left(\mathbf{r},\hat{\mathbf{\Omega}},t\right)}{\partial t}+\hat{\mathbf{\Omega}}\nabla\mathbf{L}\left(\mathbf{r},\hat{\mathbf{\Omega}},t\right)\\&\quad=-\mu_e\mathbf{L}\left(\mathbf{r},\hat{\mathbf{\Omega}},t\right)+\int_{\mathbf{\Omega}\prime}\mu_s\mathbf{P}\left(\hat{\mathbf{\Omega}}'\rightarrow\hat{\mathbf{\Omega}}\right)\mathbf{L}\left(\mathbf{r},\hat{\mathbf{\Omega}}',t\right)d\Omega'+\Sigma\left(\mathbf{r},\hat{\mathbf{\Omega}},t\right)\end{aligned} \tag{10}$$

1.2 Integral Form of VRTE

Equation (4) is a linear first-order partial integro-differential equation. It can be converted into an integral equation by methods of characteristics [14]. The general idea of this method consists of integration of VRTE along the photon trajectory of length s starting from the point $(\mathbf{r}_0, t_0)$ till the point $(\mathbf{r}, t)$, where

$$\mathbf{r} = \mathbf{r}_0 + s\hat{\mathbf{\Omega}}, t = t_0 + \frac{s}{v} \tag{11}$$

Equation (10) can be written as

$$\frac{1}{v}\frac{\partial \mathbf{L}\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right)}{\partial t} + \mathbf{\Omega}\nabla\mathbf{L}\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) + \mu_e\mathbf{L}\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) = q\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) \tag{12}$$

where $q\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) = \int\limits_{\mathbf{\Omega}\prime} \mu_s \mathbf{P}\left(\hat{\mathbf{\Omega}}' \to \hat{\mathbf{\Omega}}\right)\mathbf{L}\left(\mathbf{r}, \hat{\mathbf{\Omega}}', t\right)d\Omega' + \Sigma\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right)$ is the total rate at which photons appear at point $(\mathbf{r}, \hat{\mathbf{\Omega}}, t)$ as the result of collisions and internal and external sources. The first two terms of (12) can be written as $\left(\frac{1}{v}\frac{\partial}{\partial t} + \Omega_x\frac{\partial}{\partial x} + \Omega_y\frac{\partial}{\partial y}\Omega_z\frac{\partial}{\partial z}\right)\mathbf{L}$. Total derivative in the method of characteristics can be represented by $\frac{d\mathbf{L}}{ds} = \frac{\partial \mathbf{L}}{\partial t}\frac{dt}{ds} + \frac{\partial \mathbf{L}}{\partial x}\frac{dx}{ds} + \frac{\partial \mathbf{L}}{\partial y}\frac{dy}{ds} + \frac{\partial \mathbf{L}}{\partial z}\frac{dz}{ds}$

Identifying the terms in the last two expressions, we can find that

$$\left.\begin{array}{l} \frac{dt}{ds} = \frac{1}{v} \\ \frac{dx}{ds} = \Omega_x \\ \frac{dy}{ds} = \Omega_y \\ \frac{dz}{ds} = \Omega_z \end{array}\right. \Rightarrow \left.\begin{array}{c} t = t_0 + \frac{s}{v} \\ x = x_0 + s\Omega_x \\ y = y_0 + s\Omega_y \\ z = z_0 + s\Omega_z \end{array}\right\} \quad \begin{array}{c} t = t_0 + \frac{s}{v} \\ \mathbf{r} = \mathbf{r}_0 + s\hat{\mathbf{\Omega}} \end{array} \tag{13}$$

where t_0, x_0, y_0, z_0 are arbitrary constants. Consequently, VRTE (12) can be re-written as

$$\frac{d}{ds}\mathbf{L}\left(\mathbf{r}_0 + s\hat{\mathbf{\Omega}}, \hat{\mathbf{\Omega}}, t_0 + \frac{s}{v}\right) + \mu_e\mathbf{L}\left(\mathbf{r}_0 + s\hat{\mathbf{\Omega}}, \hat{\mathbf{\Omega}}, t_0 + \frac{s}{v}\right) = q\left(\mathbf{r}_0 + s\hat{\mathbf{\Omega}}, \hat{\mathbf{\Omega}}, t_0 + \frac{s}{v}\right) \tag{14}$$

We can integrate (14) by introducing the integrating factor

$$\begin{aligned} &\frac{d}{ds}\left[\mathbf{L}\left(\mathbf{r}_0 + s\hat{\mathbf{\Omega}}, \hat{\mathbf{\Omega}}, t_0 + \frac{s}{v}\right)\exp\int^s \mu_e\left(\mathbf{r}_0 + s'\hat{\mathbf{\Omega}}\right)ds'\right] \\ &\quad = \exp\left[\int^s \mu_e\left(\mathbf{r}_0 + s'\hat{\mathbf{\Omega}}\right)ds'\right]q\left(\mathbf{r}_0 + s\hat{\mathbf{\Omega}}, \hat{\mathbf{\Omega}}, t_0 + \frac{s}{v}\right) \end{aligned} \tag{15}$$

Integrating (15) from $s = -\infty$ till upper limit s and assuming that $\mathbf{L}\left(\mathbf{r}_0 + s\hat{\mathbf{\Omega}}, \hat{\mathbf{\Omega}}, t_0 + \frac{s}{v}\right) \to 0$ when $s \to -\infty$ we obtain for the left part of (15)

the expression $\mathbf{L}\left(\mathbf{r}_0 + s\hat{\mathbf{\Omega}}, \hat{\mathbf{\Omega}}, t_0 + \frac{s}{v}\right) \exp \int^s \mu_e \left(\mathbf{r}_0 + s'\hat{\mathbf{\Omega}}\right) ds'$. Then multiplying both sides of Eq. (15) by factor $\exp\left(-\int^s \mu_e \left(\mathbf{r}_0 + s'\hat{\mathbf{\Omega}}\right) ds'\right)$ we get

$$\mathbf{L}\left(\mathbf{r}_0 + s\hat{\mathbf{\Omega}}, \hat{\mathbf{\Omega}}, t_0 + \frac{s}{v}\right) = \int_{-\infty}^{s} \exp\left[\int_{s'}^{s} -\mu_e \left(\mathbf{r}_0 + s''\hat{\mathbf{\Omega}}\right) ds''\right] \left[q\left(\mathbf{r}_0 + s'\hat{\mathbf{\Omega}}, \hat{\mathbf{\Omega}}, t_0 + \frac{s'}{v}\right)\right] ds' \tag{16}$$

Setting $\mathbf{r} = \mathbf{r}_0 + s\hat{\mathbf{\Omega}}$, $t = t_0 + \frac{s}{v}$ and changing the sign of variables, we obtain the integral form of transport equation:

$$\mathbf{L}\left(\mathbf{r}, \hat{\mathbf{\Omega}}, t\right) = \int_0^{\infty} \exp\left[-\int_0^{s} \mu_e \left(\mathbf{r} - s''\hat{\mathbf{\Omega}}\right) ds''\right] \left[q\left(\mathbf{r}_0 - s'\hat{\mathbf{\Omega}}, \hat{\mathbf{\Omega}}, t_0 - \frac{s'}{v}\right)\right] ds' \tag{17}$$

Since $1/\mu_e$ is equal to the collision mean free path, the exponent in the attenuation factor $\exp\left[-\int_0^s \mu_e \left(\mathbf{r} - s''\hat{\mathbf{\Omega}}\right) ds''\right]$ is equal to the number of the collision mean free paths along the straight line between $\mathbf{r}$ and $\mathbf{r}$ - $s'\hat{\mathbf{\Omega}}$. It is frequently called the optical path length between the two points. If we assume that there is only external source of photons, the VRTE in its integral form can be reformulated as

$$\mathbf{L} = \mathbf{KL} + \mathbf{Q}', \tag{18}$$

where $\mathbf{K}$ is integral operator. If we consider the solution of (18) obtained by the iterations

$$\begin{aligned} &\mathbf{L}_0 = \mathbf{Q}' \\ &\mathbf{L}_1 = \mathbf{KL}_0 \\ &\cdots\cdots \\ &\mathbf{L}_{n+1} = \mathbf{KL}_n, \end{aligned}$$

it can be shown that at certain conditions on $\mathbf{K}$ and $\mathbf{Q}'$ this iterative process converges to the solution of (18). The estimation of the integral of collision $\mathbf{K}$ can be performed by the Monte Carlo technique presented in the next section.

2 Monte Carlo Solution of the VRTE

For any application, the Monte Carlo method requires a stochastic model, where the mean values of a certain random variables are the values of the physical properties (in our case, the Mueller matrix coefficients of the scattering medium)

to be determined by means of the numerical calculation [15]. These mean values are calculated by averaging over the multiple series of the independent samples.

To solve the vector RTE in turbid media with Monte Carlo technique, one needs to simulate the random walk of incident "photon packages" within the scattering and absorbing media before being detected. This approach is particularly appropriate for the modeling of real optical experiments with complex geometry (optical interfaces of arbitrary shape, various configurations for signal detection), and samples combined of different optical materials (scattering, absorbing, having different optical refractive indices, etc.).

The drawback of using Monte Carlo method for the solution of radiative transport problem is its high computational cost. As the technique is not deterministic, we have to estimate the variance of the quantities of interest— Mueller matrix coefficients of a sample. When calculated standard deviation is below the defined threshold (clearly its value influences the accuracy of the results), the calculations are stopped. As it follows from the central limit theorem, the Mueller matrix coefficients are within ±3% of the mean values with 95% confidence when the variance threshold is set at 1.5%. The decrease of standard variance and standard deviation are proportional to $1/n$ and $1/\sqrt{n}$, respectively, where n is the number of simulated photon trajectories.

2.1 Generation of Random Numbers with Given Probability Distribution

Let us define as $P(z)$, the probability distribution function of a random variable z ($z_0 \le z \le z_1$). It means that probability

$$dP\,(z \in dz) = P(z)\mathrm{d}z \tag{19}$$

Then cumulative probability function $f_P(z)$ is given by

$$f_P(z) = \int_{z_0}^{z} P(v)dv \tag{20}$$

It is clear that if $z_0 \le z \le z_1$ with the distribution probability function $P(z)$, the cumulative probability function $f_P(z)$ varies between 0 and 1 with its own distribution probability function $Q(f_P)$, which is related to $P(z)$ as

$$Q\,(f_P)\,\mathrm{d}f_P = P(z)\mathrm{d}z \tag{21}$$

It follows from (21) that cumulative probability function $f_P(z)$ is uniformly distributed between 0 and 1

$$Q\left(f_P\right) = \frac{P(z)}{\mathrm{d}f_P/\mathrm{d}z} = 1 \tag{22}$$

Hence, if we generate a random uniformly distributed variables $f_P \in [0,1]$, the solutions to (20) will obey a defined probability distribution function $P(z)$ and $z_0 \leq z \leq z_1$. When photon propagates through homogeneous absorbing and scattering medium, characterized by its extinction coefficient μ_e, the probability of photon not being scattered or absorbed over the distance z is given by

$$P(z)\mathrm{d}z = \mu_e \exp\left(-\mu_e z\right) \mathrm{d}z \tag{23}$$

Putting (23) into (20), we obtain

$$f_P(z) = \int_{z_0}^{z} \mu_e \exp\left(-\mu_e v\right) dv = \left(1 - \exp\left(-\mu_e z\right)\right) \tag{24}$$

or

$$z = -\frac{\ln\left(1 - f_P\right)}{\mu_e} = \frac{\ln\left(f'_P\right)}{\mu_e} \tag{25}$$

where f'_P is random variable uniformly distributed on [0,1].

When probability distribution function $P(z)$ cannot be integrated analytically, one can implement the so-called *acceptance-rejection* method. The random equi-distributed draws of both $z \in [z_0, z_1]$ and $x \in [0,1]$ are performed independently by a standard numerical random number generator [16]. Then $P(z)$ is calculated and compared with x. If $x \leq P(z)$ the value of z is accepted, otherwise it is rejected. When the dimensionality of the problem increases, the acceptance-rejection method can lead to many unwanted samples being taken before the accepted sample is generated.

Let us consider the case, when the probability density function depends on two variables (not necessarily independent) and can be decomposed into the sum of two positive terms

$$P\left(\theta, \varphi\right) = f_1\left(\theta, \varphi\right) + f_2\left(\theta, \varphi\right) \tag{26}$$

The functions f_1 and f_2 are not the probability density functions, because they are not normalized. Introducing the norms F_1 and F_2 (the integrals of f_1 and f_2 over the full finite range of variations of variables θ and φ), we can present the probability density function $P(\theta, \varphi)$ as a linear combination of two probability density functions $P_1(\theta, \varphi)$ and $P_2(\theta, \varphi)$:

$$P\left(\theta, \varphi\right) = F_1 P_1\left(\theta, \varphi\right) + F_2 P_2\left(\theta, \varphi\right) \tag{27}$$

We can interpret (27) as a splitting of all possible events generating the pair of (θ, φ) variables into two mutually excluding groups. The numerical procedure of generating the (θ, φ) pairs can be described as follows:

- Draw a number u, which is uniformly distributed in the interval [0; 1]
- If $u < F_1/(F_1 + F_2)$, then the pair (θ, φ) has to be drawn following the probability density function P_1
- Otherwise the probability density function P_2 has to be used

This method is of particular interest if the variables θ and φ can be considered as independent ones for both probability density functions P_1 and P_2, as it will be shown below.

2.2 Photon Random Walk

Since the Monte Carlo algorithm simulates the random trajectory of the polarized photon within the scattering medium, one must define the photon path. It is constructed as the recursive sequence of the following steps:

1. Characterize the source photon by choosing its position, direction, and initial polarization.
2. Define the site and the type of next event (scattering on particle, transmission, or reflection on the interfaces). If the photon has reached the boundaries of the domain of simulations, go to step 5.
3. Change the direction of propagation of the photon and its polarization according to the type of event.
4. Scoring, check if the photon stays within the domain of simulations. If so, back to step 2.
5. The photon path terminates whenever it reaches the boundaries of the domain of simulations which means that it has no return possibility.

The sequence of N photons (package) with the same input Stokes vector $\mathbf{S}^{\text{in}}$ (i.e., the same polarization state) is emitted from a source. Due to the statistical nature of Monte Carlo technique, each individual photon follows its own trajectory in the scattering medium, and finally contributes to the detector (if being detected) independently, carrying its unique output Stokes vector.

Hence, for the fixed $\mathbf{S}^{\text{in}}$, an output Stokes vector is found by averaging over the simulated output polarization states of the photon. This step is repeated with the same $\mathbf{S}^{\text{in}}$, until the variance threshold is reached. For the accurate estimation of the Mueller matrix of a sample, the initial polarization state of the photon package has to be varied over the surface of the Poincaré sphere widely enough. There are two possibilities for the selection of the initial polarization:

1. Four fixed polarization states with linearly independent Stokes vectors. The optimal choice corresponds to vortices of regular tetrahedron inscribed into the Poincaré sphere.
2. Statistical sampling of the input Stokes vectors over the whole surface of Poincaré sphere. This choice becomes the most effective in terms of calculation time for multiple scattering media.

To calculate the Mueller matrix of a sample when using the minimal sampling approach (1), the following matrix equation is obtained:

$$\mathbf{F} = \mathbf{M} \cdot \mathbf{W}, \tag{28}$$

where four different input Stokes vectors form the columns of matrix **W** and four corresponding output Stokes vectors form the columns of matrix **F**. When four vectors $\mathbf{S}^{\text{in}}$ are linearly independent, **W** can be inverted, and Mueller matrix is defined as

$$\mathbf{M} = \mathbf{F} \cdot \mathbf{W}^{-1} \tag{29}$$

Using the approach (2), the statistical sampling of the input Stokes vectors at the surface of the Poincaré sphere can be described by

$$\mathbf{S}^{in}(\alpha, \beta) = [1, \quad \cos\alpha, \quad \sin\alpha\cos\beta, \quad \sin\alpha\sin\beta]^{\mathrm{T}} \tag{30}$$

where $\alpha = \frac{\pi}{2} - 2\varepsilon$, $\beta = 2\theta$ are the polar and the azimuthal angles on the Poincaré sphere. These angles vary over the intervals [0, π] and [0, 2π], respectively, according to a joint probability density function $P(\alpha, \beta)$.

If we define the correlation matrix **D** of the input components of Stokes vector as

$$\mathbf{D} = \int_{\alpha=0}^{\alpha=\pi} \int_{\beta=0}^{\beta=2\pi} \mathbf{S}^{in}(\alpha, \beta) \cdot \left[\mathbf{S}^{in}(\alpha, \beta)\right]^{\mathrm{T}} P(\alpha, \beta)\, \mathrm{d}\alpha\, \mathrm{d}\beta \tag{31}$$

and similarly, the correlation matrix **G** of the input and output components of Stokes vector as

$$\mathbf{G} = \int_{\alpha=0}^{\alpha=\pi} \int_{\beta=0}^{\beta=2\pi} \mathbf{S}^{out}(\alpha, \beta) \cdot \left[\mathbf{S}^{in}(\alpha, \beta)\right]^{\mathrm{T}} P(\alpha, \beta)\, \mathrm{d}\alpha\, \mathrm{d}\beta =$$

$$\mathbf{M} \int_{\alpha=0}^{\alpha=\pi} \int_{\beta=0}^{\beta=2\pi} \mathbf{S}^{in}(\alpha, \beta) \cdot \left[\mathbf{S}^{in}(\alpha, \beta)\right]^{\mathrm{T}} P(\alpha, \beta)\, \mathrm{d}\alpha\, \mathrm{d}\beta = \mathbf{M} \cdot \mathbf{D}, \tag{32}$$

a Mueller matrix **M** can be obtained by the inversion of a matrix **D**:

$$\mathbf{M} = \mathbf{G} \cdot \mathbf{D}^{-1} \tag{33}$$

It is obvious that the existence of $\mathbf{D}^{-1}$ depends on the choice of $P(\alpha, \beta)$. We chose $P(\alpha, \beta)$ to be a uniform distribution over the intervals $[0, \pi]$ and $[0, 2\pi]$. It gives a simple diagonal matrix $\mathbf{D}$

$$D = \pi^2 \begin{bmatrix} 2\,0 & 0 & 0 \\ 0\,1 & 0 & 0 \\ 0\,0 & 1/2 & 0 \\ 0\,0 & 0 & 1/2 \end{bmatrix} \tag{34}$$

Let us now consider in detail the phenomena treated within Step 2 of Monte Carlo algorithm, namely, scattering on the spherical scatterers and the reflection/transmission on the interfaces.

2.3 Scattering by the Spheres Embedded in a Homogeneous Medium

It was already shown that for photon propagating in scattering and absorbing medium with extinction coefficient μ_e the distance d between two scattering events is a random variable defined as

$$d = \frac{\ln(z)}{\mu_e} \tag{35}$$

where z is a random number uniformly distributed between 0 and 1 (see Eqs. (23), (24), and (25)).

The probability $P(\theta, \varphi)$ that photon will be scattered at angle θ and its initial local reference frame $\left(\mathbf{e}_{\mathbf{x}}^{\mathbf{0}}, \mathbf{e}_{y}^{\mathbf{0}}, \mathbf{e}_{z}^{\mathbf{0}}\right)$ will be rotated by angle φ (see Fig. 1) is calculated as a

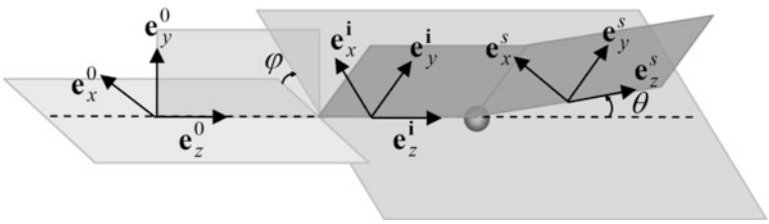


Fig. 1 Scattering angles θ and φ are not independent parameters. The initial local reference frame $\left(\mathbf{e}_{\mathbf{x}}^{\mathbf{0}}, \mathbf{e}_{y}^{\mathbf{0}}, \mathbf{e}_{z}^{\mathbf{0}}\right)$ is rotated by angle φ to obtain the incident frame $\left(\mathbf{e}_{\mathbf{x}}^{\mathbf{i}}, \mathbf{e}_{y}^{\mathbf{i}}, \mathbf{e}_{z}^{\mathbf{i}}\right)$ with unit vectors $\mathbf{e}_{\mathbf{x}}^{\mathbf{i}}$and $\mathbf{e}_{y}^{\mathbf{i}}$ parallel and perpendicular to the scattering plane defined by the incident direction and direction of scattering (which is defined by angle θ). After scattering of photon by a sphere photon's new polarization state is given by the Mie solution defined in the scattering reference frame $\left(\mathbf{e}_{\mathbf{x}}^{s}, \mathbf{e}_{y}^{s}, \mathbf{e}_{z}^{s}\right)$, which becomes a new local reference frame. (Adapted from [17])

function of the photon energy, radius of spherical scatterer and relative optical index contrast by Mie solution [13]

We need first to rotate vector $\mathbf{S^0}$ and reference frame $\left(\mathbf{e_x^0}, \mathbf{e}_y^0, \mathbf{e}_z^0\right)$ in Stokes space and real space, respectively, by angle φ about the direction of propagation defined by vector $\mathbf{e}_z^0$.

$$\mathbf{S^i} = \mathbf{R_M}(\varphi) \cdot \mathbf{S^0}, \quad \begin{cases} \mathbf{e_x^i} = \cos\varphi\ \mathbf{e_x^0} + \sin\varphi\ \mathbf{e}_y^0 \\ \mathbf{e}_y^\mathbf{i} = -\sin\varphi\ \mathbf{e_x^0} + \cos\varphi\ \mathbf{e}_y^0 \\ \mathbf{e_z^i} = \mathbf{e_z^0} \end{cases} \tag{36}$$

Scattering of photon by the angle θ is described by a matrix $\mathbf{M}(\theta)$ defined in a scattering reference frame.

$$\begin{bmatrix} I^s \\ Q^s \\ U^s \\ V^s \end{bmatrix} = \frac{1}{k^2 n_m^2 r^2} \begin{bmatrix} M_{00} & M_{01} & 0 & 0 \\ M_{01} & M_{00} & 0 & 0 \\ 0 & 0 & M_{22} & M_{23} \\ 0 & 0 & -M_{23} & M_{22} \end{bmatrix} \begin{bmatrix} I^i \\ Q^i \\ U^i \\ V^i \end{bmatrix} \tag{37}$$

$$M_{00} = \frac{1}{2}\left(|S_2|^2 + |S_1|^2\right), M_{01} = \frac{1}{2}\left(|S_2|^2 - |S_1|^2\right),$$

$$M_{22} = \frac{1}{2}\left(S_2^* S_1 + S_2 S_1^*\right), M_{23} = \frac{1}{2}\left(S_2^* S_1 - S_2 S_1^*\right)$$

Not all coefficients are linearly independent: $M_{00}^2 = M_{01}^2 + M_{22}^2 + M_{23}^2$

This reference frame is obtained by rotating the incident reference frame $\left(\mathbf{e_x^i}, \mathbf{e}_y^\mathbf{i}, \mathbf{e}_z^\mathbf{i}\right)$ by angle θ about the direction $\mathbf{e_y^i}$ perpendicular to the reference plane.

$$\mathbf{S}^s = \mathbf{M}(\theta)\,\mathbf{R_M}(\varphi) \cdot \mathbf{S^0}, \quad \begin{cases} \mathbf{e_x^s} = \cos\theta\ \mathbf{e_x^i} - \sin\theta\ \mathbf{e_z^i} \\ \mathbf{e_y^s} = \mathbf{e_y^i} \\ \mathbf{e_z^s} = \sin\theta\ \mathbf{e_x^i} + \cos\theta\ \mathbf{e_z^i} \end{cases} \tag{38}$$

The intensity $\mathrm{d}I^s$ of light scattered in a solid angle $\mathrm{d}\Omega$ around the direction $\hat{\mathbf{\Omega}}$ is given by the first component of Stokes vector of scattered light $\mathbf{S^s}$:

$$\mathrm{d}I^s = S_0^s(\theta, \varphi)\,\mathrm{d}\Omega \tag{39}$$

where

$$S_0^s(\theta, \varphi) = M_{00}(\theta)\, I + M_{01}(\theta)\cos(2\varphi)\, Q - M_{01}(\theta)\sin(2\varphi)\, U$$

$$= \frac{1}{2}\left[|S_1(\theta)|^2 (I - Q\cos(2\varphi) + U\sin(2\varphi)) + |S_2(\theta)|^2 (I + Q\cos(2\varphi) - U\sin(2\varphi))\right] \tag{40}$$

where $|S_i(\theta)|^2$ are the squared values of complex amplitude functions of a scattered electric field

$$S_1(\theta) = \sum_n \frac{2n+1}{n(n+1)} \left(a_n \pi_n(\cos\theta) + b_n \tau_n(\cos\theta)\right) \tag{41}$$

$$S_2(\theta) = \sum_n \frac{2n+1}{n(n+1)} \left(a_n \tau_n(\cos\theta) + b_n \pi_n(\cos\theta)\right) \tag{42}$$

$$\pi_n(\cos\theta) = \frac{P_n^1(\cos\theta)}{\sin\theta}, \quad \tau_n(\cos\theta) = \frac{P_n^1(\cos\theta)}{d\theta}, \quad P_n^1 = -\frac{dP_n}{d\theta} \tag{43}$$

where the functions P_n are the Legendre polynomials of degree n [13]. The total intensity of the scattered light in the space is obtained by integrating the intensity of the scattered light over all directions of propagation included in the sphere of unit volume:

$$I^s = \int S_0^s d\Omega = \int_{\theta=0}^{\theta=\pi} \int_{\varphi=0}^{\varphi=2\pi} S_0^s(\theta,\varphi) \sin\theta \, d\theta \, d\varphi = \pi I \int_{\theta=0}^{\theta=\pi} \left(|S_1(\theta)|^2 + |S_2(\theta)|^2\right) \sin\theta d\theta$$

$$= \pi I \int_{\theta=0}^{\theta=\pi} \left(|S_1(\theta)|^2\right) \sin\theta d\theta + \pi I \int_{\theta=0}^{\theta=\pi} \left(|S_2(\theta)|^2\right) \sin\theta d\theta = \pi I (T_1 + T_2) \tag{44}$$

The intensity of scattered light in any direction of propagation (characterized by the angles θ and φ) is proportional to the number of the photons scattered along that direction. The combined angular probability density function $P(\theta, \varphi)$ can be written as

$$P(\theta,\varphi) d\theta d\varphi = \frac{S_0^s(\theta,\varphi)}{I^s} \sin\theta \, d\theta \, d\varphi \tag{45}$$

This function is a joint probability density function of the random variables θ and φ, hence we can apply the acceptance–rejection method described above to generate the values of angles θ and φ. Three random numbers θ_0, φ_0, and ξ, uniformly distributed over the intervals $[0, \pi]$, $[0, 2\pi]$, and $[0, \max_{\theta,\varphi} S_0^s(\theta,\varphi)]$, respectively, are generated first. If $\xi \le S_0^s(\theta_0, \varphi_0)$ then pair (θ_0, φ_0) is accepted, otherwise random draw of θ_0, φ_0 and ξ is repeated. This probability of acceptance is always smaller than 0.5 (limit case for very small scattering particles) and decreases with the increase of scatterers size, leading to the unproductive increase of computational time.

To avoid this problem, we suggest to decompose the angular probability density function $P(\theta, \varphi)$ in a sum of two positive terms (see Eq. (27)). First we define φ_0 and W from the following expressions using the first three components of the incident Stokes vector:

$$I^2 W^2 = Q^2 + U^2 \tag{46}$$

$$\tan 2\varphi_0 = U/Q \tag{47}$$

$W \in [0, 1]$, because $I^2 \geq Q^2 + U^2 + V^2$. It follows from Eqs. (46) and (47) that

$$Q = IW\cos(2\varphi_0), \tag{48}$$

$$U = IW\sin(2\varphi_0) \tag{49}$$

Multiplying Eq. (48) by cos2φ and Eq. (49) by sin2φ and then subtracting them, we get

$$Q\cos 2\varphi - U\sin 2\varphi = IW\cos[2(\varphi - \varphi_0)], \tag{50}$$

Consequently, substituting (50) in (40), we obtain

$$\begin{aligned} S_0^s(\theta,\varphi) &= \frac{1}{2}\left[|S_1(\theta)|^2 (I - IW\cos 2(\varphi - \varphi_0)) + |S_2(\theta)|^2 (I + IW\cos 2(\varphi - \varphi_0))\right] \\ &= \frac{I}{2}\left[|S_1(\theta)|^2 \left(1 - W + 2W\sin^2(\varphi - \varphi_0)\right) + |S_2(\theta)|^2 \left(1 - W + W\cos^2(\varphi - \varphi_0)\right)\right] \end{aligned} \tag{51}$$

Using (44) and (51), the combined angular probability density function $P(\theta, \varphi)$ from (45) can be written as

$$\begin{aligned} P(\theta,\varphi) &= \frac{1}{2\pi(T_1 + T_2)}\left[t_1(\theta,\varphi) + t_2\left(\theta,\varphi\right)\right] \\ &= \frac{1}{2\pi}\left[\frac{T_1}{(T_1 + T_2)}\frac{t_1(\theta,\varphi)}{T_1} + \frac{T_2}{(T_1 + T_2)}\frac{t_1(\theta,\varphi)}{T_2}\right] \end{aligned} \tag{52}$$

where

$$t_i(\theta,\varphi) = \frac{1}{2}|S_i(\theta)|^2 \sin\theta\left[1 - W + 2W\sin^2\left(\varphi - \varphi_0 + (i-1)\frac{\pi}{2}\right)\right], i = 1, 2 \tag{53}$$

The functions $P_1(\theta,\varphi) = \frac{t_1(\theta,\varphi)}{T_1}$ and $P_2(\theta,\varphi) = \frac{t_2(\theta,\varphi)}{T_2}$ are the probability density functions. Moreover, each of the functions P_1 and P_2 is the product of $P_i(\theta)$ and $P_i(\varphi)$ (i = 1,2), meaning that the random variables θ and φ are statistically independent.

The procedure of generating the angles θ and φ can be summarized as:

1. The technique of decomposition in sum of two positive terms described below the Eq. (27) is applied first to choose either probability density function $P_1(\theta,\varphi)$ or $P_2(\theta,\varphi)$.
2. The angle θ is obtained by applying the acceptance-rejection method for the probability density function

$$P_i(\theta) = \frac{|S_i(\theta)|^2 \sin\theta}{\int_0^\pi |S_i(\theta)|^2 \sin\theta \, d\theta} \tag{54}$$

3. Then again, the technique of decomposition is applied to the probability density functions

$$P_1(\varphi) = \frac{1}{2\pi}(1 - W) + \frac{1}{\pi} W \sin^2(\varphi - \varphi_0), \tag{55}$$

$$P_2(\varphi) = \frac{1}{2\pi}(1 - W) + \frac{1}{\pi} W \cos^2(\varphi - \varphi_0), \tag{56}$$

as these probability density functions are the sum of a constant value $(1 - W)/2\pi$ and the function $g_i(\varphi) = \frac{W}{\pi}\sin^2\left(\varphi - \varphi_0 + (i-1)\frac{\pi}{2}\right)$, $i = 1,2$.

4. For the choice of constant value, the angle φ is drawn uniformly over the interval $[0, 2\pi]$, otherwise the angle φ is determined by the inversion of the probability density function $G_i(\varphi) = \frac{\pi}{W} g_i(\varphi)$ obtained by the normalization of g_i.

2.4 Reflection/Refraction at the Interfaces

When a photon with the current position $\mathbf{r_n}$ and direction of propagation $\mathbf{\Omega_n}$ continues its travel through the scattering medium, we first calculate the scattering distance d from Eq. (35) and compare it with the geometrical distance d_0 to the closest interface defined as $\mathbf{r_{n+1}} - \mathbf{r_n} = d_0 \, \hat{\mathbf{\Omega}}_\mathbf{n}$. If $d_0 < d$ then the next location of a photon will be considered at the interface, where it can be either reflected or transmitted. To make a choice between the reflection and transmission, we use the Fresnel laws for the interface between two semi-infinite media. First, we rotate the frame of reference $(\mathbf{e_x}, \mathbf{e}_y, \mathbf{e}_z)$ by the corresponding angles until the axes x and y become perpendicular and parallel, respectively, to the plane of incidence. Then the reflectivity for an incoming Stokes vector $\mathbf{S} = [I, Q, U, V]^\mathrm{T}$is defined as

$$R = \frac{1}{2}\left[\left(r_\mathrm{II}^2 + r_\perp^2\right) + (Q/I)\left(r_\mathrm{II}^2 - r_\perp^2\right)\right] \tag{57}$$

where r_II and $r_\perp$ are the complex reflectance coefficients from the reflection Jones matrix. The random number p_R uniformly distributed over [0, 1] is generated and compared to R. In case $p_R < R$, the photon is reflected at the interface, otherwise it is transmitted. The state of polarization of a reflected/transmitted photon is defined as a product of its Stokes vector and the interface reflection/transmission Mueller matrix obtained from the Fresnel laws. The new direction of propagation is obtained by using the Snell law.

3 Programmatic Example of Polarization-sensitive Monte Carlo

In this section, we illustrate a typical polarization-sensitive Monte Carlo program for modeling the propagation of polarized light through isotropic host medium with embedded spherical scatterers. To do so, we will use the MATLAB® language, a matrix-based language well known by scientists. The program shared here has an educational purpose and will not have the necessary speed for serious computational analysis. Several C or C++ based programs are available and the reader is referred to the following references for more computationally intensive work [5].

The program proposed here follows the Meridian Plane approach and its flowchart is reproduced from Ramella-Roman et al. [5] (Fig. 2).

At the start of the program, constants such as the element of the scattering matrix and geometry parameters are defined. In the Matlab implementation of this program, the scattering parameters are obtained from the MatScat Matlab library [18, 19].

```
% Calculate amplitude scattering matrix, n_s, n_m are index of
refraction of
% sphere and media, respectively, sphere_r is the sphere radius,
wavelength, and nang the number of far
field angles to evaluate
[S, C, ang] = calcmie(sphere_r, n_s, n_m, lambda, nang,
'ConvergenceFactor', 1);
% Calculate scattering matrix.
M = getMuellerMatrix(S);
s11=squeeze(M(1,1,:));
s12=squeeze(M(1,2,:));
s33=squeeze(M(3,3,:));
s43=squeeze(M(4,3,:));
```

All the steps described below will be run within a loop whose length is determined by the desired total number of photons used in the simulation.

As reported above, all Monte Carlo programs start with the definition of starting parameters and constants, specifically:

1. Define source photon by choosing its position (x,y,z), direction determined by direction cosines U, and initial polarization described in form of a Stokes vector S. We will also define the weight of a photon package W.

```
x = 0.0; y = 0.0; z = 0.0; % pencil beam
U = [0 0 1]; % Initial direction cosine
S = [1 1 0 0]; % Initial Stokes Vector
W=1; % Initialize photon package weight
```

2. Define the site and the type of the next event, this will require the calculation of a step length (s). This step is common to polarized and unpolarized Monte Carlo programs. If the photon has reached the boundaries of the value of the Stokes vector, it is updated through a rotation to the main laboratory reference frame.

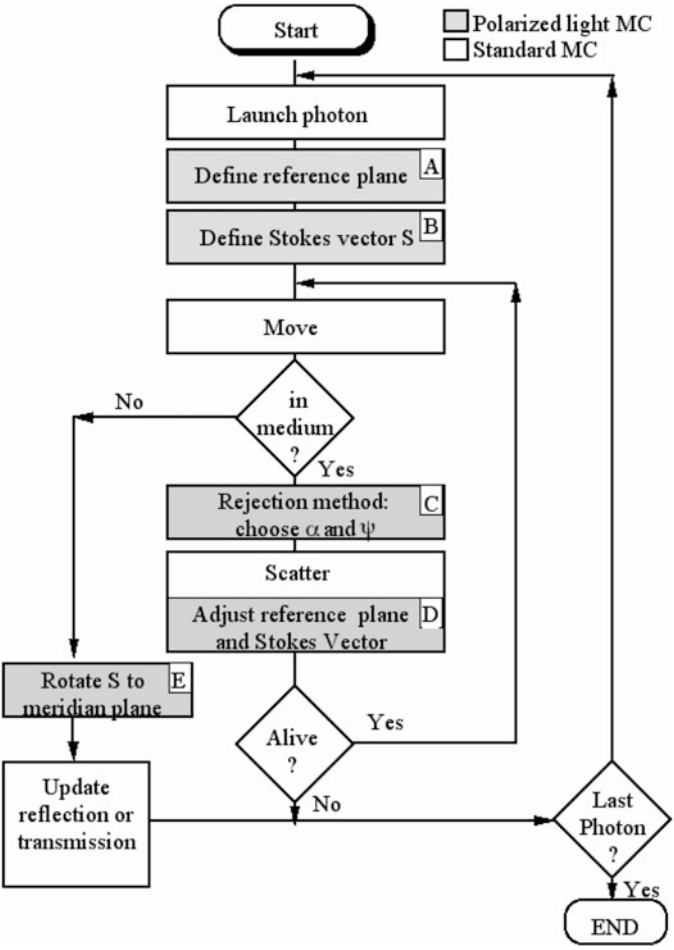


Fig. 2 Flowchart of polarized Monte Carlo program, the white cells represent the blocks that are used in standard Monte Carlo programs, the gray cells represent the blocks that are specific of polarized light Monte Carlo programs

```
s = -log(rand)/(mus+mua); % calculate step size
x = x+U(1)*s; % new photon position
y = y+U(2)*s; % new photon position
z = z+U(3)*s; % new photon position
absorb = W*(1-albedo); % update the photon weight
W = W-absorb;
if (z<=0) % check for boundary exiting the medium
     phi=atan2(U(2),U(1)); % calculate necessary angle of
     rotation to main reference frame.
     S2=rotSphi(S, phi); % use rotation matrix to bring the
     Stoke vector to the
     RH=RH+S2*W; % laboratory reference frame
     photon_status = DEAD; % photon is terminated
end
```

3. Change the direction of photon propagation and its polarization according to the type of event. This step is the core of the polarized Monte Carlo and will require first the calculation of two angles φ (phi in the code) and Θ (theta in the code). This process is conducted using the *acceptance-rejection* method.

```
condition =1; % this flag is used to step in and out of the
acceptance-rejection
        method implemented as a while loop
while condition ==1
   theta     = acos(2*rand - 1); % initial choice of theta
   phi = rand*2.0*pi; % initial choice of phi
   % s11, s12 are elements of the scattering matrix which are
   calculated at
   % initialization
   I0=s11(1)*S(1)+s12(1)*(S(2)*cos(2*phi)+S(3)*sin(2*phi));
   ithedeg = 1+floor(theta*nangles/pi); %digitization of angles
   I=s11(ithedeg)*S(1)+s12(ithedeg)*(S(2)*cos(2*phi)+S(3)
   *sin(2*phi));
   xra=rand*I0;
   % Implementation of rejection method
   if (xra>=I)
        condition= 1; % fail rejection an recalculate values of
        theta / phi
   else
        condition = 0; % pass accept values of theta and phi
   end
end
```

The second part of choosing a direction requires the rotation of the Stokes vector reference plane to a new Meridian Plane established by φ and Θ and then the multiplication of the updated Stokes vector to the local scattering matrix M.

```
U2= update (U, phi, theta); % update photon trajectory vector
*/ costheta = cos(theta);
S2 = rotSphi(S, phi); % rotate to a new Meridian plane
% Multiply the Stokes vector to the scattering matrix
S(1) = s11(ithedeg)*S2(1)+s12(ithedeg)*S2(2);
S(2) = s12(ithedeg)*S2(1)+s11(ithedeg)*S2(2);
S(3) = s33(ithedeg)*S2(3)+s43(ithedeg)*S2(4);
S(4) = -s43(ithedeg)*S2(3)+s33(ithedeg)*S2(4);
% Back rotation to new meridian plane
if ((phi>pi) && (phi<2*pi))
   cosi=(U2(3)*costheta-U(3))/temp; % first calculate the
   needed angle
else
   cosi=-(U2(3)*costheta-U(3))/temp;
                        % for speed with operate in
                        % cosines and sines rather than angles
end
sini =(sqrt(1-cosi*cosi));
cos22=2*cosi*cosi-1;
sin22=2*sini*cosi;
% finally we update the Stokes vector to the new plane
```

```
S2(1) = S(1);
S2(2) = S(2)*cos22-S(3)*sin22;
S2(3) = S(2)*sin22+S(3)*cos22;
S2(4) = S(4);
```

4. Scoring, in this step we check if the photon stays within the domain of simulations. If so, back to step 2. If not the photon is terminated. This step is the 'roulette' that gives Monte Carlo its name.

```
if (W<THRESHOLD) % THRESHOLD is set at initialization as
                 described in
                 % 6.2.3
   if (rand <= CHANCE) % CHANCE is set at initialization
   W = W/CHANCE;
else
   photon_status = DEAD;
   end
end
```

The program included in this book was tested against its C version previously published achieving errors well below statistical significance. The parallelization of this program was also performed bringing computational speed to reasonable values.

Using a 16 cores Mac pro computer and utilizing numerical values from [6] the total computational time was 40 seconds for 1e6 photons. The program can be downloaded in its entirety at https://github.com/jramella/Polarized-Monte-Carlo.

4 Applications

In the context of biophotonics, polarization-sensitive Monte Carlo models have been used primarily to gain a better understanding of how polarized light travels in biological environments. For example, Antonelli et al. used their polarized Monte Carlo program to simulate the back scattering of cellular environment of the colon impacted by colon cancer [12] (Fig. 3).

They noted that both size and index of refraction of scatterer influenced the back reflected Mueller Matrix and confirmed the relations between diagonal Mueller matrix elements $|M_{11}| = |M_{22}| > |M_{33}|$ also for biological environments. Ultimately through simulations and experimental work, they determined that colon tissue behaves as a depolarizer across the visible spectrum and furthermore that the mucosal layer in the colon cannot be described uniquely as a monodisperse layer. This investigators suggested that key model parameters could be identified with this combined experimental and computational approach for cancer staging [20].

Backscattered Mueller matrix images of an optical phantom of colon tissue was also simulated with polarized Monte Carlo [21]. The optical phantom represents a 1 cm thick disc of radius 0.5 cm filled with a scattering medium that mimics healthy colon tissue N (μ_s = 285.6 cm^{-1}, μ_a = 8.7 cm^{-1}) with two cylindrical inclusions of radius 0.1 cm, which mimic (1) an increase in vascularization – zone

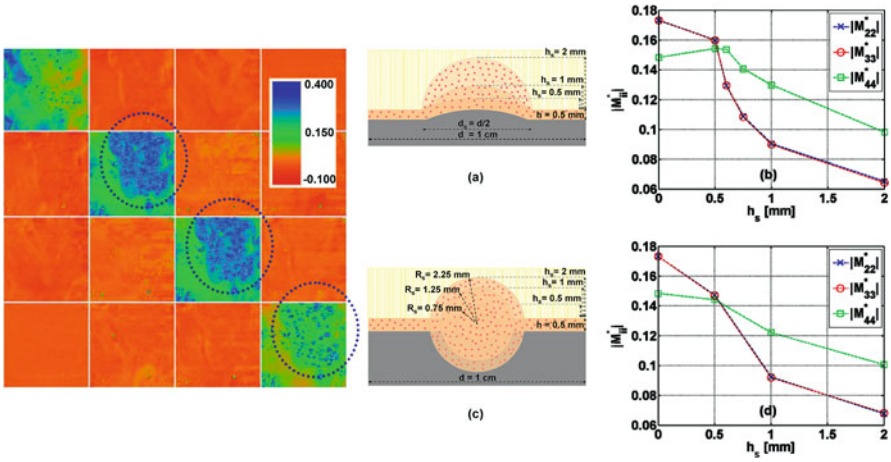


Fig. 3 Left panel: Experimental Mueller matrix images of human colon specimen for cancer diagnostics: how Monte Carlo modeling can help in the interpretation of experimental data. Right panel: (**a**, **c**) two illustrations of exophytic growth originating in the mucosa, whose thickness increases in a non-uniform fashion; (**b**, **d**) absolute values of the normalized diagonal elements |M*ii| of the backscattering Mueller matrix calculated at $\lambda = 633$ nm for a single mucosa layer ($r = 200$ nm, $\mu s = 93.1$ cm^{-1}; $m = 1.46/1.38$, $h = 0.5$ mm) on absorbing substrate (albedo $a = 0$), with (**b**) h_s varying from 0 (flat surface) to 2 mm; or (**d**) R_s from 0 to 2.25 mm. (Reprinted with permission from [20] ©The Optical Society)

A ($\mu_s = 285.6$ cm^{-1}, $\mu_a = 49.5$ cm^{-1}) and (2) an increase of cellular density – zone D ($\mu_s = 513.8$ cm^{-1}, $\mu_a = 8.7$ cm^{-1}) that are typical signatures of the early stage of epithelial colon cancer development. The simulated Mueller matrix of a phantom was diagonal as expected for the optically isotropic medium measured /simulated at normal incidence and detection angles.

The total backscattered intensity image (Fig. 4b, e) demonstrated higher intensity values for the zone D compared to zones N and A, because the photons undergo more collisions within part D of the optical phantom, and hence have higher probability to be backscattered to the detector. On the contrary, the degree of preservation of both linear (Fig. 4c, f) and circular polarization (Fig. 4d, g) is higher for the zone A, because the photons with longer trajectories undergoing larger number of scattering events that randomize polarization have less chance to survive and reach the detector in highly absorbing part of a phantom.

However, the negative sign of a normalized M_{33} element was never observed in our Mueller polarimetric experiments with a healthy colon tissue [22]. They reproduced the positive sign of a normalized M_{33} element by reducing the volume fraction of scatterers [23]. These simulations confirmed that light is backscattered by the subwavelength scatterers (Rayleigh regime) in these polarimetric experiments in reflection configuration.

Wood et al. [24] combined experimental measurements of polarized light transmission through phantoms with Monte Carlo simulations to study the effect

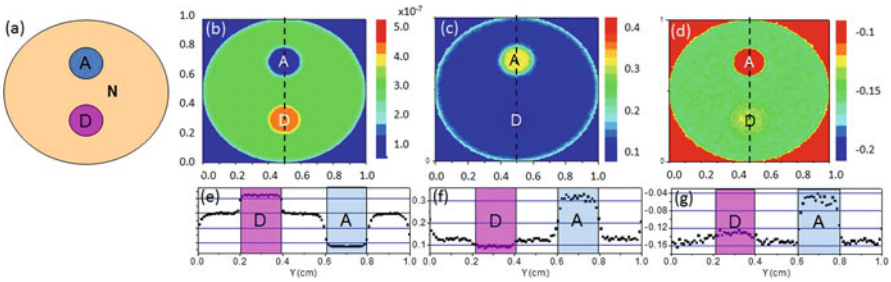


Fig. 4 Simulated diagonal Mueller matrix images of colon tissue phantom in reflection geometry, wavelength 550 nm, normal incidence and detection angles, diffuse illumination: (**a**) top view of the optical phantom (see text); (**b**) element M_{00} – total reflected intensity; (**c**) normalized element M_{11}/M_{00} ($= M_{22}/M_{00}$) – degree of preservation of the incident linear polarization; (**d**) normalized element M_{33}/M_{00} – degree of preservation of the incident circular polarization; (**e**) M_{00} values extracted from the image (**a**) along the dashed line; (**f**) M_{11}/M_{00} values extracted from the image (**b**) along the dashed line; (**g**) M_{33}/M_{00} values extracted from the image (**c**) along the dashed line

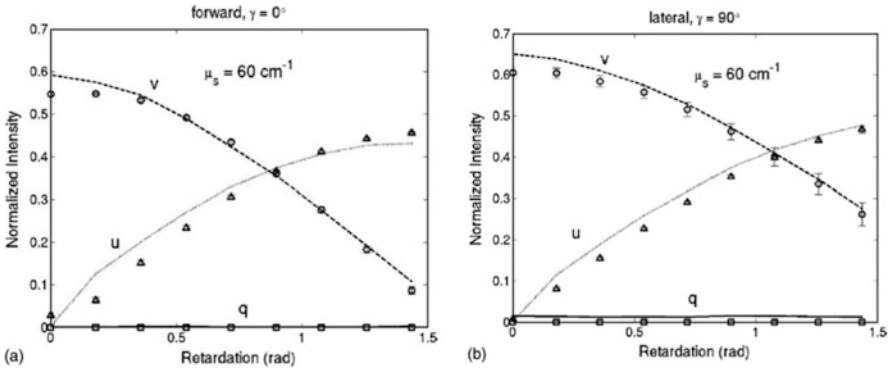


Fig. 5 Experimental measurements symbols and Monte Carlo calculations lines of the normalized Stokes parameters q squares and solid line, u triangles and dotted line, and v circles and dashed line in (**a**) forward $\gamma = 0$ degrees and (**b**) lateral $\gamma = 90$ degrees detection geometries with input circularly polarized light and a scattering coefficient of 60 cm^{-1}. Birefringence is varied from $\delta = 0$ to 1.4364 rad ($n = 0$ to 1.628×10^{-5}). Experimental values are the means of measurements on five samples

of birefringence, optical activity, and scattering on the total polarized transmission. Their well-controlled experiment utilized strain to induce birefringence in polyacrylamide optical activity was obtained with the addition of sucrose and was ultimately utilized by other investigators when studying out-of-plane polarimetry determination of directional anisotropy in turbid media [25]. (Fig. 5).

Ma et al. have used the Monte Carlo approach extensively to study Mueller Matrix coefficients and decomposition parameters as well as determining new parameters of interest with combination of Mueller Matrix elements. Their experimental work, described in Chapter "Mueller Matrix Microscopy", focuses on

polarized sensitive microscopy for multiple applications including cancer staging, skeletal muscle evaluation, algae classification, and many others [26–28].

Rehn et al. [29] studied computationally the penetration depth of polarized light and proposed that elliptical polarization allows selection of different probing depths. In further experimental work, they validated their findings on both phantoms and biological media [30]. Table 1 shows a summary of their work on depth and mean free path determination for various tissue and incident polarization showing the deep penetrating capability of elliptically polarized light in contrast with the shallower capability of linearly polarized light.

Ramella-Roman had investigated the penetration of polarized light into scattering environments (where the scatterers were spherical). In her work, the influence of the sphere size on the depolarization capability of a media was explored, an example of this is shown in Fig. 6. Co-polarized photons are collected in a three-dimensional matrix at different depth and visualized. Smaller size spheres depolarize linearly polarized light faster than larger spheres as evident in Fig. 6a. The 2 μm spheres demonstrate strongly forward directed plots while the 0.01 μm spheres have a smaller glow-ball effect for the parallel polarization that is soon lost in the background noise. The isocontour images in Fig. 6a show a similar behavior for the degree of polarization for the spheres of diameter = 0.482 μm and diameter = 2.0 μm (contour here is at a degree of polarization of 0.9).

Finally, Ghassemi et al. expanded previously developed [5, 6] Monte Carlo models [25] to include birefringence and a novel out-of-plane instrumentation geometry. In their study, the computational effort was validated with experimental findings. The modality was ultimately used to explore the arrangement of collagen in scar formation.

5 Summary and Future Directions

The application of polarized light for analysis of biological media requires new modeling approaches with increased complexity. The heterogeneity of biological media is not well represented by current models that focus primarily on monodispersed spherical scatterers except for few reports.

Here, we composed a wish list of future Monte Carlo integrations that are seen as necessary for future real-life applications.

1. Layers: an epithelium, a protective layer of cells, is omnipresent in tissues and organs both in human and animals and can be arranged as a single layer of cells or compound multilayer structure. Many applications of polarized light include noninvasive probing of the epithelium, for example in the skin, in the cervix, in the colon, and many other organs. A Monte Carlo program capable of separating the absorption and scattering properties of the epithelium from those of the underlying tissue could be extremely useful for the characterization of

Table 1 Depth-indicator ZC calculated with optical properties corresponding to different types of tissues. Depths are reported in [MFP] and corresponding values in μm. The wavelength is 633 nm

					Depth-indicator Z_C							
					Linear		Elliptical $\varepsilon = 0.1\pi$		Circular		Multiple scattering	
Tissue	Absorption coefficient $\mu_a(\text{cm}^{-1})$	Scattering coefficient $\mu_s(\text{cm}^{-1})$	Anisotropy factor $g(-)$	MFP (μm)	[MFP]	(μm)	[MFP]	(μm)	[MFP]	(μm)	[MFP]	(μm)
Example[a]	0	100	0.92	100	7.7	769.5	14.8	1474	16.7	1667	67.5	6750
Human skin	3.2	168.6	0.81	59.3	3.4	200.1	4.6	271.8	5.8	343.1	8.3	491
Pork skin	1	492.6	0.95	20.3	12.7	256.9	23.5	476.6	26.0	528.6	40	812
Aorta	0.52	316.4	0.87	31.6	4.5	142.3	7.9	249.7	9.2	291.2	22.8	719
Gray matter	2.7	357.1	0.94	28	9.1	254.9	17.1	477.9	18.8	525.3	23.1	647
White matter	2.2	534.8	0.82	18.7	3.8	70.1	5.5	102.8	6.6	123.4	13.2	246
Uterus	0.35	393.7	0.69	25.4	2.6	65.9	3.1	77.3	3.5	88.3	10.5	267

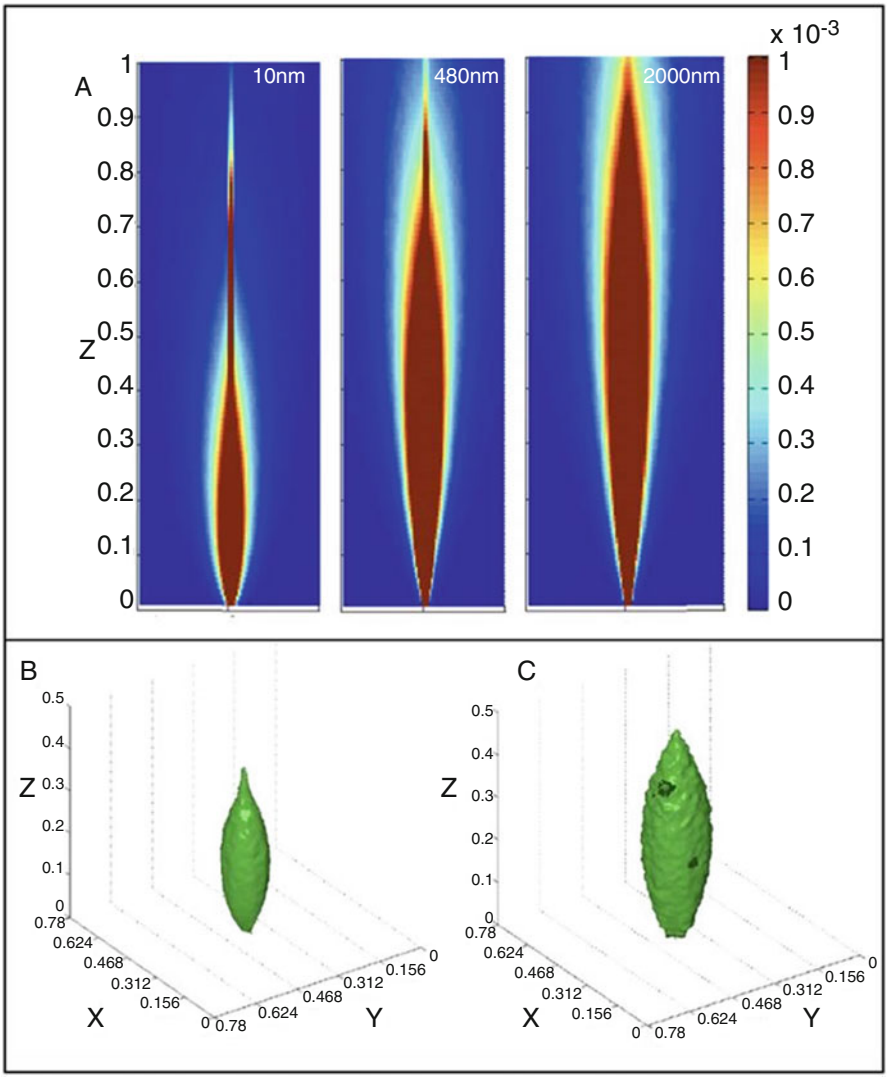


Fig. 6 Assessment of polarized light propagation in three different scattering environment (polydispersed solution of microsphere of 10 nm diameter 480 nm diameter and 2000 nm diameter). (**a**) The 2000 nm spheres demonstrate strongly forward directed plots while the 10 nm spheres have a smaller glow-ball effect for the parallel polarization that is soon lost in the background noise. (**b**) Isocontour images for the degree of polarization for the spheres of diameter 480 nm left and 2000 nm right (iscontour here is at a degree of polarization of 0.9)

the epithelium influence in the overall polarimetric response as well as for the optimization of the modalities focusing uniquely on the epithelium.

2. Nonspherical scatters: biological tissue encompasses a variety of structures from nanometer range to large microns to millimeter sized scatterers. The

approximation of spherical scatterers is acceptable within the smaller sized particles and in the Rayleigh regime, but it is not desirable when considering interaction with larger structures. An example of such structures are collagen fibers which are common in connective tissue and due to their birefringence (both intrinsic and of form) have a strong polarized signature. Recently some publications have reported the inclusion of cylindrical scatterers in Monte Carlo models [31, 32] but the programs are not available to the community at large and further work is needed to validate them.

3. Enclosures: since its inception, polarized light imaging has been used to heighten the contrast of cancerous lesions within scattering tissue. For example, Jacques et al. [6, 33] used combined co- and cross-polarization to visualize borders of skin cancer, which were not visible to the naked eye. Similarly, De Martino et al. used polarization to image colon cancer [12] while others focused on breast cancer diagnosis with polarized light [34, 35] In all these examples, one type of tissue (often cancerous) with specific scattering and polarization properties invades a different one. Some Monte Carlo programs have attempted to mimic this problem [20] but further work is needed to fully characterize this intriguing issue. Novel approaches such as the voxel-based programs could be instrumental in this context [36].
4. Roughness: the issue of tissue roughness (particularly in applications involving skin) has been largely ignored both in instrumentation and modeling of polarized light transfer. Yet it has been demonstrated that rough surface scattering can dominate the polarized light signature. Scatmech [37] —the program developed at NIST by T. Germer—offers a great way to model different type of roughness, but it was developed primarily for levels of roughness well below the ones we encounter in biomedical studies. Further work that could integrate the rough interface with the multi-scattering underlying tissue is needed.
5. Coherence: the vast majority of Monte Carlo programs developed so far utilize the Stokes–Mueller formalism, and hence cannot be used in a more extensive study of coherence within the scattering media. While some initial studies tackled this problem, they were limited to specific applications such as Optical Coherence Tomography. Although we realize the complexity of the task, future work in this context is needed. Novel mathematical approaches are appearing [38] that could be used to bridge the gap between Monte Carlo approach and electromagnetic model of field propagation.

Finally, the combination of new instrumentation, mathematical approaches, and applications make the need of strong computational approaches ever stronger. We believe that the next 10 years will see an exponential development of Monte Carlo based models and algorithms that include polarization and will tackle many of the problems discussed in this chapter.

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Polarized Light Imaging of the Sub-diffuse Regime of Tissues



Steven L. Jacques

1 Introduction

This chapter discusses polarized light imaging (PLI) using the difference of two images: a co-polarized image and a cross-polarized image. The difference image cancels the multiply scattered light from deeper tissues and isolates the light scattered by the superficial tissue layer, the so-called "sub-diffuse regime". This superficial layer includes the epithelial layer and the superficial region of the underlying connective tissue (stroma). The epithelial/stromal interface is where pathology often arises and is of medical interest.

This chapter discusses the use of polarized light to interrogate and characterize this superficial tissue layer where sub-diffuse scatter of photons occurs.

1.1 Imaging Superficial Tissue Layers

Pathology arises most often in epithelial layers like the epidermis of skin or the epithelial mucosa of the oral cavity, and at the epithelial/stromal interface (epidermal/dermal junction in skin). However, most of the light that is backscattered out of tissue as observable reflectance involves photons that have penetrated deeper than the superficial layer. The doctor's eye is blinded by the snowstorm of photons from deeper tissue, and cannot easily discern structural variation in the superficial layer. Absorbing structures (e.g., superficial blood vessels, freckles) are back-illuminated by the deeply backscattered light and hence cast shadows that

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are observable. Variations in microscopic structure create perturbations in light scattering, but these do not cast clear shadows. Therefore, imaging methods that specifically rely on superficially backscattered photons are needed as diagnostic screening tools to detect structural changes that indicate early onset of pathology in epithelia and at the epithelial/stromal interface.

Optical coherence tomography (OCT) and Confocal Scanning Laser Microscopy (CSLM) are image modalities that select superficially scattered photons, involving scanning a focused beam over a field of view. OCT and CSLM have the advantage of mapping the 3D structure of the tissue. Non-linear imaging methods like Second Harmonic Generation (SHG) can be implemented in a CSLM experimental setup and provide imaging that is also sensitive to structure.

Similar to OCT and CSLM, polarized light imaging (PLI) images superficially. PLI is a low-cost rapid-surveillance imaging modality, involving simultaneous acquisition of all pixels in a field of view. PLI integrates the superficially scattered photons over the depth of the sub-diffuse regime, and, therefore, provides a 2D image integrated over a superficial depth. Adjusting the field of view does not require any change in the PLI setup.

Figure 1 shows the PLI method considered in this chapter for use with imaging of skin. The method uses the difference between two images, a co-polarized and cross-polarized image. The camera views the tissue from above the field of view. The illumination is oblique and a smooth transparent plate (e.g., a glass flat) contacts the skin, coupled by a drop of gel. Consequently, specular reflectances off the air/glass

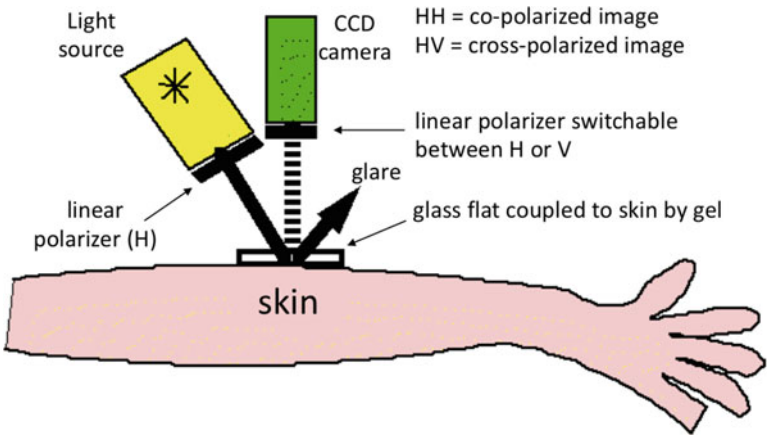


Fig. 1 Polarized light imaging (PLI) using two images: (1) a co-polarized image, and (2) a cross-polarized image. A linear polarization filter in front of the light source yields H illumination. A linear filter in front of the camera yields collection of H or V light, depending on the orientation of the filter. Hence, HH and HV images are acquired. The illumination is delivered obliquely and a glass flat is coupled by gel to the tissue, providing refractive index matching. Hence, surface glare (specular reflectancerefletance) at the air/glass/tissue surfaces is deflected away from the camera. Light scattered from within the tissue can reach the camera

and glass/gel surfaces are directed away from the camera. The specular reflectance off the gel/skin surface is mostly directed away from the camera and the few photons that scatter off the irregular skin surface and reach the camera have been minimized by the closely matched refractive index of gel and skin. The camera collects sub-surface scatter from within the skin, the photons that are scattered toward the camera. It is also possible to both deliver and collect light from an angle, such that surface glare scatters away from the camera. This approach allows for non-contact imaging.

The next subsection describes the image analysis to select the sub-surface scattered photons.

1.2 Polarization Gating to Image Sub-diffuse Regime

The PLI in this chapter illuminates with horizontal linearly polarized light (H), defined as photons whose electric field is aligned with the scattering plane specified by the light source/tissue/camera triangle. The "H" is here used to indicate horizontal linearly polarized light, and the light source is filtered by a linear polarizer to yield H illumination. Hence, the photons' electric field is parallel to the glass/gel/tissue scattering plane, which slightly lessens the specular reflectance.

The camera acquires two images, one image with a linear polarization filter in front of the camera oriented parallel to the illumination to yield a co-polarized image (HH), and a second image with the camera's filter oriented perpendicular to the illumination to yield a cross-polarized image (HV). "V" indicates a vertically oriented filter whose electric field is perpendicular to the scattering plane. The first letter indicates the orientation of polarized illumination and the second letter indicates the orientation of polarized camera acquisition.

The two measurements HH and HV can be discussed in terms of two contributing components: (1) $\frac{1}{2}$ of the fully depolarized multiply scattered light that largely comes from deeper tissue, here denoted as D, and (2) a co-polarized signal from the superficial tissue where escaping photons have not yet been full depolarized, here denoted as S:

$$\begin{aligned} co-polarized &= HH = \frac{1}{2}D + S \\ cross-polarized &= HV = \frac{1}{2}D \\ superficial &= HH - HV = S \\ total &= HH + HV = S + D \end{aligned} \tag{1}$$

2 Polarized Monte Carlo Simulations

This report has used the polarized Monte Carlo simulation program by Ramella-Roman et al. (the meridian plane simulation stok.c) [1] as a starting point for the following simulations. The program delivered an H Stokes vector of a pencil beam illuminating the tissue phantom,

$$H = \begin{vmatrix} I \\ Q \\ U \\ V \end{vmatrix} = \begin{vmatrix} 1 \\ 1 \\ 0 \\ 0 \end{vmatrix} \tag{2}$$

and the simulation returned the images of escaping I and Q flux density, denoted HI(y,x) and HQ(y,x), in units of [W/cm^2 per W delivered] or [cm^{-2}]. Also the total HI and HQ signals were recorded. The total intensity image is

$$HI = HH + HV \tag{3}$$

The difference image is

$$HQ = HH - HV \tag{4}$$

The Monte Carlo simulations used matched boundary conditions, so the effects of specular reflectance of incident light preventing light from entering a tissue and total internal reflectance preventing light from escaping the tissue were neglected.

The stok.c program uses the light scattering of Mie spheres to mimic the scattering properties of the tissue. The input parameters governing the simulation are listed in Table 1.

Table 1 Parameters for stok.c to mimic a standard tissue illuminated by red light

Program inputs for stok.c:		
Sphere diameter	Dia	0.500 μm
Wavelength	λ	0.633 μm
Sphere refractive index	n_{sph}	1.5867 (poystyrene)
Medium refractive index	n_{med}	1.3314 (water)
Sphere number density	ρ	0.0690 μm^3
Absorption coeff.	μ_a	1 cm^{-1}
To yield the optical properties:		
Absorption coeff.	μ_a	1 cm^{-1}
Scattering coeff.	μ_s	100 cm^{-1}
Anisotropy	g	0.8226
Reduced scattering coeff.	$\mu_s(1-g)$	17.74 cm^{-1}

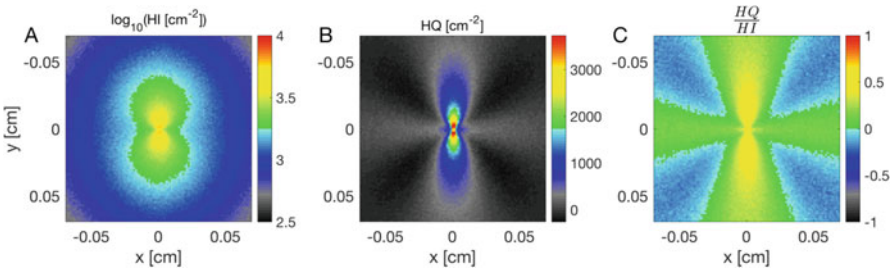


Fig. 2 (**a**) Intensity map of reflectance, HI, as $\log_{10}(\text{HI}\ [cm^{-2}])$. (**b**) Co- minus cross-polarized difference image, HQ $[cm^{-2}]$. (**c**) Ratio HQ/HI [dimensionless]. The incident photons are horizontally polarized, their electric fields aligned with the x-axis

Figure 2a shows the total intensity HI image presented as the $\log_{10}$(HI). Figure 2b shows the HQ image on a linear scale. The images are 100x100 pixels, with pixel sizes dx = dy = 0.0014 cm. Figure 2c shows the ratio $\frac{HQ}{HI}$, which cancels the units and presents a cross pattern scaled between -1 to $+1$. The electric field of the incident H photons is oriented parallel to the x axis, inducing dipoles oriented along x. The scatter pattern as a dipole collapses is equatorial around the dipole, i.e., around the dipole in the y−z plane at x = 0. Hence, the scattered reflectance in the HQ image is most strong along y at x = 0.

2.1 HQ Versus n Scatterings with Constant μ_s

This section discusses Monte Carlo simulations where μ_s was held constant while the microsphere diameter size was varied. Varying sphere diameter caused variation in the anisotropy of scatter, g. Because μ_s was constant, the mean free path between scattering events was constant. Hence, the spatial pattern and the total amount of HQ flux escaping the tissue was influenced by g. The results, therefore, inform about the relative efficiency of depolarization of the HQ signal by scatterers with different g values. All simulations used a wavelength of 633 nm (the red of a HeNe laser).

As the microsphere diameters were varied from 10–1000 nm, the scattering cross-section of the spheres changed. To keep μ_s constant, the number density of scatters, ρ $[\#/\mu m^3]$ was adjusted.

The stok.c program was modified to collect a set of HI and HQ images using photons that had only scattered once (n=1), twice (n=2), up to n = 9. These were denoted HI(y,x,n) and HQ(y,x,n), where n indicates the number of scattering events.

Figure 3 shows the $\frac{HQ}{HI}$ cross patterns for n = 1, 2, 3, and 6 scattering events before photon escape. The strong scatter along y at x=0 indicates the effect of the equatorial scatter by induced dipoles. As n increases, the pattern diminishes in intensity.

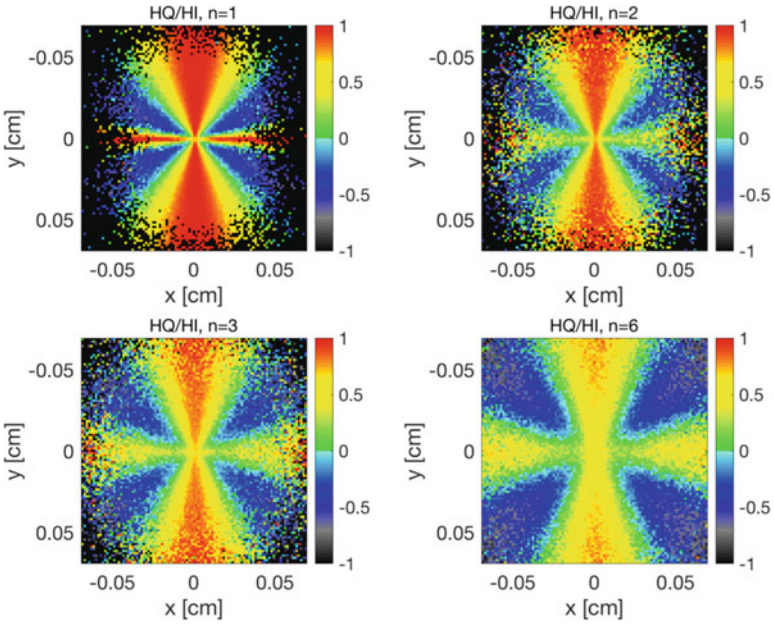


Fig. 3 Pattern of escaping flux ratio $\frac{HQ(y,x)}{HI(y,x)}$ as the number of scattering events (n) occuringoccurring before photon escape increases: n = 1, 2, 3, and 6 (See Table 1 for parameters of Mie sphere model)

Figure 4 shows the profile of HQ(y) along the y-axis at x = 0, where the HQ signal is strongest. The HQ signal from photons scattering only once (n=1) is strongest. As n increases, the HQ signal decreases, and for n>6 there remains only a broadly distributed background HQ signal.

Next, the total HQ was determined for a range of microsphere diameters, ranging from 10–1000 nm, which specified a range of g values. The dependence of HQ on the number of scattering events, i.e., $HQ_{total}(n)$, was determined. Figure 5 shows $HQ_{total}(n)$, illustrating how the total HQ signal contributed by photons experiencing n scattering events decreases approximately exponentially as n increases,

$$HQ_{total}(n) \approx Ke^{-n/n_e} \tag{5}$$

where n_e is the number of scatterings that cause the HQ signal to attenuate to 37% of an initial value. The value of n_e varied for the different microsphere diameters, i.e., for the different g. This n_e value specifies the average number of scattering events involved in the HQ_{total} signal for a particular microsphere size. The n_e increases as the anisotropy g increases. In other words, a more forward-directed scattering (high g) is less effective in attenuating the HQ signal and allows more scattering events before HQ signal escape.

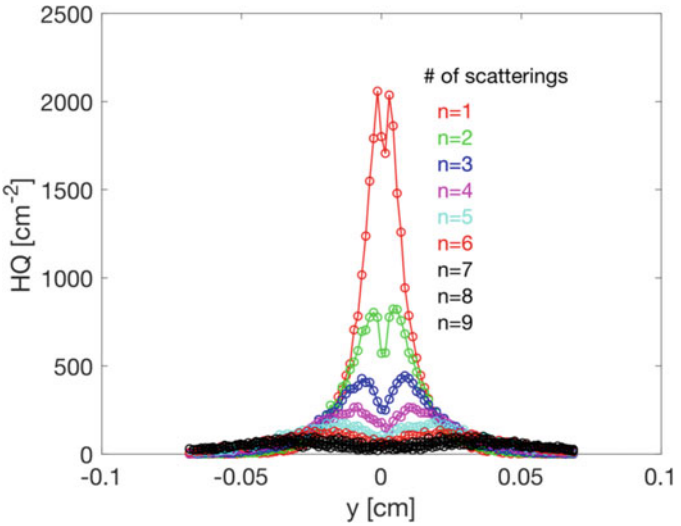


Fig. 4 The profile of escaping HQ reflectance along the y axis at x = 0, for HQ that involved 1 to 9 scattering events before escape from the tissue. The strongest contribution to HQ is for n = 1 scattering event. As n increases, the contribution of HQ(n) decreases

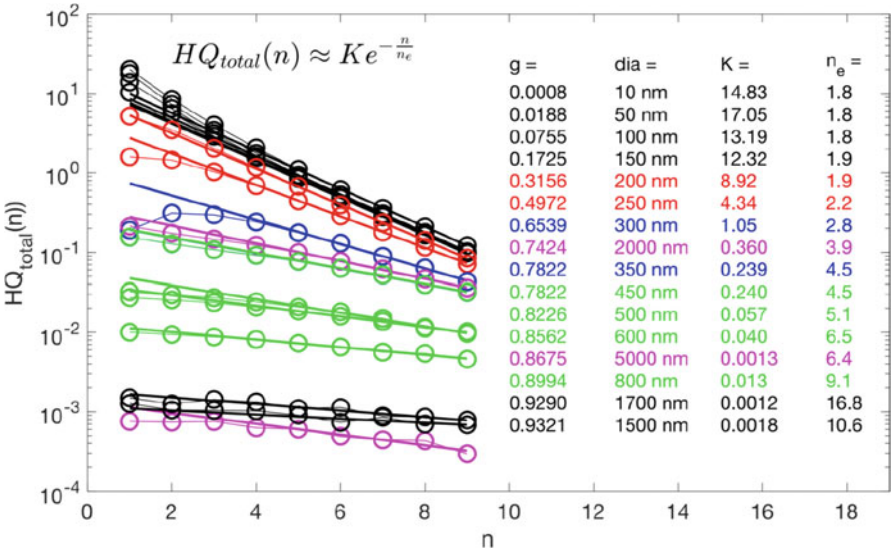


Fig. 5 The total HQ signal involving n scattering events plotted versus n, for a range of microsphere diameters and hence a range of anisotropy g values. The curves are approximately fit by $HQ_{total}(n) = Ke^{-n/n_e}$ (thick solid lines). For a particular g value, the average number of scattering events experienced by escaping HQ photons is n_e (See Table 1 for parameters)

The HQ_{total} flux escaping the tissue is plotted in Fig. 6a as a function of the anisotropy (g). The semilog plot illustrates that total HQ signal falls drastically as

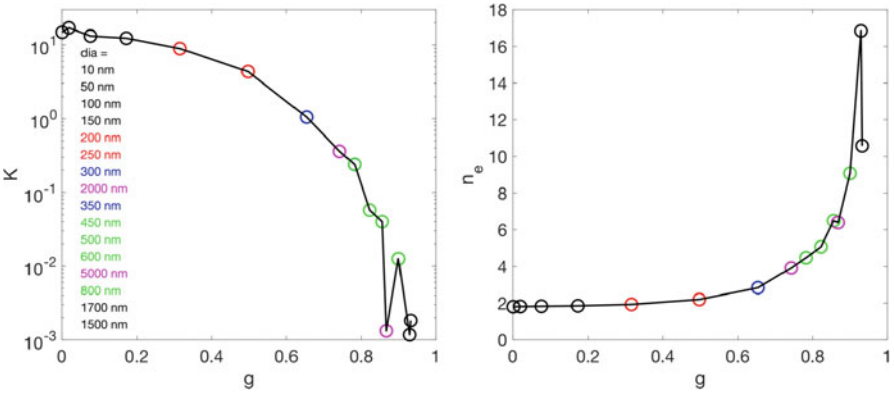


Fig. 6 (**a**) The magnitude (K) of the HQ signal falls as anisotropy of scatter g increases. (**b**) The n_e increases as g increases, indicating that the efficiency of depolarization ($\frac{1}{n_e}$) falls as g increases. The circles correspond to different microsphere diameters (dia), which yield different g values. The HQ signal begins to oscillate for the largest microspheres, as expected from Mie theory

g increases. Figure 6B plots n_e versus g, indicating that n_e increases rapidly once g exceeds ~0.7. In other words, forward-directed scatter attenuates the magnitude (K) of the HQ_{total}(n) reflectance involving n scattering events, and also decreases the efficiency ($\frac{1}{n_e}$) of depolarization.

While HQ initially falls exponentially versus the number of scattering events, there is a residual HQ component that persists and decays more slowly. Does this persistent HQ present a problem for imaging superficial tissues? To address this question, a Monte Carlo simulation of reflectance of HI and HQ from a tissue slab of increasing thickness (d = 0.0100–2.6 cm) was conducted using the parameters in Table 1. Figure 7 shows that as the slab thickness increases, the total diffuse reflectance HI increases to a maximum value, and similarly the polarized reflectance HQ also increases to a maximum value, both behaving as $a(1 - e^{-d/b})$.

The factor b is the 1/e thickness of the slab where HI or HQ have approached to within 37% of their maximum value. This b equals 2.4 mm for HI and 0.560 mm for HQ. This b value for HQ compares well with an experimental measurement of b, which used a razor blade to partially block the polarized illumination (500 nm< λ <~750 nm) to cast a shadow on the skin of a forearm. The observed HQ signal that diffused laterally into the shadow specified b to equal about 0.600 mm [2], close to the 0.560 mm value.

The factor b is also similar to the value of six mean free scattering paths, $6 \times \frac{1}{\mu_s}$ or 0.6 mm, which is the average number of scattering events times the mean free path between scattering events. The depth of the sub-diffuse regime as defined by polarized light reflectance appears to be approximately 0.5–0.6 mm. The persistent residual HQ signal does not appear to significantly increase the depth of HQ imaging.

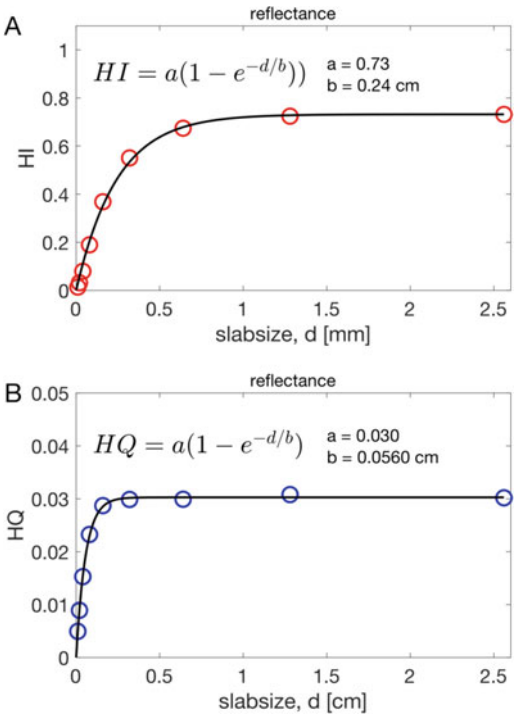


Fig. 7 Reflectance of HI and HQ from a slab of increasing thickness (d [cm]). (**a**) HI vs d. (**b**) HQ vs d. The data exponentially approach a maximum value, indicating a superficial depth (**b**) of response. The response of polarized reflectance is on average b = 0.0560 cm

2.2 HQ Versus n Scatterings with Constant $\mu_s(1-g)$

This section discusses Monte Carlo simulation results where $\mu_s' = \mu_s(1-g)$ was held constant while the microsphere diameter size was varied. In practical measurements, the total diffuse reflectance (R_d), which is the HI signal, is related to the ratio $\frac{\mu_s(1-g)}{\mu_a}$. The HQ signal is small compared to the HI value, which is largely dominated by the diffuse reflectance from deeper tissue layers. Analysis of HI may be accomplished by different methods, for example, by recording the point spread function of escaping reflectance, or by recording the reflectance at a series of wavelengths and applying spectral analysis. This discussion assumes that a measurement and an analysis of HI have been conducted to yield the average values of the absorption coefficient (μ_a [cm^{-1}]) and the reduced scattering coefficient (μ_s' [cm^{-1}]) for the tissue. This assumption is needed because the values of μ_a and μ_s' affects the Monte Carlo simulation of HI and HQ.

In the simulations, keeping $\mu_s(1-g)$ constant required adjustment of the number density ρ [#/$\mu\mathrm{m}^3$] to adjust μ_s. As dia increased, g increased, so $1-g$ decreased, which required an increase in μ_s to keep $\mu_s(1-g)$ constant. The value of μ_a was adjusted to cause HI to attenuate. The value of HI ranged from 0 to 1, i.e., HI equaled the fraction of delivered light reflected by the tissue.

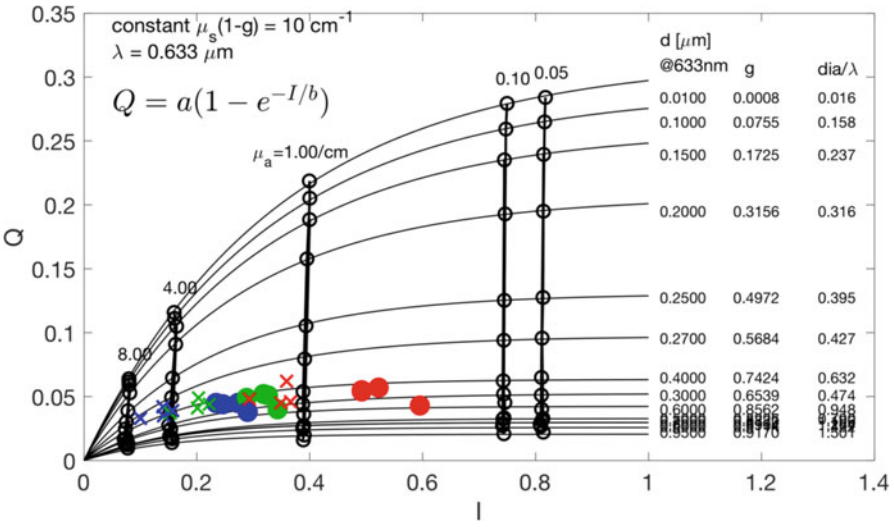


Fig. 8 Plotting HQ versus HI as an analysis grid, where the reduced scattering coefficient $\mu_s(1-g)$ is held constant, the absorption coefficient μ_a is varied, and the diameter dia of Mie spheres is varied. A measurement of HQ and HI will interpolate within this analysis grid to specify an apparent ratio $\frac{dia}{\lambda}$ and an anisotropy g. This "equivalent" Mie sphere with diameter dia and anisotropy g mimics the scattering properties of the sub-diffuse regime of the tissue. The diameter values dia assume λ is 0.633 μm. The ratio $\frac{dia}{\lambda}$ allows different wavelengths to specify an apparent dia. The colored symbols show previously reported experimental measurements on skin [3] (red 470 nm, green 524 nm, and blue 625 LEDs; o's = lightly pigmented forearm, x's = darkly pigmented forearm), indicating g = 0.6–0.7 range

Consider an experiment where the diffuse light reflectance specifies the μ_a and μ_s'. Then a grid of HQ vs HI values can be calculated using the polarized light Monte Carlo simulation. This plot is here called an analysis grid.

Figure 8 shows an analysis grid that plots HQ versus HI for a constant reduced scattering coefficient, μ_s' = 10 cm^{-1}, and polystyrene microsphere diameters (10–1000 nm) in water at a wavelength λ = 0.633 nm. The absorption coefficient, $\mu_a\ cm^{-1}$, was varied from 0.05 to 8, which caused HQ and HI to vary.

Then a measured HQ,HI pair interpolates via the analysis grid to yield an equivalent microsphere diameter dia at 0.633 μm wavelength, and an apparent anisotropy value g = <cos(θ)>, where θ is the photon deflection angle of a scattering event. The HQ reflectance, therefore, specifies g as a metric that characterizes the scatterers in the sub-diffuse regime. The g is related to an apparent ratio of microsphere diameter to wavelength ($\frac{dia}{\lambda}$), such that the microsphere's scattering would mimic the observed tissue scattering. Variation in the HQ,HI values would indicate changes in the structure of the superficial tissue layers.

A better approach would use a distribution of microsphere diameters to mimic the tissue. There is some fluctuation in the HQ response for spheres whose ratio $\frac{dia}{\lambda}$ is larger than 0.45, as expected for larger Mie spheres. A mixture of sizes

would mitigate this fluctuation, and improve the characterization of tissues with high anisotropy, which is common for tissues.

The grid of Fig. 8 shows that the greatest sensitivity of HQ to $\frac{dia}{\lambda}$, and hence to g, occurs between 100–300 nm diameters, a range that corresponds to structures from cytoskeleton to chromosomes. HQ is increasingly less sensitive to diameters less than 100 nm or greater than 500 nm. Therefore, HQ imaging may be limited in the size of scattering structures to which it responds. Also, the number density ρ of different sizes of scattering structures will vary. A large concentration of structures can compensate for a low HQ sensitivity of the structures, and yield a detectable signal. Alternatively, a low concentration will decrease the otherwise strong HQ response of a scatterer and make it difficult to detect that scatterer.

It should be emphasized that the analysis grid of Fig. 8 assumed a homogeneous tissue. In reality, tissues are often multilayered. A more accurate model of a true tissue architecture would yield a more accurate specification of any changes in the apparent size and anisotropy of scatterers in a tissue.

Another consideration is the solid angle of collection by the lens of a polarization camera. The HQ and HI of Fig. 8 considered all the escaping light at all angles of escape. A camera will collect a more narrow range of angles, which may alter the analysis grid.

The above discussion might imply to the reader that the interpretation of HQ images is fraught with complexity and difficult to interpret. However, the practical task of surveillance of a tissue to detect early onset of pathology need not rely on absolutely accurate values of dia and g. The doctor is interested in detecting changes in the structure of the epithelial layer of a tissue (or epidermis of skin). The HQ image rejects the large amount of background multi-scattered diffuse light from deeper tissue and presents a low-level but measurable sub-diffuse signal from the superficial layers of the tissue. Perturbations in an otherwise uniform HQ image may reveal early onset of pathology. With clinical experience, the sensitivity and importance of observed changes may become understood.

Figure 8 also includes experimental data previously reported [3], which used red, green, and blue LEDs as light sources. The HQ data are in the 0.046 ± 0.007 range. The HQ,HI data indicate g values in the 0.6–0.7 range. The polarized reflectance from the skin is likely dominated by the scattering of collagen fibrils and fiber bundles in the superficial papillary dermis. A process that involves remodeling of the dermis (eg.e.g., aging, cosmetic skin rejuvenation, scar formation, wound healing) may show variations in the HQ, HI response.

3 Mechanisms of Depolarization

A polarization camera acquires co-polarized (HH) and cross-polarized (HV) images and then the intensity image (HI) and the difference image (HQ) are calculated by Eq. 1. Different tissues, however, will depolarize the incident H light more or less

efficiently depending on their structure. Therefore, the rate of depolarization is a characteristic of a tissue.

Since the observed HQ signal will attenuate as the photons are scattered and as the photons cross different micro-domains of birefringence, the process of depolarization is a gradual process. Therefore, it may be useful to consider depolarization as a diffusion process.

3.1 Depolarization by Scattering

Consider the depolarization of linearly polarized light as it passes through a scattering tissue. Figure 9 illustrates the diffusion of polarization orientation in θ space. The electric field (E field) of the incident H photons is initially oriented horizontally. As a photon propagates through the tissue, it is scattered and/or retarded by birefringence, which causes its E field to migrate in the orientation angle space θ.

The electric fields of photons incident on the tissue are initially oriented horizontally (H, θ=0°). As the photons scatter, the orientations of their electric fields migrate in angle space (θ) (Fig. 9):

$$p(\theta) = \frac{e^{-\frac{\theta^2}{2\sigma^2}}}{2\sigma\sqrt{\pi/2}} = \frac{e^{-\frac{\theta^2}{4\pi\chi n}}}{\sqrt{4\pi\chi n}} \tag{6}$$

where θ is the orientation angle of the electric field of photons (θ = 0 for H light), σ is the standard deviation [radians], χ is the diffusivity [radians^2/scatter], and n is the number of scattering events experienced by the population of photons. The first expression for p(θ) [radians^{-1}] is a normal distribution. The second expression is the distribution during one-dimensional diffusion. The expressions indicate the standard deviation of θ increases with the number of scatterings (n) experienced:

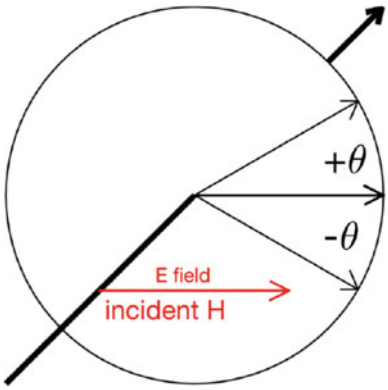


Fig. 9 The diffusion of polarization orientation in θ space. The electric fields (E field) of the incident H photons migrate in the orientation angle space θ as they propagate through a tissue

$$\sigma = \sqrt{2\chi n} \tag{7}$$

The projection of p(θ) on the horizontal axis yields the portion of a photon's electric field (E) that is still horizontally linearly polarized, which equals the expectation value for cos(θ). The square of this electric field is the HH intensity transmitted after n scattering events.

$$\begin{aligned} E &= \int_0^{\pi} p(\theta) cos(\theta) d\theta \\ HH &= E^2 \approx \frac{1}{2}(1 + e^{-\chi n}) \end{aligned} \tag{8}$$

The first H in HH(n) indicates the incident polarized intensity is horizontal. The second H indicates the transmitted polarized intensity is also horizontal. HV indicates the transmitted vertically polarized intensity: HV= 1−HH, where absorption is neglected and the total transmitted intensity is HI=HV+HH=1.

Figure 10a shows an example of p(θ) as n increases from 1 to 30 scatterings. The angular distribution of the electric fields broadens diffusively. In this example, χ equals 0.2 radians^2/scatter. Figure 10b shows the evolution of HH and HV intensities as n increases. Both intensities converge on the same value of $\frac{1}{2}$. The value of HQ approaches zero as HH and HV both approach $\frac{1}{2}$. In other words, HQ=HH−HV=HH−(1−HH)=2HH−1 =2($\frac{1}{2}(1 + e^{-\chi n})$)−1, or simply:

$$HQ = e^{-\chi n} \tag{9}$$

Therefore, the angular (θ) diffusion model for the depolarization of HQ reflectance leads to a simple exponential decay of HQ versus the number of scattering events, characterized by a factor χ. The units of χ were [radians^2/scatter] in the diffusion model, but in Eq. 9 the units of χ are regarded as dimensionless.

Equation 9 describes the diffusion of light from an impulse of H light. However, a real experiment involves the attenuation of a beam of photons (ballistic photons) by scattering events. Only when a photon is first scattered does it enter into the diffusion process. So a more exact solution would convolve the diffusive impulse response against the temporal and depth dependence of photons scattering for the first time.

3.2 Depolarization by Birefringence

The previous section discussed depolarization by scattering, with an incremental loss of HQ=HH−HV after each scattering event. However, if the tissue has birefringent fibers, such as collagen and muscle, then the transit of a photon between scattering events will also depolarize the light. In skin, for example, the collagen

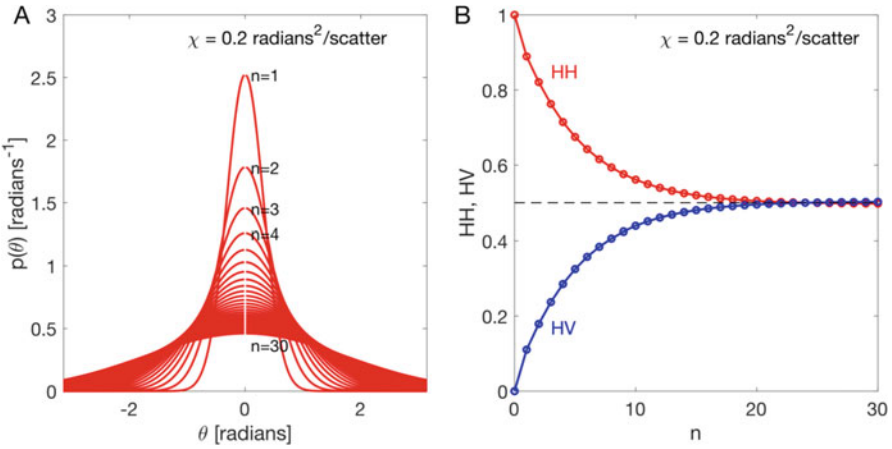


Fig. 10 Depolarization of HH for the case of $\chi = 0.2$ radians^2/scatter. (**a**) Spreading of the angular orientation of the electric field of a population of photons initially linearly polarized (H, θ=0 radians). After one scatter (n=1), the probability density function p(θ) for the E field's orientation is distributed. As n increases, the angular distribution broadens. (**b**) The loss of HH and the growth of HV as the number of scatterings (n) increases, with HH and HV approaching $\frac{1}{2}$. The HH(n) was calculated using Eq. 8 and HV(n) = 1−HH(n)

fibers are organized in small microdomains on the scale of 10–100 μm, each microdomain having its own orientation of fibers. The H and V components of the photons passing through a microdomain of oriented fibers will be retarded differentially, leading to the conversion of linearly polarized light into elliptically polarized light, which is equivalent to more equal amounts of H and V light. Hence, the difference HQ = HH-HV will incrementally decrease as light passes from one scattering site to the next through birefringent fibers.

Consequently, the HQ will fall exponentially, as in Eq. 8. A simulation study [4] reported the exponential decay of HQ as incident H light passed through sequential randomly oriented microdomains of thickness ΔL with birefringence β. The results were summarized:

$$HQ \approx e^{-(\frac{\beta \Delta L}{2})^2 n} \tag{10}$$

which suggests that $\chi \approx (\frac{\beta \Delta L}{2})^2$ due to birefringence.

In summary, the HQ signal decays exponentially as $e^{-\chi n}$ due to (1) each scattering event, and (2) each passage through a microdomain of birefringence.

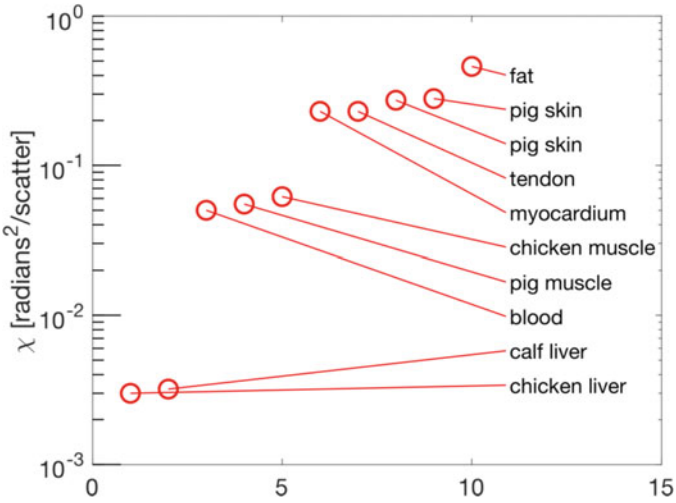


Fig. 11 The diffusivity in angle space θ, χ [radians^2/scattering], for the photon electric field in various tissues (data from [4])

4 Rates of Depolarization for Various Tissues

Figure 11 summarizes some values of χ for various tissues as reviewed by Jacques 2011 [4]. Basically, skin depolarizes much faster than muscle, and much much faster than liver, largely due to differences in birefringence, which is one of the two mechanisms of depolarization (birefringence and scattering). The depolarization in blood and fat is due to strong scattering. The rate of depolarization of light by tissue can vary over a couple orders of magnitude, and, therefore, χ may prove to be a useful metric for characterizing a tissue.

There is a balance between the efficient backscatter of photons to generate measurable reflectance and the depolarization of photons. Photons must scatter to escape as reflectance, but scattering depolarizes the photons. If the tissue also has birefringent fibers, low scattering allows longer paths before photon escape, and longer paths allow more retardation leading to depolarization. This balance is important for polarized light imaging (PLI) based on the HQ signal.

5 Conclusion

Polarized light imaging (PLI) based on the HQ signal offers a low-cost rapid method of surveillance of a tissue's superficial layers (epithelia and the epithelial/stromal interface) where much pathology arises. While there are many factors involved in interpreting the HQ signal in an absolutely accurate manner, a PLI image can detect

perturbations in a broad field of view. Hence, PLI may provide a useful method for early detection of pathology and identifying the margins of pathology to guide surgical excision.

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Mueller Polarimetric Imaging for Cervical Intraepithelial Neoplasia Detection



Angelo Pierangelo, Tatiana Novikova, Jean Rehbinder, André Nazac, and Jérémy Vizet

1 Introduction

Improving the health of populations has always been a primary concern of advanced human societies. Since the beginning of the twentieth century, considerable efforts have been invested in disease prevention and quality care access. However, reliable and early detection of pathologies, allowing faster, less invasive, and tailored treatment, is still a major societal challenge today.

Cancer continues to be a leading health problem for the human population. Approximately 19.1 million new cancer cases were discovered worldwide in 2018 [1]. Early detection of cancer and complete surgical resection of malignant lesions are the two main steps needed to increase patients' chance of recovery significantly. However, the cancer diagnosis is based on histological analysis of biopsies taken from the suspicious tissues zones, which are very difficult to identify at an early stage of disease development. In addition, the surgeon has only visual cues and, sometimes, his tactile sense to identify the pathological areas to be removed, which makes the accurate assessment of surgical margins very difficult or

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impossible in some cases. Undetectable pathological areas may thus escape surgery and give rise to the disease recurrence.

Over the last century, the considerable development of biomedical imaging modalities has led to significant improvements in diagnosing and treating many types of pathologies, including cancer. Several imaging techniques are currently used in medical practice. They can be grouped into two categories. The first one includes all imaging techniques that allow in-depth exploration of the human body such as computed tomography (CT) scan, positron emission tomography (PET) scan, magnetic resonance imaging (MRI), and others. However, they are characterized by a millimeter spatial resolution that severely limits their ability to detect cancerous lesions early. The second group includes all kinds of optical imaging techniques (endoscopy for the exploration of the internal cavities of the human body, the colposcope for the analysis of the cervix, and others), which provide images of the tissues surface with a large field of view and microscopic resolution, making them potentially well adapted for the early detection of pathological areas. However, these systems are often based on visible white light illumination and cannot detect disease early when tissue has not been significantly altered.

Many new optical imaging techniques, including fluorescence, second harmonic generation, and optical coherence tomography, have been tested over the last decades in a quest to develop new systems capable of acquiring images with micrometric resolution and increased contrast [2–4].

In recent years, optical polarimetry has received considerable attention in medical diagnosis as it provides additional information compared to the techniques based on light intensity measurements only. Polarimetric imaging holds great promise for biomedical applications for several reasons: (i) it exploits the polarization of light, which is sensitive to morphological changes in tissue structure at a microscopic scale; (ii) it can easily be implemented with a wide field of view for the analysis of large areas; (iii) it is a noninvasive technique, allowing noncontact imaging of tissue without the use of staining agents; (iv) it can be implemented employing “cold” white light sources (LEDs, halogen lamps, and others) that are harmless to patients. In particular, among the different possibilities to implement polarization-based techniques [5], the complete imaging Mueller polarimetry is the only one that allows the comprehensive polarimetric characterization of a sample. Furthermore, recent studies have shown the great potential of this technique in detecting pathological zones on a wide variety of tissues, such as skin [6, 7], stomach [8], colon [9–13], rectum [14], and cervix [15–21].

All these studies have demonstrated that Mueller polarimetric imaging modality holds great promise for the following applications: (i) the early detection of cancer and the improvement of biopsy performance [15, 17]; (ii) the staging of cancer before surgery [9, 11, 13]; (iii) the detection of residual tumors after radio- and chemotherapy treatments [14].

This chapter will focus on using wide-field multispectral imaging Mueller polarimetry to improve the early detection and surgery for precancerous cervical lesions.

2 Cervical Cancer

Each year, about 500,000 new cases of cervical cancer are diagnosed worldwide, and about 260,000 women die from it. Screening with a Papanicolaou (Pap) smear followed, if necessary, by colposcopy has reduced the incidence rate of cervical cancer in developed countries. However, because of its high incidence in low- and middle-income countries that lack the resources and infrastructure for regular, high-quality screening [22], it remains the second most common female cancer and the leading cause of death among women worldwide.

The cervix is the lower part of the uterus that protrudes into the vagina. It has a cylindrical shape and is traversed along its entire length by the endocervical canal (also called endocervix) that connects the vaginal lumen to the uterine cavity. The entrance to the endocervical canal on the vaginal side is also named the "external os." The cervix surface that directly faces the vagina is called the ectocervix, and it is lined with a stratified squamous epithelium (also called the Malpighian epithelium) about 300 microns thick. Otherwise, the endocervical canal is lined with a single layer of columnar mucous cells (also called the glandular epithelium) about 5 microns thick. The interface between the Malpighian and glandular epithelium is identified as the junction zone. The connective tissue or stroma lies beneath these two types of epithelium. It is mainly composed of collagen and accounts for 95% of the cervical volume. A thin fibrous layer, called the basement membrane, separates both epithelia from the underlying connective tissue.

The leading cause of cervical cancer (95–98% of cases) is a chronic infection caused by different types of human papillomavirus (HPV), particularly HPV-16 and HPV-18 [23], which invade the cervix through the junction zone. This infection is widespread and can lead to dysplasia or cervical intraepithelial neoplasia (CIN), i.e., the proliferation of abnormal cells into the Malpighian epithelium. The lesion is classified histologically as CIN1 (mild dysplasia) or CIN2 (moderate dysplasia) if this transformation affects one-third or two-thirds, respectively, of the Malpighian epithelial thickness. Otherwise, it is classified as CIN3 or precancerous lesion (severe dysplasia) if the entire thickness of the Malpighian epithelium is affected. Finally, the disease can progress from CIN3 to invasive cancer if the malignant cells spread into the connective tissue beneath the Malpighian epithelium (see Fig. 1).

The recently introduced HPV vaccines [23] will not change this situation in the short term. Moreover, they are not curative and can be used for prevention purposes with good but not complete efficiency. Indeed, none of these vaccines create full immunity against all types of HPV [25]. Additionally, there is a strong reluctance to use them even in some developed countries (e.g., France), where the level of vaccination in adolescent girls remains very low [26]. Finally, the high cost of these vaccines and, in some cases, political or logistical obstacles prevent their widespread use in low-resource countries [27]. For all these reasons, cervical cancer will remain a major health problem for many years to come. Therefore, improving cervical cancer detection and management is essential in both developed and low-resource countries.

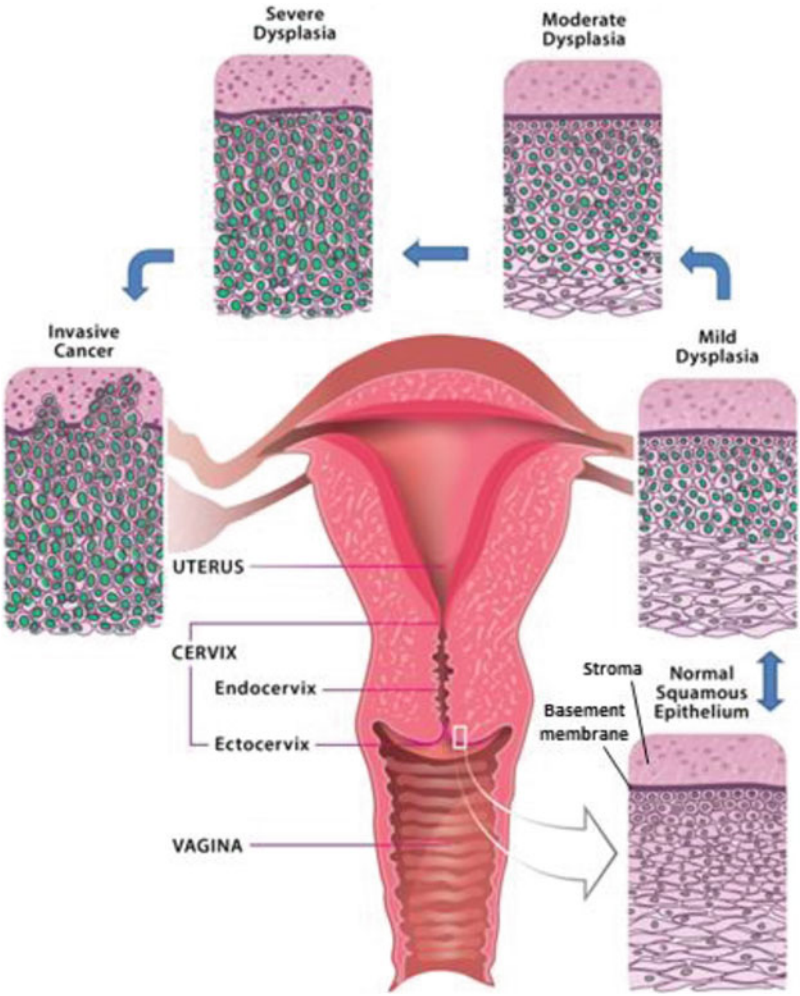


Fig. 1 Schematic illustration of the development of cervical cancer. (Reproduced with permission from [24])

In developed countries, the first step in cervical cancer screening is the Pap smear, which consists of collecting epithelial cells from the surface of the cervix and examining them under an optical microscope. Women with an abnormal smear are referred for a colposcopy test, which involves examining the cervix with a low magnification microscope (called a colposcope) after applying 3% acetic acid and iodine as contrast agents. Suspicious areas, usually bleached by the application of acetic acid and iodine negatives, are biopsied to confirm the presence of a malignant transformation by histology analysis. Finally, precancerous lesions are surgically removed by conization, i.e., the excision of a conical piece of tissue (~2 cm

in diameter) containing abnormal zones from the cervix, usually guided by the colposcope again after applying acetic acid and iodine. The conization is sufficient to prevent any recurrence of the disease. Unfortunately, the colposcopy test is highly operator-dependent, with a sensitivity of about 60–70% and a specificity of 50% to detect CIN 2-3 and moderate inter-operator reproducibility [28–40]. As a result, it is difficult for the practitioners to correctly locate the sites to be biopsied and determine the limits of the surgical resection. Currently, about 20,000 conizations are performed in France each year. Histological analysis of these specimens reveals that about half of them are not pathological or have CIN1 lesions, and their resection is not justified. The definition of the precise limits for surgery is also crucial because the size of the conization should ensure the healthy margins for the excised tissue whereas remaining small enough to minimize the risk of complications in future pregnancies [41, 42]. Current screening programs require up to three visits to a healthcare professional and several weeks in total for the diagnosis and treatment, if necessary [43]. These programs are expensive and require establishing a network of highly trained medical doctors and tissue analysis laboratories, which is prohibitive in low-resource countries. Therefore, new effective and affordable strategies are needed to perform the following:

1. Improve the colposcopy test accuracy in order to significantly reduce the cost and time of diagnosis and patient management in the developed countries
2. Design and fabricate new low-cost instruments that could be a part of an economically viable "screen and treat" approach in the low-resource countries

In the following text, we present the milestones that have been achieved in demonstrating the relevance of the imaging Mueller polarimetry for cervical cancer diagnosis.

3 Mueller Polarimetry

Mueller polarimetry is an optical characterization technique that allows measuring the Mueller matrix M of a sample, i.e., a 4×4 real-valued matrix containing information on its polarimetric properties [44]. The measurement of M can be performed using a Mueller polarimeter that consists of the following components: (i) a light source, (ii) a polarization state generator (PSG), (iii) a polarization state analyzer (PSA), and (iv) a detector. The PSG modulates the polarization of the incident light beam by producing four polarization states to probe the sample. These polarization states are analyzed at the output of the sample through four different polarization configurations of the PSA. This procedure provides 16 intensity measurements required to determine the Mueller matrix M of the sample [18], whose main polarimetric properties are then obtained using the Lu–Chipman polar decomposition [45] that allows decomposing M as:

$$M = M_{\triangle} M_{\mathrm{R}} M_{\mathrm{D}} \tag{1}$$

where M_Δ, M_R, and M_D are the Mueller matrices of a depolarizer, a birefringent medium, and a diattenuator, respectively.

The most diagnostically relevant polarimetric parameters characterizing a biological tissue are the total depolarization Δ and the linear birefringence, characterized by the linear phase retardance R and the azimuth of the optical axis α.

The parameter Δ quantifies the ability of a sample to depolarize both linearly and circularly polarized light. It is expressed as:

$$\Delta = 1 - \frac{|a| + |b| + |c|}{3} \tag{2}$$

where a, b, and c are the eigenvalues of M_Δ. It ranges between 0 and 1 for a non-depolarizing and a pure depolarizing system, respectively.

The parameter R quantifies the difference in optical phase shift between two polarization eigenaxes, which are indicated as fast and slow axes. It is expressed as:

$$R = \arccos\left(\frac{\mathrm{tr}\,(M_R)}{2} - 1\right) \tag{3}$$

where $\mathrm{tr}(M_R)$ is the trace of M_R. This parameter is measured in degrees and varies between 0° and 180°. Generally, Eq. (3) is used to calculate the total phase retardance, considering both the linear and circular contributions. In our specific case, Eq. (3) is well suited to calculate the linear phase retardance, as the circular phase retardance is negligible for cervical tissue.

Finally, the parameter α provides the direction (or azimuth) of the fast optical axis projected on the image plane. It is expressed as:

$$\alpha = \frac{1}{2}\arctan\left(\frac{M_{R24}}{M_{R43}}\right) \tag{4}$$

and varies between −45° and +45°. This range is extended between 0° and 180° to reduce the recurring jumps in the color map of the α images, thus simplifying their visual perception and interpretation [18].

4 Mueller Polarimetric Imaging of Cervix: The Proof of Principle

Mueller polarimetric imaging modality for the detection of precancerous cervical lesions was first tested ex vivo on fresh cervical specimens (conizations) immediately after surgical resection (close to in vivo condition) using a wide-field multispectral imaging Mueller polarimetric system based on nematic liquid crystals

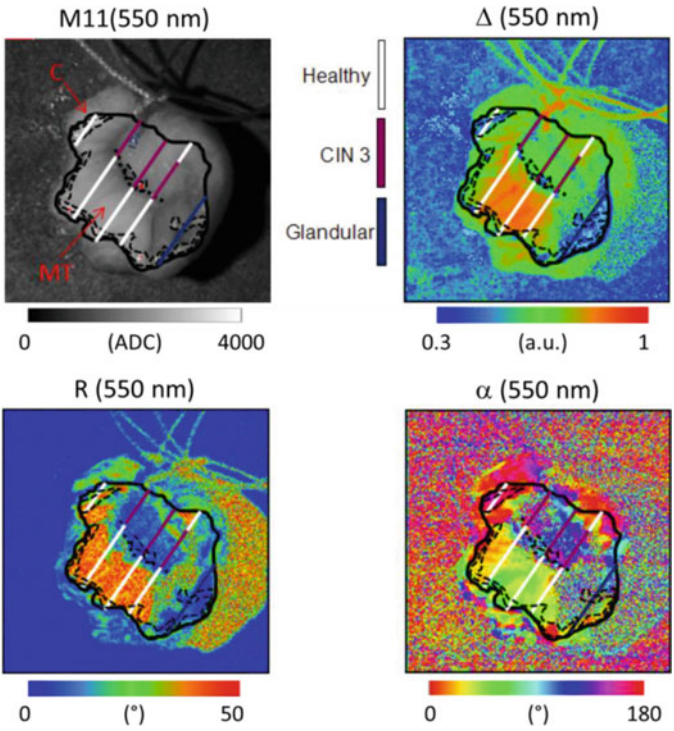


Fig. 2 Polarimetric images at 550 nm for a cervical conization specimen with histologically confirmed glandular epithelium (blue lines), healthy Malpighian epithelium (white lines), and CIN3 (purple lines). (**a**) Conventional unpolarized intensity image (non-normalized M_{11} coefficient of measured Mueller matrix), (**b**) total depolarization image Δ, (**c**) linear phase retardance image R, and (**d**) azimuth image α. Δ, R, and α images were obtained from measured Mueller matrix, using the Lu–Chipman decomposition for each pixel [45]. (Adapted with permission from [15]. ©The Optical Society)

[14, 15]. Preliminary tests on a few conization specimens were performed at the Institut Mutualiste Montsouris in Paris (France).

Figure 2 shows the polarimetric images obtained at 550 nm for a cervical conization specimen with the histologically confirmed glandular epithelium (blue lines), healthy Malpighian epithelium (white lines), and CIN3 (purple lines). In particular, the glandular epithelium, which covers the endocervical canal, should not be visible on the surface of the cervix. However, due to hormonal changes, the junction zone may shift during a woman's life (e.g., in the ovulatory phase for young women) so that the glandular epithelium becomes visible on the surface of the cervix. This completely benign modification in the cervical structure is called ectropion. In addition, the glandular epithelium can also cover almost all the surface of benign polyps that originate in the endocervical canal and protrude from the surface of the cervix.

Healthy Malighien epithelium is characterized by a value for R between 30° and 70°. In contrast, the value of R is less than 10° for the glandular epithelium and the precancerous zone. Furthermore, the azimuth α is spatially uniform for the healthy squamous epithelium, where it varies on a scale of a few millimeters. In contrast, the azimuth α appears less organized in the precancerous zone, where it varies on a scale of a few hundred microns. Finally, it varies randomly for the glandular epithelium. The total depolarization Δ is characterized by an average value of 0.9 for the healthy squamous epithelium, whereas its value drops to 0.3 for the glandular epithelium and 0.7 for the precancerous zone.

The presence of a precancerous lesion can be accurately and unambiguously detected by combining the information obtained from the maps of R, α, and Δ. Furthermore, this example clearly shows that almost two-thirds of the conization was resected unnecessarily.

5 Interpretation of the Experimental Results

To better understand the polarimetric contrasts described above, thin histological sections from a conization specimen were analyzed using a Mueller polarimetric microscope in transmission, focusing on the healthy zones only [16]. In particular, 12 regions of interest (ROIs) were selected from the microscopic images of two different histological slides. For each selected ROI, the image of α obtained by the postprocessing (see Eq. (4)) of Mueller polarimetric microscopy data was compared with the image of the collagen fiber orientation obtained with second harmonic generation (SHG) microscopy, as shown in Fig. 3a, b. In particular, we calculated the mean value and the standard deviation of α for all pixels contained within a selected ROI. Then we compared these values with those obtained for the collagen fiber orientation by SHG microscopy for the same ROI. Figure 3c shows that these two parameters are linearly correlated, demonstrating that polarimetric parameter α enables visualization and quantification the orientation of collagen fibers that compose the cervical connective tissue.

These results suggest that the strong anisotropy observed in the healthy areas of cervix is not due to the Malpighian epithelium but rather to the presence of a well-ordered and structured collagen layer in the underlying connective tissue (near the tissue surface).

In addition, we have developed and validated a polarized Monte Carlo algorithm to model the interaction of polarized light with isotropic scattering media to explain the strong depolarization characterizing the healthy zones of most tissues (cervix, rectum, colon, and others) [12, 13] measured with a wide-field imaging Mueller polarimeter in a reflection configuration. In particular, we found that thick biological tissues show a lower depolarization for the linearly polarized incident light (compared to the circularly polarized light). This effect is generally observed in the Rayleigh regime when the diameter of the scattering particles is smaller than the wavelength of probing light, which is not the case for the scattering of visible light

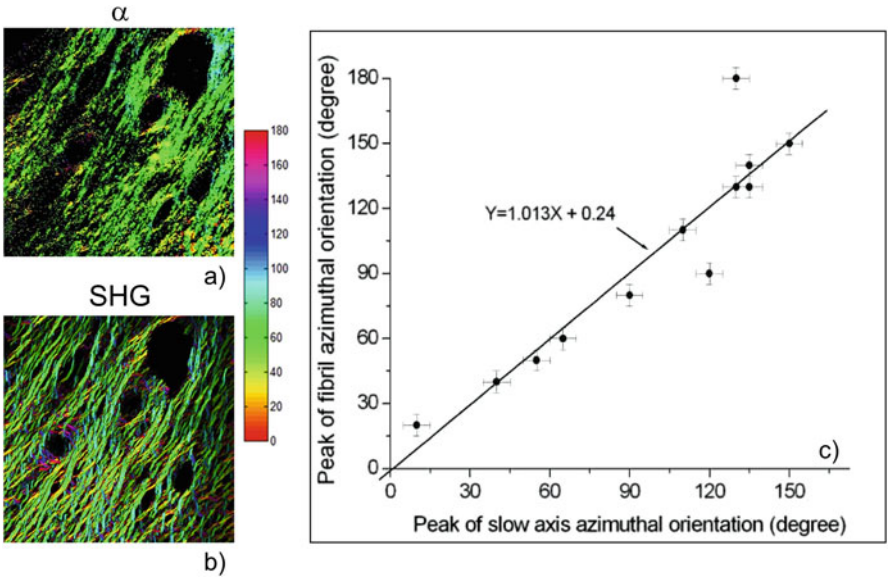


Fig. 3 Analysis of connective tissue from thin sections of a conization specimen using Mueller polarimetric microscopy and SHG microscopy. (**a**) Image of α obtained by the postprocessing (see Eq. (4)) of Mueller polarimetric microscopy data for a selected ROI; (**b**) image of collagen fiber orientation by SHG microscopy for the same ROI; (**c**) comparison between α and collagen fiber orientation for 12 ROIs selected from two different histological slides. (Adapted with permission from [16]. ©The Optical Society)

(380–750 nm) by cell nuclei (size of a few microns), as was initially thought. Thus, small particles (cell organelles, disorganized collagen fibers, etc.) present in tissue volume seem to be at the origin of the depolarization observed in backscattering configuration. In the case of cervix, the depolarization could be a signature of small scatterers contained within both Malpighian epithelium and underlying connective tissue.

Pathologists consider that cervical precancerous lesions affect the Malpighian epithelium only by increasing the cellular density and modifying the morphology of the cells. However, the polarimetric response of the cervix suggests that malignant transformation occurring within the Malpighian epithelium is a complex process that also involves the adjacent connective tissue. Thus, the detection and quantification of the connective tissue changes may be a new and promising tool to improve cervical pre-cancer diagnosis compared to conventional colposcopy.

In addition, we have modified our polarized Monte Carlo algorithm to address the need for a more realistic anisotropic scattering model that simultaneously accounts for the effects of both anisotropic and isotropic tissue scattering [46]. Finally, we are validating this model to explain the polarimetric contrasts observed in cervix and other tissues with depolarization and linear birefringence properties.

6 The First Statistical Evaluation

Within the project "PAIR Gynéco" framework funded by the National Cancer Institute of France, an extensive series of measurements were carried out ex vivo on formalin-fixed conization specimens in the Kremlin Bicêtre University Hospital (France). The main objective of these studies was to test the Mueller polarimetric imaging technique on a large number of specimens and evaluate its performance, in terms of sensitivity and specificity, for detecting precancerous lesions [17].

The measurements described in this section were performed using a multispectral wide-field imaging Mueller polarimetric system based on ferroelectric liquid crystals and operating in a reflection configuration. These crystals allowed us to build a faster and more temperature-stable polarimetric image acquisition system than the prototype based on nematic liquid crystals we used for the preliminary studies described in Sect. 4. This new system, illustrated in Fig. 3, uses a 150 Watt halogen source (Olympus CLH-CS), generating incoherent white light. A conventional silica fiber bundle is used to transport light from the source to the optical bench. A monochromatic CCD camera (f080b Allied Vision, 512 × 386 pixels in binning mode), coupled with optical zoom, is used as a detector. The zoom enables modifying the observed area's size from a minimum of 3 × 2 mm^2 to a maximum of 5 × 4 cm^2. In addition, an automated dichroic filter wheel is used to perform sequential spectral measurements between 450 and 700 nm in 50 nm steps and with a spectral bandwidth of 40 nm for each wavelength. This system is calibrated with the Eigenvalue Calibration Method (ECM) [47]. A sandblasted metallic plate (SMP), characterized by a neutral polarimetric response, is placed between the PSG and the PSA during system calibration. An automated wheel containing some optical components needed for the calibration (precisely, a first polarizer, a second polarizer, and a retarder oriented at 0°, 90°, and 30° to the laboratory frame, respectively) is placed in the position indicated as C1 in Fig. 4. The filter wheel and the calibration wheel are controlled using Field Programmable Gate Arrays (FPGA). For the experiments discussed in this section, the described imaging Mueller polarimetric system was used with a field of view of 4 × 3 cm^2 corresponding to a spatial resolution of about 100 μm/pixel (estimation of actual image resolution obtained with the optics used).

This system can measure depolarization Δ with an accuracy better than 0.5%, as well as the linear phase retardance R and the azimuth α with an accuracy of about 0.5°, under optimized conditions of signal-to-noise ratio.

The measurements described below were carried out at 450 nm, 550 nm, and 600 nm.

In order to evaluate the performance of the system to detect precancerous lesions, the polarimetric images of the fixed conization specimens were compared with the results of the gold standard histological analysis.

In general, according to the conventional protocol, after the formalin fixation of cervical conization, a pathologist cuts it into slices about 3 millimeters thick. For each slice, a histological section about 10 μm thick is taken using a microtome.

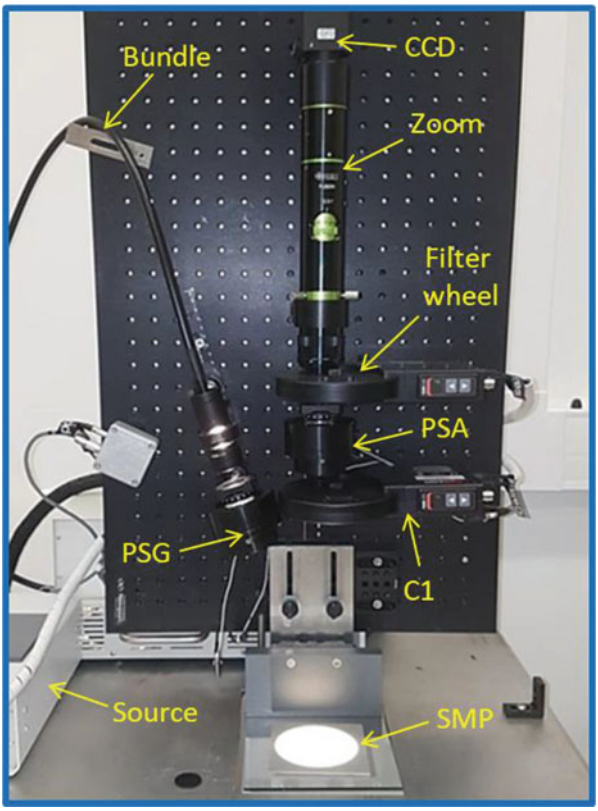


Fig. 4 Photograph of the custom-built multispectral wide-field imaging Mueller polarimetric system based on FLCs and operating in a reflection configuration

Histological sections are made in planes orthogonal to the surface of the ectocervix that is directly imaged. A pathologist then examines each histological slide with a commercial optical microscope. The final histology diagnosis is based on the analysis of the Malpighian epithelium.

In some cases, an accurate direct comparison between polarimetric contrasts and histological findings is difficult or even impossible. Therefore, an original method based on the analysis provided by pathologists has been developed to reconstruct detailed histological mapping directly projected onto the polarimetric images. This method consists of the following steps:

- Before taking the polarimetric images of a conization specimen, shallow incisions in the Malpighian epithelium of a specimen are made at a distance of about 3 mm from each other in order to approximately mark the lines of the future cuts performed by a pathologist according to the standard protocol.
- Polarimetric images of the incised Malpighian epithelium are acquired for each precut sample.

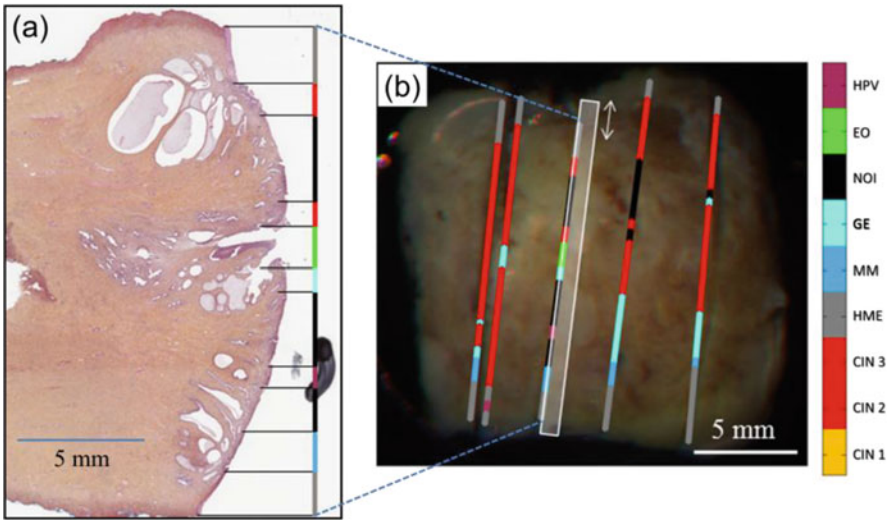


Fig. 5 Illustration of the method used to compare polarimetric images with the histological analysis. (**a**) Example of a histological section of cervical tissue annotated by a pathologist. (**b**) Final histological map (color-coded lines) superimposed on the polarimetric image of a conization specimen. The projection line, with the segments rendered in different colors according to the pathologist's analysis, is shown to the right of the histological section in (**a**) (for the color-coded legend of the histological diagnosis, see Table 1. The white box in (**b**) shows the uncertainty in the horizontal placement of the projection line of the third incision from the left boundary of the cervix. The double white arrow illustrates the uncertainty in the vertical placement. (Adapted with permission from [17]. ©SPIE)

- A thin histological section is obtained from each slice, as described above, and scanned using a commercial Leica Aperio CS scanner. A pathologist then labels the cervical epithelium on each scanned section with all identified histological types. Once a section of cervical tissue has been interpreted and labeled, the boundary points of each annotated epithelial segment are projected onto the image plane along a line parallel and close to the corresponding incision, as shown in Fig. 5a.
- All the lines obtained from the tissue section analysis are then repositioned on the polarimetric images in the proximity of the corresponding incisions, as shown in Fig. 5b.

All considered types of histological diagnosis of the Malpighian epithelium are summarized in Table 1.

There is some uncertainty in repositioning histologically annotated lines on polarimetric images, as shown in Fig. 5b. A slight shift in the positioning of a line in either vertical or horizontal direction can result in a variation of more than 10% in the calculated values of the sensitivity and specificity of the polarimetric diagnosis. For this reason, we selected sub-areas positioned between two adjacent lines marked by the same histological diagnosis to avoid these systematic effects, as shown in

Table 1 Histological types of epithelium considered in the study

Legend	Histological type
HME	Healthy Malpighian epithelium
MM	Malpighian metaplasia
GE	Glandular epithelium
HPV	Infection (Papilloma virus)
CIN 1	Cervical intraepithelial neoplasia – Grade 1
CIN 2	Cervical intraepithelial neoplasia – Grade 2
CIN 3	Cervical intraepithelial neoplasia – Grade 3
EO	Endocervical orifice
NOI	Non identified

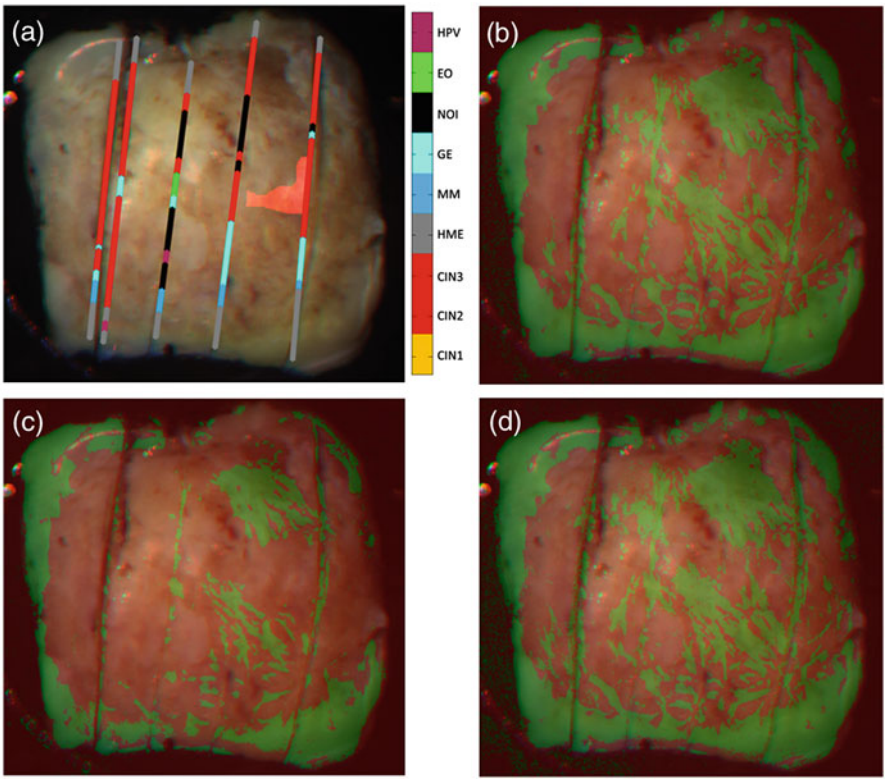


Fig. 6 (**a**) Illustration of the CIN2-3 zone selection between two adjacent lines labeled with the same histological diagnosis in the image of a conization specimen (see text). Binary image segmentation using (**b**) a threshold $R_t = 8.9°$ for the measurement performed at 550 nm, (**c**) a threshold $R_t = 10.1°$ for the measurement performed at 450 nm, (**d**) a threshold $R_t = 9.2°$ for the measurement performed at 600 nm. The pixels rendered in green correspond to $R > R_t$ and those rendered in red correspond to $R < R_t$. (Adapted with permission from [17]. ©SPIE)

Fig. 6a. Thus, we reasonably assumed that the tissue zone between two adjacent lines of the same color can be labeled with the same histological diagnosis.

This study is focused primarily on evaluating the performance of imaging Mueller polarimetry for distinguishing a precancerous lesion (CIN3) from a healthy Malpighian epithelium (HME). Therefore, the analysis does not include the different types of benign modifications of the cervix (MM and GL), HPV-infected epithelium, and low-grade dysplasia (CIN1) due to the impossibility of defining histologically confirmed image zones for these tissue types.

We choose the parameter R as the decision variable for the polarimetric diagnosis. First, we set a threshold value R_t for this parameter. The polarimetric diagnosis of each pixel in a histologically labeled precancerous zone is considered as a true positive (TP) if the value of R is lower than R_t. Otherwise, the polarimetric diagnosis is considered to be a false negative (FN). Similarly, the polarimetric diagnosis of each pixel in a histologically labeled healthy zone is considered as a true negative (TN) if the value of parameter R is higher than R_t. Otherwise, the polarimetric diagnosis is considered as a false positive (FP). The sensitivity S_e and specificity S_p of the polarimetric diagnosis are then calculated pixel-wise as:

$$S_\mathrm{e} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{5}$$

and

$$S_\mathrm{p} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}}. \tag{6}$$

A total of 50 conization specimens were analyzed. Accurate reconstruction of the histological mapping was only possible for 33 conization specimens, including a total of 83 histologically labeled image zones (on average 2–3 zones per specimen). The analysis of polarimetric diagnosis is performed pixel-wise. The detection performance of the imaging Mueller polarimetric modality is evaluated using a sliding threshold value R_t.

Figure 6b–d illustrates the binary segmentation of the selected conization specimen image (Fig. 6a) based on the polarimetric diagnosis by setting a threshold value R_t at 8.9° for 550 nm, 10.1° for 450 nm, and 9.2° for 600 nm, respectively. Pixels with linear phase retardance values above and below the threshold are rendered in green and red, respectively.

The sensitivity and specificity curves are calculated as a function of the threshold value for different measurement wavelengths (Fig. 7a, c, e). The optimal value of R_t is found at each wavelength to detect the healthy zones and precancerous lesions with maximal sensitivity and specificity. The crossover point (P_c) of the sensitivity and specificity curves provides an optimal performance threshold. For example, at 450 nm, for $R_\mathrm{t} = 10.1°$, both sensitivity and specificity of about 78% are obtained.

All pixels with a significant contribution of the specular reflection from the surface that masks the bulk polarimetric response of the cervix were removed from the analysis to improve the statistical accuracy of the technique. The specular reflection preserves the polarization of light backscattered by tissue. Thus, its contribution

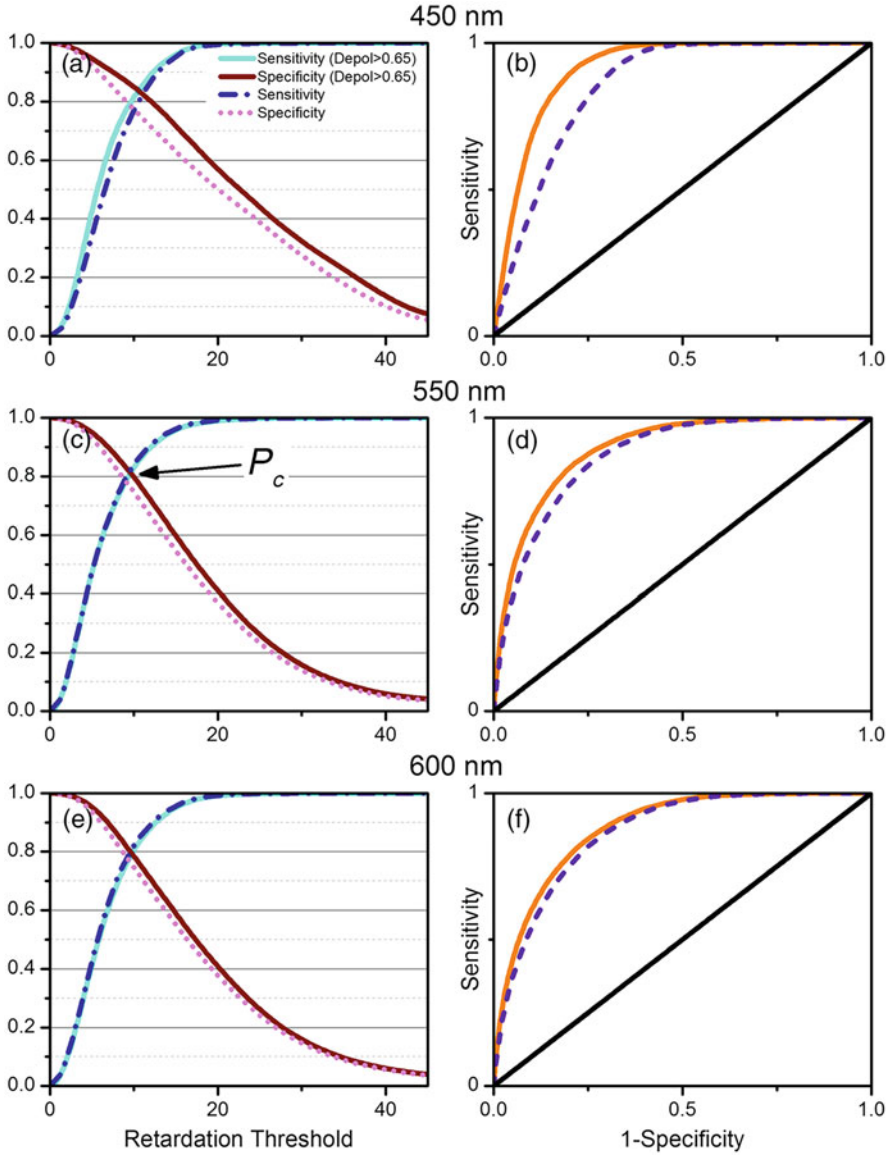


Fig. 7 Classification performance obtained by varying the threshold: (**a**), (**c**), and (**e**) sensitivity (dashed blue) and specificity (dashed magenta) curves. The solid cyan and brown lines correspond to the statistics obtained after removing the pixels affected by specular reflection. P_c is the intersection point of the S_e and S_p curves; (**b**), (**d**), and (**f**) are the resulting ROC curves (purple dashed lines for the initial statistics and orange lines for the specular reflection corrected statistics) compared to the $S_e = 1 - S_p$ line (solid black line), corresponding to a random test. (Adapted with permission from [17]. ©SPIE)

Table 2 Statistical performance of the polarimetric classification based on *R* value

	450 nm			550 nm			600 nm		
	R_{th}	Se	Sp	R_{th}	Se	Sp	R_{th}	Se	Sp
All pixels	10.1°	77.6%	77.1%	8.9 deg	78.7%	78.8%	9.2 deg	78.1%	77.8%
($\Delta > 0.65$)	10.6°	83.8%	83.6%	9.7 deg	81.7%	80.9%	9.8 deg	79.7%	79.2%

Rth threshold, *Se* sensitivity, *Sp* specificity

to the detected signal decreases the depolarization power Δ significantly for the corresponding pixels. Therefore, all pixels with a value of Δ less than 0.65 were removed from the statistical analysis.

The detection performance of imaging Mueller polarimetric modality can also be estimated by using the Receiver Operating Characteristics (ROC) curve, as shown in Fig. 7b, d, f. The ROC curve displays the values of S_e as a function of the values of $1 - S_p$. The straight line $S_e = 1 - S_p$ represents the ROC curve of a perfectly random diagnostic test.

The area under the curve (AUC) can be used to compare the different classifiers over the range of sensitivity and specificity values obtained. In contrast, the Youden index gives the best possible trade-off between a given classifier's sensitivity and specificity values. It is calculated as S_e+S_p-1 and represents the distance between the test ROC curve and the line corresponding to a perfectly random test.

Table 2 summarizes the optimal thresholds for the linear phase retardance and the corresponding sensitivity and specificity values obtained (with and without the threshold of 0.65 applied to the depolarization power) for all wavelengths used. The elimination of the pixels affected by the specular reflection significantly increases the detection performance of the technique. The best detection performance is obtained at 450 nm, with sensitivity and specificity values of about 83%.

Table 3 shows the optimal threshold values R_t obtained for the different measurement wavelengths when considering the AUC values and the maximum Youden index as the detection performance indicators for the classification test. The evolution of the AUC parameter with the wavelength gives an overall view of the test performance in relation to the evolution of the point values of S_p and S_e at the P_c crossover point. The best performance is always obtained at 450 nm, but the influence of the specular reflection is also more critical for this wavelength. In clinical practice, the classifier needs to be implemented with a preselected threshold value. The Youden index enables the evaluation of this optimal value. The thresholds for the linear phase retardance determined using the Youden index are slightly higher than the corresponding values for P_c, as shown in Table 3. By varying the threshold between the two values obtained, the increase of sensitivity is more significant than the loss of specificity. For example, at 450 nm, the values of $S_e = 92.0\%$ and $S_p = 77.6\%$ are obtained considering $R_t = 13.2°$ for the set of pixels with $\Delta > 0.65$.

The ROC curve can be obtained for any chosen classification parameter. The use of the normalized values obtained from the ROC curve is beneficial for comparing

Table 3 Area under the curve and Youden indices for the ROC curves presented in Fig. 7

	450 nm		550 nm		600 nm	
	All pixels	($\Delta > 0.65$)	All pixels	($\Delta > 0.65$)	All pixels	($\Delta > 0.65$)
AUC	0.85	0.91	0.87	0.89	0.86	0.88
Max. Youden index (associated R_{th} value)	0.59 (13.2 deg)	0.70 (13.0 deg)	0.59 (10.3 deg)	0.63 (10.4 deg)	0.57 (10.6 deg)	0.59 (10.6 deg)

the relatively simple thresholding approach based on the linear phase retardance (as well as on the depolarization power to eliminate specular reflections), with more complex approaches based on machine-learning algorithms that allow the efficient combination of several polarimetric parameters at different wavelengths.

The best sensitivity and specificity of imaging Mueller polarimetry for distinguishing CIN 3 lesions from HME are obtained at shorter wavelengths that enable a more detailed exploration of the sample surface where precancerous lesions are located.

The results of these studies are promising. Indeed, the measured linear phase retardance R makes it possible to distinguish a precancerous lesion from a healthy Malpighian epithelium of the cervix. Furthermore, these results can be considered representative of those expected in vivo because the formalin fixation does not significantly alter the anisotropy of a tissue's refractive index. On the other hand, the fixation in formalin modifies the scattering properties of tissue and significantly increases its depolarization power Δ [48]. Finally, it is worth mentioning that the azimuth α of the linear phase retardance depends strongly on mechanical stress conditions that should be very different for ex vivo and in vivo analysis of the cervix.

An in vivo study under real-life conditions of blood perfusion and mechanical stress is needed to definitively validate the ability of imaging Mueller polarimetry to detect cervical precancerous lesions accurately. In addition, all possible histological types of the cervical tissue, including benign tissue transformations (MM, GE), HPV-infected epithelium, and low-grade dysplasia (CIN1), have also to be considered in the statistical analysis for a conclusive study.

7 A Generalized Framework for Statistical Analysis

In our first statistical evaluation of the potential of imaging Mueller polarimetry to improve the early detection of cervical cancer, presented in the previous section, we used the Lu–Chipman decomposition to extract the main cervical polarimetric properties, such as the depolarization power Δ, the linear phase retardance R, and the azimuth of the optical axis α. However, the decomposition of a Mueller matrix is not unique. Many decompositions (inverse, symmetric, and others) [49–54] are available to extract these fundamental polarimetric properties from measured Mueller matrices. In addition, other algebraic approaches can be used to obtain different diagnostically relevant polarimetric parameters (purity index, entropy, and others) [55–57]. Finally, helpful information can also be obtained directly from the 16 images composing a measured Mueller matrix. A large number of polarimetric parameters are thus available for analysis. This number drastically increases if a multispectral approach is chosen. Moreover, considering that all these polarimetric parameters are obtained for each pixel of an image and that a sufficient number of different cervical images have to be acquired for reliable statistical analysis,

the amount of data produced is massive and requires the adapted management and analysis approaches.

Different histological tissue types can be found on the surface of the cervix, as described in the previous section. Therefore, unambiguous detection of precancerous lesions requires precisely identifying each histological type, which can be obtained using all the available polarimetric information.

For this purpose, we have developed a statistical data postprocessing framework for an extended multiparametric and multiclass analysis of biological tissues based on machine-learning approaches [58].

As a starting point for applying the proposed framework, an annotated database of polarimetric images (similar to the one described above) and a set of classifiers (e.g., random forest, support vector machine, naive Bayes classifiers, and others) are required. The final objective is then to predict the pixel diagnostic labels based on the polarimetric data. The complete data processing pipeline consists of three main steps described below.

- *Data preprocessing*. In this first step, the physical admissibility of the measured Mueller matrices at each wavelength is checked. A data reduction is also performed by grouping neighboring pixels, reducing data redundancy, and decreasing the impact of measurement noise during the training phase.
- *Definition of a risk metric*. The training and evaluation of learning algorithms require the definition of a unique merit function to be optimized. In our framework, the performance of a classifier is evaluated based on a user-defined loss function calculated for each pixel. The loss function represents the cost of misclassifying a pixel based on its real histological label and its predicted label. The empirical risk for a sample is obtained as a weighted sum of the losses for all pixels. The real risk used as a merit function is an average of the empirical risks over all samples.
- *Classification based on a nested cross-validation*. It consists of conducting several rounds of testing and dividing the available data into training, validation, and test groups, different for each round. This strategy is recommended in cases where the available data are limited. It ensures that the data used to evaluate a classifier are different from the training data, thus avoiding overfitting. The resulting performance evaluation for a classifier is a set of optimized risk values. This set's statistical analysis (mean, standard deviation, and others) gives a good estimate of the classifier performance on new data.

The proposed machine-learning-based approach allows automatic selection of the most diagnostically relevant polarimetric parameters across all wavelengths, extraction of the best classifier among a set of the predefined classifiers, and fine-tuning the parameters for the retained classifier. This statistical data processing framework supports binary and multiclass discrimination and will be used to analyze the data acquired in vivo in the context of a clinical trial.

8 Mueller Polarimetric Colposcope for In Vivo Analysis of Uterine Cervix

A custom-built Mueller Polarimetric Colposcope (MPC) was tested in vivo on 20 patients at the Kremlin Bicêtre University Hospital (France). The MPC was constructed by modifying a commercial colposcope (Olympus OCS-500) used for the uterine cervix analysis in routine medical practice for cervical cancer screening [18].

The standard colposcope is a binocular stereoscopic microscope that allows illuminating and analyzing the cervix in vivo with a low magnification (between 4X and 6X). The cervix can be observed directly through the binocular system or visualized on a computer screen using a CCD camera. A bandpass dichroic filter with a spectral bandwidth of 40 nm is used to select the wavelength at 550 nm. The Allied Prosilica GT1920 camera provides images with a size of 800 × 600 pixels ("binning" mode) at a framerate of 25 images per second. The spatial resolution of the acquired images is about 80 μm/pixel for a field of view of 4 × 3 cm^2 (estimation of the recorded image resolution obtained with the colposcope optics).

A miniaturized imaging Mueller polarimetric system, framed in a 30 × 50 × 100 mm^3 sealed metal box, is placed in front of the colposcope, as shown in Fig. 8a. The nonpolarized light emitted by a halogen lamp is transported to the colposcope head by a liquid light guide (Thorlabs LLG0528-6). The MPC is equipped with a collimator that images the exit tip of the liquid light guide at a distance of about 30 cm, corresponding to the position of the cervix to analyze. At this distance, the

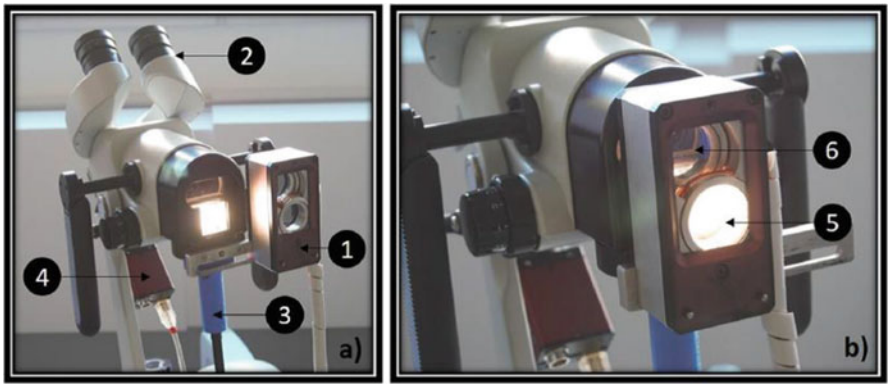


Fig. 8 Photo of the MPC system (1: polarimetric head, 2: a binocular system for stereoscopic viewing of the cervix, 3: liquid light guide, 4: CCD camera). (**a**) The polarimetric head is slid into the "OFF" position, and the MPC can be used as a conventional colposcope. (**b**) The polarimetric head is slid into the "ON" position to allow the acquisition of Mueller polarimetric images in 1.6 seconds (5: Polarization State Generator (PSG), 6: Polarization State Analyzer (PSA)). (Adapted with permission from [18]. ©Springer Nature (Creative Commons—Attribution 4.0 International—CC BY 4.0))

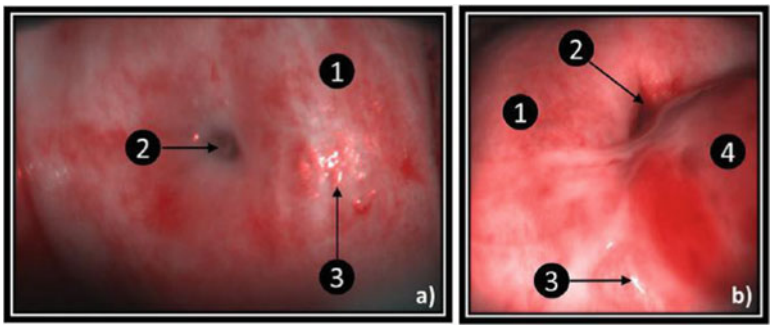


Fig. 9 In vivo unpolarized RGB images of two uterine cervices: (**a**) healthy uterine cervix (Cervix 1); (**b**) healthy uterine cervix with a polyp (Cervix 2). (1) Ectocervix, (2) cervical os, (3) bright spots of the specular reflection, (4) polyp. (Adapted with permission from [18]. ©Springer Nature (Creative Commons—Attribution 4.0 International—CC BY 4.0))

beam diameter is about 6.5 cm. First, the light delivered by the liquid light guide passes through the Polarization State Generator (PSG). Next, the light backscattered by cervical tissue is analyzed by the Polarization State Analyzer (PSA). This first MPC prototype enabled measuring in vivo 16 images of the complete Mueller matrix in about 1.6 seconds, thus reducing the artifacts related to the patient's movements during image acquisition.

As shown in Fig. 8a, the polarimetric head of the MPC can be placed in the "OFF" position and allow the standard colposcopy observations. Otherwise, it can be placed in the "ON" position and acquire Mueller polarimetric images as shown in Fig. 8b. This system can measure depolarization Δ with an accuracy better than 0.5%, as well as the linear phase retardance R and the azimuth α with an accuracy of about 0.5°, under an optimized signal-to-noise ratio condition.

The preliminary results of in vivo MPC measurements performed on the cervices of two patients are presented and discussed below. The color photos of two healthy cervices (indicated as Cervix 1 and Cervix 2, respectively) with no CIN lesions (as confirmed by previously performed Pap smear and biopsies) are shown in Fig. 9. In particular, a polyp, identified by the practitioner as a benign growth, is visible to the naked eye on the surface of the Cervix 2 (Fig. 9b).

Bright white spots in the color images of both cervixes shown in Fig. 9 correspond to the saturated camera pixels due to the specular reflections of light by the samples' surface. Figure 10a–d show the images of the Mueller matrix element M_{11} (corresponding to the non-normalized intensity of the reflected light), Δ, R, and α images, respectively, for the Cervix 1 at 550 nm. Figure 10e–h show the same images for the Cervix 2. For the first patient, a region of interest (ROI), indicated as "Zone 1" in Fig. 10a, was selected for the entire ectocervix. For the second patient, the regions of interest indicated as "Zone 2" and "Zone 3" in Fig. 10e were selected for the ectocervix and the polyp, respectively. The Malpighian epithelium entirely covers the ectocervix. On the other hand, the polyp is covered by the glandular

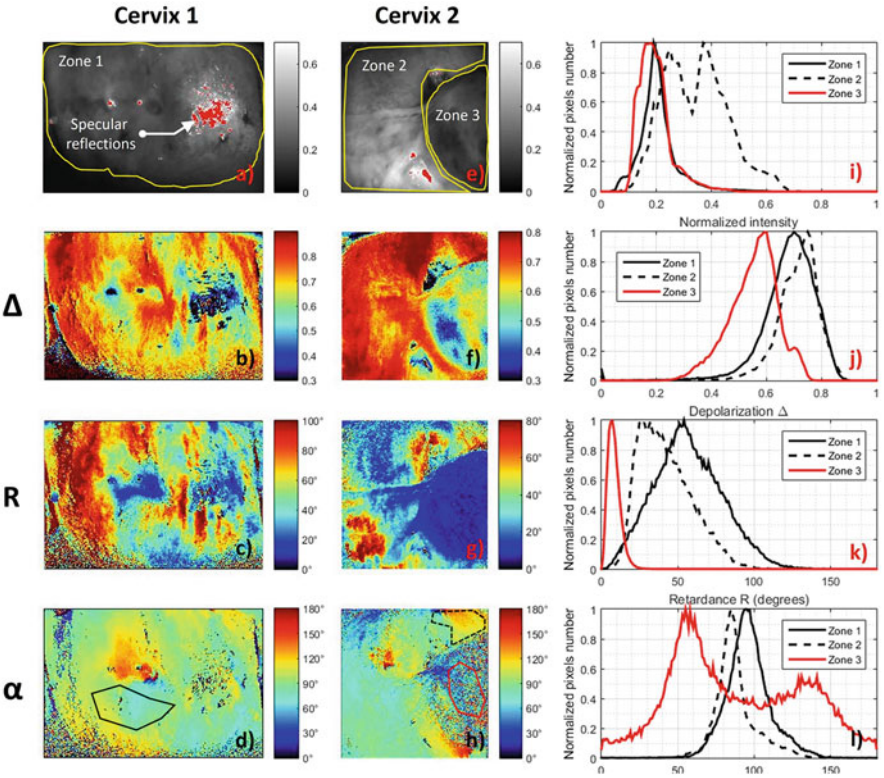


Fig. 10 (**a**–**d**), M_{11}, Δ, R, and α images, respectively, for the Cervix 1 at 550 nm. (**e**–**h**), the same images for the Cervix 2. (**i**–**l**) histograms of M_{11}, Δ, R, and α images, respectively, calculated for all pixels contained in the selected ROIs ("Zone 1" for the Cervix 1; "Zone 2" and "Zone 3" for the Cervix 2). The solid black line contour selected in Figure (**d**), as well as the dashed black line and solid red line contours selected in Figure (**h**), delineate three different areas analyzed in more detail in Fig. 11. (Adapted with permission from [18]. ©Springer Nature (Creative Commons—Attribution 4.0 International—CC BY 4.0))

epithelium. Figure 10i–l show the histograms of M_{11}, Δ, R, and α images calculated for all pixels contained in "Zone 1," "Zone 2," and "Zone 3."

The amount of the backscattered light intensity is very different for the three selected zones, as shown in Fig. 10i. A high mean value of the linear phase retardance R is observed in the zones of the healthy Malpighian epithelium for both Cervix 1 and Cervix 2. However, the mean value of R considerably decreases in the polyp zone of the Cervix 2. Indeed, the mean value of linear phase retardance is about 60° and 40° for "Zone 1" and "Zone 2," respectively, whereas it becomes less than 10° for "Zone 3," as shown in Fig. 10k. The parameter R appears to be relevant for the discrimination between the Malpighian epithelium and the glandular epithelium. The polyp zone is overall less depolarizing than the healthy zone of the ectocervix. Indeed, the parameter Δ is characterized by a mean value of 0.75

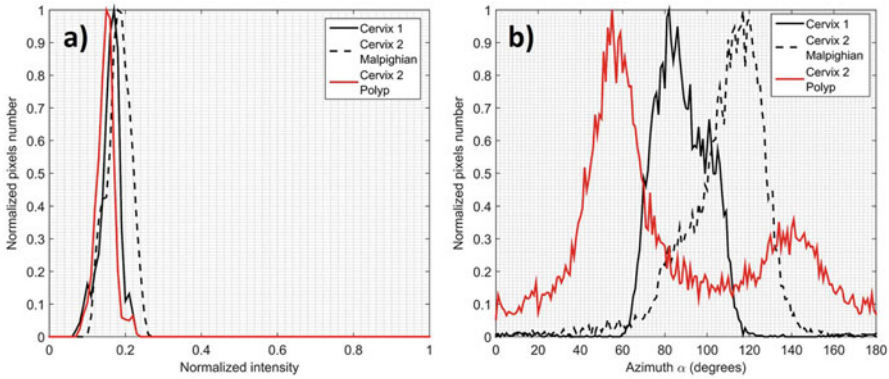


Fig. 11 Histograms of (**a**) non-normalized backscattered intensity of light, (**b**) azimuth of the fast axis for the subzones delineated in Fig. 10d, h. (Adapted with permission from [18]. ©Springer Nature (Creative Commons —Attribution 4.0 International—CC BY 4.0))

for both "Zone 1" and "Zone 2" and 0.6 for "Zone 3" as shown in Fig. 10j. The spatial variation observed for both parameters R and Δ is due to the specular reflection and the inhomogeneity of the cervical tissue vascularization. The peak of blood hemoglobin absorption at 550 nm can strongly affect the penetration depth of probing light and, consequently, the measured values of these parameters. Finally, the azimuth α of the fast optical axis is characterized by a well-defined preferential orientation in the healthy zone of the ectocervix.

On the contrary, the value of the parameter α varies pixel by pixel within the zone of the polyp. Indeed, the mean value of α is about 90° for "Zone 1" and "Zone 2," whereas it varies between 0° and 180° for "Zone 3" (with two peaks around 50° and 150°), as shown in Fig. 10l. It is worth mentioning that the spatial inhomogeneity in vascularization does not affect the measured values of the parameter α.

In order to verify that the random variation of the azimuth α observed within the polyp zone is not related to a low intensity of detected signal, the subzone delineated by the solid red line contour was selected within "Zone 3" and analyzed more in detail. For comparison, two other subzones were selected within the Malpighian epithelium of both cervixes (shown by the solid and dotted black line contours within "Zone 1" and "Zone 2," respectively). Finally, the histograms of the M_{11} coefficient (i.e., the non-normalized intensity of the reflected light) and the azimuth α were calculated for the three selected subzones. As a result, the intensity of the reflected light is approximately the same for all three selected sub-zones as shown in Fig. 11a. The distribution of the parameter α is well peaked for the ectocervix at about 80° for Cervix 1 and 120° for Cervix 2 as shown in Fig. 11b. However, the value of the parameter α varies between 0° and 180° in the subzone of the polyp for the Cervix 2 as shown on Fig. 11b. The analysis of the three selected subzones proves that the random distribution of the parameter α is not an artifact related to a loss of the signal-to-noise ratio but is the signature of the polyp microstructure that is very different from the one of the Malpighian epithelium [18].

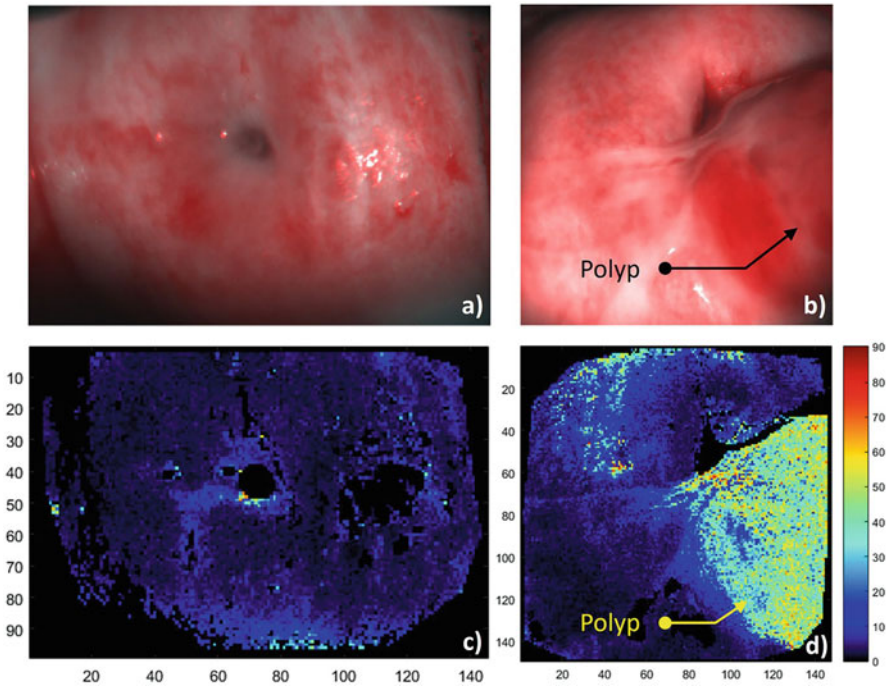


Fig. 12 Nonpolarized RGB images of (**a**) the Cervix 1 and (**b**) the Cervix 2. The standard deviation of the azimuth of the optical axis computed for each pack of 3 × 3 pixels in the images of the azimuth α of (**c**) the Cervix 1 and (**d**) the Cervix 2. (Adapted with permission from [18]. ©Springer Nature (Creative Commons—Attribution 4.0 International—CC BY 4.0))

In order to significantly increase the image contrast between the Malpighian epithelium and the glandular epithelium (polyp), the map of the standard deviation $\sigma(\alpha)$ of the azimuth is calculated for both analyzed cervices. The images of the parameter α are divided into 3 × 3 pixel patches, and the value attributed to each patch is the standard deviation of α over its 9 pixels. The pixels corresponding to the low-intensity level (less than 5% of the camera's dynamic range) were excluded from these calculations. These pixels are mainly located at the outer part of the images and in the zone of the cervical os. In addition, the saturated pixels were also excluded from the analysis. Due to the low spatial variation of the azimuth in the region of Malpighian epithelium, a low mean value of $\sigma(\alpha)$ is obtained for both "Zone 1" and "Zone 2" (about 7.2° and 6.7°, respectively). On the other hand, due to the significant spatial variation of the azimuth in the polyp zone, a higher mean value of $\sigma(\alpha)$ (about 37.2°) is obtained for "Zone 3." This method dramatically improves the visual image contrast between the Malpighian and glandular epithelium zones, as shown in Fig. 12.

9 Summary

This chapter summarizes the results and discusses the main milestones that allowed us to prove the relevance of the imaging Mueller polarimetry for improving the early detection of cervical pre-cancer.

The structure of cervix is very complex, and several histological tissue types can be found on its surface, as described in Sect. 5. Furthermore, a different polarimetric signature characterizes each histological type. Thus, detecting and delineating a precancerous lesion on the cervical surface is a complex problem that requires a multiclass and multiparametric approach. Indeed, our preliminary results obtained on a few fresh cervical specimens analyzed ex vivo have shown that the combination of the linear phase retardance R, the azimuth α, and the depolarization Δ (extracted from the measured Mueller matrices using the Lu–Chipman decomposition) should help to accurately identify the different histological types and unambiguously detect a precancerous lesion, as described in Sect. 4.

The first statistical analysis performed on a dataset of the polarimetric images recorded for ex vivo formalin-fixed cervical samples showed that the linear phase retardance R at 450 nm is an effective parameter to distinguish a precancerous lesion (CIN2-3) from the healthy squamous epithelium with a sensitivity and specificity of about 83%. In order to perform a complete analysis that considers all the histological types of the cervix, we developed a statistical postprocessing framework based on machine learning approaches for in-depth multiparametric and multiclass analysis of the cervix, as described in Sect. 6. The proposed framework makes it possible to simultaneously test several polarimetric parameters (such as the components of the measured Mueller matrices or the physical quantities obtained using different decompositions [49–54] and algebraic methods available [55–57]) at several wavelengths. In particular, it automatically selects the most diagnostically relevant polarimetric parameters, identifies the best classifier, and refines its parameters. This statistical data processing framework supports both binary and multiclass discrimination and will be tested in a new statistical study in vivo.

A first Mueller polarimetric colposcope (MPC) has been designed and built. This system has been successfully tested in vivo on several patients in a clinical setting for a feasibility study. The MPC was built by integrating a miniaturized imaging Mueller polarimetric system onto a commercial colposcope to minimize the modification of the established medical practice. Indeed, this system can be used as a conventional colposcope and provides reliable polarimetric images at 550 nm in about 1.6 seconds, which significantly reduces blurring effects caused by involuntary patient movements during the measurements. Furthermore, the preliminary in vivo results are consistent with those obtained ex vivo, paving the way for clinical use of imaging Mueller polarimetry.

Following the studies described in this chapter, a new multispectral MPC prototype has been developed for cervical analysis at different depths. This system allows acquiring images in the blue, green, and red parts of the visible spectral range simultaneously in about 1 second.

A clinical trial in a cohort of patients is being prepared to test the multispectral MPC under real-world conditions, including tissue blood perfusion and mechanical stress to confirm the capability of imaging Mueller polarimetry to improve the detection of cervical precancers.

This study will pave the way for the real-world clinical application of the imaging Mueller polarimetry in cervical cancer diagnosis.

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Polarimetric Endoscopy



Ji Qi and Daniel S. Elson

1 Introduction

Endoscopes are used for observation within the human body either through natural body orifices such as mouth, urethra, nose, and anus or minimally invasively via surgical key holes. Clinicians take advantage of endoscopes to visualize otherwise inaccessible views inside the body and identify various pathologies. As the current paradigm of diagnosis and treatment shifts further towards non- and minimally invasive methods, endoscopes are becoming an essential medical device for cancer screening, diagnostic investigation, and surgical guidance.

Currently, the imaging modality for routine endoscopy is white light imaging (WLI), which employs white light illumination and colour cameras to image tissues inside the body. WLI recreates the optical information detected by naked eye, and images are therefore intuitive to understand. Clinical inspection and decision-making are made in accordance with the colour contrast under white light illumination. A complementary modality to WLI, narrow band imaging (NBI), is a common endoscopic tool, which makes use of narrow band illumination (typically using two bands centred at 415 nm and 540 nm, respectively). As a result, NBI enhances contrast arising from hemoglobin absorption in superficial tissue layers, and therefore significantly improves visualization of mucosal and vascular patterns to facilitate lesion detection. However, not all cancers present obvious differences in colour or vascular patterns, and tissue regions with colour or abnormal patterns

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may be non-cancerous, resulting in limited detection sensitivity and specificity in practice [1, 2].

The image contrast mechanisms for both WLI and NBI are highly related to wavelength spectral characteristics of tissues, particularly from scattering and absorption. With theoretical advances in optical imaging methods and the technological progress of photonic components and devices, state-of-the-art biophotonics imaging has considerably broadened the possibilities to observe biological tissues from new perspectives beyond wavelength related WLI and NBI and has potential to strengthen the ability of endoscopes resolving conditions that are difficult to detect with the standard of care imaging techniques.

Optical polarization is a fundamental property of light that can be used to provide a unique and interesting insight into tissue. The polarization properties of tissue, namely depolarization, retardance, and diattenuation carry pathological information on the tissue morphology and structures [3–5]. The progress in polarimetry – the technique used to measure those polarization properties – enables label-free characterization of tissue changes, for example, nuclear enlargement of cells and collagen remodelling, to which polarization-blind imaging methods like WLI and NBI are not particularly sensitive [6, 7]. An increasing number of medical applications of polarimetric techniques have been explored, including detection of cervical and gastrointestinal cancer, colon cancer staging, assessment of cancer therapy, investigation of partial bladder obstruction, guiding mass spectrometry for rapid histopathology, characterization of collagen fibres, etc., as introduced in other chapters.

The development of polarimetric endoscopes can extend the image contrast beyond wavelength-related mechanisms and complement the wavelength-related information obtained from routine endoscopy. Recognizing the potential benefits of obtaining polarimetric information for internal organ diagnostics, investigators have explored a number of designs and prototypes over the past decade aiming towards polarimetric endoscopes that can be clinically validated and translated to the clinic. This chapter introduces the major types and typical designs of medical endoscopes and explains the main considerations when building polarimetric endoscopes. This chapter also describes the progress of polarimetric endoscopy made in the past decade, including a summary and outlook for this research area.

2 Endoscopes

There are two major types of endoscopes that are widely used in clinic, namely rigid and flexible endoscopes. Rigid endoscopes are primarily used in minimally invasive surgery and enter the body via surgical ‘key holes’; they are indeed the key enabling element for modern minimally invasive surgery. Due to the stiffness of the endoscope, the position of the distal tip – and therefore the inspection area and perspective – can be controlled by accurately manipulating the angle of the endoscope during surgery. Flexible endoscopes are mainly employed for

examination of internal organ systems through natural body orifices, for instance in colonoscopy and gastroscopy, where a curved path is necessary.

Endoscopes generally comprise at least one illumination and imaging channel (flexible ones often have additional air/water and biopsy/suction channels). The illumination channel delivers light from the light port at the proximal end to the tip of the endoscope and requires a high transmission to visible wavelengths. The spatial distribution of light radiance illuminating the tissue should also be sufficiently uniform across a wide area. Incoherent optical fibre bundles satisfy these requirements and are therefore widely adopted as illumination channels. A cold light source such as a filtered Xenon lamp is used for illumination and does not deliver excessive heat that could thermally damage the tissues, delivered to the illumination port via a separate incoherent fibre bundle to achieve good coupling at the light port.

The imaging channel delivers the image from the distal end to the proximal end, traditionally consisting of an objective lens, image relay system, and an eyepiece. The objective lens typically includes a negative lens to widen the field of view, followed by a positive focusing lens system which converges the rays and forms a demagnified real image to an image relay system. The objective is designed as telecentric in image space to allow better connection with the relay system [8]. The aperture stop is typically located at the back focal plane of the first negative lens. The relay optics of the imaging channel vary depending on the type of endoscope.

For rigid endoscopes, the image relay system is usually optics-based and mounted in a rigid outer tube (although at the current time, rigid video endoscopes are becoming increasingly common). The long and narrow nature of rigid endoscopes poses a challenge in the optical design of its relay optics because it is not easy to achieve a bright and high-definition image covering a large field of view with an image system of highly restricted diameter. The conventional design was based on a group of thin positive lenses mounted in an inner tube. The optical invariant of such a relay system is relatively small, leading to limited light throughput efficiency and also suffering from vignetting. An elegant solution was provided by Harold Hopkins at Imperial College London in 1960 by introducing the rod lens relay system, and this was subsequently commercialized by Karl Storz. The rod lens system could produce a larger optical invariant and negligible vignetting in addition to a simpler assembly. At the same time, it inherited the advantages of conventional systems including preeminent resolution, wide field of view, wide range of working distances, and low chromatic aberration. The image quality allowed surgeons to visualize sufficient details and enabled increasing surgical complexity. It was a milestone in the history of endoscopy and paved the way for the widespread use of minimally invasive surgery. Gradient index (GRIN) lenses are occasionally used to relay images. These can be manufactured down to 0.2 mm in diameter [8] and are therefore advantageous as image relay systems when the endoscope needs to transit through extremely small channels. It is difficult for GRIN lens-based endoscopes to produce a colour image of good quality due to the excessive dispersion of GRIN lenses, so they are only used in certain scenarios [8].

The image relay system of traditional flexible endoscopes is also optical. Traditionally since the 1960s, these used a flexible spatially coherent fibre bundle

with an identical spatial arrangement of fibres at the distal and proximal ends – hence the name 'fiberscope'. Typically, such bundles consist of 3500–18,000 fibres with the individual fibre core diameter ranging from 7 to 11 μm. An individual optical fibre in the bundle preserves the intensity and colour of light in transmission and serves as a pixel, jointly enabling relay of a colour image from the objective to the eyepiece. The resolution of the image is strongly dependant on the core diameter and the number of optical fibres comprising the bundle, as constrained by fibre manufacturing techniques and bundle flexibility. It is impossible to manufacture a practical fiberscope for high-definition imaging. The individual fibres are also fragile and easy to break and frequent bending can permanently disable pixels.

Advances in electronic technologies have allowed miniaturized image sensors at the distal end of the endoscope (videoscopes) that overcomes the problems of fiberscopes regarding the resolution and fragility. The imaging channel of the videoscope is simply an objective in conjunction with the imaging sensor with electronic image transfer. High-definition images and videos can be obtained since the resolution is not limited by the number of cores in the fibre bundle, but the image sensor size and the objective. These videoscopes have superseded fiberscopes in all but a few special narrow lumen scenarios like fetoscopy, ureteroscopy, endomicroscopy, etc.

Light sources used for endoscopy are predominantly incoherent. Xenon arc lamps are most wide due to their brightness, stability, bandwidth, and favourable colour temperature. With the rapid evolution of LED technology, endoscope sources based on LEDs have become more popular due to superior performance in lifespan, reliability, and ease of colour temperature tuning. The light from LED and Xenon arc lamps is then coupled into a flexible light cable that is either a liquid light guide or an incoherent fibre bundle. The light cable then connects the endoscope light port to deliver illumination light into the illumination channel of the endoscope.

To achieve polarimetric endoscopy, a desirable strategy is to minimally alter commercially available endoscopes (rather than building from scratch) since this can reduce efforts in designing optical systems and manufacturing processes for reliability and quality control, with more simple approval procedures from governing agencies and increased willingness from clinicians for adoption. Nevertheless, polarization is not generally considered when commercial endoscopes are designed, no matter whether they are rigid or flexible, but is an important concern when modifying these for polarimetry. Endoscopes normally have their own polarization characteristics and are not polarization-maintaining. Another challenge is the strict size restrictions, with many polarization modulation devices difficult to miniaturize to endoscopic dimensions.

It is therefore necessary to take account of the polarization properties of optical components within endoscopes. In the following section, we first summarize the polarization arising from optical materials, including strains and refraction at the lens surfaces, reflections including total internal reflection by the prisms, and thin film coatings at optical surfaces, which are normally employed in commercially available endoscopes. We then introduce the polarization characteristics of optical fibres, illumination channels of endoscopes, Hopkins rod lenses, and GRIN lenses (Fig. 1).

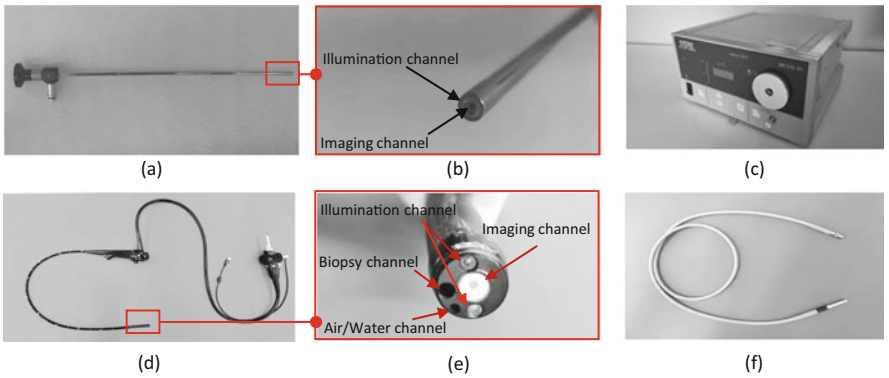


Fig. 1 (**a**) Rigid endoscope (**b**) distal end of a typical rigid endoscope (**c**) an example of an endoscope light source (**d**) flexible endoscope (**e**) distal end of a typical flexible endoscope (**f**) light guide cable

3 Instrumental Polarization

3.1 Optical Materials and Strains

Some widely used optical materials are retarding, including ruby, quartz, sapphire, calcium fluoride, and many other crystals, as well as various plastics where retardance originates from the internal strains in the manufacturing process. Although generally a normal lens made from non-retarding optical materials is highly polarization maintaining represented by an identity Mueller matrix, internal strain may arise from fire-polishing of the surfaces for some uncoated moulded glass lenses with high numerical apertures, resulting in radially symmetric, tangentially oriented retardance patterns [9]. It is possible to relieve this stress induced retardance by annealing in the lens fabrication process [9]. External strains from fastening and mounting are also a potential source for retardance due to the photo-elastic effect, which cannot be always avoided in a compact optical instrument.

3.2 Reflection and Refraction

Reflection and refraction at the optical surfaces constitute a basis for imaging systems. Reflection and refraction are not depolarization process but linear diattenuation and sometimes linear retardance depending on the incident angle and relative refractive index may arise, as inferred from Fresnel's Equation. Refraction and reflection are completely polarization maintaining for both linear and circular polarization under normal incidence. At this condition, the Mueller matrix for refraction is an identity matrix, and that for reflection is also an identity matrix

with the last two diagonal elements negative, which refers to a flip of polarization direction for the incident 45°/−45° linearly polarized light and a flip of chirality for circularly polarized light, due to the reflectional symmetric transformation of the coordinate systems during the mirror reflection. The negative signs can be adjusted to positive by a convention often seen in polarimetric imaging of tissues through different defined coordinates' positive orientations.

Under non-normal incidence, the interface reflects the parallel component of the electric field to the reflection plane (***p*** polarization) less efficiently than the perpendicular component (***s*** polarization), presenting linear diattenuation. The linear retardance depends on the attenuation coefficient of the materials represented by the imaginary part of relative refractive index. High material transparency, for example, air, glass, usually means low attenuation coefficient. Most optical materials have superior material transparency with their material attenuation negligible. Reflection and refraction at the interfaces of such materials are equivalent to linear diattenuators with the diattenuation dependant on the incidence angle. The forms of their Mueller matrices are given by,

$$
\begin{aligned}
M\text{reflect} \propto & \begin{bmatrix} \Theta1^2+\Theta2^2 & \Theta1^2-\Theta2^2 & 0 & 0 \\ \Theta1^2-\Theta2^2 & \Theta1^2+\Theta2^2 & 0 & 0 \\ 0 & 0 & -2\Theta1\Theta2 & 0 \\ 0 & 0 & 0 & -2\Theta1\Theta2 \end{bmatrix} \\
M\text{refract} \propto & \begin{bmatrix} \Theta1^2+1 & \Theta1^2-1 & 0 & 0 \\ \Theta1^2-1 & \Theta1^2+1 & 0 & 0 \\ 0 & 0 & 2\Theta1 & 0 \\ 0 & 0 & 0 & 2\Theta1 \end{bmatrix} \\
\Theta1 = & \cos(\theta i - \theta r) \\
\Theta2 = & \cos(\theta i + \theta r)
\end{aligned}
\tag{1}
$$

θ_i and θ_r denote the incidence angle and refraction angle, respectively. The polarization property of a single uncoated optical surface can be obtained according to Eq. 1. Lenses with small numerical apertures are highly polarization maintaining with negligible diattenuation and retardance as a rule of thumb. For optical systems with high numerical apertures, polarization ray tracing and polarization aberration theory are helpful to analyse diattenuation and retardance [10, 11].

When the reflective material has non-negligible attenuation coefficient like metals, reflection leads to linear retardance in addition to linear diattenuation, forming an effective diattenuated retarder. The linear diattenuation reaches to its maximum at the Brewster angle, and the linear phase retardance monotonically rise with incidence angle.

3.3 Total Internal Reflection

Prisms involving total internal reflection are frequently used in optical imaging systems including endoscopes. In total internal reflection, ***s*** and ***p*** polarized components are reflected with equal efficiency, but a phase shift arises between them. The Mueller matrix for total internal reflection has the same form to a linear retarder in which the linear retardance δ is determined by the incidence angle θ_i and the relative refractive index n,

$$\delta = 2 \arctan \frac{\cos \theta i \sqrt{n^2 \sin^2 \theta i - 1}}{n \sin^2 \theta i} \tag{2}$$

3.4 Thin Film Coating

Thin film coatings are usually applied on optical surfaces to control their reflectivity Via optical interference. The coatings normally manifest weak linear retardance and linear diattenuation stemming from the coherent superposition of light rays passing through the different layers. The retardance and diattenuation from the coatings may lead to polarization aberrations that are of the fourth and sixth orders with respect to the incidence angle [12]. The weak linear diattenuation and linear retardance from single layer antireflection coatings can sometimes be exploited to reduce effective instrument polarization in the entire optical system [12].

3.5 Optical Fibres

Single mode fibres have strong retardance and weak diattenuation. Theoretically, a single mode fibre with perfectly cylindrical symmetry would maintain the polarization state of the input light and not demonstrate birefringence or diattenuation at all, and its Mueller matrix would be represented by an identity matrix. However, in reality, there are many random symmetry-breaking imperfections including the geometric asymmetry (e.g. slightly elliptical cores) and inherent strains induced during fabrication, as well as extrinsic stresses induced by bending, twisting, squeezing, temperature changes, and other effects [13]. As a result, two orthogonal polarizations of the fundamental mode propagate at different velocities, known as polarization mode dispersion (PMD) in fibre optics [14]. For high-speed fibre optic communications, PMD causes pulse broadening and leads to an increase in bit-error rate. For polarimetry, PMD is the reason why a single mode fibre manifests strong retardance. The phase retardance of a single mode fibre may be controlled by being bent, twisted, and squeezed, which is the basis of fibre polarization controllers.

The induced phase retardance is proportional to the squeezing pressure and bending curvature, and the orientation of effective optic axis is controlled by subtle twist.

The asymmetries of fibres also lead to the different absorption of light polarized along two orthogonal axes, known as the polarization dependant loss (PDL) in fibre optics. PDL exactly corresponds to diattenuation in polarimetry and can be determined directly by Mueller polarimetry. It is worth mentioning that 'polarization maintaining' fibres (PM fibres) are not strictly polarization-maintaining. A systematic high birefringence is intentionally introduced in PM fibres so that the birefringence generated by the random asymmetries is almost negligible compared to the systematic birefringence, and the systematic birefringence prohibits coupling between the two orthogonal principal polarization directions. Therefore, a PM fibre can only maintain two linear polarization states along its two orthogonal principal polarization directions.

Multimode fibres do not only experience polarization mode dispersion but also spatial mode coupling. When light is injected into a single guided mode in the multimode fibre, part of it couples into other modes, each of which undergoes distinct polarization scrambling. Thus, the output polarization state of the individual modes varies from one mode to another. Considering that the emergent light from the multimode fibre is a superposition of all the modes of various polarization states, the multimode fibre is effectively a depolarizer without wavefront shaping at the input or mode decoupling at the output.

3.6 Illumination Channel of Endoscopes

As was introduced earlier, the illumination channel of endoscopes is the incoherent optical fibre bundle with large individual fibre cores, that is, multimode fibres. The emergent light from the fibre bundle is a superposition of all the individual multimode fibres, and the emergent light from the individual multimode fibre is a superposition of all the modes of various polarization states. Therefore, this type of fibre bundle highly approximates to an ideal depolarizer.

3.7 Imaging Channel of Rod Lens-Based Rigid Endoscopes

The polarization properties of the imaging channels of rod lens-based rigid endoscopes have been investigated experimentally [15]. By measuring the Mueller matrices of laparoscopes from a prominent endoscope manufacturers Karl Storz GmbH (Fig. 2a), it was revealed that the image channel of the Storz laparoscope was effectively a uniaxial crystal with the crystal axis parallel to the optical axis of the laparoscope shown in Fig. 2b, demonstrating spatially varying and symmetric phase retardance, with low diattenuation and depolarization. The retardance originates from the outer protective windows at both the distal and proximal ends that are

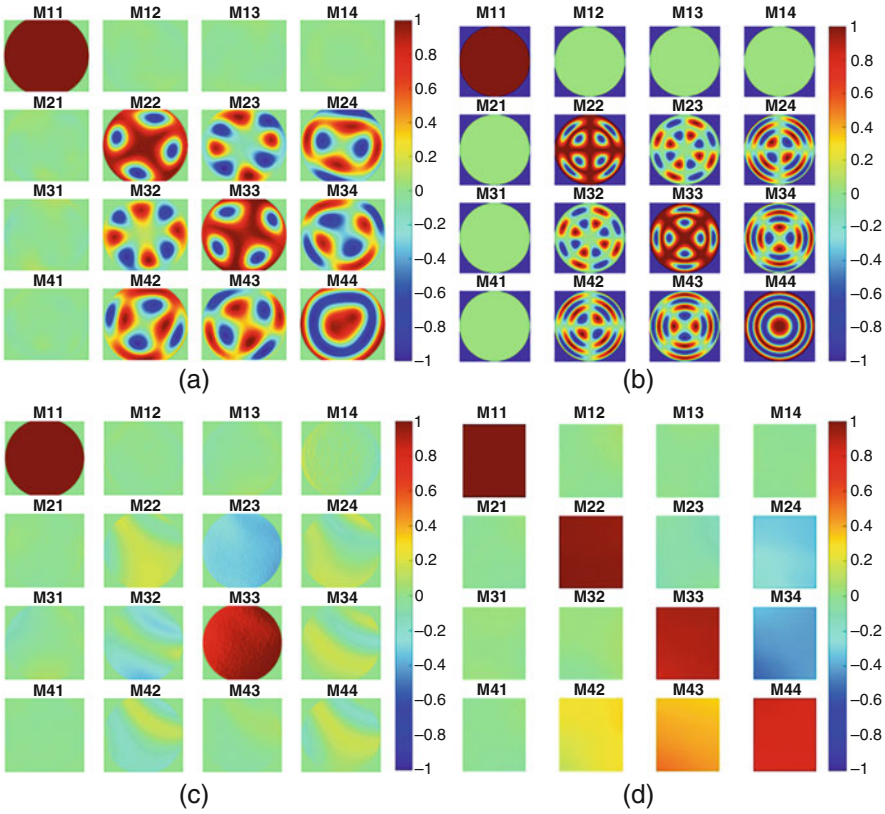


Fig. 2 The Mueller matrix images of (**a**) the imaging channel of a Karl Storz laparoscope with normal crystal windows. (Unpublished data, but similar data can be found in Wood and Elson [15]); (**b**) the simulated Mueller matrix image of a uniaxial birefringent crystal. (Reproduced from Wood and Elson [15]). The similarity between (**a**) and (**b**) indicated that the laparoscope was effectively a uniaxial crystal with the crystal axis parallel to the optical axis of the laparoscope. (**c**) the imaging channel of a Olympus laparoscope. (Unpublished data, but similar data can be found in Wood and Elson [15]); (**d**) a customized Karl Storz laparoscope by replacing the birefringent protective windows with non-birefringent-fused silica windows. (Reproduced from [16, 17])

made from a birefringent crystal sapphire. If used in polarimetry, this endoscope would require careful calibration for the highly spatially varying retardance and additional efforts to avoid any small displacement between the endoscope and the image sensor after calibration. Endoscopes with less or no spatially varying retardance are desirable for polarimetric imaging. The laparoscope from another major manufacturer Olympus (Fig. 2c) was also measured. Its imaging channel demonstrated both depolarization and retardance. The retardance comes from the same source as Storz laparoscope. The depolarization may originate from multiple birefringent materials in its optical design. The laparoscopes for which the image

channel is depolarizing like the Olympus ones are not suitable to adapt for polarimetric endoscopy.

The Mueller matrix for a customized laparoscope with the birefringent windows replaced by fused silica is displayed in Fig. 2d [16, 17]. This is equivalent to a linear retarder with low phase retardance (27°). The top left 3 × 3 sub-matrix essentially demonstrated an identity form, which implies that the polarization properties of the customized laparoscope are not significant and can be called non-polarized or polarization maintaining for linear polarizations. The polarization effects of rigid endoscopes for linearly polarized light can then be removed for linear polarization imaging by simply replacing their crystal windows with non-birefringent material. Such endoscopes would simplify and reduce the requirements for the design of the polarization state analyzer (PSA) since the PSA does not have to be miniaturized to match the dimensions of endoscopic tip.

3.8 GRIN Lenses

Gradient index (GRIN) lenses can also be used as endoscopic objectives, endoscopic relay lenses, and collimating or focusing lenses and offer advantages when working in environments requiring extremely small dimensions. The Mueller matrix of a GRIN lens was measured in [18] and is reproduced in Fig. 3. It also demonstrates a

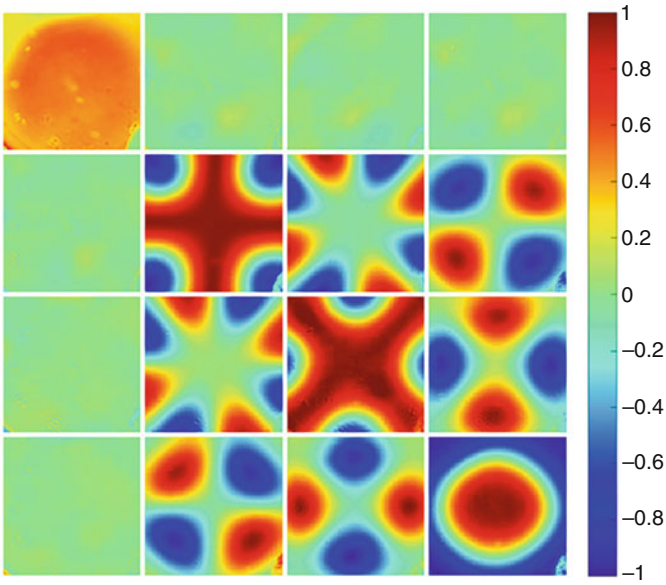


Fig. 3 The Mueller matrix image of a GRIN lens. (The figure was reproduced from Chang et al. [20])

conoscopic interference pattern, which indicates that a GRIN lens is also equivalent to a uniaxial crystal with the crystal axis parallel to the optical axis of the lens, with low diattenuation and depolarization. Such a GRIN lens has been optimized and used as a point detection-based snapshot Stokes [18] and Mueller polarimeter [19]. A GRIN lens can also be altered to become a Stokes polarimetric imaging device with the conoscopic interference pattern removed by a pre-determined matrix division procedure [20]. It is noted that the polarization properties of a GRIN lens change significantly with its working distance. Therefore, the approaches that take advantage of the spatially varying polarization properties of GRIN lenses may turn out difficult to apply without precise information about the geometry of the optical system and the scene, for example, the object distance and the 3D profile of the detected object.

4 Complete Mueller Polarimetric Endoscopy

To acquire the most detailed information about the polarized light interaction with tissue – and potentially the most useful clinical data about the tissue – it is desirable to acquire a full Mueller matrix at a single location or across an image of the sample. One option to achieve this is to adapt a suitable existing clinical device. However, since endoscopic illumination generally uses bundles of multimode fibres (or less commonly LED emitters at the tip) that do not preserve the input polarization, the polarization state generator (PSG) must still be mounted at the tip of the device.

In certain types of endoscope – for instance the Storz rigid endoscope with a lens-based imaging channel – the polarization response may be negligible or can be characterized in advance [16, 21], as introduced in the previous section. Therefore, it may be possible to mount the PSA and other optics outside the patient. Modification of the optics and imaging detector is not possible in most other types of endoscope, for example, any device with a chip-at-tip technology like flexible videoscope, where it is impractical to modify the imaging optics without completely redesigning the endoscope. Consequently, designing and building an endoscope from scratch may be preferential where endoscopes cannot be adapted, although this comes at the cost of requiring more complex regulatory and ethical approval to conduct human studies. Whilst the custom-build route would benefit from close industrial collaboration, it would be difficult to secure this without having a strong clinical and commercial case for support that goes beyond in vivo pilot studies. Another alternative is to design fibre optics-based Mueller matrix sensor that can be scanned to build up an image pixel by pixel [22].

In the following sections, we outline current methods for endoscopic Mueller matrix detection that are compatible with endoscopy.

4.1 Mueller Polarimetric Rigid Endoscope

As described above, it is desirable to adapt clinical endoscopes where possible, particularly rigid devices that have integrated lens systems. In one example, a simple miniaturized PSG for 4 × 4 Mueller polarimetry used a combination of a polarizer fixed over the ring-shaped illumination channel of a rigid endoscope together with a ring-shaped quarter waveplate attached to a rotatable rigid tube around the endoscope. The illumination state can then be adjusted electronically via a proximal rotation stage and simply requires a slight increase in the diameter of the endoscope, as determined by the thickness of the cylindrical tube walls. For rotational symmetry reasons, the rotating optic would most conveniently be a ring shape, which therefore requires use of an endoscope with a central imaging channel and surrounding fibres – a common configuration in commercial devices. By rotating this outer tube around the endoscope, the polarized light could be altered into different illumination states [21, 23]. The choice of four of these – 45°, 0°, 30°, and 60° – allowed the optimum PSG combination for this setup (see Fig. 4) [24]. Using the same endoscope as described above in Sect. 3, it was possible to use a PSA imaging detector at the proximal end to generate full Mueller polarimetric images in 30 seconds, mainly limited by the slow rotation speed of the motor controlling the outer tube.

The final instrument maintained the high field angle and working distance of the incorporated endoscope, and achieved an 'HD high-definition' spatial resolution through the use of a suitable camera. It was used to image full Mueller matrices from various polarizing and retarding test samples and a porcine bladder and information such as linear depolarization, circular depolarization, cross-polarization, birefringence, and the fast axis orientation could be extracted by per-pixel processing (see Fig. 4). In many cases, the contrast between different regions or features within the image were emphasized by the polarization-resolved signals, particularly in regions of the image that exhibited birefringence or obvious fibrous structures. The observations were confirmed by local image metrics, which were able to demonstrate structures that were invisible to the non-polarization-resolved images [21].

4.2 Single Mode Fibre-Based Laser Mueller Polarimeter

An interesting full Mueller matrix polarimeter that has potential for flexible endoscopy using a single fibre was published by Manhas et al. [22]. The method relies on recording the Mueller matrix of the single mode fibre (assuming linear and circular retardance terms only) by using a mirror as the sample and detecting through a double-pass of the fibre. The Mueller matrix data from a real sample may then be determined by matrix algebra, excepting the absolute axis orientations that remain unknown due to uncertainty about the fast axis orientation of the fibre (Fig. 5). The PSG and PSA used a polarizer – ferroelectric liquid crystal – quarter

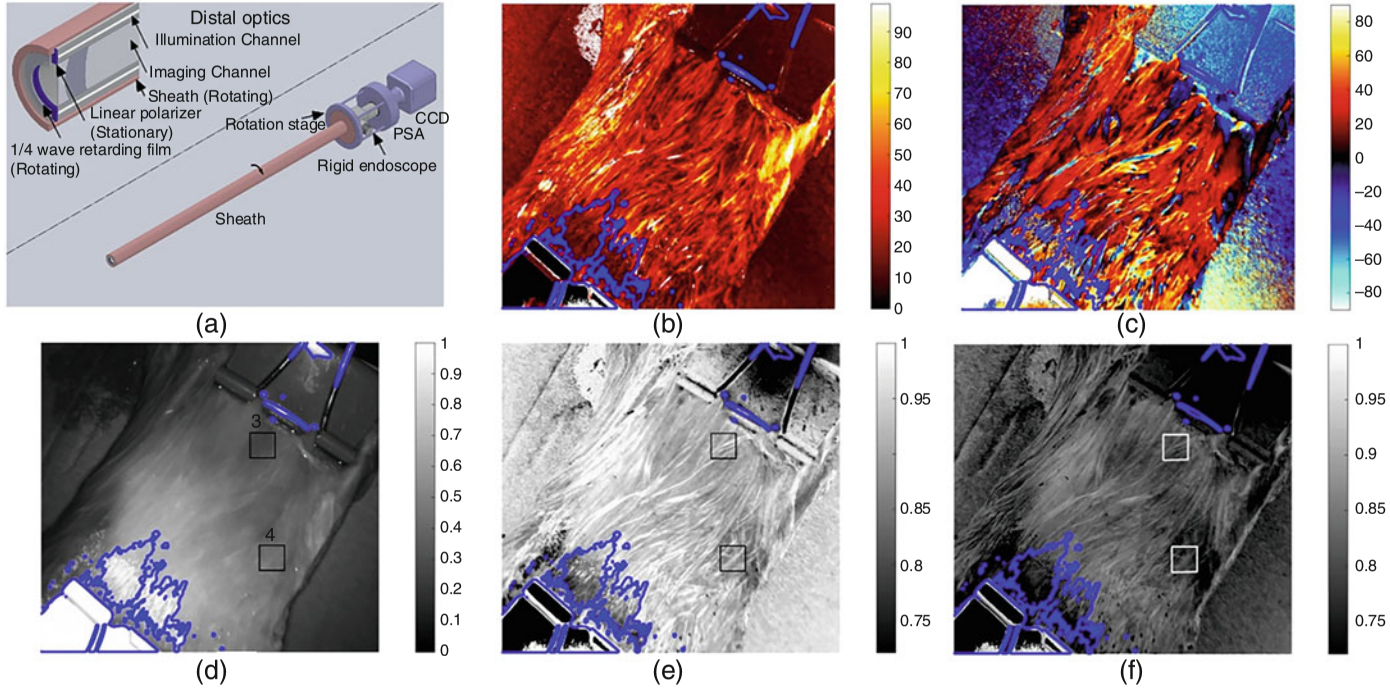


Fig. 4 Mueller polarimetric rigid endoscope. (**a**) the structures of this polarimetric endoscope, consisting of a stainless steel sheath, a motorized rotation stage, a rigid endoscope, a PSA, and a CCD image sensor. The part that rotates during acquisition is represented in red, and the stationary part is in purple. (**b–f**) the retardance image, the fast axis orientation image, polarization insensitive image for comparison, circular depolarization and linear depolarization images of a porcine bladder, obtained by decomposing the Mueller polarimetric images acquired by the Mueller polarimetric rigid endoscope. The areas enclosed by blue lines are affected by pixel saturation and are invalid. (The figure was reproduced from Qi et al. [21])

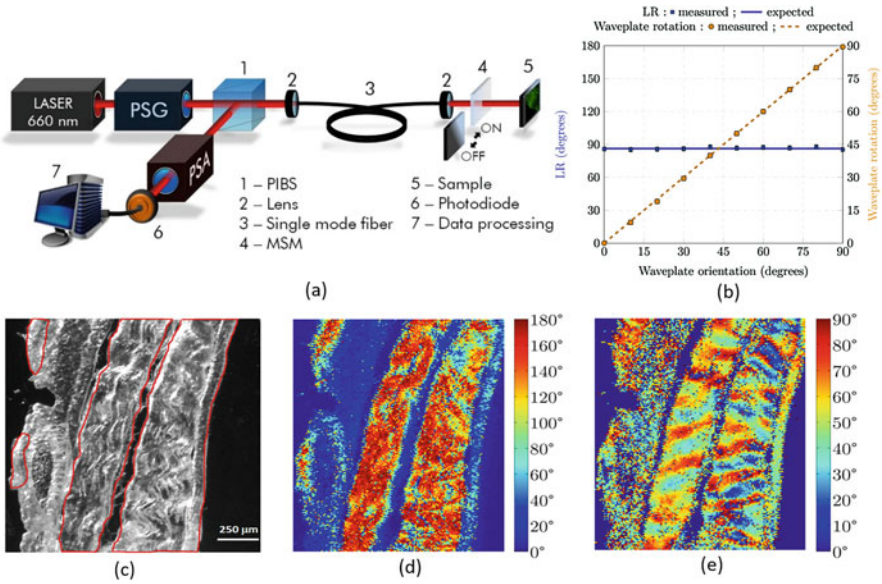


Fig. 5 Single mode fibre-based laser Mueller polarimeter. (**a**) schematic of the polarimeter. (PIBS: Polarization Insensitive Beam Splitter; MSM: Micro-Switchable Mirror; PSG: Polarization State Generator; PSA: Polarization State Analyser). (**b**) The validation results by sensing a rotating waveplate sample on the top of a mirror. The vertical axis denotes linear retardance. (**c**–**e**) The intensity, linear retardance and fast axis orientation of the 50 micron thick rat tail collagen sample. The parameters were obtained by decomposing the Mueller matrices of the sample acquired with the polarimeter. (The figure was reproduced from Manhas et al. [22])

waveplate – ferroelectric liquid crystal combination. Test samples included linear birefringence value and axis angle images for a mirror covered by a transparent birefringence adhesive tape and also a 50 micron thick rat tail collagen sample, with image data acquired by scanning of the sample across the detection site (Fig. 5).

One limitation of this approach is that a high brightness source is preferred (in this case a laser) to obtain satisfactory illumination intensity through the single mode fibre, which may lead to speckle artefacts. The single mode fibre has a small core diameter and may be smaller than a speckle size induced by a laser backscattered from tissue, and as a result, the measured degree of polarization of the emergent light from the tissue might be always close to 1 rather than being sensitive to tissue depolarization properties. The assumptions of no depolarization and diattenuation in the fibre can be realized for lengths compatible with endoscopy, but the calibration process that accounts for linear and circular retardance may also prove ineffective after fibre bending, twisting, or movement – important features of a practical system. The authors instead proposed a switchable tip-mounted mirror that can either be used to calibrate the fibre or to detect the sample light. Finally, to build up an image would require the actuation of the fibre tip and lens assembly as well as the miniaturization of the switchable mirror.

4.3 High Speed Single Mode Fibre Mueller Polarimeter

A single fibre scanning approach requires a fast Mueller matrix measurement to achieve a clinically effective imaging rate. Therefore, components that modulate state of polarization in temporal domain such as ferroelectric liquid crystals with relatively slow (10's ms) response times used in Sect. 4.2 should be avoided. Bending and twisting of the fibre are essential for endoscopic applications. The PMD of a single mode fibre is highly sensitive to bending and twisting of the fibre, which may quickly invalidate the polarimetry calibration of the fibre when the fibre polarimeter is being used for measurement. Therefore, calibration and sample measurements should be conducted simultaneously rather than sequentially as used in Sect. 4.2.

A solution based on the ideas of spectrally encoded snapshot Mueller polarimetry and coherence gating was proposed to address the two issues above, respectively, to achieve a high speed fibre Mueller polarimeter (Fig. 6a) [26]. The spectrally encoded snapshot Mueller polarimetry method employed in this work was also known as the channelled spectro-polarimetry technique [27–29]. The method takes advantage of the wavelength-dependant phase retardance of linear retarders (multiple order linear retarder), so the phase retardance of each retarder in the PSG and PSA is modulated in the wavelength domain rather than in the temporal domain. The Mueller matrix of the sample can then be reconstructed from a single snapshot by a spectrometer.

The idea of coherence gating was employed to separate calibration signals and sample measurement signals. In practice, the device comprised a Michelson-type interferometer with a low coherence source, a reference arm (reference mirror), a sample arm including partial reflector in front of the sample (partial reflector) and a mirror behind the sample (sample mirror), and a spectrometer detector (c.f. spectral-domain OCT). The length of the reference arm was then set to allow interference between the reference mirror and partial reflector light (the AC or interference component), while the light transmitted by the sample and reflected by the sample mirror was outside of the coherence window, forming part of the DC component. This design enables the sample and fibre Mueller matrices determined simultaneously in a single shot by the spectrometer, which allows for the possibility of the correction of the signal during bending or movement of the fibre during use. The Mueller matrix sampling rate of this fibre Mueller polarimeter is then determined by the acquisition rate of the spectrometer, which was 70 kHz in this case, and this allows for the possibility to raster scan a full image. The polarimeter was tested on a quarter wave plate and a linear polarizer during handling of the fibre and showed a satisfactory correction for sample fibre movement. An alternative approach using coherence gating to separate calibration and measurement signals in the fibre Mueller polarimeter can be wavelength gating (Fig. 6b), as proposed and validated in [30].

A crucial assumption of the spectrally encoded snapshot Mueller polarimetry method is that the sample is achromatic within the working wavelength band in terms of its polarization properties. For the fibre polarimeter developed based on

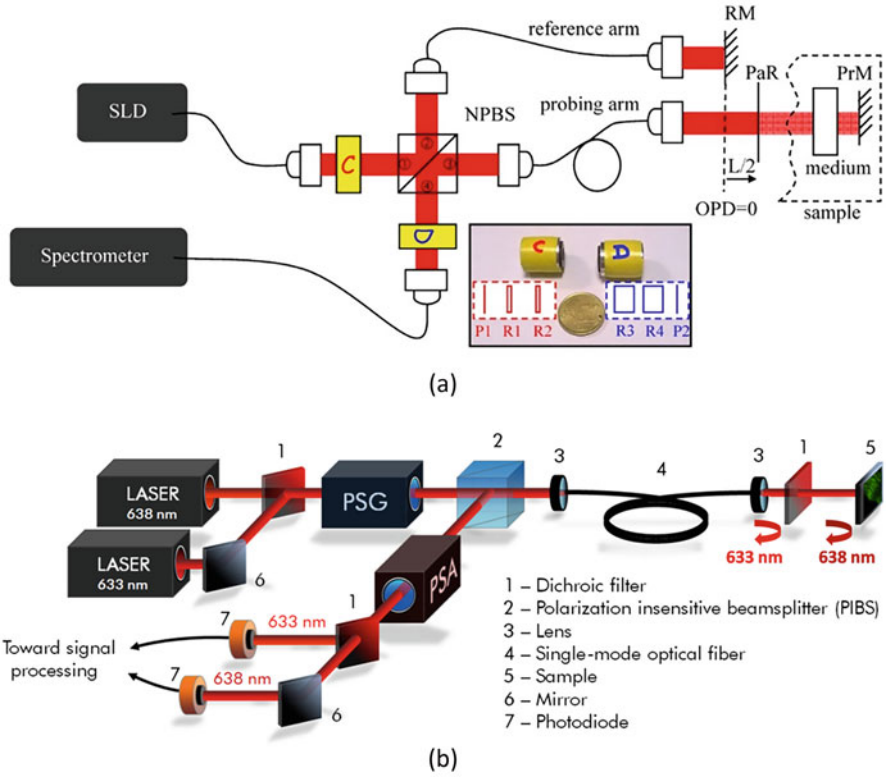


Fig. 6 High speed single mode fibre Mueller polarimeter. (**a**) Schematic of the polarimeter based on coherence gating. (SLD: super luminescent diode, C: passive polarization state coding block, D: passive polarization state decoding block, NPBS: non-polarizing beam splitter cube, RM: reference mirror, PaR: 50% partial reflector, PrM: probe mirror. or the Coding and Decoding blocks, P1, P2: linear polarizers are crossed, R1, R2, R3, R4 are retarders oriented, respectively, at 45°, 0°, 0°, and 45° according to the orientation of P1). (The figure was reproduced from [26]). (**b**) Schematic of the single mode fibre Mueller polarimeter based on wavelength gating. (The figure was reproduced from Jérémy et al. [30])

this method, it is also assumed that all the optical elements in this fibre polarimeter (except the coding and decoding retarder plates) are achromatic and that the single mode fibre is a retarder without diattenuation and depolarization in addition. The dispersion of the fibre due to the employment of a relative wide bandwidth of the low coherence light source (50 nm), however, can induce spectral depolarization of fibre and may invalidate the assumption that SMF is a pure retarder, which may induce error for depolarization assessment of the sample [26].

5 Partial Mueller Polarimetric Endoscopy

5.1 Polarimetric Endoscopes Based on Stereo-endoscopes

Since the imaging optics within an endoscope cannot be guaranteed to leave a measurable polarization state that can be traced to the input state, it is desirable to mount the PSG at the tip of the endoscope. Provided only a low number of polarization states are required, it may be possible to multiplex the number of imaging channels within the endoscope so that a fixed polarization optics can be attached to the tip rather than a controllable polarization state. Although stereo endoscopes are not commonly used outside of robotic-assisted minimally invasive surgery, they are available for certain interventions. Such endoscopes – with two sets of rod lenses in parallel – have been adapted to record a different polarization state through each channel. Provided the illumination is polarized, the detection channels can be co- and cross-polarized, thereby achieving multi-state imaging with no moving parts. In [31], the polarization states were generated in a disposable tip attachment (see Fig. 7), from which superficial or deeper image information could be resolved using simple calculations on the output images. The images of porcine tissues presented in the paper showed high contrast for superficial blood vessels due to hemoglobin absorption reducing the cross-polarized light intensity (see Fig. 7c). Video-rate single snapshot imaging could be achieved as a result of the parallel detection. To increase the polarization resolution, two different illumination states could also be generated by placing polarizers at two different orientations over the illumination channels of a dual illumination bundle endoscope. By switching light between these bundles and recording the two output images, four elements from the Mueller matrix could be obtained [32].

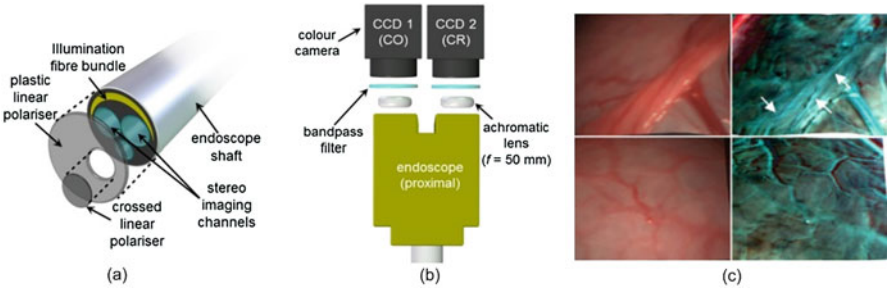


Fig. 7 Polarimetric endoscope based on a stereo-endoscope. (**a**) Distal optics. A linear polarizer placed over the endoscope illumination channel generates linearly polarized light. The co-polarized (CO) component is detected through the right channel while a separate section with its polarization axis orthogonal collects cross-polarized (CR) light, (**b**) proximal optics. Bandpass filters and colour cameras were used for in vivo narrow band imaging. (**c**) Standard colour (left column) and pseudo-colour images reconstructed from tissue polarization information (right column) of porcine bladder. White arrows indicate vascular features whose visibility has been enhanced based on Weber contrast measurement. (The figure was reproduced from Clancy et al. [31])

5.2 3 × 3 Mueller Polarimetric Endoscope Based on a Rigid Endoscope

While some endoscopes, particularly larger stereo ones, may have two input bundles that allow two different polarizers to be attached to the tip (see last section), if more elements of the Mueller matrix are to be measured, a variable PSG is required. In the case of rigid endoscopes, one way to avoid miniaturization of the state changing optic is to distally attach a polarizer over the illumination fibres and rotate the endoscope itself, using an electronically controlled proximal rotation stage. In our previous work [33], we mounted the linear polarizer to sequentially generate multiple input states (0°, 45°, and 90°) and multiple linear output images (0°, −45°, and 90°) were sequentially recorded, that is, measuring the top-left 3 × 3 elements of the Mueller matrix (see Fig. 8). Besides the polarization filtering, a mercury lamp together with a colour camera allowed multiple narrow wavelength bands to be detected simultaneously, increasing the fidelity of the polarization-resolved signal. Although a polarization-maintaining rigid endoscope was used, it maintained the typical 70° field-of-view and large depth of focus from ~1 to 15 cm.

The images acquired were decomposed into linear depolarization, retardance, and linear diattenuation for physical interpretation [34]. The employed interpretation method [34] assumes that (i) there was no circular diattenuation, (ii) the depolarization matrix was diagonal, and (iii) there was identical depolarization for 0°/90° and −45°/+45°. Other interpretation methods [35] that do not require assumptions may also be used. Testing of the device used standard polarization altering samples, and an example biological dataset of rat tissue in situ is shown in the figure. This illustrates the complementary information that is obtained from polarization-preserving absorbing tissue, such as the liver, and retarding tissues, such as stomach, with strong structural orientation, in this case from muscle. Differences in these signals due to the variation on optical absorption across the different detection wavelengths were also evident, with the polarization-preserving tissues becoming less so for higher penetrating wavelengths.

In practice, this system was affected by the very slightly kinked shaft of the rigid endoscope, which resulted in a variation of the field-of-view for the different orientations. This needed to be carefully corrected during processing to avoid any registration artefacts. Also, the assumptions mentioned for the matrix decomposition method would not apply in the case of anisotropic tissues, and there are differences between the 3 × 3 decomposed images and those derived from the more complete 4 × 4 Mueller matrices, although qualitative variations are maintained.

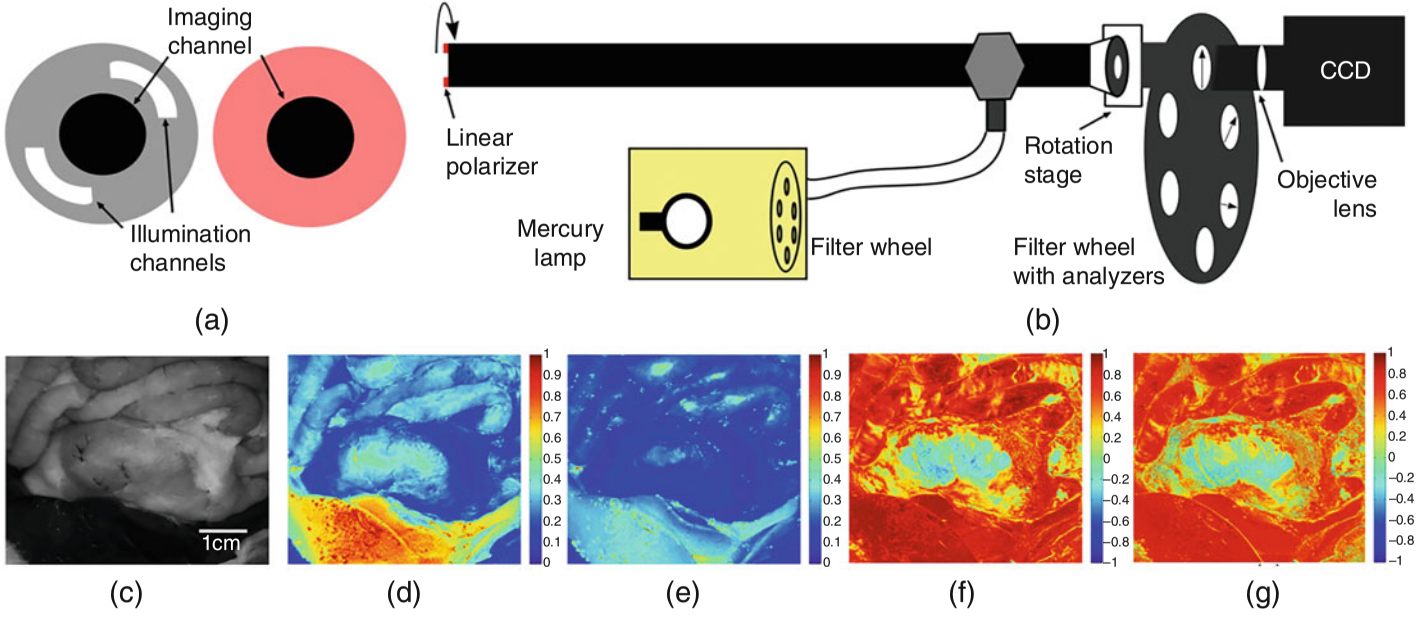


Fig. 8 3 × 3 Mueller polarimetric endoscope based on a rigid endoscope. (**a**) View of the distal end of endoscope and ring-shaped linear polarizing film covering illumination channel. (**b**) Side-view of the endoscope. (**c**) Polarization insensitive intensity image of a rat abdomen, small bowel, stomach, liver and fat. (**d**) Linear polarization maintaining power image at 546 nm (**e**) Linear polarization maintaining power image at 628 nm (**f**) Linear retardance at 546 nm (represented by the cosine of linear phase retardance value), (**g**) Linear retardance at 628 nm (represented by the cosine of linear phase retardance value). (The figure was reproduced from Qi et al. [33])

5.3 3 × 3 Modified Flexible Endoscope for Mueller Polarimetry

Altering commercial flexible endoscopes to become polarimetry compatible requires one more design step compared with rigid ones. As introduced earlier, the imaging channel of a traditional fiberscope is a coherent fibre bundle. The individual fibres in the bundle are multimode fibres (more precisely, few mode fibres) when working in the visible wavelength range of light. Considering that frequently moving and bending of the scope are also essential for a flexible endoscope, it would be very challenging to place a polarization state analyser at the proximal end of the imaging channel of the flexible endoscope and infer the polarization state of light at the distal end. For a videoscope, the entire imaging channel is simply an objective and the image sensor has been miniaturized to sit within the distal end of the scope. Therefore, altering commercial flexible endoscopes to become polarimetry compatible requires miniaturization of the polarization state analyser Fig. 9.

A solution based on an elaborate mechatronic design has been proposed [36]. The basic configuration was two rotated linear polarizers employed for generation and analysis of polarization states in front of the illumination and imaging channels, respectively. Rotation of the polarizers was achieved by a small motor, a miniaturized driving gear, a miniaturized driven gear, and a small polarizer. The motor employed was a servomotor with feedback signals to compute the rotation angle. The gears were helical teeth with small modulus to guarantee the minimal gap between the driving gear and the driven gear. The driver gear had a hollow structure with a support part attached to the base and the polarizer was mounted into the hollow part of the driven gear. The motor was controlled by a closed-loop proportional-integral-derivative (PID) system with the motor inertia corrected. Flexible electric wires were built inside the endoscope body to supply the power to the motor and transmit motor control signals. The acquisition time (excluding the Mueller matrix reconstruction) used for a 3×3 Mueller matrix image was no more than 2 seconds. The prototype was then validated with a rotating linear polarizer, and the estimated error was less than 9%. The device was subsequently demonstrated via its ability to image a nude mouse in vivo.

Electromagnetic compatibility (EMC) is essential for endoscopes that are regulated for clinical use. Integration of the motor at the distal end of the endoscope may reduce the EMC performance of the entire system. The motor itself may also become a source of electromagnetic interference source to degrade the image quality. Efforts are still needed to overcome the issues so that the design could eventually be clinically translated.

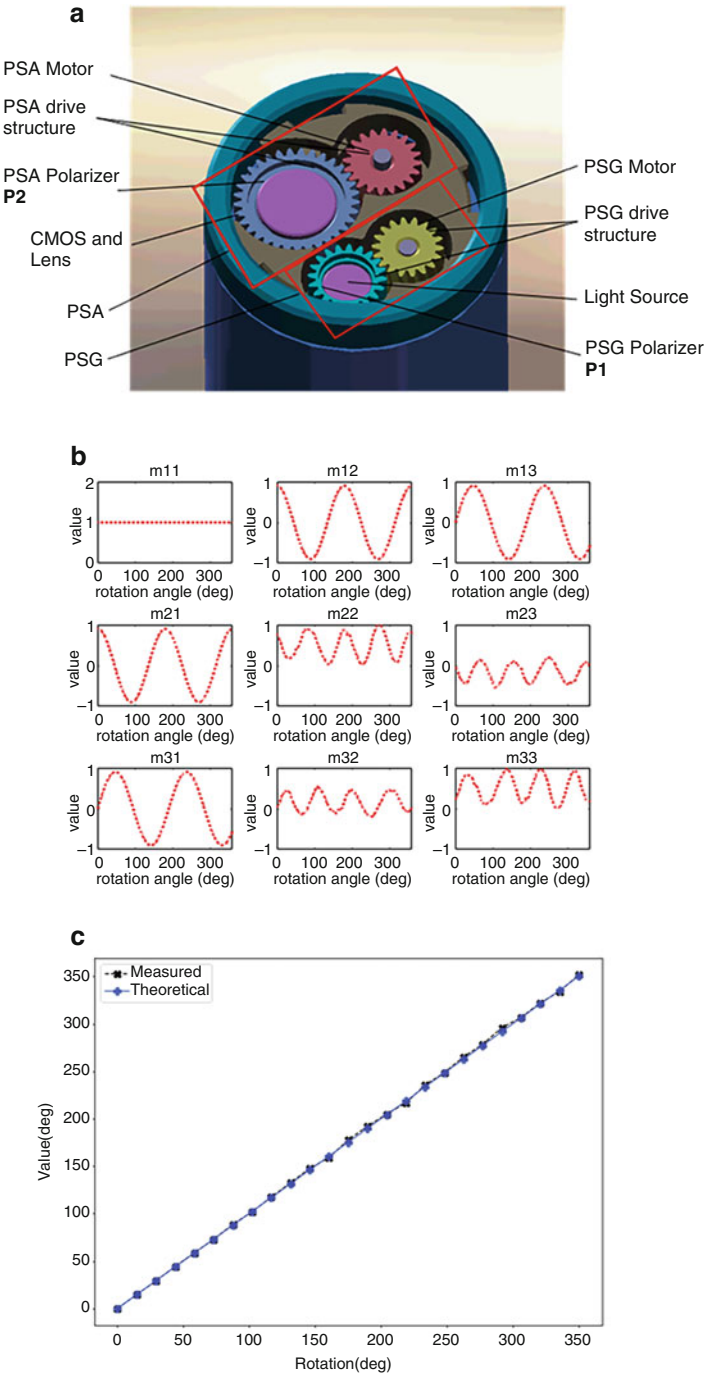


Fig. 9 3 × 3 Mueller polarimetric endoscope altered from a flexible endoscope. (**a**, **b**) 3D and sectional view of the endoscope distal structure, respectively; (**c**) the measured 3 × 3 Mueller matrices of a linear polarizer test object rotated from 0° to 360° with the polarimetric endoscope; (**d**) comparison between the theoretical rotation angle of the linear polarizer test object and the rotation angle obtained from the measured 3 × 3 Mueller matrices. (The figure was reproduced from Fu et al. [36]). Figure provided courtesy Honghui He, Jian Wu and Yifan Fu.

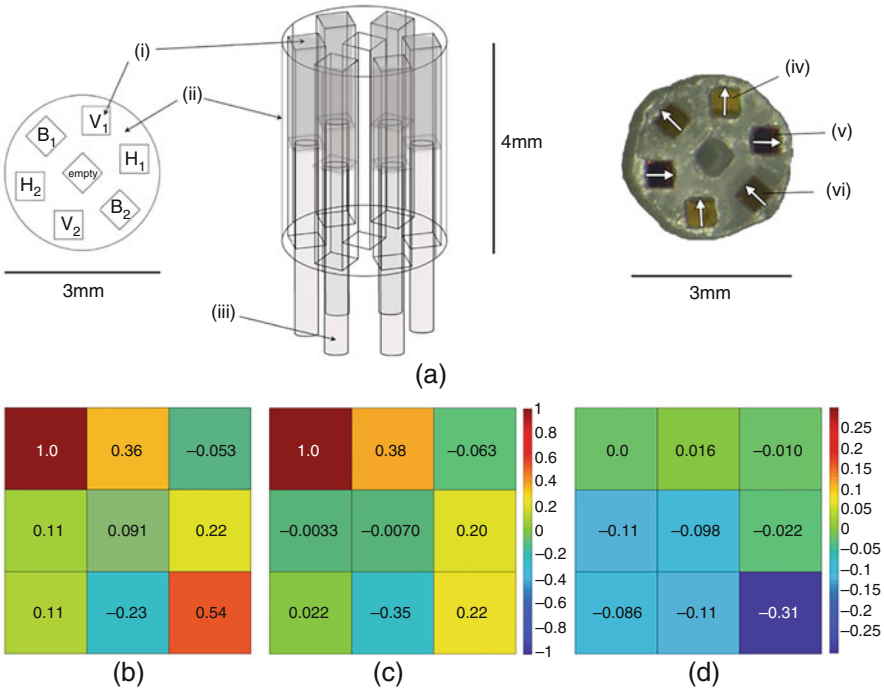


Fig. 10 3 × 3 Mueller polarimetric fibre optic probe. (**a**) Schematics and photo of the probe distal structure; heat maps of (**b**) normal, and (**c**) stretched 3 × 3 Mueller matrices for one representative chicken breast tissue, and (**d**) displays their difference. (The figure was reproduced from Forward et al. [37])

5.4 3 × 3 Mueller Polarimetric Fibre Optic Probe

A flexible optical fibre probe for 3 × 3 Mueller matrix measurements was proposed in [37] Fig. 10 to address the potential need of polarimetric information of tissue during flexible endoscopy. As introduced earlier, the inherent and varying birefringence of flexible single- and multimode optical fibres are often unpredictable when the fibre is in use, and thus make it challenging to simply use free space PSG and PSA at the proximal end of a flexible probe. Therefore, it is necessary to miniaturize the PSG and PSA to fit the distal end.

The proposed probe uses six multimode fibres for light delivery and collection. A 500 um × 500 um square-shaped linear polarizing film prepared with an automatic dicing saw to adequately cover the entire fibre cross section was fitted to the distal end of each individual fibre of the probe. Among the six polarisers, two oriented horizontally, two were vertical and the rest two oriented at −45°. The sample would then be illuminated sequentially by the horizontal, vertical and −45° polarizing fibre, respectively, and the linear output images (0°, −45°, 90°) were recorded

simultaneously. With nine measurements, the 3×3 Mueller matrix of the sample was obtained.

It is noted that none of the illumination and detection fibres is co-axial in this design. To illuminate and collect light from the same tissue volume as much as possible, the probe's geometry requires additional consideration. The probe cannot work properly if it is not positioned far enough from the tissue surface. Considering that the beam intensity profile from a multimode fire is not homogeneous, the non-co-axial issue result in overestimation of the optical anisotropy of the sample when the tissue volume sampled by the probe is also not homogeneous.

A calibration method was also proposed for this probe [37]. The method assumes nine gain coefficients obtained by fitting data for an additional rotating linear polariser to correct radiometric measurements. This method assumes that the transformation matrix between the actual and the nominal PSG/PSA matrix is diagonal, which is normally not the case. The eigenvalue calibration method for 3×3 Mueller polarimeters [38] which is an extension of the widely used eigenvalue calibration method for complete Mueller polarimeters can be used instead. This method does not require to model the PSG/PSA or precisely know the properties and orientations of the calibration samples, which makes it easy, quick, and convenient to implement.

5.5 Polarization-Resolved Coherent Fibre Bundle

As introduced in the previous section, mode coupling in an individual multimode fibre scrambles the polarization input from the distal end of the fibre. Coherent fibre bundles consisting of individual multimode (or more precisely few mode) fibres are conventionally used for amplitude-only imaging.

Advances in the transmission matrix approach in optics have enabled the study and control of light propagation in disordered media, like coherent fibre bundles. An interesting polarization-resolved holographic system based around a coherent fibre bundle (shown in Fig. 11) has been demonstrated in [39]. This system was first used to characterize the full transmission matrix of the fibre bundle, and therefore recover the input polarization information in addition to the phase from the distal end of the bundle. The system was then demonstrated for resolving diattenuation and retardance, via imaging of test samples with customized polarization properties. Since the approach is based on holography, this polarization system is effectively a Jones matrix imaging polarimeter without resolving depolarization of a target rather than a Mueller matrix imaging polarimeter. As a proof-of-concept study, the approach only works for transmission geometry and requires many optical components including a spatial light modulation at the distal end of the system that are difficult to miniaturize. Frequent bending of the fibre bundle which is required for flexible endoscopy may also bring challenges.

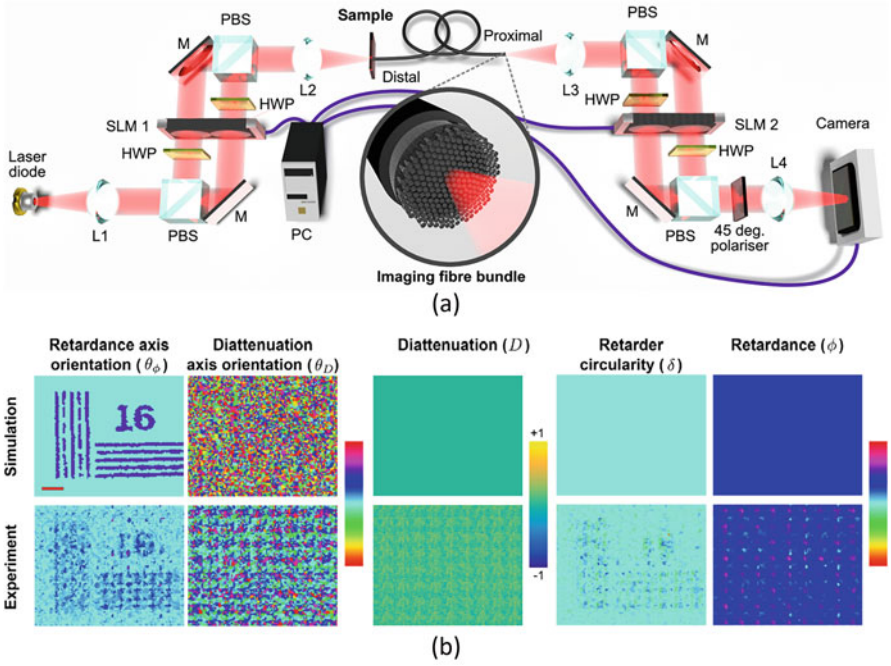


Fig. 11 Polarization-resolved coherent fibre bundle. (**a**) Schematic of the setup for full transmission matrix characterization of the coherent fibre bundle and for recovery of polarization images of the sample (spatial light modulator: SLM, polarizing beam splitter: PBS, half wave plate: HWP, lens: L, mirror: M). (**b**) Polarization-resolved properties of a birefringent USAF test target with patterns encoded in the retardance optic axis orientation. (The figure was reproduced from Gordon et al. [39])

6 Summary

There are many challenges when adapting different types of endoscope for polarimetry, and each of the technologies that were presented in this chapter represent a certain compromise. Simplicity usually comes at the cost of a reduced polarimetric data sampling, whereas full Mueller polarimetry generally reduces the spatial resolution or lengthens the acquisition time. Even illuminating the tissue with a pure polarization state and proximally detecting the scattered polarization are not straightforward given the optics that are typically found in endoscopes, and especially when considering the regulatory approval that is required to apply the endoscopes in surgery. In this chapter, we have reviewed various examples of how different groups have approached this problem. It is notable that there is a lack of in vivo human studies or sustained collaborations with industry – these two, to a certain extent, go hand in hand. Industry requires a clear exemplar application where clinical and commercial value can be realized, although it would be difficult to establish such an application without a more ergonomic and clinically compatible

endoscope system available to clinical teams of investigators. The success of non-endoscopic human applications of polarimetry such as in skin, oral, or cervical imaging can help to bridge this gap.

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Mueller Polarimetry of Brain Tissues



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1 Introduction

The Human brain is made of billions of neurons with trillions of inter-connections whose function is to receive and send nerve impulses, and different types of cells that support and protect neurons to assure their reliable and correct functionality.

The cortex (or brain gray matter) is a 2–4 mm thick outermost layer of the brain that contains cell bodies of neurons. Similar to other cells brain neurons have a nucleus, cytoplasm, and a cell membrane, but they also have axons (see Fig. 1) that conduct electrical impulses (action potential) between different neuron cell bodies located in the brain gray matter or the spinal cord.

Large number of axons are linked together, thus, creating cable-like structures that are called fiber tracts. The latter may consist of millions of axons and form the white matter of brain. The individual brain fiber tracts are responsible for different functional activities (e.g., voluntary movements, speech, memory, vision, etc.).

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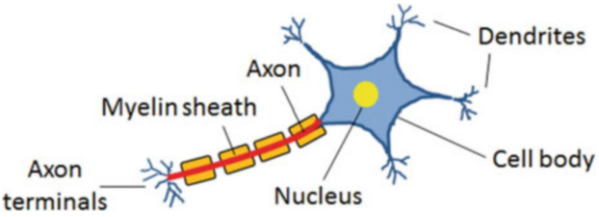


Fig. 1 Schematics of a neuron. Inter-cellular connections (or dendrites) can be considered as the neuron receptors. The axon insulated by a myelin sheath is a part of neuron that carry electrical signal and that can be quite long

The studies of structural and functional brain properties have a long history, make use of different techniques, and cover wide range starting from the fundamental studies and going to real-life clinical applications.

A non-exhaustive list of such techniques includes, the atomic force microscopy (AFM) for the detection of live neuron mechanical properties that play an important role in the neuronal growth and interactions with the extracellular matrix or other neurons [1], in vivo magnetic resonance imaging (MRI) for the diagnosis and intra-operative guidance during neurosurgery [2, 3], ultrasound brain imaging that not only provides the data on brain structure and functionality but also holds promise on therapeutic applications [4]. Scanning tomographic techniques, such as X-ray computer tomography (CT), positron emission tomography (PET) allow visualizing the entire volume of the brain with resolution down to several hundreds of microns. They show contrasts in density and allow imaging of traumatic brain injuries [5], blood flow [6], glucose metabolism activity [7], etc.

Different optical modalities have been explored for the investigation of the brain matter organization including its architecture at different spatial scale as well as brain connectivity in order to obtain diagnostic and/or functional information. Noninvasive near-infrared spectroscopy of live brain was shown to be sensitive to systemic physiological changes related to cardiac output, cerebral blood flow, and intracranial pressure [8]. The rapidly growing field of neurophotonics suggests using fiber optic-based neuroendoscopes for brain functional diagnostics, optical stimulation of single neurons, and engineering of brain cells at the molecular level intended to assist the diagnosis and therapy of neurodegenerative diseases [9, 10]. A wide range of optical techniques is used for in vivo brain imaging in animals and humans, including multi-spectral optical imaging, cortex speckle-flow imaging, two-photon microscopy, and noninvasive near-infrared imaging of living human brain [11, 12].

Polarized light imaging (PLI) technique based on the measurements of birefringence of thin (few tens of microns) unstained histological sections of formalin-fixed brain with polarized light microscopy was used for creating a brain connectome [13–17]. This technique combined with a voxel-based numerical analysis of acquired data provides both local scalar retardance values and spatial orientations of brain fibers including crossing, bending, and inclination of fiber bundles [18, 19]. The visualization of neuron circuitry helps to understand brain functional properties related to its structural organization.

The diattenuation imaging technique was also tested on thin unstained sections of formalin-fixed brains in transmission geometry. It was demonstrated that measurements of polarization-dependent attenuation may provide relevant information about the regions with out-of-plane and in-plane orientation of fiber tracts [20].

A polarization-sensitive optical coherence microscope (PS-OCM) with an axial and transverse resolution of few microns was developed and successfully used for the investigation of brain fiber organization in coronal section of formalin-fixed mouse brain [21]. PS-OCM operates in reflection geometry and detects the neuronal fibers and their in-plane orientation.

Further on in this chapter we will focus on the multi-spectral wide-field imaging Mueller matrix polarimetry [22] and discuss its application for brain studies.

2 Optical Properties of Brain Tissue

Both spectral and imaging optical techniques are used for the study of brain tissue optical properties, and brain functional characterization [23–25]. The majority of the tissue in the brain is both scattering and absorbing. That is why medium and many optical studies of brain tissue are focused on the experimental measurements and numerical modeling of brain tissue diffuse reflectance and transmittance in order to define its scattering (μ_s) and absorption (μ_a) coefficients [26] as well as the penetration depth of probing signal in the visible or near-infrared wavelength range for both diagnostic and therapeutic applications.

Pathological alterations of brain tissue change its micro-structure, vascularization, and water content that, in turn, affect the scattering and absorption properties. Hence, it suggests measuring scattering and absorption coefficients and using them as the optical markers for brain tissue diagnosis [27–29].

Apart from scattering and absorption the white matter of healthy brain demonstrates the anisotropy of refractive index, because it is composed of the bundles of densely packed axons (or brain fiber tracts) originating from the bodies of neuron cells in brain gray matter (or cortex) and connecting them with other neurons or spinal cord. Each axon is covered by a myelin sheath that represents a hollow rod with radially arranged protein and lipid components [30]. Besides the myelinated axons there are neuroglia type cells in brain white matter, but there is no cell bodies of neurons.

The lipid-rich substance of a myelin sheath has higher refractive index compared to that of axon and surrounding glia. The densely packed and highly aligned structure of myelin results in strong anisotropy of the refractive index (optical birefringence), as well as anisotropy of scattering (optical diattenuation) that both affect the polarization state of light propagating through brain tissue. It was demonstrated that at macroscopic scale (i.e., when the spatial resolution of a polarimeter is not

sufficient to resolve a single fiber) the fiber tract of healthy brain white matter represents a medium with negative birefringence and the orientation of optical axis along the direction of fiber tract bundles [31]. The so-called form birefringence of brain white matter introduces a linear phase retardance and anisotropy of scattering on the elongated brain fibers results in a linear diattenuation of a propagating polarized electromagnetic wave.

3 Mueller Polarimetry

An overview of the essential features of the complete Mueller matrix polarimetry is provided below within the framework of Stokes–Mueller formalism. We briefly recall the basic principles of Mueller polarimetry, list the optical components used for the polarization modulation, and discuss the optimal design and calibration of the complete Mueller matrix polarimetric system.

3.1 General Principles of Mueller Polarimetry

To determine the optical properties of a sample one needs to measure polarization changes occurring in the probing beam light after its interaction with a sample. The schematic of operation of a Mueller polarimetric system in transmission geometry is shown in Fig. 2 but the same description holds also for the reflection configuration. The polarization state of the light beam coming from a source is modulated by Polarization State Generator (PSG). The output state is analyzed by Polarization State Analyzer (PSA) followed by a detector (a CCD camera for imaging polarimeter), according to the general schema outlined in Fig. 2. The incident polarization states generated by the PSG and those set on the PSA vary during a given experiment.

Different types of PSG and PSA may be used, being combined in optical systems with different design features. They provide various degree of completeness of polarization state generation and analysis, accuracy, spectral range, etc. Depending on the application the polarimetric characterization may be performed in either spectral or imaging mode.

The Mueller polarimeter we have used in our studies is based on a sequential modulation and analysis of light beam polarization by using the discrete polarization basis states [33]. In general, the PSG sequentially generates N_1 different



Fig. 2 General principle of Mueller polarimetric measurement technique

polarization states, represented by different Stokes vectors $\mathbf{S}_k$ $(k = 1, \ldots, N_1)$ of the light beam incident on a sample. $\mathbf{M}$ is the Mueller matrix that describes the polarimetric properties of a sample, and the product $\mathbf{MS}_k$ is the Stokes vector that describes the polarization state of the light beam after its interaction with a sample. The light beam passes through the PSA that is characterized by the Stokes vector $\mathbf{S}'_m$, $(m = 1, \ldots, N_2)$. It means that if we will send the unpolarized light beam through the PSA in the reverse direction, the polarization state of the transmitted light beam will be described by the Stokes vector $\mathbf{S}_m$. Finally, the light beam intensity is registered on a detector and is described by the scalar product of two Stokes vectors $\mathbf{S}'_m$ and $\mathbf{MS}_k$.

We construct the modulation matrix $\mathbf{W}(4 \times N_1)$ by using the Stokes vectors $\mathbf{S}_k$ as its columns. Depending on the value of N_1 the PSG is called incomplete $(N_1 < 4)$, complete $(N_1 = 4$ and the Stokes vectors are linearly independent, hence, they form a complete basis in the polarization space and matrix $\mathbf{W}$ is invertible), and redundant $(N_1 > 4$ and the $\mathbf{S}_k$ vectors form an overcomplete basis in the polarization space, and, consequently, the modulation matrix $\mathbf{W}(4 \times N_1)$ has a pseudo-inverse).

The PSA operates as a polarization filter for the light beam emerging from a sample. Similarly to the construction of the modulation matrix $\mathbf{W}(4 \times N_1)$, we can build the analysis matrix $\mathbf{A}(N_2 \times 4)$ of the PSA. We use the Stokes vectors $\mathbf{S}'_m$ of the polarization filter PSA to define the rows of the analysis matrix $\mathbf{A}$. The PSA may also be incomplete, complete, or redundant depending on the dimension N_2 of the polarization space spanned by the row vectors of the analysis matrix $\mathbf{A}$. Generating N_1 polarization states of the incident light by PSG and projecting them on the N_2 states of the polarization filter PSA we obtain the raw intensity signals matrix that can be written as

$$\mathbf{B} = \mathbf{AMW} \tag{1}$$

To obtain the Mueller matrix $\mathbf{M}$ of a sample from the raw intensity matrix $\mathbf{B}$, one needs to know the matrices $\mathbf{A}$ and $\mathbf{W}$, i.e., the instrument has to be calibrated. Assuming that the instrument is a complete Mueller polarimeter (i.e., elements PSG and the PSA are complete or redundant), the Mueller matrix $\mathbf{M}$ of a sample can be calculated by inverting (or pseudo inverting in case of redundant PSG / PSA) the matrices $\mathbf{A}$ and $\mathbf{W}$

$$\mathbf{M} = \mathbf{A}^{-1}\mathbf{B}\mathbf{W}^{-1} \tag{2}$$

The polarization modulation and analysis can be implemented by using different optical components, e.g., rotating retardation plates (e.g., wave plates [34]), photoelastic modulators (PEMs) [35], liquid crystal variable retarders (LCVR) [36, 37], or ferroelectric liquid crystals (FLC) [38, 39]. All approaches have some pros and cons:

- Mechanical rotation of a wave plate requires more time comparing to electrically driven switching of LCVR or FLC state;

- PEM-based polarimeter is challenging to miniaturize due to the large dimensions of PEM;
- LCVR-based polarimeters are well suited for the imaging applications, but the operation performance of LCVR is affected by the variations of ambient temperature, and their response time (few milliseconds) may be a limiting factor for the biomedical applications;
- FLC-based polarimeters do not require any mechanical moving parts and allow the fast polarization modulation (few microseconds) that can be beneficial for the analysis of biological samples.

The liquid crystal-based optical components are well adapted for polarimetric imaging, since they display broad angular acceptance, can be made with wide clear apertures, and show low aberrations, but they are restricted to a visible wavelength range only [40].

3.2 Optimal Design and Calibration of Mueller Polarimeter

There is a degree of freedom on the selection of the appropriate PSG and PSA polarization states basis. The optimal design of a complete Mueller polarimeter targets the minimization of noise in the elements of Mueller matrix **M** in the presence of additive noise in the elements of raw measurement matrix **B** (see Eq. 2). The Stokes vectors that represent these polarization states should form the matrices **A** and **W** that will minimize an error propagation from **B** to **M**. The optimization of the **A** and **W** matrices has to reduce the numerical errors produced by their inversion (or pseudo-inversion).

To achieve this, the polarization basis states of the PSA (and PSG) have to be as different as possible from each other. It has been shown that for a complete or redundant PSA the optimal polarization basis states should be the vertices of a Platonic solid inscribed into the Poincaré sphere [32]. For example, for $N_2 = 4$ the optimal polarization states form the vertices of a tetrahedron, $N_2 = 6$ - the vertices of an octahedron, $N_2 = 8$ - the vertices of a cube [41]. The optimization of the matrix **A** not only minimizes the noise on Stokes vector **S** but also distributes it equally between the components of vector **S** [42].

The determination of the matrices **W** and **A** is called the calibration. It is often quite difficult to make the accurate models of these two matrices because of existing inaccuracies in the characteristics of optical components of PSG / PSA, their relative alignment, interference effects, beam divergence, or Gaussian white noise.

To overcome these problems the Eigenvalue Calibration Method (ECM) developed by E. Compain [43] is used for the calibration of the complete Mueller polarimeter. This method does not require a precise and accurate optical modeling of the components of PSG and PSA and can be applied in both transmission and reflection configurations. The ECM determines the actual **W** and **A** matrices from a set of measurements with the reference samples, namely the linear polarizers

and retarders that do not need to be perfectly characterized in advance, because their optical parameters (transmission, diattenuation, retardation) are provided by the calibration procedure itself.

The results of using the custom-built multi-spectral wide-field imaging Mueller polarimetris system based on FLCs [33, 39] for polarimetric studies of brain tissue are presented in the following section.

4 Pre-clinical Studies of Brain Tissue with Polarized Light

Extensive studies of thin unstained sections of a brain with 3D-PLI modality in transmission geometry [13–17] have provided a wealth of data on density of brain fibers and their in-plane and out-of-plane orientation that make significant contribution to building a brain connectome. However, if we are looking for clinical (or pre-clinical) applications of visible wavelength range polarized light for brain tissue imaging, the reflection configuration is the only relevant one. We cannot see through a whole brain in transmission with visible light because of its shallow penetration depth.

Polarized light microscopy of thick formalin-fixed brain specimen allows one to determine the local orientation and alignment of nerve fiber bundles in backscattering configuration by measuring the degree of linear depolarization of the probing beam at different azimuth angles [44]. PS-OCT technique operates also in reflection geometry and can be used for the real-life applications, e.g., live open brain studies. But even the full field PS OCT is a microscopy technique [45] that requires image scanning and/or stitching, thus making it less suitable for wide-field (few cm by few cm) imaging of brain tissue in clinical settings.

That is why we use the multi-spectral wide-field imaging Mueller polarimetry for *ex vivo* brain tissue studies and discuss it potential for the clinical applications.

4.1 Mueller Polarimetry for the Assessment of Alzheimer Disease Progression in Mice Model

Diagnosis and treatment of neurodegenerative diseases represent the important societal and public health challenges because of worldwide aging of population [46, 47]. The Alzheimer's disease (AD) represents the most common form of dementia with estimated contribution of up to 60–70% of all cases [48].

The most widely used types of biomarkers of AD disease include the changes in brain parts (e.g., shrinkage of hippocampus, loss of neuron cells in cortex), and the accretion of certain proteins including abnormal extracellular deposits of amyloid-beta Aβ protein and intracellular neurofibrillary tangles of tau protein [49]. The spatial resolution of the imaging techniques used for clinical diagnosis of AD or

assessment of the impact of experimental drugs treatment (computer tomography scan, magnetic resonance imaging, positron emission tomography) is limited to a few hundreds of microns [50] that affects their accuracy, especially for the detection of abnormal accumulation and deposits of proteins. Hence, a definite diagnosis of AD and/or its staging are possible *post mortem* only using a histological analysis of brain tissue specimens.

The standard protocol of histological analysis is time-consuming, (includes many steps), operator-dependent (relies on pathologist's skills) and suffers from lack of quantitative metrics (qualitative information only). Thus, for improving the diagnosis accuracy there is a need of a new fast label-free quantitative technique.

Diffraction phase microscopy studies demonstrate that tissue of AD brain exhibits strong fluctuations of refractive index combined with an increase of scattering anisotropy compared to the normal brain tissue [51].

Using multi-spectral wide-field imaging Mueller matrix (MM) polarimetry for the quantitative assessment of the progression of Alzheimer disease in mice model was reported in [52].

This technique has already demonstrated its capabilities for the diagnostic characterization of various types of tissues by exploring the alterations of tissue polarization and depolarization properties within the pathological zones of myocardium [53], uterine cervix [39, 54–56], colon and rectum [57, 58].

Wide-field imaging MM polarimetric system [33] operating in a visible wavelength range in backscattering configuration was used for the measurements of 21 formalin-fixed paraffin-embedded brains excised from the sacrificed APP-transgenic mice (APPPS-21, APPtg) [59] at different stages of AD development: 50, 75, 100, 125, 150, 175, and 200 days (three specimens per each AD stage). The photo and the white light microscopy images of thin histological sections of AD mice brains are shown in Fig. 3. To reduce the contribution of specular reflection related to the uneven specimen surface topography a layer of refractive index matching oil was spread on the imaged surface of a sample. The Mueller matrix images of the FFPE AD mouse brain (day 75) taken at 450 nm are shown in Fig. 4.

All Mueller matrix images were normalized by M_{00} values pixel-wise. Consequently, their values varied within the interval $[-1, 1]$. The Mueller matrices were post-processed with the polar Lu-Chipman decomposition algorithm [60] that represents any physically realizable Mueller matrix [61] as a product of the Mueller matrices of three basic optical elements, namely diattenuator, retarder, and depolarizer.

The lipid content of a myelin sheath is soluble in paraffin, hence, the most of myelin was dissolved in the FFPE blocks of mice brains [62]. That is why the Mueller matrix images show almost no retardance and diattenuation (except of the edge zones of paraffin block) and demonstrate the depolarization properties only that are related to light scattering within FFPE mice brain tissue.

The statistical analysis of the distribution of total depolarization values $\Delta = 1 - (|M_{11}|+|M_{22}|+|M_{33}|)/3$ within the regions of interest (ROI) in images of FFPE blocks that contain mice brains measured at different time points of AD

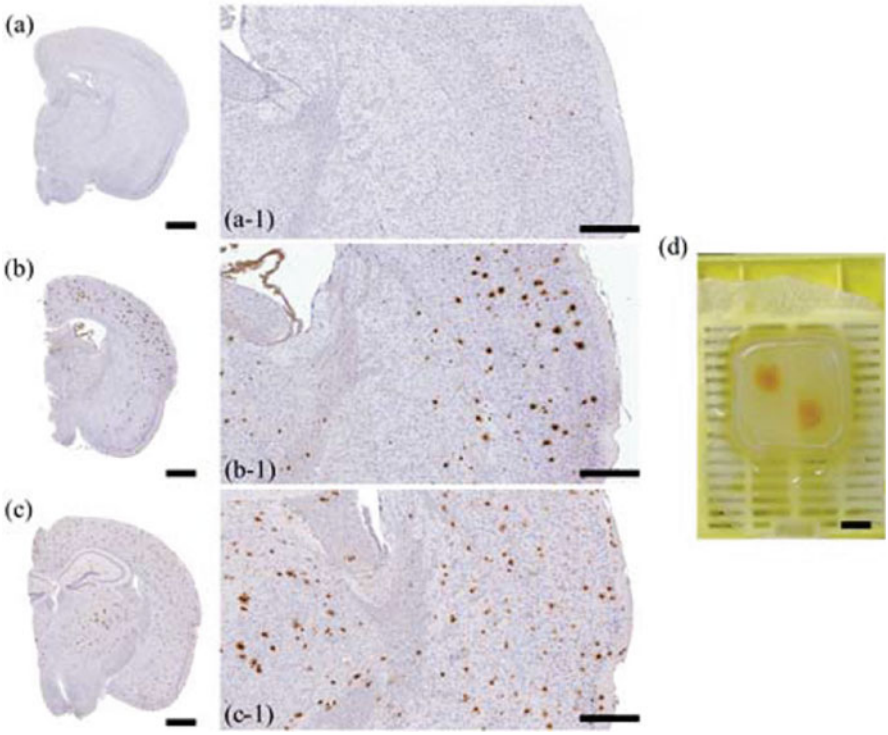


Fig. 3 AD mouse brain specimens. Histology images of thin sections: (a, a-1) 50-day-old mouse, (b, b-1) a 125-day-old mouse, and (c, c-1) a 200-day-old mouse. The panels (a-1, b-1, c-1) show the enlarged central zone in the panels (**a, b, c**), respectively; (**d**) a photo of a FFPE block. Scale bars: (**a, b, c**) 1 mm, (a-1, b-1, c-1) 500 μm, (**d**) 5 mm. Reprinted with permission from [52]. ©The Optical Society

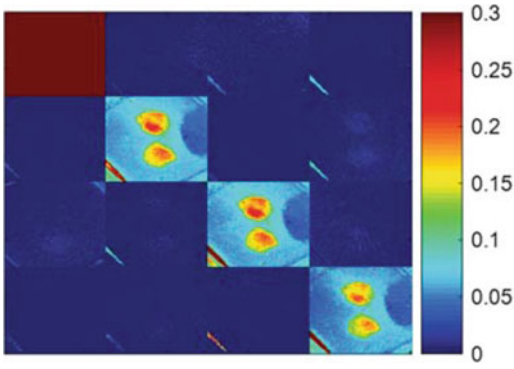


Fig. 4 4 × 4 Mueller matrix images of a FFPE mouse brain sample (75-days old) measured at 450 nm. Field of view: 12 × 14.5 mm (251 × 301 pixels). Reprinted with permission from [52]. ©The Optical Society

progression was applied for testing the capabilities of MM polarimetry to assess the advancement of AD-related structural alterations and provide the quantitative scoring to support the diagnosis of a pathologist.

The first four statistical moments of the distributions of total depolarization that include the mean value Z_1, the standard deviation Z_2, the skewness Z_3, and the kurtosis Z_4 were calculated for 21 brain samples according to [63]

$$Z_1 = \frac{1}{N}\sum_{i=1}^{N}\Delta_i \tag{3}$$

$$Z_2 = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(\Delta_i - Z_1)^2} \tag{4}$$

$$Z_3 = \frac{1}{Z_2^3}\frac{1}{N}\sum_{i=1}^{N}(\Delta_i - Z_1)^3 \tag{5}$$

$$Z_4 = \frac{1}{Z_2^4}\frac{1}{N}\sum_{i=1}^{N}(\Delta_i - Z_1)^4 \tag{6}$$

where N is the number of pixels in the ROI of the corresponding image. The calculated statistical moments of the distributions of total depolarization were averaged over each mice age group. The corresponding mean value and standard deviation for each time point of AD evolution are shown in Fig. 5.

The first two statistical moments of the total depolarization distribution are not sensitive to the progression of AD, as the results of data linear regression are almost constant with time. However, the absolute values of the statistical moments of higher orders grow exponentially with time of AD progression.

It is known that scattering of light within biological tissue is the main source of depolarization of the probing polarized light beam. Because of the accumulation of Aβ-plaques in AD brain tissue its scattering coefficient μ_s and the effective size of scattering particles increase compared to healthy brain tissue. Both factors contribute to the increase of scattering anisotropy coefficient g [64] in AD brain. Despite the fact that both μ_s and g parameters increase with AD progression [51] the reduced scattering coefficient of bulk brain tissue $\mu_s' = \mu_s(1-g)$ remains almost unchanged because the variations of both parameters compensate each other to certain extent.

The negative values of the 3rd statistical moment of the distribution of total depolarization demonstrate that it becomes more skewed towards low values with the AD progression. Simultaneously the exponential growth of kurtosis indicates that the distribution of total depolarization becomes "heavy-tailed," i.e., the contribution of outliers grows exponentially with AD time evolution.

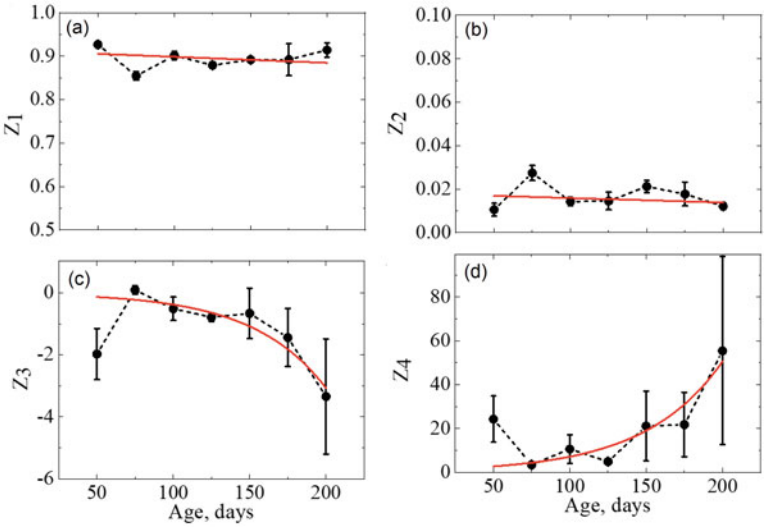


Fig. 5 Statistical moments of the 1st–4th order of the spatial distributions of total depolarization of FFPE mouse brain at different stages of AD development (50 to 200 days) measured at 450 nm: (**a**) mean values, (**b**) standard deviation, (**c**) skewness, (**d**) kurtosis. Results of the fit with (**a, b**) linear and (**c, d**), exponential functions are shown by red solid lines. Reprinted with permission from [52]. ©The Optical Society

The exponential increase of high-order statistical moments is related to the structural modifications of AD brain tissue that enhance the effective scattering cross-section due to the increase of size of scattering particles [51] and affect the angular dependence of a scattering phase function. Light scattering becomes more anisotropic compared to Rayleigh scattering on particles with a size smaller than the wavelength of probing light [64, 65]. Furthermore, the increase in spatial variations of the refractive index in AD brain tissue contributes to the spatial variations of the scattering phase function. Enhanced angular anisotropy and irregularity of the phase scattering function increase the variations of the photon trajectories in terms of its path length and directionality. The number of scattering events varies significantly along the different photon trajectories within AD brain tissue. Consequently, the variations of the degree of polarization of the detected backscattered photons increase with the AD evolution [65].

These phenomena explain the increase in both skewness and kurtosis of the distributions of total depolarization of FFPE AD brain tissue with AD progression and suggest using these parameters as the markers of β-amyloidosis severity to support the diagnosis of a pathologist.

4.2 *Visualization of Fiber Tracts of Healthy Brain White Matter with Imaging Mueller Polarimetry*

Surgery remains the crucial treatment step for most patients with brain tumors [66–68]. A clear identification of the border between a tumor and surrounding brain is essential in order to excise the tumor and preserve neurological functions. However, tumor tissue is often difficult to differentiate from infiltrated brain tissue during surgery despite the advanced intra-operative modalities such as ultrasound [69], magnetic resonance imaging [70] and 5-aminolevulinic acid (5-ALA) induced fluorescence imaging [71].

In terms of optical anisotropy the healthy brain white matter differs strongly from brain tumor that erases the optical anisotropy of brain white matter because of the random and isotropic growth of tumor cells. However, this difference in structural complexity (or entropy) is currently not detectable during surgery with a commercial operating white light microscope.

Instead of identifying brain tumor tissue one may increase the surgical site contrast by visualizing the fiber tracts of the healthy brain white matter. The visualization of fiber tracts will improve delineation of tumors from surrounding brain, due to the lack of organized fiber tracts within brain tumor tissue. In addition, the visualization of fiber tracts in real time during surgery will allow a surgeon to orient himself, and to spare essential fiber tracts. It will increase the chance of removing the lesion more completely and decrease the risks of harming surrounding brain, thus, making surgery more efficient and safe.

4.2.1 Thick Sections of Fixed Human Brain

The capability of polarized light imaging to visualize the fiber tracts of healthy brain white matter in backscattering geometry was explored in [33]. A wide-field (several cm^2) liquid crystal-based imaging Mueller polarimetric system described in Sec. 4.1 was used for the measurements of thick sections of formalin-fixed human brain from the autopsy of an anonymous donor. Two adjacent thick ($\sim$1 cm) sections of a hemisphere of formalin-fixed human brain in a coronal plane were prepared for the polarimetric measurements and histology analysis, respectively (see Fig. 6).

Thin whole-mount silver-stained section of brain tissue was analyzed with a white light microscope in transmission configuration. The polarimetric images were taken on the adjacent brain section with the wide-field imaging Mueller polarimetric system in reflection configuration and compared to the results of histology analysis.

Brain specimen was placed flat in a glass Petri dish of 14.5 cm in diameter. A sufficient amount of distilled water was poured to cover the tissue surface for partial index matching and removing the artifacts related to uneven surface topography. Mueller matrix measurements of a formalin-fixed human brain thick section were performed at 550 nm. The maps of basic polarimetric properties of a brain specimen, namely retardance, diattenuation, and total depolarization were

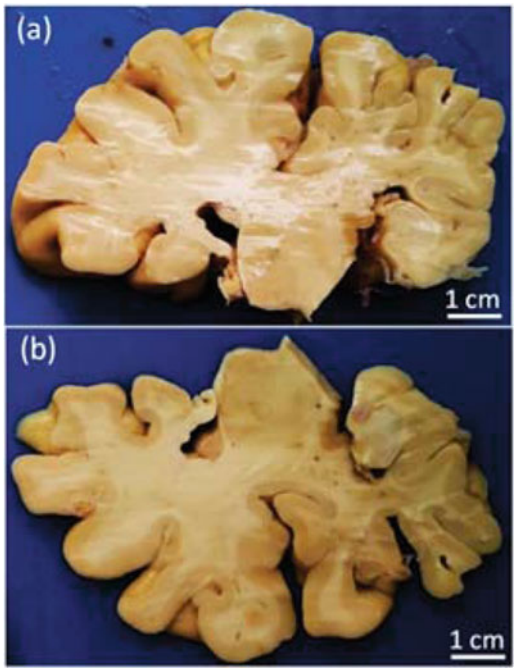


Fig. 6 Photo of two adjacent sections of a fixed human brain (one half) in the coronal plane, dimensions 9 × 6 × 1; (**a**) brain section imaged with the Mueller polarimetry, (**b**) brain section used for the histology analysis

obtained by applying pixel-wise Lu-Chipman polar decomposition of measured Mueller matrix. The corresponding images of intensity and polarimetric parameters are shown in Fig. 7.

Neither circular birefringence nor linear or circular dichroism were detected in the experiments with a formalin-fixed brain specimen. The cortex of a formalin-fixed human brain specimen is less depolarizing compared to brain white matter. The clear border between these two zones of brain tissue is present in the image of total depolarization. As expected, white matter of healthy brain tissue demonstrates measurable linear retardance that varies up to 20°. The map of the azimuth of the optical axis shows the orientation of optical axis of linear birefringent medium. To verify the correlation of the azimuth of the optical axis measured in reflection with the direction of brain fiber tracts four zones (70 × 70 pixels) corresponding to U-fibers were selected in the map of azimuth (see Fig. 8a) and the circular histograms were calculated for each zone (Fig. 8, middle row). The enlarged images of corresponding zones of the same U-fibers in the histology image of the whole-mount silver-stained brain tissue section (Fig. 8b) are presented in Fig. 8, bottom row. It is worth to note that silver-staining of a thin histological section is considered to be a gold standard technique for the ex vivo visualization of brain fiber tracts.

The statistics for the circular histograms for the zones 1–4 of U-fibers tracts is shown in Table 1. The relatively small values of standard deviation confirm that all four distributions are peaked and quite narrow. There is an obvious correlation between the orientation of brain fiber tracts seen in the histology images with the

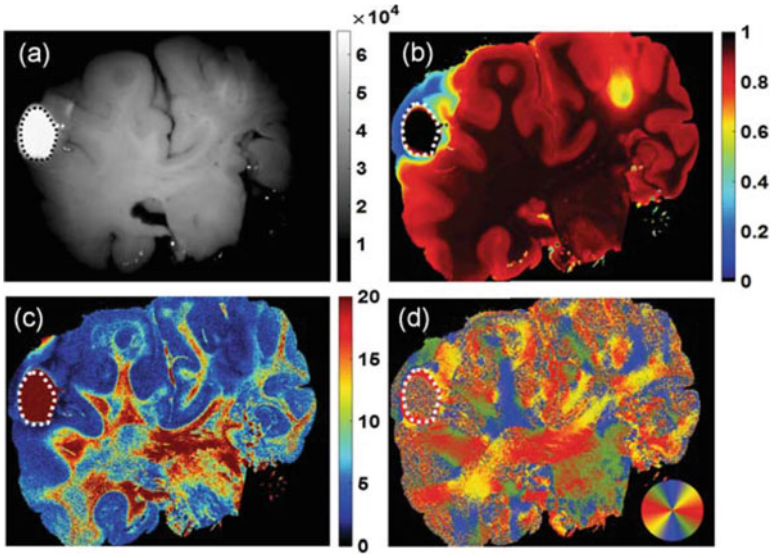


Fig. 7 Images of a thick coronal section of fixed human brain immersed in water: (**a**) total intensity, (**b**) total depolarization, (**c**) linear retardance (degrees), and (**d**) azimuth of the optical axis. The dashed line delineates the area of specular reflection on air-water interface. Measurement wavelength is 550 nm. Adapted from [33]

preferential orientation of the fibers tracts calculated from the circular histograms of the azimuth for the corresponding zones 1–4.

Figure 9 shows the enlarged zone of U-fibers in the image of the azimuth of the optical axis that is correlated with the histology analysis. Brain cortex tissue demonstrates random orientation of the azimuth that is explained by the absence of ordered fibers within gray matter of brain at the scale of spatial resolution ($\sim$100 μm) of the wide-field imaging Mueller polarimetric system.

Using polar coordinate system the in-plane normalized retardance vector $\mathbf{R}$ can be written as $\mathbf{R} = (R/R_{max}, \varphi)$, where R_{max} is the maximal value of the scalar linear retardance over its image and φ is the azimuth of the optical axis. Thus, the map of vector retardance can be considered as a merged map of two polarimetric parameters, namely the normalized scalar retardance and the azimuth as shown in Fig. 10. The pixels rendered in black within brain tissue zone in the image of retardance vector $\mathbf{R}$ correspond to scalar retardance values below the threshold of 1°. Low retardance values combined with a random orientation of azimuth serve as a cut-off filter for the cortex zones in brain tissue image recorded in backscattering configuration. When white matter image zones with low scalar linear retardance values demonstrate a narrow peaked distribution of the azimuth values, it means that fiber tracts with different in-plane orientation are crossing each other. This effect was also observed in PLI of thin brain sections in transmission configuration [72].

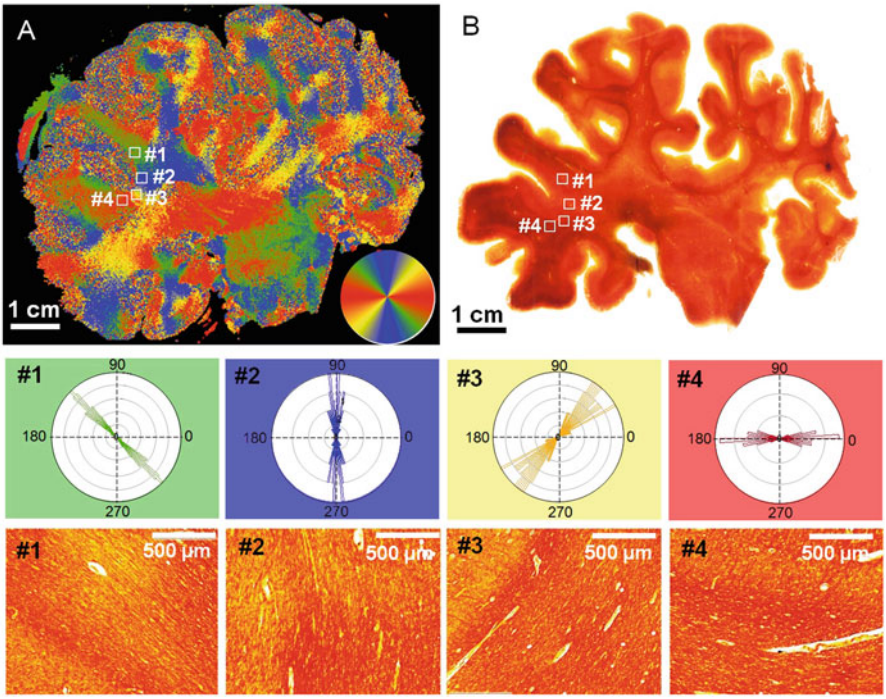


Fig. 8 (**a**) Map of the azimuth of the optical axis; (**b**) photo of a thin whole-mount brain tissue section stained with Bielschowsky silver impregnation. White boxes show the zones (#1, #2, #3, #4) of the same U-fibers in (**a**) and (**b**) with the corresponding circular histograms of the azimuth (middle row) and the enlarged histology images (bottom row). Adapted from [33]

Table 1 Mean values and standard deviation for the circular histograms of the azimuth. Reprinted with permission from [33]. ©IEEE

Zones	Mean value (°)	Standard deviation (°)
#1	136.3	5.9
#2	91.1	10.4
#3	49.1	11.8
#4	179.5	11.1

It is known that tissue fixation in formalin may change its optical properties due to linking the soluble and structural proteins of tissue [73].

4.2.2 Thick Sections of Fresh Calf Brain

To confirm the capability of the wide-field imaging Mueller polarimetry to visualize nerve fiber tracts of a fresh brain white matter the polarimetric measurements were also carried out on thick sections of a fresh animal brain in reflection geometry.

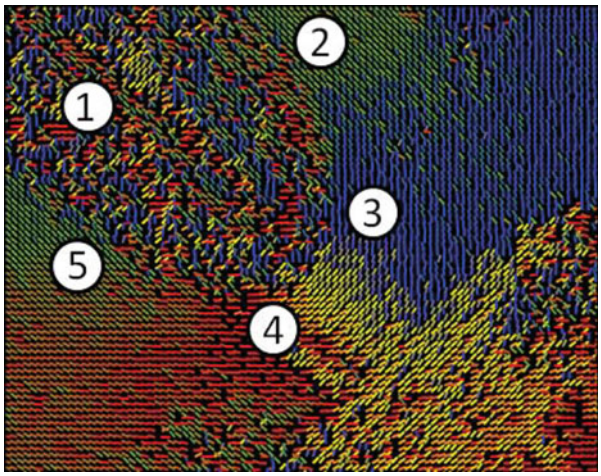


Fig. 9 Enlarged U-fibers zone in the map of the azimuth plotted in polar coordinate system with corresponding vectors of unit length: (1) zone of cortex with random azimuth orientation; (2)–(5) highlight U-fibers zone. Adapted from [33]

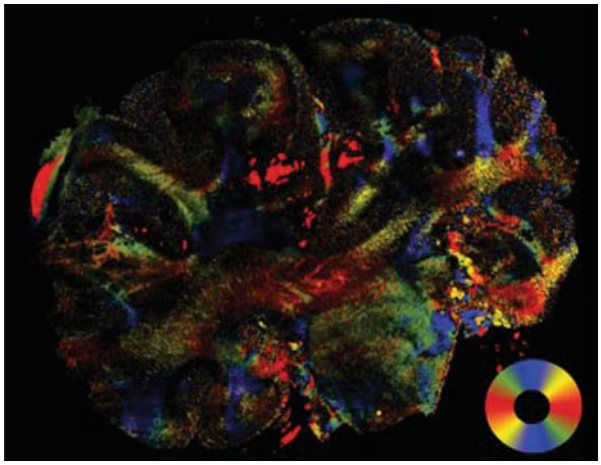


Fig. 10 Map of the normalized retardance vector **R** in polar coordinate system. Blue color corresponds to the vertical orientation of the optical axis in the image plane, red color corresponds to the horizontal orientation, yellow and green colors correspond to the ±45° orientations, respectively

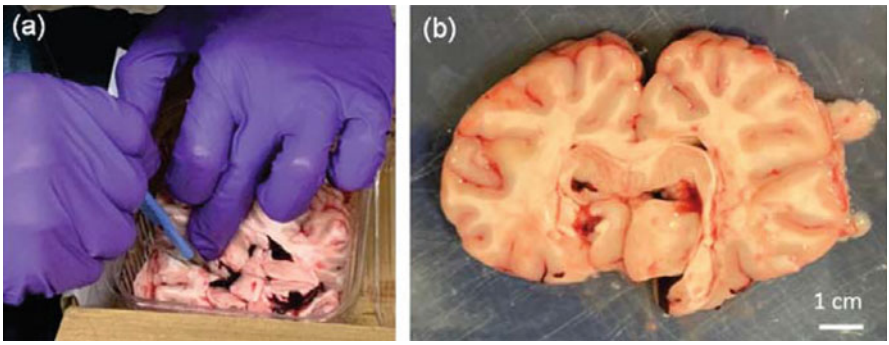


Fig. 11 Preparation of fresh calf brain section: (**a**) cut of a brain with the scalpel in a coronal plane; (**b**) 1 cm thick section of a fresh calf brain. Adapted from [33]

Polarimetric Images of Flat Sample Surface

The entire fresh calf brain was cut in a coronal plane without formalin fixation to prepare a thick tissue section (see Fig. 11). The dimensions of calf brain tissue section were approximately $9 \times 7 \times 1$ cm.

A brain section was rinsed with water, put flat in a Petri dish (no water was added) and measured immediately after the sample preparation. The recorded Mueller matrix images of a 1 cm thick fresh brain tissue section measured in reflection geometry were processed by applying Lu-Chipman polar decomposition pixel-wise.

The results of polarimetric measurements taken at 550 nm are shown in Fig. 12. The specular reflection on uneven surface of a sample results in bright zones in the gray-scale intensity image (Fig. 12a). The total depolarization values for a fresh calf brain white matter measured in air vary between 0.75 and 1 (similar to corresponding values of the total depolarization measured for the formalin-fixed human brain tissue immersed in water, see (Fig. 7b). The depolarization map of the fresh calf brain tissue is similar to that for the fixed brain tissue with clear contrast between the zones of gray and white matter.

The linear scalar retardance values of fresh brain tissue measured in air (Fig. 12c) are slightly smaller compared to the corresponding values measured for formalin-fixed brain tissue immersed in water (Fig. 7c). The azimuth map (Fig. 12d) highlights clearly the directions of fiber tracts as was seen in the azimuth maps of white matter of formalin-fixed brain tissue (Fig. 7d).

The U-fibers tracts are seen at the periphery of a fresh calf brain specimen as well as both vertical and horizontal fiber tracts in the central part of the azimuth map. It suggests that the values of azimuth of the optical axis are invariant under formalin fixation of brain tissue. The latter changes the polarimetric properties of tissue (e.g., the absolute values of both ordinary and extraordinary refractive indices), but does not change the microarchitecture of brain fiber tracts and the associated anisotropy of refractive index of brain white matter.

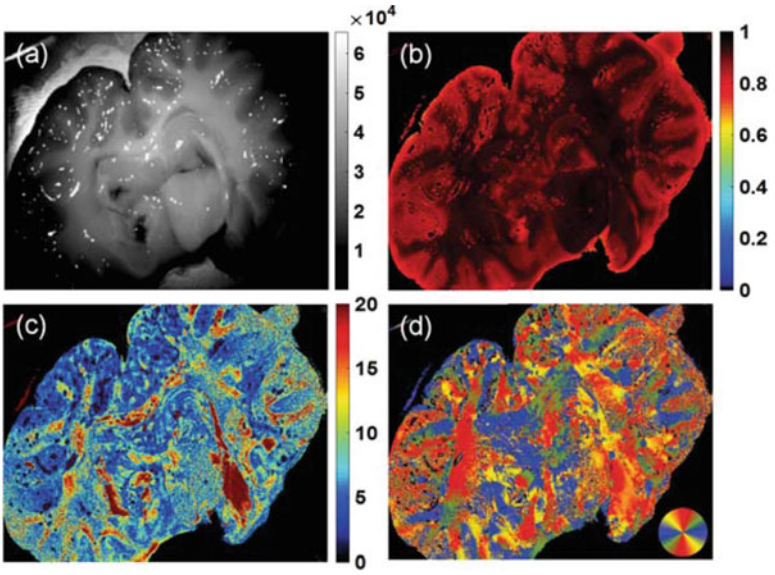


Fig. 12 Images of a 1 cm thick section of fresh cadaveric calf brain measured in air in reflection with the wide-field imaging Muller polarimeter: (**a**) gray-scale total intensity image, (**b**) depolarization, (**c**) scalar linear retardance (degrees), and (**d**) azimuth of the optical axis. Reprinted with permission from [33]. ©IEEE

The wide-field Mueller polarimetric imaging of both fixed human and fresh calf brain tissue was also performed at the wavelength of 600 and 650 nm. The values of both depolarization and linear retardance did not show pronounced spectral dependence. The image contrast for the total depolarization, linear scalar retardance, and azimuth of the optical axis were very similar to those found with the measurements taken at 550 nm.

Low spectral sensitivity of Mueller polarimetric measurements of the formalin-fixed brain specimen in a visible wavelength range is related to the absence of blood in such specimen (<1% according to the pathologist's analysis). The amount of blood in a fresh cadaveric animal brain tissue is also much smaller compared to that in live brain tissue. The absorption coefficient of hemoglobin (a primary optical absorber in blood) has local maxima at 500 and 550 nm and drops significantly within a red part of visible spectrum [74].

Polarimetric Images of Brain Specimens Under Adverse Conditions

The wide-field imaging Mueller polarimeter was also tested under adverse conditions, namely complex topography of a specimen surface and its irrigation with blood that mimic intra-operative conditions, in order to check the impact of such conditions on the capability of the imaging Mueller polarimetric system to visualize brain fiber tracts at sight at any given moment during neurosurgery [75].

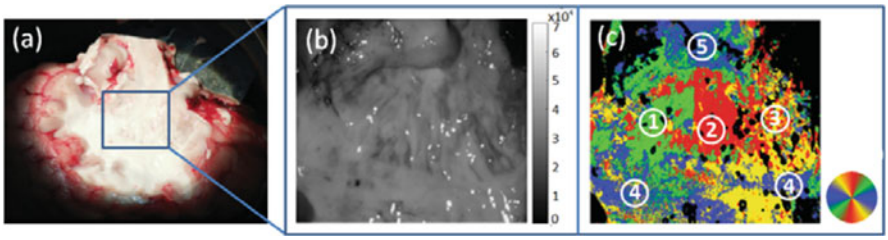


Fig. 13 Cerebral tracts of a cadaveric calf brain: (**a**) photo of a prepared cavity with brain deep white matter tracts. The imaged zone of the size of 3× 3 cm is marked by a blue box; (**b**) intensity image; (**c**) map of the azimuth of the optical axis: the parieto-spinal tract (1) is rendered in green, the centro-spinal tract (2) is rendered in red, the fronto-spinal tract (3) is rendered in yellow, the longitudinal tracts (4) and (5) are rendered in blue. Measurements were performed at 550 nm. Reprinted with permission from [75]. ©The Optical Society

Both temporal lobe and large part of the frontal lobe of a cadaveric calf brain hemisphere were removed, thus, resulting in a 2 cm deep cavity in a calf brain tissue (Fig. 13a).

The gray-scale intensity image shown in Fig. 13b does not indicate the directions of the brain deep fiber tracts. The bright spots seen in the intensity image are due to specular reflection because of complex surface topography of the specimen.

We applied the depolarization threshold of $\Delta = 0.85$ to remove less depolarizing zones of both cortex and specular reflection in the image of the azimuth of the optical axis. Contrary to the intensity image the map of the azimuth highlights both the longitudinal and the cerebro-spinal (or spino-cerebral) tracts (Fig. 13c).

Presence of blood within surgical cavity is an inherent factor of open brain surgery. Hence, one needs to test the impact of blood on the visualization of brain white matter fiber tracts with imaging Mueller polarimetric modality.

We prepared an entire fresh calf brain for the polarimetric measurements by placing it in a Petri dish (diameter of 14.5 cm) and exposing the brain intra-hemispheric corpus callosum that represents a wide thick fiber tract consisting of flat bundle of fibers, connecting both brain hemispheres (see Fig. 14a, d). Another piece of brain tissue was placed on top of the corpus callosum to create a 2 mm deep cavity that was filled with undiluted blood (Fig. 14a, d). The polarimetric images of the corpus callosum covered with blood were taken at both 550 nm and 650 nm. The zone of blood is clearly seen in the intensity image at 550 nm (Fig. 14b), whereas this zone is almost undetectable in the intensity image taken at 650 nm, (Fig. 14e). This phenomenon is related to the sharp decrease of blood hemoglobin absorption coefficient in the red part of the visible spectrum and associated increase in light penetration depth.

The depolarization threshold $\Delta = 0.85$ was applied to filter out the zones of cortex and specular reflection in the azimuth maps. The vertical orientation of fiber tracts is clearly seen in the corpus callosum zone that is not covered by blood.

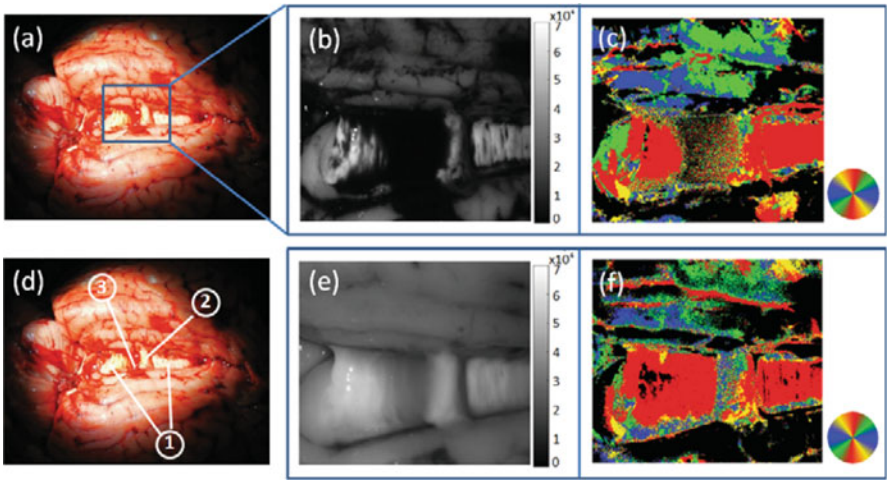


Fig. 14 Images of the corpus callosum in a cadaveric calf brain with a 2 mm deep pool of 100% blood: (**a**) color photo; (**b**) intensity image; (**c**) azimuth of the optical axis measured at 550 nm; (**d**) corpus callosum (1) with a 2 mm thick tissue specimen (2) creating a 2 mm deep pool filled with whole blood (3); (**e**) intensity image and (**f**) azimuth of the optical axis measured at 650 nm. Reprinted with permission from [75]. ©The Optical Society

However, the azimuth of the optical axis is completely random within the blood-covered zone in the map of the azimuth recorded at 550 nm (Fig. 14c). The impact of blood on the orientation of the optical axis is strongly reduced once we use the measurement wavelength of 650 nm (Fig. 14f).

The surgical cavity is regularly rinsed during neurosurgery with a saline solution. Thus, the presence of small amount of blood should not affect significantly the capability of the wide-field imaging Mueller polarimetry to visualize in-plane relative orientation of brain fiber tracts in the tissue-at-sight at any moment of neurosurgery.

5 Summary

The sensitivity of the multi-spectral wide-field imaging Mueller polarimetry operating in reflection geometry to brain tissue architecture at meso-scale (lateral resolution of about 100 μm) was demonstrated for the formalin-fixed paraffin-embedded, formalin-fixed and fresh brain specimens. The anisotropy of polarized light scattering and associated increase in total depolarization of backscattered light evolves with the advancement of Alzheimer disease in mice model and can be used as an optical marker to support quantitatively the histology analysis.

The anisotropy of refractive index of healthy brain white matter that is not visible to a naked eye was also measured with the wide-field imaging Mueller polarimeter. The difference in scattering properties of the cortex and white matter of healthy brain leads to significant contrast enhancement in the images of total depolarization of formalin-fixed and fresh brain specimens. The directions of nerve fiber tracts are clearly seen in the maps of azimuth of the optical axis calculated from the wide-field Mueller matrix images of thick formalin-fixed and fresh brain specimens measured in reflection. The presence of fiber tracts was also confirmed by a histology analysis of a formalin-fixed human brain specimen.

We demonstrated that the azimuth of the optical axis is a robust indicator of the in-plane orientation of fiber tracts as this polarimetric parameter is not strongly affected by the surface complex topography. The impact of blood on the azimuth maps can be mitigated by the proper choice of the measurement wavelength. These properties are essential for the polarimetric imaging of brain surgical cavity in clinical settings.

Wide-field imaging Mueller polarimetry does not require sample scanning or image stitching. This is an important advantage of the technique in terms of measurement time and complexity of the instrument compared to the polarization-sensitive optical coherence tomography or polarization-sensitive optical coherence microscopy, two-photon excited fluorescence, and second harmonic generation microscopy.

Using polarized light in a visible wavelength range excludes any potential harm to patients. In addition the wide-field imaging Mueller polarimetric system does not require sample staining and operates in backscattering configuration, that is the one required for any optical modality exploring visible light while targeting *in vivo* applications.

Compared to ultrasound imaging modality the visible wavelength range wide-field imaging Mueller polarimetry has quite shallow penetration depth of light within brain tissue. However, it demonstrates the capability to detect the cortex and white matter of brain in the depolarization images and provide the relative spatial in-plane orientation of brain fiber tracts using both scalar linear retardance and azimuth maps.

Taking polarimetric images of live brain in both green and red part of a visible spectrum may help to estimate the proximity of tumor to the surface of surgical cavity during neurosurgery and guide the resection of a tumor.

The perspective studies have to confirm the capability of the wide-field imaging Mueller polarimetry to visualize the border between tumor and healthy brain tissue both *ex vivo* and *in vivo*. Obviously, the transfer of the wide-field imaging Mueller polarimetric modality to clinics for brain diagnosis and assistance during neurosurgery will require further research studies, including both theoretical and instrumental development.

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Clinical Applications of Polarization-Sensitive Optical Coherence Tomography



Martin Villiger and Brett E. Bouma

1 Introduction

Optical coherence tomography (OCT) measures the subsurface microstructure of biological tissue. It is a mature imaging technique that is essential in contemporary ophthalmology. Coherence gating also has enabled compact imaging probes with the capacity to reach remote organs of the human body. OCT is routinely used for guiding percutaneous coronary interventions and in a clinical research setting for imaging of the gastrointestinal tract and the lung. Commercial clinical instruments are also available for applications in dermatology.

Polarization-sensitive (PS) OCT employs additional hardware components to evaluate the polarization state of the backscattered light in addition to its amplitude. Observing the pathlength dependence of the detected polarization states, caused by the roundtrip propagation of light into tissue, offers insight into the polarization properties of the tissue. PS-OCT can afford contrast between tissues that are indiscernible in OCT's conventional scattering signal. PS-OCT has been used for clinical research investigations of retinal disease, the anterior eye, coronary atherosclerosis, and the lung. Additional applications are in preclinical phase. While PS-OCT has evolved into a valuable research instrument, it awaits integration into commercial instruments.

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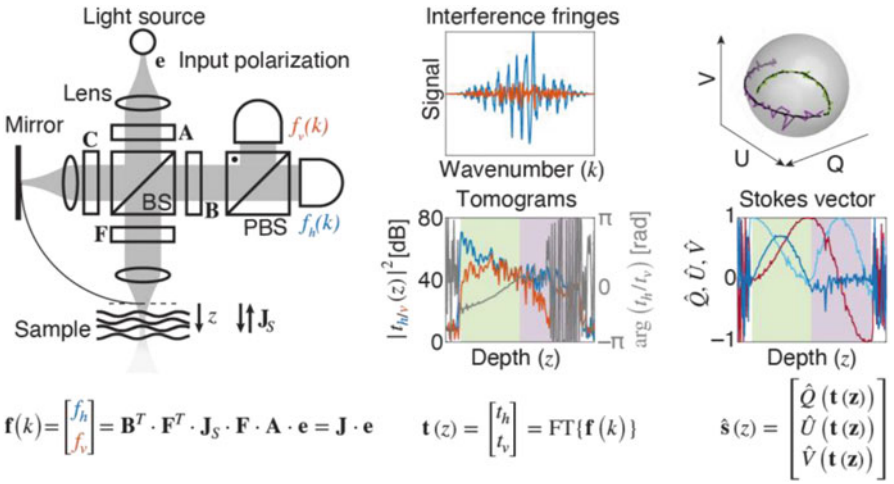
In contrast to the majority of polarimetric techniques, PS-OCT is a coherent imaging modality. Accordingly, PS-OCT fundamentally detects fully polarized light, and the dominating contrast mechanism is tissue birefringence. Coherence gating, in combination with confocal gating, excels at isolating singly back-scattered light. Multiply scattered light, which drives depolarization in many other polarimetric methods, is strongly suppressed and manifests in the randomization of the detected polarization states. Only through incoherent spatial averaging of measured polarization states in the Mueller-Stokes formalism can traditional metrics of depolarization be obtained with PS-OCT.

This chapter aims to discuss PS-OCT and its use in clinical applications. It is not intended to be an extensive review of PS-OCT, which is available in the literature [6, 19]. The basic principle of OCT is also well explained in text books [21] and not covered here in detail. Instead, the chapter provides a succinct summary of OCT polarimetry and reviews the current state of the art in terms of instrumentation and signal processing. It then provides an overview of clinical investigations that have been performed with different PS-OCT imaging geometries in various organs and diseases. It ends with an outlook of future developments and a discussion of what may be needed for PS-OCT to find more widespread and commercial adoption.

2 Polarization-Sensitive OCT

2.1 Basic Principles of OCT Polarimetry

Optical coherence tomography uses interference to measure the pathlength of light backscattered in biological tissue relative to the pathlength of a reference signal. Contemporary OCT measures the interference signal as a function of wavenumber and then reconstructs the pathlength-encoded sample reflectivity profile through the Fourier transform relation between the two domains. The pathlength accurately approximates the sample depth of a scattering event up to a few scattering pathlengths and provides efficient depth sectioning in weakly scattering tissues. Although confocal gating helps to further suppress multiply scattered light from shallower depths, OCT generally uses weakly focused beams that provide a long depth of field and enable imaging relatively large fields of view with modest lateral resolution. By virtue of recording the interference between the sample and the reference light, only sample light matching both the spatial mode profile and the polarization of the reference light contributes to the interference signal. Hence, all light detected by OCT is fully polarized. In order to determine the effective polarization state of the sample light, PS-OCT splits the sample light into two orthogonal polarization states, which then independently interfere with identically orthogonally polarized reference signals [42], as illustrated in Fig. 1. The phase difference between the two channels together with their relative signal amplitudes defines the polarization state of the detected sample light for each wavenumber,



$$\mathbf{f}(k) = \begin{bmatrix} f_h \\ f_v \end{bmatrix} = \mathbf{B}^T \cdot \mathbf{F}^T \cdot \mathbf{J}_S \cdot \mathbf{F} \cdot \mathbf{A} \cdot \mathbf{e} = \mathbf{J} \cdot \mathbf{e} \qquad \mathbf{t}(z) = \begin{bmatrix} t_h \\ t_v \end{bmatrix} = \mathrm{FT}\{\mathbf{f}(k)\} \qquad \hat{\mathbf{s}}(z) = \begin{bmatrix} \hat{Q}(\mathbf{t}(\mathbf{z})) \\ \hat{U}(\mathbf{t}(\mathbf{z})) \\ \hat{V}(\mathbf{t}(\mathbf{z})) \end{bmatrix}$$

Fig. 1 Basic system schematic and principle of PS-OCT. A polarizing beam splitter creates two polarization-diverse detection channels that independently record the interference of the sample and reference light as a function of wavenumber k. The Fourier transformation of these signals generate complex-valued tomograms that correspond to the two components of a Jones vector. Conversion to Stokes parameters facilitates visualization of the depth-dependent variation of the detected polarization states

according to the Jones formalism. Under the assumption that the relative group delays between the eigenstates of a birefringent sample medium are substantially smaller than the OCT axial resolution, the relative amplitude and phase between the tomograms reconstructed from the two detection channels encode the polarization state of the sample signal as a function of pathlength traveled into the tissue. This then enables observation of depth-dependent changes in the polarization state induced by tissue birefringence and other polarization effects.

It is important to appreciate that detected polarization states embody the cumulative effect of round-trip propagation to a given sample depth, which can be described by its corresponding Jones matrix. In order to unambiguously recover the full Jones matrix, measurements with at least a second input polarization state are in general necessary. The round-trip nature of PS-OCT measurements also complicates the reconstruction of depth-resolved tissue polarization properties. Furthermore, OCT uses identical illumination and detection paths, which results in an axially symmetric scattering geometry without a defined orientation of the scattering plane, and light is collected from within the coherent backscattering cone. This geometry has the effect of canceling out during reverse transmission any polarization rotation or circular dichroism that occurs during forward propagation through a medium.

Compared to intensity-based polarimetric techniques that require multiple combinations of polarizer and analyzer settings, OCT's coherent detection and the ability to multiplex two input states offer single-shot measurements of the tissue's polarization effects. However, reconstructed Jones vectors and matrices can be cast

into the Mueller-Stokes formalism. Incoherent spatial averaging of these Stokes vectors or Mueller matrices leads to partially polarized Stokes vectors and offers insight into depolarization effects.

2.2 PS-OCT Formalism

The two complex-conjugate interference terms f_h and f_v recorded in the horizontally (h) and vertically (v) polarized detection channels as a function of the wavenumber k define the Jones vector of the detected electromagnetic field:

$$\mathbf{f}(k) = \begin{bmatrix} f_\mathrm{h}(k) \\ f_\mathrm{v}(k) \end{bmatrix} = \mathbf{B}^\mathrm{T}(k) \cdot \mathbf{F}^\mathrm{T}(k) \cdot \mathbf{J}_\mathrm{S}(k) \cdot \mathbf{F}(k) \cdot \mathbf{A}(k) \cdot \mathbf{e} = \mathbf{J}(k) \cdot \mathbf{e}. \tag{1}$$

As illustrated in Fig. 1, $\mathbf{e}$ depicts the source polarization state, and the Jones matrix $\mathbf{J}(k)$ comprises of single-pass system transmission matrices $\mathbf{A}$ and $\mathbf{B}$, as well as round-trip transmission matrices of the sample $\mathbf{J}_\mathrm{S}$ and further system elements $\mathbf{F}$. The reference light is prepared by means of element $\mathbf{C}$ to provide equal reference signals in the two orthogonally polarized detection channels.

For OCT, most biological tissues can be pictured as a collection of randomly positioned point-like scatterers (p). A birefringent sample can then be modeled as consisting of point-like scatterers with a scattering amplitude a_p, located at positions z_p inside a medium with a mean refractive index n and birefringence Δn. The resulting interference signal is

$$\boldsymbol{f}(k) = S(k) \sum_\mathrm{p} a_\mathrm{p} \exp\left(i2knz_\mathrm{p}\right) \begin{bmatrix} \exp\left(ik\Delta n z_\mathrm{p}\right) & 0 \\ 0 & \exp\left(-ik\Delta n z_\mathrm{p}\right) \end{bmatrix} \cdot \mathbf{e}, \tag{2}$$

where $S(k)$ is the power spectral density. Birefringence of the background medium results in a differential delay between the ordinary and extraordinary waves, which have been aligned here with the horizontal and vertical polarization directions to simplify the expression. Also, the difference between the Poynting vectors of the two eigenpolarization states has been ignored. Backscattering effectively arises from spatial refractive index inhomogeneities, but we separate here the mean refractive index of the background medium, which defines the pathlength experienced by propagating light, from smaller and spatially confined index variations that are instead approximated as point-like scatterers.

Fourier transformation of $\mathbf{f}(k)$ results in the complex-valued Jones vector tomogram, termed A-line:

$$\mathbf{t}(z) = \mathrm{FT}\left\{\mathbf{f}(k)\right\} = \begin{bmatrix} t_\mathrm{h}(z) \\ t_\mathrm{v}(z) \end{bmatrix} = \mathbf{J}(z) \cdot \mathbf{e}. \tag{3}$$

The Fourier transform is taken from kn to $2z$. It is assumed that the mean refractive index n is known, enabling to convert the measured pathlength into axial distance z. Here, and throughout the remainder of this chapter, we write $\mathbf{J}(k)$ and $\mathbf{J}(z)$ without explicitly differentiating between the wavenumber and the spatial domain. In general, the differences in birefringence-induced group delays result in different pathlengths for sample signals from identical depths. In practice, the encountered sample retardance is usually considerably smaller than the coherence length. In Eq. (2) this enables to approximate $k\Delta n \approx k_\mathrm{c}\Delta n$ in the phase retardance, where k_c is the center wavenumber. This is equivalent to treating the sample retardance as independent of the wavelength and effectively removes the difference in pathlength between the eigenstates, resulting in

$$\mathbf{t}(z) \approx \sum_\mathrm{p} a_\mathrm{p} s\left(z - z_\mathrm{p}\right) \begin{bmatrix} \exp\left(ik_\mathrm{c}\Delta n z_\mathrm{p}\right) & 0 \\ 0 & \exp\left(-ik_\mathrm{c}\Delta n z_\mathrm{p}\right) \end{bmatrix} \cdot \mathbf{e}. \tag{4}$$

Here, $s(z) = \mathrm{FT}\{S(k)\}$ is the axial point spread function of OCT. In this case we can interpret the amplitudes and phase difference at a given tomogram position as encoding the polarization state at the corresponding sample depth.

The axial spacing of the point-like scatterers is, in general, much smaller than the coherence length. This effectively results in random phase terms of the individual scatterers and leads to fully developed speckle. OCT can readily differentiate tissue layers with different back-scattering properties if they are thicker than the coherence gate by evaluating their mean signal. But homogeneous layers cause a speckle pattern with a strong contrast that varies on a short scale, with the average speckle size defined by the coherence gate in the axial direction and the lateral resolution in the transverse direction.

As visualized in Fig. 1, the sample retardance is encoded in the amplitude and phase difference between the speckle patterns reconstructed from the two polarization channels. Conversion to Stokes formalism offers visualization of the depth-dependent polarization evolution on the Poincaré sphere:

$$\mathbf{s}(z) = I(z) \begin{bmatrix} 1 \\ \hat{\mathbf{s}}(z) \end{bmatrix} = I(z) \begin{bmatrix} 1 \\ \hat{Q} \\ \hat{U} \\ \hat{V} \end{bmatrix} = \begin{bmatrix} |t_\mathrm{h}|^2 + |t_\mathrm{v}|^2 \\ |t_\mathrm{h}|^2 - |t_\mathrm{v}|^2 \\ 2\,\mathrm{Re}\left\{t_\mathrm{h}\bar{t}_\mathrm{v}\right\} \\ -2\,\mathrm{Im}\left\{t_\mathrm{h}\bar{t}_\mathrm{v}\right\} \end{bmatrix}, \tag{5}$$

where $\hat{}$ indicates a normalized metric and the overbar corresponds to complex conjugation.

Because OCT measures reflected round-trip signals, it is critical to define the employed coordinate system. We follow the convention of expressing all polarization states in the fixed coordinate system of the receiver, independent of the propagation direction of the light [18, 97], which avoids a flip of one of the transverse coordinates upon reflection. According to the reciprocity theorem [84], light propagating along the reverse path between an input and output port

will experience the same transmission coefficient and phase delay as in the forward direction. This results in a transpose relationship between the Jones matrix describing forward and backward propagation, which has several implications for the reconstruction of sample polarization properties.

The sample Jones matrix of round-trip propagation to depth z and back can be described as

$$\mathbf{J}_{\mathrm{S}}(z) = \mathbf{J}_{\mathrm{fw}}^{\mathrm{T}}(z) \cdot \mathbf{R}(z) \cdot \mathbf{J}_{\mathrm{fw}}(z), \tag{6}$$

where $\mathbf{J}_{\mathrm{fw}}$ defines the single-pass forward (fw) transmission to that same depth and the transpose (T) of $\mathbf{J}_{\mathrm{fw}}$ the reverse transmission. $\mathbf{R}(z)$ describes the signal strength and a possible polarization dependence of the backscattering amplitude at depth z, with $\mathbf{R}(z) = \mathbf{R}^{\mathrm{T}}(z)$. The forward transmission is defined through the local effect of tissue properties on the polarization of propagation light by the differential equation:

$$\frac{\partial \mathbf{J}_{\mathrm{fw}}(z)}{\partial z} \cdot \mathbf{J}_{\mathrm{fw}}^{-\mathbf{1}}(z) = \mathbf{\Omega}(z) = \frac{ik\Delta n(z)}{2} \left(\hat{r}_{\mathrm{q}}(z)\, \boldsymbol{\sigma}_{\mathrm{q}} + \hat{r}_{\mathrm{u}}(z)\, \boldsymbol{\sigma}_{\mathrm{u}} \right) + ik \langle n \rangle \mathbf{I}. \tag{7}$$

Here, we ignore tissue diattenuation and model tissue to act locally as a linear retarder with birefringence Δn and an optic axis orientation θ, measured as the angle of the linear polarization experiencing the extraordinary index with the horizontal polarization direction. In Poincaré coordinates, the optic axis corresponds to the direction:

$$\hat{\boldsymbol{r}} = \begin{bmatrix} \hat{r}_{\mathrm{q}} \\ \hat{r}_{\mathrm{u}} \\ \hat{r}_{\mathrm{v}} \end{bmatrix} = \begin{bmatrix} \cos(2\theta) \\ \sin(2\theta) \\ 0 \end{bmatrix}. \tag{8}$$

$\mathbf{I}$ in Eq. (7) is the identity matrix and $ik\langle n \rangle$ is the group delay. $\boldsymbol{\sigma}_n$ are the first two Pauli matrices:

$$\boldsymbol{\sigma}_{\mathrm{q}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \boldsymbol{\sigma}_{\mathrm{u}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \tag{9}$$

In the present case, $\mathbf{J}_{\mathrm{fw}}$ belongs to the special unitary group of degree 2, SU(2), and $\mathbf{\Omega}$ is its corresponding Lie algebra. However, inclusion of diattenuation and extension to the special linear group over the complex numbers SL(2, C) is straightforward. Integration of the differential equation in Eq. (7) is difficult in the general case. However, if $\mathbf{\Omega}$ is assumed to be constant in a tissue layer from z to $z + \Delta z$, then we obtain

$$\mathbf{J}_{\mathrm{fw}}(z + \Delta z) = \exp(\mathbf{\Omega} \Delta z) \cdot \mathbf{J}_{\mathrm{fw}}(z) = \mathbf{L}(\varphi, \theta) \cdot \mathbf{J}_{\mathrm{fw}}(z), \tag{10}$$

where $\mathbf{L}$ is a linear retarder with retardance $\varphi = k_{\mathrm{c}} \Delta n \Delta z$ and an optic axis orientation of θ, obtained through matrix exponentiation of its generator $\mathbf{\Omega}$. We

can then iteratively model $\mathbf{J}_{\mathrm{fw}}(z)$ as the sequential product of transmission through individual tissue layers. Although each individual layer is modeled as a linear retarder, the combined effect of transmission to a given depth is a general retarder, with an optic axis orientation no longer confined to the QU-plane in the Poincaré sphere. However, any circular retardance component cancels out in the round-trip matrix $\mathbf{J}_{\mathrm{S}}$ owing to the product with the transpose of $\mathbf{J}_{\mathrm{fw}}$.

2.3 Instrumentation

PS-OCT can be implemented in most existing OCT imaging geometries. There exist other OCT architectures that measure lateral sample information in parallel, but the discussion here is limited to point-scanning OCT that acquires individual sample depth profiles, or A-lines, using a loosely focused beam. Assembling multiple A-lines, acquired while scanning the beam across the sample, defines a cross-sectional view of the tissue, termed B-scan. Volumetric imaging can be achieved by scanning along both transverse directions. Although these implementations of OCT require beam scanning, fast acquisition speed (typical A-line rates range from 100 kHz to several tens of MHz) and the use of lose focus enable relatively large fields of view that can reach several cm^2 and provide camera-rate imaging [88]. As depicted in Fig. 2, many OCT systems use galvanometric mirrors for beam scanning, which provides a convenient microscopic imaging geometry for tissue and organs that can be readily accessed, such as the skin [87] or the anterior eye [10]. Retinal (PS)-OCT uses the same scanning principle but uses the eye to focus the beam on the retina in combination with pupil matching (Fig. 2) [83]. OCT's compact form factor without the need for mechanical axial scanning has enabled its integration into catheters and endoscopes [33]. Side-looking probes that perform a helical scan pattern inside a protective transparent sheath are routinely used to guide percutaneous coronary interventions [11] and inspect the esophagus and the lung. OCT imaging catheters currently used in the clinic employ probes that consist of a rotating fiber interfaced through a rotary joint. Catheters that instead use a distal micromotor are being developed and have been used for pulmonary imaging [25].

The primary distinction between PS-OCT and OCT is the use of polarization-diverse detection to determine the polarization state of the scattered sample light. This can be accomplished using two cameras in spectrometer-based systems [44] or using dual-balanced polarization-diverse detection using wavelength-swept frequency domain implementations [72], as depicted in Fig. 3. Because the spectrometer embodies the most involved component of a spectral-domain system, the increase in system complexity is particularly significant for these systems, and strategies to detect two polarization states with a single line camera have been proposed [9, 14].

PS-OCT systems that employ a single input polarization state only measure a single Jones vector for each pathlength. Taking into account the intrinsic transpose symmetry of the sample Jones matrix by restricting it to be a (linear) retarder fully

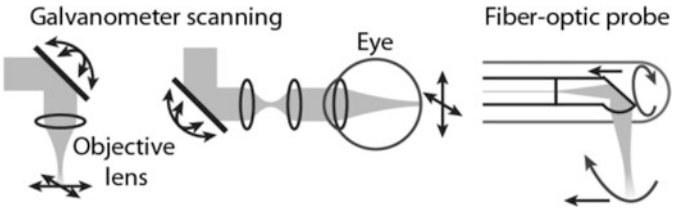


Fig. 2 OCT and PS-OCT afford different imaging geometries. Galvanometric scanners, in combination with an objective lens, allow scanning of ex vivo samples or readily accessible tissues such as the skin. Retinal imaging uses the eye for focusing the beam and relays the scanner into the pupil of the eye. Catheter-based OCT employs a fiber probe inside a protective transparent sheath. Combined rotation and pullback of the probe defines a helical scan pattern ideally suited for imaging of luminal organs

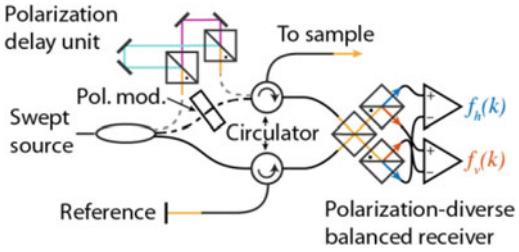


Fig. 3 Typical fiber-based Mach-Zehnder layout with various implementations of fiber-based PS-OCT using a wavelength-swept laser source. Single input state PS-OCT uses a regular fiber in the sample arm. Multiple input states can be generated sequentially with a polarization modulator. Full Jones matrices can be measured simultaneously with pathlength multiplexing of two orthogonal input states using a polarization delay unit

defines $\mathbf{J}_S$ with simple formulas if a circular input state is used [42, 43]. However, the achievable polarization sensitivity in this case depends on the alignment of the tissue's optic axis with the polarization state propagating through the tissue [65]. This can lead to the masking of sample birefringence in samples with a depth-dependent optic axis. Also, single input state PS-OCT traditionally requires careful control of the polarization state with fiber-based systems [94] or the use of polarization-maintaining fibers [3, 34] to ensure circular polarization incident on the sample.

In order to unambiguously describe the polarization effect of round-trip transmission to a given pathlength in the general case, at least two distinct input polarization states are required. Algebraically, using two distinct input states provides

$$[\mathbf{t_1}(z)\ \ \mathbf{t_2}(z)] = \mathbf{J}(z) \cdot \left[\mathbf{e_1}\ \mathbf{e_2}\right], \tag{11}$$

which then enables solving for $\mathbf{J}(z)$ [67]. Simultaneous measurement of two input states can be achieved by frequency- or depth-multiplexing [8, 59, 72]. While depth-multiplexing limits the effectively available imaging range, it offers fully

coherent, single-shot measurements of $\mathbf{J}(k)$ and hence $\mathbf{J}(z)$. Sequential modulation of the source polarization state offers an alternative solution [87]. However, two sequential measurements generally illuminate two slightly different sample locations. The presence of phase noise furthermore frustrates reliable recovery of $\mathbf{J}(z)$, especially when using physically orthogonal polarization states. Instead, using two polarization states that are orthogonal to each other on the Poincaré sphere, e.g., linear and circular polarization, enables recovery of the sample Jones matrix without reliance on the relative phase between the sequential measurements but requires constraining the sample to be a pure retarder. While $\mathbf{J}(z)$ can be reasonably approximated as being a pure retarder, this is not true for $\mathbf{J}(k)$, because summation is not a group operation of SU(2). Hence, sequential modulation does not enable accurate recovery of $\mathbf{J}(k)$, only an approximate version of $\mathbf{J}(z)$.

Using multiple input states enables the use of single-mode fiber- and catheter-based imaging systems, where the polarization state effectively incident onto the sample depends on the exact fiber configuration and is a priori unknown. The use of fiber-based imaging systems that require little control is an important practical consideration for imaging in a clinical setting. Figure 3 depicts the increasing system complexity from single input state PS-OCT to modulated input state and PS-OCT with multiplexed input states.

2.4 Polarimetric Signal Reconstruction

While fiber-based systems offer practical advantages in a clinical setting, fiber-based optical components are imperfect and can introduce undesirable polarization effects. Optical circulators in particular are known to induce differential group delay, which corresponds to a wavelength-dependent retardation. Furthermore, fiber polarization controllers are chromatic and act as first-order waveplates. As a result, the polarization state incident on the sample likely varies as a function of the wavelength. This is captured in Eq. (1) by the wavenumber dependence of all the system components. Taking the Fourier transformation of their products formally results in the convolutions of their individual transforms, which can give rise to artifacts [104]. A system element without spectral dependence is an identity element of the convolution and acts in the same way on all wavenumbers and also the pathlength-resolved Jones matrices. If the differential group delays induced by the system elements are substantially smaller than OCT's axial coherence gate, then their detrimental impact is negligible, and the elements simply rotate all polarization states. However, as the delays approach the dimension of the coherence gate, they cause increasingly severe artifacts which frustrate the reconstruction of pathlength-resolved polarization states [104].

One strategy for mitigating the resulting artifacts is to widen the coherence gate by splitting the spectral signal into several narrower bins [105]. Each bin is individually processed and then corrected for the rotation effect of the system elements relative to the central bin, before averaging across all bins. The downside

of this approach is the loss of axial resolution induced by using a narrower spectral width. In order to preserve the full axial resolution, the measurements should be compensated for the effects of the system components directly in the k-domain through left and right multiplication with $(\mathbf{B}^{\mathrm{T}}(k)\cdot\mathbf{F}^{\mathrm{T}}(k))^{-1}$ and $(\mathbf{F}(k)\cdot\mathbf{A}(k))^{-1}$, respectively. This is only possible if $\mathbf{J}(k)$ is directly measured using multiplexed input states [12, 115]. The system elements can generally be recovered in a blind fashion without dedicated calibration measurements, directly from imaging data, including polarization changes occurring due to catheter rotation [55, 100].

Correction of system-induced polarization effects has been critical for enabling catheter-based PS-OCT with established single-mode fiber-based imaging hardware. However, re-engineering of the optical system offers an alternative solution for eliminating system-induced artifacts, which is more readily achievable for bench-top and ophthalmic imaging. Both approaches ideally lead to reliable measurements of $\mathbf{J}(z) = \mathbf{B}^{\mathrm{T}}\cdot\mathbf{F}^{\mathrm{T}}\cdot\mathbf{J}_{\mathrm{S}}(z)\cdot\mathbf{F}\cdot\mathbf{A}$, where the system elements are free from group delay.

Polarization effects in PS-OCT are dominated by tissue birefringence. Although the need for more advanced reconstruction techniques to isolate depth-resolved tissue properties was recognized early on [93], many early PS-OCT efforts focused on the analysis of the cumulative round-trip signal. It is straightforward to characterize $\mathbf{J}(z)$ in terms of its retardance and possibly diattenuation. Polar decomposition can serve to separate retardation and diattenuation, although only concurrent decomposition that directly recovers the corresponding Lie generators preserves the underlying transpose symmetry [55] that confines the cumulative retardation and diattenuation to the QU-plane.

If a single circular input state is used to illuminate the sample and the combined effect of transmission through $\mathbf{F}$ and $\mathbf{B}$ corresponds to a quarter waveplate, then the amplitude ratio between t_{h} and t_{v} directly reveals the round-trip retardance, and their phase difference the corresponding optic axis orientation. Using, instead, multiple input states with possibly unknown system transmissions, it is possible to compute the polarization effects relative to a reference position in the sample, e.g., the surface, where $\mathbf{J}(z_{\mathrm{ref}}) = \mathbf{B}^{\mathrm{T}}\cdot\mathbf{F}^{\mathrm{T}}\cdot\mathbf{F}\cdot\mathbf{A}$, independent of the sample:

$$\mathbf{J}(z)\cdot\mathbf{J}^{-\mathbf{1}}(z_{\mathrm{ref}}) = \mathbf{B}^{\mathrm{T}}\cdot\mathbf{F}^{\mathrm{T}}\cdot\mathbf{J}_{\mathrm{S}}(z)\cdot\mathbf{F}^{-\mathrm{T}}\cdot\mathbf{B}^{-\mathrm{T}}. \tag{12}$$

Here, $-\mathrm{T}$ indicates the transpose inverse, and the computed product corresponds to a similarity transform of $\mathbf{J}_{\mathrm{S}}(z)$. Assuming that the system components are static and pure retarders, this simply rotates the cumulative sample retardation from the QU-plane to a different plane, which can be readily identified, although with a π ambiguity [78]. Even if correctly rotated into the QU-plane, circular retardance in the system components would cause a rotation of the detected optic axes within the QU-plane around the V-axis. Optic axis orientations measured with PS-OCT are in general relative to an unknown reference, unless a calibration procedure is performed.

Because the OCT signal is fraught with speckle, it is common to have low signal values even in strongly scattering tissue layers, with unreliable polarization state

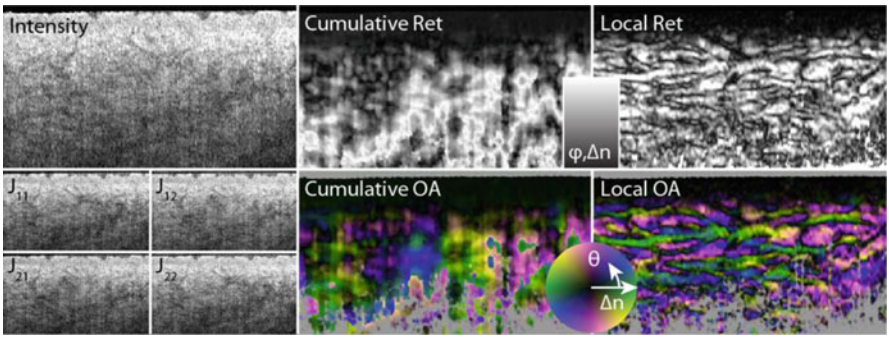


Fig. 4 Reconstruction of tissue retardance and birefringence in the skin of the forearm of a healthy volunteer. The conventional scattering tomogram is the incoherent summation of the individual polarization channels and displayed in logarithmic scale. Cumulative retardance and local retardance, which is proportional to tissue birefringence (Δn), are displayed in linear gray scale, spanning from 0 to π for cumulative retardance and from 0 to 3×10^{-3} for Δn. Optic axis orientation (OA) is encoded in color, with retardance in brightness, as indicated by the color wheel. Only local retardance on optic axis orientation provides a comprehensible view of the tissue organization in strands of interwoven collagen bundles. (Figure adapted from Li et al. [57])

measurements. Spatial averaging of the detected polarization states offers a simple but effective strategy for denoising. However, coherent averaging of Jones vectors and Jones matrices requires management of their relative phase offsets [54]. This is simplified by resorting to the Stokes formalism, which enables incoherent averaging of Stokes vectors [79], Mueller-Jones [101], or SO(3) matrices [100].

Cumulative polarization effects relate to the local tissue properties but can be difficult to interpret in samples consisting of multiple layers with distinct birefringence, as visualized in Fig. 4. In this case, it is beneficial to reconstruct local, i.e., depth-resolved tissue polarization effects by computing the differential matrix:

$$\mathbf{J}_\mathrm{S}\,(z+\Delta z)\cdot\mathbf{J}_\mathrm{S}{}^{-\mathbf{1}}(z) \propto \mathbf{J}_\mathrm{fw}^\mathrm{T}(z)\cdot\exp\left(\mathbf{2\Omega}\Delta z\right)\cdot\mathbf{J}_\mathrm{fw}^{-\mathrm{T}}(z), \tag{13}$$

which provides a similarity transform of the effective local sample property in the given tissue layer [67]. Here we have omitted the static system components, which would simply add a further similarity transformation, in line with Eq. (12). While $\mathbf{J}_\mathrm{fw}(z)$ can in principle comprise diattenuation, the presence of polarization dependence of the scattering amplitude $\mathbf{R}$ would frustrate this formulation and has been ignored here, leaving only an additional amplitude scaling. Because similarity transforms preserve eigenvalues, Eq. (13) allows assessing the local tissue birefringence (and possibly diattenuation, if homogeneous [64]) of $\exp(2\mathbf{\Omega}\Delta z)$. The round-trip retardance φ through a tissue slab of thickness Δz relates to tissue birefringence Δn through $\varphi = 2k_\mathrm{c}\Delta n\Delta z$ and enables reconstruction of depth-resolved tissue birefringence maps (Fig. 4). Similar reconstruction is possible using a geometric reasoning with Stokes vectors [44, 105] or Mueller matrices [101].

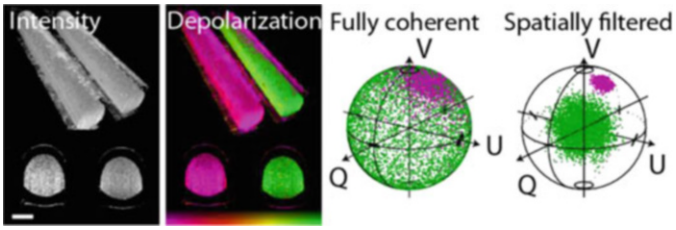


Fig. 5 Depolarization obtained from PS-OCT. Two capillaries containing intralipid (left) and a suspension of gold nanorods (right) show comparable intensity tomograms but strong depolarization (green) only in case of the nanorods. Individual, coherently measured polarization states closely localize on the Poincaré sphere for the signal of intralipid and are distributed over the entire unit sphere for the nanorods. When averaged over a few neighboring states, the resulting depolarization collapses the intensity-normalized Stokes vectors of the nanorod signal into the center of the Poincaré sphere. (Adapted from Lippok et al. [60])

In order to also recover the local optic axis orientation, which relates to the eigenvectors of $\exp(2\boldsymbol{\Omega}\Delta z)$, Eq. (13) requires correction with the single-pass matrices $\mathbf{J}_{\mathrm{fw}}(z)$. These can be recovered in an iterative approach by separating the linear retardance measured across each layer Δz into two equal parts [23, 24, 93, 100]. Figure 4 demonstrates local optic axis imaging in the skin. Chapter "Tractography Using Polarization-Sensitive Optical Coherence Tomography" in this book is dedicated to the topic of measuring depth-resolved optic axis tractography.

Incoherent spatial averaging of Stokes vectors and Mueller matrices not only increases the accuracy of derived polarization metrics but also expresses the statistics of the individual polarization measurements as depolarization. The original conversion of a Jones vector results in a fully polarized Stokes vector. Yet, spatially averaging across a few speckle results in general in a reduced degree of polarization (DOP), as illustrated in Fig. 5. Likewise, a measured Jones matrix can be converted into a Mueller-Jones matrix, which after spatial averaging becomes a general Mueller matrix including depolarization [61]. The variation of polarization states between neighboring speckle grains can be due to intrinsic sample variation, e.g., the presence of distinct polarization-dependent scattering matrices $\mathbf{R}(z)$. A similar effect is obtained by multiply scattered light, where the scattering angles influence the resulting polarization states, imitating the effect of a polarization-dependent $\mathbf{R}(z)$. To date, depolarization in PS-OCT has only been assessed as a cumulative, scalar metric.

2.5 Tissue Polarization Properties

Conventional OCT tomograms visualize the subsurface microstructure of biological tissues by providing maps of backscattering intensity in logarithmic scale. PS-OCT complements these signals of tissue structure with insight into the tissue

polarization properties and can afford contrast between tissues that are indiscernible in OCT's conventional scattering signal. PS-OCT detects changes in the polarization state as a function of depth traveled into the tissue. Birefringence is considered the dominating contrast mechanism for PS-OCT. Only a few investigations of tissue diattenuation have been reported [49], although it is possible that advanced reconstruction techniques will be able to detect small diattenuation signatures that are currently overshadowed by birefringence. Tissue birefringence is elevated in many tissues with a fibrillar architecture, such as collagen, muscle, or myelinated axons. Through a combination of intrinsic and form birefringence, homogeneous slabs of such tissue appear as linearly birefringent uniaxial media at the spatial scale investigated by PS-OCT, with their extraordinary optic axis aligned along the fiber direction [69, 113]. Possible polarization rotation induced by optical activity is negligible compared to the linear birefringence of these tissues. Furthermore, OCT's imaging geometry precludes measurement of optical rotation.

Moreover, even for linearly retarding samples, the effective birefringence experienced by PS-OCT depends on the alignment of the probing beam propagation direction with the optic axis of the extraordinary refractive index. While PS-OCT can identify the fast and slow optic axis orientation, it is challenging in general to obtain the sign of birefringence. PS-OCT only measures the "in-plane," azimuthal optic axis orientation. The birefringence of fibrillar structures that are oriented outside of a plane orthogonal to the probing beam will be underestimated. Retrieving the full three-dimensional optic axis orientation including the sign of birefringence requires additional measurements at different probing beam angles [47, 110].

The relatively low birefringence of many tissues results in modest retardance, increasing with less than 1°/μm of depth. The weak backscattering of biological tissues, exacerbated by the statistical properties of speckle, makes the detection of the resulting minute changes of the polarization states along depth challenging. The matrix differentiation along depth, which is necessary for depth-resolved measurements, is further sensitive to noise. For samples without prominent birefringence that are mostly polarization preserving, such as the retina, it can be advantageous to investigate the cumulative retardance or compute the retardance across specific anatomical layers of interest, rather than over a predefined axial layer thickness. However, depending on the sample structure, cumulative polarization signals can be difficult to interpret. The relatively strong birefringence of dermal collagen, combined with the weaving, mesh-like organization of the human skin, defines a sample with multiple layers of distinct optic axis orientation. As visualized in Fig. 4, only local birefringence and optic axis reconstruction render an intuitive "image" of this organization. Although the cumulative signal contains the same information, it is difficult to derive comparable insight from that presentation.

Depolarization has initially been linked to multiple scattering [2], where the polarization of side-scatter effectively results in a randomization of the detected polarization states. Polarization scrambling has then been observed in the retinal pigment epithelium [35], possibly also due to multiple scattering by melanin granules [7]. The strong polarization-dependent scattering of gold nanorods results

in pronounced depolarization when detected from an ensemble of randomly oriented particles, and nanorods can serve as exogenous contrast agents for PS-OCT [60] (Fig. 5). Previously, the dynamic variation of the polarization signal due to particle diffusion has been used for nanorod localization [15]. In a similar way, birefringent tissue structures with orientation that varies on a scale smaller than the resolution of OCT would lead to randomized polarization states and hence depolarization. Lastly, depolarization has a strong dependence on noise, which is completely depolarized [48]. Tomogram areas with decreased signal strength, typically deep inside the tissue, are dominated by noise and hence appear depolarizing. Hence, depolarization offers a convenient metric to identify tomogram areas carrying meaningful polarization information and mask other areas. On the other hand, this noise dependence complicates quantitative assessment of depolarization and should be corrected to unambiguously identify sample-induced depolarization [66].

3 Clinical Investigations with PS-OCT

3.1 Ophthalmic PS-OCT

Retinal imaging has been the driving application of OCT imaging and resulted in its rapid adoption into clinical practice [20, 92]. With its ability to non-invasively visualize the retinal layers at close to cellular resolution, OCT has become a mainstay of ophthalmology. Many retinal pathological conditions manifest in a change of the layer structure that can be visualized with conventional OCT. Yet, several layers of the ocular fundus also impact the polarization state of light, which can serve to obtain tissue-specific contrast and quantitative metrics that can help in the diagnosis of retinal pathology. Already the original demonstration of PS-OCT [42] proposed polarization-sensitive measurements of the retina, which has been reported a while after [13]. Despite early progress, PS-OCT remains to be implemented in commercial ophthalmic instruments and has only been performed with prototype research instruments. Retinal PS-OCT has mostly focused on measuring retardance of the retinal nerve fiber layer (RNFL) and on leveraging the intrinsic depolarization of the retinal pigment epithelium (RPE) for improved segmentation of the retinal layers [35].

The RNFL is the innermost layer of the retina and composed of retinal ganglion cell axons that exit the eye through the optic nerve. Ganglion cells receive visual information through a cascade of neuronal signals initiated by the photoreceptors and transmit visual information along their axons, which are unmyelinated within the retina, to the brain. The RNFL is a sensitive structure and is considered an extension of the central nervous system (CNS). In addition to several retinal pathologies, many CNS disorders also manifest in changes of the RNFL. The retina, thus, can serve as a window to the brain [62, 90].

Glaucoma is one of the leading causes of blindness and results from damage to the optic nerve head, usually due to high intraocular eye pressure. During

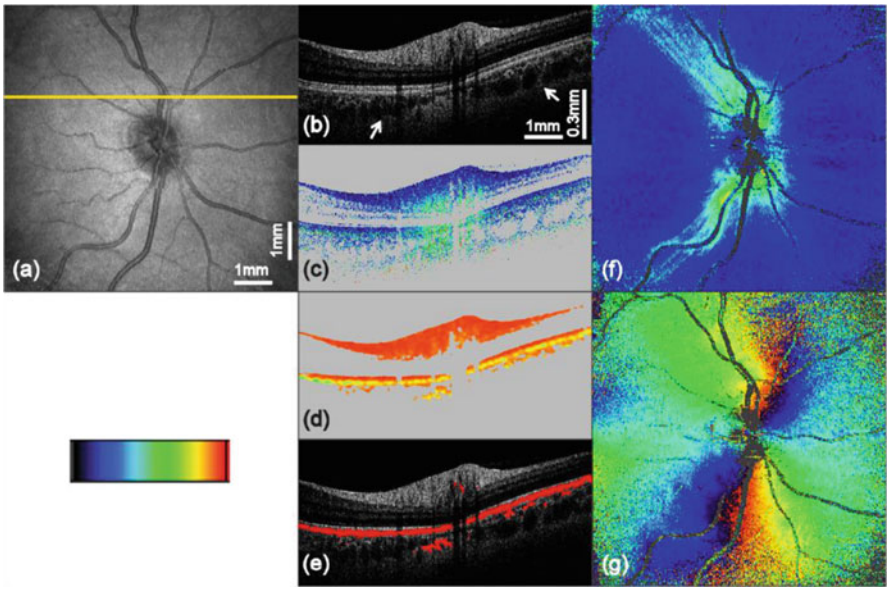


Fig. 6 Retinal PS-OCT of the optic disc region from a healthy human volunteer (scan angle 25° × 25°, scan pattern 1024 × 256 A-scans). The yellow line in (**a**) indicates the position of the corresponding B-scans in (**b**–**e**): (**a**) fundus projection image, (**b**) intensity B-scan on logarithmic gray scale, (**c**) retardation image (color scale 0° to 40°), (**d**) DOPU image (color scale 0 to 1), (**e**) overlay of segmented depolarizing structures (red) with the intensity image, (**f**) 2D en face retardation map retrieved at IS/OS and ETPR (color scale 0° to 40°), (**g**) 2D en face axis orientation map retrieved at IS/OS and ETPR (color scale −90° to +90°). (Reproduced from Trasischker et al. [94])

progression of glaucoma, the retinal ganglion cell axons degrade, which leads to a thinning of the RNFL and visual defects. While thinning of the RNFL can be measured with conventional OCT, it would be advantageous to detect early damage before irreversible axonal loss occurs. RNFL retardance depends both on RNFL thickness and its birefringence, induced by the retinal ganglion cell axons. Early axonal degeneration is thought to lead to a reduction of RNFL birefringence before measurable thinning occurs.

Figure 6 demonstrates PS-OCT of the RNFL, adapted from [94]. Here a single input state was used, and the PS-OCT cross-sectional images display cumulative retardance and optic axis orientation. The depolarization of the RPE is captured by computing the degree of polarization uniformity (DOPU), which corresponds to the DOP of normalized Stokes vectors. Because the transmission through the anterior segments of the eye also influences the polarization state of the probing light, its effect is corrected by compensating for the retardance measured at the retinal surface [82]. En face maps of RNFL retardance are then obtained by measuring the cumulative retardance from the retinal surface to posterior retinal layers that features a stronger backscattering signal than the posterior interface of the RNFL.

The optic axis image reveals the radial orientation of the axons, expanding from the optic nerve throughout the entire retina. The RNFL is thickest close to the optic nerve head. Deriving RNFL birefringence requires dividing the roundtrip RNFL retardance by its thickness or fitting the cumulative depth-dependent retardance evolution across the RNFL.

A recent study by Desissaire et al. compared the RNFL birefringence in seven patients with glaucoma and six patients with diabetic retinopathy to the RNFL birefringence of healthy controls [118]. Although glaucoma resulted in a slight reduction of the birefringence compared to healthy controls, the difference was not statistically significant. On the other hand, diabetic retinopathy showed significantly lower birefringence. Depth-resolved tissue birefringence reconstructed from measurements with two multiplexed input states has also been used to investigate birefringence of the RNFL and other structures in the eye [91]. A similar approach, including detailed correction of system polarization distortions [12], further employed the consistency of the optic axis within the RNFL for its automated segmentation [111]. Reported RNFL values are in the range of $\Delta n = \sim 1\text{-}4 \times 10^{-4}$, which results in a round-trip retardance of only 10-20°, depending on the wavelength, across the RNFL with a mean thickness of a little more than 100 μm. Reliable measurements of this small retardance from a weakly scattering sample are inherently challenging and rely on a combination of optimal instrumentation, robust reconstruction, and good measurement protocols.

In addition to glaucoma, many CNS neurodegenerative diseases, such as Alzheimer's, Parkinson's, and multiple sclerosis (MS) can lead to axonal degeneration of the retinal ganglion cell axons, which provides further clinical motivation for reliable and accurate RNFL birefringence measurements. Indeed, decreased birefringence of the peripapillary RNFL has been reported in a cohort of 66 MS patients, independent of the accompanied RNFL thinning [46]. Decreased birefringence could indicate microtubule abnormality and serve as a potential biomarker for detecting early neurodegeneration.

Beyond the retina, PS-OCT has shown promise to help in the detection of subretinal fibrosis, which results in the deposition of collagen-rich tissue with pronounced birefringence [36, 86]. PS-OCT systems employing longer wavelengths at 1 μm can also detect reliable signals from the sclera behind the retinal layers and the choroid. The sclera is a collagen-rich outer layer of the eyeball, visible at the front of the eye by its white color. Leveraging advanced, depth-resolved reconstruction of both birefringence and optic axis orientation, Willemse et al. revealed layers of the peripapillary sclera with distinct optic axis orientations [112]. The subretinal sclera plays a role in the development of glaucoma and high myopia, and there is a clinical need for tools to investigate the pathophysiology of both disorders.

In the anterior segment of the eye, PS-OCT has been used to monitor filtering blebs, which are generated during trabeculectomy surgery required in advanced cases of glaucoma to reduce the intraocular pressure. The filtering bleb is created in the sclera next to the iris and bypasses the drainage of the aqueous humor of the

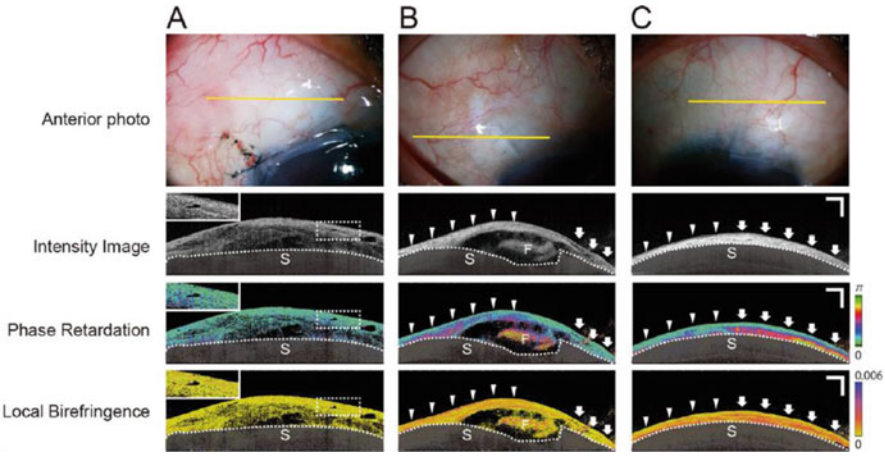


Fig. 7 Representative intensity, phase retardation, and local birefringence images. (**a**) Good bleb function. An 80-year-old woman, 1 month after trabeculectomy, had intraocular pressure (IOP) of 9 mmHg without hypotensive medications. Sclera is covered with a semitransparent black shadow. The white dotted lines delineate the boundary between the conjunctiva and sclera. (**b**) A 65-year-old woman, 5 years after trabeculectomy, had IOP of 15 mmHg with hypotensive medications. High-reflectivity areas in the intensity image show high phase retardation and local birefringence (arrowheads). Low-reflectivity areas in the intensity image show low phase retardation and local birefringence (arrows). (**c**) Flat bleb. A 76-year-old woman, 3 years after trabeculectomy, had IOP of 15 mmHg with hypotensive medications. Note the hyperreflectivity, high phase retardation, and high birefringence areas (arrows) on the right and the hyperreflectivity, mild phase retardation change, and high birefringence areas (arrowheads) on the left. S, sclera; F, scleral flap. Scale bars, 1 mm. (Reproduced from Fukuda et al. [28])

eye to reduce intraocular pressure. The healing and scaring of the bleb can impact its functionality, and PS-OCT offers a pathway to non-invasive monitoring [28, 52] (Fig. 7). While the collagen-rich sclera provides a prominent backscattering signal and clear birefringence, the cornea, which is likewise composed of collagen, appears mostly transparent and with only modest impact on the transmitted polarization state. Indeed, transmission through the cornea can be compensated from the signal at the retinal surface [82], and the extent to which the angle of incidence on the corneal surface impacts the observed retardation effects remains a topic of investigation [10]. Corneal retardance has also been investigated as a method to diagnose keratoconus [29]. Interestingly, the cornea is composed of thin lamellae of tightly arranged collagen fibrils, with subsequent lamellae having distinct fiber orientations. On a macroscopic scale, these varying orientations avoid significant retardation of the transmitted light and make the cornea a weakly retarding transparent optical element. However, at a microscopic scale, the individual layers act as relatively strong retarders, with a birefringence roughly two orders of magnitude higher than the birefringence of the RNFL [56].

3.2 Intravascular PS-OCT

Unlike retinal imaging, intravascular imaging is more invasive by nature and required the development of fiber-optic probes and imaging catheters [11]. Because of its invasiveness, intravascular (IV) OCT is exclusively used in the catheterization laboratory on patients undergoing percutaneous coronary intervention (PCI). Accompanied by antithrombotic medication, PCI is a frequent treatment method for patients suffering acute coronary events, including myocardial infarction (MI), which remains a leading cause of death. Acute cardiac events are typically caused by the rupture of a coronary atherosclerotic plaque. These plaques consist in the slow build-up of lipid-rich material within the vessel wall, accompanied by chronic inflammation. A fibrous tissue layer separates the lipid-core material from the vessel lumen. Plaques that rupture typically have a thin fibrous cap, are depleted of collagen, and are overlying a large, necrotic, lipid-rich core. When the cap ruptures, the substrate of the lipid-core enters in contact with the arterial blood, which coagulates and forms a thrombus. When the thrombus is nonocclusive, it may become organized and lead to a rapid progression of atherosclerosis which is clinically silent. Depending on the degree and duration of occlusion, there are different manifestations on the electrocardiogram and potentially irreversible damage to the myocardium. Another important mechanism for MI is plaque erosion, whereby thrombus may develop along the vessel wall in areas of damaged endothelium.

During PCI, a catheter is introduced through the femoral or radial artery to gain access to the coronary arteries. The occlusive thrombotic material is removed, and balloon angioplasty is frequently used to open the flow-limiting lesions and reestablish sufficient blood flow. Subsequent placement of a coronary stent, a mesh-like structure placed in the artery with an inflating balloon, helps maintain the gained lumen diameter. Traditionally, PCI is guided solely by coronary angiography, which uses a radiopaque contrast agent administered through the catheter to visualize 2D X-ray projections of the coronary vessel's lumen. Intravascular OCT is a powerful tool to directly visualize plaque features and complement angiographic guidance. OCT facilitates guiding PCI with better physiological outcomes than using coronary angiography alone [4, 68]. After stent implantation, OCT can assess strut apposition and coverage, neointimal hyperplasia, and neoatherosclerosis. Like other instruments used for PCI, the OCT imaging catheter is equipped with a guide-wire provision, which ensures the catheter's insertion path along a thin ($0.0014''$) guide-wire, which is initially placed in the target coronary artery. Because OCT cannot image through blood, OCT imaging is synchronized with the administration of a bolus of contrast agent, which displaces the blood and provides a clear view on the vessel wall, leaving a short time window of a few seconds to perform pullback imaging.

Despite the merits of intravascular OCT imaging, there remains an urgent clinical need for improved imaging methods to characterize plaque morphology and composition more accurately. Current PCI procedures are limited, as there

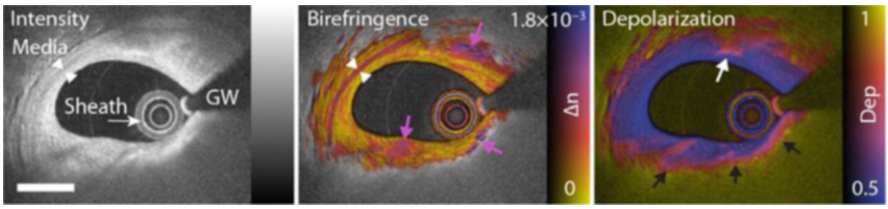


Fig. 8 Intravascular polarimetry in the left circumflex coronary artery of a 52-year-old man. Intensity of the reflection signal shows subsurface plaque morphology in a conventional logarithmic gray scale. White arrow heads indicate tunica media. Birefringence in color hue, overlaid on the reflection signal, reveals regions and layers of distinct birefringence. Birefringence is displayed only in areas of low depolarization. Pink arrows indicate birefringence features. Depolarization in color hue, overlaid on the reflection signal, indicates zones of pronounced depolarization, due to macrophages (white arrow) and lipid-core material (black arrows). Scale bar, 1 mm. (Adapted from Villiger et al. [102])

remains a high rate of major adverse coronary events (MACE) within only a few years following PCI, frequently caused by remote lesions not involved in the initial event [22]. Prospective identification of plaques that are likely to cause future events would offer a path to developing improved therapies. Impaired vascular healing is also increasingly investigated to explain disease progress and the development of acute coronary syndrome and requires refined insight into the make-up and composition of coronary atherosclerotic lesions.

The relevance of additional contrast available to PS-OCT for plaque characterization has already been suggested in the initial report of PS-OCT [42], which used a calf coronary artery as sample. Interstitial collagen and smooth muscle cells (SMC) play a critical role in plaque stability and vascular healing, and they feature birefringence. Microscopic PS-OCT has been used early on to leverage this contrast mechanism [30, 53]. Birefringence measured with PS-OCT in human aortic plaques ex vivo positively correlated with histologically measured collagen and SMC content [70]. Investigating this contrast with intravascular catheter-based imaging motivated the development of the required advanced reconstruction strategies that compensate for system and catheter transmission and recover depth-resolved tissue birefringence. Intravascular PS-OCT then enabled ex vivo validation [102] and clinical pilot studies [74, 77, 95]. Figure 8 displays an example of intravascular polarimetry in the coronary artery of a 52-year-old man. Birefringence maps reinforce several features that are already discernible in the OCT intensity image. The tunica media exhibits high birefringence, frequently bordered by fine lines of low birefringence at the locations of the internal and external elastic laminae. The contrast of the media is less pronounced and more variable in conventional OCT images. Furthermore, fibrous tissue often exhibits heterogeneous birefringence across layers that appear homogeneous in the OCT intensity images. In line with the original motivation for intravascular PS-OCT, a clinical pilot study of intravascular polarimetry found lower birefringence in the caps of unstable lesions that caused acute events than in caps of stable lesions [74].

Depolarization complements birefringence and provides the polarimetric characterization of tissues rich in lipid, cholesterol crystals, and macrophages. The precise mechanism leading to this signature remains under investigation but is likely due to multiple scattering of light by lipid droplets that approach or exceed in size the wavelength used for OCT. Whereas conventional OCT lacks the ability to identify the presence of a necrotic core, which is a critical parameter of plaque stability, depolarization offers additional insight that may enable such differentiation.

Although intravascular polarimetry is performed through a spinning catheter in a beating heart, with continued catheter and fiber motion, repeatability analysis comparing repeat pullbacks in patients showed convincing agreement [103]. Intravascular polarimetry also enabled investigation of the polarimetric signatures of vascular tissue response to drug-eluting stent implantation [77] and of coronary thrombus in patients [96].

The further development of reconstructing the depth-resolved optic axis orientation through a rotating catheter [100] provided insight into the physical orientation of the fibrillar tissue structures giving rise to birefringence. As shown in Fig. 9, the SMC in the tunica media are oriented in a circumferential direction, while the optic axis of intimal tissue aligns mostly in the longitudinal direction. In advanced atherosclerotic lesions, the optic axis orientation frequently revealed distinct tissue layers that appear continuous in conventional OCT images and that also feature remarkably uniform scalar birefringence (Fig. 9). Optic axis orientation may provide unique insight into the progression of coronary atherosclerosis and the healing of silent plaque ruptures [26, 99]. The vascular healing response may lead to a distinct orientation of the organizing thrombus and the deposited collagen, producing noticeable features in the optic axis orientation.

Intravascular polarimetry with PS-OCT provides quantitative metrics of tissue polarization properties. Its clinical prospects have recently been reviewed in detail [75, 76]. Although it requires a dedicated PS-OCT console, it employs conventional intravascular imaging catheters which facilitates clinical translation. Compared to the eye, the tissues of the vessel wall scatter more strongly and feature higher birefringence, making them more amenable to polarimetric imaging. On the other hand, imaging through a spinning catheter complicates the management of polarization states, and the position of the catheter within the lumen is difficult to control, which can lead to oblique imaging angles. Future progress with advanced processing may increase axial resolution, which is currently limited due to spectral binning [105], and would improve the assessment of the birefringence in thin fibrous caps. Improving the robustness of the reconstruction of depth-resolved optic axis orientation is also needed. Lastly, improved analysis of depolarization may be critical for better identifying plaques with large necrotic cores and improving risk stratification of patients with coronary artery disease.

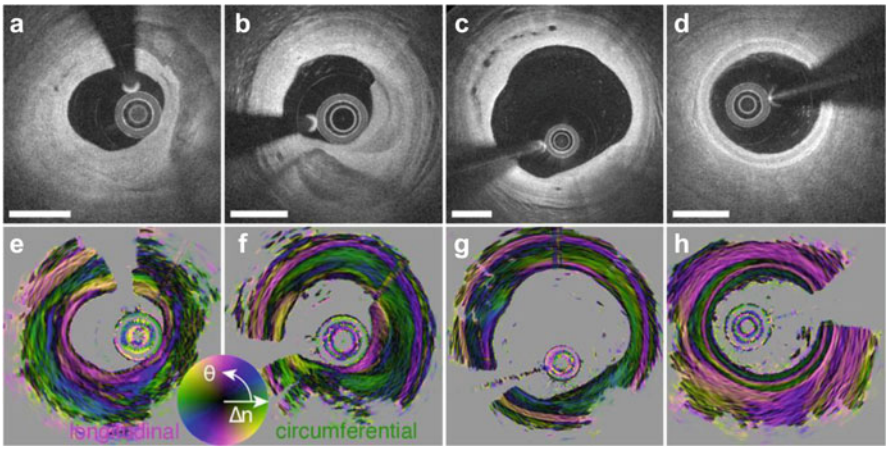


Fig. 9 Depth-resolved optic axis maps in the coronary arteries of human patients. (**a**, **e**) Right coronary artery of a 69-year-old man presenting with unstable angina. (**b**, **f**) Right coronary artery of a 73-year-old man with stable angina. (**c**, **g**) Right coronary artery of a 69-year-old man with stable angina. (**d**, **h**) Left anterior descending coronary artery in a 61-year-old man with stable angina. The three layers of the coronary arterial wall are indicated with white arrowheads in (**d**). The media layer is highlighted with purple arrowheads in (**e**–**h**). Scale bars, 1 mm. (Reproduced from Villiger et al. [100])

3.3 Other Clinical Applications of Catheter-Based PS-OCT

The geometry of helically scanning fiber probes is ideal for organs with cylindrical lumen, such as blood vessels, the esophagus, or airways in the lung. An OCT catheter can be easily deployed through the working channel of a regular bronchoscope, where OCT has been investigated to help guide transbronchial biopsy [39] and for minimally invasive diagnosis of pulmonary fibrosis [38]. The potential of PS-OCT to enhance biopsy yield by avoiding collagen-rich desmoplastic tissue surrounding tumor nodules has been investigated on ex vivo samples [37, 40]. Healthy airway is surrounded by a thin layer of airway smooth muscle (ASM). Thickening and contraction of the ASM are considered a primary cause of asthma, and the ability to image and quantify ASM could guide and improve patient therapy. However, the scattering properties of ASM are similar to those of surrounding connective tissue, and identification with conventional OCT is challenging. Adams et al. leveraged the unique birefringence properties of ASM, with an optic axis orientation circumferential to the airway, to differentiate ASM from surrounding tissue, including birefringent cartilage [1], which enabled a new avenue for researching allergic asthma [16]. Further refined reconstruction which recovers the depth-resolved optic axis orientation and employed the consistency of the optic axis orientation to automatically threshold birefringent tissues was used in a few patients [98] to assess the ASM reduction in response to bronchial thermoplasty, an emerging therapy for severe asthma.

3.4 Microscopic PS-OCT of the Skin and Other Tissues

One of the first investigated potential clinical applications of PS-OCT was the assessment of burn depth in skin [80]. Burn injuries extending to the reticular dermis heal slowly and benefit from excisional surgery and grafting. In comparison, burns limited to the superficial dermis heal spontaneously. In contrast to the normal skin, burned skin lacks birefringence, because the thermal injury denatures the collagen proteins. PS-OCT can serve to identify the depth of thermal injury. However, OCT's field of view makes it challenging to cover large surface areas, often encountered in burn victims. Imaging depth may be sufficient for the relevant partial thickness burns, but it is easily impaired by the presence of blood or other contamination on the burn wound surface. While higher-speed OCT systems can mitigate some of these challenges [50], it remains unclear if a contact imaging method is clinically acceptable.

In normal skin, highly birefringent collagen fibrils are organized in a mesh-like, interwoven arrangement. As shown in Fig. 4, this results in difficult-to-interpret cumulative retardance images. In scars, the normal organization of skin is disturbed, resulting in denser and more regularly aligned collagen, which in turn results in more pronounced cumulative retardance. Fitting the cumulative retardance across the visible depth in patients with hypertrophic or normotrophic scars resulted in a higher retardance rate in scars than in the normal skin [32]. However, fitting of cumulative retardance assumes homogeneous retardance with constant optic axis orientation, which is not the case for normal skin. Indeed, assessment of depth-resolved scalar birefringence only showed modestly higher birefringence in scars than in normal skin [45]. Furthermore, a statistically non-significant correlation between PS-OCT birefringence and collagen content assessed with histopathology was found. It is possible that the inspection of depth-resolved optic axis orientation would offer additional insight. Additional metrics related to the optic axis distribution may prove helpful in the assessment of the skin and scars [57].

Tumors are generally highly cellular without organization that would lead to notable birefringence. In contrast, many tissues surrounding tumors are highly organized and oftentimes birefringent. PS-OCT has thus been investigated to delineate tumor margins, for instance, in breast cancer [81, 101, 109]. In breast tissue, it is straightforward to identify adipose tissue due to its characteristic structural signal. However, scattering contrast between malignant tissue and the surrounding, uninvolved connective tissue is limited and greatly improved with PS-OCT. PS-OCT may also be useful in diagnosing and grading oral squamous cell carcinoma. Depth-resolved PS-OCT of the oral mucosa in healthy subjects revealed rich, highly organized birefringence features [106], which are expected to be altered by pathology. Lastly, white matter in the brain is highly birefringent and may offer useful contrast to delineate glioma and preserve critical white matter tracts [58].

PS-OCT has also been used to image cartilage and assess degeneration due to osteoarthritis. The strong birefringence of healthy cartilage results in banding

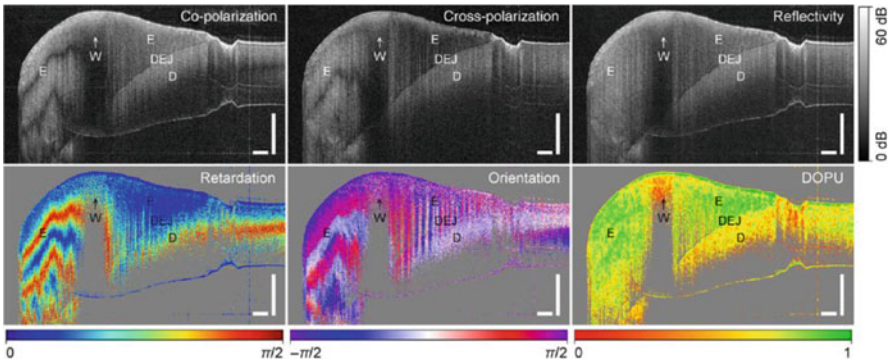


Fig. 10 OCT B-scan of the co-, cross-polarization channels and the determined reflectivity image showing enamel (E), the dentin-enamel junction (DEJ), dentin (D), and white spot (W) of an extracted human molar tooth. For polarization contrast imaging, the retardation, the fast axis orientation, and the DOPU (3 × 3 kernel) are displayed. Scale bars correspond to 500 μm. (Reproduced from Golde et al. [31])

patterns in conventional OCT imaging, whereby the polarization state of light from a given tissue depth is orthogonal to the reference beam and fails to create interference [17]. However, the detailed organization of collagen in the cartilage is involved, with an optic axis orientation that partially aligns with the beam direction, requiring advanced analysis strategies [117] or retrieval of the full 3D optic axis orientation [47].

Both dentin, the core of a tooth, and enamel, its outer surface, are birefringent. The crystalline structure of enamel is altered during the process of demineralization, occurring, e.g., as a result of dental caries. PS-OCT and cross-polarized OCT, which simply detects light with an orthogonal polarization to that incident on the sample, have long been investigated for identification of dental lesions, including some clinical pilot studies [27, 63, 89]. Depolarization has also been proposed for the detection of demineralization [31] (Fig. 10). PS-OCT could serve for screening applications in early caries diagnostics.

3.5 Future Developments

Every OCT system is "sensitive to polarization," due to its coherent measurement scheme. However, converting the detected polarization effects into meaningful tissue contrast requires the additional instrumental and algorithmic complexity of polarization-sensitive OCT. As a result, PS-OCT has been established as a valuable research instrument, but mostly in the hands of highly specialized research groups, and has not found widespread traction, despite commercially available PS-OCT instruments (Thorlabs).

Ophthalmic PS-OCT will continue to provide relevant insight into ocular tissues and their pathological changes. Detailed understanding of the eye's polarization properties can also be important to help interpret conventional OCT tomograms. There is a compelling clinical rationale for measuring RNFL birefringence. However, its optical and polarization characteristics make detailed investigation challenging. It is possible that the increased clinical need for methods to monitor neurodegenerative diseases spurs additional technical development that enables more accurate and detailed investigation of these delicate nerve fibers. RNFL polarization properties may serve as biomarkers for a range of retinal and CNS conditions. Widespread implementation of PS-OCT into commercial ophthalmic imaging platforms is hindered by the increased system complexity. The need for polarization-diverse detection increases hardware cost and can only be justified by a significant gain in clinical utility.

Compared to ophthalmic PS-OCT, intravascular PS-OCT is at the very beginning of clinical investigations. The layers of the coronary vessel wall feature pronounced polarization features that simplify the interpretation of intravascular tomograms. Furthermore, initial quantitative analysis demonstrated potential for identifying weak fibrous caps that are prone to rupture and for improving OCT's sensitivity to detecting necrotic core fibroatheromas. Furthermore, the possible insight that depth-resolved optic axis may offer into plaque healing following silent ruptures could help in addressing elusive questions regarding the interplay of vascular healing and disease progression. The intrinsic variation of the incident polarization states during probe rotation results in a continuous variation in the detected polarization states, independent of the tissue properties. To avoid the resulting polarization fading when using a single detection channel, commercial IV-OCT systems either integrate polarization-diverse detection or actively manage the polarization state of the reference arm. This simplifies the technical challenge of implementing PS-OCT and potentially facilitates its integration into commercial instruments. Recent progress demonstrated a pathway to catheter-based single input state PS-OCT that leverages the residual system-induced spectral polarization state variations to overcome the previous limitations of using a single input state in layered tissues [114], further reducing the technical barriers to integration into existing architectures. Such developments may prove critical to enable larger-scale clinical investigations that will be needed to demonstrate the clinical potential of any polarization-based metric.

Rotational, side-looking OCT probes comply well with the geometry of biopsy needles. Several preliminary ex vivo studies have confirmed the potential of PS-OCT in identifying biopsy targets. The recent progress permitting polarization-sensitive imaging with rotation probes paves the way to implementing this contrast in a practical clinical setting, e.g., in the lung, or for transperineal template mapping biopsy in prostate cancer. Other tissues and organs would benefit from forward-viewing imaging geometries, ideally compatible with the working channel of contemporary flexible white light endoscopes, which are actively being developed but remain to be translated into the clinic.

Owing to the pronounced birefringence of myelinated axons, PS-OCT has recently gained traction as a microscopy platform to image white matter fiber

tracts in ex vivo brain tissue [107, 108]. The same contrast could serve to guide stereotactic neurosurgeries with intracranial probes, using the existing side-looking probe geometry, which has previously been used in combination with dynamic OCT contrast [85]. The birefringence of myelinated axons extends to peripheral nerves and has shown promise in several areas. Bench-top PS-OCT has investigated birefringence of nerves in the prostate ex vivo [116], and animal studies have shown that PS-OCT can assess nerve myelination [71]. Such contrast could be decisive to guide nerve-sparing surgeries, especially if combined with new OCT platforms that enable fast and large field of view imaging [88].

PS-OCT's resolution is sufficient to visualize individual tissue layers, yet the observed polarization features are a compound effect of the microscopic tissue organization, averaged within the resolution volume defined by OCT's coherence gate and lateral beam width. While individual studies have investigated the association of the birefringence observed in PS-OCT with histopathological metrics for specific tissue types, the majority of PS-OCT studies merely phenomenologically evaluate polarization properties. Future PS-OCT efforts would benefit from improved understanding of the association between microscopic tissue organization and macroscopically observed birefringence. Several tissues present a combination of form and intrinsic birefringence, of which the form birefringence directly depends on the tissue arrangement of fibrillar microstructures [41]. For instance, the negative uniaxial birefringence observed in myelinated axons at OCT resolution has been shown to be the combined effect of the positively birefringent axon and equally positively birefringent myelin but with an optic axis oriented radially around the axon [69]. This model has been helpful for the interpretation of polarized light imaging of brain sections ex vivo [5]. Similar models, as well as increased comparison and validation with polarized light microscopy [73] could be important to underpin future clinical PS-OCT work.

4 Conclusions

Polarization provides a powerful contrast mechanism to complement conventional OCT tomograms of backscattering intensity. PS-OCT provides insight into the make-up and physical orientation of the tissue microstructure beyond the spatial resolution of OCT, without compromising OCT's imaging speed and large field of view that are essential for clinical applications. PS-OCT has been investigated since the inception of OCT and has become a valuable research instrument but still awaits commercial translation into clinical instruments. More widespread adoption of PS-OCT has been hindered by its increased system complexity, the need for involved reconstruction, and the lack of a unifying signal processing framework that could accommodate various hardware implementations. Because OCT is a mature and established imaging modality, additional inclusion of PS cannot compromise conventional OCT operation and has to be achieved with a reasonable trade-off between added cost and potential benefit. Meeting this challenge offers the enormous opportunity to rapidly translate PS-OCT and achieve a clinical impact.

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Tractography Using Polarization-Sensitive Optical Coherence Tomography



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1 Introduction

The term "tractography" refers to a special imaging technology for visualizing fiber organization in the tissue. This term is often used in association with "diffusion" magnetic resonance imaging (MRI) which was originally developed to measure nerve fiber orientations by imaging the anisotropic diffusion of water molecules in white matter of brain [1, 2]. Once the fiber orientation data are measured at each pixel in the image space, computational tracking algorithms can be used to connect relevant neighboring pixels to construct a "tract" representation of the fiber structure, i.e., the tractography [3]. Diffusion MRI is considered as a standard tool in the field of brain connectome that focuses on investigating the sophisticated neural network in the brain. It has also been proven valuable in visualization of fiber organization in other fibrous tissues such as the heart and skeletal muscle [4, 5].

Although the term "tractography" rarely appears outside the MRI field, fiber orientation measurements are also used to study tissue organization at microscopic levels because fiber organization plays important roles in maintaining normal functions in fibrous tissues. Although fiber orientation can be estimated using conventional staining-based histological imaging [6], image processing is necessary to automate the process to obtain 2D fiber tractography in practice [7]. Fiber orientation measurement has also been a focus of many advanced high-resolution imaging techniques such as x-ray scattering [8], light scattering [9], scanning electron microscopy (SEM) [10], and nonlinear imaging microscopy [11]. Among various technologies, polarized light microscopy (PLM) is specifically developed

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to measure fiber orientation in materials by exploring the optical polarization-dependent responses in birefringent materials [12, 13]. In addition to being a commonly used microscopic imaging tool in crystallography and materials science, PLM is a well-recognized tool in studying tissue fiber structures in particular for visualization of collagen organization [14].

Optical coherence tomography (OCT) is a high-resolution imaging technology that is well-suited for in vivo applications due to its capability to acquire depth-resolved tomography images using coherent gating [15]. It has also been explored to visualize fiber orientation in fibrous tissues such as the brain white matter [16, 17], cardiac tissue [18, 19], and collagen tissue [20]. OCT can image the fiber orientation in a volume sample without the need of sample sectioning that is usually needed in many other microscopic imaging methods. As in all intensity-based imaging systems, OCT detects fibers based on the contrast difference in image intensity between fibers and surrounding tissues. Therefore, OCT can only obtain fiber orientation when its spatial resolution is sufficient to resolve individual fibers or fiber bundles.

As a functional extension of OCT, polarization-sensitive OCT (PSOCT) incorporates optical polarization measurements to characterize optical birefringent properties of fibrous tissues in addition to the intensity contrast in traditional OCT. Similar to optical polarization-based PLM, PSOCT can obtain the phase retardation and optic axis information which can then be used to derive fiber orientation at the pixel level [21–23]. Although PSOCT shares with PLM the basic optical polarization principles in their measurements, the depth-resolved sectioning capability allows PSOCT to imaging nondestructively the fiber orientation at different depths of the sample. Recent studies demonstrated that PSOCT-based tractography can successfully visualize fiber organization in many biological tissues such as the skeletal muscle [24], heart [25], artery [26], cartilage [27], cornea [28], and skin [29]. Due to its high imaging speed, large field of view, and optical sectioning capability, PSOCT provides an effective way for high-resolution tractography imaging study of fiber organization in the tissue.

2 Optical Birefringence and Fiber Orientation Measurements

Both PLM and PSOCT rely on optical polarization to measure fiber orientation in birefringent samples. Optical birefringence describes the “double refraction” phenomenon that may happen in anisotropic materials such as crystals, where a light experiences different optical refractive indices depending on the light’s polarization and propagation direction. Therefore, an incident light may be refracted into different directions depending on polarization, i.e., the “double refraction.”

In a general birefringent material, the optical refractive index may be different along the three Cartesian coordinates. However, a fibrous tissue usually demonstrates “uniaxial” birefringence due to the cylindrical symmetry in its geometric

profile. The birefringent properties of a uniaxial sample can be described using two refractive indices: the ordinary refractive index n_o and the extraordinary refractive index n_e. A linearly polarized light experiences n_o when its polarization is perpendicular to the fibers, and n_e when its polarization is parallel to the fibers. These two lights of orthogonal polarization directions experience the maximum amount of difference in refractive index $n_e - n_o$. Such a difference in refractive index induces a phase retardation between those two lights after passing through a distance d:

$$\delta = \frac{2\pi}{\lambda} \times d \times \Delta n, \tag{1}$$

where $\Delta n = n_e - n_o$ is defined as the "birefringence" value of the uniaxial material, d is the sample thickness, and λ is the wavelength of the light.

In fibrous samples, light absorption can also become anisotropic depending on polarization. Incident lights with polarization parallel and perpendicular to the fibers may experience different absorption coefficients of μ_e and μ_o, respectively. Such an anisotropic absorption can be characterized using the diattenuation coefficient:

$$\sigma = (\mu_e - \mu_o) \times d. \tag{2}$$

When defining the birefringence, the extraordinary refractive index is aligned with the fiber orientation. The value of birefringence Δn can be either positive or negative. In samples with positive birefringence such as collagen and muscle, n_e is greater than n_o, and the "slow" optic axis (speed of light c/n_e is smaller than c/n_o) is aligned with fiber orientation. In samples with negative birefringence such as nerve fibers, the "fast" optic axis is aligned with the fibers. In either case, the fiber orientation can be determined once the optic axis is measured.

In polarization optics, the complex 2×2 Jones matrix and real number 4×4 Mueller matrix are widely used to facilitate quantitative analyses of polarization responses in a sample. Because Mueller calculus is based on optical intensity, it is commonly used in light intensity-based imaging systems such as PLM [12, 13], whereas Jones calculus is more popular in interference-based systems such PSOCT [30, 31]. Nevertheless, the Mueller matrix and the associated Stokes vector can be constructed by transforming the corresponding Jones matrix and vector.

The Jones matrix of a uniaxial birefringent sample can be described using the Jones matrix of a linear retarder $\mathbf{J}_R(\theta, \eta)$ that is oriented at θ (i.e., the optic axis) and has a complex retardation of η [32]:

$$\mathbf{J}_R(\theta, \eta) = \mathbf{R}(\theta)\,\mathbf{D}(\eta)\,\mathbf{R}(-\theta) = \begin{bmatrix} \cos\frac{\eta}{2} + i\cos 2\theta \sin\frac{\eta}{2} & i\sin\frac{\eta}{2}\sin 2\theta \\ i\sin\frac{\eta}{2}\sin 2\theta & \cos\frac{\eta}{2} - i\cos 2\theta \sin\frac{\eta}{2} \end{bmatrix} \tag{3}$$

where η is the complex retardation $\delta + i\sigma$ based on the definitions of δ and σ in Eqs. (1) and (2). The matrix $\mathbf{R}(\theta)$ is the Jones matrix of a rotator, and $\mathbf{D}(\eta)$ is a complex

diagonal matrix:

$$\begin{aligned}\mathbf{R}(\theta) &= \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \\ \mathbf{D}(\delta) &= \begin{bmatrix} \exp\left(i\frac{\eta}{2}\right) & 0 \\ 0 & \exp\left(-i\frac{\eta}{2}\right) \end{bmatrix}\end{aligned}. \tag{4}$$

Once a sample's Jones matrix is measured, its optical polarization properties including retardation, diattenuation, and optic axis can be fully determined. For example, the phase retardation δ and diattenuation σ can be calculated from the two complex eigenvalues λ_1 and λ_2 of the Jones matrix in Eq. (3) as

$$\begin{aligned}\delta &= \tan^{-1}\left|\frac{\mathrm{Im}(\lambda_1 \times \lambda_2^*)}{\mathrm{Re}(\lambda_1 \times \lambda_2^*)}\right| \\ \sigma &= \ln\left|\frac{\lambda_2}{\lambda_1}\right|\end{aligned}. \tag{5}$$

The two optic axes θ_i ($i = 1,2$) can be calculated from the two eigenvectors $\mathbf{V}_i$ ($i = 1,2$) of the Jones matrix:

$$\theta_i = \frac{1}{2}\tan^{-1}\frac{2\left|\mathbf{V}_i(2)\right|\left|\mathbf{V}_i(1)\right|\cos\varphi}{\left|\mathbf{V}_i(2)\right|^2 - \left|\mathbf{V}_i(1)\right|^2}, \tag{6}$$

where φ is the phase difference between the two elements of the eigenvector $\mathbf{V}_i(1)$ and $\mathbf{V}_i(2)$. In samples with negligible diattenuation, the complex retardation η becomes the birefringence-induced phase retardation δ of real values (Eq. 1). In such a case, the calculation can be significantly simplified, and it is usually not necessary to measure the full Jones matrix to derive retardation δ and optic axis θ.

Measuring fiber orientation using optic axis has been long utilized in PLM. A PLM can be constructed by incorporating optical polarization components in the optical path of a regular optical microscope to polarize the incident light and analyze the sample-induced polarization changes. Tissue samples have to be sectioned into thin slices for PLM imaging as in conventional histology evaluation. In such a thin sample, polarization properties can be considered homogeneous thickness-wise, although they may vary significantly across the sample in the lateral directions.

In a typical PLM system, the thin birefringent sample is placed between an optical polarizer and a polarization analyzer. After passing through the sample, the induced polarization change of the polarized incident light is assessed with the help of the polarization analyzer to derive the retardation and optic axis. As a sample PLM implementation, the illumination light is circularly polarized after passing through a linear polarizer and a quarter waveplate. The transmitted light is analyzed using a linear polarizer rotating at a constant frequency of ω [33]. The circularly polarized incident light is used because it doesn't require a prior knowledge of the fiber orientation in the sample [33]. The Jones vector of the detected light can be calculated using Jones calculus as

$$\begin{bmatrix} E_{\mathrm{H}} \\ E_{\mathrm{V}} \end{bmatrix} = \mathbf{J}_P(\omega t)\,\mathbf{J}_R(\theta, \delta)\,\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ i \end{bmatrix} \tag{7}$$

where $\mathbf{J}_R(\theta, \delta)$ is the Jones matrix of the birefringent sample assuming the diattenuation is negligible (Eq. 3). $\mathbf{J}_P(\omega t)$ is the Jones matrix of a linear polarizer oriented at angle ωt. The measured light intensity can then be derived from the Jones vector as

$$I = |E_{\mathrm{H}}|^2 + |E_{\mathrm{V}}|^2 = \frac{I_0}{2}\left[1 + \sin\delta\,\sin\left(2\omega t - 2\theta\right)\right]. \tag{8}$$

Equation (8) shows that the light intensity follows a sinusoidal modulation at two times of the rotational frequency of the analyzer, which can be precisely detected using a lock-in system. The retardation δ can be obtained as the amplitude, and the optic axis can be obtained from the phase angle in the measured intensity. An array detector such as a camera can be used to form an image in PLM. The detection algorithm such as Eq. (8) needs to be applied to the signals detected at each image pixel to construct images of the retardation and optic axis. In samples where diattenuation has to be considered, additional measurements with a different incident polarization are usually necessary to fully determine the sample Jones matrix [32].

3 Fiber Orientation Measurements in PSOCT

PSOCT measures the phase retardation and optic axis in birefringent samples based on the same polarization theories used in PLM. However, there are some major differences in implementation which make PSOCT more capable and challenging at the same time. First of all, in contrast to the intensity measurement in PLM, OCT signals are measured using low coherence interference to achieve depth gating, often in a Michelson Interferometer configuration. PSOCT often uses polarization-diverse detection where the two orthogonal polarization components of the backscattered light are detected. Because both amplitude and phase information can be obtained from interference signals, Jones vectors of the backscattered light can be conveniently obtained to construct the sample Jones matrix. However, Mueller matrix can still be obtained by transforming the measured Jones matrix and used to study depolarization effects induced by assembled average [31]. In addition, OCT works in the reflection configuration so that the light passes through the sample twice (in and out, usually referred to as "round-trip"). Because sample sectioning is not required in PSOCT imaging, the assumption of depth-wise homogeneous polarization properties used in PLM may not hold true in PSOCT measurements of thick samples.

There are a variety of PSOCT implementations which have been reviewed in details in the literature [30, 31]. One classic implementation uses circularly

polarized incident light similar to the example case of PLM described in the previous section [34]. Specifically, a quarter waveplate oriented at 45° is placed in front of the sample to convert a linearly polarized (horizontally (H) or vertically (V) polarized) incident light into a circularly polarized light, which is then focused into the sample. The "round-trip" backscattered light coming of the sample passes through the quarter waveplate again and is detected by interfering with reference lights in the OCT system. Because of the round-trip, the light experiences twice of the phase retardation inside the sample. If the birefringent sample has negligible diattenuation, its polarization response can be represented using Jones matrix $\mathbf{J}_R(\theta, 2\delta)$ as described in Eqs. (3) and (4). The Jones vector $[E_\mathrm{H}, E_\mathrm{V}]^\mathrm{T}$ of the backscattered light can be calculated using the Jones calculus as

$$\begin{bmatrix} E_\mathrm{H} \\ E_\mathrm{V} \end{bmatrix} = \mathbf{J}_R\left(45°, \frac{\pi}{2}\right)\mathbf{J}_R\left(\theta, 2\delta\right)\mathbf{J}_R\left(45°, \frac{\pi}{2}\right)\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} e^{i2\theta}\sin\delta \\ \cos\delta \end{bmatrix}, \tag{9}$$

where $\mathbf{J}_R(45°, \pi/2)$ is the Jones matrix of the quarter waveplate oriented at 45°. From Eq. (9), the phase retardation of the sample can be calculated from the amplitudes of the Jones vector, $\delta = \tan^{-1}(E_\mathrm{H}/E_\mathrm{V})$, and the optic axis can be calculated from the phase difference $\Delta\phi$ between E_H and E_V as $\theta = \Delta\phi/2$.

Similar to the case of PLM, the aforementioned approach using a single circularly polarized incident light is not able to accurately measure the birefringence and diattenuation in samples with a significant diattenuation [30]. In such cases, full depth-resolved Jones matrix of the sample needs to be measured using different incident polarizations [30, 31]. Once the Jones matrix is measured, the phase retardation, diattenuation, and optic axis can be determined, for example, from the eigenvalues and eigenvectors of the Jones matrix as described in Eqs. (5) and (6).

Due to the round-trip optical path inside the sample, it is important to note that the aforementioned PSOCT measurements are affected by the "cumulative" polarization properties from the sample surface to the measurement depth as illustrated in Fig. 1. As a quantitative illustration of a single A-line PSOCT measurement, the sample can be digitized into multiple grids (or pixels in a single A-scan) in the depth direction along the incident light with each grid size equal to the depth resolution of the PSOCT system (Fig. 1). The depth resolution in a typical PSOCT system is usually ~10 μm, similar to the thickness in a histology section. Within such a small thickness, the tissue inside each grid can be considered homogeneous so that its polarization properties can be fully described using a Jones matrix of a retarder as described previously.

The single-trip Jones matrix from the sample surface to depth d_n is the product of a series of linear retarders, which is equivalent to a general retarder:

$$\mathbf{J}_n^\mathrm{ST} = \mathbf{J}_n \cdots \mathbf{J}_2\mathbf{J}_1 = \mathbf{R}\left(\alpha_n\right)\mathbf{J}_R\left(\phi_n, \rho_n\right) = \mathbf{R}\left(\alpha_n\right)\mathbf{R}\left(\phi_n\right)\mathbf{N}\left(\rho_n\right)\mathbf{R}\left(-\phi_n\right), \tag{10}$$

where $\mathbf{J}_i$ represents the Jones matrix of the tissue in the ith grid (Fig. 1). $\mathbf{R}(\alpha_n)$ represents a general rotator (Eq. 4). The variable ρ_n represents the effective

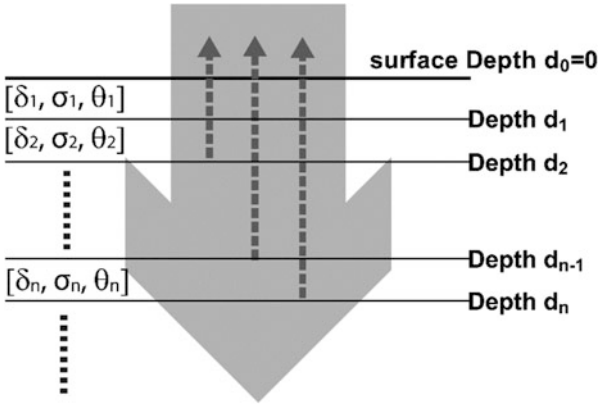


Fig. 1 An illustration of the depth-resolved birefringence measurements in PSOCT. After being incident into the sample from the surface, the light may be backscattered at different depths. The tissue can be digitized into discrete grids with the depth resolution in PSOCT as the grid size. The polarization properties of each grid can be represented using phase retardation δ, diattenuation σ, and optic axis θ

"cumulative" phase retardation, and ϕ_n represents the effective "cumulative" optic axis. The "round-trip" Jones matrix of the sample from depth d_n can be obtained from Eq. (10) as

$$\mathbf{J}_n^{\mathrm{RT}} = \left(\mathbf{J}_n^{\mathrm{ST}}\right)^{\mathrm{T}} \mathbf{J}_n^{\mathrm{ST}} = \mathbf{R}\left(\phi_n\right) \mathbf{N}\left(2\rho_n\right) \mathbf{R}\left(-\phi_n\right) = \mathbf{J}_R\left(\phi_n, 2\rho_n\right), \tag{11}$$

where the rotator $\mathbf{R}(\alpha_n)$ is canceled out. Therefore, the round-trip Jones matrix measured in a birefringent sample using PSOCT would still appear as a linear retarder with a phase retardation of $2\rho_n$ and optic axis of ϕ_n. As discussed before, ρ_n in general is a complex number with phase retardation as the real component and diattenuation as the imaginary part.

When the actual optic axis remains the same throughout the imaging depth d_n, the measured effective ϕ_n is the true optic axis, and $2\rho_n$ is twice of the true "cumulative" phase retardation in the sample (i.e., the summation of phase retardation from surface to depth n). This is shown in Fig. 2a, b where the sample has depth-dependent birefringence, but with a fixed optic axis. The PSOCT measurements of cumulative retardation and optic axis have the typical "banding" appearance (alternating high and low values) due to phase wrapping. However, the unwrapped optic axis correctly represents the true sample optic axis [35]. The cumulative retardation increases linearly with depth, and the "local" retardation can be calculated as the derivative of the cumulative results.

However, PSOCT measurements do not reveal the true optic axis in samples where the optic axis changes over the depth. The PSOCT results shown in Fig. 2c, d are calculated in a birefringent sample with a depth-dependent optic axis as $\theta = 20^\circ \sin(d\pi/4 - \pi/2)$ where d is the depth. The cumulative retardation obtained

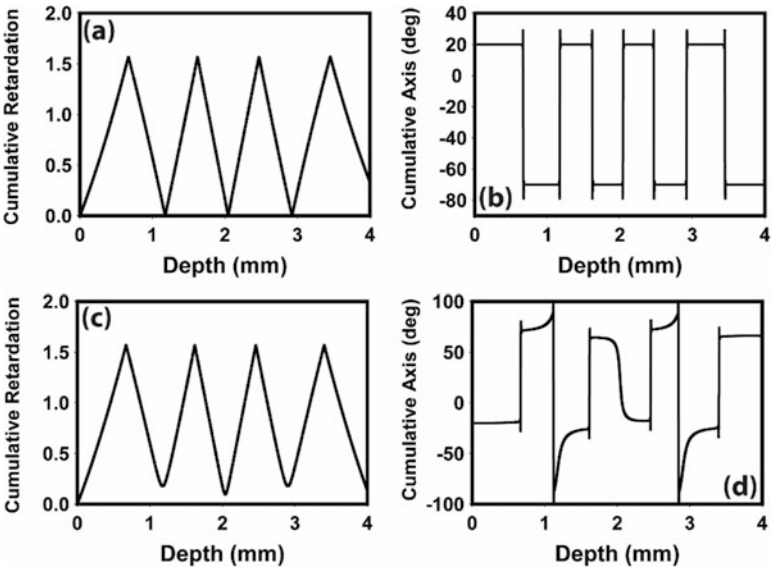


Fig. 2 Simulation results of 1D "cumulative" PSOCT measurements of depth-dependent phase retardation and optic axis in birefringent samples. The results shown in (**a**) and (**b**) are calculated from a sample with depth-independent optic axis at 20° and depth-varying phase retardation of $\delta = 0.0075[1 + \sin(d\pi/4)]$. The results shown in (**c**) and (**d**) are from a sample with the same phase retardation as in (**a**) and (**b**), but the optic axis varies with depth as $\theta = 20°\sin(d\pi/4 - \pi/2)$

(Fig. 2c) starts to deviate from a linear curve and does not reach the minimum value of 0 at phase wrapping depths. The obtained cumulative optic axis varies significantly over the depth and does not have a clear relationship with the true optic axis in the sample.

4 Measuring Local Optic Axis in PSOCT

As discussed above, the PSOCT "cumulative" measurements do not represent the actual depth-resolved optic axis when the fiber orientation varies with depth in the sample. To reveal the true fiber orientation inside a tissue, the cumulative PSOCT measurements have to be untangled to extract the true "local" optic axis for each image pixel (grid in Fig. 1). A straightforward method would be to "peel" off the cumulative PSOCT measurements depth by depth starting from the first depth grid because no cumulative effect exists in the measurement of the first tissue layer (grid from surface to depth d_1 in Fig. 1). However, early attempts of this approach were unsuccessful in obtaining reliable images of fiber orientation in tissues because inaccurate measurements can be accumulated significantly in consecutive matrix operations [36].

An improved algorithm based on Jones matrix was later developed to derive the depth-resolved local optic axis orientation from cumulative PSOCT measurements [23]. By separating the calculation of local retardation and optic axis, this method achieved good accuracy in optic axis measurements [37]. From Eq. (11), the round-trip cumulative PSOCT results can be used to derive the Jones matrix term $\mathbf{J}_R(\phi_n, \rho_n)$ in Eq. (10). When combining $\mathbf{J}_R(\phi_{n-1}, \rho_{n-1})$ obtained at depth d_{n-1} with the round-trip Jones matrix measured at depth d_n, a new Jones matrix can be derived as

$$\mathbf{N}_n = \mathbf{J}_R^{-1}(\phi_{n-1}, \rho_{n-1})\,\mathbf{J}_n^{\mathrm{RT}}\mathbf{J}_R^{-1}(\phi_{n-1}, \rho_{n-1}) = \mathbf{R}(-\alpha_{n-1})\,\mathbf{J}_n^{\mathrm{T}}\mathbf{J}_n\mathbf{R}(\alpha_{n-1}). \tag{12}$$

Equation (12) indicates that the matrix $\mathbf{N}_n$ is a similar matrix transformation of ${\mathbf{J}_n}^{\mathrm{T}}\mathbf{J}_n$ [38]. Therefore, the local complex retardance η_n of the depth grid d_n can be obtained from the eigenvalues of $\mathbf{N}_n$, and the retardance δ and diattenuation σ are the real and imaginary components of its eigenvalues.

To obtain the local optic axis, a modified "cumulative" Jones matrix is constructed from the measured round-trip $\mathbf{J}_R(\phi_n, 2\rho_n)$ in Eq. (11) by using only the real part of ρ_n:

$$\mathbf{J}_n^{\mathrm{RM}} = \mathbf{R}(\phi_n)\,\mathbf{D}(2\kappa_n)\,\mathbf{R}(-\phi_n), \tag{13}$$

where ϕ_n is the effective "cumulative" optic axis calculated from the round-trip Jones matrix and κ_n is the real part of the effective complex "cumulative" retardation ρ_n. Meanwhile, a modified single-trip cumulative Jones matrix from sample surface to depth d_{n-1} can be constructed using the calculated "local" retardation and optic axis at each depth:

$$\mathbf{J}_{n-1}^{\mathrm{SM}} = \mathbf{J}_R(\theta_{n-1}, \delta_{n-1}) \cdots \mathbf{J}_R(\theta_2, \delta_2)\,\mathbf{J}_R(\theta_1, \delta_1), \tag{14}$$

The local Jones matrix of the nth grid can be obtained by removing the single-trip Jones matrix of the $(n-1)$th grid (Eq. 14) from the round-trip Jones matrix (Eq. 13):

$${\mathbf{J}_n^{\mathrm{T}}}\mathbf{J}_n = \left[\left(\mathbf{J}_{n-1}^{\mathrm{SM}}\right)^{\mathrm{T}}\right]^{-1}\mathbf{J}_n^{\mathrm{RM}}\left[\mathbf{J}_{n-1}^{\mathrm{SM}}\right]^{-1}, \tag{15}$$

from which the local optic axis θ_n can be calculated using eigen-decomposition. The process described in Eqs. (13, 14, and 15) is carried out iteratively from the first depth grid where the local Jones matrix $\mathbf{J}_1$ is constructed directly from the cumulative Jones matrix. Before the iteration at the next depth, the local Jones matrix is constructed using the calculated θ and δ (Eq. 3), and the single-trip Jones matrix is updated as in Eq. (14).

Figure 3 shows the calculated 1D "local" birefringent properties of the same birefringent sample used in Fig. 2. The sample in Fig. 3a, b has a fixed optic axis and a depth-varying retardation, whereas both the retardation and optic axis vary with

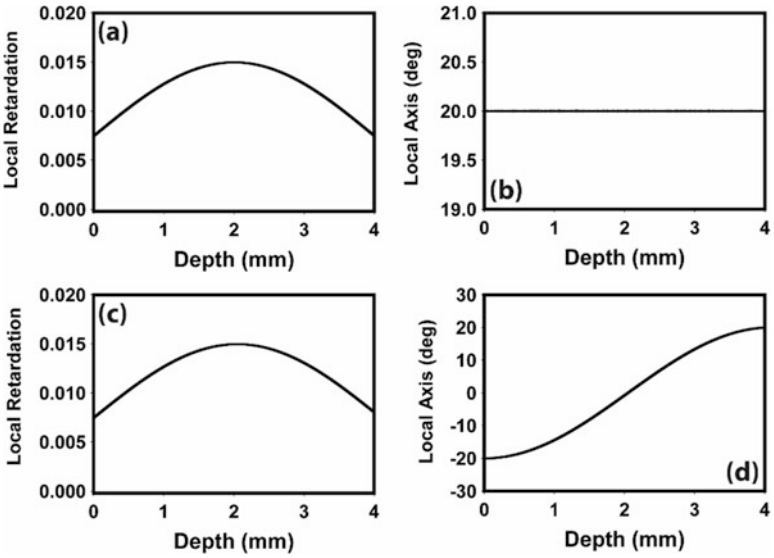


Fig. 3 An illustration of the 1D depth-resolved "local" birefringence measurements in PSOCT calculated using Eqs. (12, 13, 14, and 15). The results shown in (**a**) and (**b**) are calculated from a sample with depth-independent optic axis at 20° and depth-varying phase retardation of $\delta = 0.0075[1 + \sin(d\pi/4)]$. The results shown in (**c**) and (**d**) are from a sample with the same phase retardation as in (**a**) and (**b**), but the optic axis varies with depth as $\theta = 20°\sin(d\pi/4 - \pi/2)$

depth in the sample shown in Fig. 3c, d. In both samples, the true depth-resolved retardation and optic axis data are correctly calculated.

The advantage of "local" birefringence measurements over the conventional "cumulative" PSOCT measurements can be better illustrated in 2D imaging examples. Figure 4 shows an example of "local" fiber orientation and retardation obtained in a PSOCT image of a mouse tibialis anterior (TA) muscle. The sample was imaged using a Jones matrix PSOCT system described elsewhere [39]. Briefly, the system is 847.8-nm wavelength, bulk optics-based spectral domain PSOCT which measures the pixel-wise Jones matrix in 3D with a 5.9-μm depth resolution, and a 12.4-μm lateral resolution. The same algorithm as described in Eqs. (12, 13, 14, and 15) was applied to each 1D A-scan of the Jones matrix measurements to calculate the local retardation and optic axis at each image pixel of the entire 3D Jones matrix dataset.

Figure 4(a) shows an en face intensity image of the tissue obtained at ~300 μm below the muscle surface. The corresponding cumulative retardation and optic axis images are shown in Fig. 4b, c. The banded appearance due to phase wrapping that appears in the 1D results (Fig. 2) also exists in these 2D cumulative results, making it impossible to correctly interpret the distribution of birefringence and fiber orientation. In contrast, the local retardation image shown in Fig. 4d clearly reveals a region with low birefringence at the lower left side of the sample. The fiber orientation image obtained from the local optic axis can be visualized in either pseudo-color image (Fig. 4e) or more intuitively as tractography using streamline

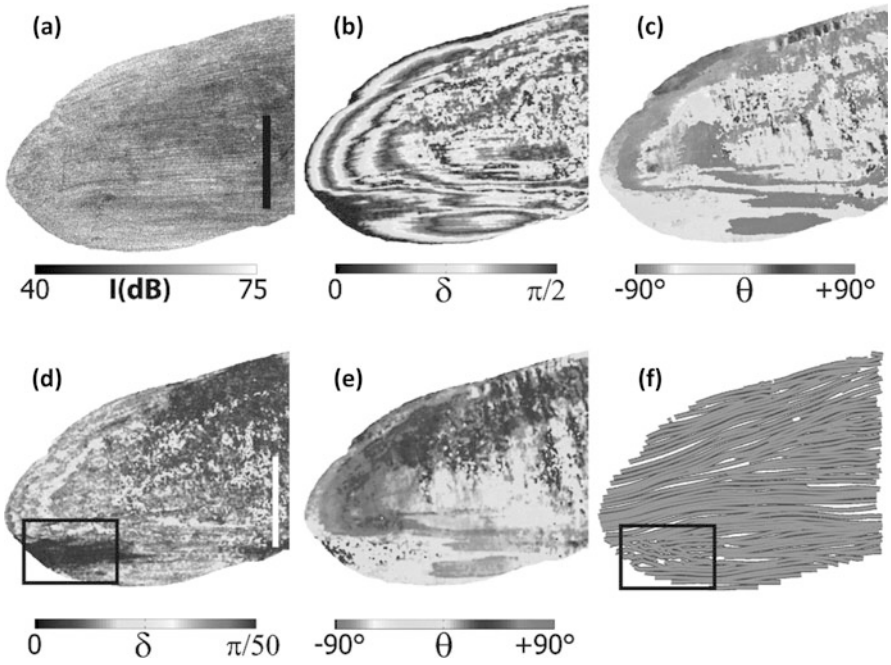


Fig. 4 An example of 2D en face measurements of retardation and optic axis in a mouse tibialis anterior (TA) muscle. (**a**) OCT intensity image, (**b**) cumulative retardation, (**c**) cumulative optic axis, (**d**) local retardation, (**e**) local optic axis, and (**f**) 2D tractography

representation (Fig. 4f). A close examination of the tractography reveals that the fiber orientation appears to be random in the lower left part of the region with low birefringence. Histology analysis confirmed muscle damage in this region [24]. Therefore, the tractography enabled by local optic axis measurements in PSOCT is valuable for evaluating microscopic-level fiber organization/damage in the tissue.

5 3D Fiber Orientation Measurements in PSOCT

It is important to note that the optic axis obtained in either PLM or PSOCT measurement is the optic axis within the evaluation plane that is perpendicular to the incident light (θ in Fig. 5). The out-of-plane inclination angle with the incident light (α in Fig. 5) cannot be obtained by the optic axis measurements. In other words, fibers with different 3D orientations cannot be differentiated by the optic axis measurements described in the previous sections as long as they are located inside the same "projection" plane.

It is known that the measured birefringence $\Delta n'$ changes with the angle between the light and the fiber axis in a uniaxial birefringent sample:

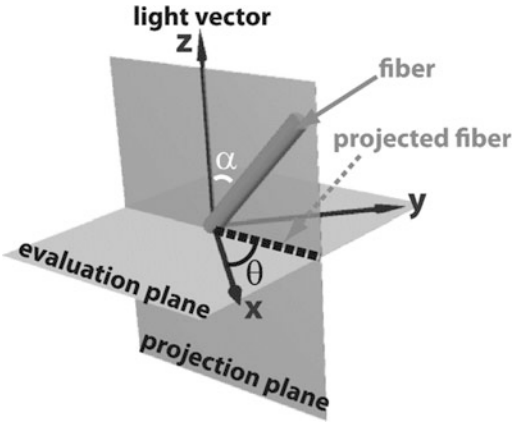


Fig. 5 An illustration of the fiber orientation measurement based on optic axis obtained in optical polarization measurements. The inclination angle α is the angle between the light vector and the 3D fiber in space. The angle θ represents the optic axis angle determined in polarization measurement. The "projection" plane is formed by the light vector and the 3D fiber direction in space, and the "evaluation" plane is perpendicular to the light vector

$$\Delta n' = (n_e - n_o)\sin^2\alpha. \tag{16}$$

The same relationship exists between the measured phase retardation δ' and the maximal retardation shown in Eq. (1). This dependency on α has been explored in PLM studies to characterize the out-of-plane fiber orientation in samples where the maximal birefringence $n_e - n_o$ is known and homogeneous [14, 40]. However, this method may be inaccurate in inhomogeneous samples with mixed tissue types or unknown birefringence.

Another method to determine the full 3D fiber orientation is to image the same sample at two different angles, i.e., the dual-angle measurement [41, 42]. Although the inclination angle α cannot be determined in optic axis measurement, the actual 3D fiber must be inside the "projection" plane formed by the light vector and projected fiber in the evaluation plane (Fig. 5). Using a second incident light direction creates a different projection plane. Therefore, the 3D fiber can be determined as the intersection of these two different projection planes. Due to optical refraction, the light direction inside the sample can be calculated as

$$\mathbf{L} = \frac{\mathbf{I}}{n} + \frac{\mathbf{N}}{n}\left(-\mathbf{N}\cdot\mathbf{I} - \sqrt{n^2 - 1 + (\mathbf{N}\cdot\mathbf{I})^2}\right), \tag{17}$$

where vector $\mathbf{I}$ represents the incident light vector and vector $\mathbf{N}$ is the normal vector of the sample surface at the point of incident. Both $\mathbf{I}$ and $\mathbf{N}$ are represented in the laboratory coordinates which can be conveniently established using the 3D imaging

scanning directions (A-, B-, and C-scans) in PSOCT. The vector corresponding to the projected fiber within the evaluation plane can be written as

$$\mathbf{P} = \begin{bmatrix} \cos\beta\cos\gamma\cos(\theta-\gamma) - \sin\gamma\sin(\theta-\gamma) \\ \cos\beta\sin\gamma\cos(\theta-\gamma) + \cos\gamma\sin(\theta-\gamma) \\ -\sin\beta\cos(\theta-\gamma) \end{bmatrix}, \tag{18}$$

where θ is the optic axis measured in PSOCT as described in the previous section. The angles (β, γ) are the polar and azimuthal angles of the refracted light vector in the laboratory coordinates. The projection plane where the true 3D fiber lies in can be represented by using its normal vector:

$$\mathbf{M}(\beta,\gamma,\theta) = \mathbf{P}\times\mathbf{L} = \begin{bmatrix} \sin\gamma\cos(\theta-\gamma) + \cos\beta\cos\gamma\sin(\theta-\gamma) \\ -\cos\gamma\cos(\theta-\gamma) + \cos\beta\sin\gamma\sin(\theta-\gamma) \\ -\sin\beta\sin(\theta-\gamma) \end{bmatrix}. \tag{19}$$

By imaging the sample with a different incident angle, a different project plane $\mathbf{M}'(\beta', \gamma', \theta')$ can be obtained. The 3D fiber vector can be calculated as

$$\mathbf{F} = \mathbf{M}(\beta,\gamma,\theta)\times\mathbf{M}'\left(\beta',\gamma',\theta'\right). \tag{20}$$

To implement this dual-angle method, the two image volumes of the same sample have to be registered together, so that the two projection planes $\mathbf{M}(\beta, \gamma, \theta)$ and $\mathbf{M}'(\beta', \gamma', \theta')$ can be calculated for each image pixel to determine the pixel-wise absolute 3D fiber orientation.

The ability to obtain 3D fiber orientation is valuable in studying samples where the fibers have large inclination angles toward the sample surface. One of such tissues is the articular cartilage, a critical tissue covering the bone surface in a joint. The collagen fibers in cartilage are known to form a unique "arcade"-like structure that divides the cartilage into three different zones [10, 43]. The collagen fibers run in parallel with the articulating surface in the "superficial" zone, bend in the "transitional" zone, and eventually become perpendicular to the bone interface in the "deep" zone. As a result, when imaging the cartilage from the surface, PSOCT signals are significantly attenuated from the transitional zone to the deep zone as the fibers become more aligned with the incident light [27, 44].

As shown in Fig. 6a, the dual-angle approach correctly images the 3D fiber organization in a piece of cartilage from the superficial zone to the deep zone, which is impossible in regular PSOCT imaging. The 3D tractography is rendered by importing the streamlines into the 3D Slice software (www.slicer.org) [45]. Once the 3D fiber orientation data are obtained, the dataset can be conveniently visualized in detail by projecting it to other 2D planes [41]. Figure 6b shows an example of projecting the 3D measurements to the sample side. This "virtual" sectioning result resembles the actual 2D measurements (Fig. 6c) from direct PSOCT imaging of the sample side.

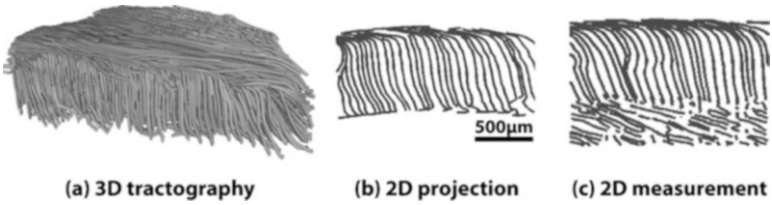


Fig. 6 3D fiber orientation measurement in a cartilage sample acquired using the dual-angle method. (**a**) 3D tractography of the sample. (**b**) 2D tractography by projecting the 3D angles onto the sample side plane. (**c**) 2D tractography measured directly from the sample side

6 Summary

Regardless of the specific implementing technology, all tractography methods rely on accurate measurements of fiber orientation at each image pixel. Therefore, the key in using PSOCT to image tissue fiber orientation is to extract the correct "local" optic axis from the original "cumulative" measurements. The method described here has shown success in transforming PSOCT into a novel tractography system. The core algorithm is based on Jones matrix calculus and can be applied to any PSOCT systems that can accurately measure depth-resolved sample Jones matrix. Such a system often uses two different incident polarizations and employs additional computation to construct the Jones matrix [31]. In tissues with negligible diattenuation, the same algorithm can be applied in PSOCT systems using a single input polarization [21, 22].

Although the example imaging results included in this chapter were obtained using a bulk optics PSOCT system, the same tractography method can be applied in optical fiber-based PSOCT systems in theory. However, despite its flexibility in practical applications, optical fibers may introduce significant artifacts and distortions in the measured Jones matrix. Fortunately, recent technology advancement has proven that such distortions can be successfully corrected using computational algorithms, and accurate local optic axis can be measured using fiber-optic PSOCT systems [46, 47].

Fibrous tissues exist in many parts of the body such as the heart, brain, skeletal muscle, blood vessel, cartilage, etc. They are essential in achieving and maintaining normal physiological functions in the body. Damage and disorganization in fibrous tissues are often associated with diseases. Tractography imaging methods can provide valuable information in our studying of these tissues. As an emerging tractography imaging technology, PSOCT has many advantages in comparison with other existing tractography methods. On the one hand, it has superior spatial resolution and can image fiber structure at the microscopic level, which is not possible in diffusion-based MRI. On the other hand, it has much better imaging depth than conventional PLM and does not require sample sectioning. PSOCT-based tractography is uniquely positioned to image fiber structure in small animal tissues, as well as thin tissues in human and large animals. With continued technology

development especially in fiber-optic implementation, PSOCT-based tractography has the potential to become a useful tool in studying fibrous tissues.

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Conflict of Interest The author holds a US patent (US15/223,624) related to the technology described in this review.

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Mueller Matrix Microscopy



Hui Ma, Honghui He, and Jessica C. Ramella-Roman

1 Introduction

Polarized light microscopy has been used on specimens with polarization optical characteristics in many different fields such as minerology material science [1–6] and biomedical science [7, 8]. The Mueller matrix (MM) microscope, as the natural generalization of the polarized light microscope, is attracting more and more attention [9–11]. The Mueller matrix contains the most comprehensive polarization description of a media [12]; furthermore, 16 MM elements may provide information on the structural and mechanical properties of a sample. With detailed analysis of MM and particularly the fast-developing machine learning-based methods, polarization parameters, such as polarimetry basis parameters (PBP) and polarimetry feature parameters (PFP), can be derived from the 16 MM elements. Those parameters tend to have clearer physical meanings and more explicit associations with specific microstructures. Therefore, the MM-derived parameters can be used for quantitative characterization of complex turbid media for different applications. With the development in polarization-related optical devices, including new light sources, polarization elements [13, 14], and polarimeters [15–17], the acquisition time of MM imaging can be greatly sorted. Meanwhile, advanced denoising, optimization, and calibration techniques may help reduce the complex errors and obtain more

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accurate MMs. The developments in hardware, information acquisition, calculation, and calibration together make MM microscopes very powerful tools in both research [18–25] and applications [26–31].

Several different types of MM microscopes have been developed in the past decade [18–38]. Similar as traditional light microscopes, MM microscopes can be classified as upright transmission, inverted transmission, and reflection modes. The polarization modulation methods used for MM microscopes include division-of-time, division-of-amplitude, and division-of-focal-plane techniques. In this chapter, a brief history of MM polarimetry will be firstly introduced. Then, different types of MM microscopes, including their working principles, hardware designs, data processing, and system calibration will be introduced chronologically, together with examples of potential application. In the concluding part of this chapter, a prospect about MM microscopy will be presented.

2 Mueller Matrix Polarimetry

Before the very first Mueller matrix microscope, MM measurement and imaging instrumentations utilizing this modality had been developed and applied in different fields [39–43].

The measurement approaches of MM polarimetry, the calculation, error analysis, and calibration methods were introduced by different researchers [32, 44–49]. Based on those studies, and the fact that a MM polarimetry can be realized by adding the polarization states generator (PSG) and polarization states analyzer (PSA) into traditional optical equipment including microscopes, the MM microscope was developed gradually.

Early reported MM microscopes were evolution of ellipsometers [32]. In 2004, Laude-Boulesteix et al. reported a new multispectral Mueller polarimetric microscope [32, 33, 35] based on nematic liquid crystal modulators. The group also introduced for the first time the eigenvalue calibration method that will be discussed later in this chapter. Following that study, Ben Hatit et al. [36] and Curiel et al. [37] used a similar system to study the angle-resolved Mueller matrix images. Mujat et al. used a similar system for investigating the anisotropic properties of the refractive index in biological samples [38]. In 2014, Arteaga et al. reported a transmissive MM microscopic imaging system based on two continuous rotating compensators [18]. A series of MM microscopes based on two rotating compensators at multiple fixed orientations were reported by Ma et al., including a transmissive monochromatic MM microscope [19, 20], a transmissive multiwavelength MM microscope, and a collinear backscattering MM microscope [21, 23]. There are also novel MM microscopes based on division-of-focal-plane polarimeters being developed [24, 25].

Generally, the approaches to measure a complete MM can be classified according to the instrumental setups, including the division-of-time method using elements such as rotating polarized optical components, photo-elastic modulators, liquid crystal variable retarders; the division-of-amplitude method using division-of-amplitude

photopolarimeter and the division-of-focal-plane method using division-of-focal-plane polarimeters.

Here we will summarize the main optical principles

3 Dual Rotating Retarder Based MM Microscopy

The dual-rotating retarder method for MM measurement was firstly proposed by Azzam in 1978 [43]. In the basic configuration of dual-rotating retarder method, a polarization state generator (PSG) and a polarization state analyzer (PSA) are used to modulate the polarization states of the light before and after the interaction with sample, respectively. The PSG consists of a fixed polarizer and a retarder, while the PSA consists of a retarder and a fixed polarizer symmetrically along the direction of light propagation. Thus, the output light intensity $\mathbf{I}_{\text{out}}$ can be written as

$$\mathbf{I}_{\text{out}} = c\,(1\ \ 0\ \ 0\ \ 0)\,M_{\text{P2}}M_{\text{R2}}M_{\text{sample}}M_{\text{R1}}M_{\text{P1}}\mathbf{S}_{\text{in}} \tag{1}$$

where c represents the detector sensitivity, M represents the MM, $\mathbf{S}_{\text{in}}$ stands for the Stokes vector of the incident light. In this measurement scheme, the two retarders synchronously rotate at certain angular speeds; hence, $\mathbf{I}_{\text{out}}$ is a periodic signal whose fundamental frequency is related to the rotating angular speeds of the retarders without sample. $\mathbf{I}_{\text{out}}$ can also be written as

$$\mathbf{I}_{\text{out}} = a_0 + \sum_{n=1}^{12}\left(a_n \cos n\omega t + b_n \sin n\omega t\right) \tag{2}$$

where a_n and b_n are Fourier coefficients, ω is the rotating angular rate. The MM elements of the sample can be decoded with Fourier coefficients as

$$\begin{aligned}
m_{11} &= a_0 - \tfrac{1}{2}m_{12} - \tfrac{1}{2}m_{21} - \tfrac{1}{4}m_{22}\\
m_{12} &= 2a_2 - \tfrac{1}{2}m_{22}\\
m_{13} &= 2b_2 - \tfrac{1}{2}m_{23}\\
m_{14} &= b_1 - \tfrac{1}{2}m_{24}\\
m_{21} &= 2a_{10} - \tfrac{1}{2}m_{22}\\
m_{22} &= 4a_8 + 4a_{12}\\
m_{23} &= 4b_{12} - 4b_8\\
m_{24} &= 4b_{11} = -4b_9\\
m_{31} &= 4b_{10} - 4m_{32}\\
m_{32} &= 4b_{12} + 4b_8\\
m_{33} &= 4a_8 - 4a_{12}\\
m_{34} &= -4a_{11} = 4a_9\\
m_{41} &= -b_5 - \tfrac{1}{2}m_{42}\\
m_{42} &= -4b_7 = -4b_3\\
m_{43} &= 4a_7 = -4a_3\\
m_{44} &= -2a_4 = 2a_6
\end{aligned} \tag{3}$$

Here the MM calculation process is based on the assumption that all polarization components are ideal.

3.1 *Dual Continuous Rotating Retarder Based Transmissive MM Microscopic Imaging System*

As introduced above, Arteaga et al. reported a dual rotating retarder based transmissive MM microscopic imaging system shown in Fig. 1 [18]:

This MM microscopic imaging system is a prototype set up on an anti-vibration optical table vertically consisting of four parts: the light source is a white LED illuminator that emits light in the visible range. The light beam is collimated by a lens and then goes through a bandpass interference filter to select the light wavelength. A 45° mirror redirects the beam in the vertical direction. Each of the

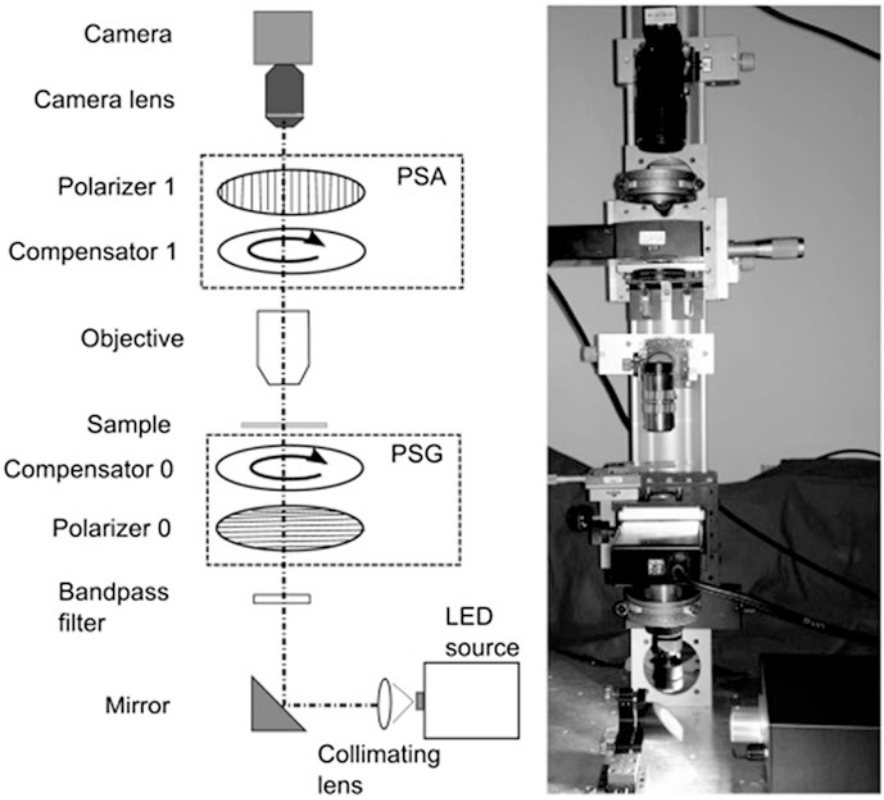


Fig. 1 Scheme and photo of the optical setup of the MM microscopic imaging system by Arteaga et al. [18]

PSG and PSA is composed of an achromatic compensator (Compensator 0 and 1, Quarter-wave polymer retarder film) and a polarizer (Polarizer 0 and 1, High-contrast linear polarizing film). Both Compensators 0 and 1 are mounted in a precision, motorized rotation stage, and both Polarizers 0 and 1 are mounted in a manual rotation stage. The objective is an infinite-corrected long working distance objective (Edmund Optics). Finally, the detection part is composed of a camera with a resolution of 752 × 480 pixels and a telecentric camera objective. Equation (1) of each pixel of the detector can be written as

$$\mathbf{I}(t) = \begin{bmatrix} 1 \\ C_{2\theta_0}^2 + C_{\delta_0} S_{2\theta_0}^2 \\ C_{2\theta_0} S_{2\theta_0} \left(1 - C_{\delta_0}\right) \\ S_{\delta_0} S_{2\theta_0} \\ -\left(C_{2\theta_1}^2 + C_{\delta_1} S_{2\theta_1}^2\right) \\ -\left(C_{2\theta_0}^2 + C_{\delta_0} S_{2\theta_0}^2\right)\left(C_{2\theta_1}^2 + C_{\delta_1} S_{2\theta_1}^2\right) \\ -\left(C_{2\theta_0} S_{2\theta_0} \left(1 - C_{\delta_0}\right)\right)\left(C_{2\theta_2}^2 + C_{\delta_1} S_{2\theta_1}^2\right) \\ -\left(S_{\delta_0} S_{2\theta_0}\right)\left(C_{2\theta_1}^2 + C_{\delta_1} S_{2\theta_1}^2\right) \\ -C_{2\theta_1} S_{2\theta_1} \left(1 - C_{\delta_1}\right) \\ -\left(C_{2\theta_0}^2 + C_{\delta_0} S_{2\theta_0}^2\right)\left[C_{2\theta_1} S_{2\theta_1} \left(1 - C_{\delta_1}\right)\right] \\ -\left[C_{2\theta_0} S_{2\theta_0} \left(1 - C_{\delta_0}\right)\right]\left[C_{2\theta_1} S_{2\theta_1} \left(1 - C_{\delta_1}\right)\right] \\ -S_{\delta_0} S_{2\theta_0}\left[C_{2\theta_1} S_{2\theta_1} \left(1 - C_{\delta_1}\right)\right] \\ S_{\delta_1} S_{2\theta_1} \\ S_{\delta_1} S_{2\theta_1}\left[C_{2\theta_0}^2 + C_{\delta_0} S_{2\theta_0}^2\right] \\ S_{\delta_1} S_{2\theta_1}\left[C_{2\theta_0} S_{2\theta_0} \left(1 - C_{\delta_0}\right)\right] \\ S_{\delta_1} S_{2\theta_1} S_{\delta_0} S_{2\theta_0} \end{bmatrix}^{\mathrm{T}} \begin{bmatrix} m_{11} \\ m_{12} \\ m_{13} \\ m_{14} \\ m_{21} \\ m_{22} \\ m_{23} \\ m_{24} \\ m_{31} \\ m_{32} \\ m_{33} \\ m_{34} \\ m_{41} \\ m_{42} \\ m_{43} \\ m_{44} \end{bmatrix} = \mathbf{B}^{\mathrm{T}}(t)\mathbf{A} \tag{4}$$

where $C_x = \cos(x)$, $S_x = \sin(x)$, $\mathbf{A}$ is a 16-element vector containing all the MM elements. The method is used to calculate the MM, which is based on the light intensity value of a single-point detector. Generalize Eq. (4) with all intensities recorded over time.

$$\begin{aligned} \mathbf{I} &= \mathbf{B}^{\mathrm{T}}\mathbf{A} \\ \mathbf{A} &= \left(\mathbf{B}\mathbf{B}^{\mathrm{T}}\right)^{-1}\mathbf{B}\mathbf{I} = \mathbf{K}\mathbf{B}\mathbf{I} \end{aligned} \tag{5}$$

The camera is capable of acquiring images at approximately 50 frames per second (fps); this is adjustable according to the exposure time. Compensator 0 rotates at 5°/s while compensator 1 rotates at 25°/s from known orientations, which makes the ratio of the angular speeds of the two retarders is set to be 5, as one of the optimum choices. Continuous frames are captured to make sure the slowest compensator rotates a complete turn. The 16-element vector $\mathbf{A}$ of an MM for each

pixel is the same function **KB** of all the measured intensities once the number of the captured frame is determined.

In the calibration process, first, a four-error model is built, including one orientation error and one retardation error of each compensator. Samples and the objective lens are removed and the camera is used as a single-point detector. Considering those errors are relatively small that they can be expanded to first order, then Eq. (1) can be written as

$$\mathbf{I}(t) = \mathbf{I}'(t) + \mathbf{I}_{\varphi 0}(t)\Delta_{\varphi 0} + \mathbf{I}_{\varphi 1}(t)\Delta_{\varphi 1} + \mathbf{I}_{\delta 0}(t)\Delta_{\delta 0} + \mathbf{I}_{\delta 1}(t)\Delta_{\delta 1} + \boldsymbol{O}\left(\Delta_i^2\right) + \cdots \tag{6}$$

The errors of the model have different effects on MM elements, and $m12$, $m13$, $m21$, and $m31$ are selected to solve the error model for calibration. For a further refinement, the MM of the objective of each pixel is premeasured using air as the standard sample, and then the results can be used to correct the measured MMs of samples as

$$M_{\text{corrected}} = M_{\text{obj}}^{-1} M_{\text{measured}} \tag{7}$$

Using the developed MM microscopic imaging system, a mosquito wing sample was measured by at 610 nm illumination with a 10× objective. The MM image and standard deviations are shown in Fig. 2.

The measurement was repeated several times to improve the signal-to-noise ratio of the averaged MM images and compute the deviation as an error estimator. The MM image of Fig. 2 shows relatively high linear birefringence and linear dichroism (wing margin, right part of the images). Linear dichroism image from Ref. [18] is shown in Fig. 3.

The contrast of linear dichroism image is higher than the light intensity image, which is likely due to the presence of some oriented dye molecule in the scales. Thin polycrystalline layer of benzil crystallized and LCD display samples were also imaged to demonstrate that the MM microscopic imaging system can be used for a wide range of popular applications especially for anisotropic samples imaging.

3.2 Dual Rotating Retarder Based Transmissive MM Microscope

Wang et al. reported the first MM microscope based on a commercial transmission optical microscope shown in Fig. 4 [19, 20].

In this MM microscope, a PSG and a PSA are added into a low-cost commercial transmitted-light microscope. Collimated by a lens, light from an LED is used as the source. Both the PSA and the PSG are composed by a fixed linear polarizer (P1, P2) and a rotatable quarter-wave plate (R1, R2). PSA and PSG are set to 30 different

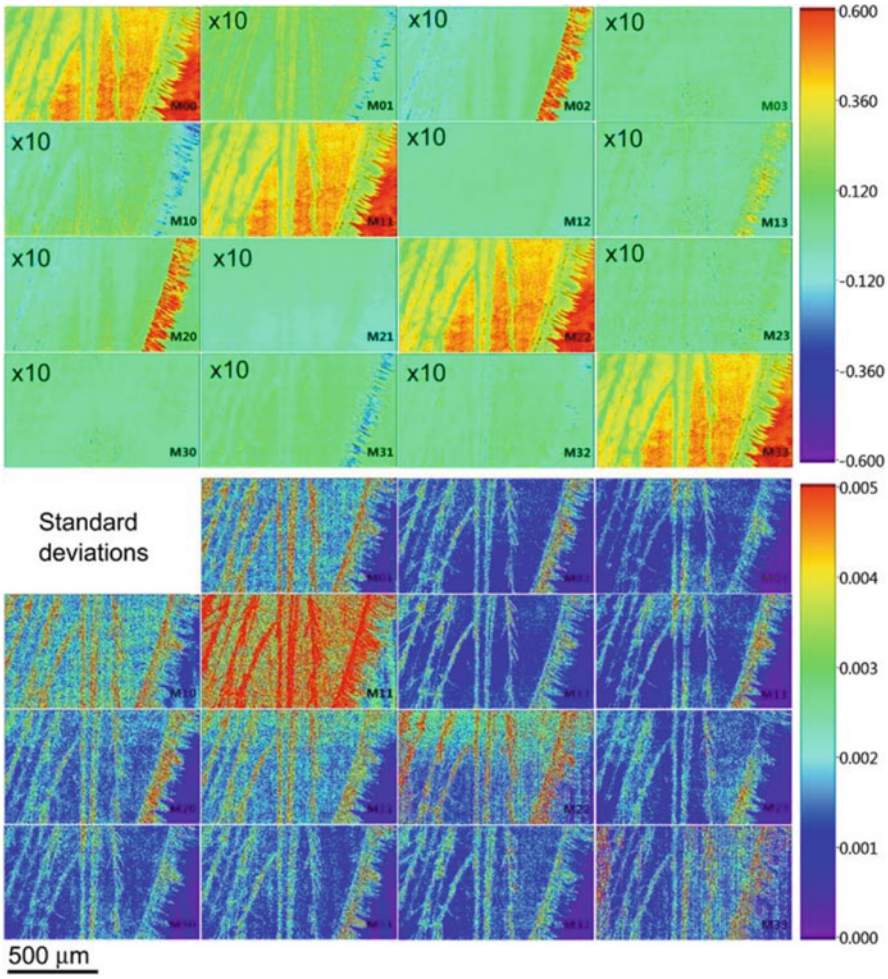


Fig. 2 Details of an MM measurement of a mosquito wing at 610 nm. The numbers appearing in some MM elements indicate the multiplying factors used to enhance the values of these elements for better visibility [18]

polarization states corresponding to 30 different angles of R1 and R2. The emergent light is detected by a 12-bit CCD camera (QImaging 74-0107A, Canada).

For each measurement process based on the dual-rotating retarder method, first, R1 and R2 started at the 0° angle so that their fast axes are parallel to the transmission axes of P1 and P2. Then R1 and R2 are rotated with different angular speeds to take light intensity images of different polarization states. To be more specific, the R1 rotated in steps of 6° while the R2 rotated in steps of 30°. The system took 30 polarization images totally, and then the MM images can be calculated. Unlike the microscopic imaging system introduced in Sect. 3.1,

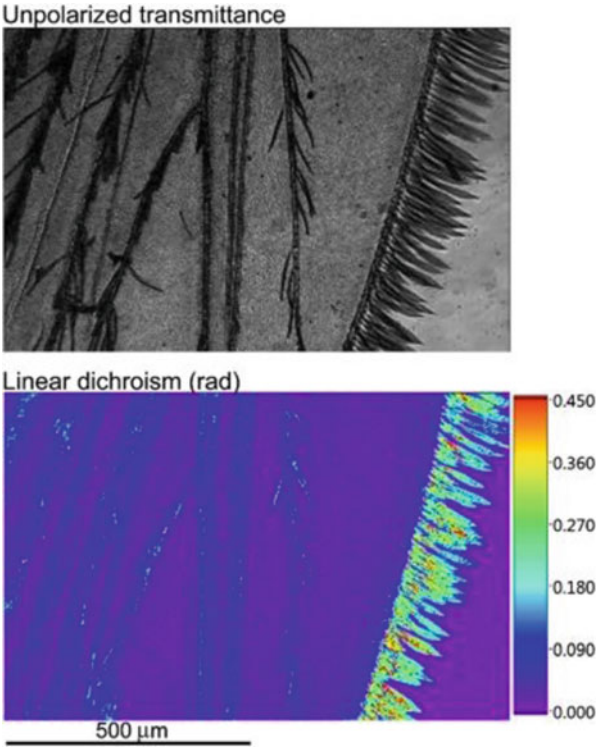


Fig. 3 Unpolarized transmittance (M11) and linear dichroism image of a part of the mosquito wing. A great contrast is obtained in the fringes located in the outer part of the wing [18]

whose retarders were rotated continuously, the measurement of this system takes 30 polarization images at fixed positions of the two retarders. Fourier analyses are used here to calculate the MM elements from the 30 intensity images. With the predetermined rotating speeds of the two retarders, each MM element is related to linear combination of different Fourier coefficients as shown in Eq. (3).

As for the calibration process, a five-error model is built to simulate the error propagation in the system [45]. They are initial azimuthal alignment errors of R1, R2, and P2; retarder errors of R1, R2, respectively. Those five errors can also be calculated from the Fourier coefficient in the following equations:

$$\begin{aligned}
\delta_1 &= \cos^{-1}\left(\frac{a_{10}\cos\alpha_9 - a_8\cos\alpha_{11}}{a_{10}\cos\alpha_9 + a_8\cos\alpha_{11}}\right)\\
\delta_2 &= \cos^{-1}\left[\frac{a_2\cos\alpha_9 - a_8\cos(4\varepsilon_3 - 2\varepsilon_5)}{a_2\cos\alpha_9 + a_8\cos(4\varepsilon_3 - 2\varepsilon_5)}\right]\\
\varepsilon_3 &= \tfrac{1}{4}\tan^{-1}\left(\frac{b_8}{a_8}\right) - \tfrac{1}{4}\tan^{-1}\left(\frac{b_{10}}{a_{10}}\right)\\
\varepsilon_4 &= \tfrac{1}{2}\tan^{-1}\left(\frac{b_2}{a_2}\right) - \tfrac{1}{2}\tan^{-1}\left(\frac{b_6}{a_6}\right) + \tfrac{1}{4}\tan^{-1}\left(\frac{b_8}{a_8}\right) - \tfrac{1}{4}\tan^{-1}\left(\frac{b_{10}}{a_{10}}\right)\\
\varepsilon_5 &= \tfrac{1}{2}\tan^{-1}\left(\frac{b_2}{a_2}\right) + \tfrac{1}{2}\tan^{-1}\left(\frac{b_8}{a_8}\right) - \tfrac{1}{2}\tan^{-1}\left(\frac{b_{10}}{a_{10}}\right)
\end{aligned} \tag{8}$$

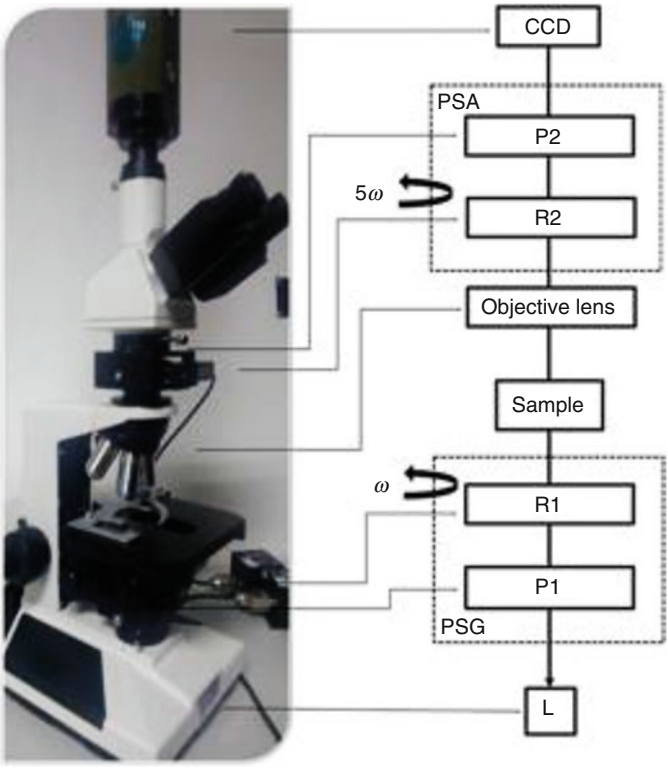


Fig. 4 Photo and schematic of the MM microscope. P: polarizer; R: quarter-wave plate; L: light source; PSG: polarization states generator; and PSA: polarization states analyzer [20]

The calibration method here is called analytical calibration method (ACM), and more details of the method can be found in Refs. [44, 45]. The MM microscope was used to examine some human cancerous tissues. The samples were 8-μm thick unstained, dewaxing sections of human cancerous liver tissue slices in different fibrosis stages (F1 to F4) provided by a local hospital, with corresponding 4-μm thick hematoxylin-eosin (HE)-stained slices as gold standard. Figure 5 shows the F4 stage MM microscopic images of the cancerous liver tissue sample under 4× objective observation. Figure 5a, b shows the images of MM-derived parameters δ and t for the cancerous liver tissue, in which the boundaries of the fibrotic regions are clear, demonstrating that the fibrosis degree of these regions is very high. The images of MM-derived parameters θ and x are shown as Fig. 5c, d, which reveal that there are a large portion of fibers with distinct orientations in the cancerous liver tissue.

Here we only show the MM-derived parameter images of human cancerous liver tissue samples in stage F4. Results for the samples in other stages and other

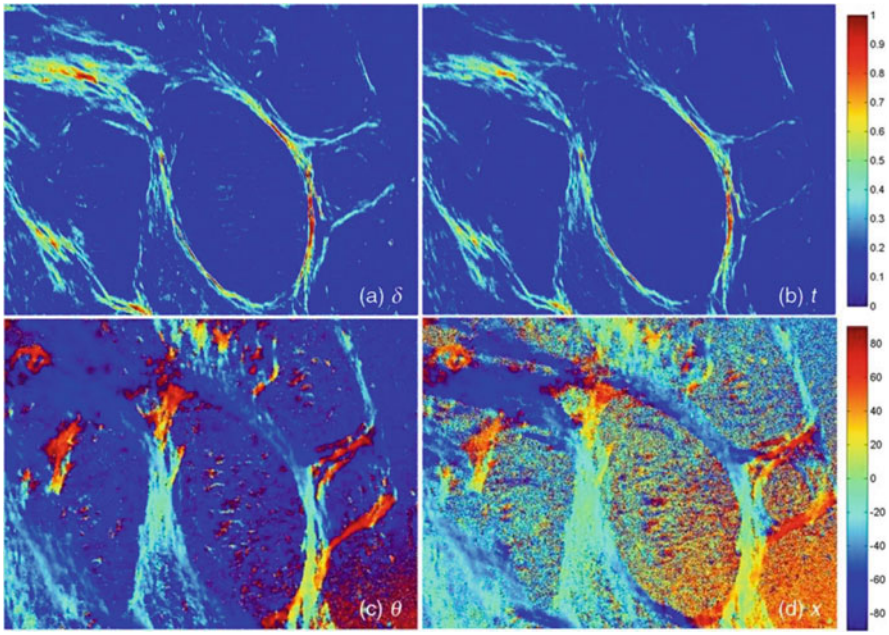


Fig. 5 Images of the 8-μm-thick nonstained dewaxing slice of liver cirrhosis tissue in F4 stage: (**a**) MM polar decomposition (MMPD) δ, (**b**) MM transformation (MMT) t, (**c**) MMPD θ, and (**d**) MMT x [20]

samples can be found in Refs. [19, 20], along with the images of HE-stained tissue slices, results of quantitative analysis, and Monte Carlo simulations. All the results demonstrated that the MM microscope can clearly observe cancerous tissues and be used as a potential label-free tool for quantitative diagnosis.

3.3 Dual Rotating Retarder Based Transmissive Multiwavelength MM Microscope

Based on the MM microscope introduced in Sect. 3.2, Zhou et al. reported a multiwavelength version of MM microscope with a light source consisted of six-monochromatic collimated LEDs and several optimization methods [21].

Figure 6a shows the schematic of the microscope for transmission multiwavelength MM imaging, which consists of four major modules: (1) collimated light source (CLS) module, (2) PSG module, (3) PSA module, and (4) image receiving module. In the CLS module, the diverging light from a monochrome LED (one of the six LEDs of different colors) is collimated by a set of convex lenses and turned to the upright direction by a steering mirror M. The six center wavelengths of the LEDs are 455 nm, 475 nm, 490 nm, 535 nm, 590 nm, and 630 nm, respectively, with

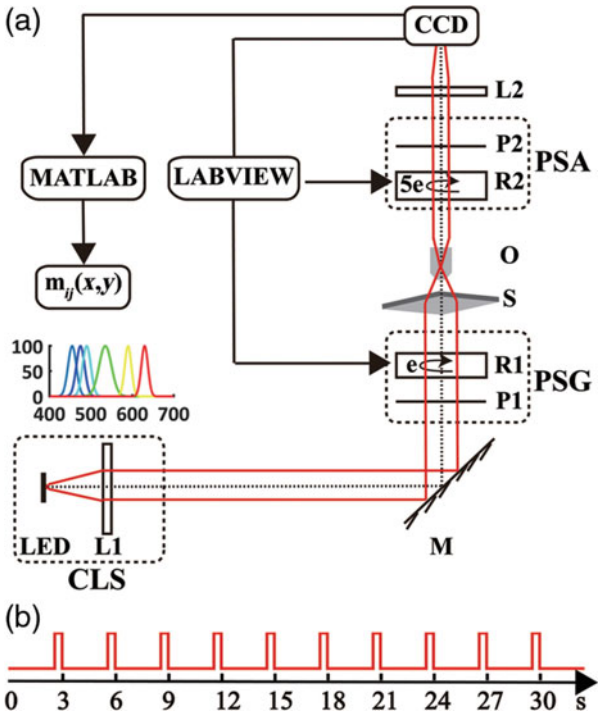


Fig. 6 (**a**) The schematic diagram of polarization microscope. CLS, monochromatic light source; P1 and P2, polarizers; R1 and R2, quarter-wave plates; L1 and L2, lenses; O, 4×, 10×, 20×, 40× objectives. (**b**) The time for image acquisition. The protuberant red lines represent the exposure time range from 0.1 to 1 ms; the following red horizontal lines represent the rotation time of R1 and R2, which is about 3 s [21]

full-widths at half-maximum close to 15 nm. The parallel beam passes through the PSG module, sample, objective lens O, and PSA module successively. Both the PSG and the PSA are consisted of a fixed polarizer (P1, P2) and an achromatic quarter-wave plate (R1, R2) mounted on a computer controlled motorized rotational stage. Finally, the emergent light is focused and detected by a 12-bit CCD detector.

The measurement and calibration processes are the same as those in Sect. 3.2. However, three optimization methods are also proposed for the multiwavelength MM microscope. The first optimization method is proposed to minimize the error introduced by the instability of the light source and the random noise of optoelectronic detectors. 10 images are taken for every positions of the PSG and PSA, and the average is used to calculate the MM. The optimization process significantly decreases the error bar of the light intensity values of pixels. Second, except for the calibration for five error sources of polarization optics, the polarization properties of objective lens are also reduced by introducing the real MM of objective lens into the calculation process as Eq. (7) in Sect. 3.1. The MM of objective lens can be evaluated using the eigenvalue calibration method [32, 47]. Finally, since the

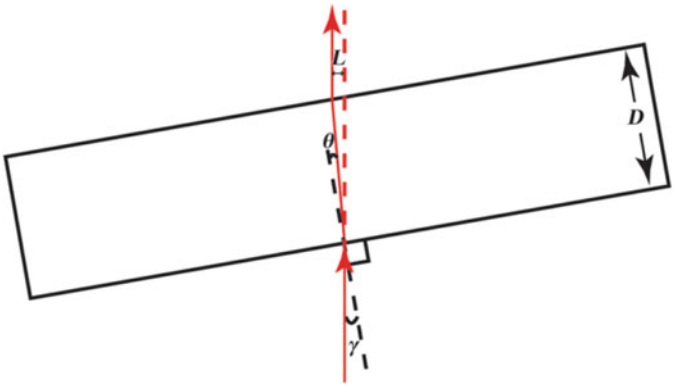


Fig. 7 The illustration of light drift. When polarization components are unparallel, the beam of light drifts and the measurement regions form dislocations. D, the thickness of the wave plate; L, the distance of deviation; γ, the angle between the optical path and the central axis of the rotating optical component in PSA; and θ, the angle of refraction [21]

measurement process consists of 30 imaging corresponding to different polarization states of PSG and PSA, images received by the detector may have dislocations due to rotating components and nonparallelism of components' surfaces. The relationship between the offset of the image L and the angle γ between the optical path and the central axis of the rotating optical component in PSA can be written as

$$\mathrm{L} = \frac{D \times \sin(\gamma - \theta)}{\cos \gamma} \tag{9}$$

where D is the thickness of the rotating optical component in PSA and θ is the angle of refraction as shown in Fig. 7. A registration of polarization component images before the calculation is required for accurate MM measurement of every pixel. The imaging results show less noise and a clearer boundary of the sample.

The transmission multiwavelength MM microscope was used to measure some unstained, dewaxing slides of human liver cancer tissues. The MMPD retardance δ images of human liver tissue with different wavelengths are shown in Fig. 8 [51].

The high δ values shown in Fig. 8 indicate the abnormal inflammatory (red regions) of liver fibroses and cirrhosis during the development of liver cancer; 455 nm light (Fig. 8a) proves to be a better choice for retardance δ as a feature parameter to distinguish between normal and abnormal tissues since shorter wavelength is more sensitive to retardance change. Other wavelengths may be more suitable for other samples and other feature parameters. The results indicate that multiwavelength MM microscope can provide more polarization information and versatile choices to achieve a wide range of applications.

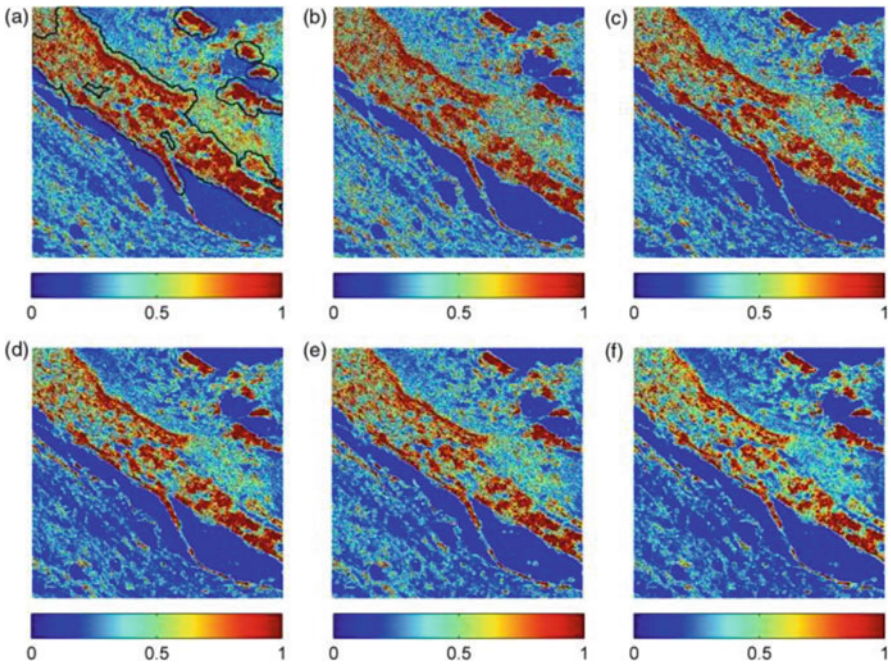


Fig. 8 (**a**–**f**) The images of the MMPD parameter δ of human liver tissue slices with probing light at 455 nm, 475 nm, 490 nm, 535 nm, 590 nm, and 630 nm, respectively; the abnormal regions in these images are indicated by black lines in (**a**) [21]

3.4 Dual Rotating Retarder Based Collinear Backscattering MM Microscope

The transmission MM microscopes mentioned above are suitable for taking MM images of thin samples, such as insect wings or pathological tissue slices with low depolarization properties. For thick samples and in situ applications, a reflection microscope is required to collect backscattering photons. The first collinear backscattering MM microscope was reported by Chen et al. [22] in 2018 as shown in Fig. 9. Later they reported a developed calibration method for the microscope in 2020 [23].

As shown in Fig. 9, the collinear backscattering Mueller matrix (CBMM) microscope is built by adding the PSG and PSA into a commercial metallurgical microscope. A 50:50 nonpolarizing plate beam splitter cube is placed in the center of the microscope, to ensure that the illumination and detection directions are collinear with the rotation axis of the sample. The CLS module consists of a monochrome LED source (center wavelength 632 nm, FWHM Full width half max 3 nm) and a set of collimating convex lenses. Within this microscope, the PSG and the PSA both consist of a fixed linear polarizer (P1 in PSG and P2 in PSA) and a rotating quarter-wave plate (R1 in PSG and R2 in PSA). Quarter-wave plates R1 and R2 are

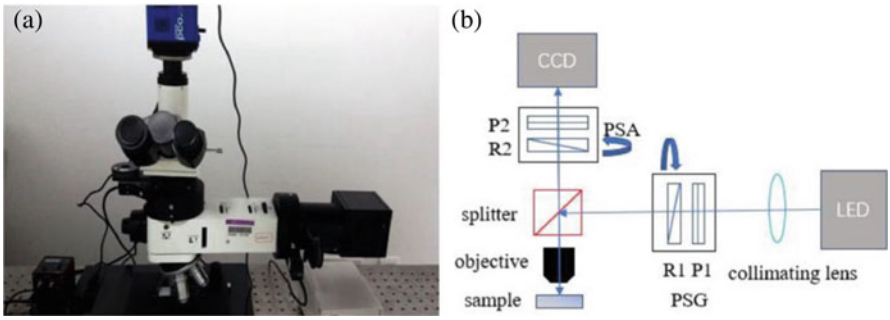


Fig. 9 Collinear reflection Mueller matrix microscope: (**a**) photo and (**b**) optical layout [22]

driven by servo motor drivers and accurately controlled by a computer. The detector used in the microscope is a CCD with a 12-bit resolution.

The CBMM microscope is also based on the dual-rotating retarders method and the 30-step imaging process for Mueller matrix measurement the same as the transmission MM microscope introduced in Sect. 3.2. However, to reduce the artifacts induced by the intrinsic polarizations of the optical system, particularly the dichroism due to the 45° beam splitter, a novel calibration method called numerical calibration method (NCM) is adopted [49]. Except for the five errors mentioned in Sect. 3.2, the background noise of CCD, the reflection MM, and the transmission MM of the splitter are also considered. Hence Eq. (1) can be rewritten as

$$\mathbf{I}_{\text{out}} = \mu + c\begin{pmatrix}1&0&0&0\end{pmatrix} M_{\text{P2}} M_{\text{R2}} M_{\text{sample}} M_{\text{R1}} M_{\text{P1}} \mathbf{S}_{\text{in}} \tag{10}$$

In where μ stands for irradiance background noise and c stands for S^T which means taking the first element of the output Stokes vector.

Three standard samples were used to measure the two MMs of the splitter, including (1) D_1 a mirror, (2) D_2 a mirror and a quarter-wave plate with the fast axis at 30° relative to the x-axis, and (3) D_3 a mirror and a quarter-wave plate with the fast axis at 120° relative to the x-axis. Here, two additional sets of reference light intensity data should be measured, that is, the light intensity data measured by replacing the standard sample D_1 with an optical absorption baffle, and the light intensity data measured by replacing the mirror in the standard samples D_2 or D_3 with an optical absorption baffle to exclude the light reflected into the CCD by the surfaces of the beam splitter cube and the quarter-wave plate in the standard samples. Therefore, three set of light intensity data were measured based on standard samples shows as

$$\begin{aligned} D_1 &= M_{\text{mirror}} \\ D_2 &= M_{\text{R}}\left(\delta, \varepsilon + \tfrac{\pi}{6}\right) M_{\text{mirror}} M_{\text{R}}\left(\delta, \varepsilon + \tfrac{7\pi}{6}\right) \\ D_3 &= M_{\text{R}}\left(\delta, \varepsilon + \tfrac{2\pi}{3}\right) M_{\text{mirror}} M_{\text{R}}\left(\delta, \varepsilon + \tfrac{5\pi}{3}\right) \end{aligned} \tag{11}$$

Equation (11) can be rewritten as

$$\begin{aligned} \mathbf{I}_i(q) &= \mu + \mathbf{A}(q)\mathbf{D}_i(q) \\ \mathbf{I} - \mathbf{J}\mu &= \mathbf{F}\text{vec}\left(\mathbf{D}_i\right) \\ \mathbf{J} &= \begin{bmatrix} 1 \\ 1 \\ \cdots \\ 1 \\ 1 \end{bmatrix} \end{aligned} \tag{12}$$

where $\mathbf{F}$ is the instrument matrix of measuring system. Equation (12) is a nonlinear equation. Using the Levenberg–Marquardt (LM) algorithm, errors can be solved, and the instrument matrix $\mathbf{F}$ can be reconstructed. With the instrument matrix $\mathbf{F}$, we can calculate MM of real samples as

$$\text{vec}\left(M_{\text{sample}}\right) = \text{pinv}\left(\mathbf{F}\right)\left(\mathbf{I} - \mathbf{J}\mu\right) \tag{13}$$

As a derivative of the previous calibration method, NCM solves the systematic errors numerically with a more complex error model instead of analyzing the relationship among systematic errors, Mueller matrix, and Fourier coefficients. It can also be used on the transmissive system. In fact, it was proved to have better performance than the ACM [31]. More detailed information of NCM can be found in [49].

Chen et al. have demonstrated the predominant performance of CBMM microscope on electrospinning fiber sample. The results are shown in Fig. 10.

The MM images of the same electrospinning fiber sample at four different azimuth orientations (0°, 90°, 180°, 270°) were detected by the CBMM microscope and a noncollinear backscattering Mueller matrix (NBMM) imaging system. Two azimuth invariant parameters (q_L and r_L) from Ref. [52] are calculated from the MMs, and their frequency distribution histograms (FDH) are shown in Fig. 10. Invariant parameters from CBMM microscope (Fig. 10c, d) show more stable distributions than parameters from NBMM system with the change of azimuth orientations of sample.

CBMM microscope has greatly broadened the application prospects of MM microscope. Compared with transmission MM microscope, backscattering MM microscope can be used on thick sample, which is crucial for imaging living samples. Compared to NBMM imaging system, CBMM microscope can reduce the influence of the azimuth orientation within the system. Noting that polarization information is sensitive to anisotropy of samples, CBMM microscope can provide better quantitative characteristics.

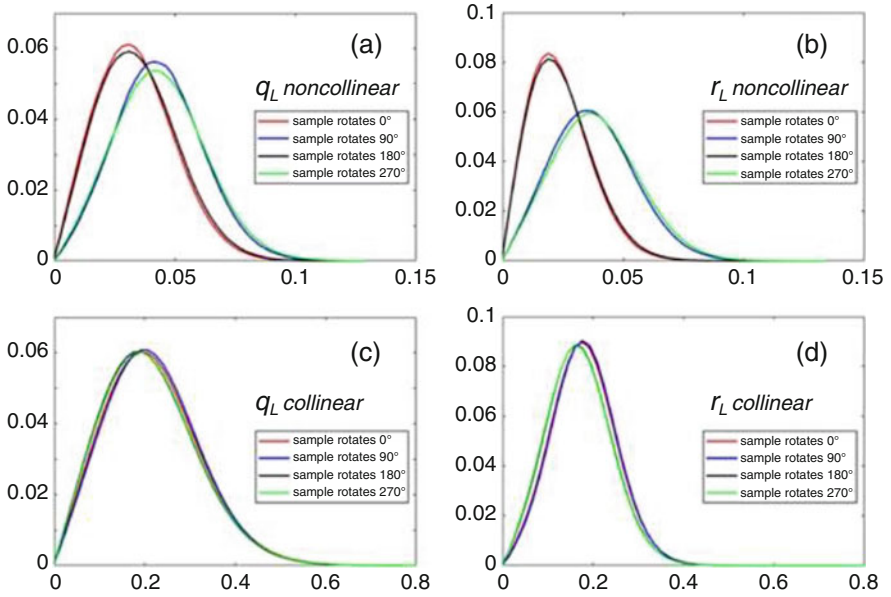


Fig. 10 Rotation invariant parameters of electrospinning fibers from the non-collinear backscattering Mueller matrices (**a**) q_L and (**b**) r_L and the collinear reflection Mueller matrix microscope (**c**) q_L and (**d**) r_L [23]

4 Rotating Retarder/DoFP Based Microscope

In recent years, the division-of-focal-plane (DoFP) polarimeter has attracted more and more attention. As a simultaneous polarization modulation method, the DoFP polarimeter is equipped with a pixelated micropolarizer array in front of the ordinary CCD sensor, and each super pixel consists of four adjacent pixels with different polarization orientations, which is capable of measuring the linear polarization states in a single shot. The combination of DoFP polarimeter and microscope can image sample's linear polarization properties. When operated with a liquid crystal variable retarder (LCVR) or a rotatable retarder and performed with at least two acquisitions, it can also be applied for measuring full Stokes vector. Ma et al. have reported some works on the combination of DoFP polarimeters and microscopes to measure sample's MM, including a DoFP polarimeter-based transmission 3×4 MM microscope, and a dual DoFP polarimeter-based transmission MM microscope.

4.1 DoFP Polarimeter Based Transmissive 3 × 4 MM Microscope

The diagram of the pixelated micropolarizer array of the DoFP polarimeter is shown in Fig. 11. Four pixels with different polarizer orientations comprise a super pixel,

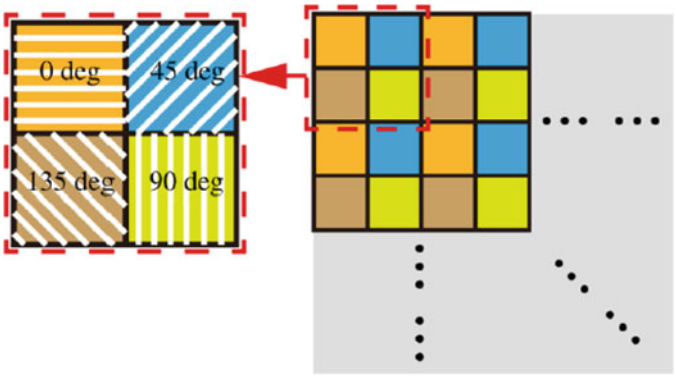


Fig. 11 Diagram of the pixelated micropolarizer array [24]

which are combined together to calculate the linear part of the Stokes vector as

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_0 \\ \mathbf{S}_1 \\ \mathbf{S}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{I}_0 + \mathbf{I}_{90} \\ \mathbf{I}_0 - \mathbf{I}_{90} \\ \mathbf{I}_{45} - \mathbf{I}_{135} \end{bmatrix} \tag{14}$$

where $\mathbf{I}_0$, $\mathbf{I}_{45}$, $\mathbf{I}_{90}$, and $\mathbf{I}_{135}$ are the intensities detected through the linear polarization components at four different orientations of the transmission axis. For a DoFP polarimeter's image sensor, the extinction ratio and the transmission axis of each pixel inherently have significant nonuniformities, calculating the Stokes vector using Eq. (14) would introduce errors. For accurate Stokes vector measurement, the nonuniformities for all pixels should be calibrated. We introduce light with different known State of Polarization (SoP) and measure the instrument responses to obtain the instrument matrix. We can calculate the Stokes vector using the polarimetric data reduction method. For each super pixel as shown in Fig. 11, the polarimetric measurement can be expressed as

$$\mathbf{I} = \mathbf{A} \cdot \mathbf{S} \tag{15}$$

where $\mathbf{I}$ is the light intensity of the four pixels and $\mathbf{A}$ is a 4×3 instrument matrix for a super pixel. Then $\mathbf{S}$ can be calculated by $\mathbf{S} = \text{pinv}(\mathbf{A}) \times \mathbf{I}$, where pinv($\mathbf{A}$) represents the pseudoinverse of $\mathbf{A}$. In addition, we reduce the edge artifacts and increase the spatial resolution of the Stokes vector images by bilinear interpolation on the raw intensity images.

Chang et al. [24] reported the DoFP polarimeter-based transmission 3×4 MM microscope, the photo and schematic are shown in Fig. 12. The original light source and the imaging sensor are replaced with the PSG and the DoFP polarimeter, respectively. In the PSG, the polarizer is fixed, thus no matter what polarization state of the light emitting diode irradiance is, the light passing through the polarizer will

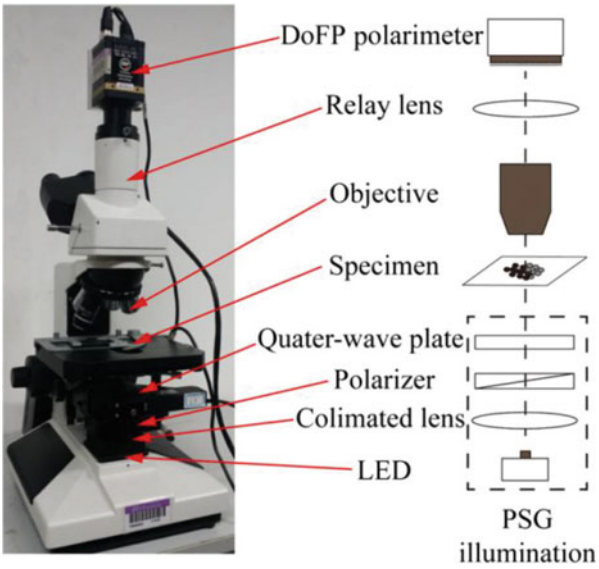


Fig. 12 Photo and schematic of the DoFP polarimeter-based transmission 3 × 4 MM microscope [24]

be linearly polarized. To measure the MM of the sample, at least four illuminations with independent polarization states are required. For each illumination, the $i^{'}$ th Stokes vector measurement is described as

$$\mathbf{S}_{\text{out}}(i) = M_{\text{sample}} \cdot M_{\text{W}} \cdot M_{\text{P}} \cdot \mathbf{S}_{\text{LED}} \tag{16}$$

where M_{sample}, M_{W}, and M_{P} are the MMs of the sample, the wave plate, and the polarizer, respectively. $\mathbf{S}_{\text{LED}}$ is the Stokes vector of the LED irradiance and $\mathbf{S}_{\text{out}}$ (i) is the Stokes vector of the specimen for the i'th illumination. Since DoFP polarimeter records only the linear part of the Stokes parameters, only the first three rows of M_{sample} are measurable, taking all the four Stokes vector measurements into account, the following equation can be obtained:

$$[\mathbf{S}_{\text{out}}] = M_{\text{sample}} \cdot [\mathbf{S}_{in}], \begin{bmatrix} \mathbf{S}_0(1) & \mathbf{S}_0(2) & \mathbf{S}_0(3) & \mathbf{S}_0(4) \\ \mathbf{S}_1(1) & \mathbf{S}_1(2) & \mathbf{S}_1(3) & \mathbf{S}_1(4) \\ \mathbf{S}_2(1) & \mathbf{S}_2(2) & \mathbf{S}_2(3) & \mathbf{S}_2(4) \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix}$$
$$\times \begin{bmatrix} 1 & \dots & 1 \\ \cos^2 2\theta(1) + \sin^2 2\theta(1) \cos\delta & \dots & \cos^2 2\theta(4) + \sin^2 2\theta(4) \cos\delta \\ 0.5 \sin 4\theta(1)\,(1 - \cos\delta) & \dots & 0.5 \sin 4\theta(4)\,(1 - \cos\delta) \\ \sin 2\theta(1) \sin\delta & \dots & \sin 2\theta(4) \sin\delta \end{bmatrix} \tag{17}$$

where δ is the retardance of the wave plate and θ (i) is the i'th fast axis angle of the wave plate. The MM can then be calculated using $M_{\text{sample}} = [\mathbf{S}_{\text{out}}] \times \text{inv}([\mathbf{S}_{\text{in}}])$, where inv($[\mathbf{S}_{\text{in}}]$) means the inverse of $[\mathbf{S}_{\text{in}}]$. Here, condition number (CN) of $[\mathbf{S}_{\text{in}}]$ should be minimized to reduce the errors transmitted from $[\mathbf{S}_{\text{in}}]$ to M_{sample}. When the retardance of the wave plate is 131.8°, the CN of $[\mathbf{S}_{\text{in}}]$ reaches to minimum 1.732. However, a 131.8° wave plate is not commercially available and requires a custom order. Thus, a commonly used quarter-wave plate is adopted and the minimum CN is searched using the genetic algorithm integrated in the MATLAB optimization toolbox. Calculation shows that when the retardance of the wave plate is 90°, and the fast axes of the wave plate are set to (−51.69°, −15.12°, 15.12°, 51.69°) or (−74.88°, −38.31°, 38.31°, and 74.88°), the CN of $[\mathbf{S}_{\text{in}}]$ reaches a minimum 3.40. The above angle scheme yields four elliptical SoPs, which are not commonly used and studied. Thus, we set two fast axes θ (1) and θ (2) to −45° and 45° to generate the left-hand and right-hand circularly polarized light, respectively, and optimize the remaining fast axes, θ (3) and θ (4) by minimizing the CN of $[\mathbf{S}_{\text{in}}]$. As shown in Fig. 13, when θ (3) =19.6° and θ (4) = −19.6°, the CN reaches to a local minimum 3.677, which is close to the optimal value 3.40.

After the optimization of $[\mathbf{S}_{\text{in}}]$, the propagation of errors in the MM construction is minimized. However, for accurate measurement, the nonideal nature of the polarizing elements should be calibrated. The nonideal nature of the polarizing elements includes the rotation errors of the polarizer and the quarter-wave plate. In addition, it also includes the retardance error of the quarter-wave plate. The calibration of the DoFP polarimeter-based MM system includes the calibration of

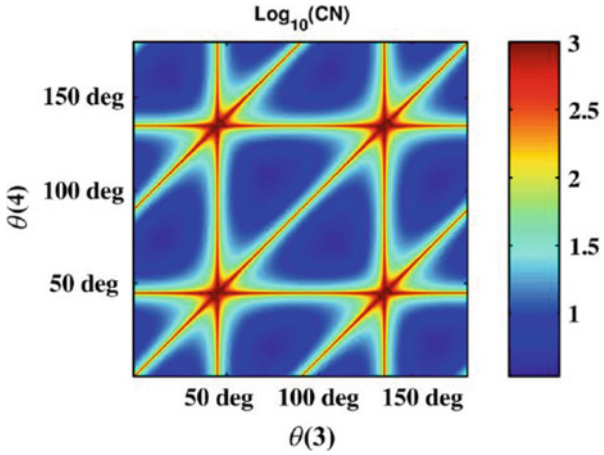


Fig. 13 CN map of $[\mathbf{S}_{\text{in}}]$. Color bar represents the common logarithm of the CN for better data visualization [24]

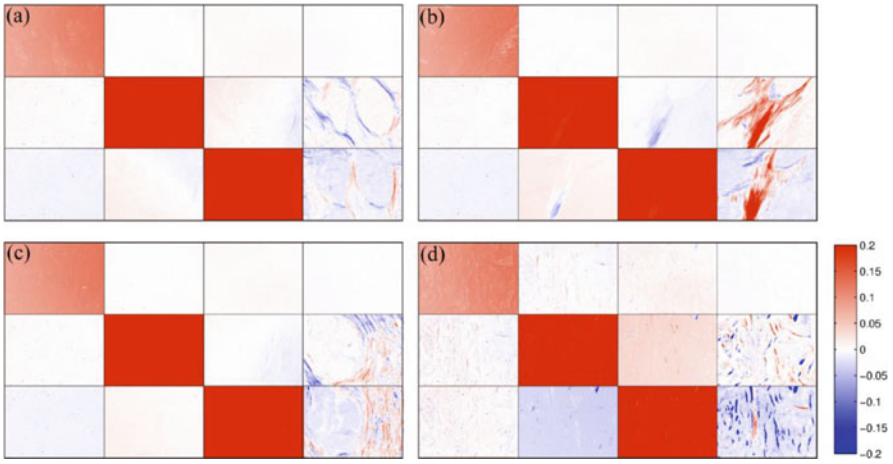


Fig. 14 The 3 × 4 Mueller matrix images of unstained human pathological specimens: (**a**) and (**b**) two regions on the liver carcinoma tissue slice and (**c**) and (**d**) two regions on the cervical carcinoma tissue slice. The Mueller matrix elements are all normalized by $m11$ except $m11$ [24]

DoFP polarimeter and calibration of the matrix $[\mathbf{S}_{\text{in}}]$. When the sample is removed from the polarization microscope and the wave plate is rotated to four different angles (−45°, −19.6°, 19.6°, and 45°), the first three rows of $[\mathbf{S}_{\text{in}}]$ will be directly measured by the DoFP polarimeter, and the three errors can be calculated. The calibration shows that the real retardance of the quarter-wave plate used in this study is 90.8°. After the optimization and calibration of the matrix $[\mathbf{S}_{\text{in}}]$, the mean errors for all the MM elements reach to about 0.3% and 0.6% when measuring the MMs of the air and a linear polarizer, respectively.

In order to test the potential diagnostic applications of the DoFP polarimeter-based MM microscope, we measure the 3 × 4 MM images of two different regions of unstained, dewaxed 12-μm slices of human liver cancer tissues specimen as shown in Fig. 14a, b, and human cervical carcinoma tissues specimen as shown in Fig. 14c, d. The fibrous structures can hardly be seen in the intensity images shown in Fig. 15a1–d1, but they are significantly enhanced in the polarization staining images as shown in Fig. 15a2–d2. The polarization staining images of the unstained dewaxed human liver and cervical carcinoma tissue specimens are calculated based on polarization staining using circularly polarized light illumination. The polarization staining images reveal much more detailed information of the microstructures, especially the pathological-related fibrosis with strong birefringence. The alignment orientations of the microfibers can also be clearly revealed from polarization staining images.

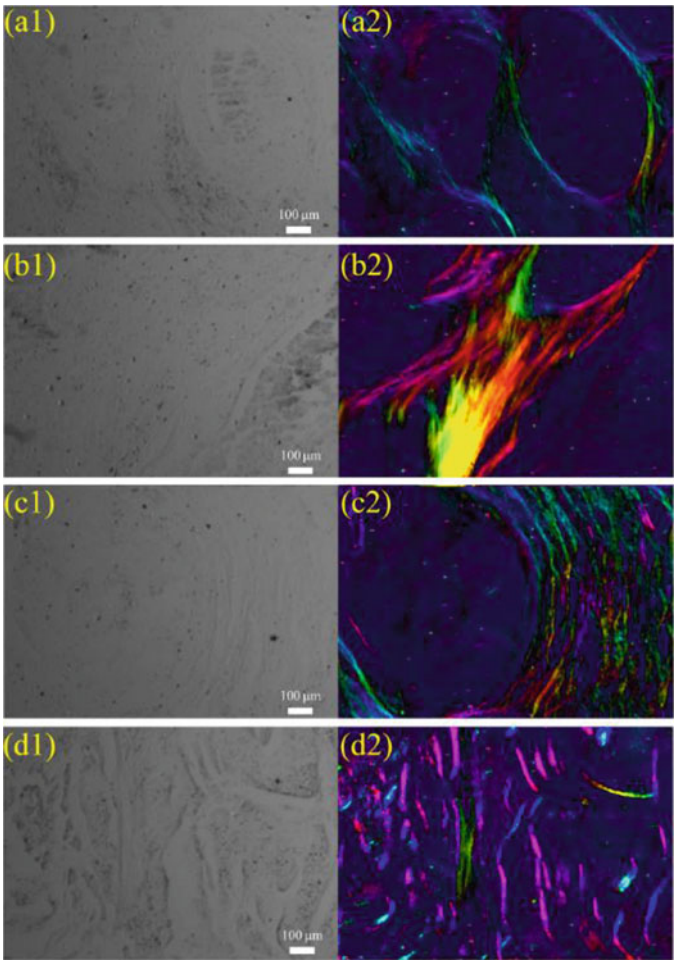


Fig. 15 (a1) and (b1) intensity images and (a2) and (b2) polarization staining images of two regions on the unstained human liver carcinoma slices. (c1) and (d1) intensity images and (c2) and (d2) polarization staining images of two regions on the unstained human cervical carcinoma slice [24]

4.2 Dual DoFP Polarimeter Based Transmissive MM Microscope

Huang et al. [25] reported an MM microscope based on dual DoFP polarimeters (DoFPs-MMM), the photo and schematic are shown in Fig. 16, which enables fast acquisition of Mueller matrix images. The DoFPs-MMM is established by adding PSA and PSG on a commercial transmission microscope. Light from the LED is collimated by a lens and modulated by PSG and then passed through the

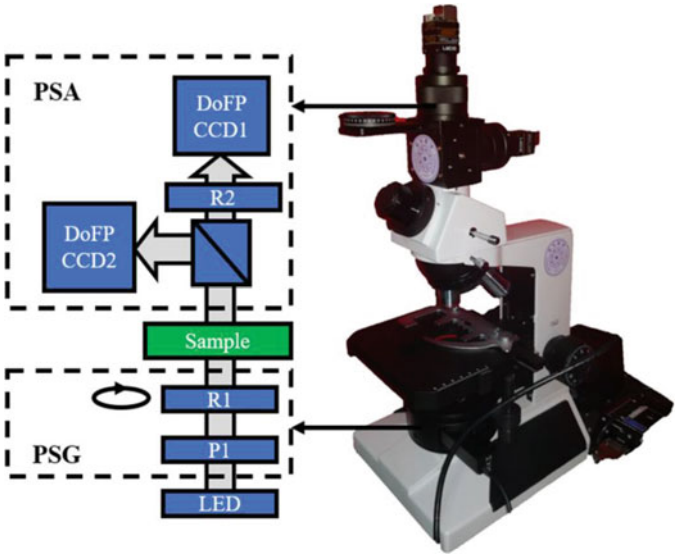


Fig. 16 Schematic of the MM microscope based on dual DoFP polarimeters [25]

objective lens and is finally detected by PSA. The PSG includes a fixed-angle linear polarizer (P1) and a rotatable zero order quarter-wave plate (R1). R1 is installed in a motorized rotation stage. The PSA includes two DoFP polarimeters (DoFP-CCD1, DoFP-CCD2) with resolution of 2048 × 2448 pixels and a frame rate of 21 fps. Two DoFP polarimeters are fixed to the transmission end and the reflection end of the 50:50 nonpolarized beam splitter prism, respectively, and a fixed-angle phase retarder (R2) is installed between the transmission end of the prism and DoFP-CCD1. In fact, similar functions can be achieved when R2 is installed between the reflection end of the prism and DoFP-CCD2. Without loss of generality, only the first case will be introduced.

In the design of the PSA, DoFP-CCD1, and DoFP-CCD2 with the same field of view (FOV), resolution, and exposure time are fixed to two ends of the prism and are aligned in the direction of 0° polarization orientation. Sub-pixel level image registration is achieved based on affine transformation in order to reduce polarization artifacts. Mutual information (MI) is used as the criterion of registration accuracy. Since R2 is installed in front of DoFP-CCD1, the 8 × 4 instrument matrix $\mathbf{A}_{\mathrm{PSA}}$ of the PSA consists of the instrument matrix of the DoFP polarimeter's instrument matrix $\mathbf{A}_{\mathrm{DoFP}}$ and the matrix product of $\mathbf{A}_{\mathrm{DoFP}}$ and M_{R2}

$$\mathbf{A}_{\mathrm{PSA}} = \begin{bmatrix} \mathbf{A}_{\mathrm{DoFP}} M_{\mathrm{R2}}\left(\theta_{\mathrm{R2}}, \delta_{\mathrm{R2}}\right) \\ \mathbf{A}_{\mathrm{DoFP}} \end{bmatrix} \tag{18}$$

where $M_{\mathrm{R2}}\left(\theta_{\mathrm{R2}}, \delta_{\mathrm{R2}}\right)$ is the Mueller matrix of R2 with fast axis orientation θ_{R2} and the retardance δ_{R2}. As for the optimization of $\mathbf{A}_{\mathrm{PSA}}$, the condition number (CN) is

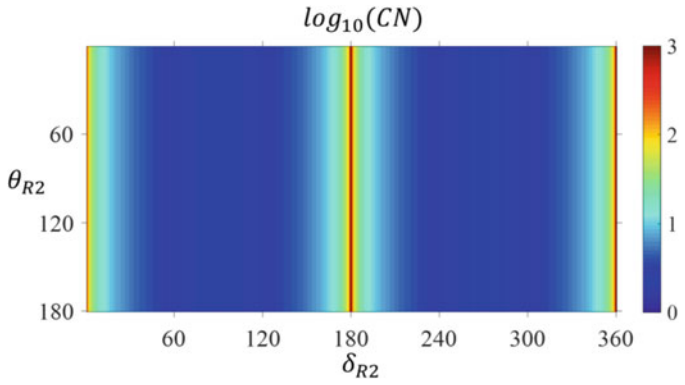


Fig. 17 CN map of $\mathbf{A}_{\mathrm{PSA}}$. Color bar represents the common logarithm of the CN for better visualization [25]

used for describing the sensitivity of the system to measurement errors. As shown in Fig. 17, under different θ_{R2} and δ_{R2}, CN of $\mathbf{A}_{\mathrm{PSA}}$ reaches to the minimum 2 when δ_{R2} is 90° or 270°. Therefore, the PSA can achieve the optimal state by simply using a commercial quarter-wave plate as R2. In addition, CN is constant under different θ_{R2}, which indicates that θ_{R2} is not critical in the optimization of the PSA.

However, since both ends of the prism may introduce nonnegligible parasitic dichroism and retardance artifacts, the measured Stokes vector may contain systematic errors. Therefore, the Mueller matrix of the prism must be considered in the calibration. We consider the relationship between the Stokes vector of incident light when using air as a standard sample

$$\mathbf{S}_{\mathrm{out}}^{\mathrm{trans\&R2}} = M_{\mathrm{trans\&R2}}\mathbf{S}_{\mathrm{in}} \tag{19}$$

$$\mathbf{S}_{\mathrm{out}}^{\mathrm{reflect}} = M_{\mathrm{reflect}}\mathbf{S}_{\mathrm{in}} \tag{20}$$

where $M_{\mathrm{trans\&R2}}$ stands for the equivalent Mueller matrix of both R2 and the transmission end of the prism, and M_{reflect} stands for the Mueller matrix of the reflection end of the prism. $\mathbf{S}^{\mathrm{trans\&R2}}{}_{\mathrm{out}}$ and $\mathbf{S}^{\mathrm{reflect}}{}_{\mathrm{out}}$ are the Stokes vector of the outgoing light from both ends, respectively. Note that since the objective lens is included in the optical path, its polarimetric response is also considered in $M_{\mathrm{trans\&R2}}$ and M_{reflect}. When the PSG generates at least four independent polarization states $[\mathbf{S}_{\mathrm{in}}]$ of the incident light, $M_{\mathrm{trans\&R2}}$ and M_{reflect} can be calculated by matrix division. Since the DoFP polarimeters can only detect the linear Stokes vectors, the $M_{\mathrm{trans\&R2}}$ and M_{reflect} measured are both 3×4 in size. However, because the elements in the fourth column of $\mathbf{A}_{\mathrm{DoFP}}$ are all zero, the fourth rows of $\mathbf{A}_{\mathrm{DoFP}}$ $M_{\mathrm{trans\&R2}}$ and M_{reflect} have no effect on the results of $\mathbf{A}_{\mathrm{DoFP}}$ $M_{\mathrm{trans\&R2}}$ and $\mathbf{A}_{\mathrm{DoFP}}$ M_{reflect}. Finally, $\mathbf{A}_{\mathrm{PSA}}$ can be modified as

$$\mathbf{A}_{\mathrm{PSA}} = \begin{bmatrix} \mathbf{A}_{\mathrm{DoFP}} M_{\mathrm{trans\&R2}} \\ \mathbf{A}_{\mathrm{DoFP}} M_{\mathrm{reflect}} \end{bmatrix} \tag{21}$$

$\mathbf{A}_{\mathrm{PSA}}$ is calculated pixel by pixel. Then, the complete Stokes vector of the outgoing light can be obtained according to

$$\mathbf{S} = \mathbf{A}_{\mathrm{PSA}}^{-1} \mathbf{I} \tag{22}$$

where $\mathbf{I}$ denotes a column vector containing intensity images corresponding to the four polarization orientations of DoFP-CCD1 and DoFP-CCD2. Since R2 has an attenuation effect on light detected by DoFP-CCD1, the light intensity detected by DoFP-CCD1 should be modified by dividing the transmittance of R2 to reduce the impact. Note that after calibration, the PSA can be used as a full Stokes-CCD for real time Stokes vector imaging.

In the design of PSG, we adopt a fixed-angle polarizer P1 and a rotatable quarter-wave plate R1 to modulate the polarization state of incident light. The polarization orientation of P1 and initial fast axis orientation of R1 are set to 0° and parallel to the 0° polarization orientations of DoFP-CCD1 and DoFP-CCD2. The PSG is precalibrated by a commercial polarimeter. When the PSG generates at least four independent polarization states $[\mathbf{S}_{\mathrm{in}}]$, the Mueller matrix of the sample can be calculated according to the equation

$$M_{\mathrm{sample}} = [\mathbf{S}_{\mathrm{out}}] [\mathbf{S}_{\mathrm{in}}]^{-1} \tag{23}$$

Since faster and more accurate Mueller matrix imaging is important to probe dynamic processes in living cells, we tested the performance of DoFPs-MMM on fresh red blood cell smear samples. Figure 18 shows images of two polarization parameters derived from the Mueller matrix, that is, the diattenuation parameter D and the azimuth α_D calculated from $m12$ and $m13$. Images of almost perfect spherical cells provide sensitive tests on the signal-to-noise ratio of the measurements. The diattenuation images are clearly shown against the background despite weak signals, while the direction of diattenuation also follows nicely with the shape of red blood cells. Both images prove that DoFPs-MMM can be applied to the quantitative characterization of cells.

5 MM Based on Birefringent Nematic Liquid Crystals

Mueller matrix imaging systems are often built with the use of variable retarder constructed of birefringent nematic liquid crystals (ferroelectric liquid crystal *FLC* and liquid crystal variable retarders *LCVR*) as they have the clear advantage of not requiring moving elements speeding up the acquisition process. These systems can be used in clinical application where speed is needed to limit motion artifacts and

increase the patient experience. The layout of this type of system is shown in Fig. 19. This assembly and its calibration have been described in detail in Chap. 3; here we will cover only the most salient portions.

The system consists of a white light source followed by a filter wheel with spectral filters of center wavelengths 500, 550, 600, 650, and 700 nm each with a 10 nm bandwidth.

Polarization State Generator (PSG) is obtained with of one linear polarizer and two nematic liquid crystals LC1 and LC2 whose retardations ($\delta 1$, $\delta 2$) depend on

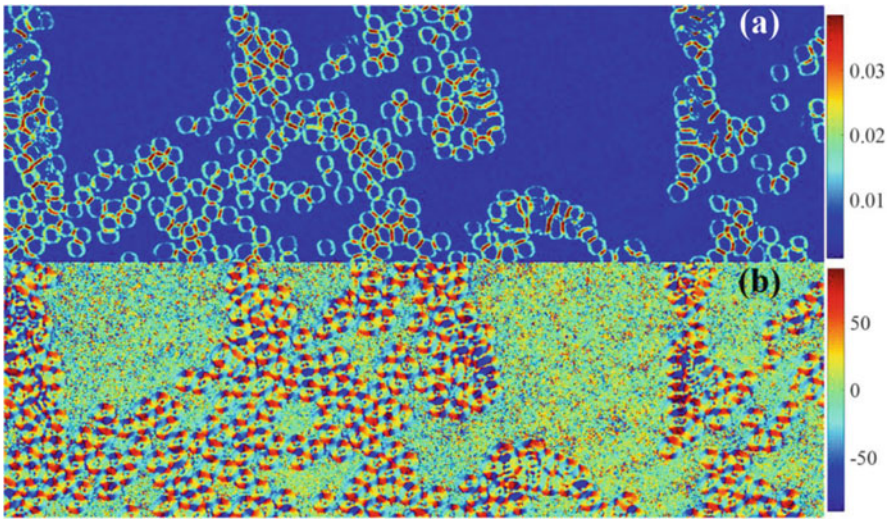


Fig. 18 Polarization parameters images of a red blood cell smear. (**a**) D, the diattenuation. (**b**) α_D, the azimuth of $m12$ and $m13$ [25]

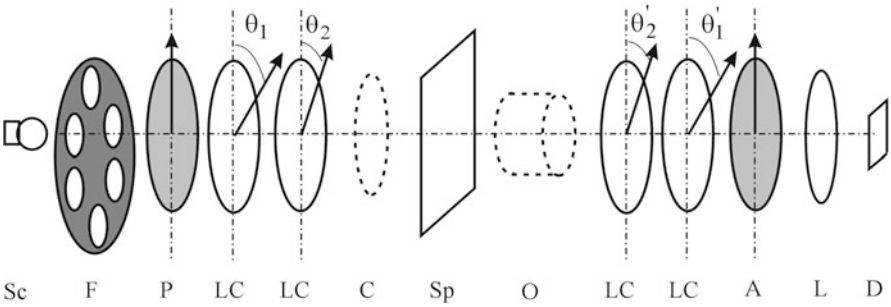


Fig. 19 Schematic of the imaging polarimeter. Sc: white light source, F: filter wheel, P linear polarizer (the arrow indicates the direction of the transmission axis), LCs: liquid crystal variable retarders (for each LC, the arrow indicates the direction of its variable index axis), Sp: sample, A: linear analyzer; L: imaging lens (achromat), D: detector (CCD camera). Higher magnifications can be achieved by inserting a condenser (C) and a microscope objective (O) (dotted lines). (Figure and description of system are from Laude-Boulesteix et al. [35])

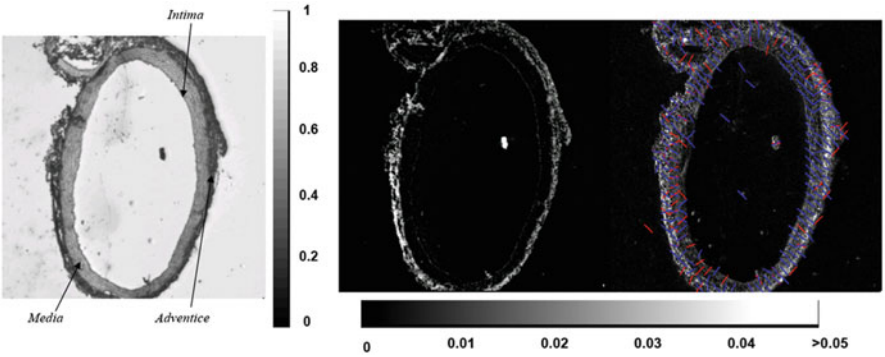


Fig. 20 Left coefficient $m11$ for a rat artery histological slide. Wavelength used was 650 nm, field of view 1 × 1 mm. In the center, depolarization power and right panel - dichroism are shown for the same sample. The fast and slow axis of dichroism are shown in red and blue, respectively [35]

their driving voltages. Similarly, the Polarization State Analyzer (PSA) consists of two LCs followed by a linear polarizer. All the LCs are driven by the same sequence in the PSA and the PSG.

The mechanical assembly of the LC and choice of retardation are critical in obtaining accurate MM images; hence, the arrangement of these elements has been the focus of previous optimization studies [14, 33].

The PSG and the PSA are operated by defining sequentially four pairs of retardation values δ_1, δ_2 for each pair of LCs. The optimal retardations are $\Delta_1 = 315° + \text{p90}$, $\Delta_2 = 135° + \text{p90}$ and the following sequence is utilized $(\delta_1, \delta_2) = (\Delta_1, \Delta_1), (\Delta_1, \Delta_2), (\Delta_2, \Delta_1), (\Delta_2, \Delta_2)$.

The optimal angular positions of the LCs are $\theta_1 = \varepsilon\ 27.4 + \text{q90}$, $\theta_2 = \varepsilon\ 72.4 + \text{q90}$ with respect to the system reference frame and input polarizer axis, where ε is ± 1 and q and p are integers not necessarily equal.

Finally, a microscope objective and a condenser (dashed lines in the figure) are used in this transmissive assembly to take images as shown in Fig. 20.

The calibration of the system was conducted with the eigenvalue calibration method which is described in Chap. 12.

6 MM Transformation Parameters for Polarization Feature Extraction

Mueller matrix polarimetry has some unique advantages. It is compatible with the existing nonpolarization optical methods, but can provide abundant information about sample's microstructure and optical features. In addition, it is very sensitive to the changes of sub-wavelength microstructures. An MM can be used as a comprehensive description of polarization features of the media. However, the individual

MM elements usually lack of explicit connections to distinctive microstructures. At present, the main challenge of MM polarimetry in various applications lies in how to analyze the obtained polarization data, that is, how to separate and extract the specific polarization parameters which reflect the specific microstructure. Mueller matrix transformation (MMT), as a polarization parameter extraction method, will be introduced in this section.

The development of MMT starts with the study of rotation, mirror transformation, and symmetry properties of the sample, which lead to polarization parameters that are invariant to the azimuth orientation of the sample or can be used to indicate breaking of mirror symmetries in the sample structure. In fact, MMT can be considered as a more general concept which encompassed Mueller matrix decomposition (MMD), since both MMD and MMT aim at "transforming" MM data into new polarization parameters which characterize certain physical properties of the media [52–54].

6.1 Mueller Matrix Rotation Invariant Parameters

There have been many studies on MM invariant parameters. Pravdin et al. [55] studied the rotation invariants in both transmission and reflection MM and mapped the orientation of the anisotropy. Gil studied both rotation transformation and retarder transformation of the MM on one side or dual sides and proposed corresponding set of invariants [56]. Jiang et al. developed the RLPI technique to reduce the orientation influence of the experimental data and proposed a few invariant and orientation parameters. To extend rotating linear polariztion imaging (RLPI) technique to the full MM, He et al. studied the Monte Carlo (MC) simulation and experimental data of concentrically aligned silk fiber and proposed the MMT theory [53]. Li et al. [54] compared these studies together with rotation and mirror transformation theory of MM and proposed a set of commonly used rotation invariants and orientation parameters as the following:

Given the experimental setup, if the illumination beam and the detection orientation are collinear, then we have the following MM parameters as invariants under azimuth rotation

$$m_{11} \tag{24}$$

$$k_C = m_{44} \in [-1, 1] \tag{25}$$

$$D_C = m_{14} \in [-1, 1] \tag{26}$$

$$P_C = m_{41} \in [-1, 1] \tag{27}$$

$$P_L = \sqrt{{m_{21}}^2 + {m_{31}}^2} \in [-1, 1] \tag{28}$$

$$D_L = \sqrt{{m_{12}}^2 + {m_{13}}^2} \in [-1, 1] \tag{29}$$

$$q_L = \sqrt{{m_{42}}^2 + {m_{43}}^2} \in [-1, 1] \tag{30}$$

$$r_L = \sqrt{{m_{24}}^2 + {m_{34}}^2} \in [-1, 1] \tag{31}$$

Denote the central 2×2 block of the MM as $\boldsymbol{B}$, we also have its trace, determinant, and Frobenius norm as rotation invariants

$$\mathrm{tr}\boldsymbol{B} = m_{22} + m_{33} = 2b \tag{32}$$

$$|\ \boldsymbol{B}\ | = m_{22}m_{33} - m_{23}m_{32} \tag{33}$$

$$\|\boldsymbol{B}\| = \sqrt{{m_{22}}^2 + {m_{33}}^2 + {m_{23}}^2 + {m_{32}}^2} \tag{34}$$

However, the set $\{\mathrm{tr}\boldsymbol{B}, |\boldsymbol{B}|, \|\boldsymbol{B}\|\}$ is not as widely used as another equivalent set $\{b, \beta, t_1\}$ by experimentalists [46]

$$b = \frac{1}{2}(m_{22} + m_{33}) \tag{35}$$

$$\beta = \frac{1}{2}(m_{23} - m_{32}) \tag{36}$$

$$t_1 = \frac{1}{2}\sqrt{(m_{22} - m_{33})^2 + (m_{23} + m_{32})^2} \tag{37}$$

Parameter b is sensitive to sub-wavelength particle scattering. β is related to optical rotation and coexistence of multiple anisotropic effects. t_1 represents the overall degree of anisotropy.

6.2 Mueller Matrix Azimuth Orientation Parameters

When the sample has a mirror symmetry plane (such as a sample with only one type of anisotropy effect), the Mueller matrix of the sample can be block-diagonized at specific azimuth orientation

$$M_{\text{mirsym}} = \begin{pmatrix} m_{11} & m_{12} & 0 & 0 \\ m_{21} & m_{22} & 0 & 0 \\ 0 & 0 & m_{33} & m_{34} \\ 0 & 0 & m_{43} & m_{44} \end{pmatrix} \tag{38}$$

Thus, for a randomly oriented sample, we can extract the information of the azimuth orientation α of the anisotropy from the measured M'_{mirsym}

$$M_{\text{mirsym}} = \begin{pmatrix} m_{11} & m_{12}c_2 & m_{12}s_2 & 0 \\ m_{21}c_2 & b + \tilde{b}\, c_4 & \tilde{b}\, s_4 & -m_{34}s_2 \\ m_{21}s_2 & \tilde{b}\, s_4 & b - \tilde{b}\, c_4 & m_{34}c_2 \\ 0 & -m_{43}s_2 & m_{43}c_2 & m_{44} \end{pmatrix} \tag{39}$$

$$c_n = \cos(\mathrm{n}\alpha)\,,\ \mathrm{s}_n = \sin(\mathrm{n}\alpha)\,,\ \tilde{b} = \frac{1}{2}\left(m_{22} - m_{33}\right) \tag{40}$$

By using the following formulas:

$$\alpha_1 = \frac{1}{4}a\tan 2\left(m'_{23} + m'_{32}, m'_{22} - m'_{33}\right) \text{ if } t_1 \neq 0 \tag{41}$$

$$\alpha_P = \frac{1}{2}a\tan 2\left(m'_{31}, m'_{21}\right) \text{ if } P_L \neq 0 \tag{42}$$

$$\alpha_D = \frac{1}{2}a\tan 2\left(m'_{13}, m'_{12}\right) \text{ if } D_L \neq 0 \tag{43}$$

$$\alpha_q = \frac{1}{2}a\tan 2\left(m'_{42}, -m'_{43}\right) \text{ if } q_L \neq 0 \tag{44}$$

$$\alpha_r = \frac{1}{2}a\tan 2\left(-m'_{24}, m'_{34}\right) \text{ if } r_L \neq 0 \tag{45}$$

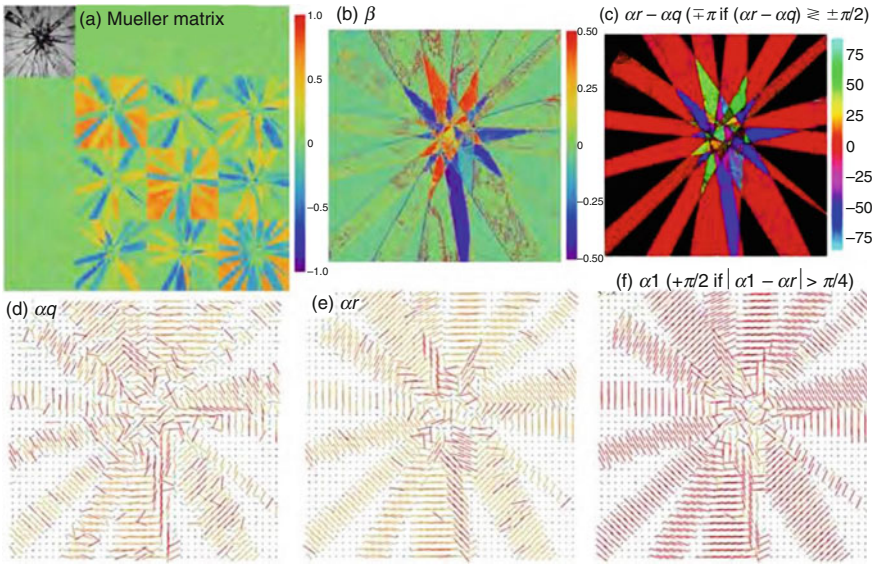


Fig. 21 Measurement of transparent sticky tapes in transmission configuration. (**a**) The Mueller matrix, $m11$ is plotted in [0, 1] grayscale, other elements are normalized by m_{11}. (**b**) Invariant parameter β can show the overlapped zone of the sample. (**c**) The difference of orientation parameters (α_r, α_q) can also show the overlapping of different anisotropies. (**d**–**f**) Orientation maps predicted by α_q, α_r, and α_1, length and color of the small bars represent q_L, r_L, t_1 [52]

Here $a$tan2 is the arctan function that takes in two parameters (dy, dx) (thus able to return the angle in full 2π range). The sign convention in Eqs. (41, 42, 43, 44, and 45) is based on the assumption that $\tilde{b}, m_{21}, m_{12}, m_{34} > 0$ and $m_{43} < 0$ when $\alpha = 0$. If the truth is the opposite, then both input parameters (dy, dx) should flip their signs (or add π to the result of $a$tan2). Experimental result is illustrated with a transparent sticky tapes sample, as shown in Fig. 21.

6.3 Mueller Matrix Symmetry Breaking Parameters

From the symmetries study of Mueller matrix, one can discover that a symmetric sample can result in a transpose symmetric or anti-symmetric Mueller matrix [68]. Therefore, when analyzing a Mueller matrix whose physical structure is unknown, we can predict the types of preserved and broken symmetry in the sample by checking the indicators proposed in Table 1.

An example of the Mueller matrix of a turbid media which contains cylinder scatterers or linear birefringence effect in one or two layers is given by Monte Carlo (MC) simulation, as shown in Fig. 22.

Table 1 Mueller matrix indicators for breaking of symmetries

Indicator	DD	RR	DR2	DR1
$m22 \neq m33$	T	T	T	T
$m23 = m32 \neq 0$				T
$\|m23\| \neq \|m32\|$	T	T	T	
$m14 = -m41 \neq 0$				T
$\|m14\| \neq \|m41\|$			T	
$P_L \neq D_L$			T	
$q_L \neq r_L$			T	

"DD/RR" means mixing pure LD/LR - linear diattenuation / linear retardance effects with a nontrivial difference angle of azimuth orientation in separated layers, "DR2/1" means LD and LR are mixed in two layers or one layer. "T" means true [54]

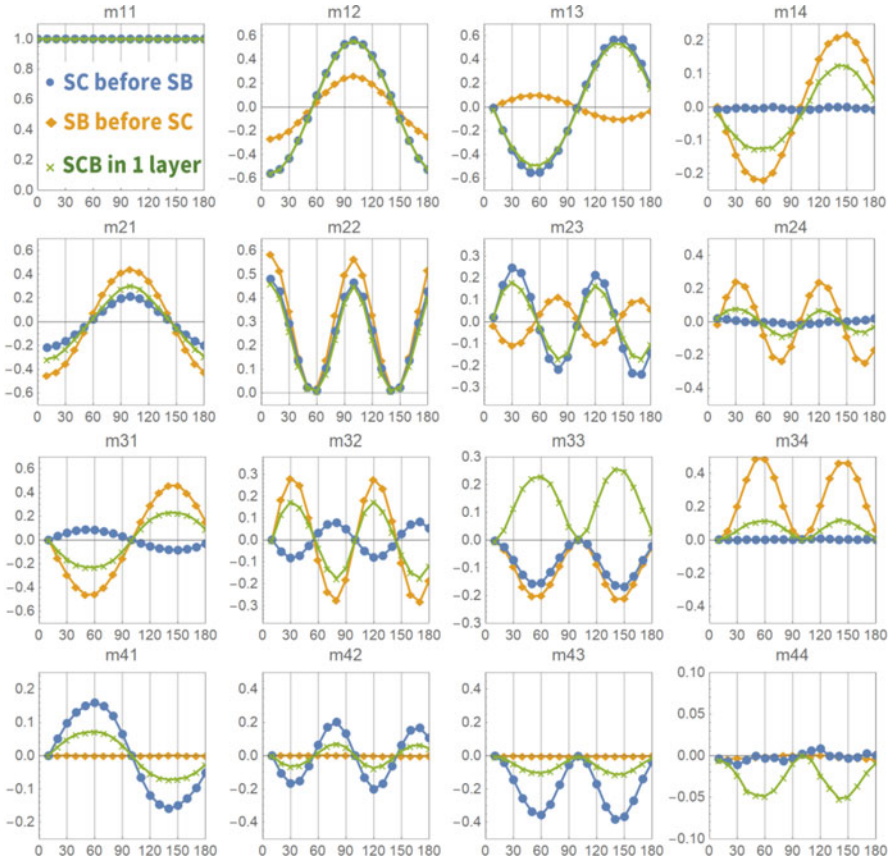


Fig. 22 MC simulation of the transmission Mueller matrix of two-layer scattering medium [54]. Layer "SC" contains well-aligned infinite long cylinder scatterers to produce LD effect, whose orientation is rotated in xOy plane from 0° (the x-axis orientation) to 180°. Layer "SB" has uniaxial birefringence to produce LR effect (the optical axis is fixed at x-axis orientation) and sphere scatterers which produce only depolarization effect. Blue dot line shows the simulation that layer SC is placed before layer SB, and orange diamond line shows that layer SB is before layer SC. The third simulation, which mixed these two layers into one layer, named "SCB," is shown as the green cross line [54]

7 Machine Leaning for Polarization Feature Extraction

So far in this chapter, we summarized the consistent efforts to derive sets of polarization parameters which are sensitive to the intrinsic physical properties of complex samples and are either insensitive or related explicitly to the orientation angle of the sample. These parameters, denoted as polarimetry basis parameters (PBPs), can be derived by MMD methods including Lu-Chipman decomposition, differential or logarithm decomposition, and Cloude decomposition, and MMT methods including derivation of rotation invariants and orientation parameters. Furthermore, taking advantages of the fast developing big-data technology, new methods based on machine learning can derive specific polarimetry feature parameters (PFPs) from the high-dimensional Mueller matrix data or from the PBPs, to identify target microstructures quantitatively and automatically or conduct classifications. The method of deriving polarization parameters on pixel level has great potential to correlate specific features with the domain knowledge, especially for these features which are difficult to characterize using the existing polarization parameters.

The process of digitizing the Whole-Slide Images (WSI) of pathological tissues has led to the advent of machine learning tools in digital pathology to help with various tasks [57], including object recognition problems, counting problems, and predicting disease diagnosis and prognosis of treatment. In histopathology, biological or medical samples are usually processed by various staining techniques to selectively display specific pathological features. An experienced pathologist examines staining samples and identifies areas of interest for diagnosis. By measuring Mueller matrix and labeling pixels in the segmentation region, polarization features are extracted from Mueller matrices of all the labeled pixels using supervised learning methods. Therefore, the combination of polarimetry method and big-data technology can be used as a powerful tool in histopathological digitalization and computer-aided diagnosis. Here, we give two examples of using machine learning methods to extract polarimetry features from Muller matrix elements and PBPs for pathological diagnosis in clinic.

In a preliminary study on the identification of ocular scar tissue after glaucoma drainage device implantation, it was found that MMPD or MMT parameters were not sufficient to distinguish sclera, conjunctiva, and scar tissue. However, machine learning algorithms can extract more structural information from the 16-dimensional Mueller matrix and clearly separate different tissues in the virtual stained image shown in Fig. 23 [58]. In this study, random forest (RF) and Gaussian kernel support vector machine (SVM) were used to quantitatively analyze and discriminate scar tissue. The feature selection based on RF is usually used to enhance the generalization ability of the model, thereby improving the inference performance. SVM can be easily extended to nonlinear classifiers using kernel techniques to handle high-dimensional/nonlinearly distributed data sets. The statistical analysis of the discrimination results shows that the algorithm has strong clinical diagnosis ability, and the feature selection result indicate that birefringence may play a decisive role in the recognition of eye scar tissue, which is consistent

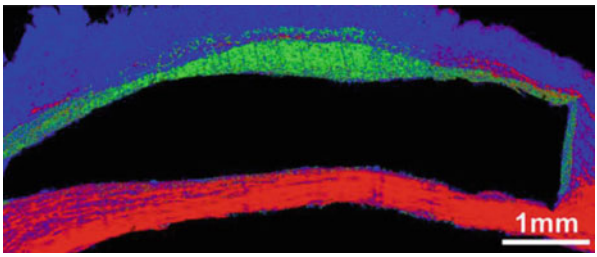


Fig. 23 The pseudo-color image of recognition result: sclera (red), scar tissue (green), and conjunctiva (blue) [58]

with other eye fiber classification studies. The polarization features derived by machine learning methods from a 4×4 Mueller matrix may quantitatively identify target microstructures on pixel level, which shows promising future on new feature extraction methods and pathological diagnosis applications.

In a recent attempt, PBPs were used to develop a pixel-based PFP extraction method using a linear discriminant analysis (LDA) classifier and a feature selection method [59]. In this work, microscopic Mueller matrix images of H&E pathological sections are taken for several typical breast tissues: healthy breast tissue, breast fibroma, breast ductal carcinoma, and breast mucinous carcinoma. The LDA-based training method learns to find the most simplified linear combination from PBPs constrained under the accuracy remains constant to characterize the specific microstructural feature quantitatively in typical breast tissues. The algorithm architecture in this study involves four widely used machine learning classifiers: linear discriminant analysis (LDA) [60], support vector machine (SVM) [61], extreme random tree (ET) [62], and artificial neural network (ANN) [63] and is constructed by using the open-source library scikit learn in Python 3.7.0. Figure 24 provides the overview of deriving PFPs algorithm for quantitative characterization of target microstructure features: First, we calculate the accuracy of two-class LDA classifiers for target feature recognition with a series of PBPs of training samples as input data, and the accuracy is denoted as Y_8. Then the number of PBPs, named n ($n = 8 - i$, $i = 1, 2, \ldots, 7$), was reduced one by one. The LDA model is trained by optimizing coefficients of n input PBP combinations to maximize the separation between different classes. Therefore, the output of LDA model is a PFP, that is, a linear combination of n PBPs with optimization coefficients, and the accuracy of PFP for pathological feature characterization can be evaluated by the score function of LDA imported from scikit learn library in Python 3.7.0. Therefore, when the number of inputs PBP is n, all possible parameter combinations are traversed, and accuracy is calculated. Select the corresponding combination of the maximum accuracy value (Y_n) and n PBPs with optimization coefficient (x_n) to enter the next step of the flowchart. If the difference between Y_8 and Y_n is less than 0.01, it indicates that the characterization ability of the new output PFP remains unchanged for the target microstructure characteristics, although the number of PBPs used is

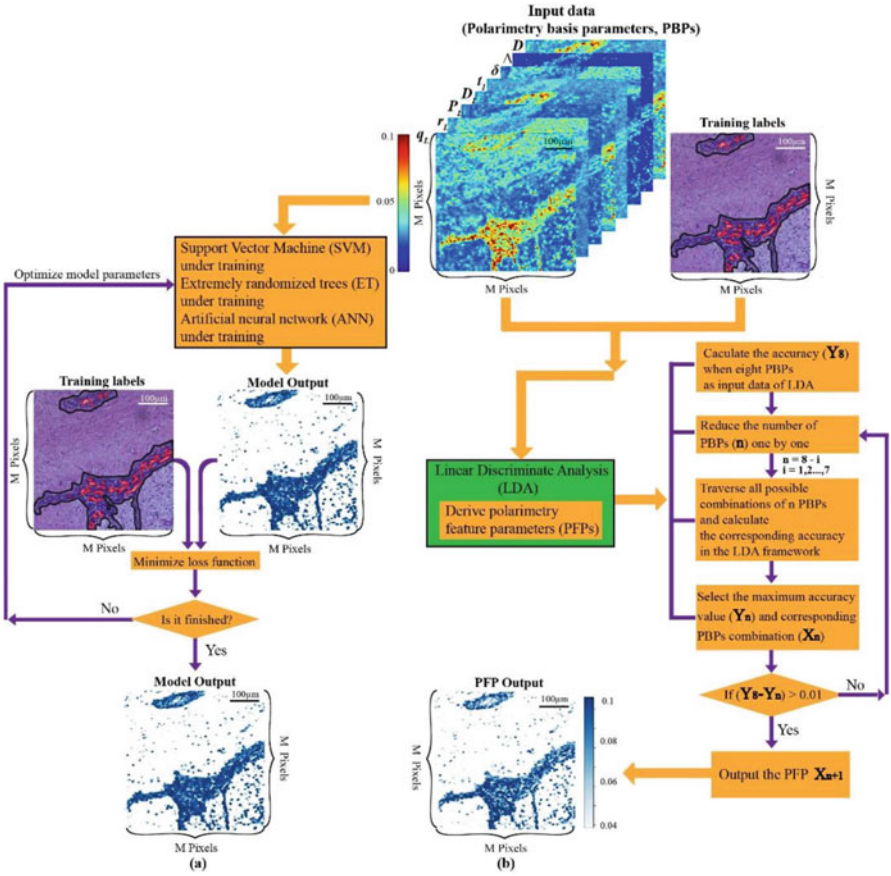


Fig. 24 Schematics of the machine learning classifiers trained for the recognitions of target microstructures in breast tissues. Two approaches were evaluated: (**a**) nonlinear models, SVM, ET, and ANN, trained for the microstructure recognitions; (**b**) the LDA-based training algorithm for the extraction of PFPs which can be used for the quantitative characterization of microstructure [59]

reduced. Otherwise, it means that the recognition ability of LDA model to target microstructure feature is compromised. Therefore, the program stops the cycle and sets x_{n+1} in the previous cycle as the output PFP, which is the simplest linear combination to represent the target pathological feature when Y_{n+1} and Y_8 maintain the same accuracy. Figure 24a shows that when the PFP is obtained to characterize the target microstructure, the same input data and labels are used to train other nonlinear models SVM, ET, and ANN.

Here, 12 PFPs are obtained to describe the three cancer-related pathological features in each typical breast pathological tissue, that is, cell nuclei, aligned collagen, and disorganized collagen, as shown in Fig. 25. The areas inside the black

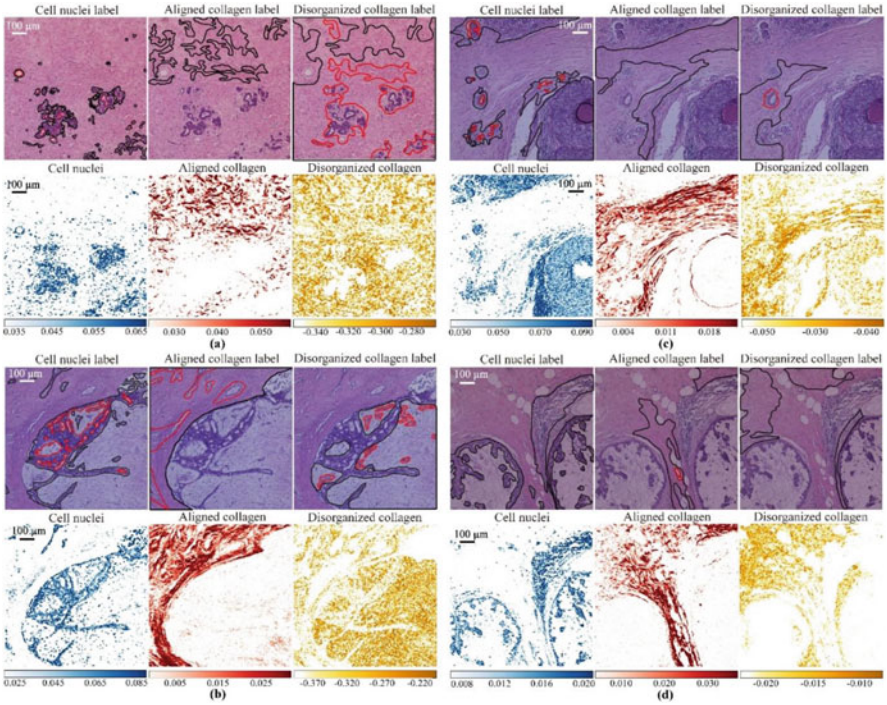


Fig. 25 Quantitative characterization of cell nuclei (blue pixels) aligned collagen (red pixels), and disorganized collagen (orange pixels) using the PFPs in H&E pathological sections of different breast tissues: (**a**) healthy breast tissue; (**b**) breast fibroma; (**c**) breast ductal carcinoma; (**d**) breast mucinous carcinoma [59]

solid line and outside the red solid line in H&E images are the target microstructural features labeled by the breast pathologist. The image size is 800 × 800 pixels. The training and testing of PFPs were completed in 224 regions of interests. By the validation of the PFPs performance with the corresponding labeled area in H&E images as ground truth, it can be concluded that the performance of PFPs in identifying the carcinogenesis-related microstructures is satisfactory (AUC 0.87–0.94, accuracy 0.82–0.91, precision 0.81–0.95, and recall 0.80–0.98). It indicates that the derived PFPs may automate the diagnosis process and predict patient survival and prognosis. Compared with other imaging processing methods in digital pathology, the proposed model has some unique advantages. First, PFP is the simplified linear function of the PBPs with physical meanings, providing quantitative characterization for target pathological feature and allowing in-depth analysis of physical interpretation. In addition, taking advantages of polarization imaging, such as sensitive to sub-wavelength microstructures and less sensitive to imaging resolution, the output PFPs with high sensitivity have the potential to rapidly scan and quantitatively analyze the whole pathological section in the low-resolution and wide-field systems.

8 Prospect About MM Microscopy

Mueller matrix microscopy is a promising technique for life science. Changes in polarization states due to propagation and scattering in the complex biological specimen are sensitive to the anisotropic and nonuniform optical properties in the medium at the scale even lower than the wavelength of illumination light. Therefore, Mueller matrix imaging provides ample microstructural information of complex medium at subcellular level. Such characterization process can be label-free, noninvasive, cross-scale, and quantitative, which make MM microscopy an attractive and powerful tool for charactering the structural features of complex samples, even living cells and tissues. MM microscopy is expected to continue its rapid advances in two frontiers. For the hardware, new instrumentations are being developed to achieve higher specifications, such as accuracy and speed, and to increase the number of information channels for solving inverse problems in a more stable manner. For the software, novel data techniques are emerging continuously for mining the abundant microstructural information encoded in polarization data.

An MM microscope measures the polarization properties of tissue and cell structures and decodes the information on the microstructural features. In pathological applications, samples are usually thin slices of tissues or single layer cells which cause minimal changes in the polarization of light. The capability of extracting detailed microstructure features of such samples depends on directly the quality of the measured polarization parameters. There will be consistent efforts to develop novel optimization methods for stable and precise polarization measurements or calibration methods for higher accuracy by removing polarization artifacts of the system.

Increasing the speed of MM imaging is also a basic need in MM measurements, particularly in dealing with living samples or dynamic process. New polarization optical components or devices, such as PSG based on faster polarization modulation or PSA based on dual DoFP polarimeter, can reduce the acquisition time. Using machine learning, one can recover an MM by taking a snapshot Stokes image, which significantly reduces the time for taking an MM image and, more importantly, eliminates errors due to system variations during the multipole exposures for taking the MM.

Light field contains a variety of different information channels, including intensity, wavelength, phase, propagation angle, polarization, and coherence. A Fourier-Mueller microscope combines the classical techniques of white-light interferometry, Mueller polarimetry, Fourier scatterometry, and nonlinear microscopy into one apparatus, allowing for the acquisition of fully angle- and wavelength-resolved Mueller matrices, which leads to additional spectral properties of the sample and spatial orientations of anisotropy as well as validation through second harmonic generation [64]. Such increase in the number of information channels may also result in more stable solutions in solving the inverse problem. Polarized light confocal microscopy is also finding more applications and will be discussed in Chap. 12.

The fast development of the AI techniques, especially machine learning, is generating a great impact on MM polarimetry. Since polarization properties represented by the Mueller matrix at each pixel are closely linked to the local microstructure which the pathologists have labeled, polarization feature extraction corresponding to specific histopathology feature is the key for the success of polarization-based digital pathology [59, 65]. New data mining techniques will be developed to decode the microstructural information from pixels of an MM image.

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Confocal Laser Scanning Polarimetry



Mariacarla Gonzalez, V. N. Du Le, and Jessica C. Ramella-Roman

1 Introduction

Confocal microscopy was first introduced in 1957 by Minsky [1], although its use in research was not established until the 1980s [2]. This method has been used in multidisciplinary fields, from analyzing biological samples to quality control of manufactured materials [3–5] and since its introduction has been improved in terms of illumination, scanning, contrast and speed, and spanning its capabilities [6–9].

The utilization of polarization in microscopy initiated in the 1970s for its contrast capabilities. The addition of polarimetry in laser scanning microscopes was first used to look at chromatin, blood cells, and other macromolecules [10–13]. The implementation of polarization in confocal microscopy started with the addition of cross-polarization components (whether stand-alone polarizers or liquid-crystal variable retarders (LCVRs), enhancing imaging quality [14–16]. Mueller matrix polarimetry (MMP) was added to the confocal approach to provide structural details and small features that were not available in confocal microscopy alone [5, 16, 17]. Mueller matrix confocal microscopy can characterize a sample based on its polarization response in three dimensions, providing depth resolution. The implementation of this method was first seen in ophthalmology, to better visualize the blood vessels in the retina [18]; however, only a subset of Mueller matrix coefficients were measured. In 2005, Lara and Dainty introduced an experimental confocal Mueller

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matrix polarimeter to obtain the 16 Mueller matrix elements at various depths [19]. Further iterations with enhanced capabilities were later introduced in 2008 to image glaucoma [20]. Modifications of non-linear microscopes with added Mueller matrix capabilities have been explored as well, increasing the amount of output information available [21–23]. A review on confocal Mueller matrix microscopy has been presented covering a wide range of applications [24]. Most recently, machine learning approaches have been introduced to further enhance the quantitative nature of confocal polarimetry [25].

The use of fluorescence combined with confocal polarimetry has been well explored as well. Confocal fluorescence polarization imaging (also known as fluorescence anisotropy) combines the high sensitivity of fluorescence along with the structural power of polarization-sensitive imaging. Applications ranging from monitoring membrane changes in human astrocytoma cells [26] to orientation of liquid crystals [27] and biological samples [28] have been demonstrated.

This chapter covers polarization-sensitive confocal microscopy and its applications. The basic principles of the technique (including a deeper look into polarization and confocal imaging), instrumentation, and signal processing will be discussed.

2 Polarization-Sensitive Confocal Imaging

2.1 Basic Principles of Polarization-Sensitive Confocal Imaging

In 1957, Marvin Minsky patented the first working confocal microscope and the principle of confocal imaging, driven by the desire to image biological events as they occur in the living tissue (in vivo) [29, 30]. In his original document, Minsky described a reflected light version of the microscope, having a single-objective lens and a dichromatic mirror along with two pinholes, one in front of the light source and one at the detector; the latter is the key to the confocal approach. As demonstrated in Fig. 1a, the pinhole serves as a spatial filter-eliminating out-of-focus light in the specimen [31]. The same principle was employed by all the modern confocal imaging system, in particularly when imaging either fixed or living cells and tissues that have usually been labeled with one or more fluorescent probes. Unlike the conventional widefield optical microscope, in which the secondary fluorescence often interferes with the resolution of features in focus, confocal approach excludes the out-of-focus fluorescence occurring in thick labeled specimens.

If the pinhole is misaligned or if its size is reduced beyond that corresponding to a diffraction-limited spot on the sample, the number of photons reaching the detector is dramatically reduced, while only a marginal improvement in xy- or z-resolution is achieved [32, 33]. On the other hand, if the pinhole is larger than the diameter of the

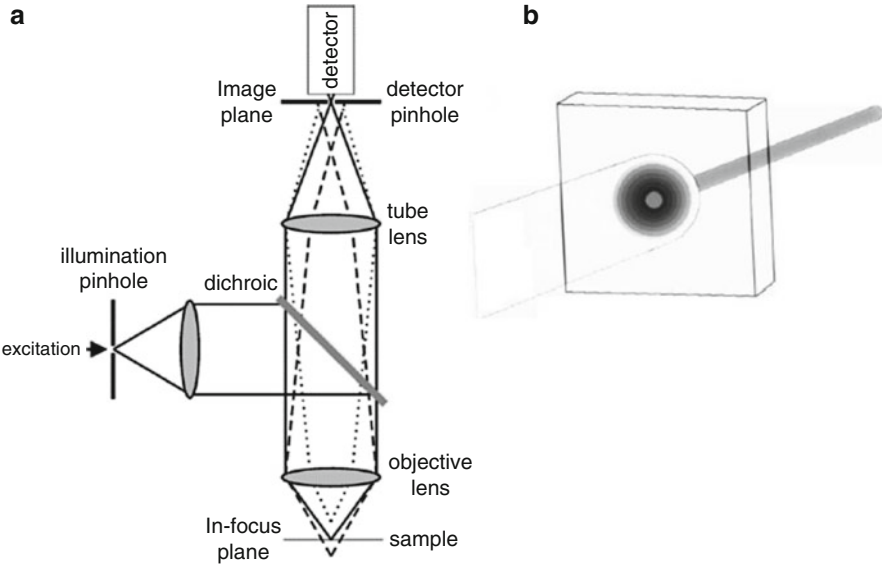


Fig. 1 (**a**) The light path in a typical LSCM: The pinhole at the light source, the focused point in the specimen, and the pinhole in front of the detector are all confocal with one another. (**b**) The relationship between the hole in the pinhole and the Airy disk focused at this plane. (Reproduced from Centonze and Pawley [31])

diffraction-limited spot, most of the additional signal comes from adjacent planes, reducing the z-resolution [32]. In practice, the size of the aperture in front of the detector is adjustable to match the size of the image of the Airy disk being projected from the excited point in the specimen to the pinhole plane by the objective lens (Fig. 1b). Through the Abbe equation, the pinhole radius r is proportional to wavelength λ and inversely proportional to the numerical aperture (NA) of the objective lens so that

$$r = 0.61\frac{\lambda}{\mathrm{NA}} \tag{1}$$

Limited by the technology in 1957, Minsky was not able to demonstrate the potential of confocal microscope in imaging biological samples. To form an image, the focal point must be scanned across the specimen. In Minsky's original microscope, the beam was stationary, while the sample was mounted on a moving stage resulting in the loss of image resolution. In addition, Minsky was not able to produce a real image but rather a signal buildup on an oscilloscope. Thus, Minsky's original microscope did not catch on with biological community until the invention of photomultiplier tube (PMT) and digital charge-coupled device (CCD) camera.

In addition, the introduction of motorized z-axis stage in all modern confocal microscopes permits more precise optical sectioning of thick specimens, enabling

the three-dimensional (3-D) reconstruction of the sample to localize the fluorescence. Modern confocal microscope can be classified as stage scanning confocal microscope and laser scanning confocal microscope (LSCM). The latter is more practical to image biological samples and is the basis of most modern confocal light microscopes.

2.2 Polarization Sensing Principles

Cross-polarization, Mueller matrix, and fluorescence polarization imaging are the most common implementation of polarization-sensitive confocal microscopy. The sensitivity of polarization to birefringent media has led to its use in imaging biological tissues, especially to visualize the extracellular matrix (ECM) [34], which has been accomplished with cross-polarization and Mueller matrix confocal microscopy to image tissues such as the cornea, tendon, and ligaments [21–23, 28]. Confocal fluorescence polarization imaging has been used to measure fluorescence anisotropy with applications in cancer sensing and staging.

The use of cross-polarization in confocal microscopy started as a way to limit the noise from backscattered signals by using quarter-wave plates and polarizing beam-splitters [7]. Another experimental setup used a half-wave plate in a transmission arm that could be crossed to an analyzer (i.e., shifted 90 degrees) to allow either the transmission or reflected signal to be measured [14]. Although this implementation did not add quantitative information, it improved the signal quality received by the detector.

The first introduction of MMP in confocal microscopy was also utilized to reduce noise and increase contrast (in structural details) in a microscopic and ophthalmoscopic application. The samples imaged consisted of a US Air Force resolution chart and a black image on a white paper [5]. Further work from the same author extended toward the use of Stokes vectors for contrast improvement in visualizing blood vessels in the retina [18].

The evolution of polarization-sensitive confocal microscopy includes using the quantitative aspect of MMP. A Mueller matrix describes the transfer function of a medium and its interaction with polarized light. It is capable of doing so through three fundamental properties of polarization: depolarization, retardance, and diattenuation [35]. Depolarization occurs primarily due to multiple scattering. Biological tissues tend to be highly depolarizing due to structures such as collagen fibrils and organelles, among others. Retardance arises from a phase shift between two orthogonal polarization eigenstates and is exhibited commonly in fibrous and anisotropic structures (e.g., tendon, cornea, sclera). Diattenuation refers to the polarization dependent absorption some molecules exhibit, such as amino acids and glucose, however it does not commonly affect bulk biological tissues [36].

A Mueller matrix is calculated by illuminating four incident polarization states, known as the polarization state generator (PSG), and analyzing by the same four incident polarization states using a polarization state analyzer (PSA). The PSGs

commonly used in the field include voltage modulation with Pockels cells [19, 37], liquid crystal retarders [21, 22, 25], and rotating elements (polarizer and wave plate combinations) [28, 38]. There are two main experimental setups for the creation of a PSA. The first one utilizes a single detector where the analysis of the polarized beam is achieved sequentially through switching or rotating polarizing elements [21–23, 25]. The second setup consists of four detectors that simultaneously acquire four images for each generated state of polarization [28, 37, 38].

Confocal fluorescence polarization microscopy or fluorescence anisotropy refers to the sensitivity of fluorescent excitation and emission to polarization and its ability to separate signals from fluorophores with overlapping spectra, yielding information on the targeted molecule and its surroundings. The emission beam is polarized with respect to the linearly polarized excitation beam; after its interaction with the sample, it is split into a parallel and perpendicular polarization vector [39]. Anisotropy is related to the difference between these two polarization components and defined as

$$r = \frac{I_{\parallel} - I_{\perp}}{I_{\parallel} + 2I_{\perp}} \tag{2}$$

where $I_{\parallel}$ is the polarized emission intensity for an electromagnetic field parallel to the reference frame and $I_{\perp}$ is the polarized emission intensity for an electromagnetic field perpendicular to the reference frame.

The sample has high anisotropy when $I_{\perp}$ is low, and the anisotropy is unity when $I_{\perp} = 0$. The latter does not frequently occur in practice. Low anisotropy is often related to the depolarizing effect of a sample, often caused by changes in the direction of the transition moments of the fluorophore [40, 41].

Confocal fluorescence polarization imaging has been previously used to capture the changes in membrane structure during processes of cell injury in human astrocytoma cells. The anisotropy, r, was calculated by acquiring polarized fluorescence emission in two separate detectors, where the experimental result agreed well with previous theoretically predicted values [26]. Another common use for this technique is to obtain orientation information. Smalyukh et al. imaged two low-birefringence liquid crystals to understand their density and decomposition by measuring their 3D orientation [27].

Mazumder et al. used confocal fluorescence polarization imaging to characterize the polarization properties of type I collagen from the fluorescence signal. The setup introduced by the group consisted of a PSG formed of a polarizer, a half-wave plate, and a quarter-wave plate, and the PSA consisted of a four-channel detector array where the fluorescence signal was acquired simultaneously for four separate states of polarization. Using Mueller matrix polar decomposition (from the autofluorescence signal), the polarimetric parameters of depolarization, retardance, and optical rotation are calculated to show the collagen fiber's strong scattering, as well as its highly anisotropic and chiral nature [28]. Further relevant literature on confocal fluorescence polarization has been published [42–45].

Most recently, combined systems using non-linear microscopy methods and polarimetry have been introduced. Okoro et al. used the signal from the second-harmonic generation (SHG) as a mask to highlight collagenous tissues in the polarization-sensitive confocal mode [10]. Saytashev et al. introduced a Self-vAlidating Mueller matrix Micro-Mesoscope (SAMMM) that allows for back-reflected Mueller matrix imaging, as well as confocal Mueller matrix and non-linear microscopy [22]. This system was used to gather information from optical phantoms [22], such as titanium dioxide and nylon fibers in milk, as well as rat cornea [21]. Most recently, the SAMMM was used in machine learning to discriminate between fibrillary collagen fibers and elastic fibers [25].

3 Instrumentation

3.1 Confocal Fluorescence Polarizing Microscopy

Confocal fluorescence polarizing microscopy offers structural information from fluorescence as well as polarization sensitivity. Fluorescent tracers are used to image sample structure; moreover, the addition of polarization elements allows further extraction of information. The addition of polarizers in the excitation and the detection arm can provide information on the orientation of the sample, since the fluorescence signal will be highest when planes of polarization are parallel (or perpendicular) to the reference frame. This information can be further used to calculate the anisotropy as shown in the chapter "Mueller Matrix Polarimetry in Biomedicine: Enabling Technology, Biomedical Applications, and Future Prospects". Bigelow [45] proposed the experimental setup to image and calculate the anisotropy of EMT6 cells expressing green fluorescent protein (GFP) and meta-tetra(hydroxyphenyl)chlorin (mTHPC). The full experimental setup is shown in Fig. 2. Highlighting the polarization addition of the confocal fluorescence microscope, a polarizer in the incident arm sets the incident state of polarization; after the interaction with the sample, a polarizing beam-splitter (PBS) combined with two PMTs is used to collect the polarization-sensitive emission signal. PMT1 and PMT2 collect the parallel and perpendicularly polarized light, respectively. Using the ratio taken from these two measurements (previously defined as r), sample anisotropy can be defined.

3.2 Mueller Matrix Polarimetry

MMP provides the full polarimetric response of a sample through its 4×4 Mueller matrix so that at least 16 intensity measurements, obtained by a set of 16 coding and decoding polarization states, are performed to determine the real independent Mueller matrix elements. Figure 3 shows the main components of a MMP confocal microscopy system [46].

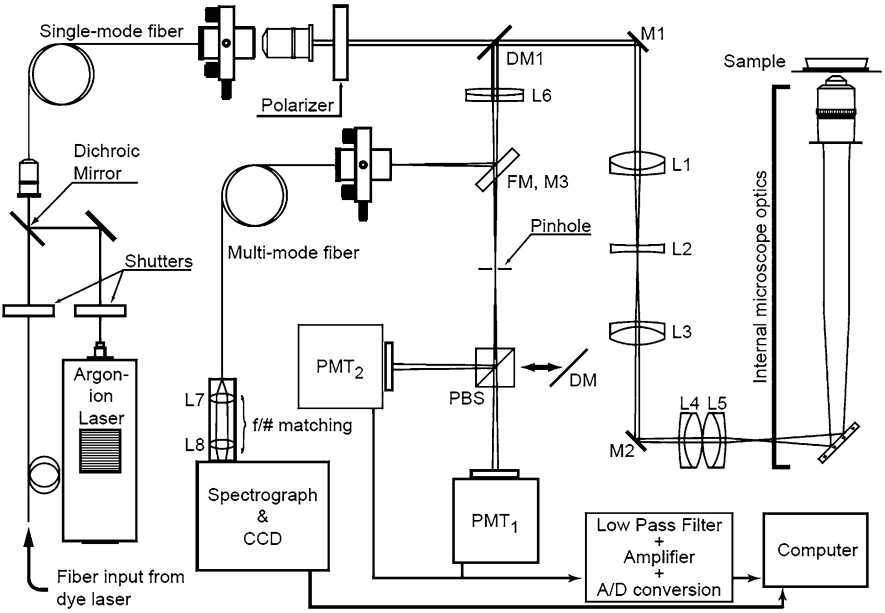


Fig. 2 Confocal fluorescence polarization microscope system. L1–L8, lenses; DM, dichroic mirror; FM, flip mirror; PBS, polarizing beam-splitter; M1–M2, mirrors; PMT1–PMT2, photomultiplier tubes. (Reproduced from Bigelow [45])

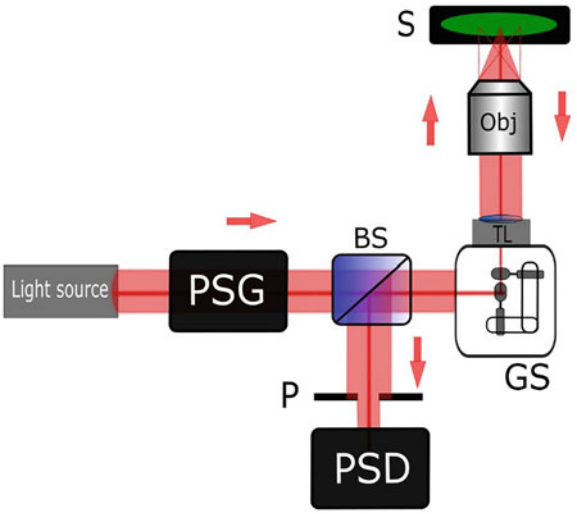


Fig. 3 Block diagram for a general Mueller matrix polarimetry confocal microscopy. PSG, polarization state generator; Obj, microscope objective; GS, galvanometric scanner; TL, tube lens; S, sample; BS, beam-splitter; P, pinhole; PSD, polarization state detector. (Reproduced from Le Gratiet et al. [46])

The main challenges in implementing MMP in confocal scanning laser microscope are encoding and decoding the polarization states to retrieve the full Mueller matrix pixel by pixel and to account for inaccuracy of PSG and PSA introduced by the different optical paths and optical elements. Some of the dedicated calibration methods include Eigenvalue calibration method (most simple, involving the use of a set of rotated polarizers and wave plate to encode polarization states) [19, 47, 48], equally weighted variance (used with Pockels cells or photoelastic modulators) [49, 50], and double-pass-through sample (used with reflectance confocal microscope) [51]. In an earlier Muller matrix confocal microscope developed by Lara and Dainty, two Pockels cells were used to generate the polarization states in few MHz [19]. Commonly, PSG and PSA consisting of a pair of LCVRs or other electro-optics devices were used to speed up sequential measurement [21, 22, 52–54]. Mueller matrix images are then extracted pixel by pixel based on the measurements of different PSA states for each of the PSG states.

In Mueller matrix laser scanning microscopy, the speed of encoding and decoding polarization states is the bottleneck and much slower than the dwell time of the galvanometric scanners. The most recent advance to get full Mueller matrix images at pixel dwell time rate is the snapshot approach [55], initially used in optical coherence tomography (OCT). In a snapshot Mueller matrix microscope, a micropolarizer array is coupled with a CCD or CMOS camera so that the simultaneous encoding and decoding of polarization states is achieved in the sensor. Each pixel of the camera is linearly polarization encoded, allowing the reconstruction of the four Stokes coefficients through a field-programmable gate array (FPGA) card in real time [56]. Although the snapshot technique does not require any sophisticated mathematical model and widely exists in numerous commercial polarization-resolved cameras, the main limit of this approach in confocal microscopy is the crosstalk between pixels and the loss of spatial resolution due to the definition of the super-pixels [57].

Modulators As mentioned above, light polarization can be modulated by rotating retardation plates [58], rotating compensators [59], Pockels cells [60], photoelastic modulators [61], or liquid-crystal variable retarders (LCVR) [52]. In case of Pockels cells, a pair of Pockels cell modulators is driven by sinusoidal high voltages and is placed at 45-degree and 0-degree azimuthal angles relative to the horizontal polarizer [60]. To generate four polarization states (16 Mueller matrix elements), two Pockels cell modulators are placed at two different azimuthal angles and driven at two frequencies. In the case of LCVR, the two modulators can produce only three independent states of polarization – linear horizontal, linear vertical, and right circular, while two additional quarter-wave plates in the entrance and exit optical paths produce a fourth state (approximately 45 $\pm$ linear) [52].

Objectives The performance of the objective is determined by analyzing its point spread function (PSF). With the ability of ray-tracing software to evaluate lens performance and to calculate wave-front aberration and PSF, most of modern objectives can perform close to diffraction limit at the center of imaging field. The smallest resolvable distance between two points equals the radius of the Airy

disk, $r = 0.21\lambda/\text{NA}$. Giving the mean rod spacing of 2.0–2.5 μm in the human retina [62], the useful magnification of the objective is about 500–1000 times the NA, while the Airy disk is enlarged four to five times the detector pixel dimension [63]. To preserve spatial resolution in polarimetry microscope, numerical aperture of polarized light objectives is usually 10–25% higher than those for ordinary microscopes. Furthermore, objectives designed for polarized light microscopy must be stress- and strain-free to prevent image quality loss [64]. In general, there are two sources of birefringence in microscope objectives: glass or material made-up lenses and strain birefringence when multiple lenses are mounted in proximity. To prevent these problems, it is advisable to choose strain-free optical glass or isotropic crystals to construct lens elements.

Dichroic beam-splitter (BS) Dichroic BS is often used in microscopy to steer beam and to split beam at a specific ratio, depending on the wavelength of the incident light. Unfortunately, the multi-layered dielectric dichroic BS can alter the polarization because its *s*- and *p*-axes have different transmission (or reflection) coefficients and phase retardations. To compensate for the dichroic effect, one common solution is to use light polarized along one of the axes of the dichroic, and the polarization angle can be modified by rotating half-wave plates in front of the objective or rotating the sample [65, 66]. Recently, Belanger et al. proposed a more practical method to compensate for dichroic effect by introducing an additional dichroic in the light path. The idea is to remove the amplitude factor and phase shift from the first dichroic by orientating the second one along *s*- or *p*-axes.

3.3 Combined Systems: Non-linear and Linear

Although conventional (single-photon) confocal microscopy can provide submicron spatial resolution of chemical dynamics within living cells, the spatial resolution is often limited by tissue scattering from the background. In principle, confocal microscopy is highly vulnerable to tissue scattering because this technique relies on collecting and re-directing fluorescent photons through a detector pinhole. In addition, scattered photons also excite diffuse fluorescence and cause photo-bleaching. Furthermore, fluorescent probes must be introduced to the specimen to produce useful images with conventional confocal microscopy. These probes include fluorescent dyes, fluorescent proteins, and quantum dots [67, 68]. The limits in conventional confocal microscopy can be overcome by introducing the non-linear optical fluorescence including two-photon excitation (TPE) and second and third harmonic generation (SHG and THG).

TPE fluorescence refers to the excitation of a fluorophore when two photons arrive within a time window of an attosecond (10–18 s) to excite the molecule. Although this is satisfied by any combination of photons that give the total energy equals the energy difference between ground state and excited state in Jablonski diagram, two photons of equal wavelength are usually used. TPE requires extremely

high photon flux, typically 1020–1030 photons/(cm^2s). The Ti:sapphire mode-locked laser is a favored light source because it can generate picoseconds to ∼100 fs pulses of high peak power but at a low repetition rate (∼100 MHz) and low average power (typically some 10 mW) [67, 68]. Unlike linear fluorescence, TPE utilizes longer excitation wavelengths in infrared region, and TPE fluorescence intensities are proportional to the square of the excitation intensity. The major advantage of TPE is the minimal loss of contrast due to photon attenuation at long excitation wavelength [67, 69, 70]. Infrared light is subject to less absorption and scattering than UV or visible light and thus can penetrate more deeply into a sample. In addition, photobleaching and photodamage are limited to the zone of TPE and do not occur above or beyond the focus [71].

SHG and THG can be incorporated in Mueller matrix laser scanning microscopy to increase the contrast based on organization and orientation of collagen fiber in the specimen. Like TPE, SHG signal is created in proportion to the square of the instantaneous local intensity and provides intrinsic axial resolution. Unlike TPE, SHG does not require actual absorption but is based on two non-linearly backscattered low-energy photons to produce one photon with double energy. In contrast to fluorescence processes, harmonically generated signals involve only virtual states. Thus, with the lack of energy deposition in the specimen, SHG produces no photodamage or bleaching. The wavelength of SHG is half, and THG is one third that of the incident wavelength, while TPE has an emission wavelength more than half of the incident wavelength. In addition, harmonic generation is not isotropic and highly sensitive to phase, making it unique in measuring molecular orientation [72]. Thus, only few biological structures that are ordered and not isotropic can produce harmonic signals. Collagen, a major protein of the ECM, is a known SHG-sensitive structures in biology [73–75]. After being secreted into the extracellular space, collagen molecules assemble into high-order polymers called collagen fibrils with diameter of 10–300 nm [68].

Figure 4 shows a complete system of Mueller matrix and non-linear microscopy used to study the cornea and cervix [21, 22, 25]. In such configuration, SHG and TPE signals in reflectance were used as ground truths to evaluate polarimetric contrast, while both the PSG and PSA are composed of a pair of LCVRs.

3.4 Polarimetric Signal Reconstruction

Mueller matrix confocal imaging systems require calibration. This can be conducted in multiple ways. Lara et al. used a double-pass eigenvalue calibration [76] on their confocal Mueller matrix system.

The double-pass method includes three main steps of the original single-pass methodology, specifically:

First, five calibration measurements are taken. These are B0, air (no sample); B1, a linear horizontal polarizer; B2, a linear vertical polarizer; and B3, a 633-nm third-

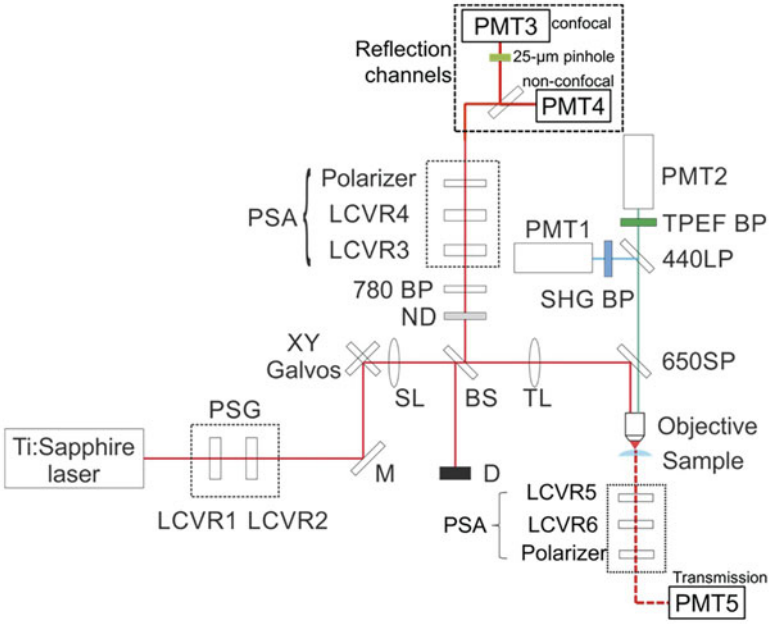


Fig. 4 A combined system of Mueller matrix microscopy and non-linear microscopy to study the cornea and cervix. (Reproduced from Du Le et al. [21])

order quarter-wave plate at the operating wavelength of this system (533 nm) and in double-pass implementation, so this last element becomes -0.26λ seventh-order wave plate, and the matrix of a mirror.

In the double-pass reflectance configuration, each measurement is then

$$B_{\text{sample}} = A \cdot \text{MM}^{-}_{\text{sample}} \cdot \text{Mirror} \cdot \text{MM}^{+}_{\text{sample}} \cdot W \tag{3}$$

The plus and minus signs in the Mueller matrix refer to the direction of propagation of light through the samples.

Second, the Mueller matrix of the samples is quantified from the eigenvalues of the product of the inverse of B0 and the corresponding non-calibrated matrix of the sample as shown in Eq. 4:

$$C_i = B_0^{-1} B_i = (A \cdot W)^{-1} \cdot (A \cdot \text{MM}_i \cdot W) \tag{4}$$

where $i = 1,2,3$.

If the sample behaves identically in both transmission directions and accounting for the orientation of the calibration sample $MM^{+}_{0^{\circ},i}$, we obtain

$$\text{MM}_i^{-} \cdot \text{Mirror} \cdot \text{MM}_i^{+} = \text{Mirror} \cdot \text{Rot}(\theta_i) \cdot \text{MM}^{+}_{0^{\circ},i} \cdot \text{MM}^{-}_{0^{\circ},i} \cdot \text{Rot}(-\theta_i) \tag{5}$$

Calculating the eigenvalues of C, we then obtain

$$\text{eig}\,(C_i) = \text{eig}\left(\text{MM}_i^{+} \cdot \text{MM}_i^{-}\right) = \text{eig}\left(\text{MM}_i^{\text{doublepass}}\right) \tag{6}$$

The third and final step of the eigenvalue calibration is conducted identically to the single-pass version, i.e., the eigenvalues of C are used to characterize the calibration samples, and W and A (PSG and PSA matrix, respectively) are found from a solution of linear equations [47].

Saytashev et al. used an external polarimeter (Thorlabs Inc., Newton, NJ) to directly measure the PSG at the sample plane as shown in Fig. 4 followed by a measurement of PSA at the pinhole and PMT location.

During data acquisition, the intensity measurement matrix M_i relates to the sample Mueller matrix M_s as

$$\text{M}_\text{i} = \text{PSA} \cdot \text{M}_\text{s} \cdot \text{PSG} \tag{7}$$

To attain Mueller matrix of a sample, first the Kronecker tensor product is calculated:

$$W = \text{PSA} \otimes \text{PSG}^\text{T} \tag{8}$$

By reshaping the M_i into vector form M_iv, we can relate intensity to sample Mueller matrix in vector form Msv with the following equations:

$$\text{M}_\text{iv} = \text{WM}_\text{sv} \tag{9}$$

$$\text{M}_\text{sv} = \text{M}_\text{iv} \times W^{-1} \tag{10}$$

Once LCVRs of PSG and PSA are set to calibrated values, the system's Mueller matrix M_sys is measured using a mirror placed at the focal plane of an objective. Since input polarization states are calibrated at the focal plane of an objective, an arbitrary sample's Mueller matrix M_s is related to a total measured Mueller matrix by an equation (Fig. 5):

$$M_\text{tot} = M_\text{sys} M_\text{s} \tag{11}$$

and therefore

$$M_\text{s} = \text{inv}\left(M_\text{sys}\right)\ M_\text{tot} \tag{12}$$

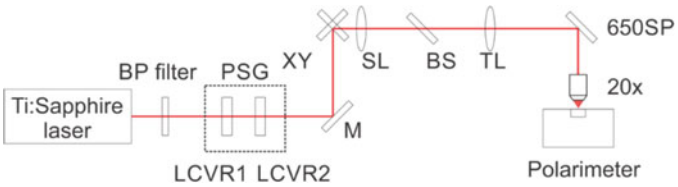


Fig. 5 Schematic diagram of PSG calibration

3.5 Lu-Chipman Decomposition

The polar decomposition or Lu-Chipman decomposition [77] has been used in polarization-sensitive confocal microscopy to extract information from the formed Mueller matrix.

The Lu-Chipman decomposition has been used to acquire information about optical phantoms like Titanium dioxide particles [3], nylon fibers submerged in milk [3], glycerol [28], among others, and biological tissues, such as the tendon, ligament, and cornea [21, 23, 28].

Du Le et al. proposed a polarization-sensitive confocal microscope capable of performing depth-resolved polarimetric and non-linear imaging to capture corneal structure in depth. The confocal Mueller matrix setup included LCVRs to control the PSG and PSA, which permits the acquisition of 16 images in order to build a Mueller matrix. By decomposing the Mueller matrix with the Lu-Chipman decomposition, the authors were able to gather information on the cornea's depolarization and retardance and compare the behavior with the non-linear signals for TPEF and SHG. Using the SHG signal as the "ground truth" for collagen presence, they observed that as the collagen signal increases in depth, the depolarization and retardance also increase in depth. The authors offer a volume reconstruction of the cornea delineating the signal changes in depth and the visualization power of polarimetric properties by Mueller matrix decomposition [21].

3.6 Coherency Matrix

Le Gratiet et al. have [38] noted, in the context of polarimetric scanning microscopy, that the structural complexity of the samples causes the scattering of polarized light to follow different diffusion regimes, from the mesoscale to the microscale. The use of the coherency matrix formalism rather than other forms of decomposition can then provide a better way to discriminate pathological and healthy tissue with laser scanning microscopy (although the concept has not been shown in confocal microscopy). The coherency matrix is the Hermitian matrix C defined as

$$C = \sum_{\mu,\upsilon} M_{\mu\upsilon}\Gamma_{\mu\upsilon} \tag{13}$$

where M is the Mueller matrix and Γ is the 16 (4×4) unitary coherency basis matrix

$$\Gamma_{\mu\upsilon} = \Lambda^{\dagger}\left(\sigma_{\mu} \otimes \sigma_{\upsilon}^{*}\right)\Lambda \tag{14}$$

which is defined as the Kronecker product of the Pauli matrices and the unitary matrix [78].

By observing the physical meaning of the terms of the coherency matrix as well as its eigenvalues, they showed the origin of the imaging contrast induced by different scattering regimes and tissue organizations of the zebrafish [38].

4 Application of Confocal-Sensitive Polarimetry

4.1 Ophthalmic: Cornea

The polarization properties of the cornea arise from the structural organization of its constituents, such as fibrils and lamellae [79–81]. Imaging the cornea in reflectance mode is crucial in the diagnosis of eye diseases. The corneal structure is depth dependent, consisting of five main layers: the superficial epithelium, the basal epithelium, the Bowman layer, the anterior and posterior stroma, and the endothelium [82].

MMP generally has advantages over polarized microscopy, PS-OCT, and scanning laser polarimetry (SLP). Polarized microscopy has limited access to the orientation of collagen fibers in the plane of the sample and only works well for systems where retardance is the only significant polarimetric effect [83]. SLP may give inaccurate result of retardance in systems with high depolarization and diattenuation effects [20, 81], while PS-OCT is not capable of measuring the degree of depolarization of partially polarized light due to the use of the Jones formalism [84, 85]. Using non-linear signals as ground truths (Fig. 6), it has been shown that Mueller matrix microscopy can reveal layered-specific birefringence information of the cornea [21].

In addition, polarimetric confocal scanning laser ophthalmoscopes were extensively used to study the living human eye [18, 86]. In such configuration, the analyzer unit was made of a linear polarizer and a quarter-wave plate. It was shown that the technique can improve the contrast of retinal blood vessel and structural details in the early diagnosis of ocular pathologies (Fig. 7).

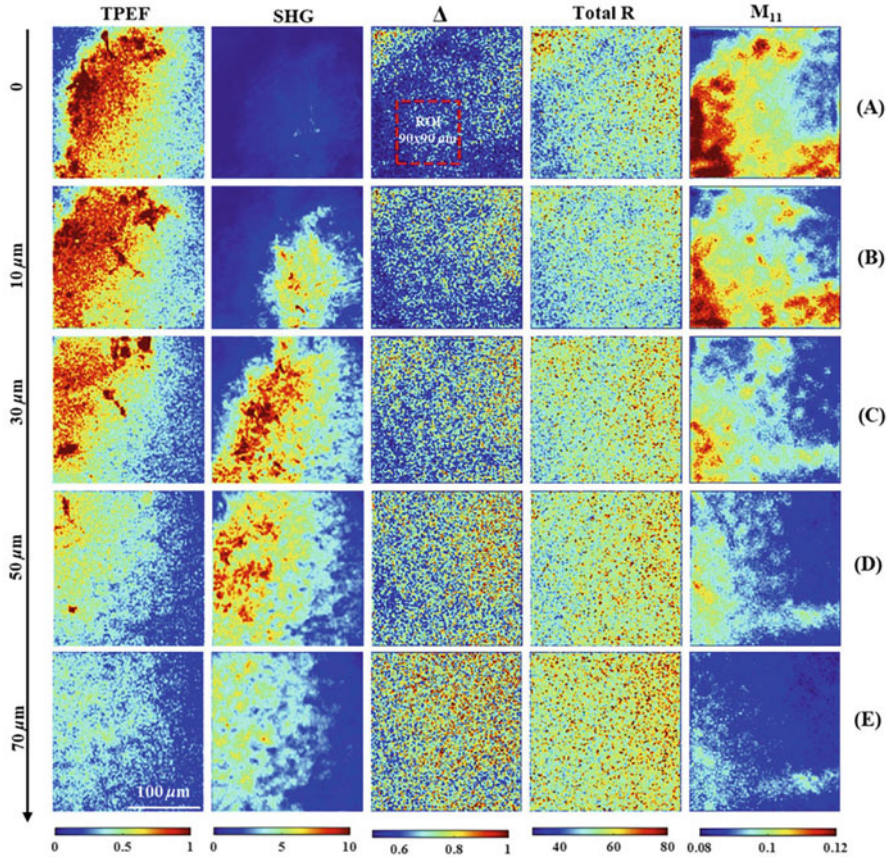


Fig. 6 Imaging of rat cornea without staining using a Self-vAlidating Mueller matrix Micro-Mesoscope (SAMMM): TPEF reveals the Bowman layer with subepithelial nerve plexus at 0–10 μm depth (**a**–**c**), and SHG reveals the epithelium-stroma interface at 10 μm (**b**). Polarimetry confocal reflectance was shown here. (Reproduced from Du Le et al. [21])

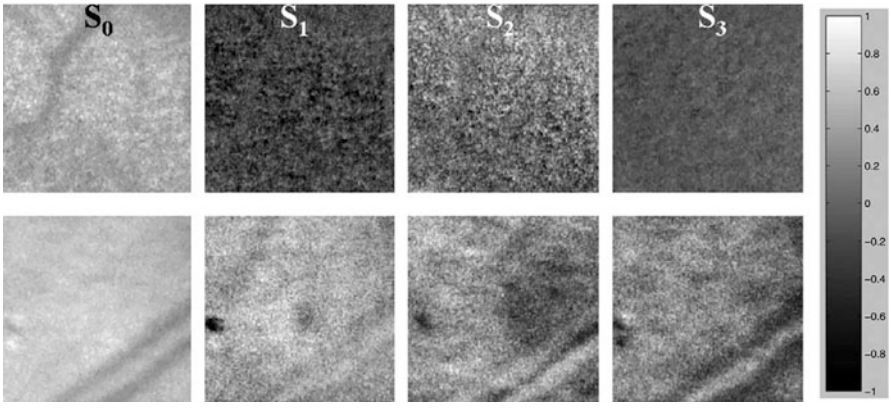


Fig. 7 Spatially resolved Stokes vector for two human subjects (top and bottom) from a confocal scanning laser ophthalmoscope (CSLO) developed by Bueno et al. [18]. A capillary is clearly visible in all the elements (bottom set). (Reproduced from Bueno et al. [18])

4.2 *Embryonic Confocal (Regenerative)*

Le Gratiet et al. used a commercial confocal system modified to achieve a complete Mueller matrix to study the structural development of the zebrafish [53], which is a frequently used model for understanding biological processes (Fig. 8). These authors used both the Lu-Chipman decomposition and the novel coherence matrix method described in Sect. 3.6 to characterize the sample. The study focused particularly on identifying different parts of the zebrafish using appropriate encoding polarimetric values. Ultimately, this could be used to track the changes of the biological organization without a priori knowledge of the sample.

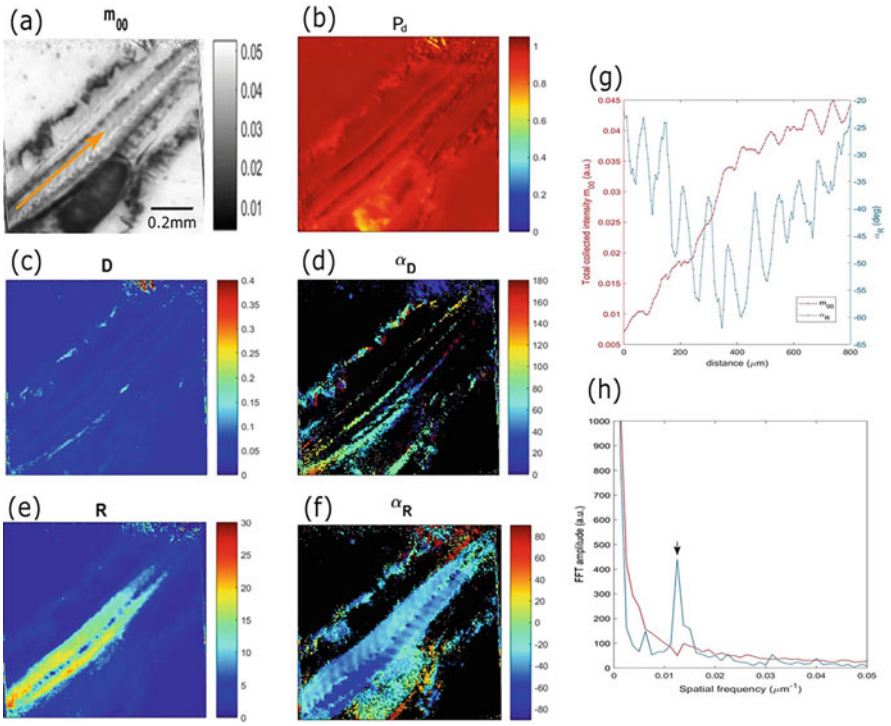


Fig. 8 (**a**) m00 image of fixed larvae tail at 72 hpf for a 10×/0.4NA objective zoom 1.1. (**b–f**) Polarimetric images from the Lu Chipman decomposition coded in (b) depolarization index P_d, (**c**) diattenuation D and (**d**) its orientation α_D, (**e**) retardance R, and (**f**) its orientation α_R. (**g**) Intensity profiles extracted along the orange arrows in figures (**a–e**), where the red and blue lines correspond to the m00 and the α_R intensity, respectively. (**h**) 1D fast Fourier transform (FFT) of the intensity profiles of (**g**). (Reproduced from Le Gratiet [53])

4.3 Other Clinical Applications

Figure 9 shows an image of porcine tendon from Okoro et al. [23], where the SHG signal (represented in green) was used to delineate the presence of collagen fibers to mask the polarization-sensitive confocal image. The outcome of the masking can be seen in gray for a stack of three images (layers ascend in z direction from the left to the right image). The author used a multi-modal imager integrating a Mueller matrix and SHG imager to increase sensitivity to collagenous tissue. The Lu-Chipman decomposition parameters of depolarization, retardance, and diattenuation as well as the degree of polarization were utilized to describe the results.

Mazumder et al. [28] imaged a collagen type I fiber to extract polarization properties using the Lu-Chipman Mueller matrix decomposition [77]. Figure 10 shows the polarimetric parameters of depolarization, retardance, and optical rotation obtained from the decomposition.

Recent attempts were made by Okoro et al. to integrate SHG measurement with a confocal microscope to paternalize polarimetric properties of human breast tissue microarray (TMA) [88]. Their studies showed that stromal collagen within the tumor has a lower depolarization and higher degree of linear polarization (Fig. 9 in Chap. 3), demonstrating the potential of confocal polarimetry approach in studying breast tissue in vivo.

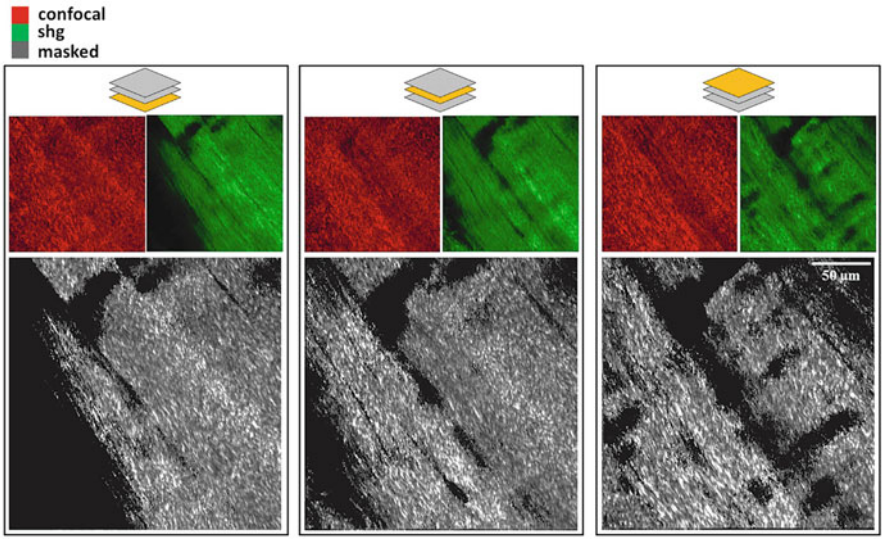


Fig. 9 SHG and polarization-sensitive image of porcine tendon. The image in red represents the confocal data, the image in green represents the SHG data, and the gray-scale image represents the masked outcome of using the SHG to delineate the collagen fibers. (Reproduced from Okoro et al. [23])

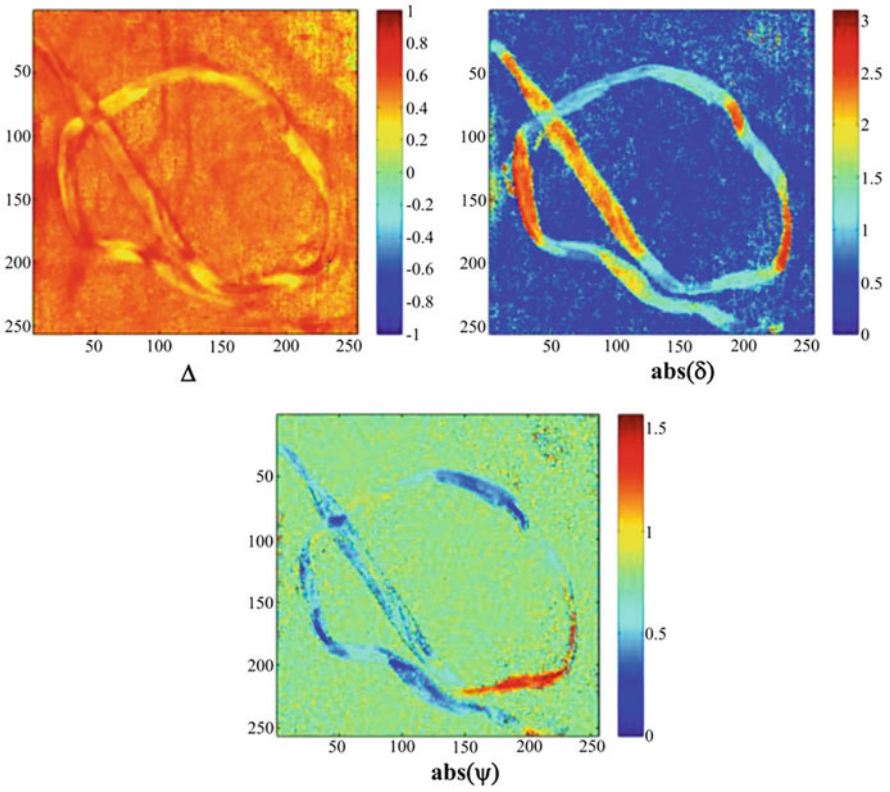


Fig. 10 Decomposition data for collagen type I showing the parameters of depolarization (top left), retardance (top right), and optical rotation (bottom). (Reproduced from Mazumder et al. [28])

4.4 Polarized Fluorescence

Entwistle and Noble [43] stained the sample with fluorescein isothiocyanate (FITC) in order to visualize the network of filaments found in a human glioblastoma multiforme cell sample. The images originating from their work can be seen in Fig. 11. The experimental setup is composed of a confocal fluorescence microscope employing cross-polarization by using a polarization-selective filter and a rotating QWP [42]. The figure shows a depth profile of the sample starting from the cover slip (A) in 1.6-micrometer intervals (vertically).

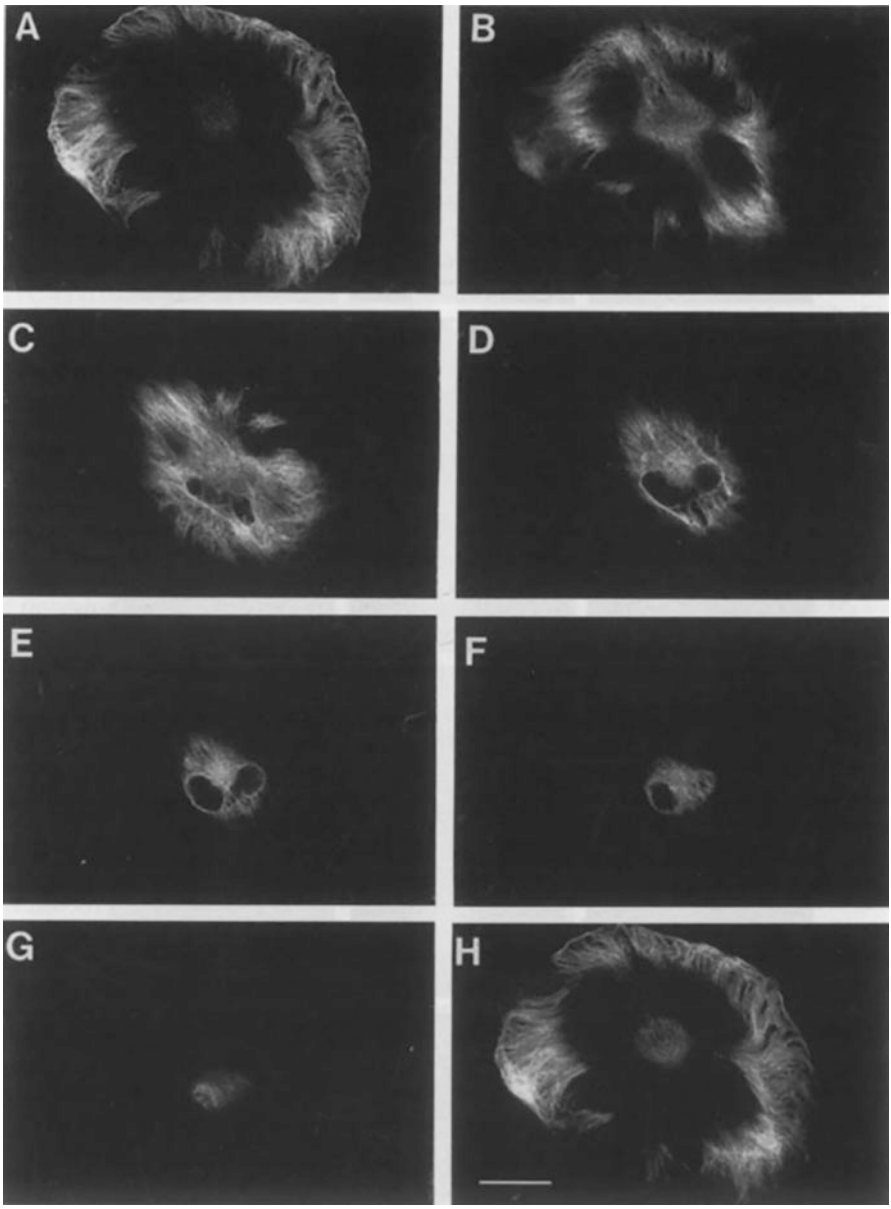


Fig. 11 Human glioblastoma multiforme cell sample stained with fluorescein isothiocyanate (FITC) in order to visualize the network of filaments. Images **a**–**h** show a depth profile every 1.6 micrometers, starting with the cover slip (**a**). The scale bar is 25 μm. (Reproduced from Entwistle and Noble [43])

5 Conclusions

Confocal laser scanning polarimetry was introduced two decades ago but is experiencing a new life with groups developing new Mueller Matrix systems and exploring a variety of biological environments. The improvements in image quality offered by this modality combined with its sensitivity to anisotropic and birefringent element are driving forces behind this new wave of research. Systems that combine non-linear microscopy and polarization-sensitive linear microscopy are also appearing leveraging the strength of each modality. Beyond image formation, the modality could offer quantitative opportunities when measuring the depolarization of an incident field at different depths, the in -situ retardation properties of the environment explored, or the local orientation of anisotropic structures.

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Improvement of Resolution and Polarisation Measurement Precision in Biomedical Imaging Through Adaptive Optics



Chao He and Martin J. Booth

1 Status and Challenges in Tissue Polarimetry

Polarimetry has many applications ranging from thin film measurement to remote sensing [1–15]. One of the most important applications of polarimetric imaging, though is biomedical sample characterisation, especially using polarisation microscopic/endoscopic systems [14–18]. For example, the polarimetric measurements can reveal in a label-free manner pathologies, such as cancer, that would not be otherwise detectable without staining. When all four elements of the Stokes vector of the light beam are calculated from the recorded intensity patterns, the method is known as imaging Stokes polarimetry. Likewise, imaging Mueller matrix polarimetry measures and characterises the polarisation properties of the sample. Both methods can be used for tissue polarimetry, which has recently played an important role in biomedical characterisation and clinical diagnosis. A recent development trend has moved from full Stokes-Mueller polarimetry to partial Stokes-Mueller polarimetry, in which only a selected portion of the Stokes vector or Mueller matrix elements from the sample are measured [19]. Such an approach is adequate for specific research topics [19]. For any such vector imaging system, there are two important criteria for evaluation of imaging quality – one is the image resolution, another one is the vectorial measurement precision. In this chapter, we focus on the use of advanced adaptive optics correction methods that enhance both the imaging resolution and vector precision, hence benefiting the characterisation accuracy for biomedical samples.

The resolution of a microscope depends on the quality of its point spread function (PSF). The PSF is affected by aberrations that cause blurring and contrast reduction

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[20, 21]. Aberrations affect all optical systems of different types and functionalities. Adaptive optics (AO) is a technology used to improve the performance of optical systems by reducing the effect of aberrations. AO uses active device such as a deformable mirror (DM) or a liquid crystal spatial light modulator (SLM), and in most cases it is used to correct phase aberrations that commonly affect a wide range of imaging applications [20–22].

The AO concept was first introduced for military applications and then for astronomical telescopes [23]. AO was required in these applications to overcome the problematic effects of atmospheric turbulence, which introduced time-varying aberrations into light passing through the atmosphere [23]. More recently, AO has also been deployed to microscopic imaging systems to enhance the resolution and image quality through compensation of specimen induced aberrations [20, 21]. For this reason, the majority of AO applications have focused on the phase domain.

There exists another type of aberration – polarisation aberration – which has been less well considered, but it can have similar effects through blurring and reducing image contrast [19, 24–34]. Moreover, they affect the measurement precision for polarisation states. These aberrations led to extra phase and polarisation distortion that can be introduced, for example, when focusing through stressed optical elements, due to Fresnel's effects at surfaces or induced via polarising effects in materials or tissues. These effects directly alter the state of polarisation (SOP) of the light field and the focal quality [19, 24–34], hence degrading vectorial information analysis and the spatial resolution of optical systems that compound the effects caused by phase aberrations. Considered together, we refer to these combined polarisation and phase effects as 'vectorial aberrations'.

Perturbed vector states in the illumination or detection beams are greatly detrimental for polarisation-sensitive microscopes, including Stokes/Mueller confocal microscopes, second/third harmonic generation microscopes, and super-resolution fluorescence polarisation microscopy [19, 35, 36]. Such effects are vital, for example, in label-free cancer detection using Stokes/Mueller microscopes, for which the correctness of the vectorial state is essential [19]. Furthermore, incorrect polarisation disrupts the interference at the focus and hence affects the efficiency and resolution of super-resolution microscopes, for example, in the creation of the zero-intensity centre of the ring-shaped STED microscope or MINFLUX microscope beams [37–39]; in the interference light fields of the structured illumination microscope or in 4pi microscopes [40, 41]. By their nature, these systems require adaptive compensation of phase and polarisation errors to compensate for these effects that vary throughout a specimen [20, 21]. We refer to these methods as 'vectorial adaptive optics' or V-AO.

Similar to phase AO, V-AO requires methods for compensation of the vectorial aberrations. This requires the use of polarisation-sensitive devices. Liquid crystal SLMs are good candidates to be used as vectorial correctors, as they are controllable birefringent devices [32, 33]. As several parameters are required to control the vectorial state, the SLMs must be used in combination. Indeed, up to four devices are required for full vectorial compensation, as explained later in this chapter. These

multiple SLMs provide the toolbox for vectorial adaptive compensation in various imaging scenarios in bio-polarimetry-based sensing or imaging.

2 Vectorial Aberrations and the Interaction of Light with Specimens

2.1 Phase Aberrations, Polarisation Aberrations, and Adaptive Correction

Phase aberrations are usually described via a scalar phase function that varies across the beam profile [42]. The aberrations can be introduced via refractive index mismatch in specimens, misalignment of the beams through the optics, or other sources of optical path length difference. Both the state of the light beam and the aberrating action of the object are represented in a similar format. We consider the input phase field as $\Phi_{\text{in}}(r, \theta)$ and wavefront aberration induced by an object as $\Phi_{ab}(r, \theta)$ in which r and θ is the radius and azimuthal angle of the polar coordinate system of the pupil of the microscopic system. The AO correction phase is defined as $\Phi_c(r, \theta)$. The output phase profile $\Phi_{\text{out}}(r, \theta)$, can then be described via

$$\Phi_{\text{out}}(r, \theta) = \Phi_{ab}(r, \theta) + \Phi_c(r, \theta) + \Phi_{\text{in}}(r, \theta) \tag{1}$$

It can be shown that the Zernike polynomials $Z_i(r, \theta)$ form a complete basis for representation of the aberration, if the system has circular pupils [20, 21]. The output aberration function can be expressed as

$$\Phi_{\text{out}}(r, \theta) = \sum_{i=1}^{N} a_i Z_i(r, \theta) \tag{2}$$

where the aberration is defined over the pupil with normalised radius (for $r \leq 1$). Each of the aberration functions in Eq. 1 could similarly be represented by the coefficients of its respective Zernike series.

To illustrate the parallels between phase AO and polarisation AO, we can consider the operation of a simple focusing system with aberrations present at its input pupil. The quality of a focus is often represented by the Strehl ratio, S, which is the ratio of on-axis intensity at the focal spot when aberrated, to the equivalent intensity when the system is aberration free. It is readily shown that for small phase aberrations

$$S = \frac{M}{M_0} = \left(1 - \sum_{i=2}^{N} a_i^2\right) \tag{3}$$

where M is the aberrated intensity, M_0 is the unaberrated intensity, a_i are the coefficients of the Zernike modes, and these coefficients are considered to be small. The first Zernike mode, piston (index $i = 1$), is omitted from the summation as it represents a constant phase offset and has no effect on the measured intensity signal, so does not influence the analysis of this system. This means that there may be an arbitrary constant phase shift between the input and output phase, but this will have no effect on the system performance. In other words, the goal of the AO system is to find a correction Φ_c that makes $\Phi_{\text{out}} - \Phi_{\text{in}} = \text{constant}$.

This analysis is correct but incomplete, wherein it assumes that the correction goal is that $\Phi_{\text{out}}(r, \theta) - \Phi_{\text{in}}(r, \theta) = \chi$, where χ represents an arbitrary constant, whereas in fact perfect correction would be achieved if $\Phi_{\text{out}}(r, \theta) - \Phi_{\text{in}}(r, \theta) = f(r, \theta)$, where the function $f(r, \theta) \bmod 2\pi = \chi$. In other words, aberrations can contain steps of an integer multiple of a wavelength and yet still provide perfect focusing.

These properties are made more obvious if we express the aberrations and system operation in terms of complex amplitudes, rather than phase, so that the Eq. 1 would be written as

$$e^{j\Phi_{\text{out}}(r,\theta)} = e^{j\Phi_{ab}(r,\theta)} e^{j\Phi_c(r,\theta)} e^{j\Phi_{\text{in}}(r,\theta)} \tag{4}$$

where $j = \sqrt{(-1)}$. In this formulation, each of the complex functions can be thought of as a complex multiplicative operator that operates on the function on its right. This formulation is useful, as we can use it to show parallels between the operation of phase AO and polarisation AO systems.

Polarisation aberrations have been defined by Chipman [24–29] via the Jones matrix (JM) or the Mueller matrix (MM). The polarisation aberration can be introduced in different ways, as mentioned in previous section, and it is also linked with different physical properties, such as retardance, diattenuation, and so on [24–29].

Either a JM or a MM can be used to describe the vectorial optical properties of an object. While the JM contains the information of absolute phase, it cannot represent depolarising effects of objects [19]. The MM however can represent depolarisation, but not the overall phase. Hence, the MM is used if comprehensive polarisation properties need to be considered. It is worth noting that, the SOP difference – which represents the polarisation distortion of the light beam – can also at some level represent the polarisation aberration. However, as the Stokes vector is a 4-element vector, its intrinsic dimension is lower than a multi-element matrix (with more than four free parameters); hence, a full polarisation aberration is more properly described via a JM or a MM.

In this chapter, we focus for simplicity on retardance-like polarisation aberrations; hence, we use the JM formulism throughout the following sections. If we assume that the object introducing the polarisation aberration is a single layer of retarder, then the JM of the sample (J_s) could be written as

$$J_s = \begin{bmatrix} \cos^2\theta + e^{i\phi}\sin^2\theta & \left(1 - e^{i\phi}\right) e^{-i\eta}\cos\theta\sin\theta \\ \left(1 - e^{i\phi}\right) e^{i\eta}\cos\theta\sin\theta & \sin^2\theta + e^{i\phi}\cos^2\theta \end{bmatrix} \tag{5}$$

where θ is the fast axis orientation with respect to the x-axis, ϕ is the retardance value, and η is the circularity [2, 43–46]. We first consider the linear retarder case for which $\eta = 0$.

As the JM representation includes the absolute phase information, we could also extract the phase aberration associated with the JM, by taking the determinant of the matrix (according to Refs. [33, 47]). We refer to the combined effect of an overall polarisation aberration J_{ab} with the conventional phase aberration as 'vectorial aberrations'.

When we consider a complex system that imparts various polarisation aberrations, each described through JMs: $J_1, J_2, J_3, \ldots J_n$, in a particular sequence (here we assume the light beam passes in order through J_1 to J_n), the overall polarisation aberration J_{ab} can be represented via the equation:

$$J_{ab} = J_n \cdot J_{n-1} \cdots J_3 \cdot J_2 \cdot J_1 \tag{6}$$

We consider the input Jones vector field as $P_{\text{in}}(r, \theta)$ and polarisation aberration induced by an object as $J_{ab}(r, \theta)$ in which r and θ are the radius and azimuthal angle of the polar coordinate system of the pupil of the microscopic system as we used before. The AO correction polarisation and phase is defined as $J_c(r, \theta)$. The output vectorial profile $P_{\text{out}}(r, \theta)$, can then be described via

$$P_{\text{out}}(r, \theta) = J_n(r, \theta) \cdot J_{n-1}(r, \theta) \cdots J_3(r, \theta) \cdot J_2(r, \theta) \cdot J_1(r, \theta) \cdot P_{\text{in}}(r, \theta) \tag{7}$$

As the multiplication of JMs is in general non-commutable, it is necessary to consider whether a system performs pre- or post-correction of the polarisation aberration. Here we describe a pre-correction AO system (Eq. 8) and a post-correction counterpart (Eq. 9), for which we want the $P_{\text{out}}(r, \theta)$ to feature a flat wavefront and uniform polarisation state after AO system. The concept of the before- and after-correction AO systems can be illustrated via

$$P_{\text{out}}(r, \theta) = J_{ab}(r, \theta) \cdot J_c(r, \theta) \cdot P_{\text{in}}(r, \theta) \tag{8}$$

$$P_{\text{out}}(r, \theta) = J_c(r, \theta) \cdot J_{ab}(r, \theta) \cdot P_{\text{in}}(r, \theta) \tag{9}$$

where we take the primary matrix reciprocity problems [19] into consideration. As the matrix multiplication is in general non-commutable, the $J_c(r, \theta)$ matrices in Eqs. 8 and 9 are in general different. An example of a pre-correction vectorial aberration correction AO system is illustrated in Fig. 1.

A parallel can be drawn between these JM operators in the above equations and the representation used in Eq. 4. A lossless JM can be expressed in 'generator' form as [26, 48, 49]

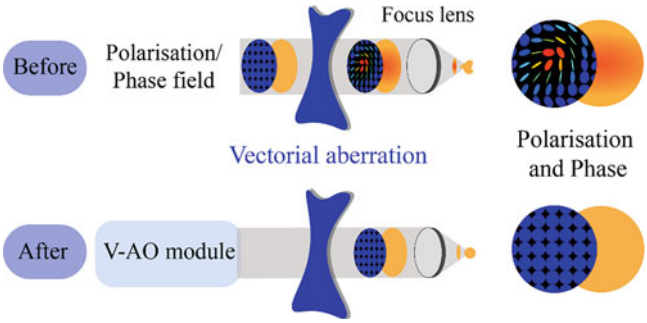


Fig. 1 Vectorial aberration correction system using AO. 'Before' and 'after' illustrate the potential state of polarisation and phase correction functions. The blue and red circles in the polarisation plot represent the orientation and ellipticity/circularity of the SOP as it varies across the beam; left-hand circular SOPs are red, right-hand circular SOPs are blue, and green lines represent linear SOPs. Other shades represent different elliptical SOPs. The yellow to red colours in the phase plot represent the phase value

$$J = e^{j(b_0\sigma_0 + b_1\sigma_1 + b_2\sigma_2 + b_3\sigma_3)} \tag{10}$$

where σ_0 is the identity matrix and where σ_1, σ_2, and σ_3 represent the three Pauli spin matrices. Concisely, the first term $b_0\sigma_0$ describes polarisation-independent effects on phase, which effectively means the aberration corrected by phase AO. The σ_1 term represents retardation effects aligned with the x- and y-axes; σ_2 retardation effects at 45° to the axes; and σ_3 circular retardation effects. Although expressed here in the form of a JM, it is clear that the first term in the exponent is equivalent to the exponential phase operator of Eq. 4. Hence, the JM representation incorporates not only the polarisation states but also the effects of the phase aberrations.

2.2 *Combined Effects of Polarisation and Phase Modulation*

In the context of vectorial light, normally two types of geometric phase will be involved – Rytov-Vladimirskii-Berry (RVB) phase and Pancharatnam-Berry (PB) phase [50–55]; the former is related to variation in the propagation direction of the light beam (such as when propagating in inhomogeneous media), while the latter is associated with polarisation manipulation of the light beam (such as when propagating in crystals that exhibit spatial variation of retardance across the beam profile) [50–55]. What is more, these two phases are also directly related to the process of spin-orbital interaction of light – an interaction of spin angular momentum, the intrinsic and extrinsic orbital angular momentum of light.

In this chapter, we are mainly concerned with PB-phase and retarder like systems (such as encountered in propagation through birefringent media). This extra phase shift is due to changes in SOP via the effects of the polarisation aberrations. We

hence consider geometric phase as a part of the polarisation aberration, rather than a separate phenomenon. Based on this observation and the above discussion about JM, we can see that phase and polarisation aberrations are intrinsically intertwined.

3 From Phase to Vectorial Adaptive Optics for Polarisation Imaging

3.1 Concepts for Phase, Polarisation, and Vectorial Adaptive Optics

This section provides a brief summary of how AO techniques have developed from phase AO to vectorial AO for polarisation-resolved imaging. Some preliminary experimental results have shown how phase AO can be used to enhance the resolution of polarisation images [56]. However, the operating mechanisms and possible applications have not yet been fully investigated. Full polarisation and phase control have been previously implemented using SLMs in various configurations for generation purposes such as vector vortex beam manipulation, beam shaping, and beam multiplexing [9, 57, 58]. However, development of the ability to control vectorial states fully to compensate vectorial beam errors has a relatively short history. Polarisation and phase control for compensation of intrinsic errors in beam generation systems have been implemented [4, 9, 10], as has dynamic control of the full vectorial field to enable focusing through highly scattering media [59–61]. However, these works have focused mostly on complex vectorial beam generation or partial compensation of polarisation errors without full adaptive feedback correction. A systematic method for full adaptive compensation of vectorial aberrations has not yet been implemented.

Systematic adaptive polarisation control to compensate externally induced polarisation disturbances in pupil domain was proposed in 2019 [34], where it was achieved with an SLM-based correction system. The system could convert fixed SOP into arbitrary SOP [34], hence featuring the ability to perform feedback AO compensation with pre-correction of the SOP state across the pupil. That work relied upon the three fundamental components – measurement, correction, and feedback control – that are required for a full polarisation AO system. Full Stokes vector polarimetry was used to measure the polarisation state across the beam profile. A dual-pass SLM system was used to control the input polarisation state. A strategy for feedback control was developed to enable SOP compensation from a sequence of polarimeter measurements. The methods were demonstrated on a range of specimens that perturbed the SOP of an input beam. The system restored the desired output state, even in the presence of strong polarisation errors. Although the first sensor-based adaptive polarisation control was presented in this work, there were still certain limitations: (1) The polarisation aberration (assumed to be in the form of a spatially varying retardance) was reconstructed via prior knowledge of the

aberration and the vectorial field measured by Stokes polarimetry. (2) Only pupil domain correction was implemented rather than focal domain. (3) The method was only suitable for pre-compensation of the SOP and is not readily applicable to more complex optical systems. (4) There was no phase correction.

In 2020, more general arbitrary vectorial state conversion approaches using SLMs for polarisation adaptive optics were theoretically proposed [32]. In this work, the authors validated through JM analysis that conversion from arbitrary SOP to fixed SOP using two passes SLM is also possible, considering that in a pure retardation system the JM is fully invertible. This point is important, as it also provides the possibility of using a single SLM (with two passes through side-by-side pupils) to act as polarisation AO device for either pre- or post-correction of polarisation aberrations. Furthermore, it was shown that through using an additional SLM pass, giving three passes in total, it was possible to convert between a fixed vectorial state (polarisation and phase) to an arbitrary state or vice versa. This work also dealt with the dynamic and geometric phases arising from the polarisation effects of the SLMs. Here we briefly summarise the mathematical representation of the phase effects for a three SLM configuration; for more details one can refer to Ref. [32].

The action of each of the SLMs can be described by its JM J_n, where n is the index of the SLM, which we will refer to by the subscripts A, B, and C. The output field P_{out} is then determined by the input field vector P_{in} via equations (see Fig. 2 for the schematic system):

$$P_{\text{out}} = J_C \; J_B \; J_A P_{\text{in}} \tag{11}$$

where the phase is defined according to the Pancharatnam connection between the SOP before and after the SLMs. $\overrightarrow{\mathbf{Q}}$ is the first eigenpolarisation state of the SLM. $\overrightarrow{\mathbf{A}}$ is the SOP of the input polarisation state. The parameters μ_1 and μ_2 are the corresponding eigenvalues of these two states. The vector $\overrightarrow{\mathbf{Q}}$ corresponds to the modulation axis of the SLM [47].

$$\phi = \arg\left[\mu_1 + \mu_2 + (\mu_1 - \mu_2)\,\overrightarrow{\mathbf{Q}} \cdot \overrightarrow{\mathbf{A}}\right] \tag{12}$$

$$\mu_1 = \exp\left(i\psi_1\right) \tag{13}$$

$$\mu_2 = \exp\left(i\psi_2\right) \tag{14}$$

where the overall phase is represented via ϕ. The ψ_1 is the induced varying phase, which depends upon the SLM pixel voltage, while ψ_2 is the fixed phase induced along the non-modulating axis of the SLM pixel. $\Delta\psi = \psi_1 - \psi_2$ is the retardance of the SLM at certain voltage configuration. If we choose the orientations of SLM axes to be 0°, 45°, and 0°, the corresponding eigenstates can be given by

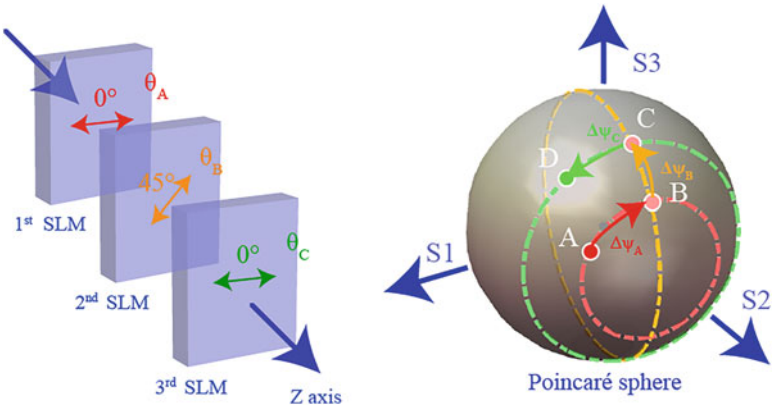


Fig. 2 Schematic of the three SLM system (with modulation axis at a chosen angle θ_A, θ_B, θ_c). SOP changes on the Poincaré sphere for the three-SLM system, as the polarisation state evolves through the first SLM (A to B), the second SLM (B to C), and the third SLM (C to D)

$$\vec{\mathbf{Q}}_A = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \vec{\mathbf{Q}}_B = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \vec{\mathbf{Q}}_C = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \tag{15}$$

If we have a general SOP input $\vec{\mathbf{A}} = (\, S_1, S_2, S_3 \,)^{\mathrm{T}}$, the scalar products are given by $\vec{\mathbf{Q}}_A \cdot \vec{\mathbf{A}} = S_1$, $\vec{\mathbf{Q}}_B \cdot \vec{\mathbf{A}} = S_2$, and $\vec{\mathbf{Q}}_C \cdot \vec{\mathbf{A}} = S_1$. We can therefore calculate the phases ϕ induced by each SLM pass via

$$\begin{aligned} \phi_A &= \arg\left[e^{i\psi A_1} + e^{i\psi A_2} + \left(e^{i\psi A_1} - e^{i\psi A_2}\right) A_1\right] \\ \phi_B &= \arg\left[e^{i\psi B_1} + e^{i\psi B_2} + \left(e^{i\psi B_1} - e^{i\psi B_2}\right) B_1\right] \\ \phi_C &= \arg\left[e^{i\psi C_1} + e^{i\psi C_2} + \left(e^{i\psi C_1} - e^{i\psi C_2}\right) C_1\right] \end{aligned} \tag{16}$$

The intrinsic interrelationships are elaborated in Ref. [32]. Following the analysis in Ref. [32], the total phase introduced by the three SLM passes equals to the sum of phase introduced individually by each pass ($\phi_A + \phi_B + \phi_C$) and the geometric phase, which is equivalent to half the area of the quadrilateral enclosed by the three passes on the Poincaré sphere (ϕ'), such that

$$\phi_{\text{Total}} = \phi_A + \phi_b + \phi_c + \phi' \tag{17}$$

where ϕ' is equal to half of the area enclosed by the path DCBA, as shown in Fig. 2. In summary, the conclusion of this work showed that an extra degree of freedom is still needed to control the phase to realise full vectorial control of the beam.

From the above validations, we know that if one is not concerned about the phase, but only the correctness of the SOP, then with a three-pass SLM configuration,

conversion between two spatially variant SOPs can be realised. While three-pass SLM configurations had been proposed and experimentally validated before, they were used for fixed SOP to arbitrary SOP and full phase control [62]. At the end of the work in Ref. [32], the authors also suggested a structure with four SLMs (or three SLMs plus an alternative phase correction device, like a deformable mirror) to fully control the polarisation and phase field from any arbitrary state to another arbitrary state. This would provide a versatile vectorial field manipulation configuration with wider applications. In principle, an appropriate combination of three hypothetical retarder modulators could perform similar functionalities with less redundancy than the four-element system: for example, one linear retarder oriented at 0°, one linear retarder at 45°, and one circular retarder (or an equivalent combination).

As we have mentioned before, while there are several different SLM configurations that can manipulate the vectorial states of light, such systems are still not necessarily enough to represent the full effects of a polarisation aberration of the type that could be represented by an arbitrary unitary JM. In 2021, it was proposed to use a sequence of SLMs in a way that could reproduce the behaviour of a spatially variant waveplate with any polarisation eigenmode and arbitrary retardance distribution [33]. Hence, it provided a controllable physical system that could act as arbitrary polarisation aberration, and hence perform as the feedback correction device in a full vectorial adaptive optics system [33]. Later, a similar multi-SLM system that was equivalent to a linear phase retarder with arbitrary phase retardation was experimentally presented [63].

3.2 Vectorial Adaptive Optics Through Sensor-Based and Sensorless Methods

Traditional phase AO requires a method of phase measurement, through a wavefront sensor or indirect optimisation methods (often termed ‘wavefront sensorless AO’ or in short ‘sensorless AO’), to determine the input aberration and a method of phase compensation; whereas V-AO requires the sensing and correction of the vectorial aberration. There are significant challenges in extending existing AO methods to this higher-dimensional analogue to conventional phase correction, as now we are dealing with high-dimensional vector parameters rather than single scalar parameter of phase. In general, the V-AO correction can also be implemented through different methods: sensor-based and modal-sensorless; conceptual sketches of the measurement/correction process are shown in Fig. 3.

Building upon previous knowledge from mathematical modelling and simulation, in 2021, the first systematic V-AO compensation technique was presented, encompassing sensor-based and sensorless approaches [64]. The work detailed sensor-based V-AO with full aberration sensing ability and a feedback correction strategy, both in the pupil and focus domains. A sensorless adaptive optics method was also implemented with a sophisticated compensation strategy. These methods

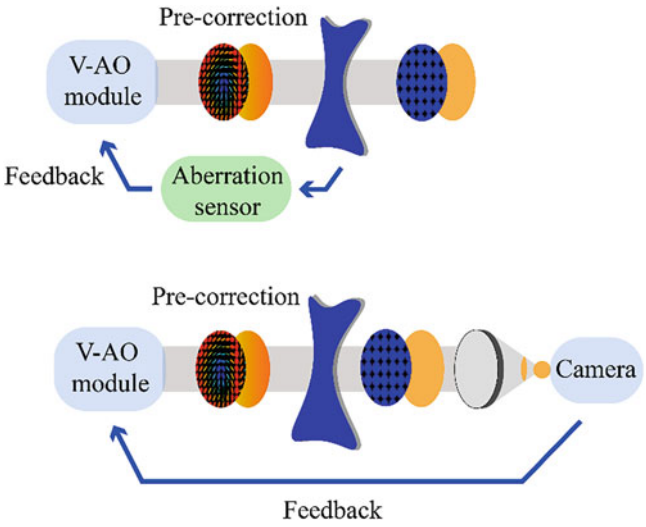


Fig. 3 Illustrations of pre-correction V-AO mechanisms in the format of sensor-based (top) and sensorless (bottom) approaches

were all validated via compensation of induced vectorial aberrations from various sources.

In contrast to conventional AO, V-AO requires a method to determine the full vectorial properties, encompassing polarisation and phase, and a mechanism for control of these properties. Conventional MM polarimetry plays a role here, as the 'sensor' for determination of the MM as well as the output SOP [19, 30]. Note here the MM and JM provide equivalent SOP conversation information as we are dealing with non-depolarising system, and for the measurement convenience, we usually measure the MM instead of the JM as the MM can be determined solely from intensity-based measurements. Sensorless V-AO methods can also be employed, for example allowing model-based optimisation of focal quality through adjustment of polarisation and phase states. Sensorless AO is useful, as it does not require any additional sensor and can permit efficient optimisation of vectorial correction through sequence of images. Such methods are widely used in phase AO, where modes such as Zernike polynomials are applied and optimised. The principle can be extended to vectorial AO, as both polarisation and phase aberrations affect image quality. An easy-to-explain example is when there is prior knowledge of, for example, the retardance axis, but unknown retardance magnitude. Appropriate vectorial aberration modes can be applied to enable image optimisation. Correction of more complex vectorial aberrations with more unknown parameters could be implemented with more complex aberration modes.

More description of the methods and their properties, with demonstrations of the improvement of both vector field and focus after correction of commonplace vectorial aberrations, can be found in Ref. [64].

4 Prospects for Vectorial Adaptive Optics

4.1 Vectorial Image Enhancement: Label-Based and Label-Free

Vectorial adaptive optics has not yet been applied into areas of vectorial imaging, which includes fluorescence-based approaches as well as the label-free polarisation imaging methods.

Advanced fluorescence polarisation microscopy (FPM) [65–68], which harnesses vectorial information about dipole emitters, has recently been attracting scientific attention. Vectorial information is encoded in the SOP of the emitted light, which is normally linearly polarised. The polarimetric detection of dipole orientation and subsequent analysis plays an important role in biomedical sample analysis. For instance, the dipole behaviours that are revealed via FPM techniques can be used to study cytoskeleton components such as actin, myosin, and microtubules, as well as septin. They have also been used to analyse the nuclear pore complex subcomplexes and their relative orientations [69–73]. Recent advances also include fluorescent dipoles super-resolution imaging via polarised illumination [74, 75]. As any of these high-resolution microscopes can suffer from polarisation aberrations, these areas have potential to be benefit from V-AO techniques.

For label-free biomedical polarimetry, cancer detection is an important application [14, 15, 17–19, 30]. Recently, such polarimetric techniques have supported the diagnosis of various cancerous tissues, such as human skin cancer, colon cancer, liver cancer, and breast cancer [19]. Polarimetric data have been used to analyse the fibrosis process among different stages of cancer development, which can be implemented via evaluation of polarimetric parameters such as retardance value [19]. Similarly, the fast axis orientation of the retardance is also used to describe the distribution of features in the fibrous regions, which has also assisted pathological diagnosis [19]. In these applications, any polarisation aberration within the measurement process, such as intrinsic polarisation aberrations from the high numerical aperture lenses [76], would also significantly affect the resolution, as well as the correctness of those quantitative vectorial metrics that are used for clinical diagnoses.

4.2 Compensation of Other Vectorial Aberrations: Diattenuator and Depolariser

There are many other possible polarisation perturbations besides the pure retardance aberrations that have mostly been discussed in this chapter. These other types of polarisation perturbations include diattenuation and depolarisation. In this section, we also give a brief explanation of strategies for their estimation and compensation.

For diattenuation-based polarisation aberrations, which arise due to intrinsic anisotropic absorption properties, some intensity, and hence information, must be lost from the original light beam [19]. It is not possible to retrieve this lost power through existing AO methods. Post-acquisition image processing can be taken into consideration as a partial solution. However, diattenuation can cause a change in the SOP, which is possible to compensate using the V-AO methods proposed before. For small diattenuation aberrations, SOP errors can generally be fully compensated using V-AO. On the other hand, if there exists a large diattenuation aberration, the consequences could be more serious. If the extinction ratio of the diattenuation is very high (in the extreme case, infinite), then any incident SOP (possibly pre-compensated with V-AO) will be converted into a fixed SOP (with varying intensity) after the diattenuation aberration. Hence, any V-AO pre-compensation would have limited use. Similar problems would occur in systems where post-compensation is required.

Depolarisation is a very important vectorial parameter that can affect various techniques and their corresponding applications [19]. For example, in biomedical applications, depolarisation can be used as an indicator to monitor the condition of muscle in vivo or for differentiating normal/cancerous tissue ex vivo [77]; it is also an important parameter to assess the performance in the waveguide chips for quantum computers [78].

There exist two possible parts of the distortion introduced by depolarisation – SOP distortion and degrees of polarisation (DOP) distortion. The former can be possibly corrected via the V-AO tools that were explained earlier in this chapter, while the DOP errors might be corrected via new devices featuring polarisance properties (the ability to enhance the DOP). In conclusion, these cutting-edge V-AO techniques would not only provide possibilities for higher image resolution but also improved vectorial information measurement precision in biomedical imaging and clinical applications.

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